

For M/V XIANG RUI KOU only

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COSCO
50000 DWT SSHLV

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1 General particulars

1.1 Main data

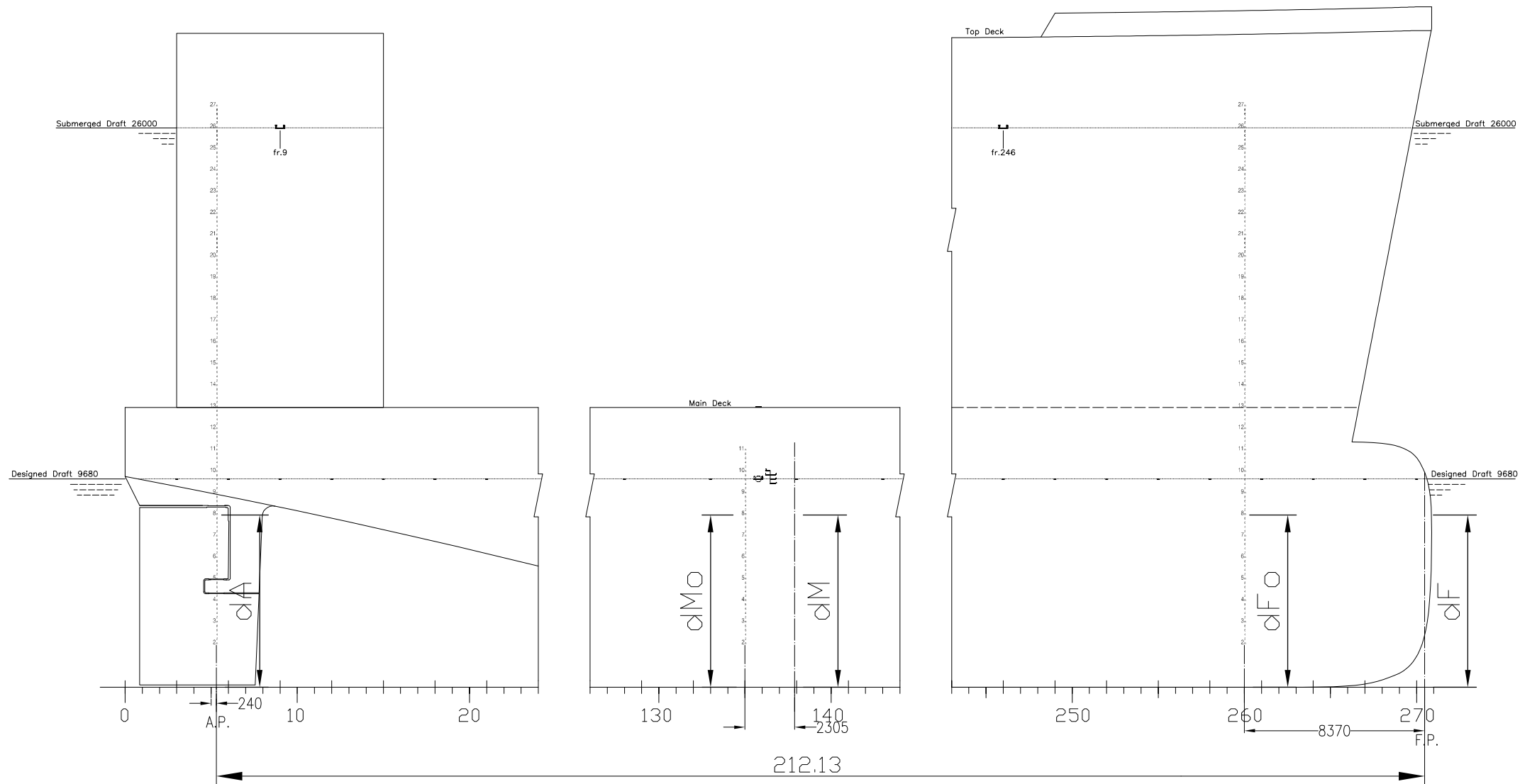
Ship's name:	XIANG RUI KOU
Ship Type:	Semi-Submersible Heavy Lift Vessel
Call Sign:	BOEZ
Official Number:	
MMSI Number:	414017000
IMO Number:	9483102
Year Built:	2011
Port of Registry:	GUANG ZHOU
Owner	
Name:	COSCO Shipping Co. Ltd.
Address:	412, Huan Shi Road East, Guangzhou, China
Tel:	
Fax:	
Builder	
Name:	Guangzhou Shipyard International Co. Ltd.
Address:	40 South Fang Cun Road, Guangzhou, China
Tel:	
Fax:	
Yard Number:	07130002
Classification Particulars:	DNV, 1A1, Ice-C, DK(+), E0, RPS, BIS CSA, Semi-submersible Vessel, ICE CLASS B CSM, AUT-0
Main Dimensions:	
Length, O. A.	216.70m
Length, P. P.	212.13m
Breadth, moulded	43.00m
Depth, moulded	13.00m
Summer load draught, mld:	9.68m

Max. submerged draught, mld:	26.00m
Summer load displacement:	69164.7t
Summer load deadweight:	48293.3t
Lightship weight:	20871.4t
Gross Tonnage:	35542
Net Tonnage:	10662

1.2 Draught marks

Location of draught marks are shown below.

PLAN SHOWING THE POSITIONS OF DRAFT MARKS

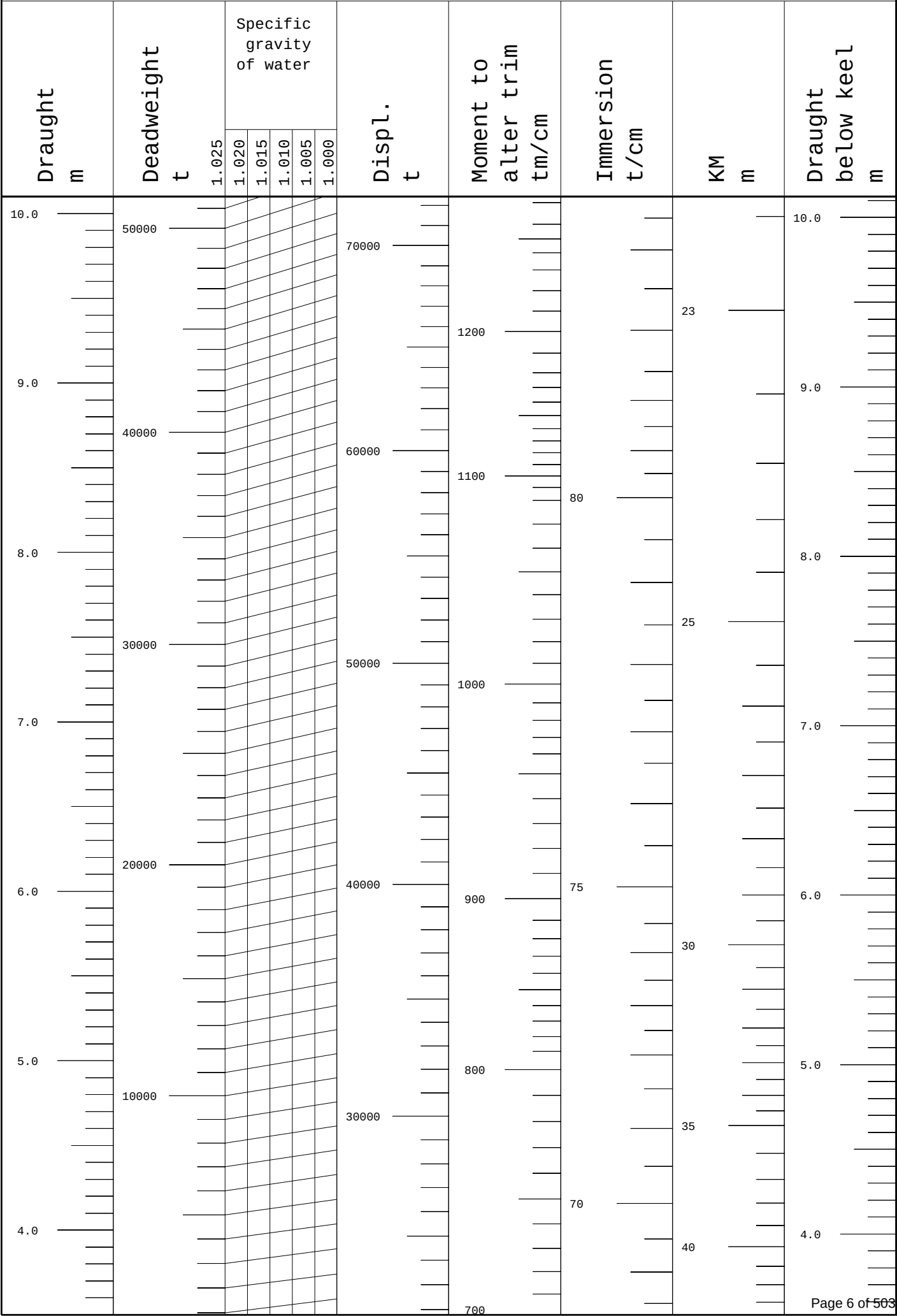


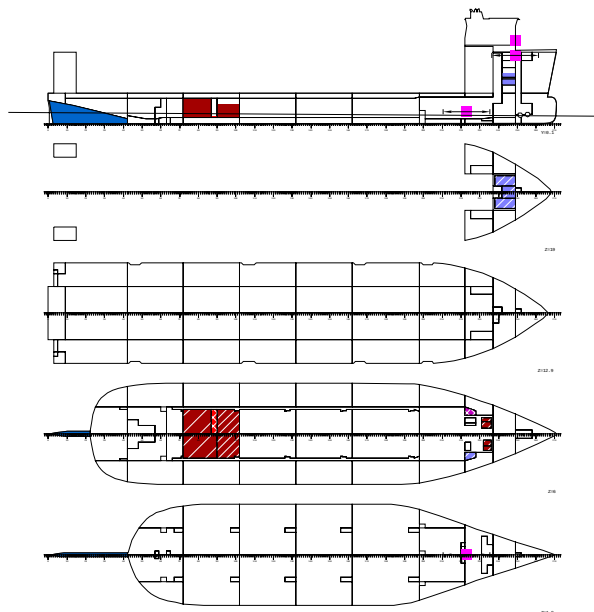
1.3 Deadweight data

Following deadweight information is included in this chapter:

- deadweight scale
- consumables in departure condition (100% cons.)
- consumables in arrival condition (10% cons.)

LOADING SCALE





LOADING CONDITION DEPARTURE: Departure consumables

FLOATING POSITION

Mean draught (moulded)	3.98 m	KM above the moulded base	39.29 m
Draught at AP (moulded)	4.60 m	KG above the moulded base	9.86 m
Draught at FP (moulded)	3.35 m	GM0 (solid)	29.43 m
Trim (by stern)	-1.25 m	Free surface correction	-0.09 m
Heeling to starboard side	-0.1 deg	GM (fluid)	29.34 m
Mean draught (heel -0.1)	3.98 m		
Trim (heel -0.1)	-1.25 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	4421.1	82.89	-0.25	7.27	2319.4
Lightweight	20871.4	113.44	-0.01	10.41	
Displacement (1.025 t/m3)	25292.5	108.10	-0.05	9.86	2319.4

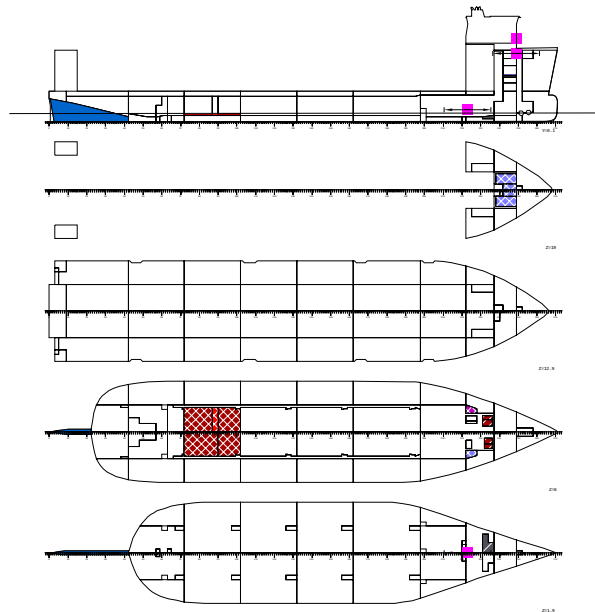
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DEPARTURE

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LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Heavy Fuel Oil density=0.991 t/m3							
HF01	HFO Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HFO Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HFO Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HFO Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Stores							
(ST0)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							
Deadweight		4421.1		82.89	-0.25	7.27	2319.4
Lightweight		20871.4		113.44	-0.01	10.41	
Displacement (1.025 t/m3)		25292.5		108.10	-0.05	9.86	2319.4
—							



LOADING CONDITION ARRIVAL: Arrival consumables

FLOATING POSITION

Mean draught (moulded)	3.53 m	KM above the moulded base	43.07 m
Draught at AP (moulded)	3.49 m	KG above the moulded base	10.08 m
Draught at FP (moulded)	3.58 m	GM0 (solid)	33.00 m
Trim (by head)	0.09 m	Free surface correction	-0.26 m
Heeling	0.0 deg		
		GM (fluid)	32.74 m
Mean draught (heel 0)	3.53 m		
Trim (heel 0)	0.09 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	1264.2	96.42	0.35	4.56	5741.8
Lightweight	20871.4	113.44	-0.01	10.41	
Displacement (1.025 t/m3)	22135.6	112.47	0.01	10.08	5741.8

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Loading Condition
ARRIVAL

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LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HFO Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HFO Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HFO Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HFO Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores (ST0)							
		10.0		199.20	0.00	29.05	0.0
Crew (CREW)							
		6.0		199.20	0.00	35.70	0.0

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Loading Condition
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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
	Miscellaneous density=1.000 t/m3 (MIS)	87.0		178.40	0.00	5.00	0.0
—							—
	Deadweight	1264.2		96.42	0.35	4.56	5741.8
	Lightweight	20871.4		113.44	-0.01	10.41	
	Displacement (1.025 t/m3)	22135.6		112.47	0.01	10.08	5741.8
—							—

1.4 Frame distance table

FRAME DISTANCE TABLE

DISTANCE FROM FRAME STATION TO AP

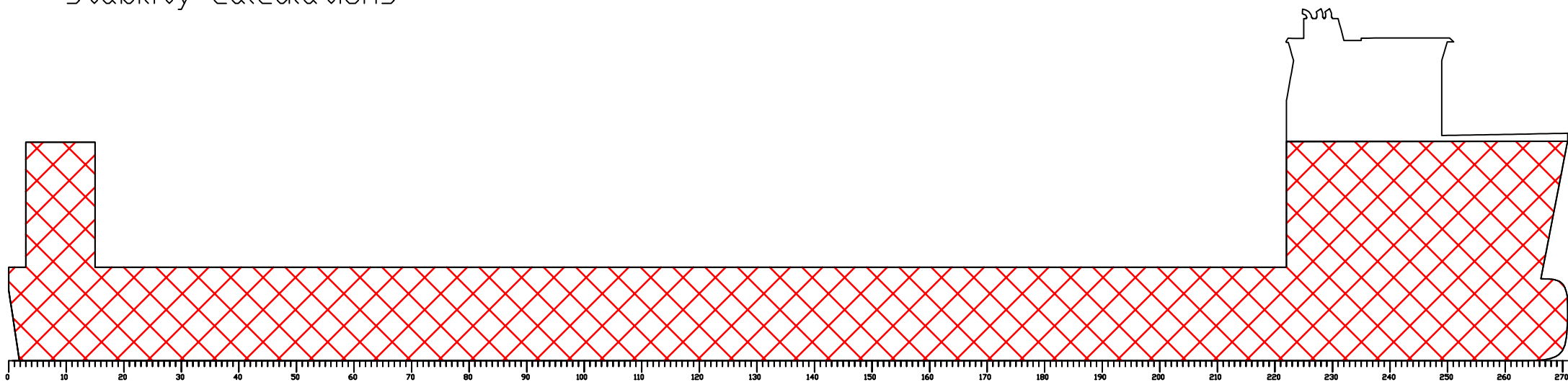
Frame No.	From AP (m)	Frame No.	From AP (m)	Frame No.	From AP (m)	Frame No.	From AP (m)	Frame No.	From AP (m)
0	-4.240	56	40.560	112	85.360	168	130.160	224	174.960
1	-3.440	57	41.360	113	86.160	169	130.960	225	175.760
2	-2.640	58	42.160	114	86.960	170	131.760	226	176.560
3	-1.840	59	42.960	115	87.760	171	132.560	227	177.360
4	-1.040	60	43.760	116	88.560	172	133.360	228	178.160
5	-0.240	61	44.560	117	89.360	173	134.160	229	178.960
6	0.560	62	45.360	118	90.160	174	134.960	230	179.760
7	1.360	63	46.160	119	90.960	175	135.760	231	180.560
8	2.160	64	46.960	120	91.760	176	136.560	232	181.360
9	2.960	65	47.760	121	92.560	177	137.360	233	182.160
10	3.760	66	48.560	122	93.360	178	138.160	234	182.960
11	4.560	67	49.360	123	94.160	179	138.960	235	183.760
12	5.360	68	50.160	124	94.960	180	139.760	236	184.560
13	6.160	69	50.960	125	95.760	181	140.560	237	185.360
14	6.960	70	51.760	126	96.560	182	141.360	238	186.160
15	7.760	71	52.560	127	97.360	183	142.160	239	186.960
16	8.560	72	53.360	128	98.160	184	142.960	240	187.760
17	9.360	73	54.160	129	98.960	185	143.760	241	188.560
18	10.160	74	54.960	130	99.760	186	144.560	242	189.360
19	10.960	75	55.760	131	100.560	187	145.360	243	190.160
20	11.760	76	56.560	132	101.360	188	146.160	244	190.960
21	12.560	77	57.360	133	102.160	189	146.960	245	191.760
22	13.360	78	58.160	134	102.960	190	147.760	246	192.560
23	14.160	79	58.960	135	103.760	191	148.560	247	193.360
24	14.960	80	59.760	136	104.560	192	149.360	248	194.160
25	15.760	81	60.560	137	105.360	193	150.160	249	194.960
26	16.560	82	61.360	138	106.160	194	150.960	250	195.760
27	17.360	83	62.160	139	106.960	195	151.760	251	196.560
28	18.160	84	62.960	140	107.760	196	152.560	252	197.360
29	18.960	85	63.760	141	108.560	197	153.360	253	198.160
30	19.760	86	64.560	142	109.360	198	154.160	254	198.960
31	20.560	87	65.360	143	110.160	199	154.960	255	199.760
32	21.360	88	66.160	144	110.960	200	155.760	256	200.560
33	22.160	89	66.960	145	111.760	201	156.560	257	201.360
34	22.960	90	67.760	146	112.560	202	157.360	258	202.160
35	23.760	91	68.560	147	113.360	203	158.160	259	202.960
36	24.560	92	69.360	148	114.160	204	158.960	260	203.760
37	25.360	93	70.160	149	114.960	205	159.760	261	204.560
38	26.160	94	70.960	150	115.760	206	160.560	262	205.360
39	26.960	95	71.760	151	116.560	207	161.360	263	206.160
40	27.760	96	72.560	152	117.360	208	162.160	264	206.960
41	28.560	97	73.360	153	118.160	209	162.960	265	207.760
42	29.360	98	74.160	154	118.960	210	163.760	266	208.560
43	30.160	99	74.960	155	119.760	211	164.560	267	209.360
44	30.960	100	75.760	156	120.560	212	165.360	268	210.160
45	31.760	101	76.560	157	121.360	213	166.160	269	210.960
46	32.560	102	77.360	158	122.160	214	166.960	270	211.760
47	33.360	103	78.160	159	122.960	215	167.760		
48	34.160	104	78.960	160	123.760	216	168.560		
49	34.960	105	79.760	161	124.560	217	169.360		
50	35.760	106	80.560	162	125.360	218	170.160		
51	36.560	107	81.360	163	126.160	219	170.960		
52	37.360	108	82.160	164	126.960	220	171.760		
53	38.160	109	82.960	165	127.760	221	172.560		
54	38.960	110	83.760	166	128.560	222	173.360		
55	39.760	111	84.560	167	129.360	223	174.160		

1.5 Stability model

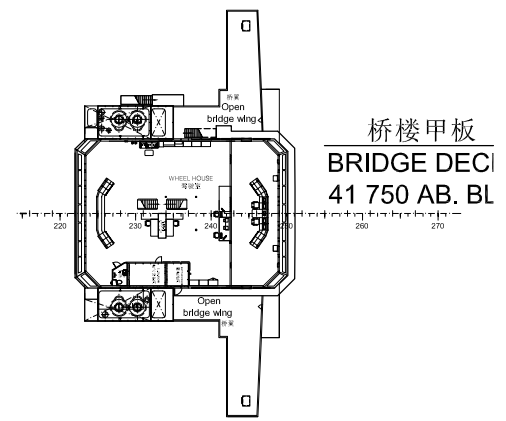
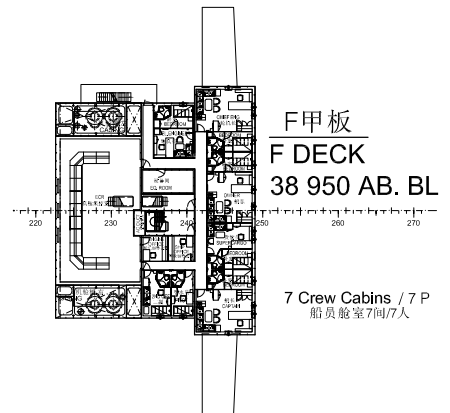
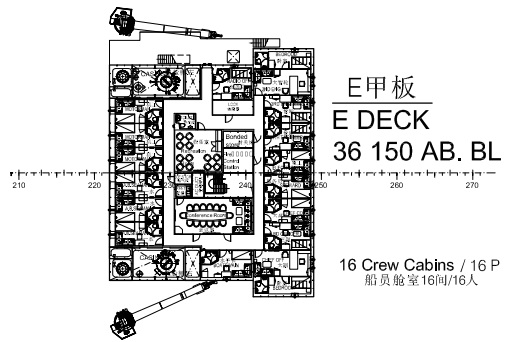
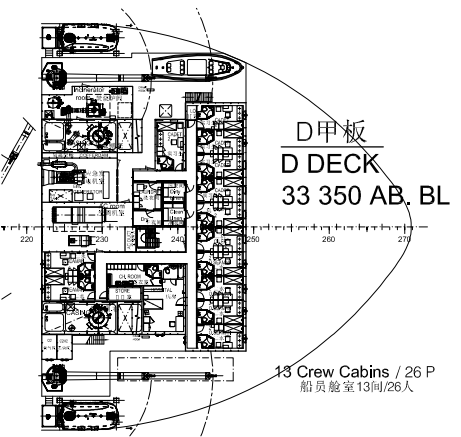
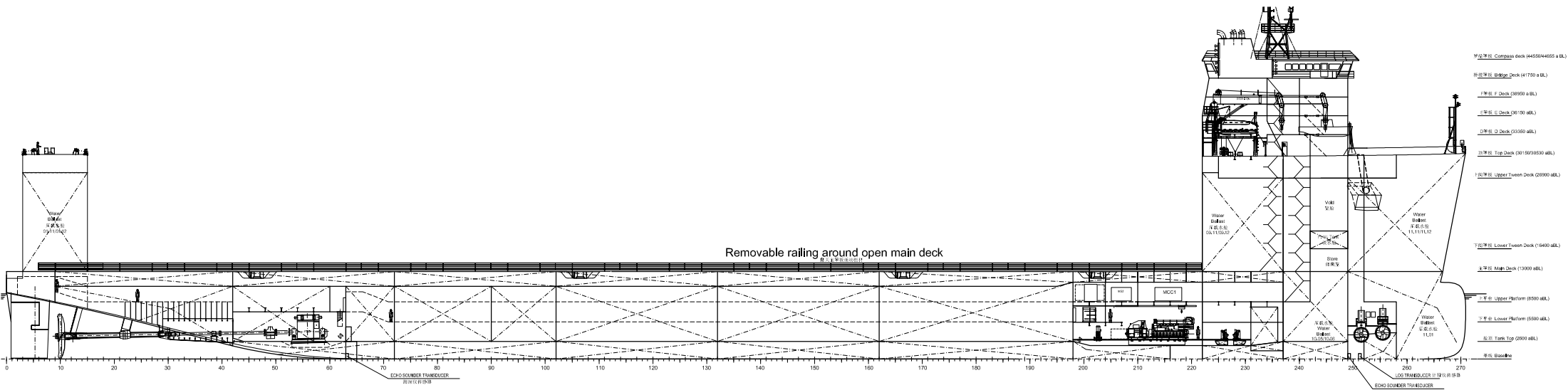
Volumes contributing to buoyancy in intact and damage stability calculations are shown in next page. Buoyancy casings are in aft position.

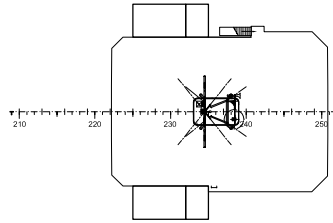
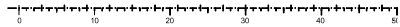
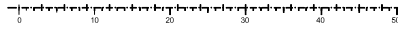
STABILITY MODEL

Volumes contributing to buoyancy in intact and damage stability calculations

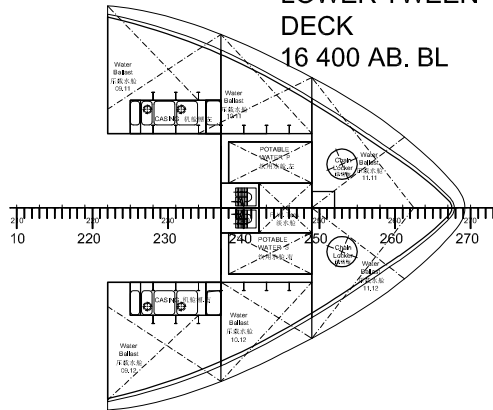


1.6 General arrangement plan

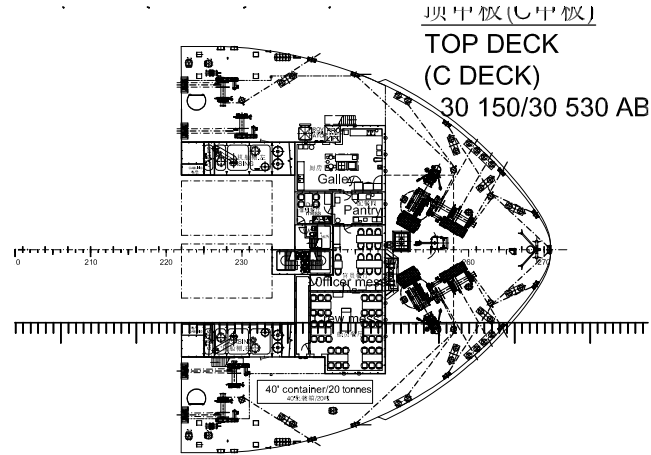




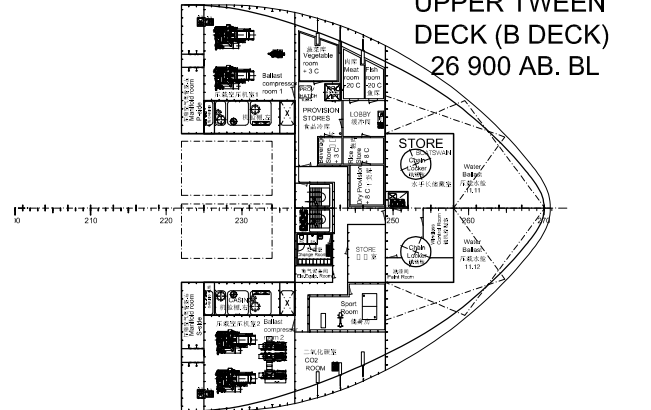
罗经甲板
COMPASS DECK
44 550/44 655 AB. BL



下间甲板
LOWER TWEEN
DECK
16 400 AB. BL

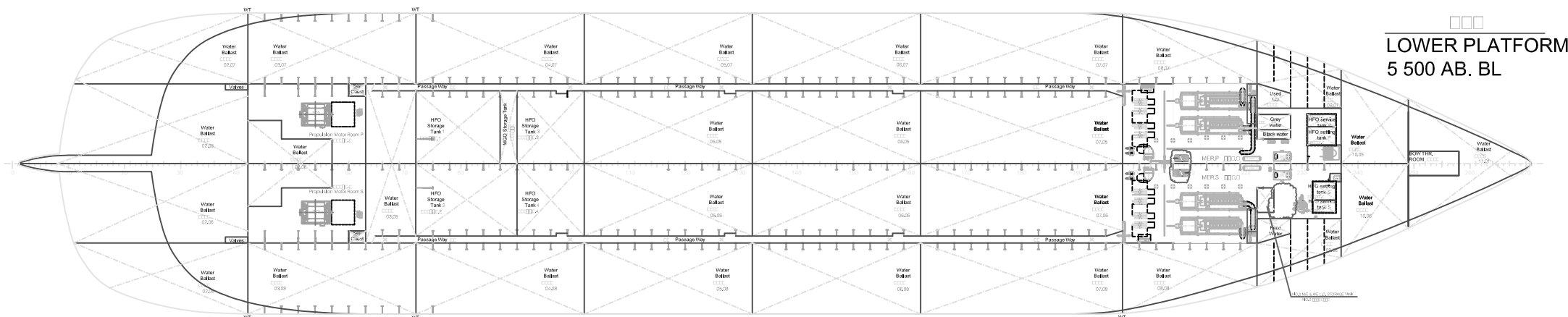
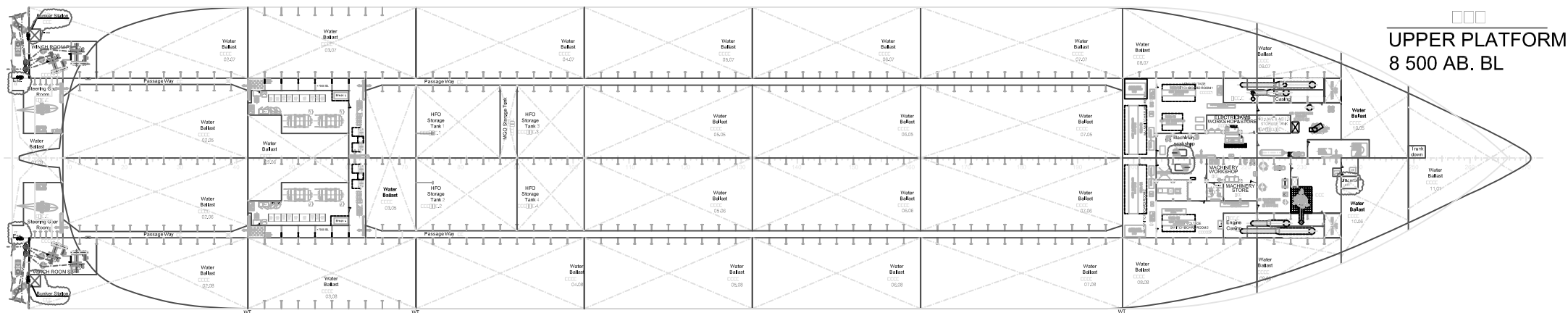


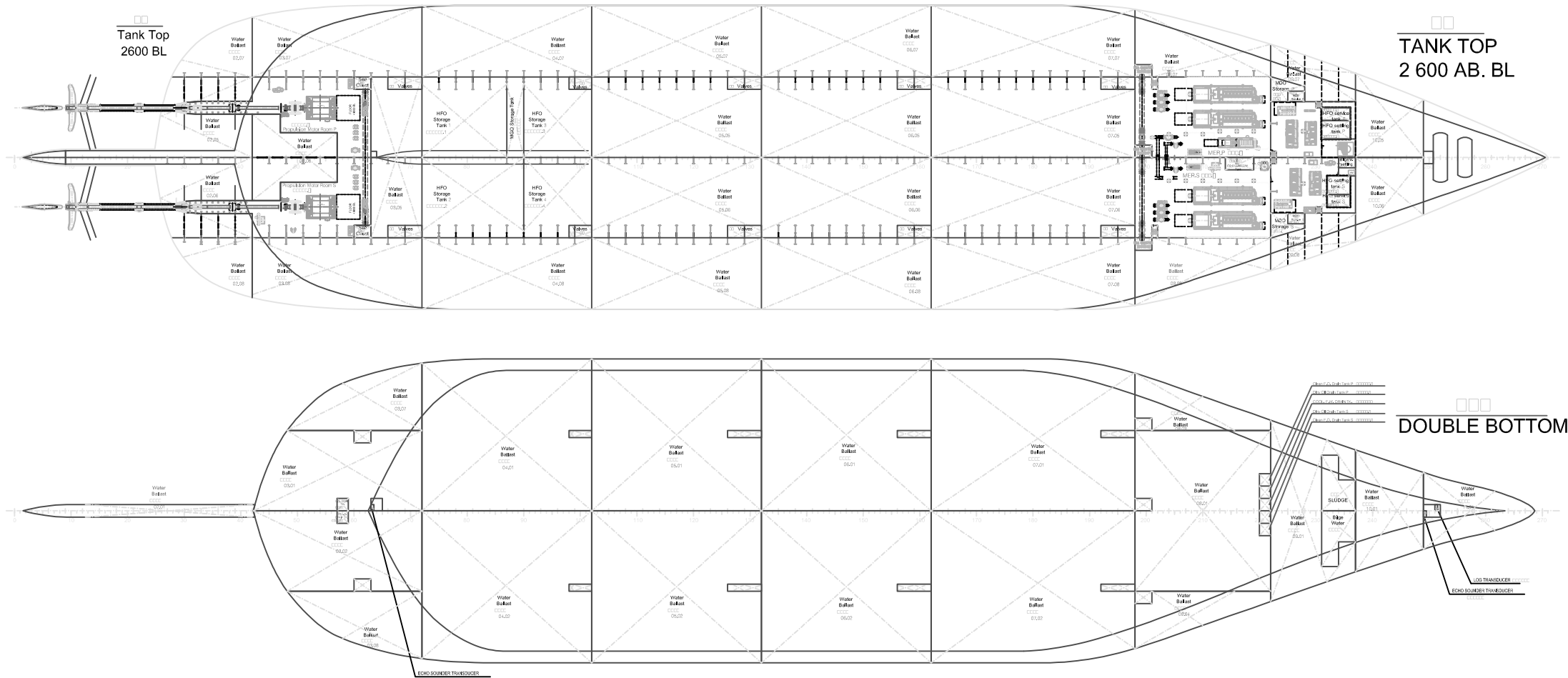
顶甲板(C甲板)
TOP DECK
(C DECK)
30 150/30 530 AB



上间甲板(B甲板)
UPPER TWEEN
DECK (B DECK)
26 900 AB. BL

MAIN DECK
(A DECK)
13 000 AB. BL





1.7 Description of buoyant cargos used in stability calculations

1.7.1 MLT 116C

Buoyancy of MLT 116C has not been taken into account in stability calculations.

Weight of cargo: 8654t
Wind profile area including legs: 3584.4m²

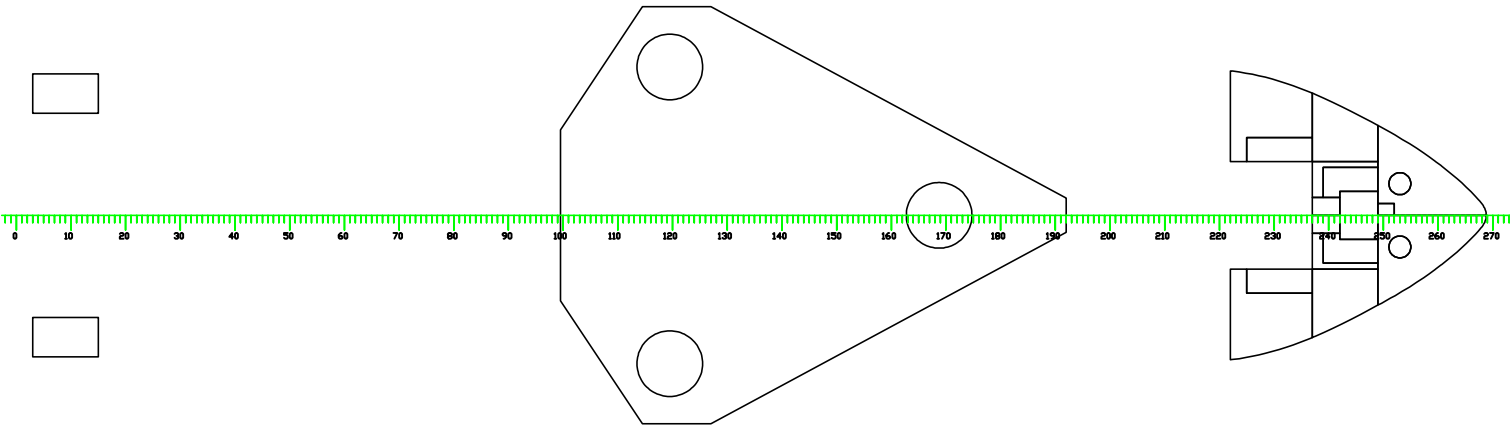
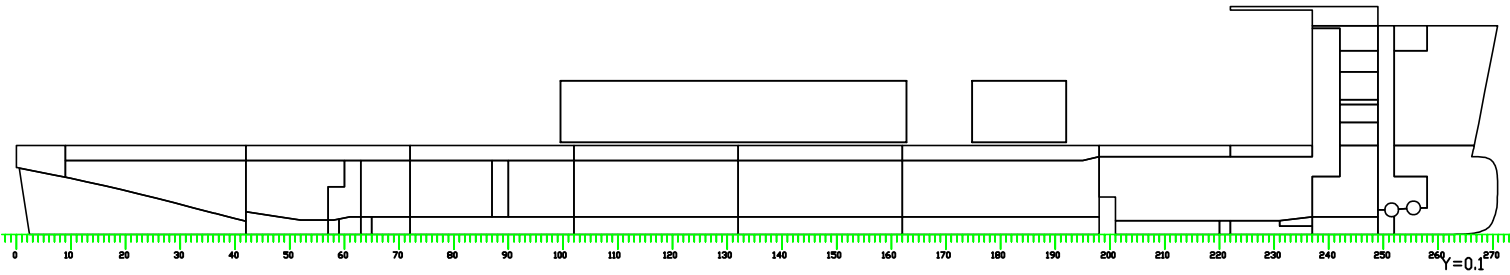
NAME	VOL	CGX	CGY	CGZ	XMIN	XMAX	YMIN	YMAX	ZMIN	ZMAX
MLT 116C	23659.0	109.06	0.00	18.00	79.60	153.60	-30.50	30.50	13.50	22.50

Coordinates are in ship coordinate system (ship origo 0,0,0).

EXPLANATION OF SYMBOLS:

VOL	volume	m ³
CGX	x-coordinate of center of gravity	m
CGY	y-coordinate of center of gravity	m
CGZ	z-coordinate of center of gravity	m
XMIN	minimum x	m
XMAX	maximum x	m
YMIN	minimum y	m
YMAX	maximum y	m
ZMIN	minimum z	m
ZMAX	maximum z	m

VESSEL CARRYING MLT 116C



1.7.2 Super Gorilla Rig 16000t

Buoyancy of Super Gorilla Rig 16000t has not been taken into account in intact stability calculations. However, buoyancy has been taken into account in damage stability calculations.

Weight of cargo: 16000t
Wind profile area including legs: 3815.8m²

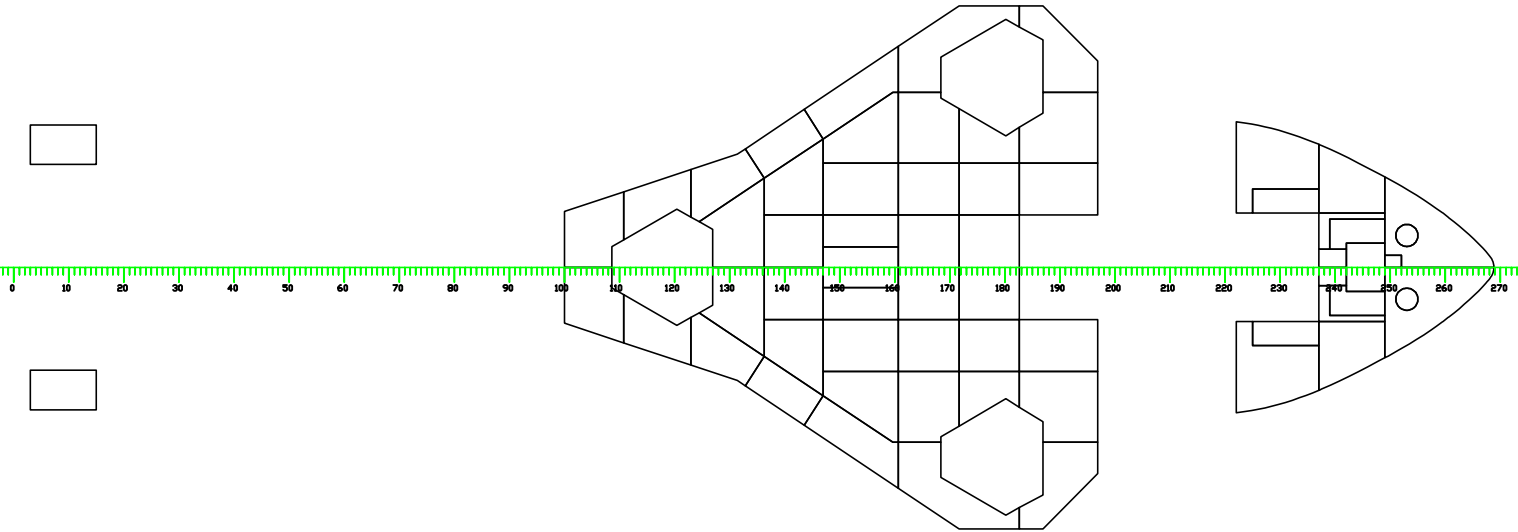
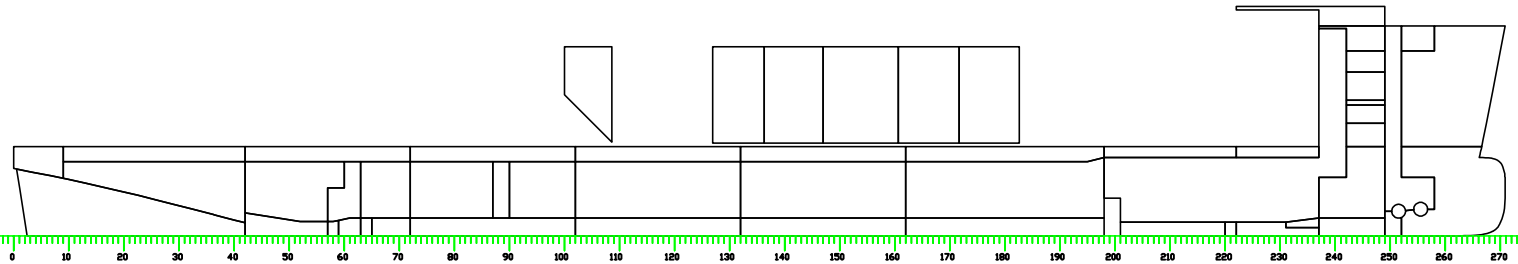
NAME	VOL	CGX	CGY	CGZ	XMIN	XMAX	YMIN	YMAX	ZMIN	ZMAX
SG 16000	43293.2	126.97	0.00	20.55	80.01	157.50	-38.00	38.00	13.50	27.50

Coordinates are in ship coordinate system (ship origo 0,0,0).

EXPLANATION OF SYMBOLS:

VOL	volume	m ³
CGX	x-coordinate of center of gravity	m
CGY	y-coordinate of center of gravity	m
CGZ	z-coordinate of center of gravity	m
XMIN	minimum x	m
XMAX	maximum x	m
YMIN	minimum y	m
YMAX	maximum y	m
ZMIN	minimum z	m
ZMAX	maximum z	m

VESSEL CARRYING SUPER GORILLA 16000T



1.7.3 Super Gorilla Rig 30000t

Buoyancy of Super Gorilla Rig 30000t has been taken into account in stability calculations. These calculations are shown in Loading Manual Addendum 1.

Weight of cargo: 30000t
Wind profile area including legs: 7846.8m²

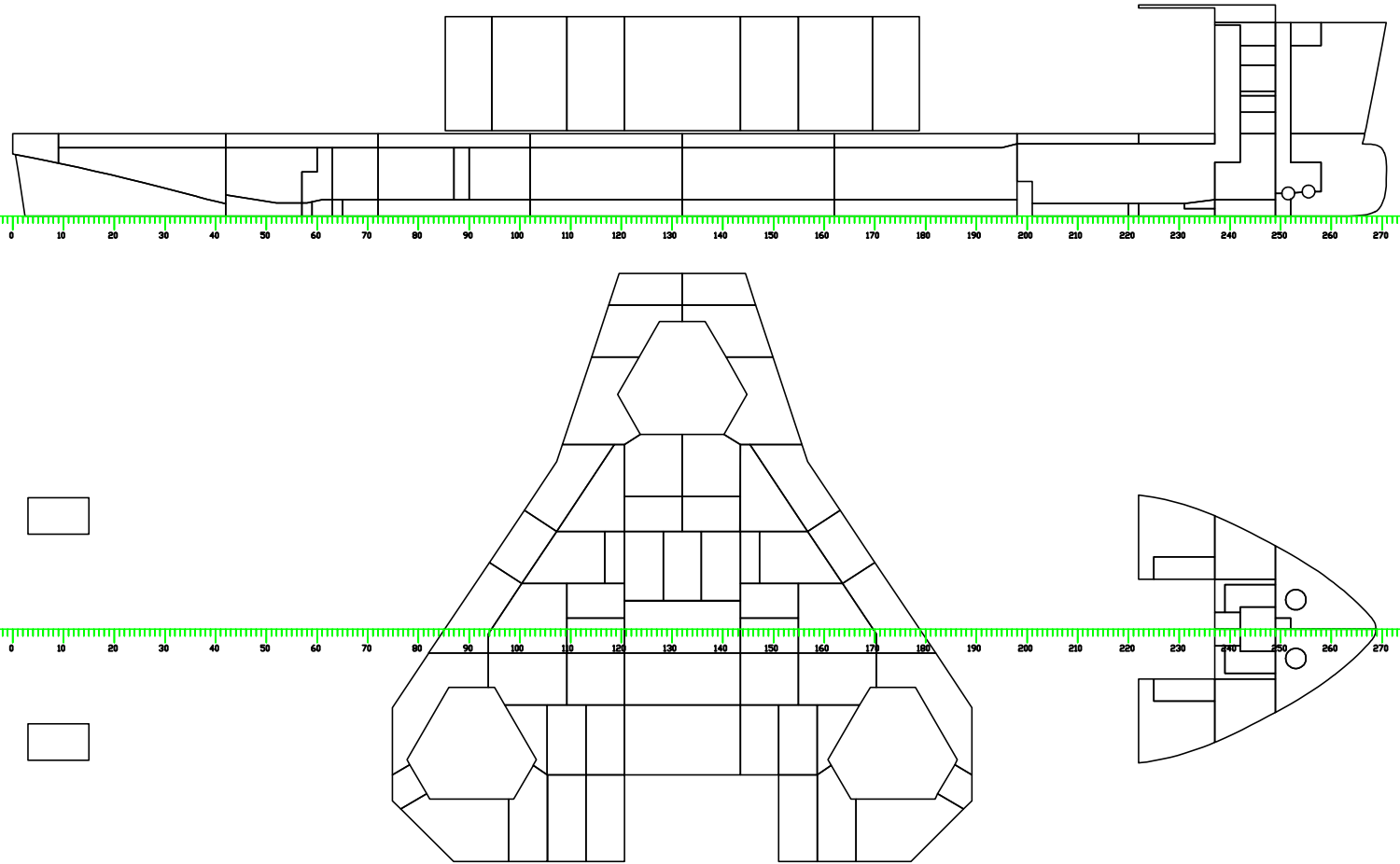
NAME	VOL	CGX	CGY	CGZ	XMIN	XMAX	YMIN	YMAX	ZMIN	ZMAX
SG 30000	80561.6	105.60	0.00	22.57	59.90	151.30	-36.68	56.62	13.00	31.50

Coordinates are in ship coordinate system (ship origo 0,0,0).

EXPLANATION OF SYMBOLS:

VOL	volume	m ³
CGX	x-coordinate of center of gravity	m
CGY	y-coordinate of center of gravity	m
CGZ	z-coordinate of center of gravity	m
XMIN	minimum x	m
XMAX	maximum x	m
YMIN	minimum y	m
YMAX	maximum y	m
ZMIN	minimum z	m
ZMAX	maximum z	m

VESSEL CARRYING SUPER GORILLA 30000T



2 Use of the manual

2.1 Introduction

The general purpose of the manual is:

1. To provide all necessary information to the Master for the proper loading and ballasting of the ship and for the control of stability and draught.
2. To satisfy the requirements of the flag state Administration with regard to approval according to regulation II-2/22 of SOLAS
3. To stress that compliance with the requirements as regards maximum KG(fluid) information shown in Chapter 3 does not ensure immunity against capsizing regardless of the circumstances or absolve the Master of his responsibilities.

2.2 Explanations to the manual

The origo (0, 0, 0) is situated at frame 0, at centreline and at baseline. Port side coordinates are positive and starboard negative.

Negative trim indicates a trim by stem and positive trim indicates a trim by stern.

Frame spacing is 800mm and web frame spacing 2400mm.

Calculations are performed by Deltamarin Ltd by utilizing results of NAPA program (NAPA Release 2009.1)

2.3 Abbreviations, symbols and definitions

Ak	total overall area of bilge keels
AP	aft perpendicular
AREA	profile area above waterline
B	moulded breadth of the ship
BL	baseline of hull body
CASE	loading case
CB	block coefficient
CL	centreline of hull body
CM	midship section coefficient
CP	prismatic coefficient
CW	waterplane coefficient
DESC	descriptive name
DISP	total displacement
dGM	free surface correction of GM
DW	deadweight

FILL	filling degree of tank
FP	forward perpendicular
FRA	total frame area
FRAM	moulded frame area
FRMIN	minimum frame
FRMAX	maximum frame
Frs.mom.	free surface moment
FSM	free surface moment
GM	corrected metacentric height
GMO	initial metacentric height
GZ	righting lever
H	height from tank bottom
IY	transverse moment of inertia of surface
KG	vertical centre of gravity corrected
KG _F	vertical centre of gravity corrected for free surfaces
KM	transverse metacentric height
KML	longitudinal metacentric height
KMT	transverse metacentric height
KN	cross curve ordinate
LCA	longitudinal centre of flotation
LCB	longitudinal centre of buoyancy
LCG	longitudinal centre of gravity
LEVER	vertical distance of lever
Loa	length over all
Lpp	length between perpendiculars
LWL	length of waterline
m	meters
MASS	mass of load in tank
MCT	moment to change trim
P	port side
S	starboard side
S.G.	specific gravity
T	draught, moulded
t	tonnes
TCB	transverse centre of buoyancy
TCG	transverse centre of gravity
TCP	change of displacement/change of draught
tm	tonnes meters
TR	trim
VCB	vertical centre of buoyancy
VCG	vertical centre of gravity
VNET	net volume
VOLM	volume moulded
VOLT	total volume
WLA	waterline area
WSA	wetted surface area

2.4 Locations of symbols and conventions

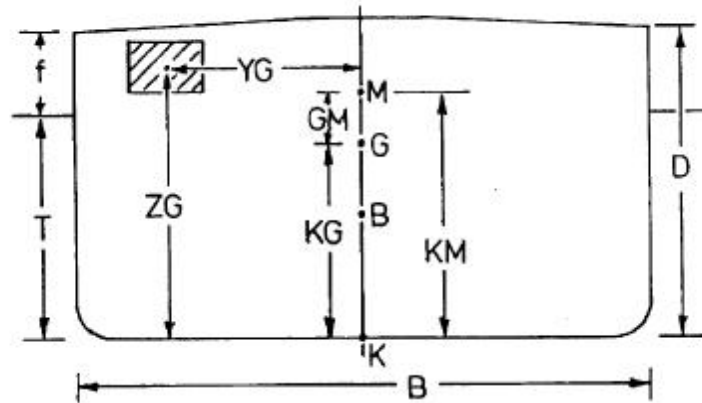


Figure 1

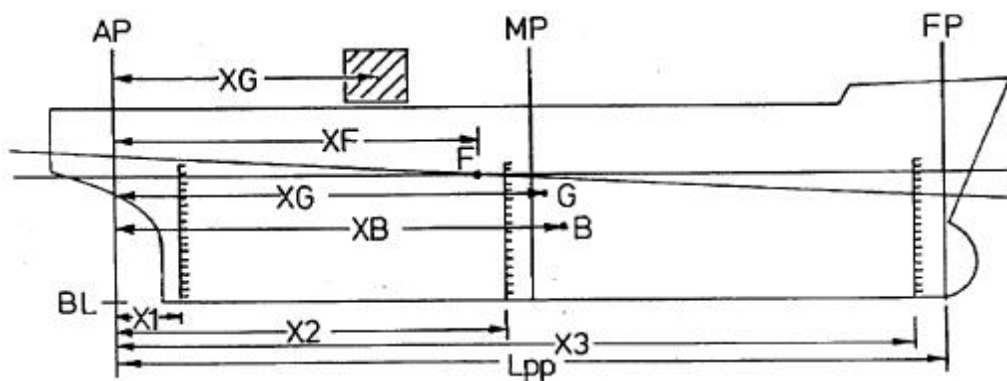
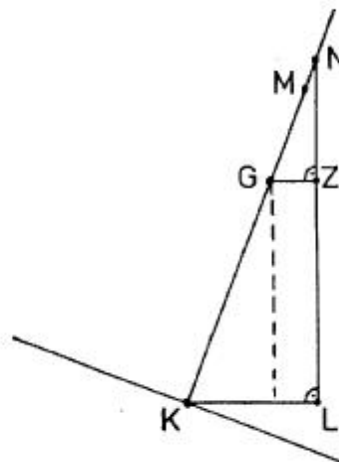


Figure 2



2.5 Conversion table

MULTIPLY BY BELOW	TO CONVERT FROM	TO OBTAIN	
0.03937	Millimeters	Inches	25.4
0.3937	Centimeters	Inches	2.54
3.2808	Meters	Feet	0.3048
2.2046	Kilograms	Pounds	0.45359
0.0009842	Kilograms	Tons (i.e. 2240 lbs)	1016.047
0.9842	Metric tons (i.e. 1000 kilos)	Tons (i.e. 2240 lbs) per inch (immersion)	1.016
2.4998	Metric tons per 1 cm (of immersion)	Tons (i.e. 2240 lbs) per inch (immersion)	0.400
8.2014	Moment to change trim 1cm (t-m units)	Tons (i.e. 2240 lbs) per inch (immersion)	0.1219
187.9767	Meter radians	Feet degrees	0.0053
	TO OBTAIN	TO CONVERT FROM	MULTIPLY BY ABOVE

Relation between weight and volume:

1000 Cubic millimeters	= 1.000 Cubic centimeters
1 Cubic centimeter of fresh water (S.G. 1.0)	= 1.000 Grammes
1000 Cubic centimeter of fresh water (S.G. 1.0)	= 1.000 Kilogram
1 Cubic meter of fresh water (S.G. 1.0)	= 1.000 Tonne
1 Cubic meter of salt water (S.G. 1.025)	= 1.025 Tonnes
1 Tonne of S.W. (S.G. 1.025)	= 0.975 Cubic metres
1 Cubic meter	= 35.315 Cubic feet
1 Cubic foot	= 0.0283 Cubic meters

3 Guidance to the master

3.1 Notes on stability

3.1.1 General precautions against capsizing

Compliance with the required minimum stability criteria does not insure immunity against capsizing, regardless of the circumstances, or absolve the Master from his responsibilities. Masters should therefore exercise prudence and good seamanship, having regard to the season and should take appropriate action as to speed and course warranted by the prevailing circumstances.

Care should be taken to ensure that cargo, allocated to the ship is capable of being stowed, so that compliance with the stability criteria can be achieved. If necessary, the amount of cargo should be limited to the extent that ballast weight may be required.

In determining the sequence of tanks from which oil fuel and fresh water is to be consumed and those into which water ballast may be admitted during the voyage the Master must ensure, prior to departure, that the required minimum stability criteria will be maintained throughout the voyage after making due allowance for free surface effect as may be appropriate.

Before a voyage commences, care should be taken to ensure that the cargo and sizeable pieces of equipment have been properly stowed or lashed, so as to minimize the possibility of both longitudinal and lateral shifting at sea, under the effect of acceleration caused by rolling and pitching.

Masters should note that stability can be adversely affected by such influences as beam wind on ships with large windage area, icing on top sides and deck cargo, water trapped on deck and in deck cargo, rolling characteristics and following seas.

For any loading condition not shown herein, or in case of doubt, the master should perform a detailed calculation to ensure sufficient stability. This Loading Manual contains Blank forms and instructions on how to use these, as well as the information necessary to perform such a calculation.

3.1.2 Submerging and emerging

Each submerging operation must be made according to a pre-engineered procedure, which will take full consideration of filling, stability and longitudinal strength. All pre-engineered procedures must be submitted and approved by the relevant warranty surveyor.

During submerging or emerging, it is of greatest importance that ballast tanks are filled to the maximum extent or empty to minimize the free surface effects.

Masters should be aware that the following measurements are necessary to achieve a high filling percentage of ballast tanks:

- a) Ballasting and submerging with considerable trim fore.
- b) Before the main deck is submerged, the vessel should be properly rolled to port side and then to starboard side for several times.

Masters should be aware that the minimum GM would appear while the main deck is just submerged or emerged from the water. At this stage it is very important to minimize the number of partial filled tanks to minimize the free surface effects.

Ballasting and de-ballasting with cargo has to be thoroughly designed taking into account the weight, center of gravity, buoyancy and wind area of the cargo. The tanks above waterline are to be filled first and emptied last as far as possible.

More information can be found in the Submerging Procedure (SC-101-03JS).

3.1.3 Free surface effect

Provided a tank is completely filled with liquid no movement of the liquid is possible and the effect on the ship's stability is precisely the same as if the tank contained solid material.

Immediately a quantity of liquid is withdrawn from the tank the situation changes completely and the stability of the ship is adversely affected by what is known as the 'free surface effects'. This adverse effect on the stability is referred to as a 'loss in GM' or as a 'virtual rise in VCG'.

Free surface effect to be calculated as following principle:

- a) In transit condition

The free surface effect of all slack tanks including ballast and all consumables, are to be taken into account. Calculation of free surface effect is based on real moments at real filling.

- b) In semi-submerged condition

The free surface effect of all slack tanks including ballast and all consumables, are to be taken into account. Whenever free liquid surface exists in a tank, the maximum effect of free surface in the tank should be taken into account.

3.1.4 Sea fastenings

Sea fastening plan and corresponding strength analysis should be made for each cargo transport case to satisfy requirements set by relevant authorities.

3.2 Ship design conditions

3.2.1 Draught limits

Minimum draught (forward/aft) in heavy weather should be at least 8.00m due to risk of slamming.

Minimum draught (forward/aft) in ice conditions should be 8.00m. Maximum draught (forward/aft) in ice conditions is 9.68m.

Minimum recommendable draught forward/aft in good weather conditions is 7.00m. In that case the aft propellers are sufficiently immersed and the performance of bow thrusters will not be reduced.

3.2.2 Design loads

The scantlings and arrangements of the hull structure and hatch covers have been approved on the basis of the following design loads:

Uniform deck load	25t/m ²
Point load at any transverse web frame	100t
Point load at any longitudinal frame	25t

3.2.3 Weather limitations

Loading and unloading of cargo in submerged condition shall take place in calm sea condition with $H_s < 0.5\text{m}$.

Loading and unloading of cargo in specific aft load conditions shall take place inside the protective circumstances of a harbour with $H_s < 0.5\text{m}$. Loading of any cargo should in no case induce dynamic loads to the hull girder.

3.2.4 Pressure limitations

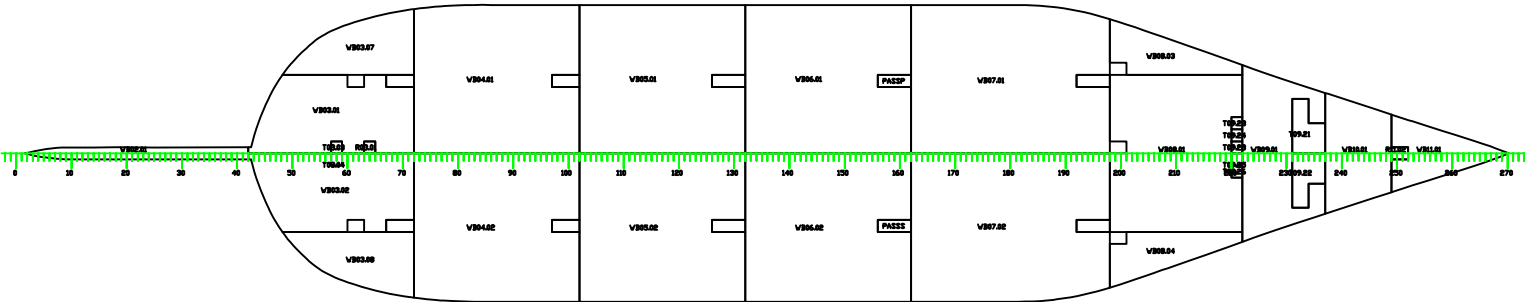
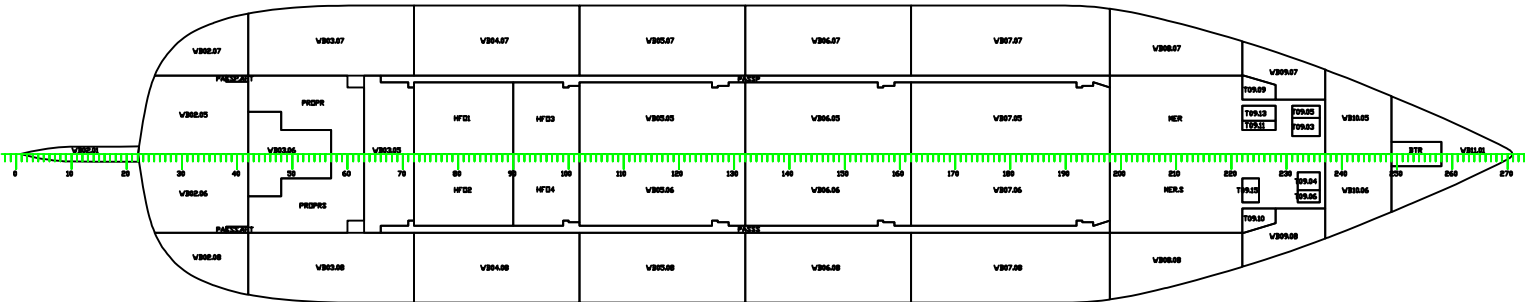
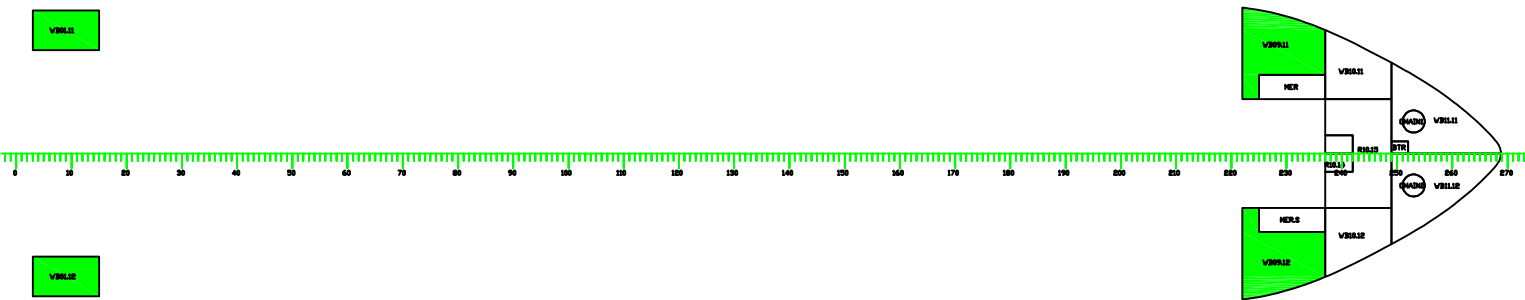
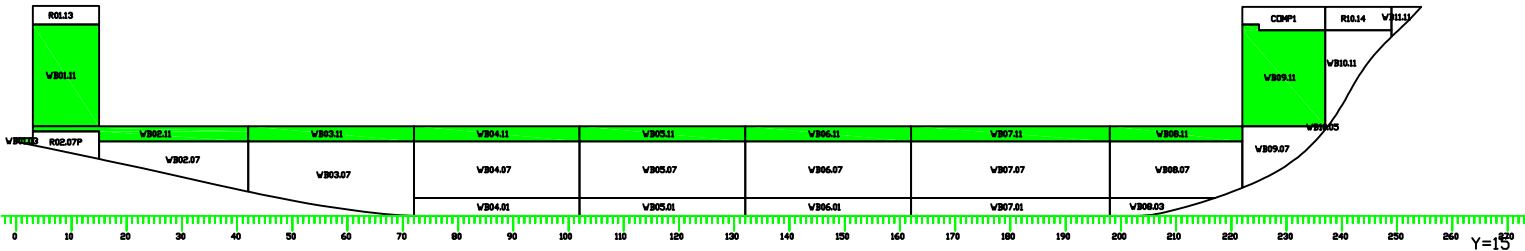
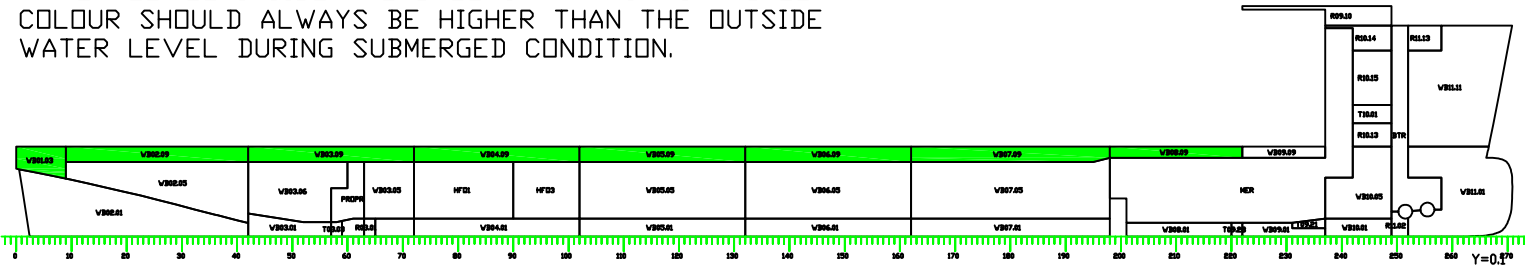
Under pressure of 0.5 bar, for pressurized ballast tanks, are only to be added for draughts $T \leq 13.0\text{m}$.

3.2.5 Water level inside the ballast tanks

The water level inside the buoyancy casings, tanks above main deck (WB09.11 and WB09.12) and top deck tanks (tanks just below the main deck), in submerged condition, shall always be higher than the outside water level, due to damage stability calculations. These tanks are shown on next page.

However, the water level inside the movable buoyancy casings should not be higher than 3 meters above sea level.

WATER LEVEL INSIDE THE BALLAST TANKS WITH GREEN COLOUR SHOULD ALWAYS BE HIGHER THAN THE OUTSIDE WATER LEVEL DURING SUBMERGED CONDITION.



3.3 Stability criteria

3.3.1 Intact stability criteria for transit condition

When calculating the vessel's fulfillment to the intact stability criteria requirements, the stability criteria of IMO resolution A.749 Ch.4.5.6 are fulfilled.

The names of the criteria used here are identical to the list output of NAPA.

Noble Denton stability criteria (only for reference) are presented in appendix 1.

V.AREA15-30

The area under the curve of righting levers (GZ curve) should not be less than 0.070 metre-radians up to an angle of 15° when the maximum righting lever (GZ) occurs at 15° and 0.055 metre-radians up to an angle of 30° when the maximum righting lever (GZ) occurs at 30° or above. Where the maximum righting lever (GZ) occurs at angles of between 15° and 30°, the corresponding area under the righting lever curve should be:

$$0.055 + 0.001 (30^\circ - \theta_{\max}) \text{ metre-radians}$$

$\theta_{\max}$ is the angle of heel in degrees at which the righting lever curve reaches its maximum.

V.AREA3040

The area under the righting lever curve (GZ curve) between the angles of heel of 30° and 40°, or between 30° and 0° if this angle is less than 40°, should be not less than 0.03 metre-radians.

V.GZ0.2

The righting lever (GZ) should be at least 0.20 m at an angle of heel equal to or greater than 30°.

V.POSMAX15

The maximum righting lever (GZ) should occur at an angle of heel not less than 15°.

V.GM0.15

The initial transverse metacentric height (GMO) should not be less than 0.15 m.

IMOWEATHER

This criterion reflects the "Severe wind and rolling criterion (weather criterion)" as defined in IMO resolution A.749 chapter 3.2.

3.3.2 Intact stability criteria for semi-submerged condition

3.3.2.1 Criteria from DNV

The stability of the vessel during semi-submerged condition should comply with the DNV criteria below.

The names of the criteria used here are identical to the list output of NAPA.

MINGM0.3-SUB

The minimum GM should not be less than 0.3m.

RANGE-SUB

The range of the GZ-curve should be minimum 7 degrees.

MAXGZ-SUB

The height of the GZ curve should be not less than 0.05m.

RSRV-SUB

The minimum distance from water surface to opening through which down-flooding could occur should not be less than 1 m.

3.3.2.2 Criteria from CCS

The stability in all semi-submerged condition shall be in compliance with Reg.3.13 in amendments (2006) of "the Statutory Surveys of Sea-going Ships Engaged on Non International Voyages (2004)" The criteria is presented below.

The names of the criteria used here are identical to the list output of NAPA.

MINGM1.0-SUB

The minimum GM should not be less than 1.00m.

MINGM0.5-SUB

The minimum GM should not be less than 0.50m.

MINGM0.15-SUB

The minimum GM should not be less than 0.15m.

MINGM0.05-SUB.

The minimum GM should not be less than 0.05m.

The usage of these criteria (what criteria to be used in which situation) is explained below.

During the submerging and emerging with deck cargo load, the stability is to meet one of the following criteria:

At the normal calm sea area with wind force not exceeding Beaufort scale 3, the initial metacentric height after correction for free surface effect, GM, should be:

- more than 1.00m at the beginning of submerging

- more than 0.50m while the ship is submerged to the maximum submerged draught
- more 0.05m while main deck is just submerged or emerged from the water

At the sea area with wind force not exceeding Beaufort scale 6 and significant wave height less than 0.5m, the initial metacentric height after correction for free surface effect, GM, should be:

- more than 1.00m at the beginning of submerging
- more 0.50m while the ship is submerged to the maximum submerged draught
- more than 0.15m while main deck is just submerged or emerged from the water

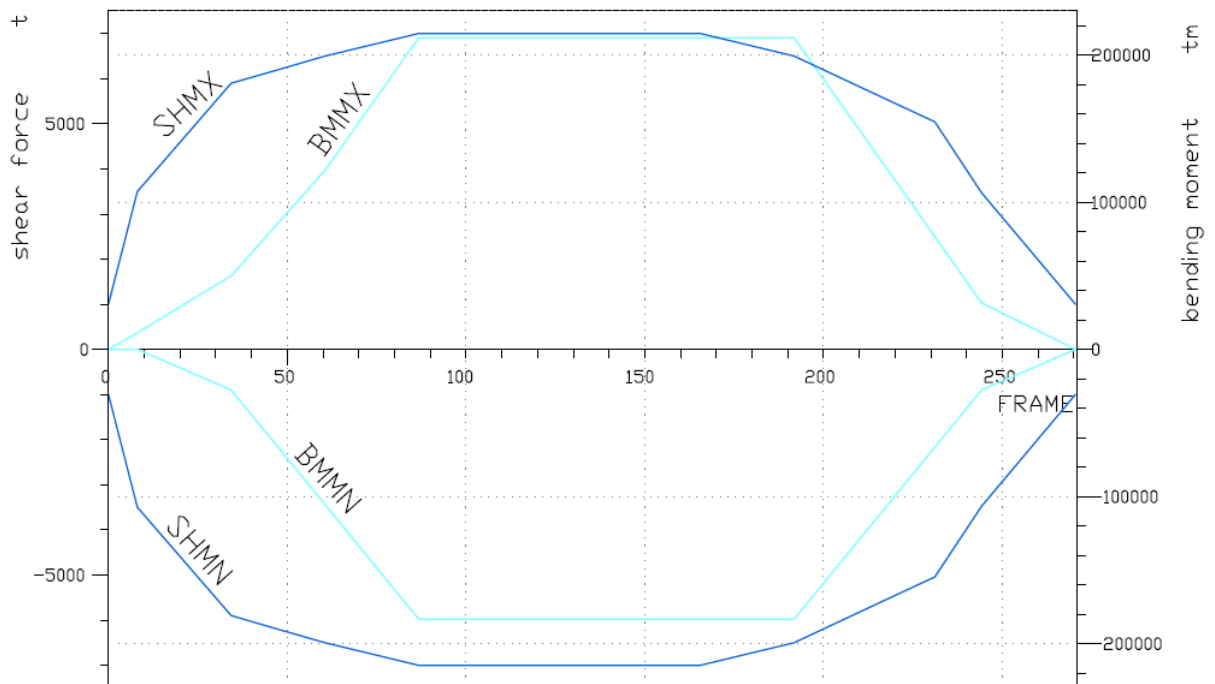
During the submerging and emerging without deck cargo load, the initial metacentric height after correction for free surface effect, GM, should be:

- more than 1.00m at the beginning of submerging
- more than 0.50m while the ship is submerged to the maximum submerged draught
- more than 0.15m while main deck is just submerged or emerged from the water

3.4 Longitudinal strength

3.4.1 Transit condition

The limits for bending moment and shear force in transit condition are shown in figure and tables.



Bending moment and shear force limits

Bending moment limits

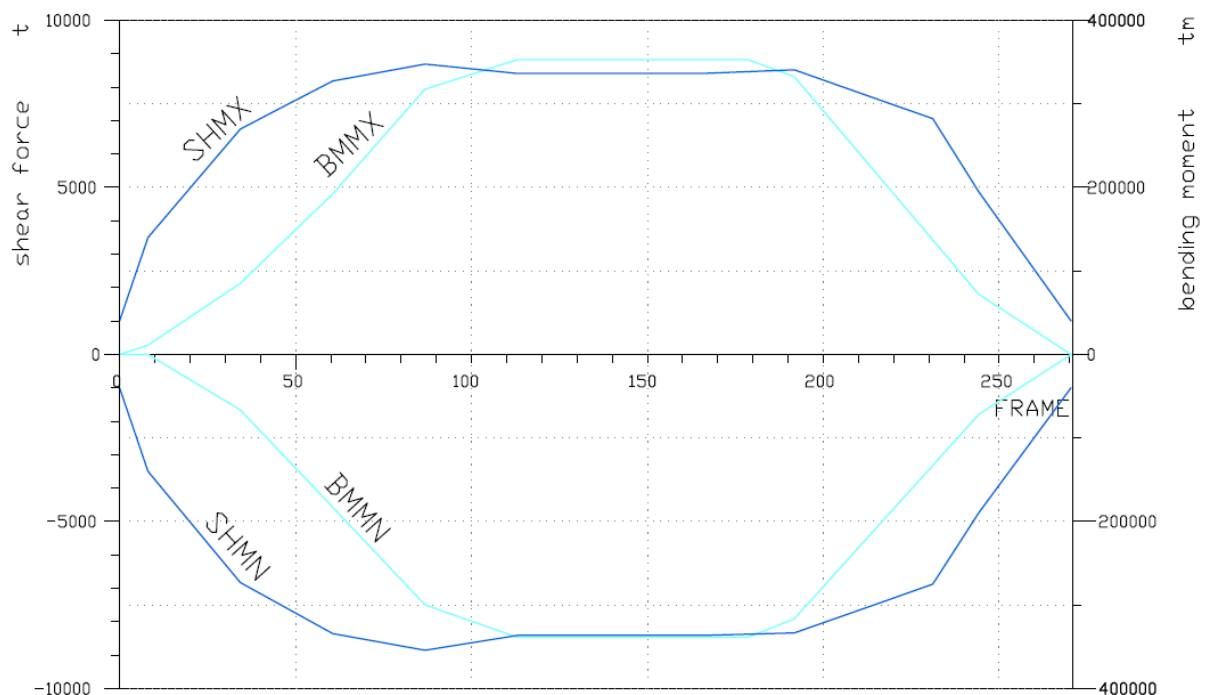
Location	L from FR0	sag	hog	sag	hog
	[m]	[kNm]	[kNm]	[tm]	[tm]
FR 0	0	0	0	0	0
AP_rule	6.49	0	109928	0	11206
0.1L	27.48	-269891	490500	-27512	50000
0.2L	48.47	-1034582	1194503	-105462	121764
0.3L	69.45	-1799273	2077397	-183412	211763
0.4L	90.44	-1799273	2077397	-183412	211763
0.5L	111.43	-1799273	2077397	-183412	211763
0.6L	132.42	-1799273	2077397	-183412	211763
0.65L	142.91	-1799273	2077397	-183412	211763
0.7L	153.41	-1799273	2077397	-183412	211763
0.8L	174.39	-1034582	1194503	-105462	121764
0.9L	195.38	-269891	311610	-27512	31765
FP	216.37	0	0	0	0

Shear force limits

Location	L from FR0	hog	sag	hog	sag
	[m]	[kN]	[kN]	[t]	[t]
FR 0	0.0	-9810	9810	-1000	1000
AP_rule	6.49	-34335	34335	-3500	3500
0.1L	27.48	-57879	57879	-5900	5900
0.15L	37.98	-60822	60822	-6200	6200
0.20L	48.47	-63765	63765	-6500	6500
0.3L	69.45	-68669	68669	-7000	7000
0.4L	90.44	-68669	68669	-7000	7000
0.5L	111.43	-68669	68669	-7000	7000
0.5L	111.43	-68669	68669	-7000	7000
0.6L	132.42	-68669	68669	-7000	7000
0.7L	153.41	-63764	63764	-6500	6500
0.8L	174.39	-54232	54232	-5528	5528
0.85L	184.89	-49471	49471	-5043	5043
0.9L	195.38	-34083	34083	-3474	3474
FP	216.37	-9810	9810	-1000	1000

3.4.2 Submerged condition

The limits for bending moment and shear force in submerged condition are shown in figure and tables.



Bending moment and shear force limits

Bending moment limits

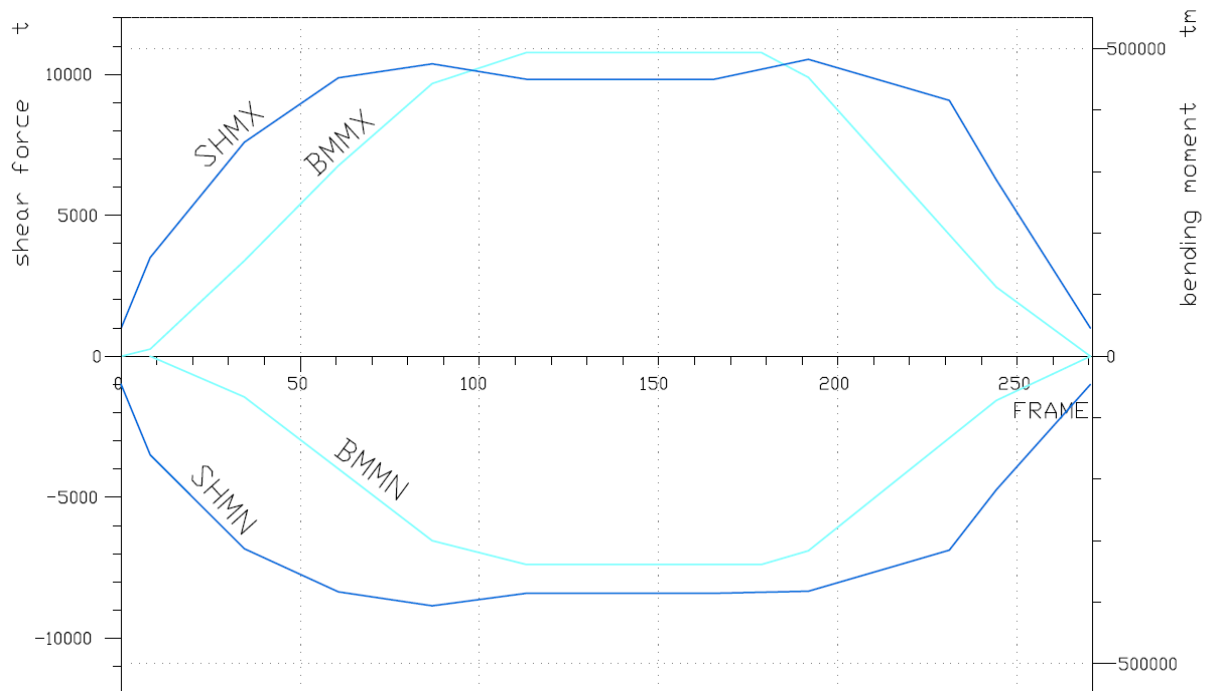
Location	L from FR0	sag	hog	sag	hog
	[m]	[kNm]	[kNm]	[tm]	[tm]
FR 0	0	0	0	0	0
AP_rule	6.49	0	109928	0	11206
0.1L	27.48	-650507	836350	-66311	85255
0.2L	48.47	-1795813	1886204	-183059	192274
0.3L	69.45	-2941120	3114948	-299808	317528
0.4L	90.44	-3321736	3460798	-338607	352783
0.5L	111.43	-3321736	3460798	-338607	352783
0.6L	132.42	-3321736	3460798	-338607	352783
0.65L	142.91	-3321736	3460798	-338607	352783
0.7L	153.41	-3104241	3263169	-316436	332637
0.8L	174.39	-1904561	1985018	-194145	202346
0.9L	195.38	-704881	706868	-71853	72056
FP	216.37	0	0	0	0

Shear force limits

Location	L from FR0	hog	sag	hog	sag
	[m]	[kN]	[kN]	[t]	[t]
FR 0	0.0	9810	-9810	1000	-1000
AP_rule	6.49	34335	-34335	3500	-3500
0.1L	27.48	66153	-66980	6743	-6828
0.15L	37.98	73233	-74473	7465	-7591
0.20L	48.47	80313	-81966	8187	-8355
0.3L	69.45	85217	-86870	8687	-8855
0.4L	90.44	82518	-82518	8412	-8412
0.5L	111.43	82518	-82518	8412	-8412
0.5L	111.43	82518	-82518	8412	-8412
0.6L	132.42	82518	-82518	8412	-8412
0.7L	153.41	83547	-81769	8517	-8335
0.8L	174.39	74016	-72237	7545	-7364
0.85L	184.89	69254	-67476	7060	-6878
0.9L	195.38	47712	-46486	4864	-4739
FP	216.37	9810	-9810	1000	-1000

3.4.3 Aftload and sideload condition

The limits for bending moment and shear force in aft/side load condition are shown in figure and tables.



Bending moment and shear force limits

Bending moment limits

Location	L from FR0	sag	hog	sag	hog
	[m]	[kNm]	[kNm]	[tm]	[tm]
FR 0	0	0	0	0	0
AP_rule	6,49	0	115797	0	11804
0.1L	27,48	-650507	1523078	-66311	155258
0.2L	48,47	-1795813	3039407	-183059	309827
0.3L	69,45	-2941120	4347686	-299808	443189
0.4L	90,44	-3321736	4844199	-338607	493802
0.5L	111,43	-3321736	4844199	-338607	493802
0.6L	132,42	-3321736	4844199	-338607	493802
0.65L	142,91	-3321736	4844199	-338607	493802
0.7L	153,41	-3104241	4448941	-316436	453511
0.8L	174,39	-1904561	2775533	-194145	282929
0.9L	195,38	-704881	1102125	-71853	112347
FP	216,37	0	0	0	0

Shear force limits

Location	L from FR0	hog	sag	hog	sag
	[m]	[kN]	[kN]	[t]	[t]
FR 0	0,0	9810	-9810	1000	-1000
AP_rule	6,49	34335	-34335	3500	-3500
0.1L	27,48	74427	-66980	7587	-6828
0.15L	37,98	85644	-74473	8730	-7591
0.20L	48,47	96861	-81966	9874	-8355
0.3L	69,45	101765	-86870	10374	-8855
0.4L	90,44	96366	-82518	9823	-8412
0.5L	111,43	96366	-82518	9823	-8412
0.5L	111,43	96366	-82518	9823	-8412
0.6L	132,42	96366	-82518	9823	-8412
0.7L	153,41	103331	-81769	10533	-8335
0.8L	174,39	93799	-72237	9562	-7364
0.85L	184,89	89038	-67476	9076	-6878
0.9L	195,38	61341	-46486	6253	-4739
FP	216,37	9810	-9810	1000	-1000

3.5 Wind profile data

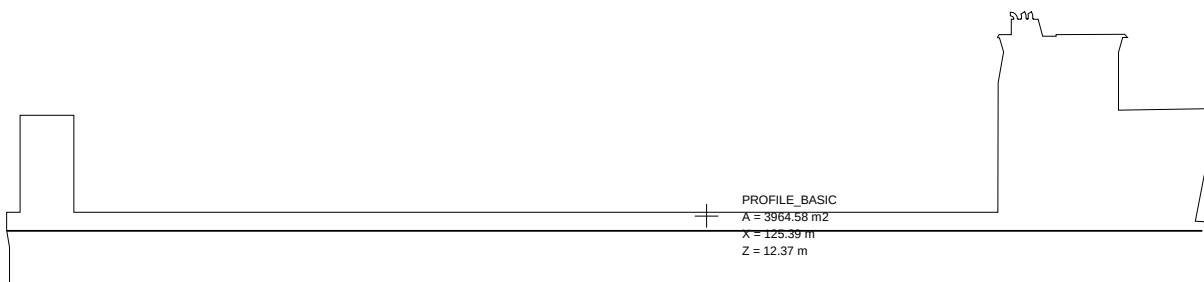
Data for wind profile without deck load is presented in next page. Total overall area of bilge keels $A_k = 110 \text{ m}^2$.

Profile Curve: PROFILE_BASIC

AREA= 3964.6 m2, LCG= 125.39 m, ZCG= 12.37 m

T	TR	AREA	ZCG	LEVER	MOM
m	m	m2	m	m	m3
4.000	0.000	3103.2	15.243	13.24	41082.4
4.100	0.000	3081.6	15.321	13.27	40883.9
4.200	0.000	3059.9	15.400	13.30	40685.4
4.300	0.000	3038.3	15.480	13.33	40486.9
4.400	0.000	3016.7	15.559	13.36	40288.5
4.500	0.000	2995.1	15.640	13.39	40090.0
4.600	0.000	2973.5	15.720	13.42	39891.6
4.700	0.000	2951.9	15.801	13.45	39693.1
4.800	0.000	2930.3	15.883	13.48	39494.7
4.900	0.000	2908.7	15.965	13.51	39296.2
5.000	0.000	2887.1	16.047	13.54	39097.8
5.100	0.000	2865.4	16.130	13.58	38899.4
5.200	0.000	2843.8	16.214	13.61	38700.9
5.300	0.000	2822.2	16.298	13.64	38502.5
5.400	0.000	2800.6	16.382	13.68	38304.1
5.500	0.000	2779.0	16.467	13.71	38105.6
5.600	0.000	2757.4	16.553	13.75	37907.2
5.700	0.000	2735.7	16.639	13.78	37708.8
5.800	0.000	2714.1	16.726	13.82	37510.4
5.900	0.000	2692.5	16.813	13.86	37312.0
6.000	0.000	2670.9	16.901	13.90	37113.5
6.100	0.000	2649.3	16.989	13.93	36915.1
6.200	0.000	2627.6	17.079	13.97	36716.7
6.300	0.000	2606.0	17.168	14.01	36518.3
6.400	0.000	2584.4	17.259	14.05	36319.9
6.500	0.000	2562.8	17.350	14.09	36121.6
6.600	0.000	2541.2	17.442	14.14	35923.2
6.700	0.000	2519.6	17.535	14.18	35724.8
6.800	0.000	2497.9	17.628	14.22	35526.4
6.900	0.000	2476.3	17.722	14.27	35328.0
7.000	0.000	2454.7	17.817	14.31	35129.6
7.100	0.000	2433.1	17.913	14.36	34931.2
7.200	0.000	2411.4	18.009	14.40	34732.8
7.300	0.000	2389.8	18.107	14.45	34534.4
7.400	0.000	2368.2	18.205	14.50	34336.0
7.500	0.000	2346.6	18.304	14.55	34137.5
7.600	0.000	2324.9	18.404	14.60	33939.1
7.700	0.000	2303.3	18.505	14.65	33740.6
7.800	0.000	2281.6	18.607	14.70	33542.1
7.900	0.000	2260.0	18.710	14.75	33343.7
8.000	0.000	2238.4	18.814	14.81	33145.2
8.100	0.000	2216.7	18.919	14.86	32946.7
8.200	0.000	2195.1	19.025	14.92	32748.2
8.300	0.000	2173.4	19.132	14.98	32549.7
8.400	0.000	2151.8	19.241	15.03	32351.2
8.500	0.000	2130.2	19.351	15.09	32152.7
8.600	0.000	2108.5	19.461	15.15	31954.2
8.700	0.000	2086.9	19.574	15.22	31755.7
8.800	0.000	2065.2	19.687	15.28	31557.1
8.900	0.000	2043.6	19.802	15.35	31358.6
9.000	0.000	2021.9	19.918	15.41	31160.1
9.100	0.000	2000.3	20.036	15.48	30961.6
9.200	0.000	1978.6	20.155	15.55	30763.1
9.300	0.000	1957.0	20.275	15.62	30564.6

T	TR	AREA	ZCG	LEVER	MOM
m	m	m2	m	m	m3
9.400	0.000	1935.3	20.398	15.69	30366.1
9.500	0.000	1913.7	20.521	15.76	30167.7
9.600	0.000	1892.0	20.647	15.84	29969.2
9.700	0.000	1870.4	20.774	15.92	29770.8
9.800	0.000	1848.8	20.903	16.00	29572.3
9.900	0.000	1827.1	21.034	16.08	29373.9
10.000	0.000	1805.5	21.167	16.16	29175.6
10.100	0.000	1783.8	21.302	16.24	28977.2
10.200	0.000	1762.2	21.439	16.33	28778.9
10.300	0.000	1740.6	21.578	16.42	28580.7
10.400	0.000	1719.0	21.719	16.51	28382.5
10.500	0.000	1697.3	21.862	16.60	28184.4
10.600	0.000	1675.7	22.008	16.70	27986.3
10.700	0.000	1654.1	22.157	16.80	27788.3
10.800	0.000	1632.6	22.308	16.90	27590.3
10.900	0.000	1611.0	22.461	17.00	27392.5
11.000	0.000	1589.4	22.617	17.11	27194.8
11.100	0.000	1567.8	22.776	17.22	26997.2
11.200	0.000	1546.3	22.938	17.33	26799.8
11.300	0.000	1524.8	23.103	17.45	26602.7
11.400	0.000	1503.4	23.270	17.56	26406.3
11.500	0.000	1482.1	23.440	17.68	26210.9
11.600	0.000	1460.8	23.613	17.81	26015.5
11.700	0.000	1439.5	23.790	17.94	25820.0
11.800	0.000	1418.2	23.971	18.07	25624.4
11.900	0.000	1396.9	24.156	18.20	25428.8
12.000	0.000	1375.6	24.345	18.34	25233.1
12.100	0.000	1354.3	24.539	18.49	25037.3
12.200	0.000	1333.0	24.737	18.64	24841.5
12.300	0.000	1311.7	24.940	18.79	24645.6
12.400	0.000	1290.3	25.148	18.95	24449.7
12.500	0.000	1269.0	25.361	19.11	24253.7
12.600	0.000	1247.7	25.580	19.28	24057.7
12.700	0.000	1226.4	25.805	19.46	23861.6
12.800	0.000	1205.1	26.036	19.64	23665.4
12.900	0.000	1183.7	26.273	19.83	23469.2
13.000	0.000	1162.4	26.518	20.02	23273.0



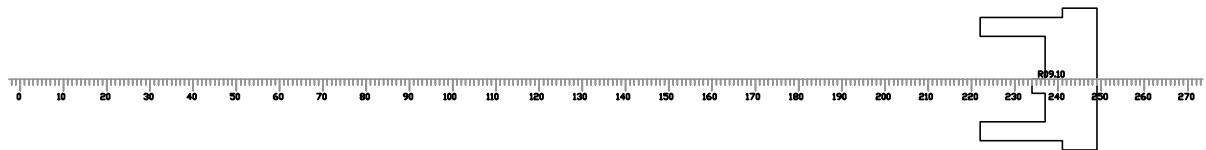
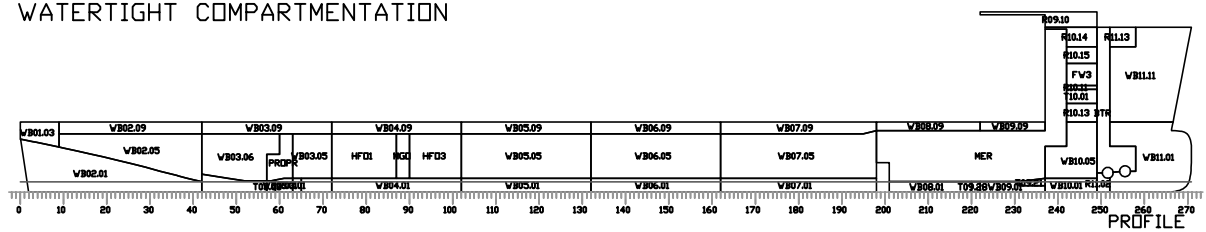
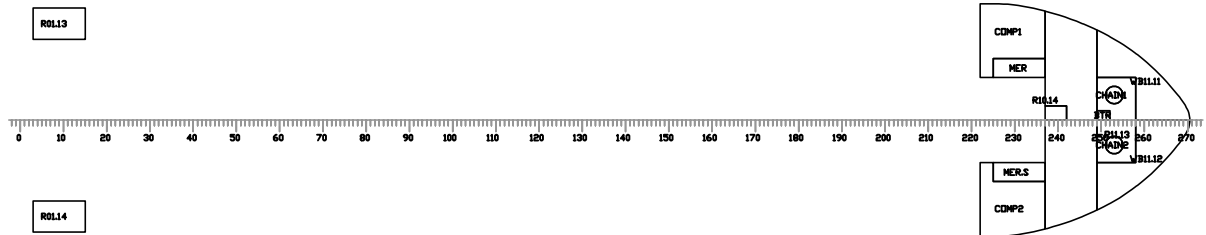
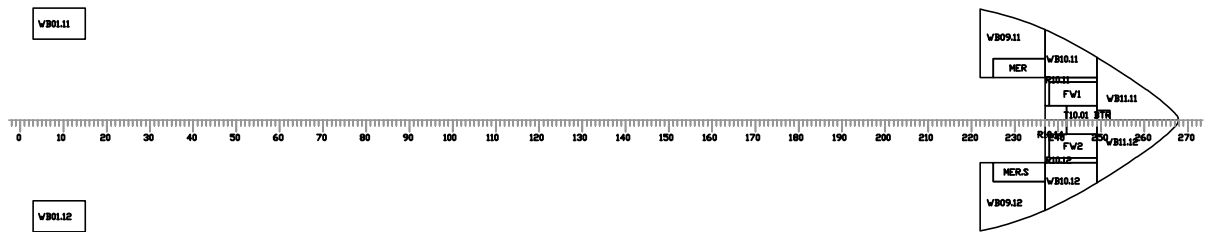
PROF	profile curve	
COEF	coefficient	
AREA	profile area above wl	m2
ZCG	ZCG of prof. area above wl	m
LEVER	Vertical distance of lever	m
MOM	mom. from center of underwater part	m3

3.6 Watertight integrity

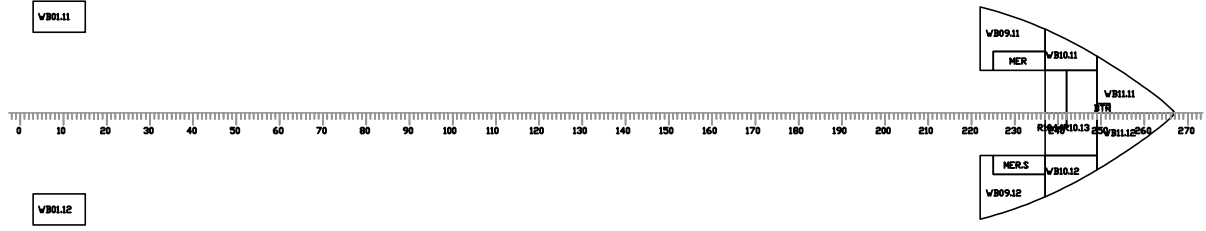
Watertight compartmentation and unprotected openings used in the intact stability calculations are shown in this chapter.

It is assumed that no progressive flooding can take place through internal openings, pipes, ducts, etc., or through external openings situated at more critical positions than those included in damage stability calculations (B3598.1173.101 Damage Stability Calculation, Chapter 1.1.3 Openings).

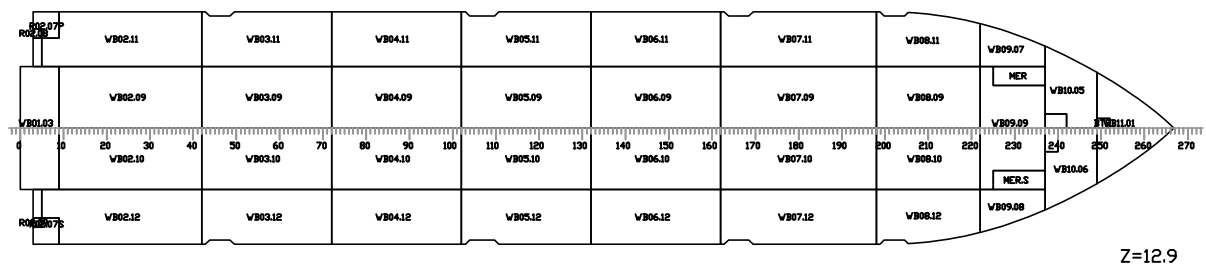
WATERTIGHT COMPARTMENTATION

 $Z=32$  $Z=29$ 

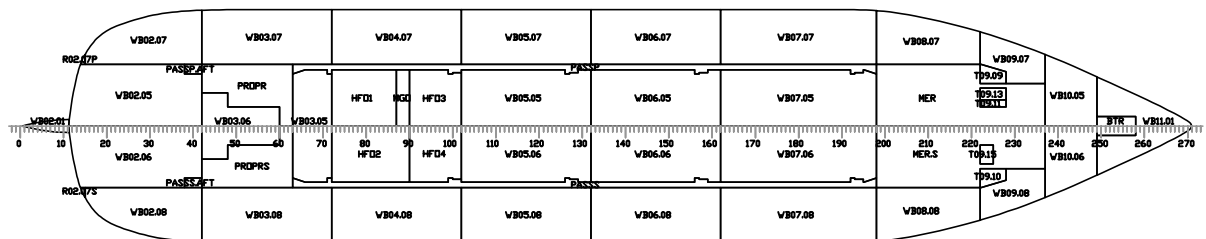
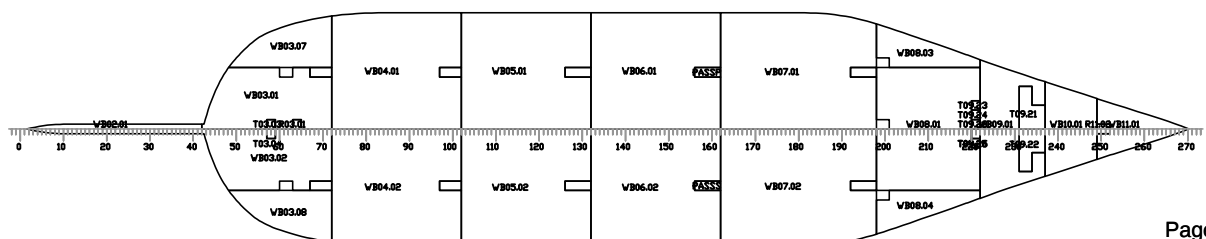
Z=18



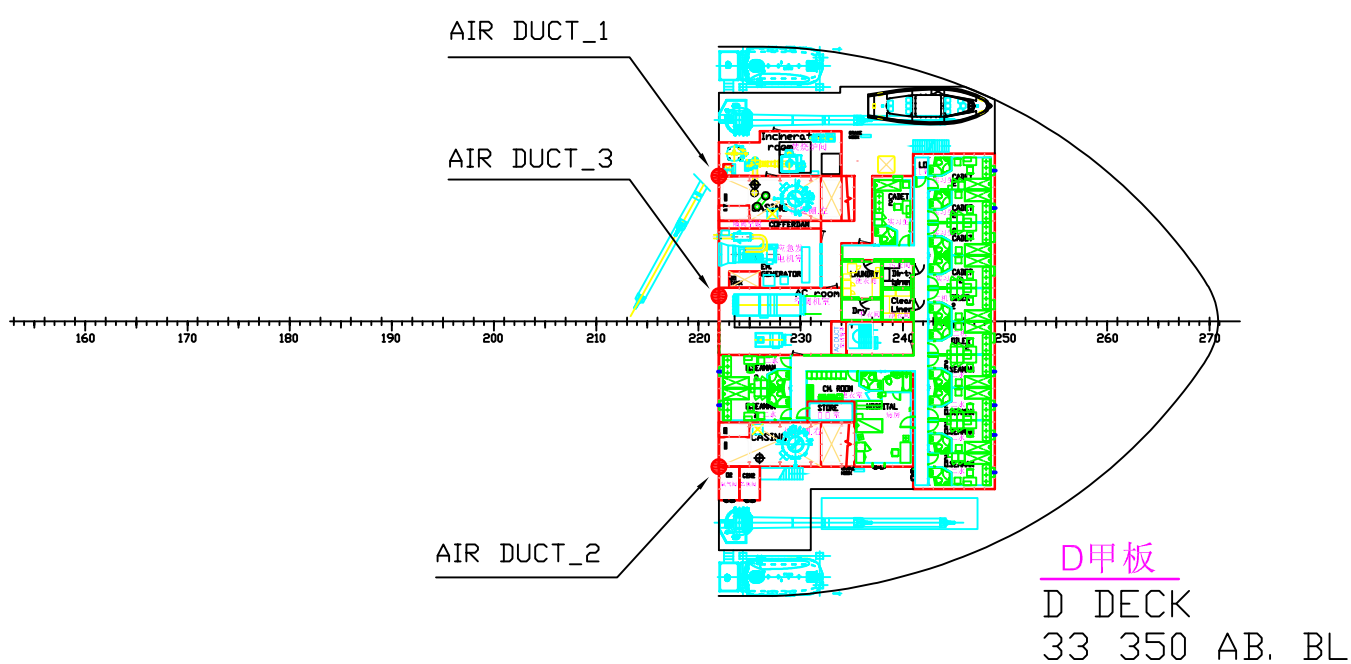
Z=14



Z=12.9

 $Z=8$ 

UNPROTECTED OPENINGS USED IN INTACT STABILITY CALCULATIONS



NAME	TEXT	FR #	Y m	Z m
AIR_DUCT_1	air duct to BW comp. syst. tanks	222.00	11.375	33.350
AIR_DUCT_2	air duct to BW comp. syst. tanks	222.00	-11.375	33.350
AIR_DUCT_3	air duct / ventilation of MER	230.00	0.000	33.350

3.7 Displacement calculation from the draught readings

- 1). Read the draught at fore, midship and aft draught marks.

dFo at frame 260
dMo at frame 135
dAo at frame 5 + 0.240m

Item	At Aft Draft Mark dAo (m)	At Mid Draft Mark dMo (m)	At Fwd Draft Mark dFo (m)
Port side			
Starboard side			
Average			

- 2). Calculate the draught at "FP" (dF).

$dF = dFo - 8.37 (dAo - dFo) / 203.76$
 $dA = dAo$
 $dM = dMo - 2.305 (dAo - dFo) / 203.76$

- 3) Trim (t), mean draught above base line (dM) and deflection (δ_d) are obtained as follows:

$t = dA - dF$ _____ in meter

$dMa = (dA + dF) / 2$ _____ in meter

$\delta_d = dM - dMa$ _____ in meter

Where:

dM : draught at midship
t > 0 : trim by the stern
t < 0 : trim by the bow
 $\delta_d > 0$: sagging condition
 $\delta_d < 0$: hogging condition

- 4). Displacement (D_1), moment to change trim per cm (MCT), tons per cm immersion (TPC) and center of flotation (LCA) corresponding to "dM" should be read from "Hydrostatic Tables".

- 5). Displacement correction due to trim and deflection.

a. Correction due to trim (D_2)

$D_2 = (LCA - 106.065) \times t \times TPC \times 100 / Lpp$ _____ in tonne

b. Correction due to deflection (D_3)

$$D_3 = (TPC - (dA + dF)/2 \times MCT) \times \delta_d \times 100 \text{ in tonne}$$

- 6). Displacement correction due to specific gravity of sea water (D_4)
when the specific gravity of sea water differs from standard of 1.025, displacement should also be corrected by following formula:

$$D_4 = (D_1 + D_2 + D_3) \times (S.G. - 1.025) / 1.025$$

- 7). Final displacement will be determined as follows.

$$D = D_1 + D_2 + D_3 + D_4 \text{ in tonne}$$

3.8 Determination of stability on board

3.8.1 General

Using the stability forms 0, 1 and 2, together with the tank tables, the hydrostatic tables, tank capacity tables and the cross curves of stability tables, it is possible to carry out stability calculations for other than the pre-calculated loading conditions.

The tables, which are used, contain all necessary basic data for the calculations that can be done in forms 0, 1 and 2.

Following tables and forms are used in the calculations:

Tank Information Tables (Ch 5.3.1)

Tank Capacities Tables (Ch 5.3.2)

GZ Correction Tables (Ch 5.3.3)

Hydrostatic Tables (Ch 5.1)

Cross-Curves of Stability (KN-Values) Tables (Ch. 5.2)

Form 0: Blank form tank capacity calculation (Ch 3.10)

Form 1: Blank form weight calculation (Ch 3.10).

Form 2: Blank form trim and stability calculation (Ch 3.10).

3.8.2 Instructions for stability calculations using the blank forms 0, 1 and 2

To get correct results of these calculations it is recommended to read the instructions very carefully and follow them step by step.

3.8.2.1 Tank capacity calculation, FORM 0

Enter the individual weight items of each tank. It's not necessarily required to use this form. The purpose of this form is to help filling of FORM 1. If FORM 0 is not used, please start the stability calculations from FORM 1.

3.8.2.2 Weight and moment calculations, FORM 1

- Enter the individual weight items (Crew, Passengers, Provisions, Tank capacities, Cargo and other miscellaneous weights) in the column *WEIGHT*. If Form 0 has

been used, copy the summary of tank capacity of each purpose from FORM 0 to FORM1. Lightweight item is based on the lightweight calculation.

- Determine the Vertical Centres of Gravity above baseline of the individual weights and enter those in the column *VCG*.
- Multiply the *Weight* of each individual weight item with its corresponding *VCG*. Enter the results in column *MV*.
- Summate the *Weight* column and enter the result in the *TOTAL SUM WEIGHT* cell (1).
- Copy the *SUM WEIGHT* (1) to the *Displacement* (1) result.
- Summate the *MV* column and enter the result in the *TOTAL SUM MV* cell (3).
- Divide the *TOTAL SUM MV* by *Displacement* (3) / (1) and enter the result in the *VCG* calculation result (5).

The result in of this calculation *VCG* is the Vertical Centre of Gravity above baseline of the loaded vessel.

Note: The next 4 steps are required only to be considered if a trim calculation is to be carried out

- Determine the Longitudinal Centres of Gravity from frame 0 of the individual weights and enter those in column *LCG*.
- Multiply the *Weight* of each individual weight item with its corresponding *LCG* and enter the result in column *ML*.
- Summate the *ML* column and enter the result in the *TOTAL SUM ML* (2).
- Divide the *TOTAL SUM ML* by *Displacement* (2) / (1) and enter the result in the *LCG* calculation result (6).

The result of this calculation *LCG* is the Longitudinal Centre of Gravity from frame 0 of the loaded vessel.

- Determine the Free Surface Moment of all partially filled tanks and enter those in column *FSM*.
- Summate the *FSM* column and enter the result in the *TOTAL SUM FSM* (4).
- Divide the *TOTAL SUM FSM* by *Displacement* (4) / (1) and enter the result in the *free Surf. correction* calculation result (7).

The result of this calculation *free Surf. correction* is the loss of stability due to free surfaces, the GM value is corrected with this value.

3.8.2.3 Explanation on determination of the weights, centres of gravity and free surface moments of tanks

Completely filled tanks

Weight, Vertical and Longitudinal Centres of Gravity are taken directly from the *Tank Information Tables*, the Free Surface Moments are taken as 0 tm.

Partially filled tanks

Weight, Vertical and Longitudinal Centres of Gravity and Free Surface Moments are taken from the *Tank Capacity Tables* of the individual tanks based on the filling percentage of the tank. Intermediate values of the required items can be derived by linear interpolation.

OR

Weight, Vertical and Longitudinal Centres of Gravity and Free Surface Moments are derived from soundings and the Sounding Tables documents. The Free Surface Moments have to be calculated by multiplying the Transversal Moment of Inertia (TMY) with the density of the contents for each individual tank.

3.8.3 Stability and trim calculation, FORM 2

3.8.3.1 Stability calculation

- Transfer necessary values from FORM 1 to FORM 2; (1), (6), (5) and (7).

Note: The *Displacement* which is used to determine the values as described in the following step is in seawater (density = 1.025 t/m.) If the vessel is floating in water with a different density the difference of densities needs to be taken into account. If, for example, the vessel is in freshwater with a density of 1.000 t/m, the *Displacement* as used in the remaining steps needs to be multiplied with a factor of $1.000 / 1.025 = 0.976$.

- Determine the *Draught* (8), *Transverse Metacentric Height KMT* (9), *Longitudinal Centre of Buoyancy LCB* (11), *Moment to Change Trim MCT* (13) and *Longitudinal Centre of Flotation LCA* (16) from the *Hydrostatic Tables* using the (corrected) *Displacement* (1). Intermediate values can be calculated by linear interpolation.
- Subtract the *VCG* from the *KMT* (9) – (5) and enter the result in the *GM₀* (Solid Metacentric Height) calculation result (10).
- Subtract *dGM* from the *GM₀* (10) – (7) and enter the result in the *GM_f* (Fluid Metacentric Height) calculation result.
- Add *VCG* to *dGM* (5) + (7) and enter the result in the *KG_f* (Corrected Vertical Height of Centre of Gravity) calculation result (19).

Based on the calculated values of KG_f an assessment of the stability can be made by checking the values in the corresponding limiting curve (Chapter 3.11). The calculated KG_f has to be below the KG-limit curves.

Note: To make a correct assessment of the KG_f one has to use the limiting curves corresponding with the correct trim.

3.8.3.2 Trim calculation

Note: Care needs to be taken to use the correct sign (+) and (-) for all values derived and calculated.

- Subtract LCG from LCB (11) – (6) and enter the result in the *trim arm* calculation result (12).
- Multiply the *trim arm* with the *Displacement* (12) * (1) and enter the result in the *trim Moment* calculation result (14).
- The *trim* is calculated by dividing the *trim Moment* by the MCT (14) / (13) and then divided by 100 to obtain metres.
- The *Draught at Aft Perpendicular* T_a is calculated as follows

$$\text{Draught} + \text{Longitudinal Centre of Buoyancy} * \frac{\text{trim}}{\text{Length between Perpendiculars}}$$

$$\Rightarrow T + LCA * \frac{\text{trim}}{L_{pp}} \Rightarrow (8) + (16) * \frac{(15)}{212.13}$$

- The *Draught at Fore Perpendicular* T_f is calculated by adding the *trim* to the calculated *Draught at Aft Perpendicular* T_a + (15) (Be aware of the sign of the individual values).

3.8.3.3 Calculation of the righting lever curve (GZ curve)

- The *Cross Curve ordinate* KN values (18) for each heeling angle are found in the *Cross Curves of Stability Tables* using the derived *Draught* (8). In case a trim calculation has been made the KN values need to be taken from the corresponding table for this trim (KN values for different trims are found in document SC-100-07 Hydrostatic data). Linear interpolation to be used for intermediate values of trim.
- Calculate intermediate values in row (20), $KG_f * \sin(\theta)$, (19) * (17).
- For a more accurate GZ-curve, calculate the *GZ Correction from FSM*:

Derive, of all partially filled tanks, the correction factor from the *GZ Corrections Table* at each heeling angle.

Summate these values for each heeling angle.

Divide these summations by the *Displacement* (1) and enter the results in the *GZ Correction from FSM* row (21).

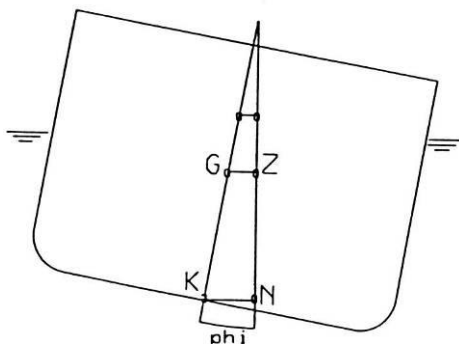
- Calculate the *righting lever GZ* for each heeling angle by deducting the $KG_f \cdot \sin(\theta)$ and the *GZ Correction from FSM* from the *Cross Curve ordinate KN* value, (18) – (20) – (21).

3.8.3.4 Drawing of the righting lever curve (GZ curve)

The calculated KG_f value is to be drawn from the baseline on the dashed KG-line and the final point is to be connected with the zero-point (KG-straight line).

The righting lever values GZ calculated are to be drawn on the degree-lines of the diagram and to be connected by a smooth curve. The KG-straight line must form a tangent to the GZ -curve through the origin.

The following figure shows the different components for the stability.



3.8.4 Notes on stability determination

The ship is to comply with the rules laid out in SOLAS chapter II-1, MSC 216(82). It must be ensured that any loading condition complies with this rule.

3.8.4.1 General

For a summary of the intact stability rules reference is made to Chapter 3.3. The drawn GZ -curve for the calculated loading case needs to be evaluated according to these regulations.

The ship is calculated to fulfil damage stability calculations according to SOLAS chapter II-1, MSC 216(82).

3.8.4.2 Assessment of stability

To make an assessment of the calculated loading case one needs the following figures;

T Draught [m],
trim [m],

KG_f Corrected Vertical Height of Centre of Gravity [m],

and limiting curves (Ch. 3.11).

- Choose the limiting curves with the correct trim value. For intermediate trim values take the next 'highest' graph.

The calculations are limited to 2.0 m forward and 2.0 m aft trim, this mean that any loading case exceeding these trim values can not be assessed with this document. The loading cases should therefore be limited to these trim values

- The loading case can be presented as a point at the intersection of Draught (T) and KG (KG_f) (T-KG point).

For the T-KG point one uses the Draught axis, the horizontal x-axis, and the KG axis, the left y1-axis.

- When the T-KG loading case point is below the limiting curves the loading case is fulfilling the requirements for intact and damage stability as mentioned SOLAS chapter II-1, MSC 216(82).

3.9 Worked example on stability

Worked example on calculation and checking the stability of a loading condition 'LC01 Light ballast departure' is presented in this chapter. Instructions given in Ch. 3.8 and blank stability forms given in Ch. 3.10 are utilized in the calculations.

FORM

1

[illegible]

Calculation of loaded ship

Displacement	(1)		47516,5 t
LCG	(6)	= (2) / (1)	109,52 m
VCG	(5)	= (3) / (1)	6,64 m
Free Surface Correction	(7)	= (4) / (1)	0,26 m

Loading condition: LC01 Light ballast dep.

FORM

2

Displacement (1) [t]	Draught (8) [m]		KMT (9) [m]	LCB (11) [m]		MCT (13) [t•m/cm]		LCA (16) [m]
47516,5	7,00		26,10	109,52		995,6		105,52

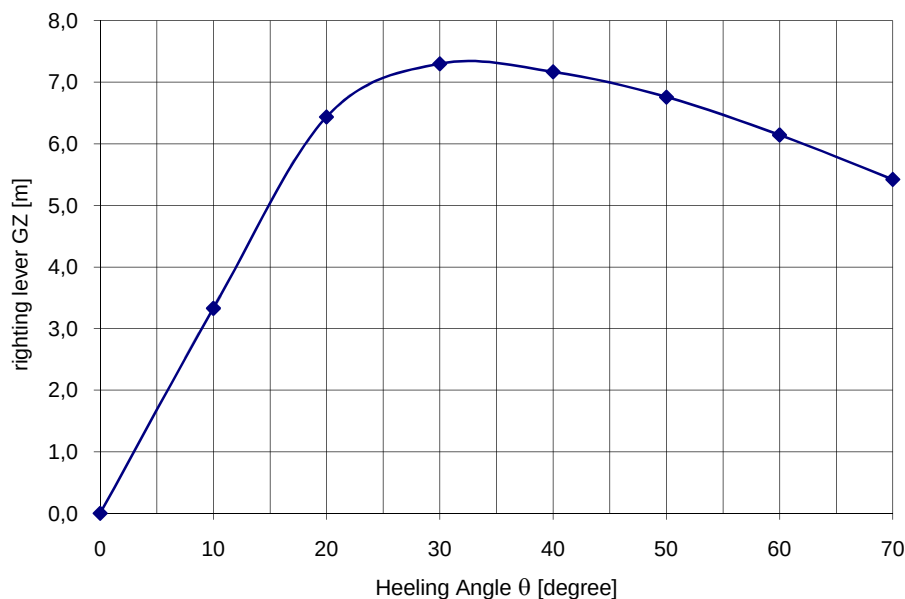
Calculations

KMT	(9)	26,10 m	LCB	(11)	109,52 m
VCG	(5)	6,64 m -	LCG	(6)	109,52 m -
GMo	(9) - (5)	(10) 19,58 m	tr arm	(11) - (6)	(12) 0,00 m
dGM	(7)	0,26 m	tr Mom	(12) * (1)	(14) 0,00 m
GMf	(10) - (7)	19,32 m -	trim	(14) / (13) / 100	(15) 0,00 m
KGf	(5) + (7)	(19) 6,9 m			

$$Ta \ (8) \pm (16) * (15) / 212.13 \quad 7,00 \text{ m}$$

$$Tf \ (8) \pm (16) * (15) / 212.13 \quad 7,00 \text{ m}$$

Heeling angle θ		10 degree	20 degree	30 degree	40 degree	50 degree	60 degree	70 degree
sin (θ)	(17)	0,174	0,342	0,500	0,643	0,766	0,866	0,940
KN	(18)	4,567	8,878	10,872	11,759	12,228	12,320	11,746
KGf * sin (q)	(19) * (17)	(20) 1,198	2,360	3,450	4,435	5,286	5,976	6,324
GZ corr. from FSM	(21)	0,039	0,080	0,121	0,154	0,181	0,201	0,215
GZ (righting lever)	(18)-(20)-(21)	3,330	6,438	7,301	7,170	6,761	6,144	5,422



3.10 Blank stability forms

This chapter contains blank forms for calculating and checking the stability for actual loading conditions.

Loading condition:	FORM
	0

ITEMS	Volume [m3]	Density [t/m3]	Weight [t]	LCG [m]	L.mom [t*m]	VCG [m]	V.mom [t*m]	FSM [t*m]
Heavy Fuel Oil								
HFO Storage Tank 1								
HFO Storage Tank 2								
HFO Storage Tank 3								
HFO Storage Tank 4								
HFO Settling Tank P								
HFO Settling Tank S								
HFO Service Tank P								
HFO Service Tank S								
Total								

Diesel Oil								
MDO Storage P								
MDO Storage S								
MDO Service P								
MDO Service S								
MGO Storage Tank								
Total								

Lubricating Oil								
L.O. DRAIN TK.								
ST L.O. DRAIN TK.								
Used LO								
NO.1 M/E & A/E L.O. STO TK								
NO.2 M/E & A/E L.O. STO TK								
Total								

Technical Water								
Feed Water								
Total								

Fresh Water								
Fresh Water P								
Fresh Water S								
Fresh Water C								
Potable water								
Total								

Ballast Water								
Water Ballast 01.03								
Water Ballast 01.11								
Water Ballast 01.12								
Water Ballast 02.05								
Water Ballast 02.06								
Water Ballast 02.07								
Water Ballast 02.08								
Water Ballast 02.09								
Water Ballast 02.10								
Water Ballast 02.11								
Water Ballast 02.12								
Water Ballast 03.01								

ITEMS	Volume [m3]	Density [t/m3]	Weight [t]	LCG [m]	L.mom [t*m]	VCG [m]	V.mom [t*m]	FSM [t*m]
Water Ballast 03.02								
Water Ballast 03.05								
Water Ballast 03.06								
Water Ballast 03.07								
Water Ballast 03.08								
Water Ballast 03.09								
Water Ballast 03.10								
Water Ballast 03.11								
Water Ballast 03.12								
Water Ballast 04.01								
Water Ballast 04.02								
Water Ballast 04.07								
Water Ballast 04.08								
Water Ballast 04.09								
Water Ballast 04.10								
Water Ballast 04.11								
Water Ballast 04.12								
Water Ballast 05.01								
Water Ballast 05.02								
Water Ballast 05.05								
Water Ballast 05.06								
Water Ballast 05.07								
Water Ballast 05.08								
Water Ballast 05.09								
Water Ballast 05.10								
Water Ballast 05.11								
Water Ballast 05.12								
Water Ballast 06.01								
Water Ballast 06.02								
Water Ballast 06.05								
Water Ballast 06.06								
Water Ballast 06.07								
Water Ballast 06.08								
Water Ballast 06.09								
Water Ballast 06.10								
Water Ballast 06.11								
Water Ballast 06.12								
Water Ballast 07.01								
Water Ballast 07.02								
Water Ballast 07.05								
Water Ballast 07.06								
Water Ballast 07.07								
Water Ballast 07.08								
Water Ballast 07.09								
Water Ballast 07.10								
Water Ballast 07.11								
Water Ballast 07.12								
Water Ballast 08.01								
Water Ballast 08.03								
Water Ballast 08.04								
Water Ballast 08.07								
Water Ballast 08.08								
Water Ballast 08.09								
Water Ballast 08.10								

ITEMS	Volume [m3]	Density [t/m3]	Weight [t]	LCG [m]	L.mom [t*m]	VCG [m]	V.mom [t*m]	FSM [t*m]
Water Ballast 08.11								
Water Ballast 08.12								
Water Ballast 09.01								
Water Ballast 09.07								
Water Ballast 09.08								
Water Ballast 09.09								
Water Ballast 09.11								
Water Ballast 09.12								
Water Ballast 10.01								
Water Ballast 10.05								
Water Ballast 10.06								
Water Ballast 10.11								
Water Ballast 10.12								
Water Ballast 11.01								
Water Ballast 11.11								
Water Ballast 11.12								
Total								
Fixed FW ballast								
Water Ballast 02.01								
Total								
Gray Water								
Black Water								
Gray Water								
Total								
Bilge Water								
Bilge Water Settling								
Bilge Water								
Total								
Sludge								
Sludge								
Total								

Loading condition

FORM

2

Displacement (1) [t]	Draught (8) [m]		KMT (9) [m]	LCB (11) [m]		MCT (13) [t•m/cm]		LCA (16) [m]

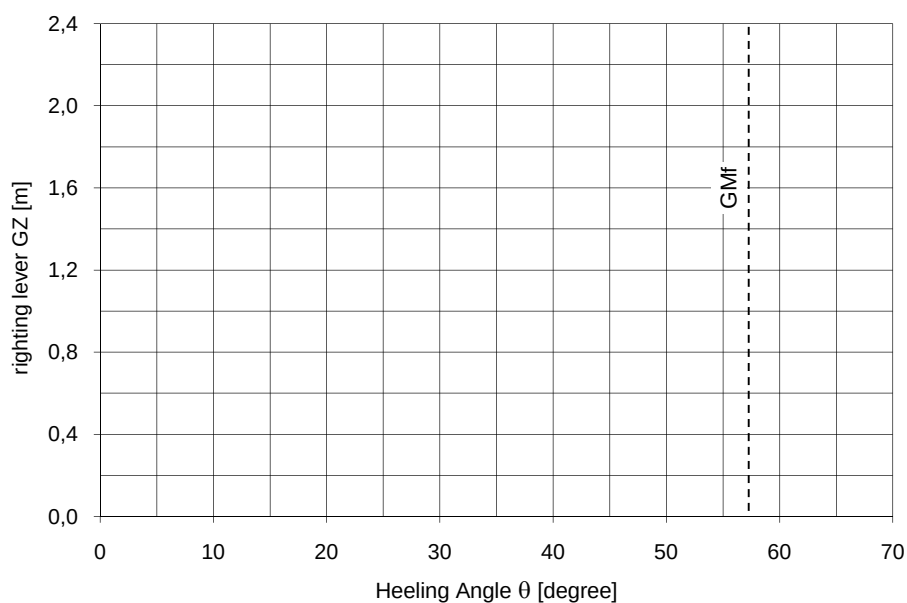
Calculations

KMT	(9)	_____ m	LCB	(11)	_____ m
VCG	(5)	_____ m -	LCG	(6)	_____ m -
GMo	(9) - (5)	(10) _____ m	tr arm	(11) - (6)	(12) _____ m
dGM	(7)	_____ m	tr Mom	(12) * (1)	(14) _____ m
GMf	(10) - (7)	_____ m -	trim		
			(14) / (13) / 100	(15)	_____ m
KGf	(5) + (7)	(19) _____ m			

$$Ta \text{ (8) } \pm (16) * (15) / 194.8 \text{ _____ m}$$

$$Tf \text{ (8) } \pm (16) * (15) / 194.8 \text{ _____ m}$$

Heeling angle θ	10 degree	20 degree	30 degree	40 degree	50 degree	60 degree	70 degree
sin (θ)	(17) 0,174	0,342	0,500	0,643	0,766	0,866	0,940
KN	(18)						
KGf * sin (θ)	(19) * (17)	(20)					
GZ (righting lever)	(18) - (20)						



3.11 Max. permissible KG_F curves

The stability of the vessel should comply with the intact stability criteria as specified in this booklet and the damage stability criteria contained in SOLAS chapter II-1, MSC 216(82).

When the vessel does not carry any deck cargo, these stability criteria will be complied with provided that the actual height of gravity of the vessel after correction for free surface effect, i.e. KG_F , is not more than the max. permissible value determined by the graphs on the following pages.

3.11.1 Max. KG_F

Maximum permissible KG_F curves from intact stability as well as damage stability are presented for different trims.

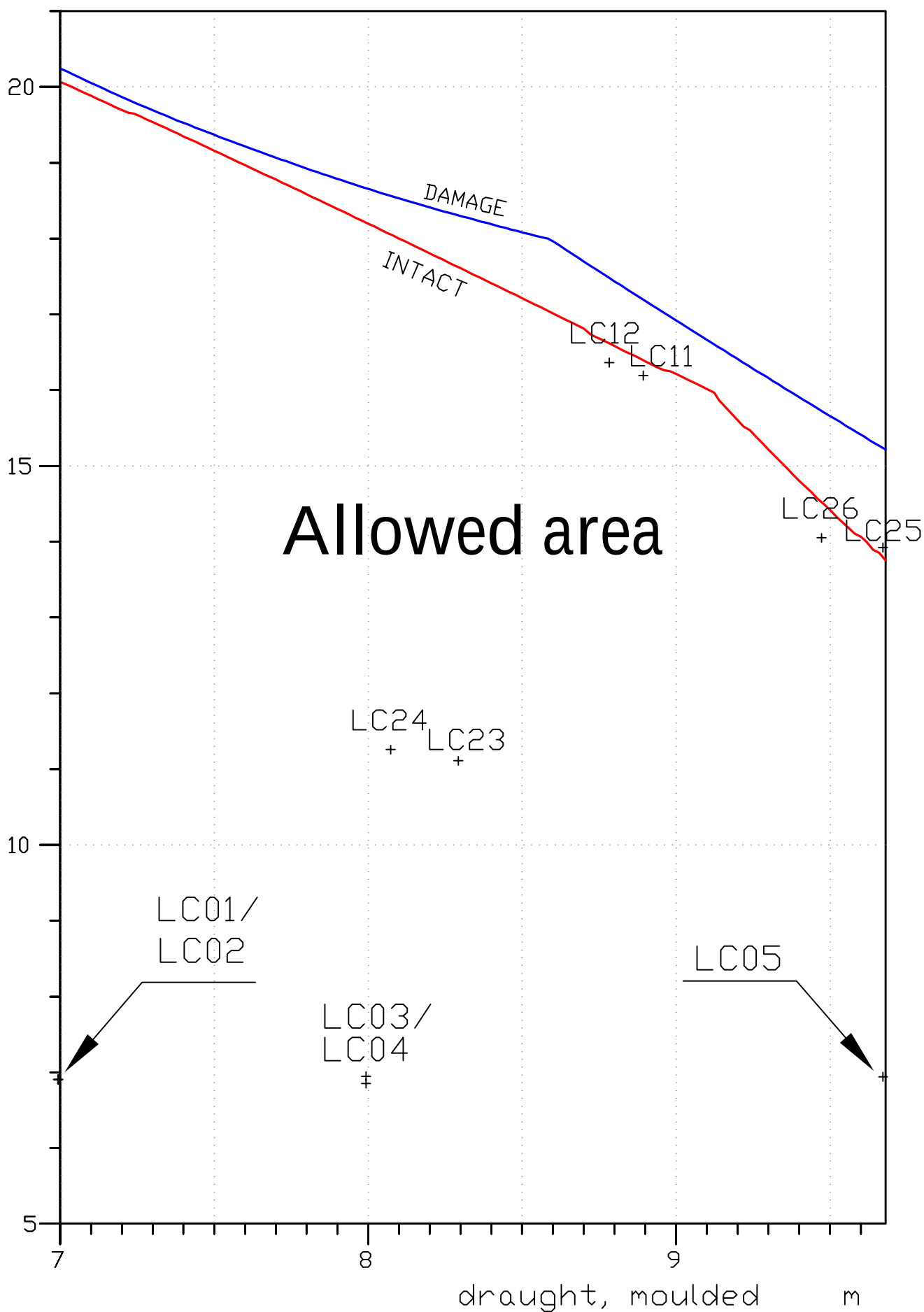
KG-T Summary, Trim -2
No effect of cargo buoyancy

Proj P6047/0

Date 2011-06-29

Time 11.18

Sign MUA



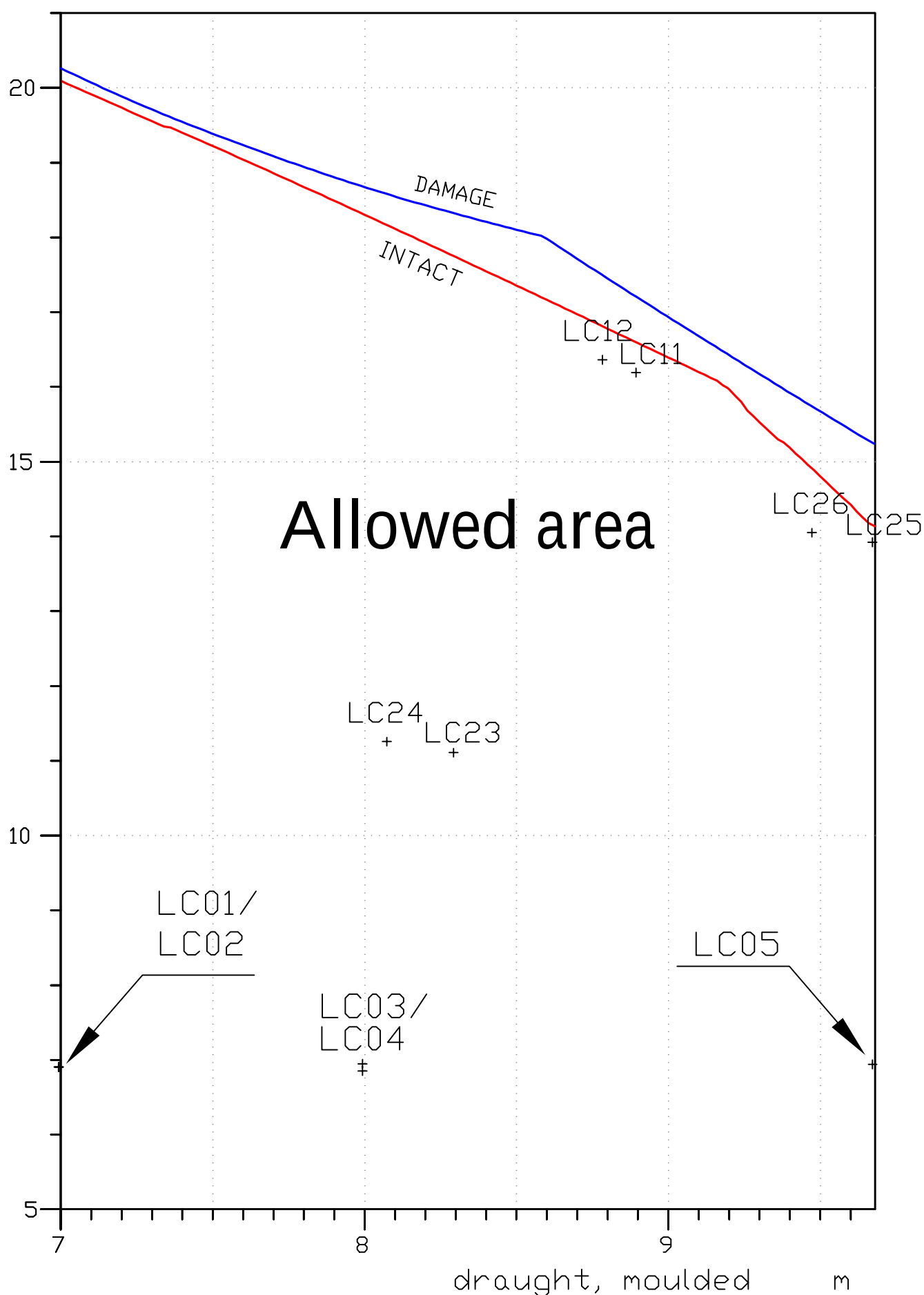
KG-T Summary, Trim -1
No effect of cargo buoyancy

Proj P6047/0

Date 2011-06-29

Time 11.09

Sign MUA



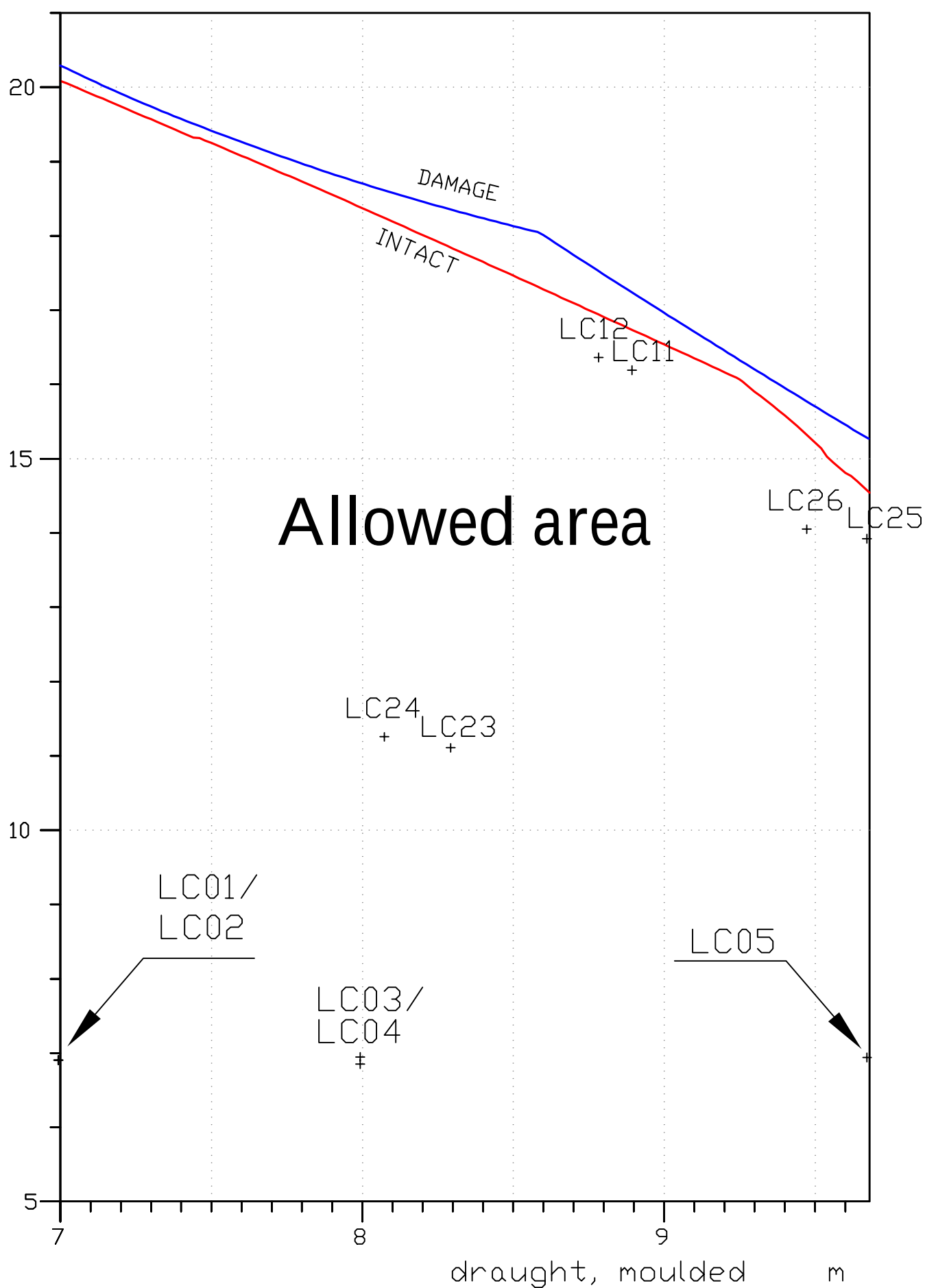
KG-T Summary, Trim 0
No effect of cargo buoyancy

Proj P6047/0

Date 2011-06-29

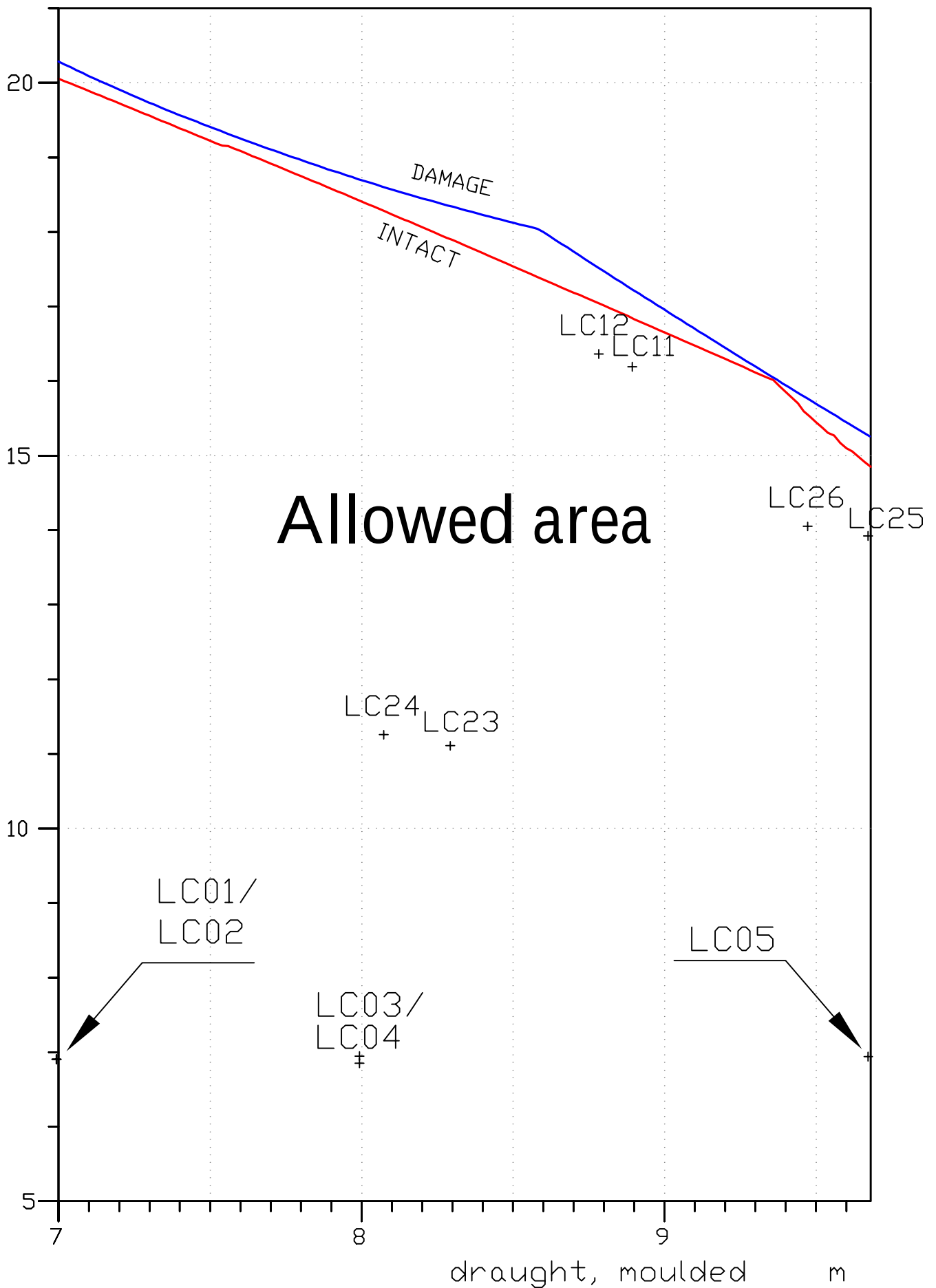
Time 10.31

Sign MUA



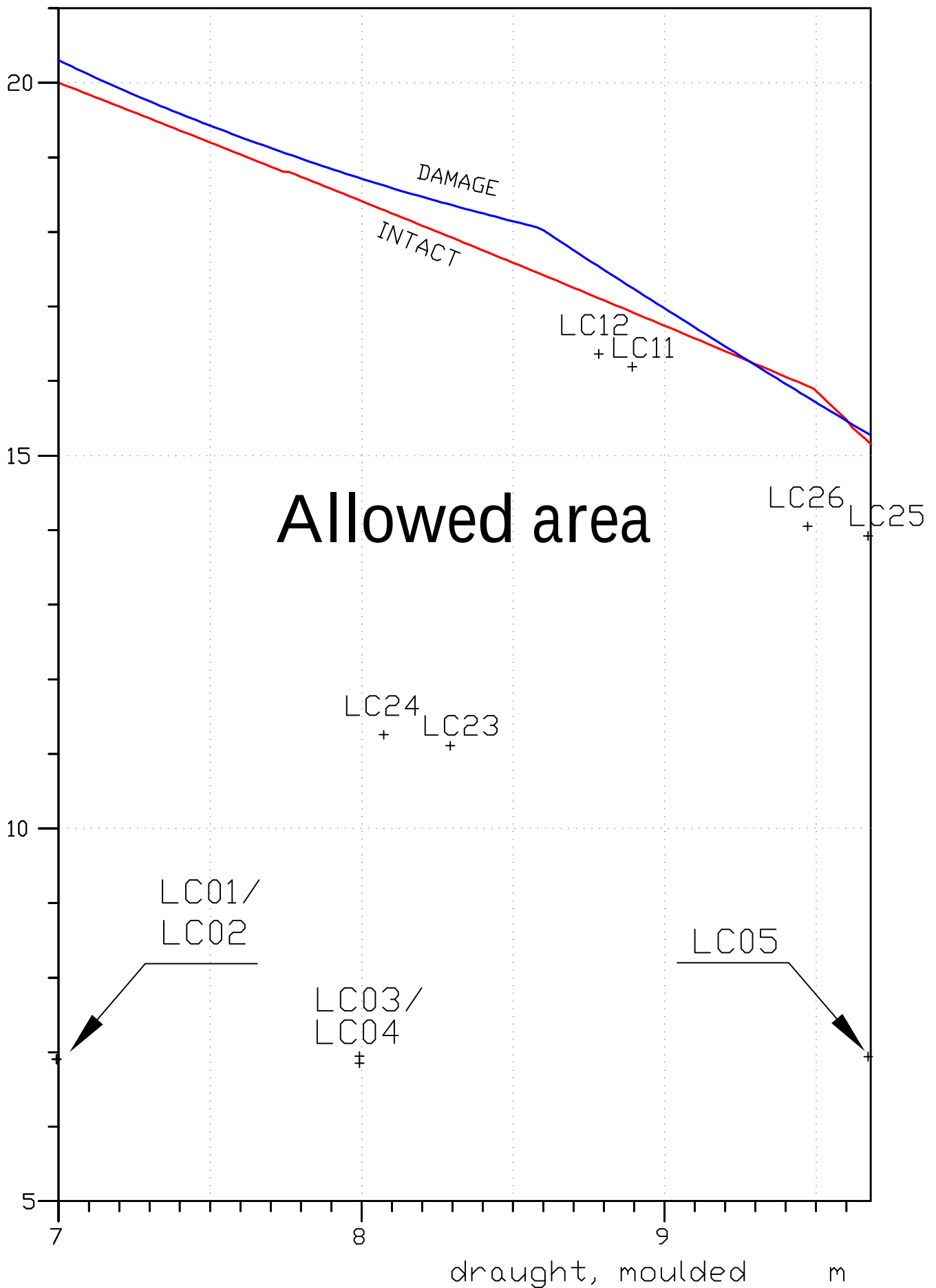
KG-T Summary, Trim 1
No effect of cargo buoyancy

Proj	P6047/0
Date	2011-06-29
Time	10.53
Sign	MVA



KG-T Summary, Trim 2
No effect of cargo buoyancy

Proj	P6047/0
Date	2011-06-29
Time	11.03
Sign	MVA

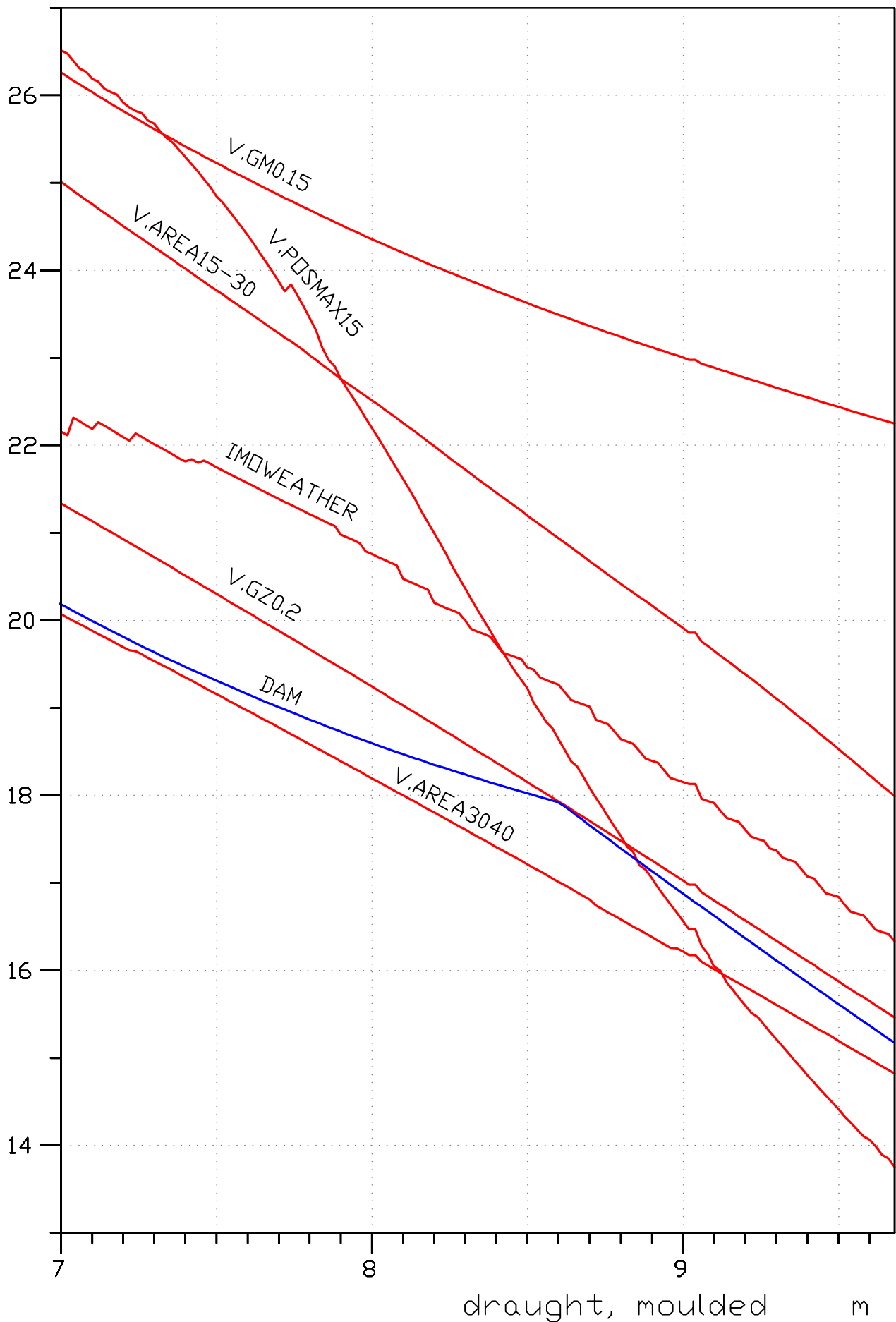


3.11.2 Max. KG_F per intact stability criterion

Maximum permissible KG_F curves are presented separately for each intact stability criterion. Also total permissible KG_F curve from damage stability is presented. Curves are shown for different trims.

KG-T Summary, Trim -2
No effect of cargo buoyancy

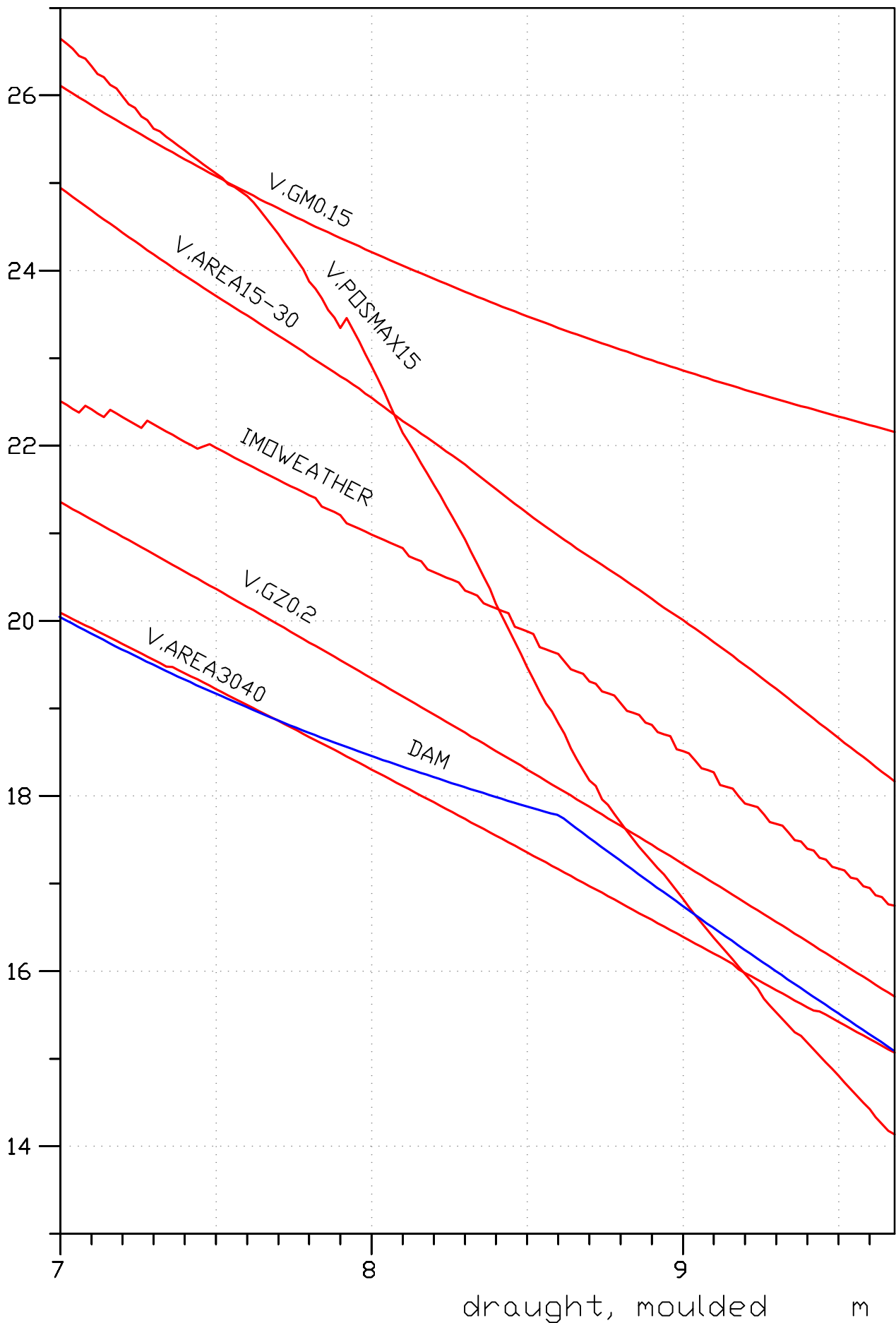
Proj	P60470
Date	2011-06-30
Time	9.55
Sign	MJA



KG-T Summary, Trim -1

No effect of cargo buoyancy

Proj	P60470
Date	2011-06-30
Time	9.53
Sign	MJA



KG-T Summary, Trim 0

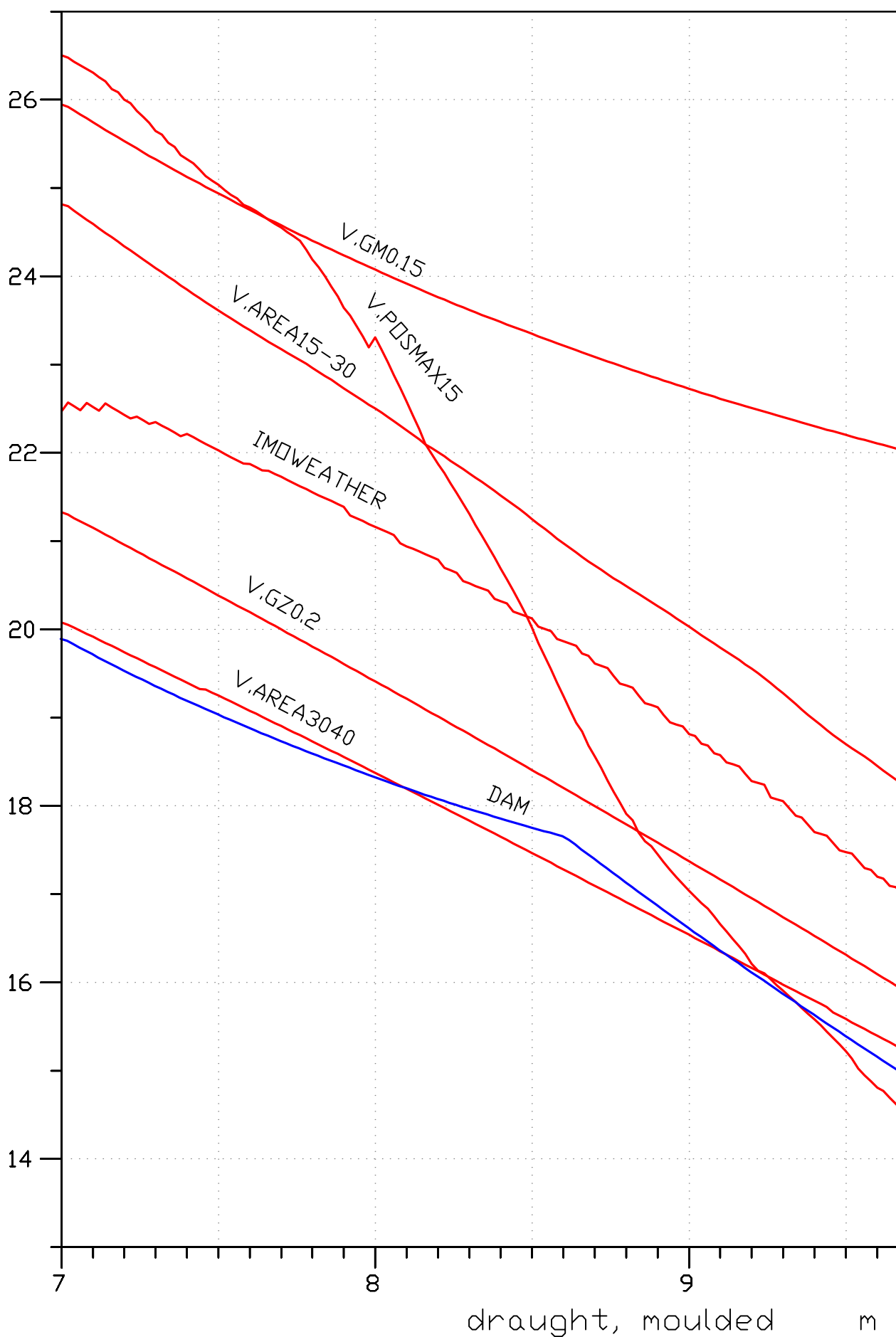
No effect of cargo buoyancy

Proj P60470

Date 2011-06-30

Time 9.48

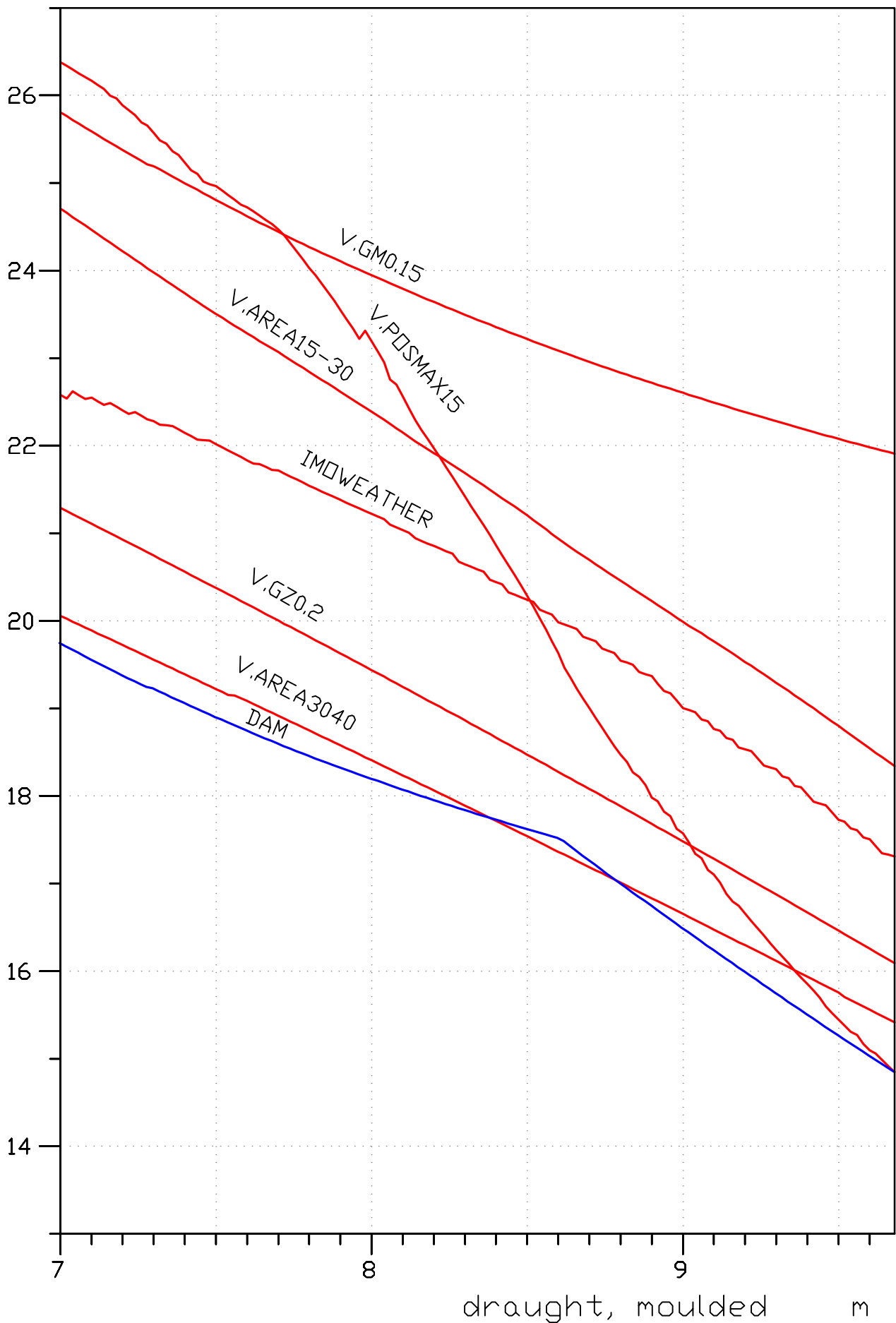
Sign MJA



KG-T Summary, Trim 1

No effect of cargo buoyancy

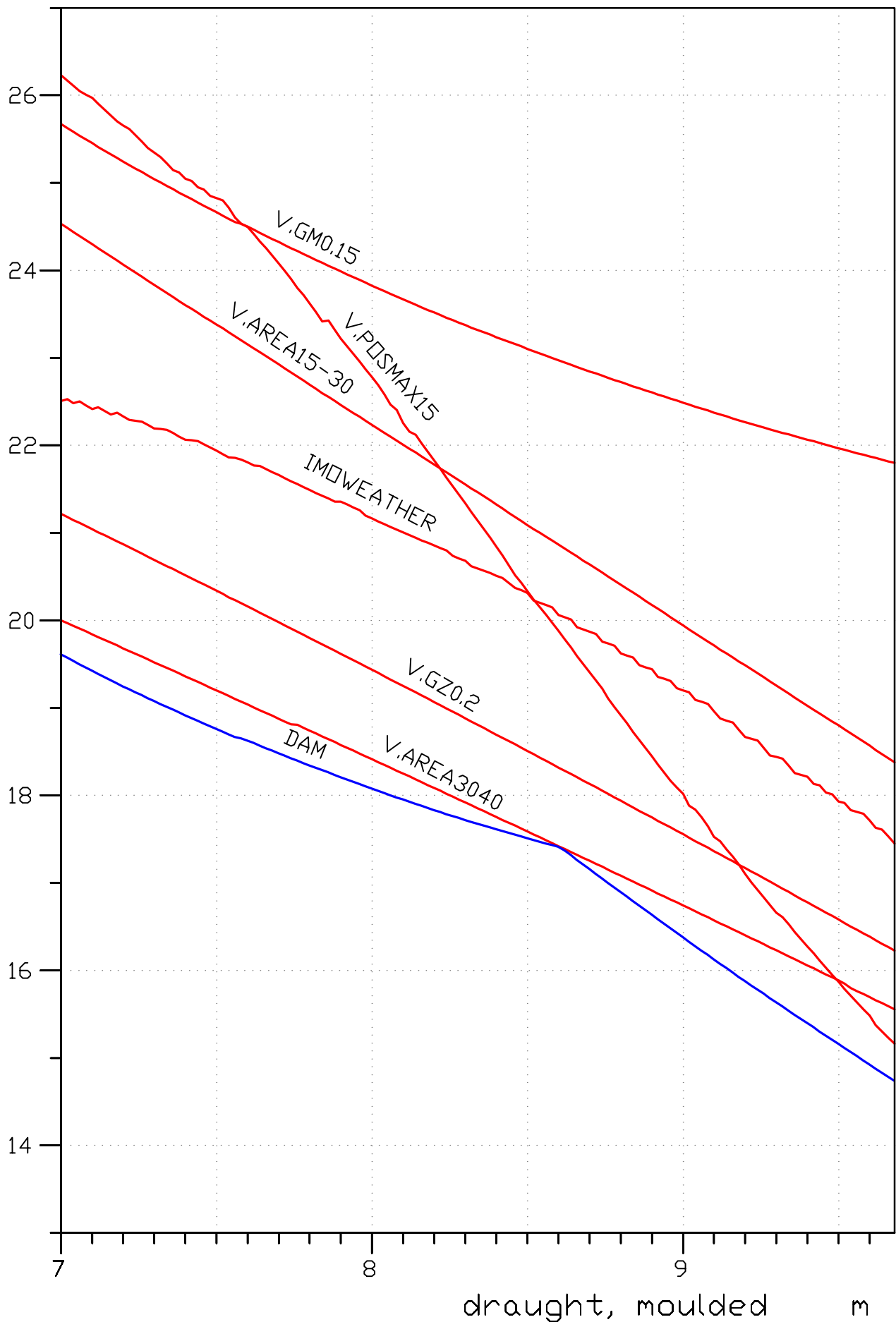
Proj	P60470
Date	2011-06-30
Time	9.51
Sign	MJA



KG-T Summary, Trim 2

No effect of cargo buoyancy

Proj	P60470
Date	2011-06-30
Time	9.55
Sign	MJA



4 Loading conditions

4.1 *Summary of the loading conditions*

CASE	TEXT	DISP t	DWT t	BW t	CARGO t	T m	TR m	HEEL degree	KMT m
LC00	Lightship, NOT A SAILING CONDITION	21192.5	321.1	0.0	0.0	3.394	-0.112	-0.0	44.581
LC01	Light ballast, departure	47516.5	26645.1	22223.9	0.0	7.000	0.000	0.0	26.097
LC02	Light Ballast, arrival	47521.4	26650.0	25385.8	0.0	7.000	0.000	0.0	26.116
LC03	Heavy ballast / min. ice draught, departure	55395.0	34523.6	30102.5	0.0	8.000	0.000	0.0	24.225
LC04	Heavy ballast / min. ice draught, arrival	55395.1	34523.7	33259.6	0.0	8.000	0.000	0.0	24.225
LC05	End of de-ballasting	68934.2	48062.8	46798.6	0.0	9.680	0.600	0.0	22.110
LC11	Non-Buoyant Cargo, departure	62691.4	41820.0	17398.9	20000.0	8.900	0.000	0.0	22.991
LC12	Non-Buoyant Cargo, arrival	61791.7	40920.3	19656.2	20000.0	8.790	-0.000	0.0	23.125
LC23	Buoyant Cargo (MLT 116C jackup), departure	57805.5	36934.1	23777.6	8654.0	8.300	0.000	0.0	23.769
LC24	Buoyant Cargo (MLT 116C jackup), arrival	56036.1	35164.7	25246.5	8654.0	8.080	0.000	0.0	24.098
LC25	Buoyant Cargo Super Gorilla 16000t, dep.	69161.3	48289.9	27787.4	16000.0	9.680	0.000	0.0	22.183
LC26	Buoyant Cargo Super Gorilla 16000t, arr.	67493.4	46622.0	29357.9	16000.0	9.480	0.000	0.0	22.372
LC31	Docking condition	37640.0	16768.6	15504.5	0.0	5.700	0.000	0.0	29.820

CASE	KGF m	GM m	GMCORR m	BMMIN tm	BMMAx tm	RELBM %	XRELBM m	SHMIN t	SHMAX t	RELSH %	XRELSH m
LC00	10.31	34.276	0.000	-1	174664	81.2	103	-2991	2925	49.8	166
LC01	6.90	19.193	0.260	-9	69381	43.6	27	-1874	1987	35.0	178
LC02	6.90	19.212	0.341	-4	68685	43.5	27	-1845	2021	34.6	179
LC03	6.95	17.276	0.137	-50174	26696	39.7	8	-3599	1349	53.6	58
LC04	6.85	17.371	0.224	-8894	26857	39.9	8	-1273	1214	32.1	199
LC05	6.94	15.170	0.601	-7950	45399	42.4	8	-1402	1568	25.8	34
LC11	16.19	6.801	0.129	-140355	34562	78.2	98	-4075	5368	77.5	135
LC12	16.36	6.763	0.200	-107671	42404	61.7	103	-4134	4993	72.1	135
LC23	11.11	12.661	0.166	-12007	90476	40.1	8	-2167	2093	40.4	195
LC24	11.26	12.842	0.226	-2	92614	39.9	8	-2272	1348	40.3	198
LC25	13.92	8.260	0.148	-3417	48415	43.9	8	-1233	1514	21.7	16
LC26	14.05	8.321	0.137	-5611	93613	47.3	48	-1913	1646	29.5	20
LC31	6.93	22.886	0.280	-3	91965	53.1	37	-1811	1873	32.9	23

CASE	name	
TEXT	descriptive name	
DISP	total displacement	t
DWT	mass of load	t
BW	mass of load	t

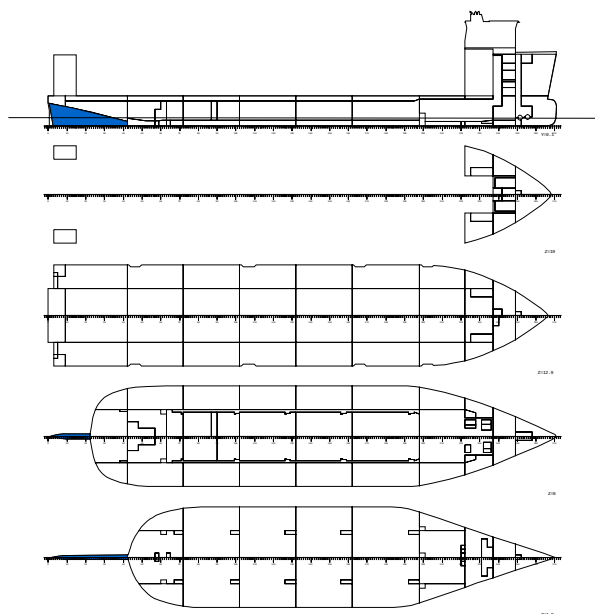
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P6047

LOADING CONDITIONS

DATE 2011-06-29
TIME 9.06
USER MVA
Page 2

CARGO	mass of load	t
T	draught, moulded	m
TR	trim	m
HEEL	heeling angle	degree
KMT	transv. metac. height	m
KGF	cgz of load	m
GM	metacentric height	m
GMCORR	GM correction	m
BMMIN	minimum bending moment	tm
BMMAX	maximum bending moment	tm
RELBM	relative bending moment	%
XRELBM	x where max. relative bending mom.	m
SHMIN	minimum shear force	t
SHMAX	maximum shear force	t
RELSH	relative shear force	%
XRELSH	x where max. relative shear force	m

4.2 LC00 Lightship, NOT A SAILING CONDITION



LOADING CONDITION LC00: Lightship, NOT A SAILING CONDITION

FLOATING POSITION

Mean draught (moulded)	3.39 m	KM above the moulded base	44.58 m
Draught at AP (moulded)	3.45 m	KG above the moulded base	10.34 m
Draught at FP (moulded)	3.34 m	GM0 (solid)	34.24 m
Trim (by stern)	-0.11 m	Free surface correction	0.00 m
Heeling	-0.0 deg	GM (fluid)	34.24 m
Mean draught (heel 0)	3.39 m		
Trim (heel 0)	-0.11 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Deadweight	321.1	14.73	0.00	3.55	0.0
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	21192.5	111.94	-0.01	10.34	0.0

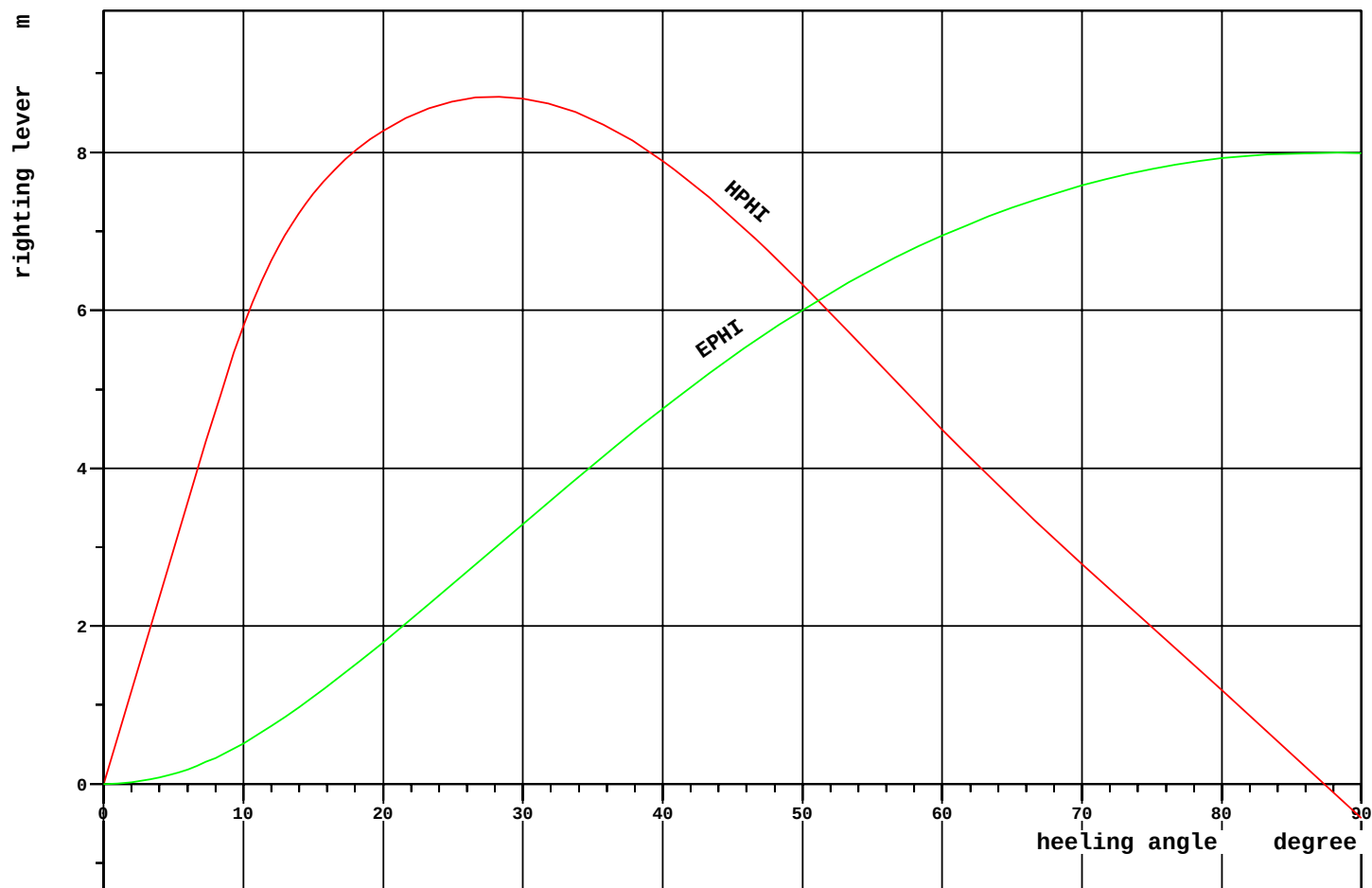
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 P6047

Loading Condition
 LC00

DATE 2011-06-30
 TIME 14.46
 USER MVA
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LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
—							—
Deadweight		321.1		14.73	0.00	3.55	0.0
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		21192.5		111.94	-0.01	10.34	0.0
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.057	3.002	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.461	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	8.683	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	28.077	deg	OK
V.GM0.15	GM > 0.15 m	0.150	34.241	m	OK
IMOWEATHER	IMO weather criterion	1.000	1.433		OK
ND-RANGE	Range of Stability acc. ND	36.000	87.315	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	25.719		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	89.308	deg	OK

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 P6047

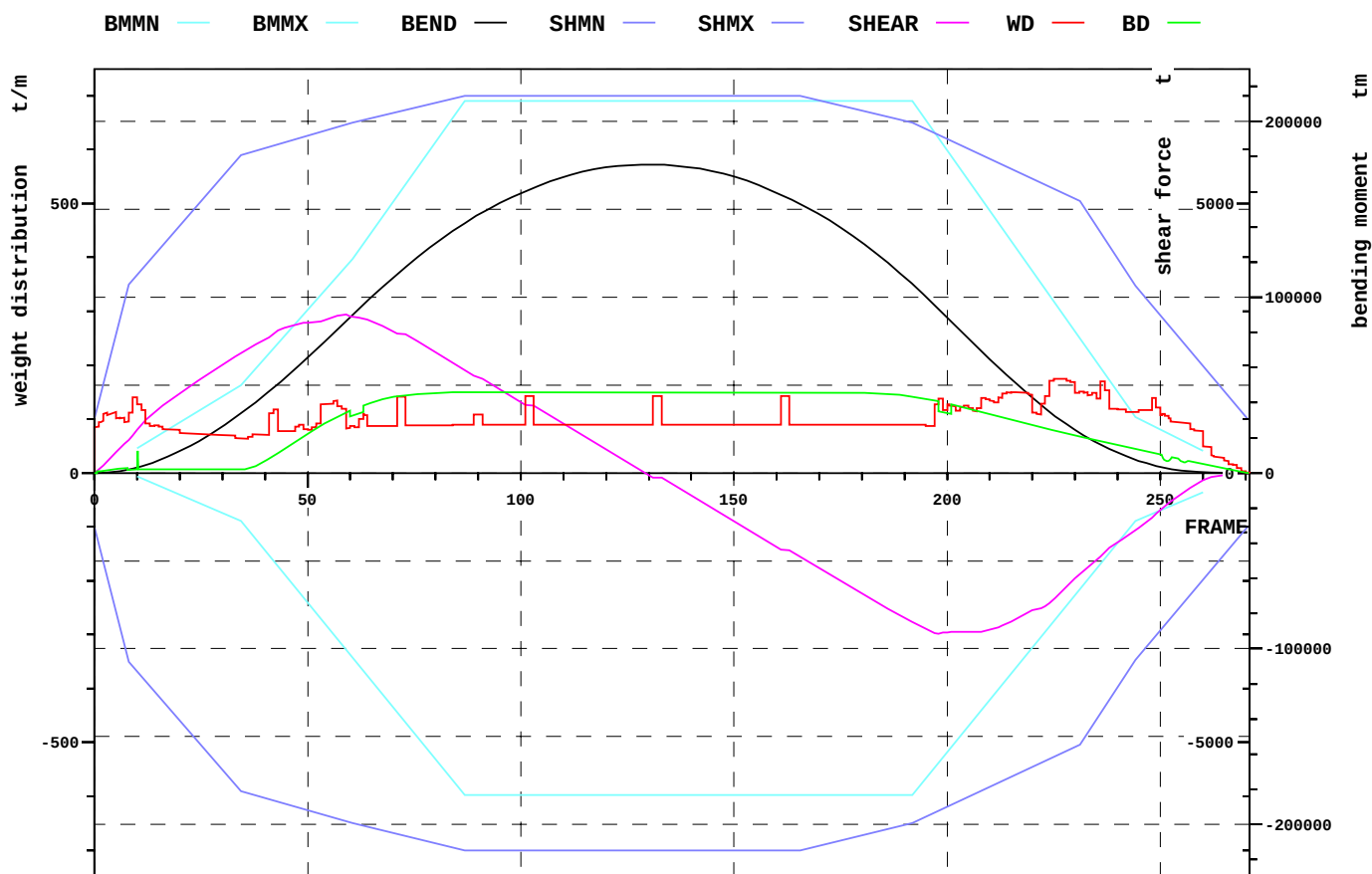
Loading Condition
 LC00

DATE 2011-06-30
 TIME 14.46
 USER MVA
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_1	UNPROTECTED	-	222.00	11.375	33.350	29.994
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	29.995
AIR_DUCT_2	UNPROTECTED	-89.3	222.00	-11.375	33.350	29.988

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

			x	frame
Shear force (min)	-2980.7 t	position:	158.4 m	198
Shear force (max)	2935.4 t		47.2 m	59
Max. rel. shear force	50.0 %		166.4 m	208
Sagging moment	-0.8 tm		215.2 m	269
Hogging moment	175487.9 tm		102.6 m	128
Max. rel. bending moment	81.6 %		102.6 m	128

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LC00

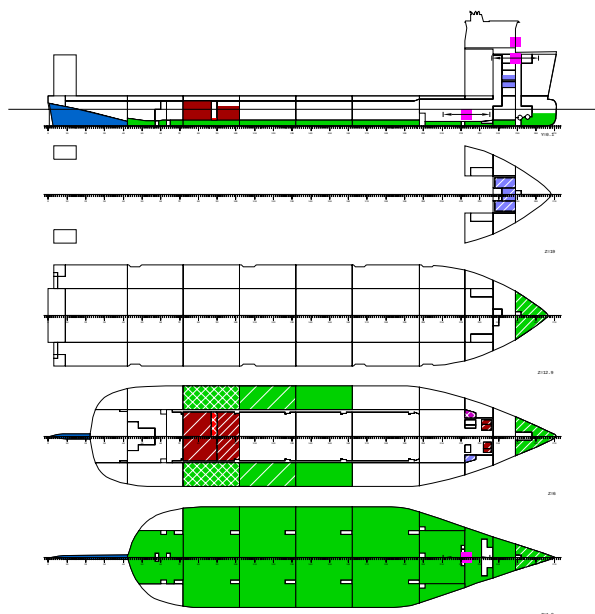
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LOADING CONDITION LC00, Lightship, NOT A SAILING CONDITION

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	13655	-13513	30261	24.1
42.00	33.60	48413	-50240	70924	62.8
57.00	45.60	81676	-94804	111952	70.7
72.00	57.60	115225	-139384	160929	69.6
87.00	69.60	142447	-183412	211763	64.9
102.00	81.60	161329	-183412	211763	74.5
117.00	93.60	172627	-183412	211763	80.2
132.00	105.60	175386	-183412	211763	81.6
147.00	117.60	170548	-183412	211763	79.1
162.00	129.60	157243	-183412	211763	72.4
180.00	144.00	131175	-183412	211763	59.2
198.00	158.40	93217	-164872	190357	45.3
210.00	168.00	64914	-129204	149176	39.5
222.00	177.60	38898	-93541	108000	31.4
237.00	189.60	14802	-48977	56548	20.9
249.00	199.20	4055	-22505	25984	9.5

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1534	-4679	4679	32.8
42.00	33.60	2576	-6075	6075	42.4
57.00	45.60	2914	-6418	6418	45.4
72.00	57.60	2579	-6718	6718	38.4
87.00	69.60	1908	-7000	7000	27.3
102.00	81.60	1259	-7000	7000	18.0
117.00	93.60	585	-7000	7000	8.4
132.00	105.60	-87	-7000	7000	1.2
147.00	117.60	-757	-7000	7000	10.8
162.00	129.60	-1425	-7000	7000	20.4
180.00	144.00	-2233	-6724	6724	33.2
198.00	158.40	-2981	-6269	6269	47.5
210.00	168.00	-2904	-5824	5824	49.9
222.00	177.60	-2506	-5380	5380	46.6
237.00	189.60	-1469	-4339	4339	33.9
249.00	199.20	-752	-3024	3024	24.9

4.3 LC01 Light ballast departure



LOADING CONDITION LC01: Light ballast, departure

FLOATING POSITION

Mean draught (moulded)	7.00 m	KM above the moulded base	26.10 m
Draught at AP (moulded)	7.00 m	KG above the moulded base	6.66 m
Draught at FP (moulded)	7.00 m	GM0 (solid)	19.44 m
Trim	0.00 m	Free surface correction	-0.26 m
Heeling	0.0 deg	GM (fluid)	19.18 m
Mean draught (heel 0)	7.00 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	22223.9	111.74	0.05	2.98	10050.8
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0

Deadweight 26645.1 106.95 0.00 3.69 12370.2

Lightweight 20871.4 113.44 -0.01 10.44
Displacement (1.025 t/m3) 47516.5 109.80 -0.00 6.66 12370.2

LOADS

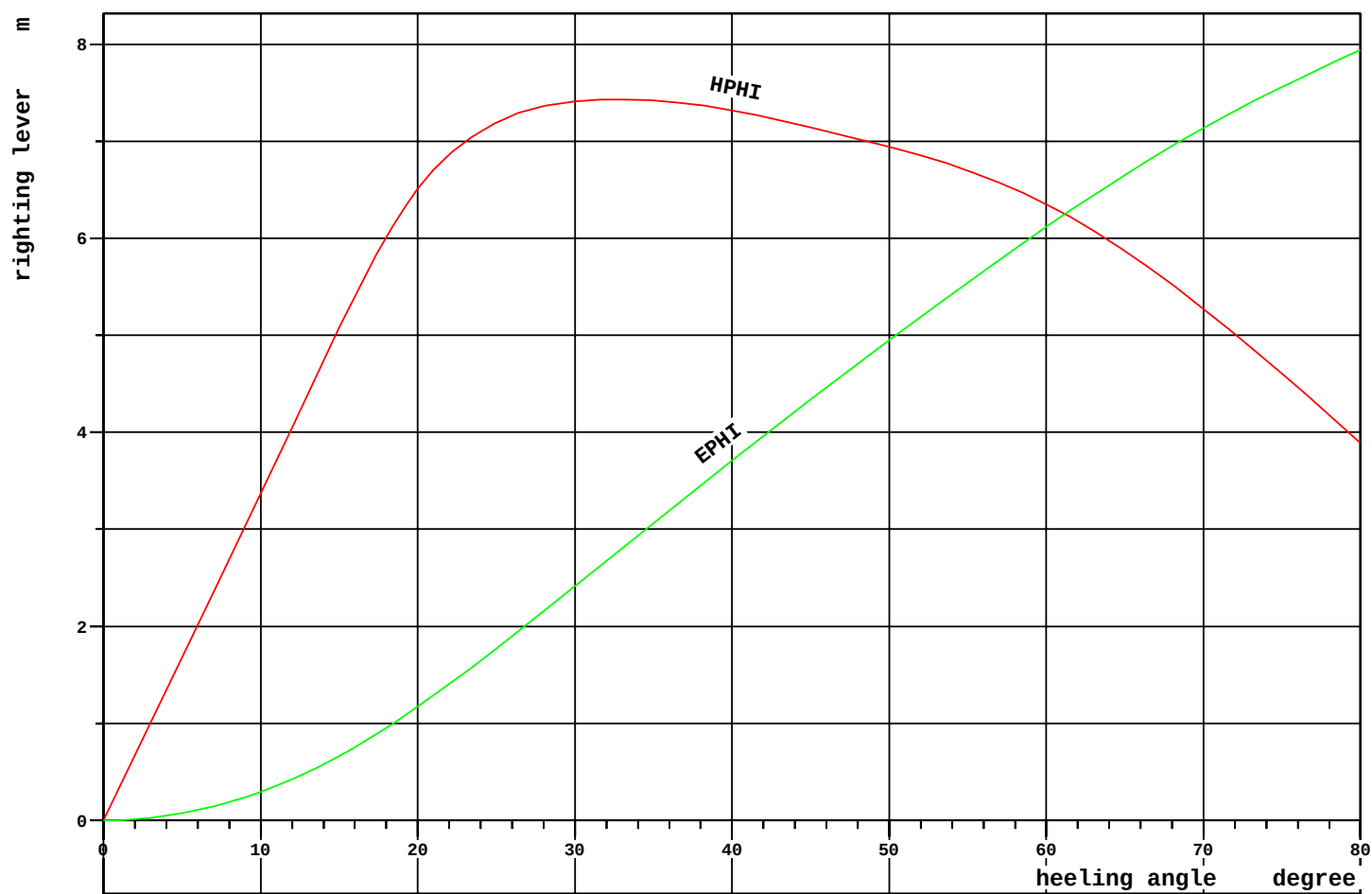
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HF0 Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HF0 Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HF0 Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	376.3	18.8	69.60	16.44	3.37	2085.3
WB04.08	Water Ballast 04.08	376.3	18.8	69.60	-16.44	3.37	2085.3
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	1550.0	77.4	93.60	16.44	5.78	2085.3
WB05.08	Water Ballast 05.08	1478.5	73.9	93.60	-16.44	5.63	2085.3
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	471.0	28.7	205.72	-0.00	3.04	1709.6
Total of Ballast Water		22223.9		111.74	0.05	2.98	10050.8

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Stores (ST0)		100.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		26645.1		106.95	0.00	3.69	12370.2
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		47516.5		109.80	-0.00	6.66	12370.2
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	2.414	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.292	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	7.431	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	32.853	deg	OK
V.GM0.15	GM > 0.15 m	0.150	19.178	m	OK
IMOWEATHER	IMO weather criterion	1.000	3.667		OK
ND-RANGE	Range of Stability acc. ND	36.000	80.000	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	66.563		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	71.636	deg	OK

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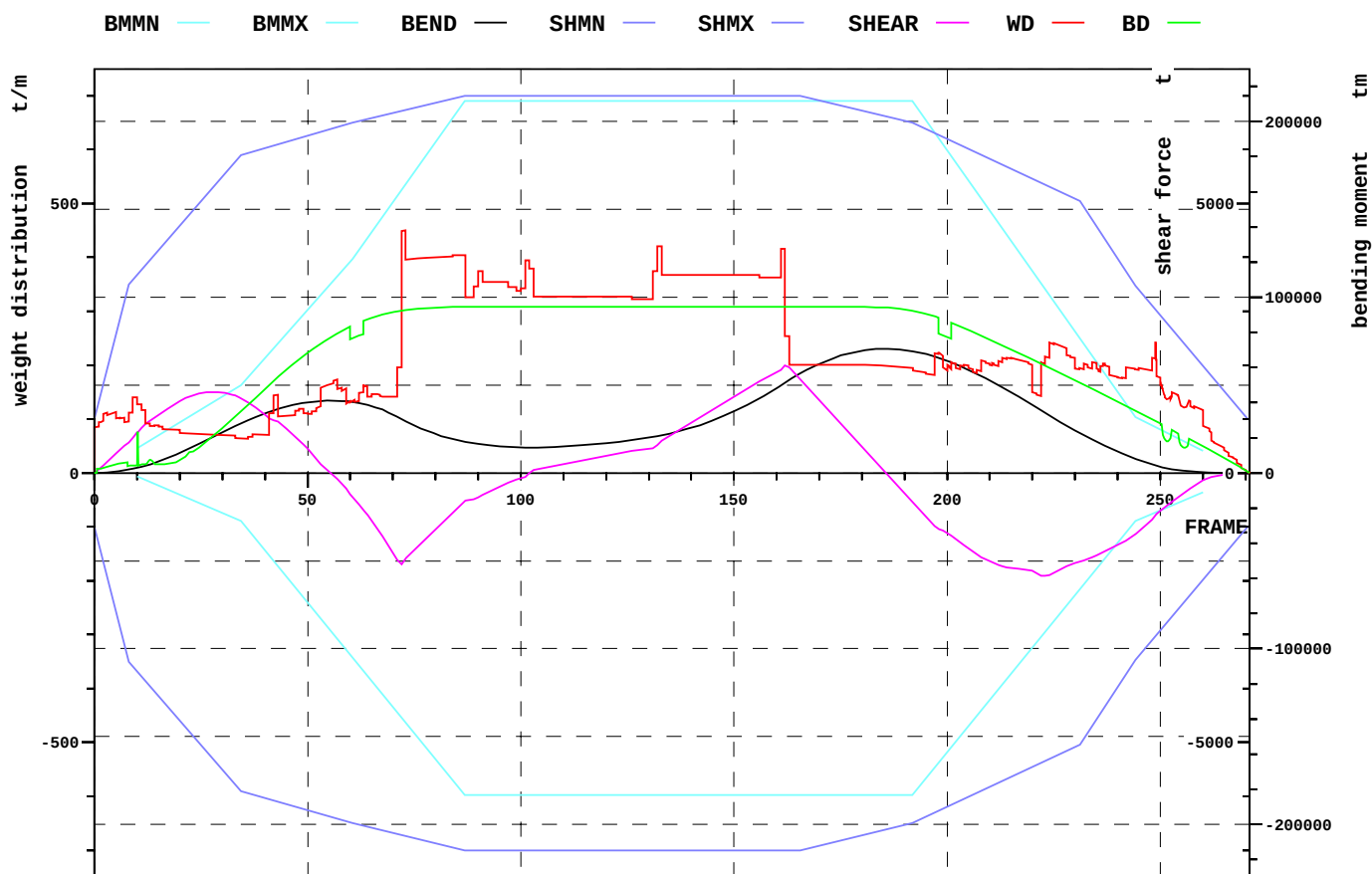
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	26.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	26.350
AIR_DUCT_1	UNPROTECTED	71.6	222.00	11.375	33.350	26.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

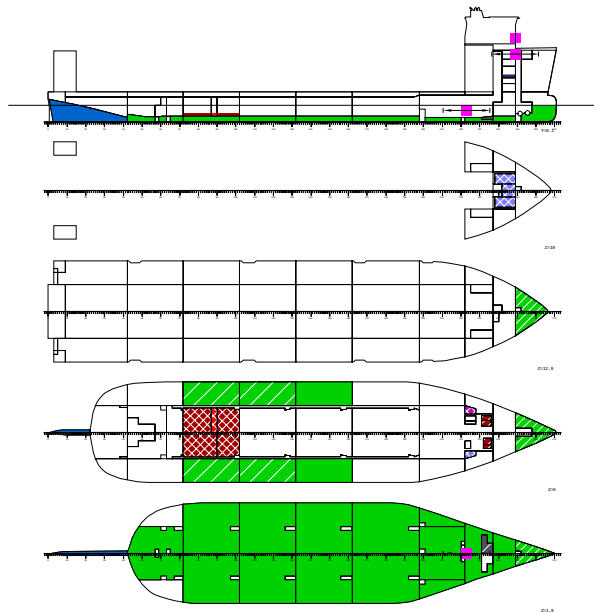
			x	frame
Shear force (min)	-1908.6 t	position:	177.6 m	222
Shear force (max)	1997.2 t		129.6 m	162
Max. rel. shear force	35.7 %		178.4 m	223
Sagging moment	-6.7 tm		214.4 m	268
Hogging moment	70668.2 tm		148.9 m	186
Max. rel. bending moment	43.8 %		27.5 m	34

LOADING CONDITION LC01, Light ballast, departure

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	12574	-13513	30261	19.2
42.00	33.60	35530	-50240	70924	41.6
57.00	45.60	41111	-94804	111952	31.5
72.00	57.60	30773	-139384	160929	13.3
87.00	69.60	17774	-183412	211763	1.8
102.00	81.60	14448	-183412	211763	0.1
117.00	93.60	16321	-183412	211763	1.1
132.00	105.60	20895	-183412	211763	3.4
147.00	117.60	31857	-183412	211763	8.9
162.00	129.60	51331	-183412	211763	18.8
180.00	144.00	69516	-183412	211763	28.0
198.00	158.40	65452	-164872	190357	29.7
210.00	168.00	52653	-129204	149176	30.7
222.00	177.60	35700	-93541	108000	28.3
237.00	189.60	15345	-48977	56548	21.9
249.00	199.20	4143	-22505	25984	9.9

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1373	-4679	4679	29.3
42.00	33.60	972	-6075	6075	16.0
57.00	45.60	-112	-6418	6418	1.7
72.00	57.60	-1692	-6718	6718	25.2
87.00	69.60	-517	-7000	7000	7.4
102.00	81.60	0	-7000	7000	0.0
117.00	93.60	268	-7000	7000	3.8
132.00	105.60	514	-7000	7000	7.3
147.00	117.60	1268	-7000	7000	18.1
162.00	129.60	1997	-7000	7000	28.5
180.00	144.00	488	-6724	6724	7.3
198.00	158.40	-1038	-6269	6269	16.6
210.00	168.00	-1636	-5824	5824	28.1
222.00	177.60	-1909	-5380	5380	35.5
237.00	189.60	-1451	-4339	4339	33.4
249.00	199.20	-776	-3024	3024	25.7

4.4 LC02 Light Ballast, arrival



LOADING CONDITION LC02: Light Ballast, arrival

FLOATING POSITION

Mean draught (moulded)	7.00 m	KM above the moulded base	26.12 m
Draught at AP (moulded)	7.00 m	KG above the moulded base	6.58 m
Draught at FP (moulded)	7.00 m	GM0 (solid)	19.54 m
Trim	0.00 m	Free surface correction	-0.34 m
Heeling	0.0 deg	GM (fluid)	19.20 m
Mean draught (heel 0)	7.00 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	25385.9	107.48	-0.02	3.50	10484.2
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	26650.1	106.95	0.00	3.55	16226.0
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	47521.5	109.80	-0.00	6.58	16226.0

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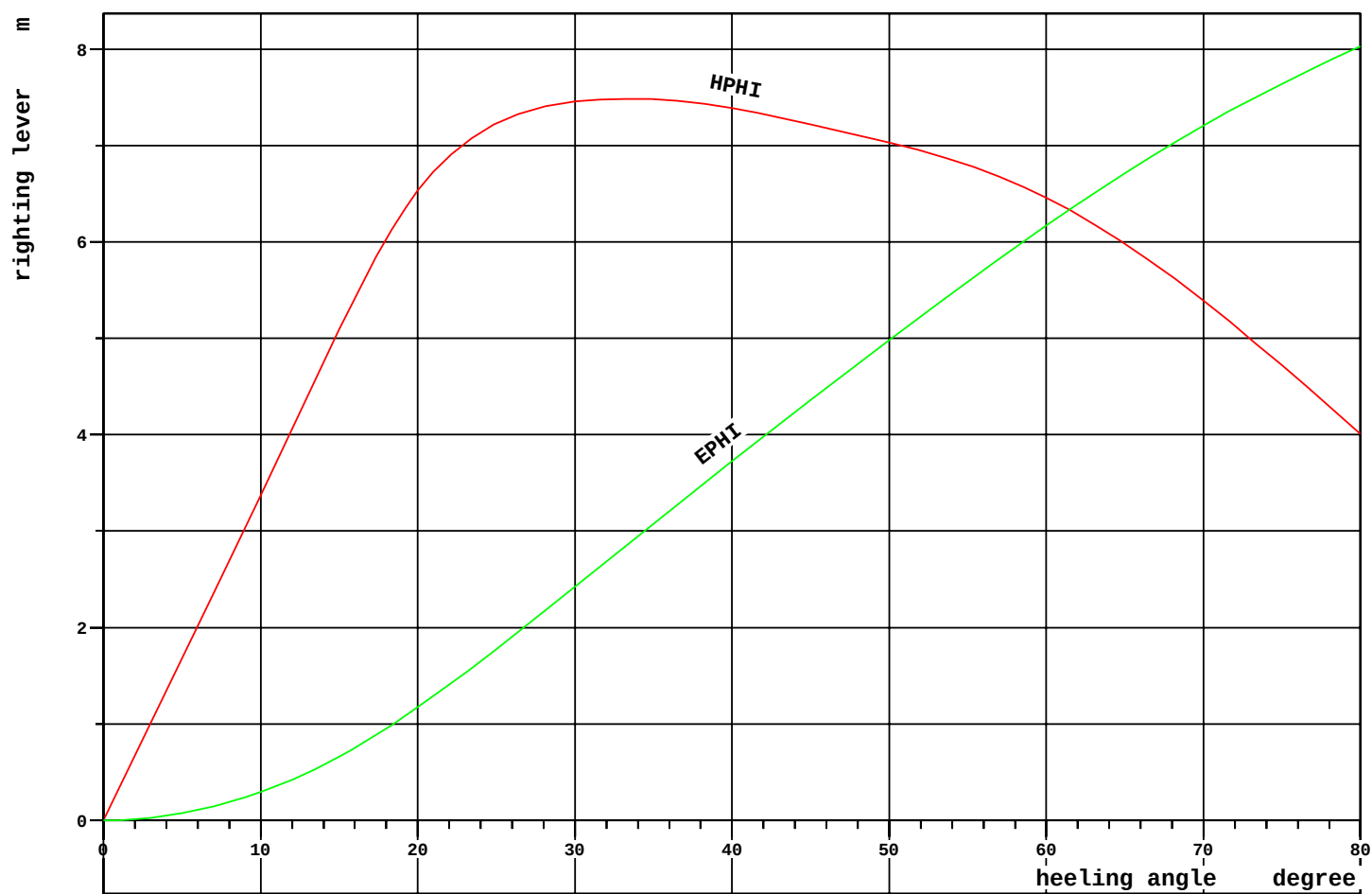
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	1865.1	93.2	69.60	16.44	6.42	2085.3
WB04.08	Water Ballast 04.08	1887.5	94.3	69.60	-16.44	6.47	2085.3
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	1501.2	75.0	93.60	16.44	5.67	2085.3
WB05.08	Water Ballast 05.08	1501.2	75.0	93.60	-16.44	5.67	2085.3
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	659.1	40.2	205.78	0.00	3.92	2143.1
Total of Ballast Water		25385.9		107.48	-0.02	3.50	10484.2
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores (ST0)		10.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							
Deadweight		26650.1		106.95	0.00	3.55	16226.0
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		47521.5		109.80	-0.00	6.58	16226.0
—							



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	2.422	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.302	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	7.485	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	33.342	deg	OK
V.GM0.15	GM > 0.15 m	0.150	19.197	m	OK
IMOWEATHER	IMO weather criterion	1.000	3.718		OK
ND-RANGE	Range of Stability acc. ND	36.000	80.000	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	67.223		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	71.632	deg	OK

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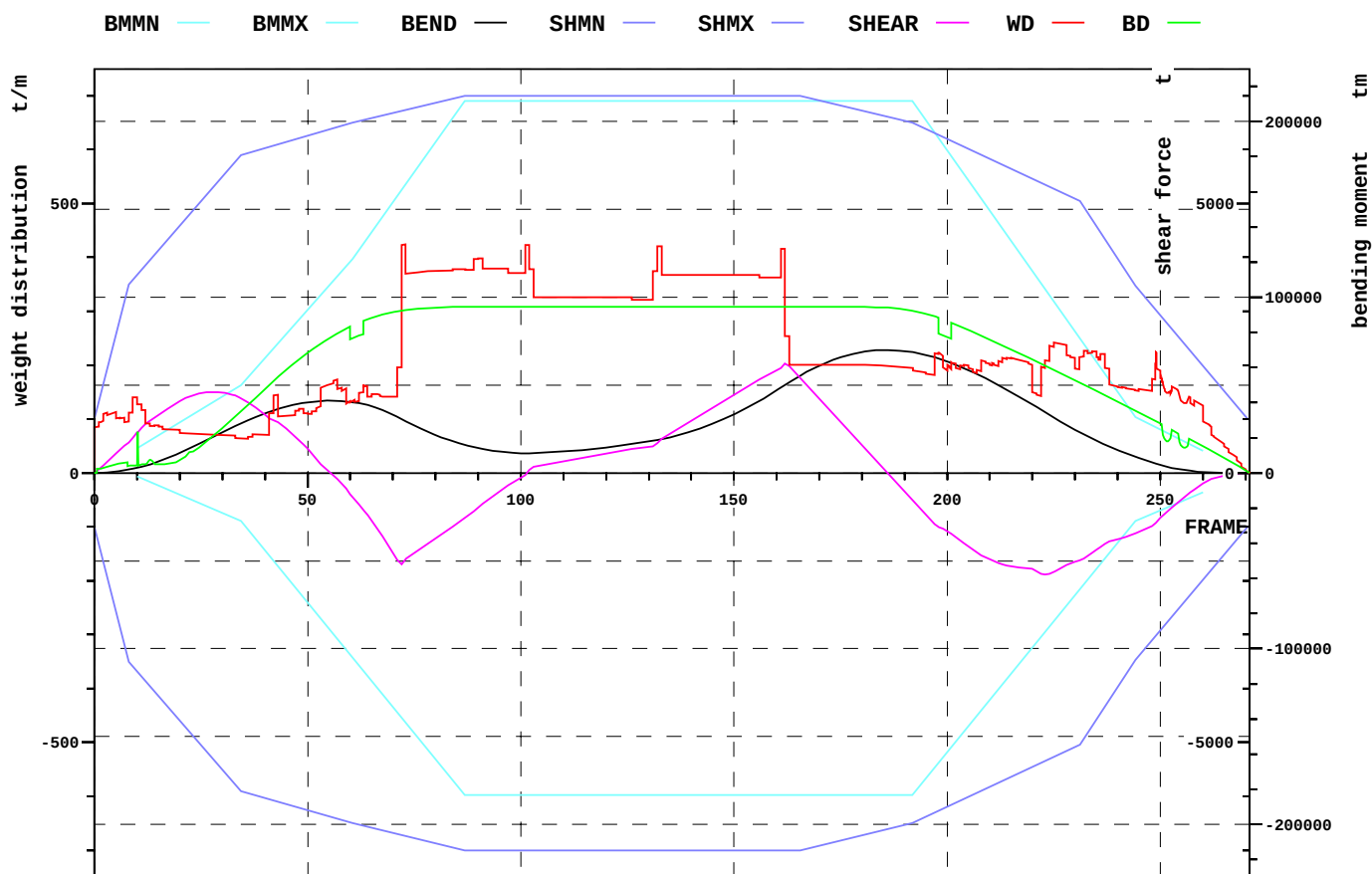
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	26.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	26.350
AIR_DUCT_1	UNPROTECTED	71.6	222.00	11.375	33.350	26.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

			x	frame
Shear force (min)	-1879.4 t	position:	178.4 m	223
Shear force (max)	2031.1 t		129.6 m	162
Max. rel. shear force	35.3 %		179.2 m	224
Sagging moment	-2.9 tm		215.2 m	269
Hogging moment	69974.4 tm		148.9 m	186
Max. rel. bending moment	43.8 %		27.5 m	34

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Loading Condition
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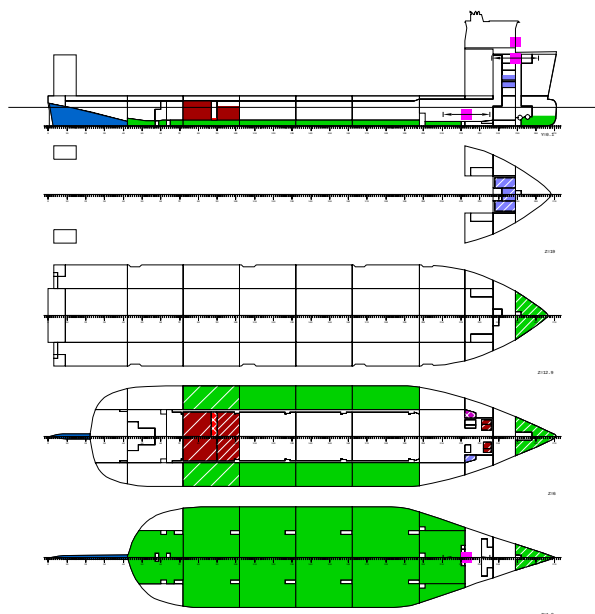
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LOADING CONDITION LC02, Light Ballast, arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	12558	-13513	30261	19.1
42.00	33.60	35494	-50240	70924	41.5
57.00	45.60	41057	-94804	111952	31.4
72.00	57.60	30697	-139384	160929	13.3
87.00	69.60	15817	-183412	211763	0.8
102.00	81.60	11214	-183412	211763	1.5
117.00	93.60	13723	-183412	211763	0.2
132.00	105.60	18776	-183412	211763	2.3
147.00	117.60	30136	-183412	211763	8.1
162.00	129.60	50006	-183412	211763	18.1
180.00	144.00	68662	-183412	211763	27.6
198.00	158.40	65067	-164872	190357	29.5
210.00	168.00	52578	-129204	149176	30.6
222.00	177.60	35935	-93541	108000	28.5
237.00	189.60	16129	-48977	56548	23.4
249.00	199.20	5169	-22505	25984	14.1

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1373	-4679	4679	29.3
42.00	33.60	972	-6075	6075	16.0
57.00	45.60	-113	-6418	6418	1.8
72.00	57.60	-1693	-6718	6718	25.2
87.00	69.60	-828	-7000	7000	11.8
102.00	81.60	59	-7000	7000	0.8
117.00	93.60	316	-7000	7000	4.5
132.00	105.60	548	-7000	7000	7.8
147.00	117.60	1302	-7000	7000	18.6
162.00	129.60	2031	-7000	7000	29.0
180.00	144.00	522	-6724	6724	7.8
198.00	158.40	-1004	-6269	6269	16.0
210.00	168.00	-1602	-5824	5824	27.5
222.00	177.60	-1875	-5380	5380	34.9
237.00	189.60	-1316	-4339	4339	30.3
249.00	199.20	-917	-3024	3024	30.3

4.5 LC03 Heavy ballast / min. ice draught, departure



LOADING CONDITION LC03: Heavy ballast / min. ice draught, departure

FLOATING POSITION

Mean draught (moulded)	8.00 m	KM above the moulded base	24.23 m
Draught at AP (moulded)	8.00 m	KG above the moulded base	6.82 m
Draught at FP (moulded)	8.00 m	GM0 (solid)	17.40 m
Trim	0.00 m	Free surface correction	-0.14 m
Heeling	0.0 deg	GM (fluid)	17.26 m
Mean draught (heel 0)	8.00 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	30102.5	109.85	0.04	4.25	5285.9
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	34523.6	106.40	0.00	4.64	7605.3
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	55395.0	109.05	-0.00	6.82	7605.3

LOADS

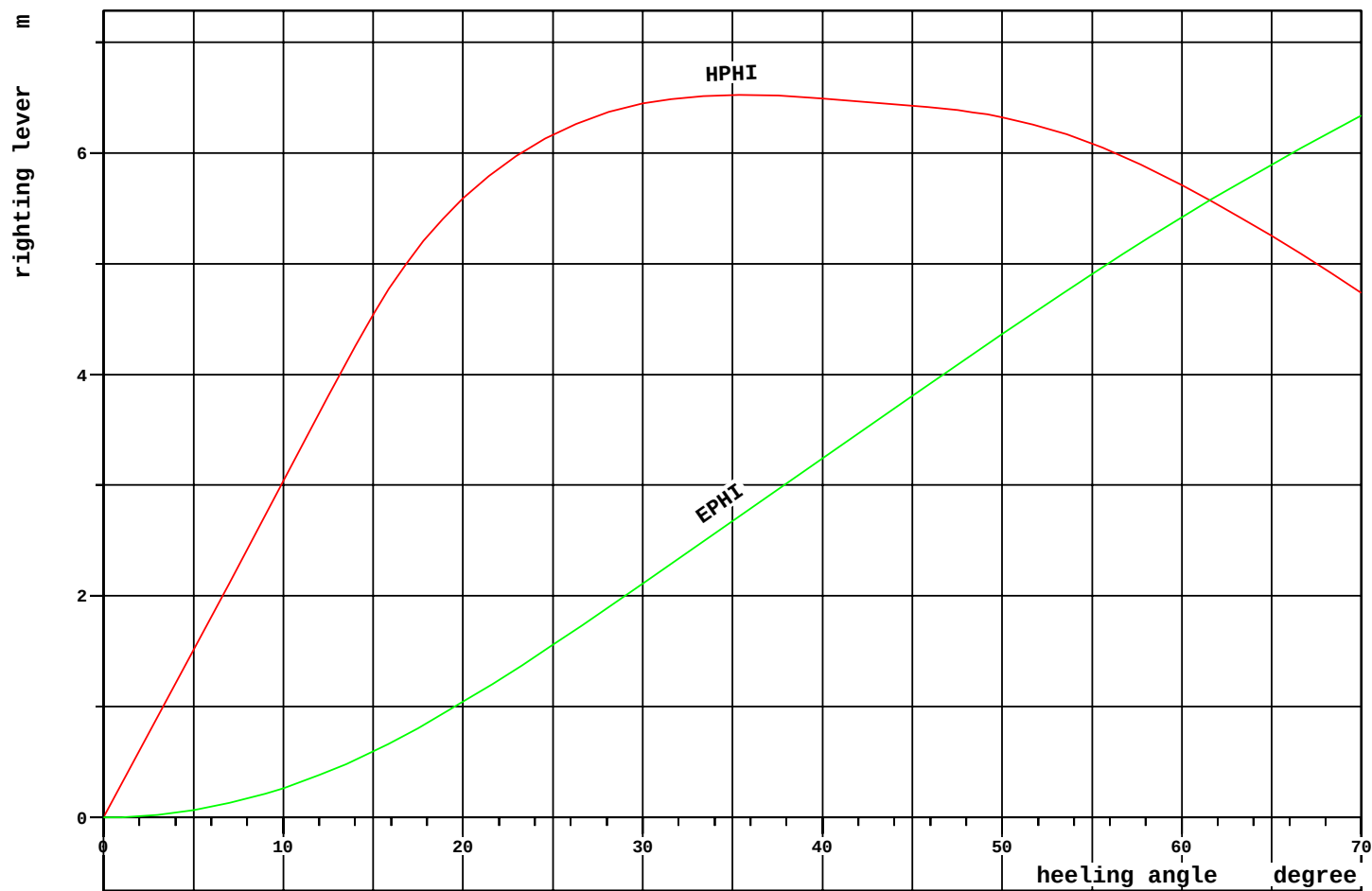
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HF0 Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HF0 Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HF0 Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	1904.0	95.1	69.60	16.44	6.50	2085.3
WB04.08	Water Ballast 04.08	1833.0	91.6	69.60	-16.44	6.35	2085.3
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.08	Water Ballast 05.08	2001.6	100.0	93.60	-16.44	6.70	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB11.01	Water Ballast 11.01	362.2	22.1	205.66	0.00	2.45	1115.3
Total of Ballast Water		30102.5		109.85	0.04	4.25	5285.9

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Stores (ST0)		100.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		34523.6		106.40	0.00	4.64	7605.3
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		55395.0		109.05	-0.00	6.82	7605.3
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	2.109	mmrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.135	mmrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	6.522	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	35.710	deg	OK
V.GM0.15	GM > 0.15 m	0.150	17.263	m	OK
IMOWEATHER	IMO weather criterion	1.000	3.986		OK
ND-RANGE	Range of Stability acc. ND	36.000	70.000	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	71.011		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	67.463	deg	OK

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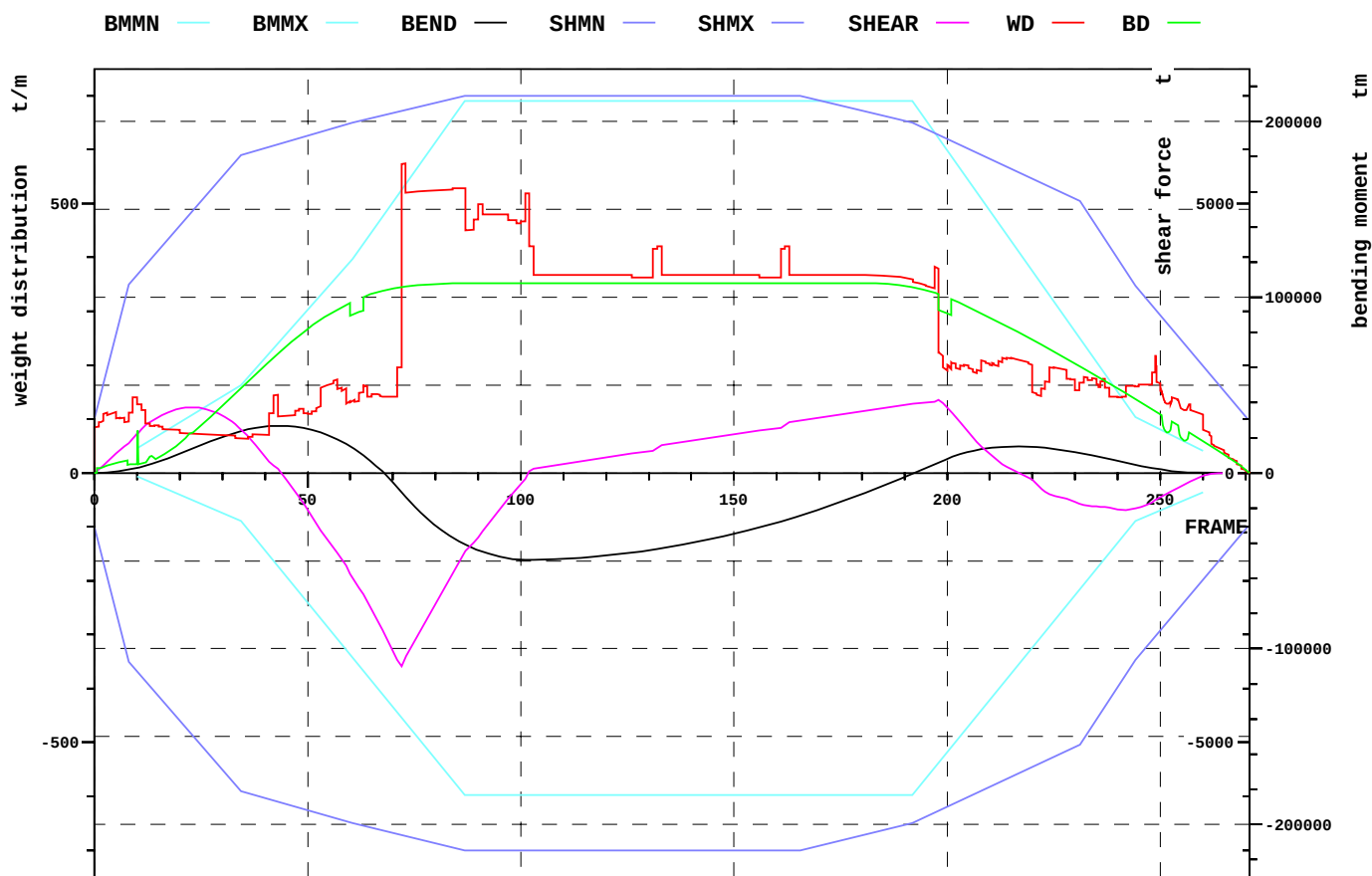
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	25.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	25.350
AIR_DUCT_1	UNPROTECTED	67.5	222.00	11.375	33.350	25.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-3588.7 t		57.6 m	72
Shear force (max)	1358.8 t		158.4 m	198
Max. rel. shear force	53.4 %		57.6 m	72
Sagging moment	-49562.2 tm		81.6 m	102
Hogging moment	26846.5 tm		34.9 m	44
Max. rel. bending moment	39.6 %		8.0 m	10

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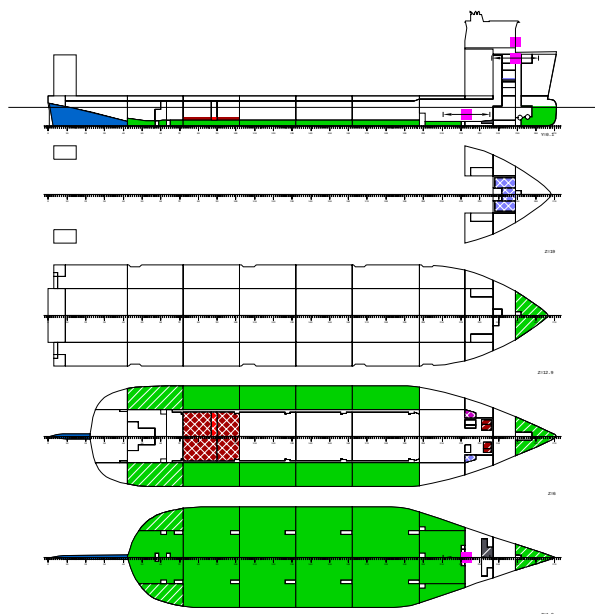
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LOADING CONDITION LC03, Heavy ballast / min. ice draught, departure

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	11962	-13513	30261	16.4
42.00	33.60	26737	-50240	70924	27.1
57.00	45.60	19085	-94804	111952	10.2
72.00	57.60	-10834	-139384	160929	14.4
87.00	69.60	-40812	-183412	211763	27.8
102.00	81.60	-49562	-183412	211763	32.3
117.00	93.60	-47587	-183412	211763	31.3
132.00	105.60	-43412	-183412	211763	29.1
147.00	117.60	-36277	-183412	211763	25.5
162.00	129.60	-26982	-183412	211763	20.8
180.00	144.00	-12059	-183412	211763	13.3
198.00	158.40	5912	-164872	190357	3.8
210.00	168.00	14151	-129204	149176	3.0
222.00	177.60	14537	-93541	108000	7.3
237.00	189.60	8437	-48977	56548	8.8
249.00	199.20	2332	-22505	25984	2.4

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1212	-4679	4679	25.9
42.00	33.60	133	-6075	6075	2.2
57.00	45.60	-1479	-6418	6418	23.0
72.00	57.60	-3589	-6718	6718	53.4
87.00	69.60	-1451	-7000	7000	20.7
102.00	81.60	28	-7000	7000	0.4
117.00	93.60	256	-7000	7000	3.7
132.00	105.60	459	-7000	7000	6.6
147.00	117.60	684	-7000	7000	9.8
162.00	129.60	884	-7000	7000	12.6
180.00	144.00	1142	-6724	6724	17.0
198.00	158.40	1359	-6269	6269	21.7
210.00	168.00	356	-5824	5824	6.1
222.00	177.60	-281	-5380	5380	5.2
237.00	189.60	-635	-4339	4339	14.6
249.00	199.20	-492	-3024	3024	16.3

4.6 LC04 Heavy ballast / min. ice draught, arrival



LOADING CONDITION LC04: Heavy ballast / min. ice draught, arrival

FLOATING POSITION

Mean draught (moulded)	8.00 m	KM above the moulded base	24.23 m
Draught at AP (moulded)	8.00 m	KG above the moulded base	6.64 m
Draught at FP (moulded)	8.00 m	GM0 (solid)	17.58 m
Trim	0.00 m	Free surface correction	-0.22 m
Heeling	0.0 deg		
		GM (fluid)	17.36 m
Mean draught (heel 0)	8.00 m		
Trim (heel 0)	0.00 m		

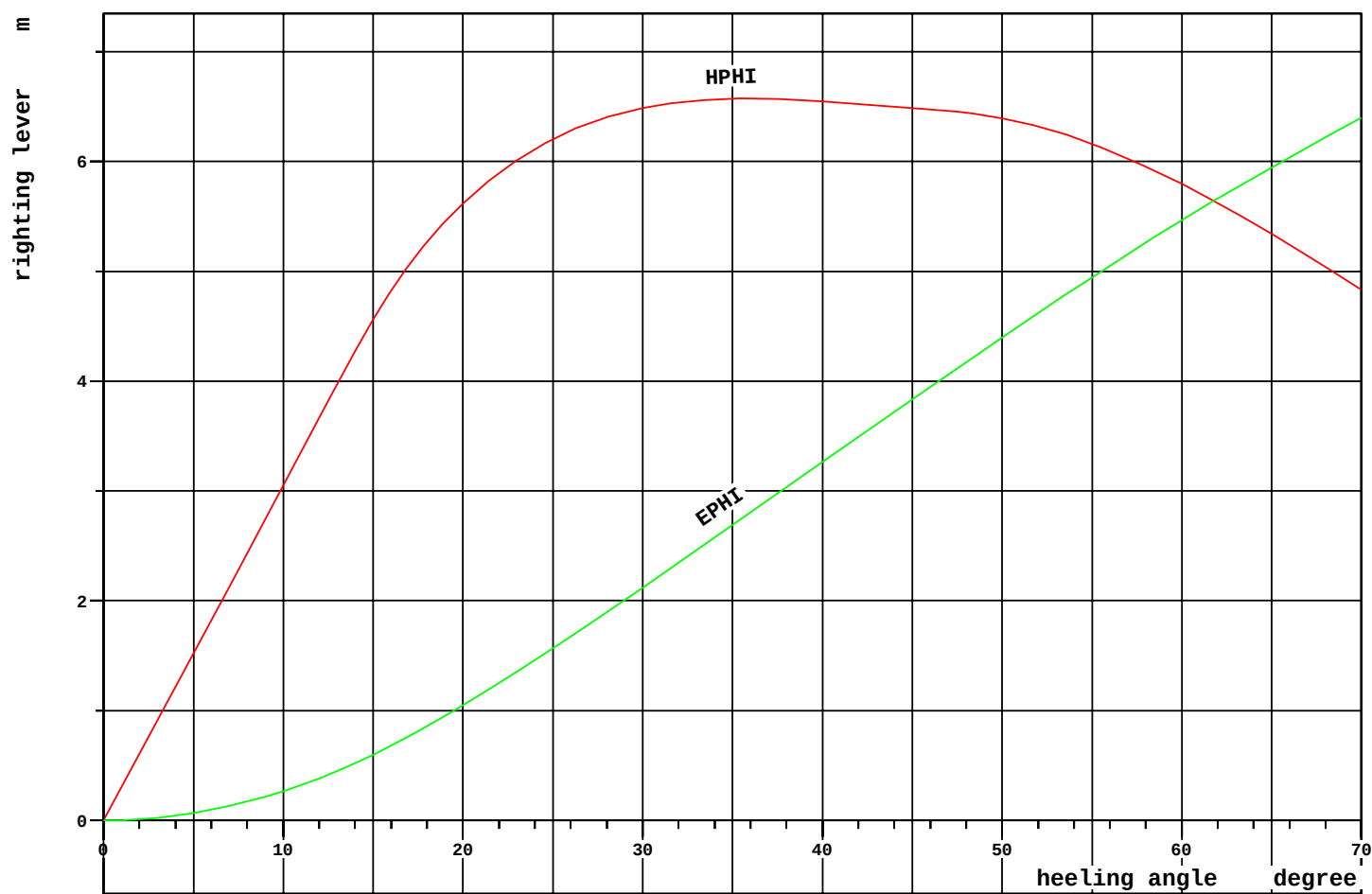
LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	33259.6	106.78	-0.01	4.34	6680.0
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	34523.7	106.40	0.00	4.34	12421.7
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	55395.1	109.05	-0.00	6.64	12421.7

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	1152.1	54.1	47.25	15.84	4.17	2039.3
WB03.08	Water Ballast 03.08	1175.0	55.2	47.22	-15.85	4.23	2044.9
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.08	Water Ballast 05.08	2001.6	100.0	93.60	-16.44	6.70	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB11.01	Water Ballast 11.01	926.0	56.5	205.82	-0.00	5.02	2595.7
Total of Ballast Water		33259.6		106.78	-0.01	4.34	6680.0
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores							
(ST0)		10.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							
Deadweight		34523.7		106.40	0.00	4.34	12421.7
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		55395.1		109.05	-0.00	6.64	12421.7
—							



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	2.120	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.144	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	6.572	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	36.016	deg	OK
V.GM0.15	GM > 0.15 m	0.150	17.358	m	OK
IMOWEATHER	IMO weather criterion	1.000	4.069		OK
ND-RANGE	Range of Stability acc. ND	36.000	70.000	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	71.657		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	67.467	deg	OK

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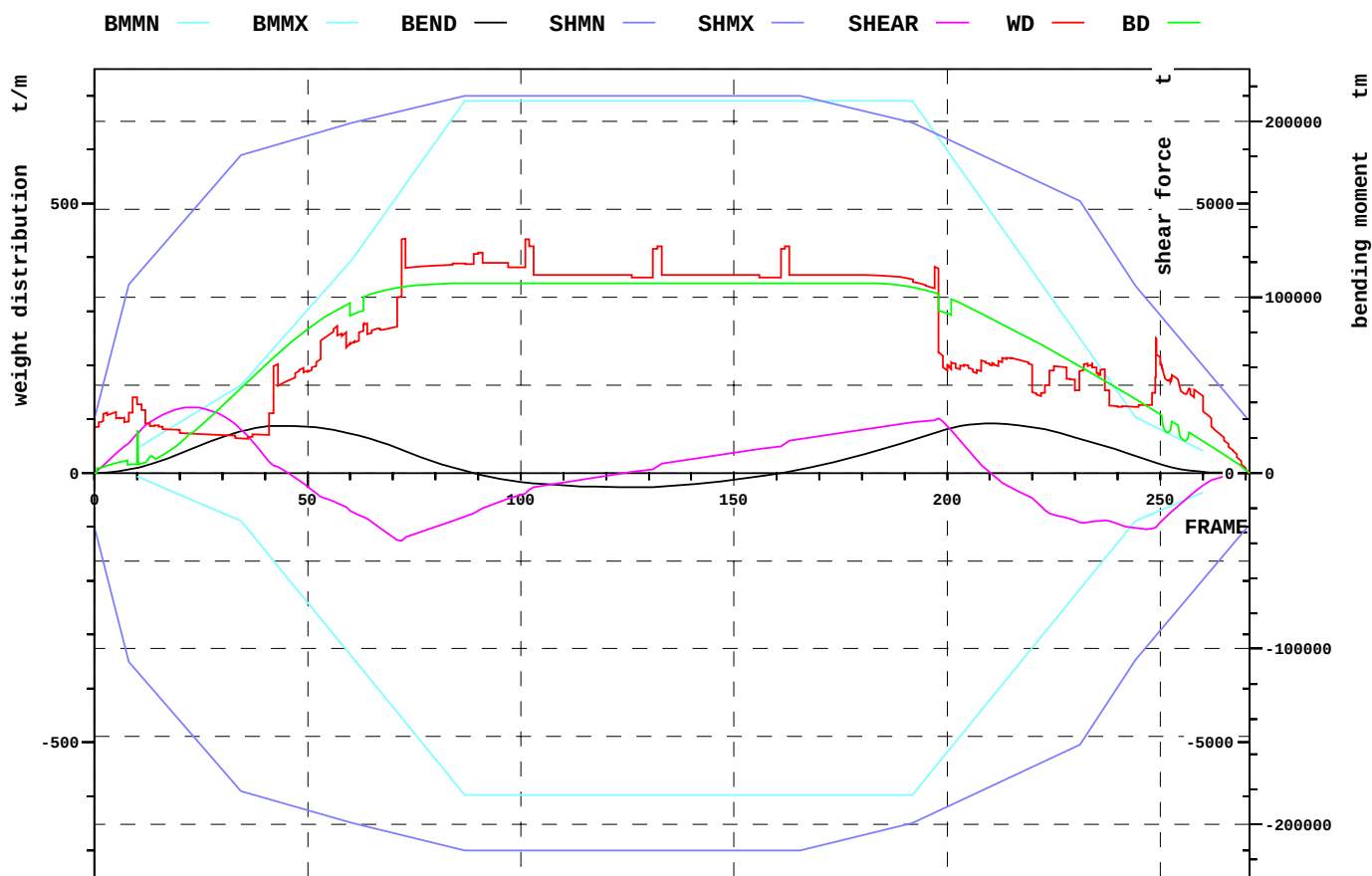
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 LC04

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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	25.350
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	25.350
AIR_DUCT_1	UNPROTECTED	67.5	222.00	11.375	33.350	25.350

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-1262.7 t		57.6 m	72
Shear force (max)	1218.9 t		17.8 m	22
Max. rel. shear force	33.4 %		199.1 m	249
Sagging moment	-8097.3 tm		98.4 m	123
Hogging moment	28247.0 tm		168.0 m	210
Max. rel. bending moment	39.8 %		8.0 m	10

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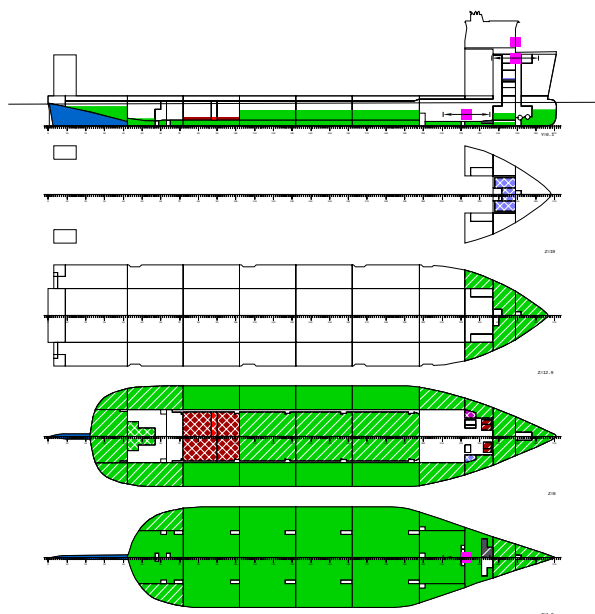
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LOADING CONDITION LC04, Heavy ballast / min. ice draught, arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	11932	-13513	30261	16.3
42.00	33.60	26678	-50240	70924	27.0
57.00	45.60	24003	-94804	111952	14.9
72.00	57.60	13176	-139384	160929	1.6
87.00	69.60	1032	-183412	211763	6.7
102.00	81.60	-5674	-183412	211763	10.0
117.00	93.60	-7856	-183412	211763	11.2
132.00	105.60	-7833	-183412	211763	11.1
147.00	117.60	-4843	-183412	211763	9.6
162.00	129.60	314	-183412	211763	7.0
180.00	144.00	10279	-183412	211763	2.0
198.00	158.40	23302	-164872	190357	5.9
210.00	168.00	28247	-129204	149176	13.1
222.00	177.60	25341	-93541	108000	18.0
237.00	189.60	15296	-48977	56548	21.8
249.00	199.20	5840	-22505	25984	16.9

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1212	-4679	4679	25.9
42.00	33.60	133	-6075	6075	2.2
57.00	45.60	-555	-6418	6418	8.6
72.00	57.60	-1263	-6718	6718	18.8
87.00	69.60	-801	-7000	7000	11.4
102.00	81.60	-317	-7000	7000	4.5
117.00	93.60	-88	-7000	7000	1.3
132.00	105.60	115	-7000	7000	1.6
147.00	117.60	341	-7000	7000	4.9
162.00	129.60	542	-7000	7000	7.7
180.00	144.00	800	-6724	6724	11.9
198.00	158.40	1017	-6269	6269	16.2
210.00	168.00	15	-5824	5824	0.3
222.00	177.60	-621	-5380	5380	11.6
237.00	189.60	-873	-4339	4339	20.1
249.00	199.20	-1007	-3024	3024	33.3

4.7 LC05 End of de-ballasting



LOADING CONDITION LC05: End of de-ballasting

FLOATING POSITION

Mean draught (moulded)	9.68 m	KM above the moulded base	22.11 m
Draught at AP (moulded)	9.38 m	KG above the moulded base	6.35 m
Draught at FP (moulded)	9.98 m	GM0 (solid)	15.76 m
Trim (by head)	0.60 m	Free surface correction	-0.60 m
Heeling	0.0 deg	GM (fluid)	15.16 m
Mean draught (heel 0)	9.68 m		
Trim (heel 0)	0.60 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	46798.6	106.96	-0.00	4.57	35706.0
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	48062.8	106.68	0.01	4.57	41447.8
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	68934.2	108.73	0.00	6.35	41447.8

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	914.9	60.0	24.80	5.42	6.51	2551.5
WB02.06	Water Ballast 02.06	916.0	60.0	24.81	-5.43	6.51	2608.3
WB02.07	Water Ballast 02.07	571.3	60.0	25.82	15.38	7.31	1545.6
WB02.08	Water Ballast 02.08	571.3	60.0	25.82	-15.38	7.31	1545.6
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.06	Water Ballast 03.06	85.5	8.9	40.60	0.00	2.90	945.3
WB03.07	Water Ballast 03.07	1487.8	69.9	46.89	15.97	4.92	2083.5
WB03.08	Water Ballast 03.08	1487.8	69.9	46.89	-15.97	4.92	2083.5
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.05	Water Ballast 05.05	1079.4	50.0	93.47	5.48	4.62	2209.8
WB05.06	Water Ballast 05.06	1079.4	50.0	93.47	-5.48	4.62	2209.8
WB05.07	Water Ballast 05.07	2001.6	100.0	93.60	16.44	6.70	0.0
WB05.08	Water Ballast 05.08	2001.6	100.0	93.60	-16.44	6.70	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB06.05	Water Ballast 06.05	1079.4	50.0	117.47	5.48	4.62	2209.8
WB06.06	Water Ballast 06.06	1079.4	50.0	117.47	-5.48	4.62	2209.8
WB06.07	Water Ballast 06.07	2001.6	100.0	117.60	16.44	6.70	0.0
WB06.08	Water Ballast 06.08	2001.6	100.0	117.60	-16.44	6.70	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB07.05	Water Ballast 07.05	1299.9	50.0	143.85	5.49	4.62	2638.2
WB07.06	Water Ballast 07.06	1299.9	50.0	143.85	-5.49	4.62	2638.2
WB07.07	Water Ballast 07.07	2390.6	100.0	143.94	16.42	6.71	0.0
WB07.08	Water Ballast 07.08	2390.6	100.0	143.94	-16.42	6.71	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	599.8	50.0	166.93	15.10	5.05	947.5
WB08.08	Water Ballast 08.08	599.8	50.0	166.93	-15.10	5.05	947.5
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	411.7	65.0	183.50	11.89	6.42	152.4
WB09.08	Water Ballast 09.08	411.7	65.0	183.50	-11.89	6.42	152.4
WB10.01	Water Ballast 10.01	157.1	50.0	193.98	0.00	0.81	2028.3
WB10.05	Water Ballast 10.05	261.5	25.0	194.09	4.65	4.14	888.3
WB10.06	Water Ballast 10.06	261.5	25.0	194.09	-4.65	4.14	888.3
WB11.01	Water Ballast 11.01	698.5	42.6	205.78	0.00	4.09	2222.0
Total of Ballast Water		46798.6		106.96	-0.00	4.57	35706.0

Fixed FW ballast density=1.000 t/m3

WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
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Gray Water density=1.000 t/m3

T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0

Bilge Water density=1.000 t/m3

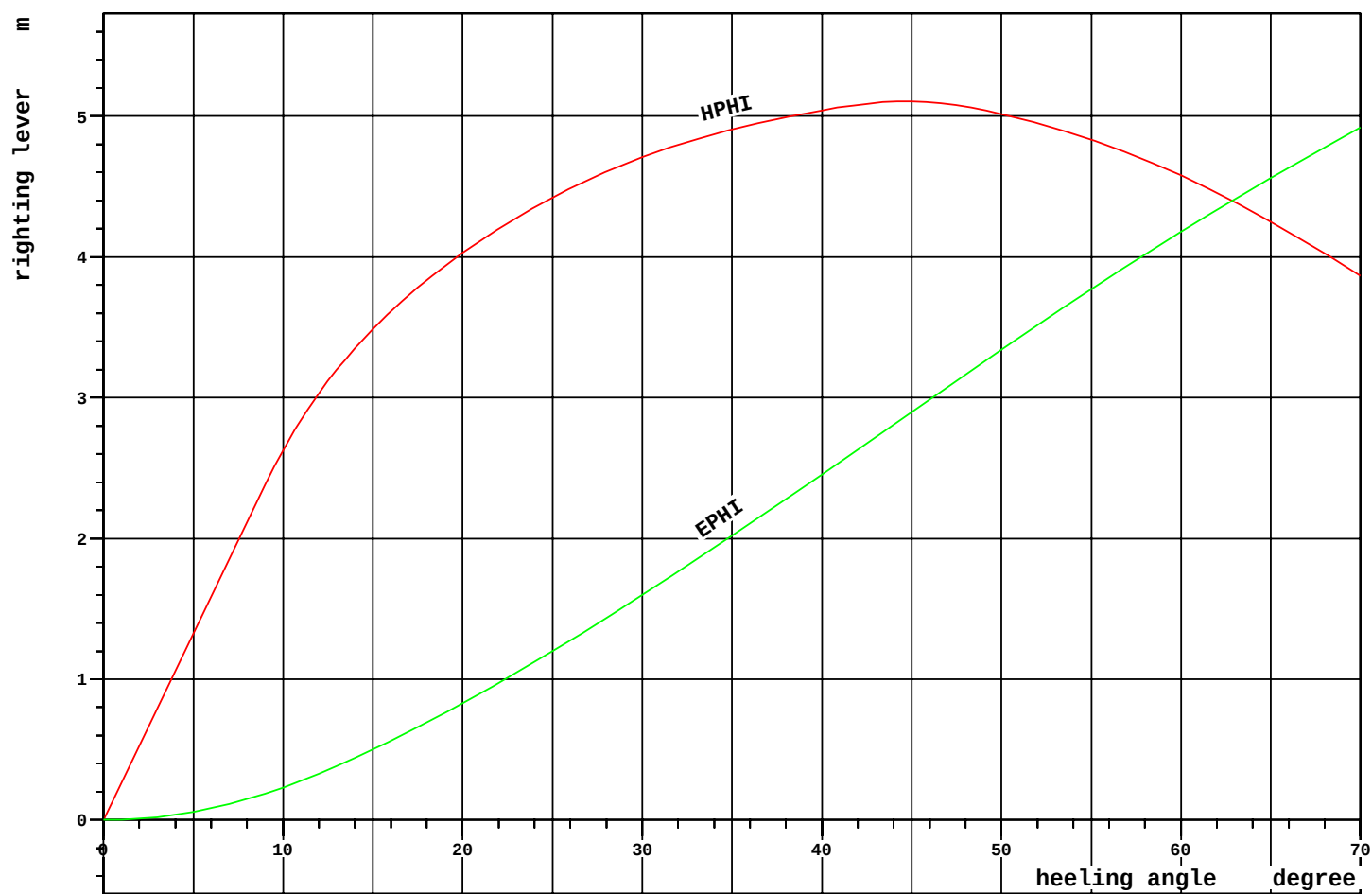
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0

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 LC05

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Sludge density=2.380 t/m3 T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores (ST0)		10.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3 (MIS)		87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		48062.8		106.68	0.01	4.57	41447.8
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		68934.2		108.73	0.00	6.35	41447.8
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	1.599	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.854	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	5.106	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	44.739	deg	OK
V.GM0.15	GM > 0.15 m	0.150	15.160	m	OK
IMOWEATHER	IMO weather criterion	1.000	4.586		OK
ND-RANGE	Range of Stability acc. ND	36.000	69.989	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	72.386		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	60.627	deg	OK

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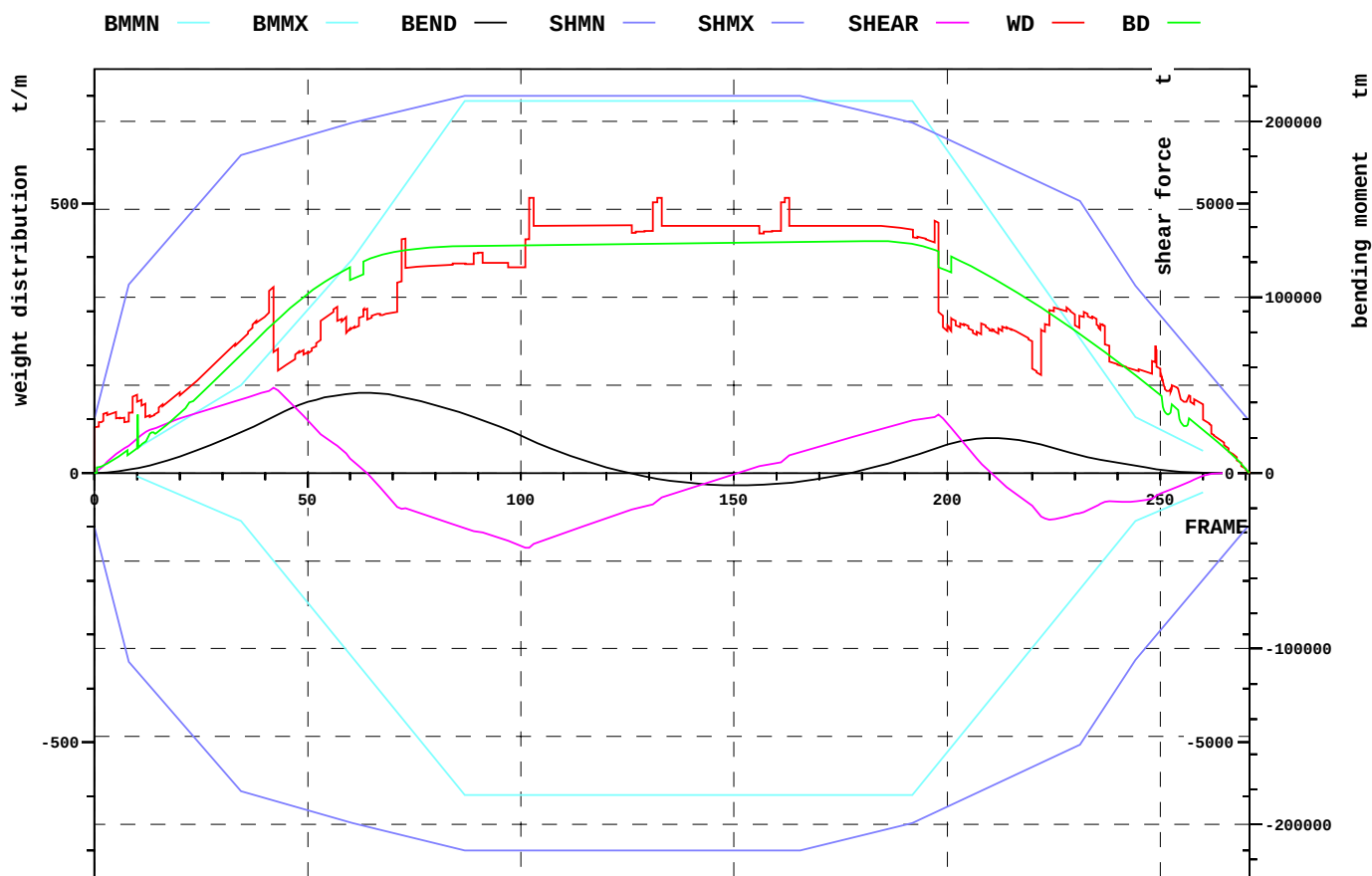
Loading Condition
LC05

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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	23.482
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	23.461
AIR_DUCT_1	UNPROTECTED	60.6	222.00	11.375	33.350	23.477

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

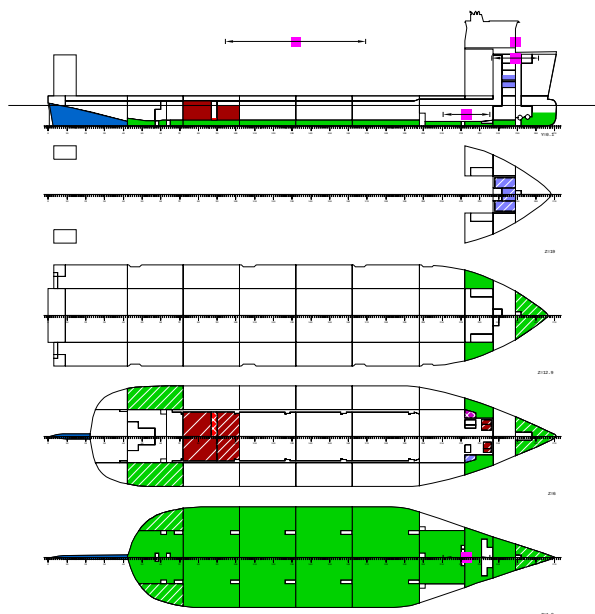
		position:	x	frame
Shear force (min)	-1392.3 t		80.8 m	101
Shear force (max)	1576.3 t		33.6 m	42
Max. rel. shear force	25.9 %		33.6 m	42
Sagging moment	-6949.2 tm		120.2 m	150
Hogging moment	45710.8 tm		51.2 m	64
Max. rel. bending moment	42.3 %		8.0 m	10

LOADING CONDITION LC05, End of de-ballasting

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	10266	-13513	30261	8.6
42.00	33.60	32073	-50240	70924	35.9
57.00	45.60	44335	-94804	111952	34.6
72.00	57.60	43430	-139384	160929	21.7
87.00	69.60	33449	-183412	211763	9.8
102.00	81.60	18966	-183412	211763	2.4
117.00	93.60	5382	-183412	211763	4.5
132.00	105.60	-3321	-183412	211763	8.9
147.00	117.60	-6799	-183412	211763	10.6
162.00	129.60	-5865	-183412	211763	10.1
180.00	144.00	1460	-183412	211763	6.4
198.00	158.40	14464	-164872	190357	1.0
210.00	168.00	19806	-129204	149176	7.1
222.00	177.60	16177	-93541	108000	8.9
237.00	189.60	7193	-48977	56548	6.5
249.00	199.20	2199	-22505	25984	1.9

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1040	-4679	4679	22.2
42.00	33.60	1576	-6075	6075	25.9
57.00	45.60	509	-6418	6418	7.9
72.00	57.60	-672	-6718	6718	10.0
87.00	69.60	-1030	-7000	7000	14.7
102.00	81.60	-1383	-7000	7000	19.8
117.00	93.60	-921	-7000	7000	13.2
132.00	105.60	-522	-7000	7000	7.5
147.00	117.60	-98	-7000	7000	1.4
162.00	129.60	261	-7000	7000	3.7
180.00	144.00	714	-6724	6724	10.6
198.00	158.40	1081	-6269	6269	17.2
210.00	168.00	27	-5824	5824	0.5
222.00	177.60	-810	-5380	5380	15.1
237.00	189.60	-538	-4339	4339	12.4
249.00	199.20	-428	-3024	3024	14.2

4.8 LC11 Non-Buoyant Cargo, departure



LOADING CONDITION LC11: Non-Buoyant Cargo, departure

FLOATING POSITION

Mean draught (moulded)	8.90 m	KM above the moulded base	22.99 m
Draught at AP (moulded)	8.90 m	KG above the moulded base	16.07 m
Draught at FP (moulded)	8.90 m	GM0 (solid)	6.92 m
Trim	0.00 m	Free surface correction	-0.13 m
Heeling	0.0 deg	GM (fluid)	6.79 m
Mean draught (heel 0)	8.90 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	17398.9	112.02	0.07	2.15	5766.9
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Solid cargo	20000.0	105.40	0.00	36.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	41820.0	105.77	0.00	18.88	8086.3
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	62691.4	108.33	-0.00	16.07	8086.3

LOADS

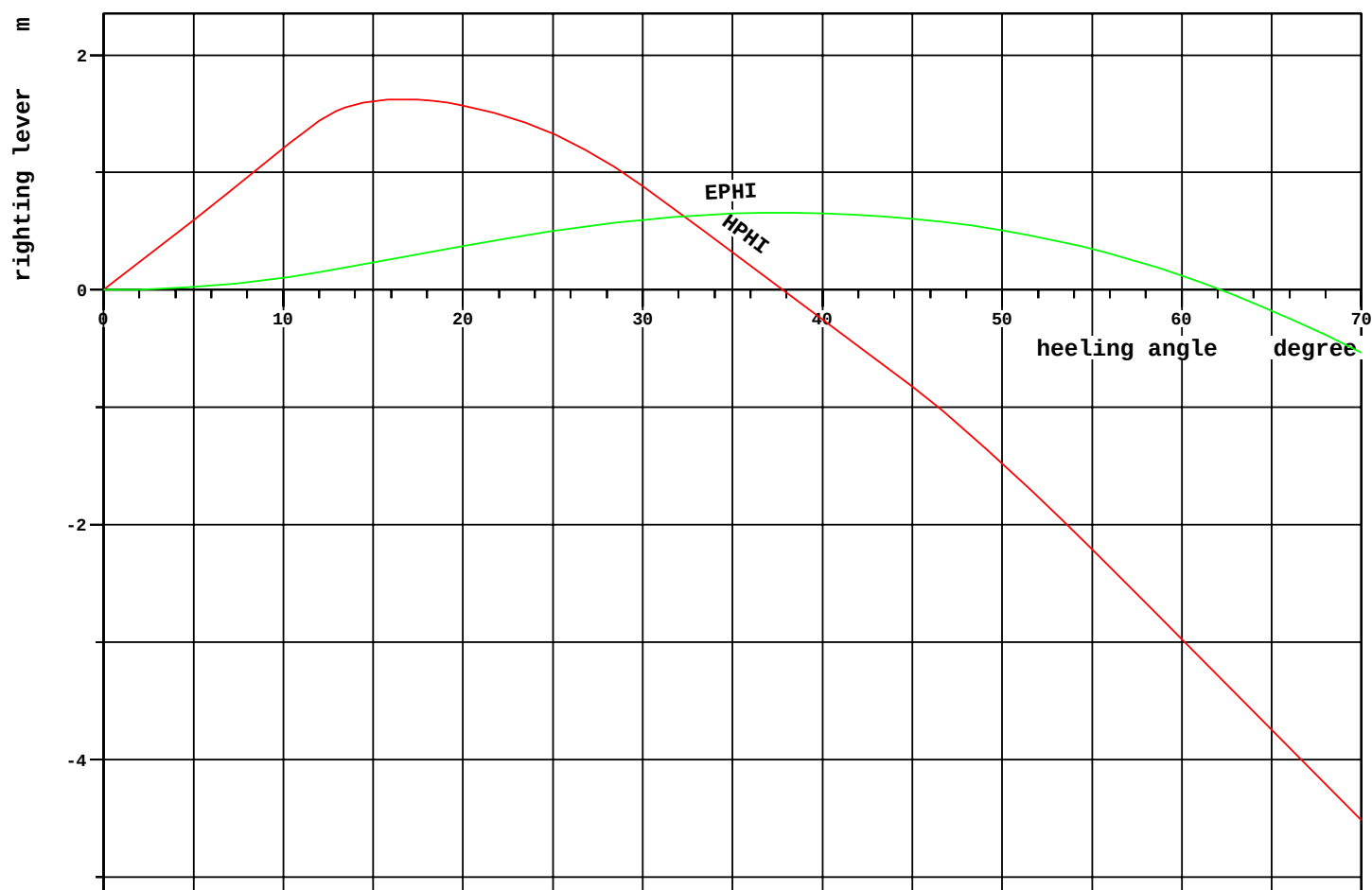
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HFO Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HFO Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HFO Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HFO Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	1060.7	49.8	47.39	15.79	3.96	2008.8
WB03.08	Water Ballast 03.08	982.0	46.1	47.52	-15.74	3.78	1974.6
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	500.0	30.5	205.73	-0.00	3.18	1783.4
Total of Ballast Water		17398.9		112.02	0.07	2.15	5766.9
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Solid cargo (CAS)	density=1.000 t/m3 NON-BUOYANT CARGO	20000.0		105.40	0.00	36.00	0.0
Stores (ST0)		100.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous (MIS)	density=1.000 t/m3	87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		41820.0		105.77	0.00	18.88	8086.3
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		62691.4		108.33	-0.00	16.07	8086.3
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.068	0.280	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.061	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	0.886	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	16.722	deg	OK
V.GM0.15	GM > 0.15 m	0.150	6.790	m	OK
IMOWEATHER	IMO weather criterion	1.000	1.395		OK
ND-RANGE	Range of Stability acc. ND	36.000	37.825	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	13.932		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	64.080	deg	OK

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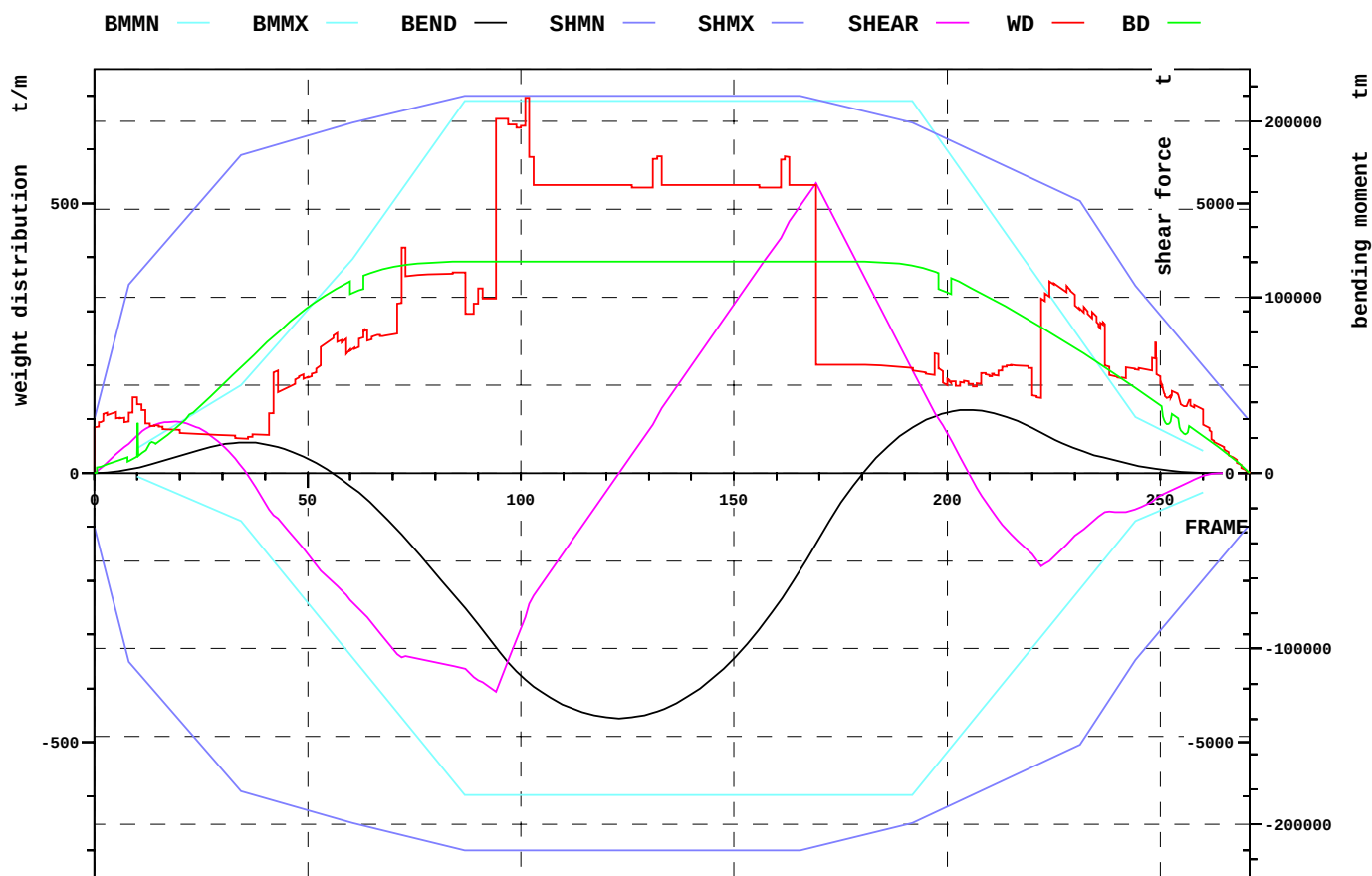
Loading Condition
 LC11

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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	24.450
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	24.450
AIR_DUCT_1	UNPROTECTED	64.1	222.00	11.375	33.350	24.450

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-4065.5 t		75.4 m	94
Shear force (max)	5377.5 t		135.4 m	169
Max. rel. shear force	77.6 %		135.4 m	169
Sagging moment	-139573 tm		98.4 m	123
Hogging moment	35978.0 tm		164.0 m	205
Max. rel. bending moment	77.8 %		98.4 m	123

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Loading Condition
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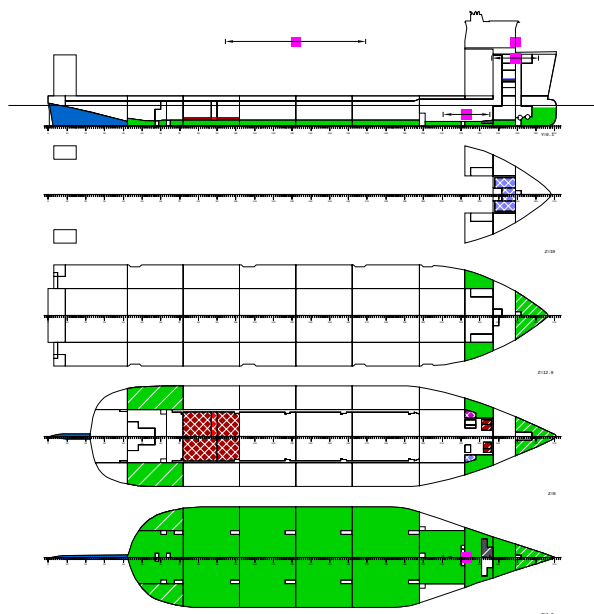
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LOADING CONDITION LC11, Non-Buoyant Cargo, departure

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	10656	-13513	30261	10.4
42.00	33.60	15452	-50240	70924	8.4
57.00	45.60	-1954	-94804	111952	10.2
72.00	57.60	-34953	-139384	160929	30.5
87.00	69.60	-77010	-183412	211763	46.1
102.00	81.60	-119574	-183412	211763	67.7
117.00	93.60	-137961	-183412	211763	77.0
132.00	105.60	-135889	-183412	211763	75.9
147.00	117.60	-112594	-183412	211763	64.2
162.00	129.60	-68877	-183412	211763	42.0
180.00	144.00	-947	-183412	211763	7.7
198.00	158.40	33027	-164872	190357	11.4
210.00	168.00	34680	-129204	149176	17.7
222.00	177.60	23196	-93541	108000	15.8
237.00	189.60	8553	-48977	56548	9.0
249.00	199.20	2218	-22505	25984	2.0

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	932	-4679	4679	19.9
42.00	33.60	-789	-6075	6075	13.0
57.00	45.60	-2095	-6418	6418	32.6
72.00	57.60	-3424	-6718	6718	51.0
87.00	69.60	-3631	-7000	7000	51.9
102.00	81.60	-2431	-7000	7000	34.7
117.00	93.60	-681	-7000	7000	9.7
132.00	105.60	1044	-7000	7000	14.9
147.00	117.60	2791	-7000	7000	39.9
162.00	129.60	4513	-7000	7000	64.5
180.00	144.00	3730	-6724	6724	55.5
198.00	158.40	998	-6269	6269	15.9
210.00	168.00	-640	-5824	5824	11.0
222.00	177.60	-1726	-5380	5380	32.1
237.00	189.60	-722	-4339	4339	16.6
249.00	199.20	-467	-3024	3024	15.4

4.9 LC12 Non-Buoyant Cargo, arrival



LOADING CONDITION LC12: Non-Buoyant Cargo, arrival

FLOATING POSITION

Mean draught (moulded)	8.79 m	KM above the moulded base	23.12 m
Draught at AP (moulded)	8.79 m	KG above the moulded base	16.17 m
Draught at FP (moulded)	8.79 m	GM0 (solid)	6.95 m
Trim	0.00 m	Free surface correction	-0.20 m
Heeling	0.0 deg	GM (fluid)	6.75 m
Mean draught (heel 0)	8.79 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	19656.3	106.92	-0.00	2.83	6588.6
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Solid cargo	20000.0	105.40	0.00	36.00	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	40920.4	105.85	0.01	19.10	12330.4
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	61791.8	108.42	0.00	16.17	12330.4

LOADS

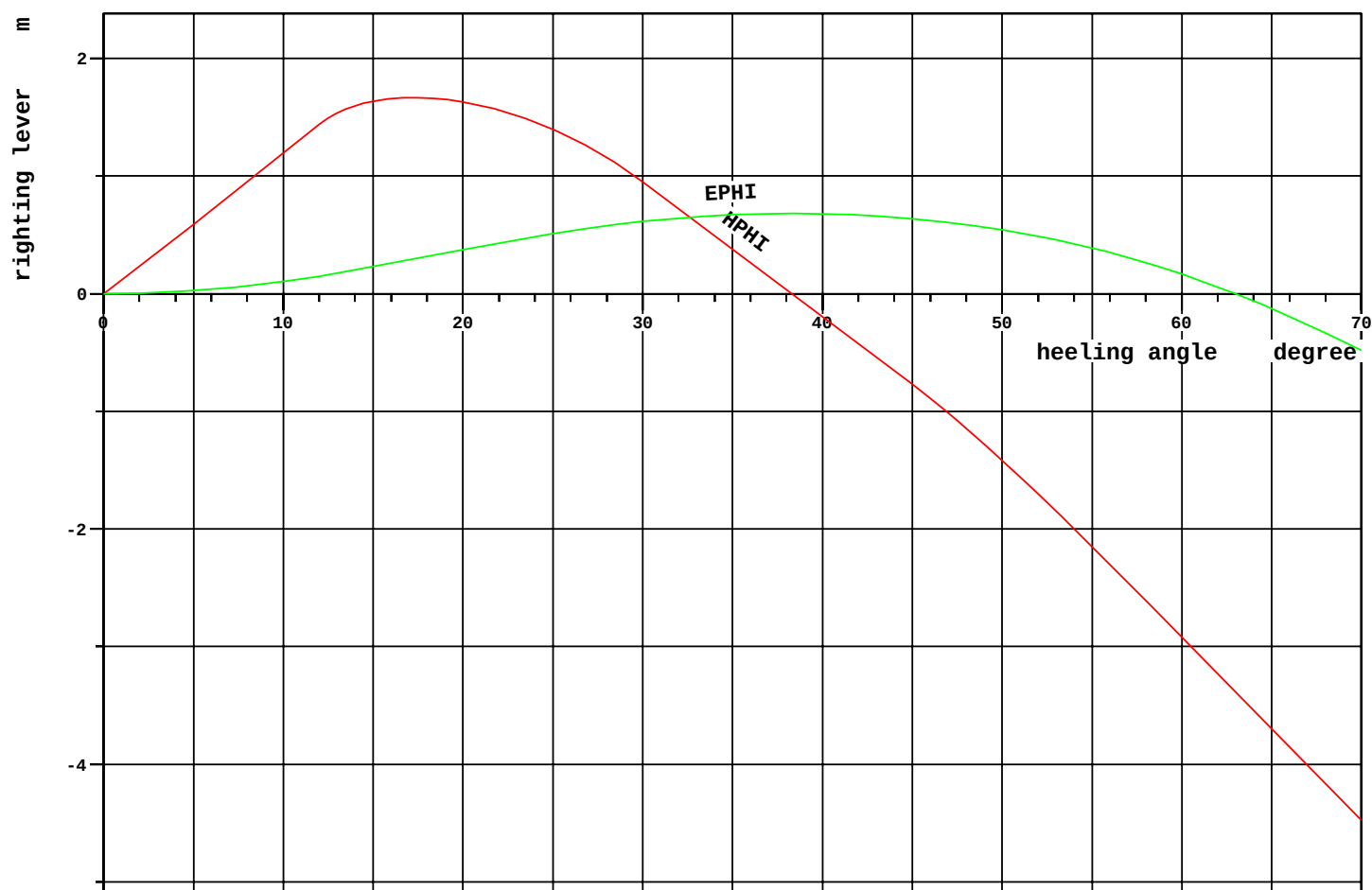
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HFO Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HFO Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HFO Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HFO Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	1995.0	93.7	46.56	16.09	6.01	2085.3
WB03.08	Water Ballast 03.08	1995.0	93.7	46.56	-16.09	6.01	2085.3
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	810.1	49.4	205.80	0.00	4.56	2418.1
Total of Ballast Water		19656.3		106.92	-0.00	2.83	6588.6
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Solid cargo density=1.000 t/m3							
(CAS)	NON-BUOYANT CARGO	20000.0		105.40	0.00	36.00	0.0
Stores							
(ST0)		10.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		40920.4		105.85	0.01	19.10	12330.4
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		61791.8		108.42	0.00	16.17	12330.4
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.068	0.297	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.069	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	0.951	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	17.283	deg	OK
V.GM0.15	GM > 0.15 m	0.150	6.751	m	OK
IMOWEATHER	IMO weather criterion	1.000	1.410		OK
ND-RANGE	Range of Stability acc. ND	36.000	38.257	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	13.970		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	64.470	deg	OK

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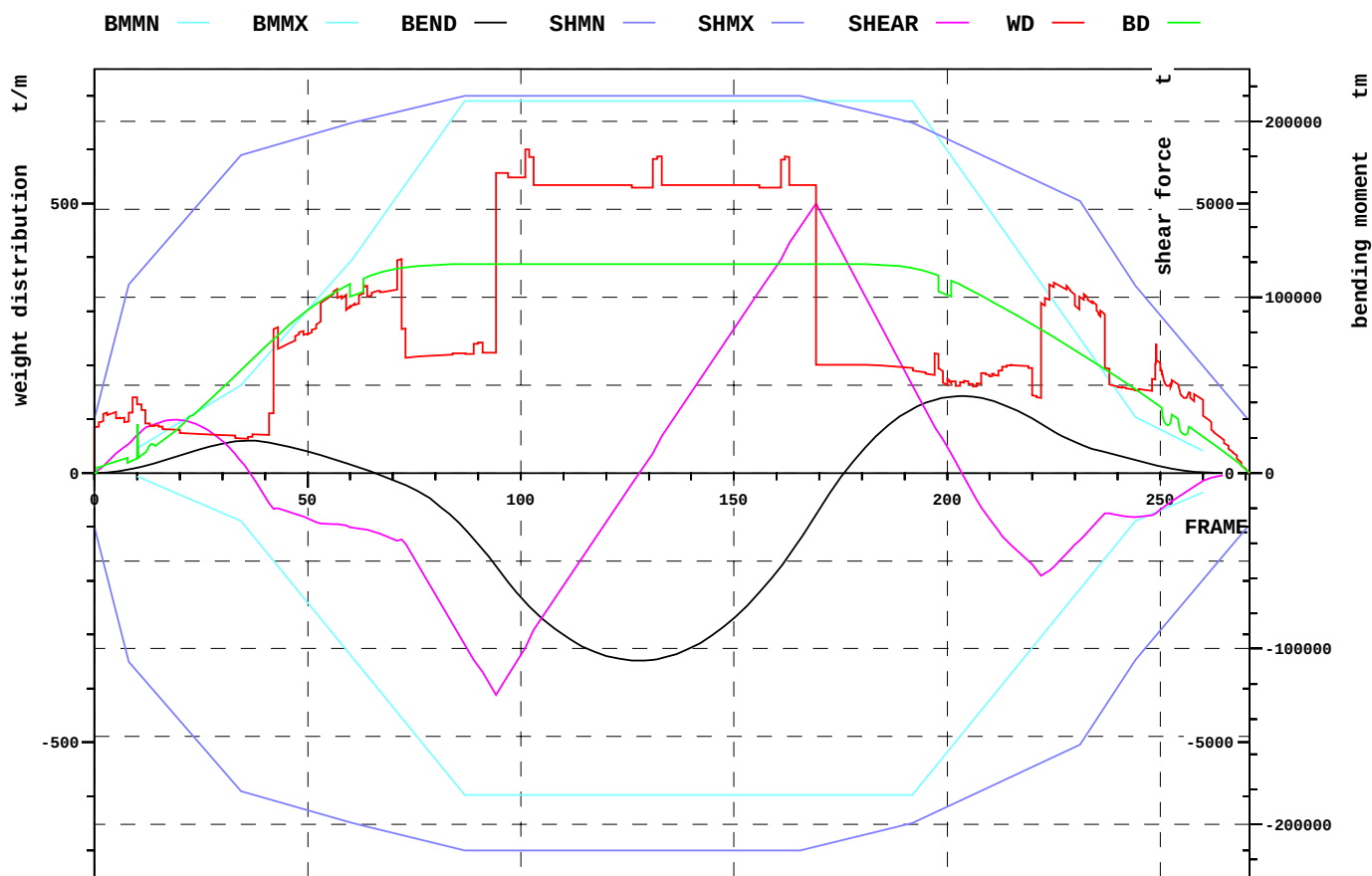
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	24.565
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	24.560
AIR_DUCT_1	UNPROTECTED	64.5	222.00	11.375	33.350	24.554

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-4124.2 t		75.4 m	94
Shear force (max)	5002.6 t		135.4 m	169
Max. rel. shear force	72.2 %		135.4 m	169
Sagging moment	-106847 tm		102.6 m	128
Hogging moment	43819.0 tm		162.4 m	203
Max. rel. bending moment	61.2 %		102.6 m	128

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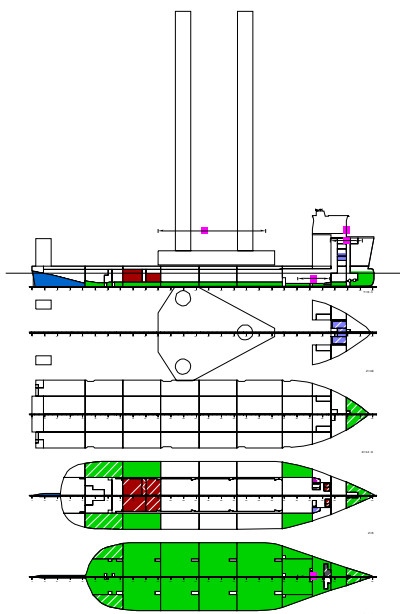
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LOADING CONDITION LC12, Non-Buoyant Cargo, arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	10849	-13513	30261	11.3
42.00	33.60	16974	-50240	70924	10.9
57.00	45.60	7159	-94804	111952	1.4
72.00	57.60	-5850	-139384	160929	11.1
87.00	69.60	-32218	-183412	211763	23.5
102.00	81.60	-75786	-183412	211763	45.5
117.00	93.60	-101481	-183412	211763	58.5
132.00	105.60	-106015	-183412	211763	60.8
147.00	117.60	-88628	-183412	211763	52.0
162.00	129.60	-50119	-183412	211763	32.5
180.00	144.00	12490	-183412	211763	0.9
198.00	158.40	42146	-164872	190357	16.6
210.00	168.00	41478	-129204	149176	22.6
222.00	177.60	28090	-93541	108000	20.7
237.00	189.60	11745	-48977	56548	15.1
249.00	199.20	4176	-22505	25984	10.1

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	973	-4679	4679	20.8
42.00	33.60	-668	-6075	6075	11.0
57.00	45.60	-945	-6418	6418	14.7
72.00	57.60	-1239	-6718	6718	18.4
87.00	69.60	-3197	-7000	7000	45.7
102.00	81.60	-3067	-7000	7000	43.8
117.00	93.60	-1259	-7000	7000	18.0
132.00	105.60	524	-7000	7000	7.5
147.00	117.60	2329	-7000	7000	33.3
162.00	129.60	4109	-7000	7000	58.7
180.00	144.00	3397	-6724	6724	50.5
198.00	158.40	735	-6269	6269	11.7
210.00	168.00	-858	-5824	5824	14.7
222.00	177.60	-1902	-5380	5380	35.4
237.00	189.60	-754	-4339	4339	17.4
249.00	199.20	-750	-3024	3024	24.8

4.10 LC23 Buoyant cargo MLT 116C, departure



LOADING CONDITION LC23: Buoyant Cargo (MLT 116C jackup), departure

FLOATING POSITION

Mean draught (moulded)	8.30 m	KM above the moulded base	23.77 m
Draught at AP (moulded)	8.30 m	KG above the moulded base	10.96 m
Draught at FP (moulded)	8.30 m	GM0 (solid)	12.81 m
Trim	0.00 m	Free surface correction	-0.17 m
Heeling	0.0 deg		
		GM (fluid)	12.65 m
Mean draught (heel 0)	8.30 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	23777.6	109.25	0.04	3.27	6806.7
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	24.7	180.00	5.25	6.25	5.0
Bilge Water	20.4	187.24	-2.65	1.91	133.0
Sludge	36.2	186.86	3.31	1.46	315.1
Solid cargo	8654.0	109.00	0.00	35.25	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	36934.1	106.20	0.00	11.24	9579.2
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	57805.5	108.82	-0.00	10.96	9579.2

LOADS

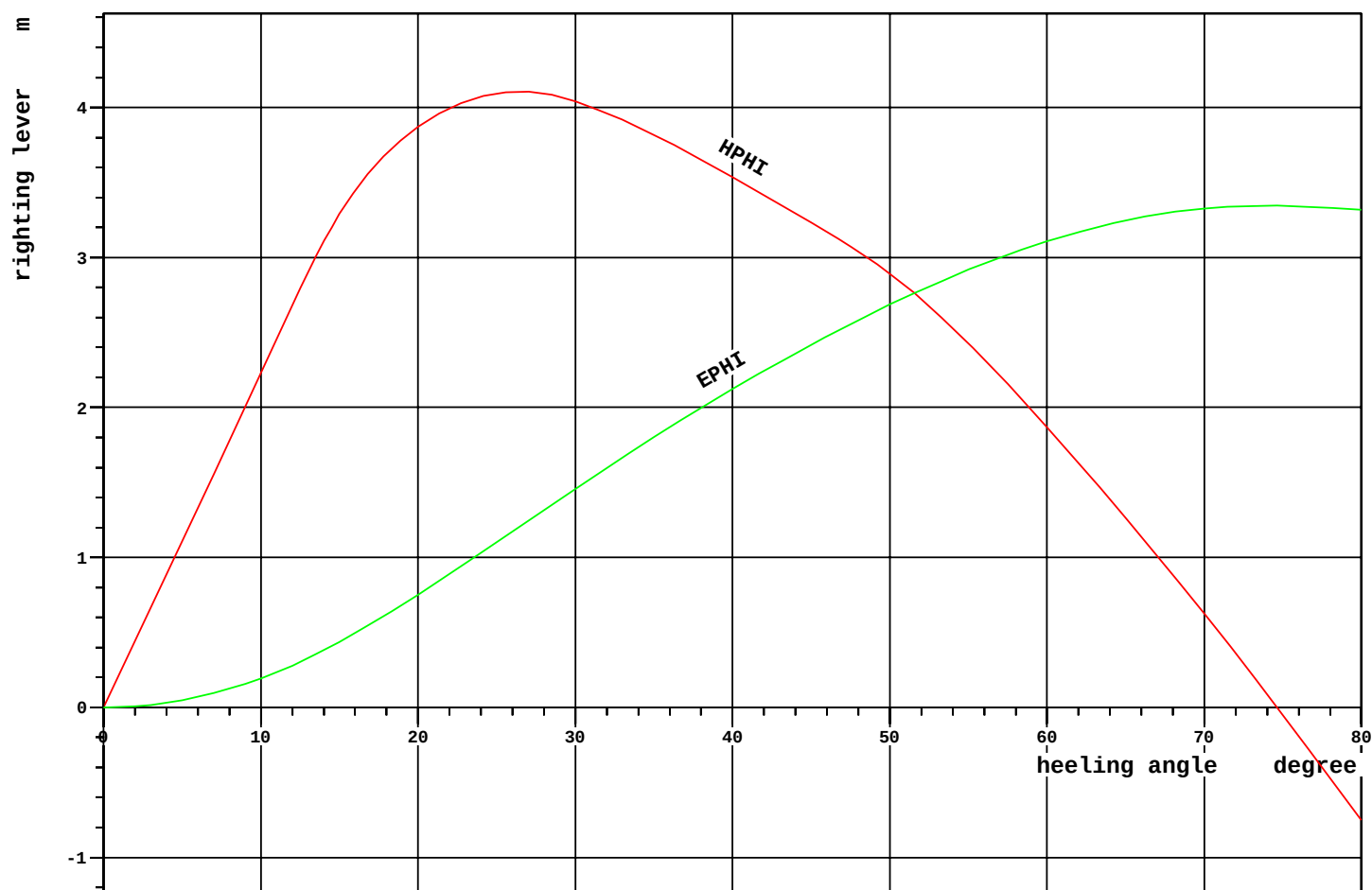
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HFO Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HFO Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HFO Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HFO Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HFO Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HFO Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HFO Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HFO Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	1174.8	55.2	47.22	15.85	4.22	2044.9
WB03.08	Water Ballast 03.08	1112.8	52.3	47.31	-15.82	4.08	2027.1
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	1119.0	68.2	205.76	-0.00	5.74	2734.7
Total of Ballast Water		23777.6		109.25	0.04	3.27	6806.7

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	9.3	50.0	180.00	4.16	6.25	0.9
T09.13	Gray Water	15.4	50.0	180.00	5.91	6.25	4.1
Total of Gray Water		24.7		180.00	5.25	6.25	5.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	5.2	50.0	188.36	-0.73	3.21	0.6
T09.22	Bilge Water	15.2	50.0	186.86	-3.31	1.46	132.4
Total of Bilge Water		20.4		187.24	-2.65	1.91	133.0
Sludge density=2.380 t/m3							
T09.21	Sludge	36.2	50.0	186.86	3.31	1.46	315.1
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	8654.0		109.00	0.00	35.25	0.0
Stores							
(ST0)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							
Deadweight		36934.1		106.20	0.00	11.24	9579.2
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		57805.5		108.82	-0.00	10.96	9579.2
—							



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.059	1.208	mrad	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.664	mrad	OK
V.GZ0.2	Min. GZ > 0.2	0.200	4.042	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	26.488	deg	OK
V.GM0.15	GM > 0.15 m	0.150	12.648	m	OK
IMOWEATHER	IMO weather criterion	1.000	2.211		OK
ND-RANGE	Range of Stability acc. ND	36.000	74.634	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	4.174		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	66.331	deg	OK

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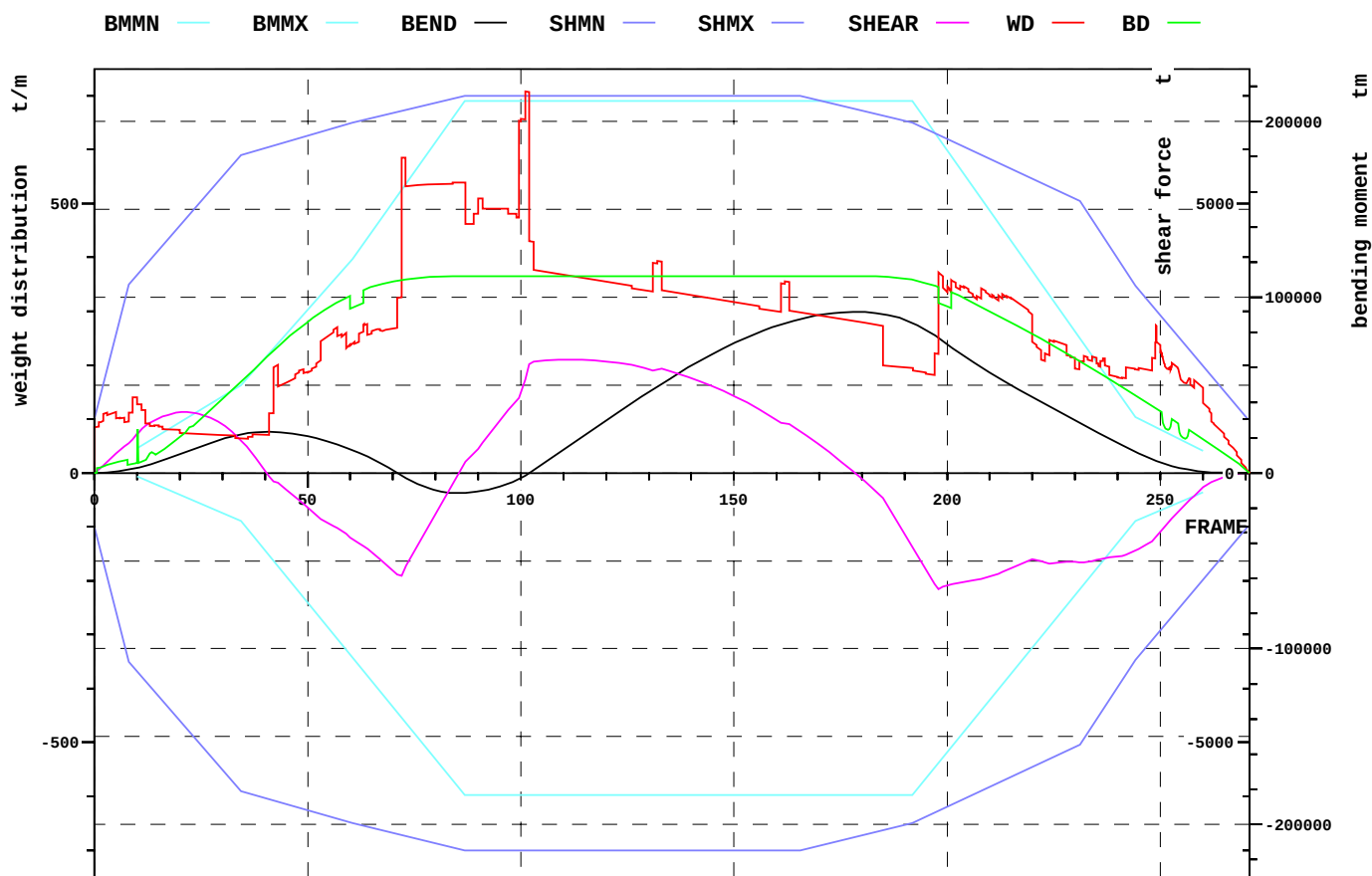
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	25.050
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	25.050
AIR_DUCT_1	UNPROTECTED	66.3	222.00	11.375	33.350	25.050

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-2156.5 t		158.4 m	198
Shear force (max)	2103.2 t		89.3 m	112
Max. rel. shear force	41.5 %		195.4 m	244
Sagging moment	-11524.2 tm		68.4 m	86
Hogging moment	91695.0 tm		142.3 m	178
Max. rel. bending moment	40.0 %		8.0 m	10

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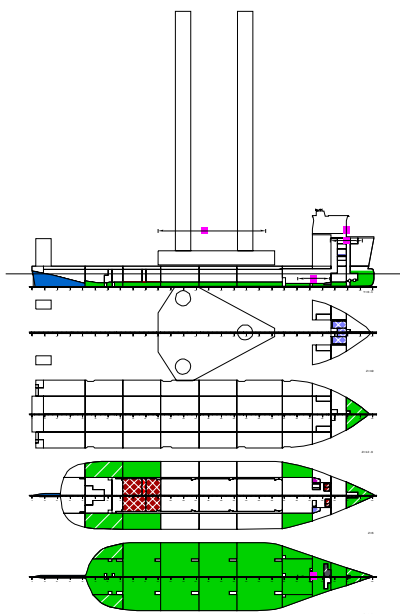
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LOADING CONDITION LC23, Buoyant Cargo (MLT 116C jackup), departure

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	11608	-13513	30261	14.8
42.00	33.60	23315	-50240	70924	21.4
57.00	45.60	16102	-94804	111952	7.3
72.00	57.60	-1408	-139384	160929	8.1
87.00	69.60	-11404	-183412	211763	12.9
102.00	81.60	-117	-183412	211763	7.2
117.00	93.60	24975	-183412	211763	5.5
132.00	105.60	49094	-183412	211763	17.7
147.00	117.60	70315	-183412	211763	28.4
162.00	129.60	85144	-183412	211763	35.9
180.00	144.00	91595	-183412	211763	39.2
198.00	158.40	76716	-164872	190357	36.0
210.00	168.00	57305	-129204	149176	34.0
222.00	177.60	40592	-93541	108000	33.1
237.00	189.60	20853	-48977	56548	32.3
249.00	199.20	6936	-22505	25984	21.4

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1134	-4679	4679	24.2
42.00	33.60	-156	-6075	6075	2.6
57.00	45.60	-1023	-6418	6418	15.9
72.00	57.60	-1909	-6718	6718	28.4
87.00	69.60	204	-7000	7000	2.9
102.00	81.60	2014	-7000	7000	28.8
117.00	93.60	2089	-7000	7000	29.8
132.00	105.60	1916	-7000	7000	27.4
147.00	117.60	1541	-7000	7000	22.0
162.00	129.60	916	-7000	7000	13.1
180.00	144.00	-112	-6724	6724	1.9
198.00	158.40	-2157	-6269	6269	34.4
210.00	168.00	-1916	-5824	5824	32.9
222.00	177.60	-1633	-5380	5380	30.3
237.00	189.60	-1581	-4339	4339	36.4
249.00	199.20	-1191	-3024	3024	39.4

4.11 LC24 Buoyant cargo MLT 116C, arrival



LOADING CONDITION LC24: Buoyant Cargo (MLT 116C jackup), arrival

FLOATING POSITION

Mean draught (moulded)	8.08 m	KM above the moulded base	24.10 m
Draught at AP (moulded)	8.08 m	KG above the moulded base	11.04 m
Draught at FP (moulded)	8.08 m	GM0 (solid)	13.05 m
Trim	0.00 m	Free surface correction	-0.23 m
Heeling	0.0 deg	GM (fluid)	12.83 m
Mean draught (heel 0)	8.08 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	25246.5	105.94	-0.00	3.57	6926.2
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Solid cargo	8654.0	109.00	0.00	35.25	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	35164.7	106.35	0.01	11.40	12668.0
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	56036.1	108.99	0.00	11.04	12668.0

LOADS

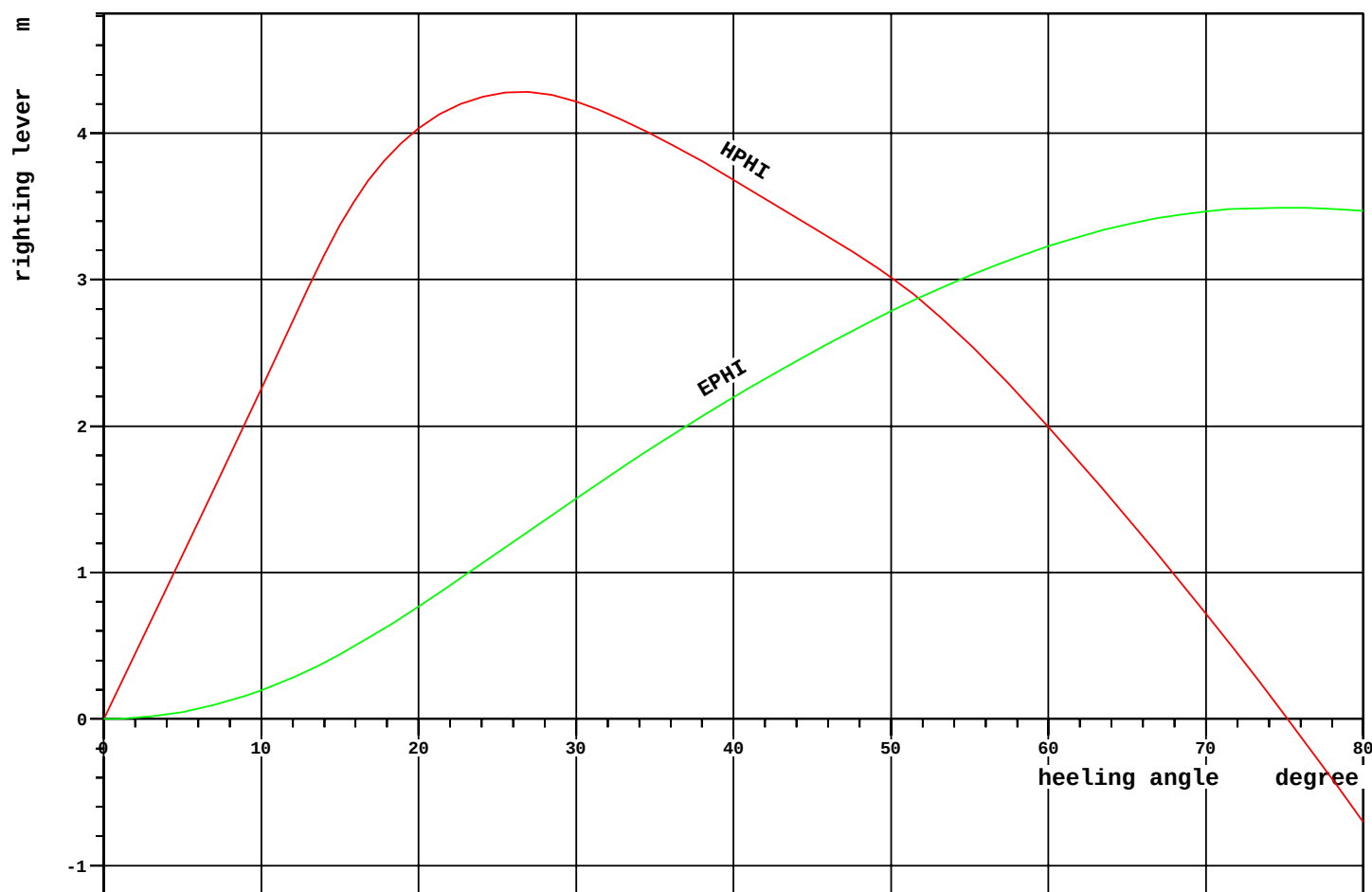
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	1847.5	86.8	46.64	16.06	5.70	2085.3
WB03.08	Water Ballast 03.08	1847.5	86.8	46.64	-16.06	5.70	2085.3
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1199.6	100.0	167.13	15.42	7.04	0.0
WB08.08	Water Ballast 08.08	1199.6	100.0	167.13	-15.42	7.04	0.0
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
WB11.01	Water Ballast 11.01	1180.6	72.0	205.73	-0.00	5.95	2755.7
Total of Ballast Water		25246.5		105.94	-0.00	3.57	6926.2

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	8654.0		109.00	0.00	35.25	0.0
Stores							
(ST0)		10.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3							
(MIS)		87.0		178.40	0.00	5.00	0.0
—							
Deadweight		35164.7		106.35	0.01	11.40	12668.0
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		56036.1		108.99	0.00	11.04	12668.0
—							



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.059	1.245	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.693	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	4.217	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	26.483	deg	OK
V.GM0.15	GM > 0.15 m	0.150	12.829	m	OK
IMOWEATHER	IMO weather criterion	1.000	2.180		OK
ND-RANGE	Range of Stability acc. ND	36.000	75.149	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	4.195		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	67.167	deg	OK

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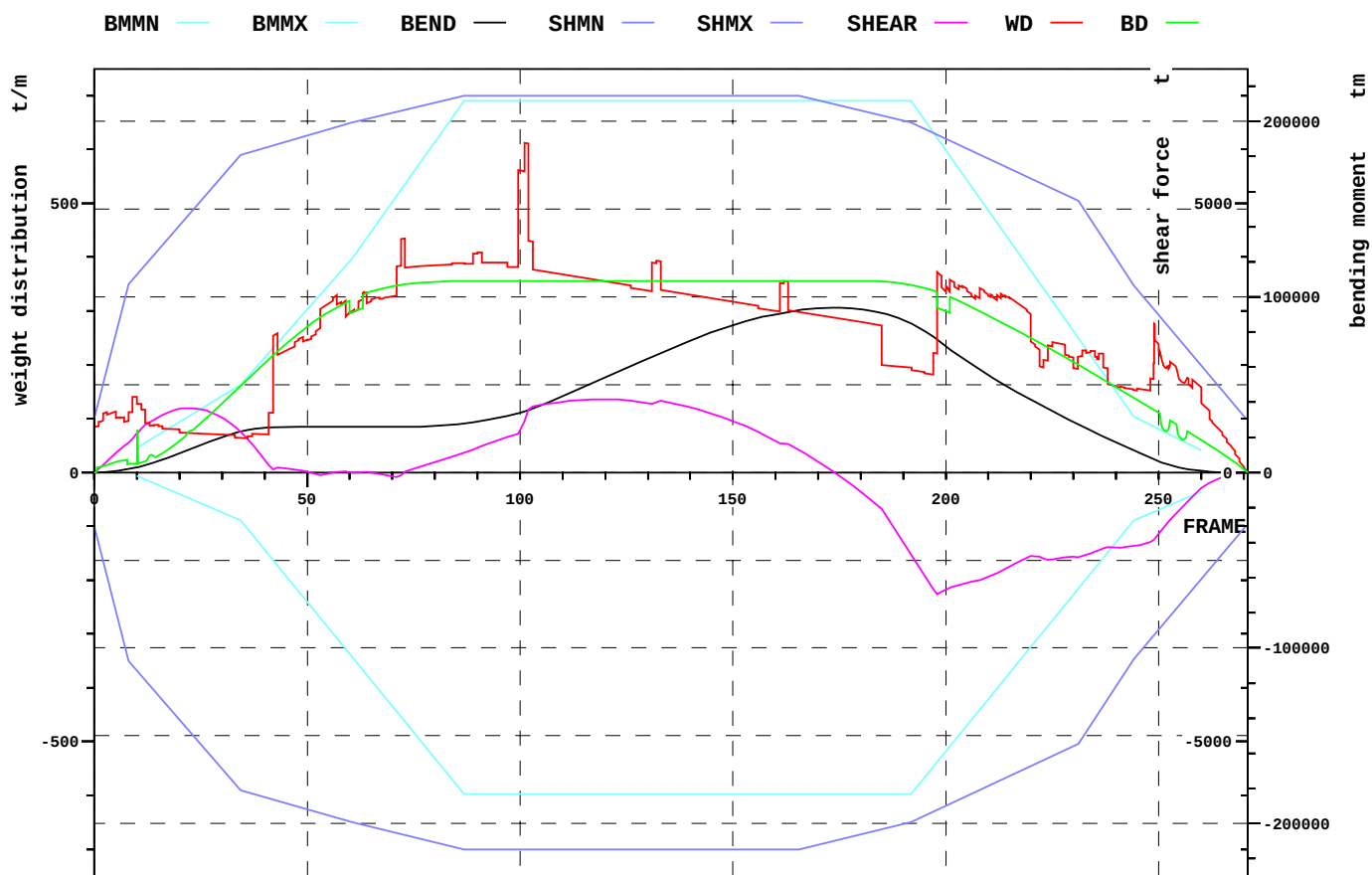
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	25.273
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	25.270
AIR_DUCT_1	UNPROTECTED	67.2	222.00	11.375	33.350	25.267

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-2261.7 t		158.4 m	198
Shear force (max)	1358.4 t		95.9 m	120
Max. rel. shear force	41.5 %		198.4 m	248
Sagging moment	-0.9 tm		216.0 m	270
Hogging moment	93797.8 tm		138.6 m	173
Max. rel. bending moment	40.3 %		138.6 m	173

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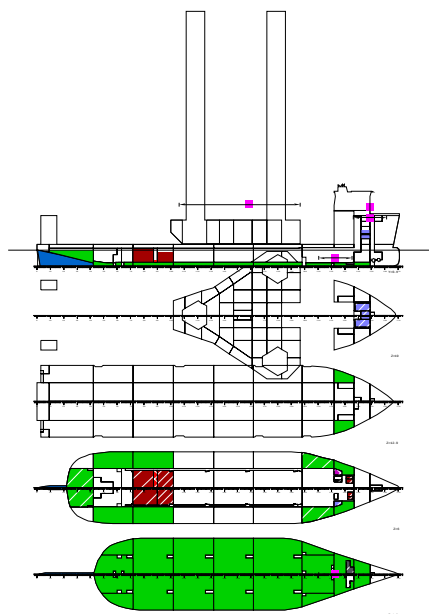
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LOADING CONDITION LC24, Buoyant Cargo (MLT 116C jackup), arrival

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	11858	-13513	30261	15.9
42.00	33.60	25827	-50240	70924	25.6
57.00	45.60	26088	-94804	111952	16.9
72.00	57.60	25902	-139384	160929	10.1
87.00	69.60	27999	-183412	211763	7.0
102.00	81.60	35383	-183412	211763	10.7
117.00	93.60	50964	-183412	211763	18.6
132.00	105.60	66972	-183412	211763	26.7
147.00	117.60	81479	-183412	211763	34.1
162.00	129.60	90991	-183412	211763	38.9
180.00	144.00	92912	-183412	211763	39.8
198.00	158.40	75507	-164872	190357	35.3
210.00	168.00	55522	-129204	149176	32.7
222.00	177.60	39060	-93541	108000	31.6
237.00	189.60	20418	-48977	56548	31.5
249.00	199.20	7344	-22505	25984	23.1

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1193	-4679	4679	25.5
42.00	33.60	58	-6075	6075	1.0
57.00	45.60	11	-6418	6418	0.2
72.00	57.60	-55	-6718	6718	0.8
87.00	69.60	365	-7000	7000	5.2
102.00	81.60	1163	-7000	7000	16.6
117.00	93.60	1355	-7000	7000	19.4
132.00	105.60	1299	-7000	7000	18.6
147.00	117.60	1040	-7000	7000	14.9
162.00	129.60	532	-7000	7000	7.6
180.00	144.00	-357	-6724	6724	5.3
198.00	158.40	-2262	-6269	6269	36.1
210.00	168.00	-1931	-5824	5824	33.2
222.00	177.60	-1567	-5380	5380	29.1
237.00	189.60	-1411	-4339	4339	32.5
249.00	199.20	-1247	-3024	3024	41.2

4.12 LC25 Buoyant cargo Super Gorilla 16000t, departure



LOADING CONDITION LC25: Buoyant Cargo Super Gorilla 16000t, dep.

FLOATING POSITION

Mean draught (moulded)	9.68 m	KM above the moulded base	22.18 m
Draught at AP (moulded)	9.68 m	KG above the moulded base	13.79 m
Draught at FP (moulded)	9.68 m	GM0 (solid)	8.40 m
Trim	0.00 m	Free surface correction	-0.15 m
Heeling	0.0 deg	GM (fluid)	8.25 m
Mean draught (heel 0)	9.68 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	3500.0	71.26	-0.31	6.16	1683.1
Diesel Oil	80.0	152.31	1.43	3.22	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	20.0	179.80	-9.33	6.25	10.1
Fresh Water	300.0	195.55	0.00	18.24	343.4
Ballast Water	27787.4	95.86	0.04	4.00	7440.2
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	24.7	180.00	5.25	6.25	5.0
Bilge Water	20.4	187.24	-2.65	1.91	133.0
Sludge	36.2	186.86	3.31	1.46	315.1
Solid cargo	16000.0	127.00	0.00	37.00	0.0
Stores	100.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	48289.9	105.14	0.00	15.23	10212.7
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	69161.3	107.65	-0.00	13.79	10212.7

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	1031.0	98.0	63.60	5.51	6.61	0.0
HF02	HF0 Storage Tank 2	1237.2	98.0	64.80	-5.51	6.61	0.0
HF03	HF0 Storage Tank 3	574.3	69.5	76.72	5.32	5.47	834.7
HF04	HF0 Storage Tank 4	574.4	69.5	76.72	-5.32	5.47	834.7
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		3500.0		71.26	-0.31	6.16	1683.1
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	20.1	10.5	70.80	5.69	3.02	259.6
T09.07	MDO Storage P	24.4	78.6	179.30	9.58	3.38	6.4
T09.08	MDO Storage S	24.4	78.6	179.30	-9.58	3.38	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		80.0		152.31	1.43	3.22	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	20.0	49.8	179.80	-9.33	6.25	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	80.0	36.5	194.80	4.81	17.46	60.2
FW2	Fresh Water S	80.0	36.5	194.80	-4.81	17.46	60.2
FW3	Fresh Water C	70.0	44.7	196.40	0.00	20.64	156.9
T10.01	Potable Water	70.0	92.6	196.40	0.00	17.61	66.2
Total of Fresh Water		300.0		195.55	0.00	18.24	343.4

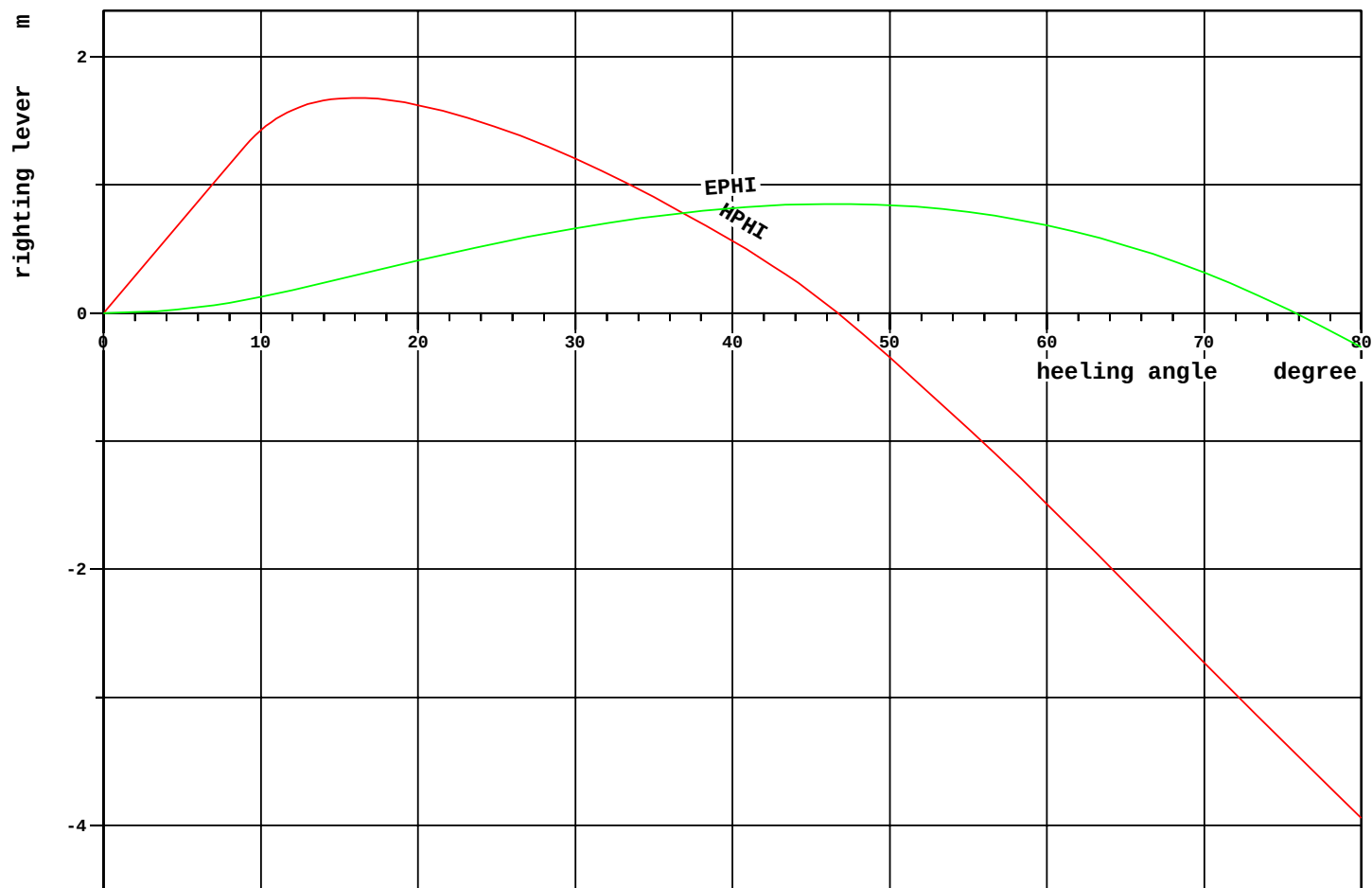
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	1192.3	78.2	23.74	5.38	7.11	2567.5
WB02.06	Water Ballast 02.06	1192.3	78.1	23.76	-5.39	7.11	2564.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.07	Water Ballast 03.07	2128.4	100.0	46.50	16.11	6.30	0.0
WB03.08	Water Ballast 03.08	2128.4	100.0	46.50	-16.11	6.30	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	987.3	82.3	167.06	15.32	6.36	1171.4
WB08.08	Water Ballast 08.08	920.1	76.7	167.04	-15.29	6.15	1137.3
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
Total of Ballast Water		27787.4		95.86	0.04	4.00	7440.2
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	9.3	50.0	180.00	4.16	6.25	0.9
T09.13	Gray Water	15.4	50.0	180.00	5.91	6.25	4.1
Total of Gray Water		24.7		180.00	5.25	6.25	5.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	5.2	50.0	188.36	-0.73	3.21	0.6
T09.22	Bilge Water	15.2	50.0	186.86	-3.31	1.46	132.4
Total of Bilge Water		20.4		187.24	-2.65	1.91	133.0
Sludge density=2.380 t/m3							
T09.21	Sludge	36.2	50.0	186.86	3.31	1.46	315.1
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	16000.0		127.00	0.00	37.00	0.0
Stores							
(ST0)		100.0		199.20	0.00	29.05	0.0
Crew							
(CREW)		6.0		199.20	0.00	35.70	0.0

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
	Miscellaneous density=1.000 t/m3 (MIS)	87.0		178.40	0.00	5.00	0.0
—							—
	Deadweight	48289.9		105.14	0.00	15.23	10212.7
	Lightweight	20871.4		113.44	-0.01	10.44	
	Displacement (1.025 t/m3)	69161.3		107.65	-0.00	13.79	10212.7
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.069	0.296	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.156	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	1.205	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	16.062	deg	OK
V.GM0.15	GM > 0.15 m	0.150	8.250	m	OK
IMOWEATHER	IMO weather criterion	1.000	1.123		OK
ND-RANGE	Range of Stability acc. ND	36.000	46.683	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	1.984		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	61.282	deg	OK

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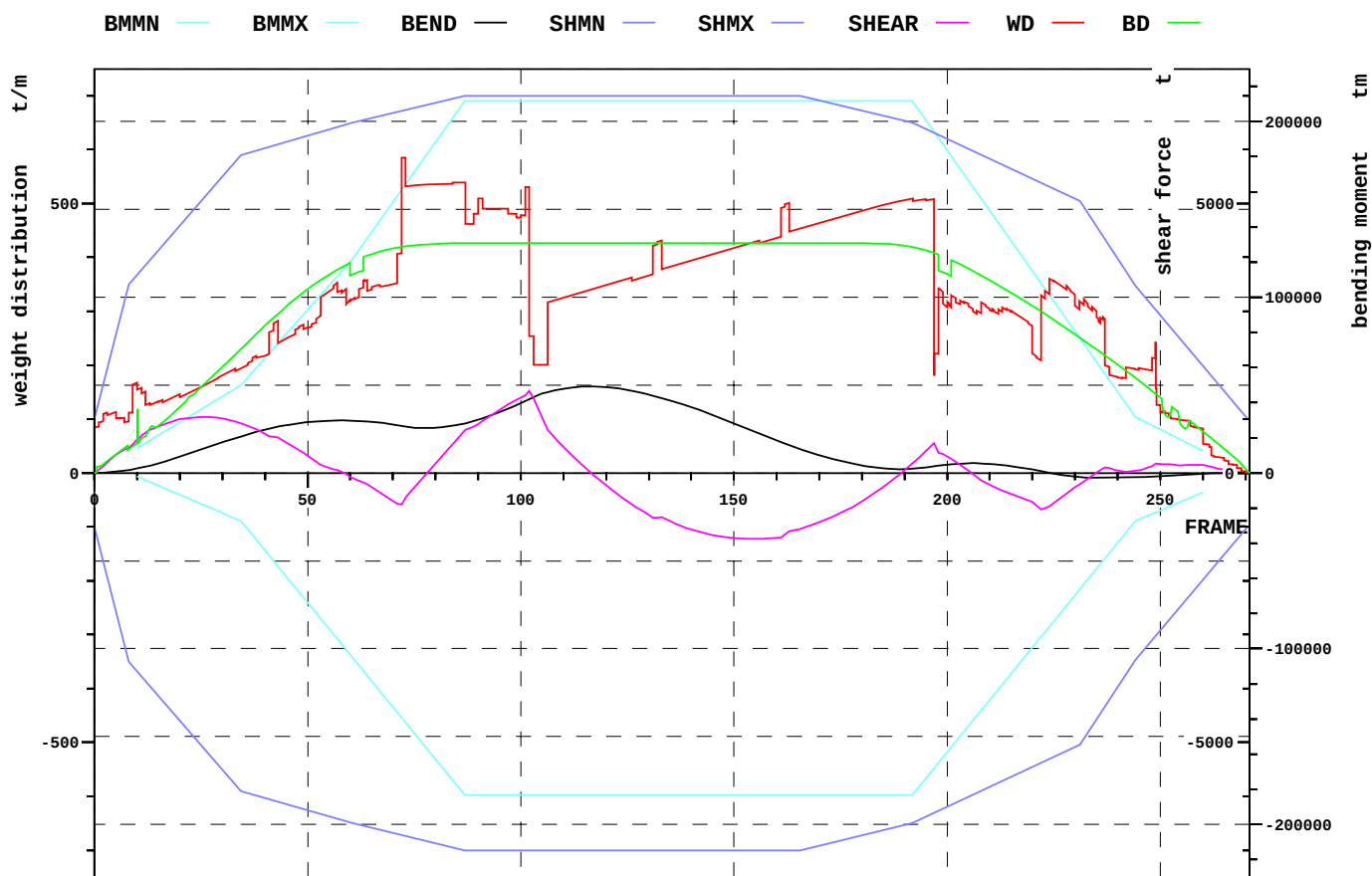
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	23.670
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	23.670
AIR_DUCT_1	UNPROTECTED	61.3	222.00	11.375	33.350	23.670

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-1223.2 t		122.4 m	153
Shear force (max)	1523.9 t		81.6 m	102
Max. rel. shear force	21.8 %		81.6 m	102
Sagging moment	-2615.3 tm		188.0 m	235
Hogging moment	49150.2 tm		93.6 m	117
Max. rel. bending moment	43.8 %		8.0 m	10

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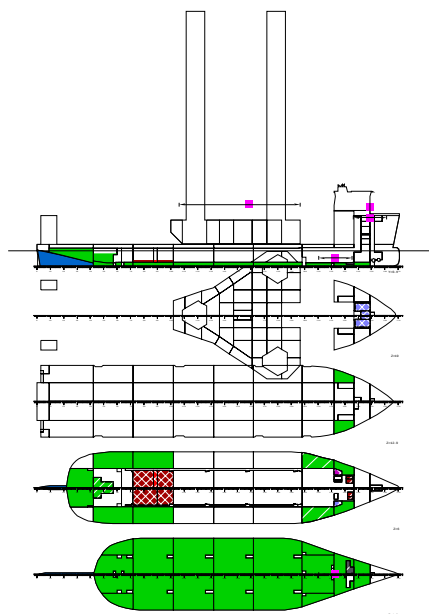
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LOADING CONDITION LC25, Buoyant Cargo Super Gorilla 16000t, dep.

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	10110	-13513	30261	7.9
42.00	33.60	25743	-50240	70924	25.4
57.00	45.60	29873	-94804	111952	20.6
72.00	57.60	26760	-139384	160929	10.6
87.00	69.60	28308	-183412	211763	7.2
102.00	81.60	42012	-183412	211763	14.1
117.00	93.60	49150	-183412	211763	17.7
132.00	105.60	43399	-183412	211763	14.8
147.00	117.60	31168	-183412	211763	8.6
162.00	129.60	16671	-183412	211763	1.3
180.00	144.00	4262	-183412	211763	5.0
198.00	158.40	4267	-164872	190357	4.8
210.00	168.00	5136	-129204	149176	3.5
222.00	177.60	1082	-93541	108000	6.1
237.00	189.60	-2533	-48977	56548	12.0
249.00	199.20	-1895	-22505	25984	15.0

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1011	-4679	4679	21.6
42.00	33.60	664	-6075	6075	10.9
57.00	45.60	51	-6418	6418	0.8
72.00	57.60	-581	-6718	6718	8.6
87.00	69.60	801	-7000	7000	11.4
102.00	81.60	1524	-7000	7000	21.8
117.00	93.60	-37	-7000	7000	0.5
132.00	105.60	-829	-7000	7000	11.8
147.00	117.60	-1181	-7000	7000	16.9
162.00	129.60	-1142	-7000	7000	16.3
180.00	144.00	-525	-6724	6724	7.8
198.00	158.40	380	-6269	6269	6.1
210.00	168.00	-215	-5824	5824	3.7
222.00	177.60	-675	-5380	5380	12.5
237.00	189.60	103	-4339	4339	2.4
249.00	199.20	179	-3024	3024	5.9

4.13 LC26 Buoyant cargo Super Gorilla 16000t, arrival



LOADING CONDITION LC26: Buoyant Cargo Super Gorilla 16000t, arr.

FLOATING POSITION

Mean draught (moulded)	9.48 m	KM above the moulded base	22.37 m
Draught at AP (moulded)	9.48 m	KG above the moulded base	13.93 m
Draught at FP (moulded)	9.48 m	GM0 (solid)	8.45 m
Trim	0.00 m	Free surface correction	-0.14 m
Heeling	0.0 deg	GM (fluid)	8.31 m
Mean draught (heel 0)	9.48 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	29357.9	93.87	-0.00	4.23	3485.4
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Solid cargo	16000.0	127.00	0.00	37.00	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	46622.0	105.31	0.01	15.48	9227.2
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	67493.4	107.82	0.00	13.93	9227.2

LOADS

Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4

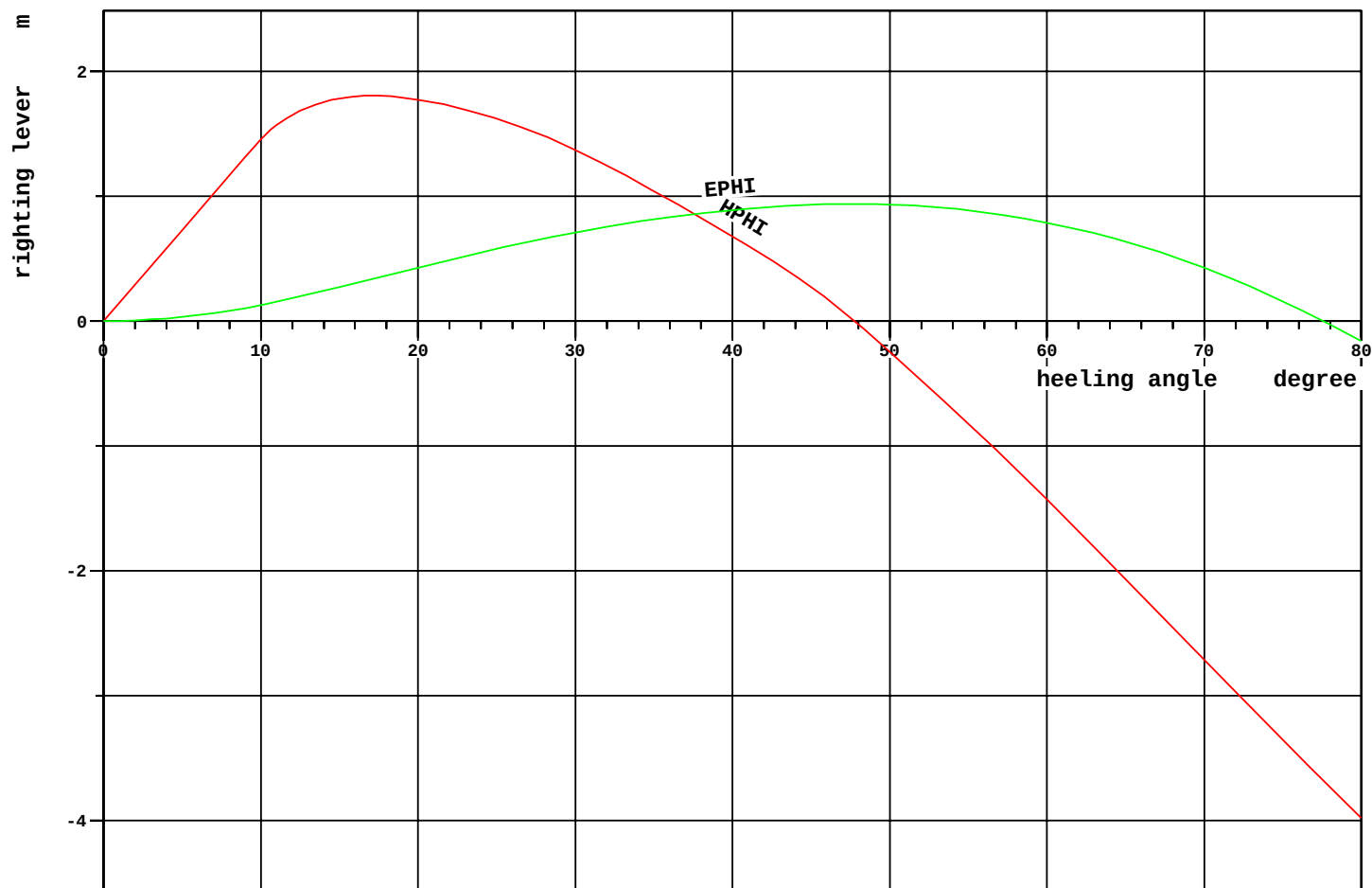
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Ballast Water density=1.025 t/m3							
WB02.05	Water Ballast 02.05	1524.8	100.0	23.01	5.35	7.78	0.0
WB02.06	Water Ballast 02.06	1526.6	100.0	23.02	-5.35	7.78	0.0
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB03.06	Water Ballast 03.06	567.8	59.1	39.21	0.00	5.18	1014.3
WB03.07	Water Ballast 03.07	2128.4	100.0	46.50	16.11	6.30	0.0
WB03.08	Water Ballast 03.08	2128.4	100.0	46.50	-16.11	6.30	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB04.07	Water Ballast 04.07	2001.6	100.0	69.60	16.44	6.70	0.0
WB04.08	Water Ballast 04.08	2001.6	100.0	69.60	-16.44	6.70	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB08.07	Water Ballast 08.07	1121.6	93.5	167.11	15.39	6.80	1235.6
WB08.08	Water Ballast 08.08	1121.6	93.5	167.11	-15.39	6.80	1235.6
WB09.01	Water Ballast 09.01	438.6	100.0	182.90	-0.00	1.12	0.0
WB09.07	Water Ballast 09.07	633.4	100.0	183.26	12.72	8.16	0.0
WB09.08	Water Ballast 09.08	633.4	100.0	183.26	-12.72	8.16	0.0
WB10.01	Water Ballast 10.01	314.2	100.0	194.02	0.00	1.44	0.0
Total of Ballast Water		29357.9		93.87	-0.00	4.23	3485.4
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Solid cargo density=1.000 t/m3							
(CAS)	SG (31.2 0 37.2)	16000.0		127.00	0.00	37.00	0.0
Stores (ST0)		10.0		199.20	0.00	29.05	0.0
Crew (CREW)		6.0		199.20	0.00	35.70	0.0

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
	Miscellaneous density=1.000 t/m3 (MIS)	87.0		178.40	0.00	5.00	0.0
—							—
	Deadweight	46622.0		105.31	0.01	15.48	9227.2
	Lightweight	20871.4		113.44	-0.01	10.44	
	Displacement (1.025 t/m3)	67493.4		107.82	0.00	13.93	9227.2
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.068	0.338	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	0.181	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	1.370	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	17.113	deg	OK
V.GM0.15	GM > 0.15 m	0.150	8.310	m	OK
IMOWEATHER	IMO weather criterion	1.000	1.184		OK
ND-RANGE	Range of Stability acc. ND	36.000	47.713	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	2.075		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	61.969	deg	OK

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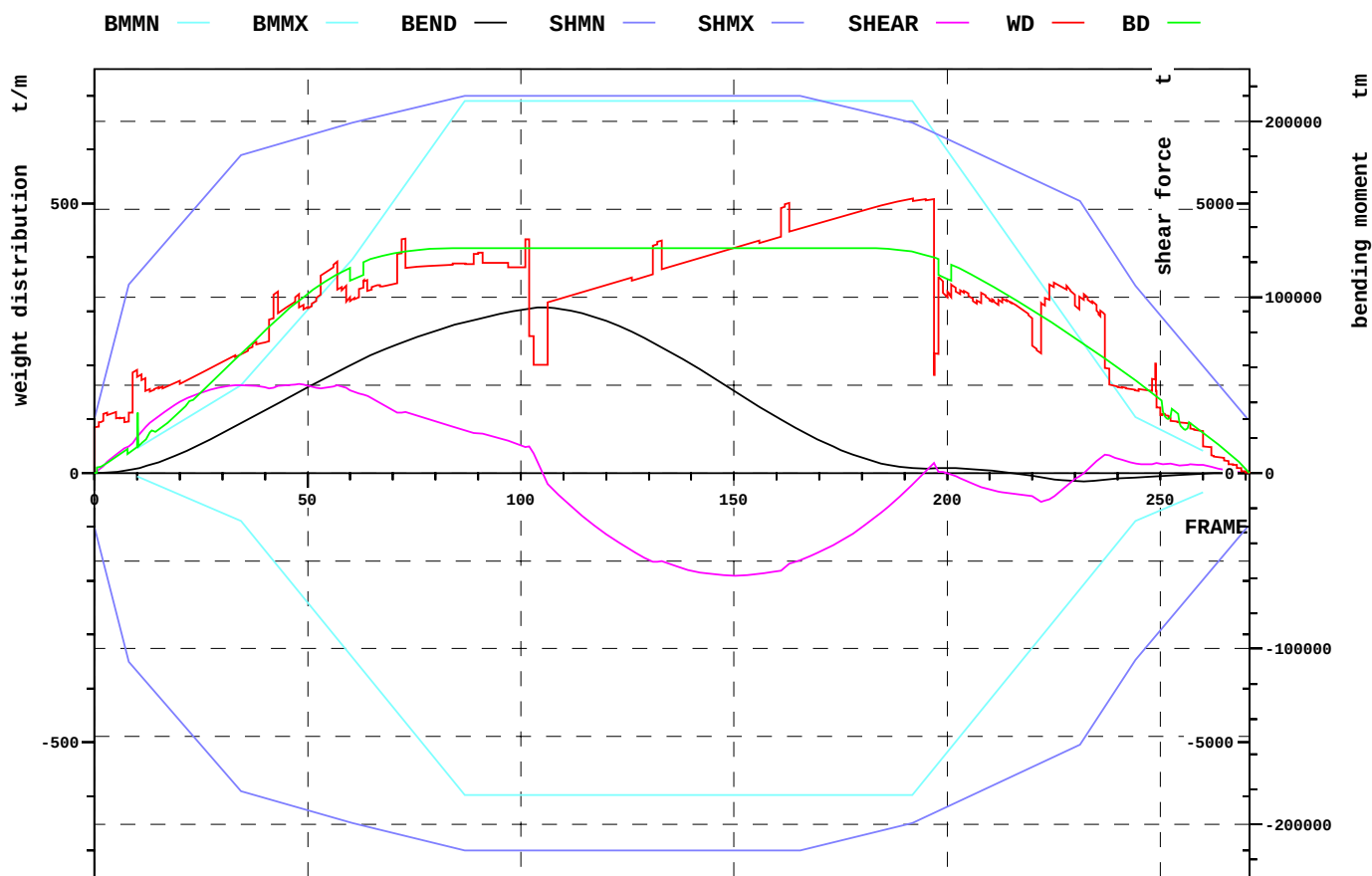
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	23.874
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	23.870
AIR_DUCT_1	UNPROTECTED	62.0	222.00	11.375	33.350	23.866

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-1903.5 t		120.0 m	150
Shear force (max)	1655.3 t		38.4 m	48
Max. rel. shear force	29.6 %		20.4 m	26
Sagging moment	-4729.4 tm		185.6 m	232
Hogging moment	94251.4 tm		84.0 m	105
Max. rel. bending moment	47.5 %		48.0 m	60

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Loading Condition
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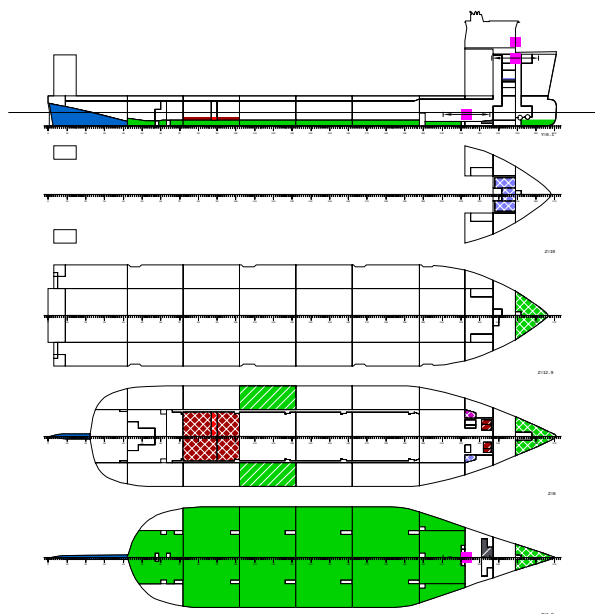
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LOADING CONDITION LC26, Buoyant Cargo Super Gorilla 16000t, arr.

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELBM %
21.00	16.80	11967	-13513	30261	16.4
42.00	33.60	38219	-50240	70924	46.0
57.00	45.60	57629	-94804	111952	47.5
72.00	57.60	74221	-139384	160929	42.3
87.00	69.60	85898	-183412	211763	36.3
102.00	81.60	93607	-183412	211763	40.2
117.00	93.60	89011	-183412	211763	37.9
132.00	105.60	72794	-183412	211763	29.7
147.00	117.60	51364	-183412	211763	18.8
162.00	129.60	28935	-183412	211763	7.5
180.00	144.00	8685	-183412	211763	2.8
198.00	158.40	2665	-164872	190357	5.7
210.00	168.00	1425	-129204	149176	6.2
222.00	177.60	-2351	-93541	108000	9.5
237.00	189.60	-4023	-48977	56548	14.8
249.00	199.20	-1902	-22505	25984	15.0

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1357	-4679	4679	29.0
42.00	33.60	1581	-6075	6075	26.0
57.00	45.60	1629	-6418	6418	25.4
72.00	57.60	1115	-6718	6718	16.6
87.00	69.60	793	-7000	7000	11.3
102.00	81.60	493	-7000	7000	7.0
117.00	93.60	-962	-7000	7000	13.7
132.00	105.60	-1648	-7000	7000	23.5
147.00	117.60	-1895	-7000	7000	27.1
162.00	129.60	-1750	-7000	7000	25.0
180.00	144.00	-1007	-6724	6724	15.0
198.00	158.40	25	-6269	6269	0.4
210.00	168.00	-304	-5824	5824	5.2
222.00	177.60	-536	-5380	5380	10.0
237.00	189.60	341	-4339	4339	7.9
249.00	199.20	187	-3024	3024	6.2

4.14 LC31 Docking condition



LOADING CONDITION LC31: Docking condition

FLOATING POSITION

Mean draught (moulded)	5.70 m	KM above the moulded base	29.82 m
Draught at AP (moulded)	5.70 m	KG above the moulded base	6.67 m
Draught at FP (moulded)	5.70 m	GM0 (solid)	23.15 m
Trim	0.00 m	Free surface correction	-0.28 m
Heeling	0.0 deg	GM (fluid)	22.87 m
Mean draught (heel 0)	5.70 m		
Trim (heel 0)	0.00 m		

LOADS

Item	Weight (t)	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil	559.2	86.85	-0.17	3.17	5105.4
Diesel Oil	41.5	119.85	3.14	2.87	273.6
Lubricating Oil	7.0	179.80	9.33	5.79	9.1
Technical Water	4.0	179.80	-9.33	5.65	10.1
Fresh Water	67.1	195.35	0.00	17.43	343.4
Ballast Water	15504.5	108.11	-0.00	1.77	4801.1
Fixed FW ballast	321.1	14.73	0.00	3.55	0.0
Gray Water	49.4	180.00	5.25	7.00	0.0
Bilge Water	40.9	187.40	-2.57	2.31	0.0
Sludge	71.0	187.03	3.22	1.73	0.0
Stores	10.0	199.20	0.00	29.05	0.0
Crew	6.0	199.20	0.00	35.70	0.0
Miscellaneous	87.0	178.40	0.00	5.00	0.0
Deadweight	16768.6	107.23	0.02	1.98	10542.9
Lightweight	20871.4	113.44	-0.01	10.44	
Displacement (1.025 t/m3)	37640.0	110.67	0.01	6.67	10542.9

LOADS

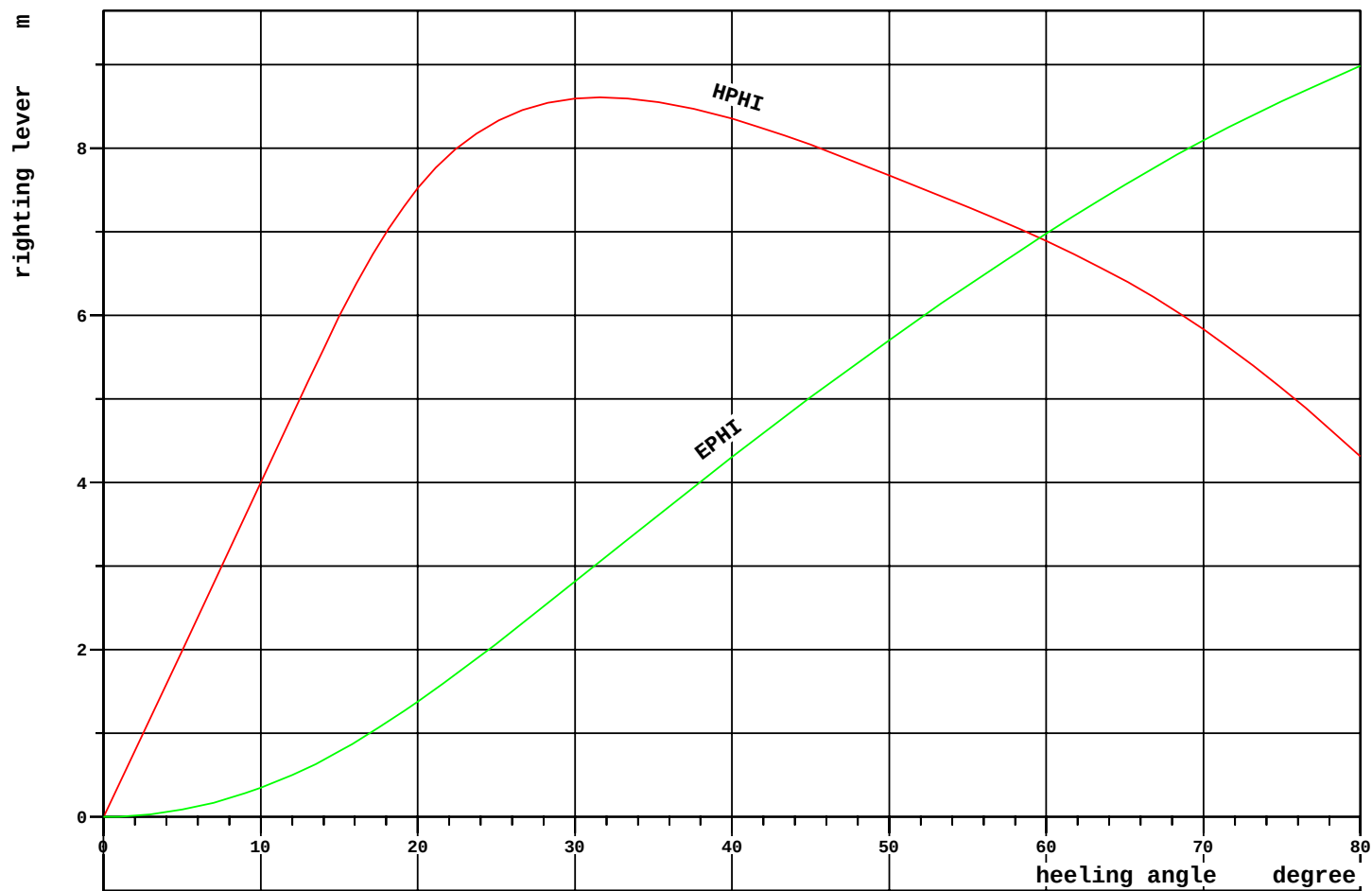
Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
Heavy Fuel Oil density=0.991 t/m3							
HF01	HF0 Storage Tank 1	126.2	12.0	63.60	5.69	3.08	1429.4
HF02	HF0 Storage Tank 2	151.5	12.0	64.80	-5.69	3.08	1715.3
HF03	HF0 Storage Tank 3	99.2	12.0	76.61	5.36	3.10	973.6
HF04	HF0 Storage Tank 4	99.2	12.0	76.61	-5.36	3.10	973.6
T09.03	HF0 Settling Tank P	27.8	50.0	186.74	3.94	3.61	5.9
T09.04	HF0 Settling Tank S	22.0	50.0	187.16	-3.94	3.65	4.7
T09.05	HF0 Service Tank P	18.5	50.0	186.74	6.13	3.61	1.7
T09.06	HF0 Service Tank S	14.7	50.0	187.16	-6.13	3.65	1.4
Total of Heavy Fuel Oil		559.2		86.85	-0.17	3.17	5105.4
Diesel Oil density=0.900 t/m3							
MGO	MGO Storage Tank	22.9	12.0	70.80	5.69	3.08	259.6
T09.07	MDO Storage P	3.7	12.0	179.29	9.58	2.21	6.4
T09.08	MDO Storage S	3.7	12.0	179.29	-9.58	2.21	6.4
T09.18	MDO Service P	5.6	50.0	181.20	8.63	2.88	0.6
T09.19	MDO Service S	5.6	50.0	181.20	-8.63	2.88	0.6
Total of Diesel Oil		41.5		119.85	3.14	2.87	273.6
Lubricating Oil density=0.900 t/m3							
T09.09	Used LO	7.0	19.4	179.80	9.33	5.79	9.1
Technical Water density=1.000 t/m3							
T09.10	Feed Water	4.0	10.0	179.80	-9.33	5.65	10.1
Fresh Water density=1.000 t/m3							
FW1	Fresh Water P	21.9	10.0	194.80	4.81	16.69	60.2
FW2	Fresh Water S	21.9	10.0	194.80	-4.81	16.69	60.2
FW3	Fresh Water C	15.7	10.0	196.40	0.00	19.93	156.9
T10.01	Potable Water	7.6	10.0	196.40	0.00	16.53	66.2
Total of Fresh Water		67.1		195.35	0.00	17.43	343.4
Ballast Water density=1.025 t/m3							
WB03.01	Water Ballast 03.01	482.3	100.0	47.42	5.24	1.46	0.0
WB03.02	Water Ballast 03.02	489.9	100.0	47.48	-5.17	1.46	0.0
WB04.01	Water Ballast 04.01	1308.9	100.0	69.62	10.53	1.31	0.0
WB04.02	Water Ballast 04.02	1308.9	100.0	69.62	-10.53	1.31	0.0
WB05.01	Water Ballast 05.01	1320.3	100.0	93.51	10.63	1.30	0.0
WB05.02	Water Ballast 05.02	1320.3	100.0	93.51	-10.63	1.30	0.0
WB05.07	Water Ballast 05.07	1039.5	51.9	93.60	16.44	4.73	2085.3
WB05.08	Water Ballast 05.08	1039.5	51.9	93.60	-16.44	4.73	2085.3
WB06.01	Water Ballast 06.01	1320.3	100.0	117.51	10.63	1.30	0.0
WB06.02	Water Ballast 06.02	1320.3	100.0	117.51	-10.63	1.30	0.0
WB07.01	Water Ballast 07.01	1555.7	100.0	143.70	10.44	1.31	0.0
WB07.02	Water Ballast 07.02	1555.7	100.0	143.70	-10.44	1.31	0.0
WB08.01	Water Ballast 08.01	854.0	100.0	167.59	-0.04	1.03	0.0
WB08.03	Water Ballast 08.03	189.6	100.0	165.55	14.11	1.49	0.0
WB08.04	Water Ballast 08.04	189.6	100.0	165.55	-14.11	1.49	0.0
WB11.01	Water Ballast 11.01	209.6	12.8	205.03	-0.00	1.61	630.6
Total of Ballast Water		15504.5		108.11	-0.00	1.77	4801.1
Fixed FW ballast density=1.000 t/m3							
WB02.01	Water Ballast 02.01	321.1	100.0	14.73	0.00	3.55	0.0

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Location	Description	Weight (t)	Filling %	L.C.G. (m)	T.C.G. (m)	V.C.G. (m)	Frs.mom. (tm)
—							
Gray Water density=1.000 t/m3							
T09.11	Black Water	18.5	100.0	180.00	4.16	7.00	0.0
T09.13	Gray Water	30.9	100.0	180.00	5.91	7.00	0.0
Total of Gray Water		49.4		180.00	5.25	7.00	0.0
Bilge Water density=1.000 t/m3							
T09.20	Bilge Water Settling	10.4	100.0	188.38	-0.73	3.97	0.0
T09.22	Bilge Water	30.5	100.0	187.07	-3.20	1.74	0.0
Total of Bilge Water		40.9		187.40	-2.57	2.31	0.0
Sludge density=2.380 t/m3							
T09.21	Sludge	71.0	98.0	187.03	3.22	1.73	0.0
Stores (ST0)							
		10.0		199.20	0.00	29.05	0.0
Crew (CREW)							
		6.0		199.20	0.00	35.70	0.0
Miscellaneous density=1.000 t/m3 (MIS)							
		87.0		178.40	0.00	5.00	0.0
—							—
Deadweight		16768.6		107.23	0.02	1.98	10542.9
Lightweight		20871.4		113.44	-0.01	10.44	
Displacement (1.025 t/m3)		37640.0		110.67	0.01	6.67	10542.9
—							—



RCR	TEXT	REQ	ATTN	UNIT	STAT
V.AREA15-30	Area depending on GZ curve top	0.055	2.815	mrاد	OK
V.AREA3040	Area under GZ curve between 30 and 40 deg	0.030	1.489	mrاد	OK
V.GZ0.2	Min. GZ > 0.2	0.200	8.611	m	OK
V.POSMAX15	Top of GZ curve at least 15 degrees	15.000	31.704	deg	OK
V.GM0.15	GM > 0.15 m	0.150	22.867	m	OK
IMOWEATHER	IMO weather criterion	1.000	3.036		OK
ND-RANGE	Range of Stability acc. ND	36.000	79.986	deg	OK
ND-WO	Wind Overturning acc. ND	1.400	57.584		OK
ND-DWNFLD	Angle of Downflooding acc. ND	20.000	77.422	deg	OK

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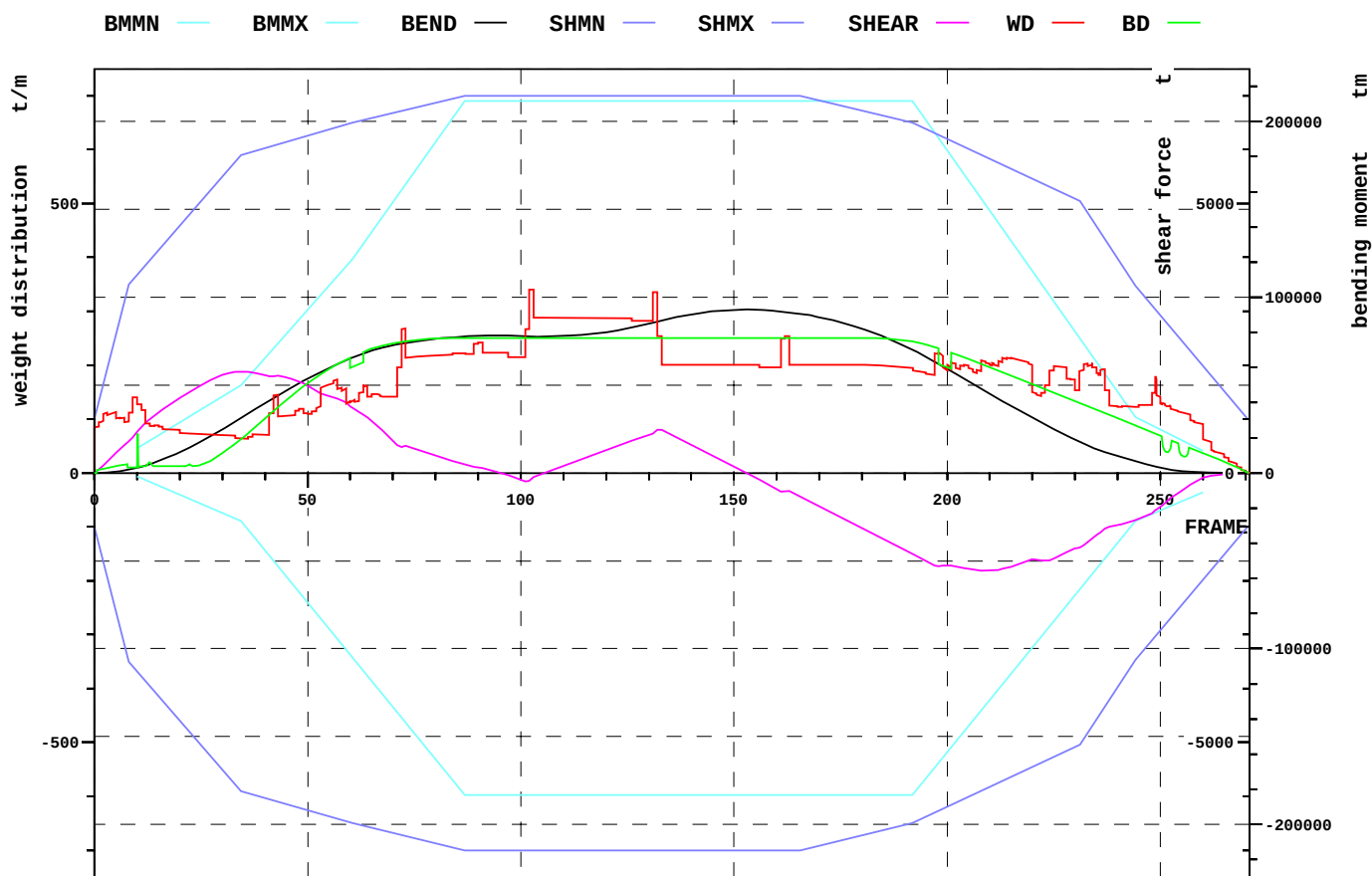
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Relevant openings

NAME	WT	IMMA degree	FR #	Y m	Z m	IMMR m
AIR_DUCT_2	UNPROTECTED	-	222.00	-11.375	33.350	27.653
AIR_DUCT_3	UNPROTECTED	-	230.00	0.000	33.350	27.650
AIR_DUCT_1	UNPROTECTED	77.4	222.00	11.375	33.350	27.647

NAME	name	
WT	watertightness of opening	
IMMA	immersion angle	degree
FR	frame number	#
Y	y-coordinate	m
Z	z-coordinate	m
IMMR	reserve to immersion	m



EXTREME SHEAR FORCE AND BENDING MOMENT

Sum of lightweight elements weight: 20871.4 t LCG: 113.44 m

		position:	x	frame
Shear force (min)	-1818.2 t		166.4 m	208
Shear force (max)	1879.8 t		27.2 m	34
Max. rel. shear force	33.1 %		23.2 m	29
Sagging moment	-1.9 tm		214.1 m	268
Hogging moment	92983.6 tm		122.4 m	153
Max. rel. bending moment	53.4 %		36.7 m	46

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LOADING CONDITION LC31, Docking condition

FR #	X m	BEND tm	BMMN tm	BMMX tm	RELB %
21.00	16.80	12963	-13513	30261	21.0
42.00	33.60	42568	-50240	70924	53.2
57.00	45.60	61983	-94804	111952	51.7
72.00	57.60	73610	-139384	160929	41.8
87.00	69.60	77679	-183412	211763	32.1
102.00	81.60	77771	-183412	211763	32.2
117.00	93.60	79209	-183412	211763	32.9
132.00	105.60	85912	-183412	211763	36.3
147.00	117.60	92403	-183412	211763	39.6
162.00	129.60	91647	-183412	211763	39.2
180.00	144.00	82067	-183412	211763	34.4
198.00	158.40	62082	-164872	190357	27.8
210.00	168.00	45156	-129204	149176	25.3
222.00	177.60	28775	-93541	108000	21.4
237.00	189.60	11975	-48977	56548	15.5
249.00	199.20	3434	-22505	25984	7.0

FR #	X m	SHEAR t	SHMN t	SHMX t	RELSH %
21.00	16.80	1442	-4679	4679	30.8
42.00	33.60	1791	-6075	6075	29.5
57.00	45.60	1384	-6418	6418	21.6
72.00	57.60	481	-6718	6718	7.2
87.00	69.60	159	-7000	7000	2.3
102.00	81.60	-142	-7000	7000	2.0
117.00	93.60	341	-7000	7000	4.9
132.00	105.60	800	-7000	7000	11.4
147.00	117.60	241	-7000	7000	3.4
162.00	129.60	-343	-7000	7000	4.9
180.00	144.00	-1026	-6724	6724	15.3
198.00	158.40	-1729	-6269	6269	27.6
210.00	168.00	-1809	-5824	5824	31.1
222.00	177.60	-1623	-5380	5380	30.2
237.00	189.60	-1036	-4339	4339	23.9
249.00	199.20	-683	-3024	3024	22.6

5 Technical information

5.1 *Hydrostatics*

Following quantities are shown in hydrostatics:

DISP, LCB, KMT, CB, WLA, MCT, TCP, VOLM, VOLT, LCA, VCB, WSA, CW, CP, CM

Explanations of symbols are in the beginning of the hydrostatic list.

Trim range (-2,0m...+6,0m; step 1,0m) covers all loading conditions in loading manual and submerging procedure.

Draught range is from Base Line to 32,0m. 0,2m step has been used.

Hydrostatics with more trims and draughts is shown in a separate document (B3598.1192 Hydrostatic data).

MAIN CHARACTERISTICS OF THE VESSEL:

Length betw. perpendiculars	212.13	m
Breadth, moulded	43.00	m
Design draught	9.68	m
X-coord. of after perpendicular	4.24	m
X-coord. of reference point	110.31	m
X-coord. of midship section	110.31	m
X-coord. of building frame 0	0.00	m
Thickness of keelplate	0.023	m
Mean thickness of shell plating	0.018	m
Density of water	1.0250	ton/m3

Calculations are based on STABHULL date 2009-04-30 time 13.55

X-coord. of aft end of DWL	0.05	m
X-coord. of fore end of DWL	216.48	m

Parts:

BASEHULL	110023.0 m3	2009-01-21	plth 18.0 mm	107 sections
BTHR1	-39.0 m3	2009-01-15	plth 0.0 mm	9 sections
BTHR2	-32.5 m3	2009-01-15	plth 0.0 mm	9 sections
SC-1	-23.1 m3	2009-01-13	plth 0.0 mm	0 sections
SC-2	-23.1 m3	2009-01-13	plth 0.0 mm	0 sections
SC-3	-23.1 m3	2009-01-13	plth 0.0 mm	0 sections
SC-4	-28.0 m3	2009-01-13	plth 0.0 mm	3 sections
SC-5	-28.0 m3	2009-01-13	plth 0.0 mm	0 sections
BOLLARDS	-26.5 m3	2009-01-21	plth 0.0 mm	0 sections
AFT-B.P	960.5 m3	2007-07-26	plth 0.0 mm	0 sections
AFT-B.S	960.5 m3	2007-07-26	plth 0.0 mm	0 sections
RUDDER	52.7 m3	2009-01-21	plth 0.0 mm	3 sections
SHAFT	42.9 m3	2009-01-21	plth 0.0 mm	15 sections
PROP	5.3 m3	2009-01-21	plth 0.0 mm	9 sections

EXPLANATION OF SYMBOLS:

DISP	total displacement	t
LCB	longitudinal centre of buoyancy	m
KMT	transv. metac. height	m
CB	block coefficient	
WLA	waterline area	m2
MCT	moment to change trim	tm/cm
TCP	change of displacement/change of draught	t/cm
VOLM	volume moulded	m3
VOLT	total volume	m3
LCA	long. centre of flotation	m
VCB	vertical center of buoyancy	m
WSA	wetted surface area	m2
CW	waterplane coefficient	
CP	prismatic coefficient	
CM	midship section coeff.	

----- total displacement t -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	814.4	408.1	90.3	526.1	1038.2	1577.0	2137.6	2717.0	3313.1
0.200	1471.5	1144.7	1144.1	1317.0	1742.0	2260.0	2815.2	3394.7	3991.9
0.400	2324.3	2183.1	2246.7	2381.0	2636.8	3072.5	3591.3	4152.1	4734.6
0.600	3349.6	3318.0	3382.6	3505.6	3701.4	4014.7	4465.7	4987.8	5541.8
0.800	4483.6	4483.9	4546.1	4661.0	4832.3	5079.9	5438.6	5900.2	6414.3
1.000	5669.7	5674.7	5733.0	5841.2	5997.2	6216.5	6505.7	6889.2	7352.6
1.200	6881.8	6887.3	6941.6	7042.8	7188.4	7385.8	7641.2	7954.0	8356.6
1.400	8115.7	8118.8	8169.4	8263.3	8400.5	8580.0	8808.3	9083.3	9425.7
1.600	9368.3	9368.4	9414.5	9501.9	9630.1	9795.3	9998.4	10250.2	10553.6
1.800	10638.1	10634.9	10674.8	10756.4	10876.0	11026.3	11208.9	11441.5	11723.6
2.000	11923.1	11916.2	11950.5	12025.5	12135.1	12270.8	12438.9	12653.2	12919.3
2.200	13223.6	13211.9	13240.5	13307.7	13405.5	13529.2	13685.8	13884.8	14138.6
2.400	14541.8	14520.2	14543.5	14601.7	14686.8	14800.5	14947.5	15134.7	15378.0
2.600	15876.6	15842.0	15859.4	15906.2	15979.0	16084.3	16224.0	16403.5	16634.6
2.800	17226.1	17180.1	17184.6	17220.4	17283.6	17380.3	17513.3	17689.6	17906.8
3.000	18587.0	18533.8	18520.3	18545.1	18599.5	18688.7	18817.6	18988.9	19193.0
3.200	19956.1	19900.0	19868.2	19879.4	19927.1	20009.8	20135.8	20299.8	20493.3
3.400	21335.8	21274.9	21229.9	21225.1	21266.6	21344.5	21466.3	21621.7	21805.7
3.600	22725.5	22654.5	22603.9	22584.3	22615.9	22693.3	22808.1	22953.3	23129.1
3.800	24124.5	24042.1	23987.8	23959.3	23979.5	24053.6	24160.0	24296.1	24462.7
4.000	25532.0	25438.8	25379.2	25347.2	25357.5	25423.0	25522.2	25649.5	25805.7
4.200	26947.8	26845.5	26779.5	26747.8	26752.3	26803.5	26893.7	27012.4	27160.8
4.400	28374.4	28262.8	28189.4	28158.1	28159.7	28194.4	28273.8	28385.9	28527.0
4.600	29812.5	29690.1	29611.3	29578.5	29576.6	29599.4	29665.1	29769.9	29902.3
4.800	31260.7	31128.8	31046.0	31007.4	31001.0	31016.7	31068.3	31162.3	31287.0
5.000	32720.1	32580.8	32491.9	32445.6	32432.4	32445.6	32484.4	32565.1	32679.8
5.200	34189.4	34045.0	33948.7	33893.6	33873.0	33882.0	33912.9	33977.1	34081.1
5.400	35670.8	35522.2	35416.8	35351.5	35322.8	35325.8	35350.4	35400.7	35492.4
5.600	37166.6	37009.3	36895.5	36819.6	36782.0	36775.4	36795.2	36836.9	36912.3
5.800	38674.9	38507.0	38385.2	38298.6	38249.1	38233.3	38247.8	38282.0	38342.6
6.000	40195.5	40018.1	39883.5	39786.7	39724.8	39700.2	39705.0	39734.3	39785.3
6.200	41727.0	41538.8	41390.3	41284.0	41211.3	41174.7	41170.2	41194.6	41236.4
6.400	43270.1	43069.3	42908.6	42789.8	42706.6	42657.2	42644.2	42659.0	42694.5
6.600	44825.9	44608.3	44436.2	44303.2	44210.5	44150.2	44125.3	44130.1	44160.6
6.800	46391.6	46157.8	45973.2	45827.6	45722.9	45652.1	45613.8	45609.4	45631.0
7.000	47966.8	47719.8	47518.8	47361.6	47242.3	47162.0	47112.3	47095.7	47106.9
7.200	49551.0	49291.1	49074.4	48904.1	48772.1	48680.1	48619.1	48588.6	48590.6
7.400	51147.5	50872.0	50641.5	50456.3	50311.7	50205.0	50133.5	50091.3	50080.9
7.600	52754.1	52460.9	52217.9	52016.7	51859.4	51738.7	51655.9	51602.1	51577.3
7.800	54370.9	54061.3	53803.5	53588.1	53417.2	53282.2	53184.8	53119.9	53083.0
8.000	55996.6	55671.9	55397.2	55169.2	54982.0	54834.5	54721.4	54645.6	54596.4
8.200	57634.7	57292.4	57000.3	56758.6	56555.9	56395.5	56268.1	56177.4	56116.1
8.400	59282.8	58921.5	58614.6	58356.3	58139.9	57963.7	57823.8	57715.8	57642.8
8.600	60941.1	60561.3	60237.9	59961.4	59731.7	59539.2	59386.9	59264.8	59175.0
8.800	62610.5	62212.2	61870.4	61577.4	61331.9	61124.9	60957.1	60821.4	60713.8
9.000	64289.1	63872.5	63510.9	63202.4	62938.8	62718.5	62533.3	62384.2	62263.5
9.200	65975.1	65542.8	65162.8	64836.3	64555.2	64319.8	64118.9	63954.0	63821.6
9.400	67667.2	67223.0	66823.8	66477.8	66181.2	65927.9	65712.3	65530.0	65386.9
9.600	69364.3	68911.2	68493.8	68129.0	67815.1	67542.9	67312.0	67115.3	66959.9
9.800	71065.7	70605.6	70173.9	69790.1	69457.2	69167.7	68919.4	68710.0	68539.4
10.000	72771.1	72305.0	71862.6	71459.5	71106.1	70799.6	70533.4	70312.0	70126.5
10.200	74479.8	74008.9	73558.1	73137.7	72765.4	72439.6	72157.6	71922.0	71723.6
10.400	76191.6	75716.6	75258.8	74825.1	74432.2	74086.9	73790.9	73538.6	73328.3
10.600	77906.2	77427.6	76963.8	76519.7	76106.7	75743.8	75432.2	75163.5	74940.2
10.800	79623.3	79141.6	78672.6	78219.4	77790.8	77410.7	77081.6	76798.3	76558.1
11.000	81342.0	80858.1	80384.3	79923.1	79483.7	79085.7	78738.3	78439.9	78168.6
11.200	83054.0	82576.9	82098.7	81630.9	81183.7	80769.4	80404.8	80089.1	79763.1

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	84766.6	84297.5	83814.8	83342.6	82889.2	82462.0	82079.4	81724.7	81313.0
11.600	86479.4	86015.3	85533.0	85057.8	84599.0	84162.1	83759.4	83326.7	82808.8
11.800	88192.4	87728.1	87253.7	86776.0	86312.2	85867.6	85417.8	84866.1	84250.1
12.000	89904.1	89440.2	88976.9	88495.7	88027.9	87571.4	87011.7	86340.0	85636.7
12.200	91531.1	91152.1	90689.2	90216.8	89746.8	89223.3	88520.4	87747.3	86970.6
12.400	92973.7	92863.2	92401.9	91939.1	91452.3	90761.7	89941.9	89088.4	88250.0
12.600	94229.5	94540.6	94115.4	93659.2	93032.4	92178.6	91272.9	90363.0	89476.0
12.800	95298.7	95908.4	95829.3	95306.1	94428.2	93474.0	92515.1	91568.2	90647.2
13.000	96183.1	96904.0	97543.6	96618.1	95639.7	94647.8	93666.9	92703.2	91763.9
13.200	96881.2	97527.3	97840.5	97557.9	96658.6	95698.0	94728.1	93767.3	92825.1
13.400	97394.3	97794.6	97995.0	98125.5	97490.7	96625.8	95698.1	94760.1	93829.2
13.600	97722.4	97948.4	98150.2	98354.9	98137.0	97430.2	96577.3	95681.4	94775.2
13.800	97902.5	98103.0	98306.2	98512.4	98597.6	98104.0	97365.6	96531.8	95663.5
14.000	98056.4	98258.2	98463.0	98670.6	98875.8	98654.0	98062.5	97309.4	96494.0
14.200	98210.9	98414.2	98620.4	98829.5	99041.6	99081.3	98661.3	98014.3	97264.4
14.400	98366.2	98571.0	98778.6	98989.2	99202.9	99385.1	99165.4	98646.1	97974.8
14.600	98522.3	98728.5	98937.6	99149.7	99364.9	99583.4	99577.8	99205.0	98625.4
14.800	98679.1	98886.7	99097.3	99310.9	99527.7	99747.8	99897.6	99682.8	99215.4
15.000	98836.6	99045.7	99257.8	99473.0	99691.4	99913.1	100130.8	100088.0	99745.9
15.200	98994.8	99205.4	99419.0	99635.8	99855.8	100079.3	100306.3	100419.2	100212.0
15.400	99153.8	99365.9	99581.1	99799.4	100021.1	100246.3	100475.1	100677.8	100613.2
15.600	99313.6	99527.2	99743.9	99963.9	100187.2	100414.1	100644.8	100879.3	100953.1
15.800	99474.1	99689.2	99907.5	100129.1	100354.2	100582.8	100815.3	101051.8	101231.8
16.000	99635.3	99852.0	100071.9	100295.2	100522.0	100752.5	100986.9	101225.3	101458.1
16.200	99797.4	100015.6	100237.2	100462.2	100690.7	100923.0	101159.3	101399.7	101644.5
16.400	99960.2	100180.1	100403.3	100630.0	100860.3	101094.5	101332.7	101575.2	101822.1
16.600	100123.8	100345.3	100570.2	100798.6	101030.8	101266.9	101507.1	101751.6	102000.7
16.800	100288.3	100511.4	100737.9	100968.2	101202.2	101440.2	101682.5	101929.1	102180.4
17.000	100453.5	100678.3	100906.6	101138.6	101374.5	101614.6	101858.9	102107.7	102361.3
17.200	100619.5	100846.0	101076.1	101310.0	101547.8	101789.9	102036.3	102287.3	102543.2
17.400	100786.4	101014.6	101246.5	101482.3	101722.1	101966.2	102214.7	102468.0	102726.2
17.600	100954.1	101184.1	101417.8	101655.5	101897.3	102143.5	102394.2	102649.8	102910.4
17.800	101122.7	101354.4	101590.0	101829.7	102073.5	102321.8	102574.8	102832.7	103096.6
18.000	101292.1	101525.7	101763.2	102004.8	102250.8	102501.3	102756.5	103016.7	103283.8
18.200	101462.4	101697.9	101937.3	102181.0	102429.0	102681.7	102939.3	103202.4	103472.0
18.400	101633.7	101871.0	102112.4	102358.1	102608.3	102863.3	103123.2	103389.5	103661.2
18.600	101805.8	102045.0	102288.5	102536.3	102788.7	103045.9	103308.3	103577.7	103851.5
18.800	101978.8	102220.0	102465.5	102715.5	102970.1	103229.7	103495.4	103766.9	104042.8
19.000	102152.8	102396.0	102643.6	102895.7	103152.7	103414.6	103683.5	103957.1	104235.1
19.200	102327.7	102573.0	102822.7	103077.1	103336.3	103601.3	103872.6	104148.3	104428.6
19.400	102503.7	102751.0	103002.9	103259.5	103521.1	103789.3	104062.7	104340.6	104623.0
19.600	102680.6	102930.0	103184.1	103443.0	103707.2	103978.3	104253.8	104533.9	104818.5
19.800	102858.5	103110.1	103366.4	103627.7	103895.2	104168.3	104446.0	104728.2	105015.0
20.000	103037.4	103291.2	103549.8	103813.5	104084.1	104359.4	104639.3	104923.6	105212.6
20.200	103217.4	103473.4	103734.4	104001.1	104274.1	104551.5	104833.5	105120.1	105411.3
20.400	103398.4	103656.7	103920.0	104190.0	104465.1	104744.7	105028.9	105317.6	105611.0
20.600	103580.5	103841.1	104107.1	104379.8	104657.1	104938.9	105225.3	105516.2	105811.8
20.800	103763.7	104026.6	104295.9	104570.8	104850.2	105134.2	105422.7	105715.8	106013.6
21.000	103947.9	104213.3	104485.7	104762.8	105044.3	105330.5	105621.2	105916.5	106216.6
21.200	104133.4	104401.9	104676.6	104955.8	105239.5	105527.8	105820.7	106118.3	106420.7
21.400	104319.9	104591.6	104868.5	105149.8	105435.7	105726.2	106021.4	106321.2	106625.8
21.600	104507.9	104782.4	105061.4	105344.9	105633.0	105925.7	106223.1	106525.2	106832.0
21.800	104697.6	104974.2	105255.4	105541.0	105831.3	106126.2	106425.9	106730.2	107039.3
22.000	104888.3	105167.1	105450.4	105738.2	106030.7	106327.9	106629.7	106936.3	107247.8
22.200	105080.0	105361.0	105646.4	105936.5	106231.2	106530.6	106834.7	107143.5	107457.3
22.400	105272.8	105555.9	105843.5	106135.8	106432.7	106734.3	107040.7	107351.9	107667.9
22.600	105466.6	105751.9	106041.7	106336.2	106635.3	106939.2	107247.9	107561.3	107879.7
22.800	105661.5	105948.9	106240.9	106537.6	106839.0	107145.1	107456.1	107771.9	108092.5
23.000	105857.4	106147.0	106441.2	106740.1	107043.8	107352.2	107665.4	107983.5	108306.5

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	106054.3	106346.1	106642.6	106943.8	107249.7	107560.4	107875.9	108196.3	108521.6
23.400	106252.3	106546.3	106845.0	107148.4	107456.6	107769.6	108087.5	108410.2	108737.9
23.600	106451.4	106747.6	107048.5	107354.2	107664.7	107979.9	108300.1	108625.2	108955.3
23.800	106651.5	106950.0	107253.1	107561.1	107873.8	108191.4	108513.9	108841.3	109173.8
24.000	106852.7	107153.4	107458.8	107769.0	108084.1	108404.0	108728.8	109058.6	109393.4
24.200	107055.0	107357.9	107665.6	107978.1	108295.4	108617.7	108944.9	109277.0	109614.2
24.400	107258.3	107563.5	107873.5	108188.3	108507.9	108832.5	109162.0	109496.6	109836.1
24.600	107462.7	107770.2	108082.5	108399.5	108721.5	109048.4	109380.3	109717.2	110059.1
24.800	107668.3	107978.0	108292.5	108611.9	108936.2	109265.5	109599.8	109939.0	110283.3
25.000	107874.9	108186.8	108503.7	108825.4	109152.1	109483.7	109820.3	110161.9	110508.6
25.200	108082.5	108396.8	108716.0	109040.0	109369.0	109703.0	110042.0	110386.0	110735.1
25.400	108291.3	108607.9	108929.4	109255.8	109587.1	109923.5	110264.8	110611.2	110962.6
25.600	108501.2	108820.1	109143.9	109472.6	109806.3	110145.0	110488.8	110837.5	111191.4
25.800	108712.1	109033.4	109359.5	109690.6	110026.7	110367.7	110713.8	111065.0	111421.2
26.000	108924.2	109247.8	109576.3	109909.7	110248.1	110591.6	110940.1	111293.6	111652.2
26.200	109137.4	109463.3	109794.1	110130.0	110470.8	110816.6	111167.4	111523.3	111884.3
26.400	109351.8	109680.0	110013.1	110351.3	110694.5	111042.6	111395.9	111754.2	112117.5
26.600	109567.2	109897.7	110233.3	110573.8	110919.3	111269.9	111625.4	111986.1	112351.8
26.800	109783.7	110116.6	110454.5	110797.4	111145.3	111498.2	111856.2	112219.2	112587.3
27.000	110001.4	110336.7	110676.9	111022.1	111372.4	111727.7	112088.0	112453.4	112823.9
27.200	110220.2	110557.8	110900.4	111248.0	111600.6	111958.3	112321.0	112688.7	113061.6
27.400	110440.1	110780.0	111125.0	111474.9	111829.9	112190.0	112555.0	112925.2	113300.4
27.600	110661.1	111003.4	111350.7	111703.0	112060.4	112422.8	112790.2	113162.8	113540.1
27.800	110883.3	111227.9	111577.6	111932.3	112292.0	112656.7	113026.6	113401.4	113747.9
28.000	111106.5	111453.5	111805.5	112162.6	112524.7	112891.8	113264.0	113641.2	113910.9
28.200	111330.9	111680.2	112034.6	112394.0	112758.5	113128.0	113502.5	113862.6	114013.0
28.400	111556.4	111908.1	112264.8	112626.6	112993.4	113365.3	113742.1	114030.5	114059.0
28.600	111783.0	112137.1	112496.2	112860.3	113229.4	113603.7	113979.4	114124.4	114083.4
28.800	112010.7	112367.1	112728.6	113095.1	113466.6	113843.2	114147.7	114160.1	114106.0
29.000	112239.6	112598.3	112962.2	113331.0	113704.9	114083.6	114232.2	114183.7	114128.7
29.200	112469.5	112830.6	113196.8	113568.0	113944.2	114264.8	114261.3	114206.3	114151.3
29.400	112700.4	113064.0	113432.6	113806.1	114184.7	114335.8	114283.9	114228.9	114173.9
29.600	112913.3	113298.6	113669.4	114045.4	114372.8	114361.6	114306.6	114251.6	114196.6
29.800	113123.9	113534.2	113907.4	114285.7	114438.9	114384.2	114329.2	114274.2	114219.2
30.000	113335.6	113761.2	114146.5	114482.8	114461.8	114406.8	114351.8	114296.8	114241.8
30.200	113548.5	113976.4	114386.2	114539.5	114484.5	114429.5	114374.5	114319.5	114264.5
30.400	113762.5	114192.7	114574.6	114562.1	114507.1	114452.1	114397.1	114342.1	114287.1
30.600	113977.6	114405.7	114617.1	114584.7	114529.7	114474.7	114419.7	114364.7	114309.7
30.800	114193.8	114559.6	114617.1	114607.4	114552.4	114497.4	114442.4	114387.4	114332.4
31.000	114398.5	114622.0	114617.1	114617.1	114575.0	114520.0	114465.0	114410.0	114355.0
31.200	114542.5	114617.1	114617.1	114617.1	114597.6	114542.6	114487.6	114432.6	114377.6
31.400	114614.5	114617.1	114617.1	114617.1	114616.9	114565.3	114510.3	114455.3	114400.3
31.600	114617.1	114617.1	114617.1	114617.1	114617.1	114587.9	114532.9	114477.9	114422.9
31.800	114617.1	114617.1	114617.1	114617.1	114617.1	114610.5	114555.5	114500.5	114445.5
32.000	114617.1	114617.1	114617.1	114617.1	114617.1	114617.1	114578.2	114523.1	114468.2

----- long. centre of buoy. m -----

trim draught	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
0.000	66.742	68.432	115.466	159.620	161.292	162.233	162.921	163.475	163.943
0.200	74.176	83.269	114.760	143.906	153.252	156.838	158.869	160.236	161.233
0.400	81.201	94.589	114.503	132.765	145.645	151.679	154.965	157.098	158.574
0.600	87.521	100.330	114.309	127.532	138.771	146.676	151.168	154.026	155.959
0.800	92.476	103.242	114.138	124.592	134.081	141.886	147.449	150.993	153.391
1.000	95.969	104.973	113.989	122.694	130.846	137.947	143.819	147.991	150.864
1.200	98.357	106.105	113.841	121.349	128.474	134.955	140.513	145.044	148.372
1.400	100.057	106.897	113.700	120.334	126.673	132.603	137.780	142.252	145.922
1.600	101.325	107.469	113.564	119.525	125.259	130.666	135.525	139.805	143.565
1.800	102.298	107.889	113.438	118.861	124.105	129.049	133.625	137.718	141.402
2.000	103.067	108.208	113.309	118.301	123.127	127.693	131.990	135.931	139.505
2.200	103.677	108.450	113.180	117.821	122.278	126.529	130.575	134.370	137.853
2.400	104.147	108.638	113.054	117.381	121.533	125.520	129.348	132.993	136.399
2.600	104.515	108.778	112.929	116.975	120.876	124.636	128.273	131.779	135.095
2.800	104.808	108.863	112.800	116.597	120.282	123.856	127.324	130.703	133.916
3.000	105.054	108.906	112.660	116.247	119.747	123.163	126.487	129.737	132.844
3.200	105.274	108.913	112.494	115.925	119.262	122.533	125.746	128.865	131.860
3.400	105.461	108.901	112.312	115.621	118.822	121.970	125.074	128.072	130.959
3.600	105.611	108.881	112.124	115.327	118.419	121.464	124.455	127.349	130.133
3.800	105.725	108.852	111.944	115.036	118.051	121.001	123.884	126.679	129.373
4.000	105.805	108.812	111.780	114.753	117.709	120.574	123.353	126.057	128.671
4.200	105.864	108.763	111.632	114.494	117.377	120.169	122.860	125.478	128.017
4.400	105.898	108.705	111.489	114.264	117.053	119.785	122.400	124.937	127.405
4.600	105.912	108.647	111.354	114.057	116.746	119.405	121.963	124.433	126.833
4.800	105.914	108.581	111.226	113.862	116.459	119.036	121.550	123.959	126.292
5.000	105.906	108.511	111.107	113.671	116.191	118.688	121.147	123.507	125.784
5.200	105.891	108.442	110.987	113.484	115.941	118.360	120.752	123.076	125.304
5.400	105.869	108.374	110.861	113.305	115.701	118.050	120.375	122.657	124.843
5.600	105.840	108.305	110.735	113.128	115.466	117.760	120.018	122.245	124.403
5.800	105.808	108.231	110.609	112.948	115.239	117.481	119.677	121.851	123.976
6.000	105.772	108.149	110.481	112.771	115.018	117.210	119.359	121.475	123.555
6.200	105.733	108.062	110.354	112.594	114.796	116.948	119.053	121.113	123.149
6.400	105.682	107.970	110.219	112.419	114.578	116.694	118.756	120.774	122.760
6.600	105.619	107.874	110.082	112.248	114.364	116.441	118.469	120.448	122.384
6.800	105.550	107.773	109.942	112.073	114.154	116.194	118.190	120.131	122.029
7.000	105.474	107.661	109.802	111.897	113.949	115.952	117.914	119.824	121.688
7.200	105.393	107.546	109.660	111.721	113.743	115.714	117.643	119.527	121.357
7.400	105.302	107.427	109.510	111.545	113.537	115.482	117.379	119.233	121.036
7.600	105.204	107.307	109.359	111.370	113.333	115.252	117.118	118.944	120.724
7.800	105.101	107.180	109.206	111.192	113.128	115.021	116.864	118.661	120.415
8.000	104.994	107.047	109.052	111.011	112.926	114.792	116.614	118.383	120.112
8.200	104.877	106.910	108.897	110.831	112.723	114.564	116.363	118.112	119.815
8.400	104.756	106.772	108.736	110.650	112.518	114.339	116.114	117.846	119.522
8.600	104.629	106.628	108.573	110.470	112.314	114.117	115.868	117.578	119.234
8.800	104.497	106.480	108.408	110.286	112.111	113.892	115.625	117.312	118.952
9.000	104.361	106.328	108.242	110.099	111.909	113.669	115.385	117.048	118.671
9.200	104.225	106.171	108.069	109.911	111.706	113.446	115.144	116.788	118.393
9.400	104.091	106.011	107.895	109.724	111.500	113.225	114.902	116.533	118.121
9.600	103.961	105.850	107.718	109.533	111.293	113.006	114.662	116.279	117.853
9.800	103.833	105.690	107.537	109.339	111.086	112.782	114.424	116.026	117.590
10.000	103.708	105.534	107.354	109.143	110.879	112.557	114.190	115.777	117.331
10.200	103.588	105.380	107.173	108.946	110.668	112.333	113.955	115.530	117.072
10.400	103.471	105.231	106.994	108.745	110.455	112.111	113.720	115.288	116.816
10.600	103.358	105.086	106.818	108.544	110.240	111.887	113.486	115.047	116.563
10.800	103.249	104.945	106.647	108.345	110.024	111.663	113.255	114.805	116.312
11.000	103.144	104.808	106.479	108.148	109.809	111.439	113.026	114.565	116.054
11.200	103.054	104.675	106.316	107.956	109.595	111.216	112.794	114.325	115.772

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	102.967	104.547	106.156	107.770	109.385	110.992	112.562	114.071	115.456
11.600	102.885	104.427	106.001	107.589	109.181	110.768	112.330	113.784	115.105
11.800	102.806	104.317	105.851	107.413	108.980	110.548	112.072	113.450	114.725
12.000	102.731	104.210	105.706	107.242	108.786	110.326	111.763	113.079	114.324
12.200	102.745	104.109	105.576	107.077	108.596	110.067	111.404	112.678	113.901
12.400	102.914	104.011	105.453	106.917	108.401	109.740	111.008	112.253	113.465
12.600	103.192	103.952	105.334	106.767	108.123	109.362	110.588	111.813	113.021
12.800	103.537	104.146	105.221	106.580	107.763	108.953	110.156	111.366	112.572
13.000	103.908	104.491	105.112	106.235	107.362	108.532	109.720	110.920	112.122
13.200	104.264	104.823	105.187	105.869	106.968	108.116	109.291	110.481	111.678
13.400	104.569	104.996	105.282	105.639	106.614	107.721	108.878	110.055	111.244
13.600	104.784	105.091	105.377	105.665	106.337	107.366	108.491	109.649	110.825
13.800	104.900	105.186	105.473	105.762	106.173	107.073	108.138	109.267	110.425
14.000	104.995	105.282	105.569	105.859	106.155	106.850	107.826	108.915	110.047
14.200	105.090	105.377	105.666	105.956	106.248	106.712	107.572	108.599	109.697
14.400	105.186	105.474	105.763	106.054	106.346	106.675	107.381	108.324	109.378
14.600	105.282	105.570	105.860	106.152	106.445	106.739	107.258	108.094	109.092
14.800	105.378	105.667	105.958	106.250	106.544	106.839	107.211	107.924	108.844
15.000	105.475	105.765	106.056	106.349	106.643	106.939	107.246	107.810	108.636
15.200	105.572	105.862	106.154	106.448	106.743	107.040	107.339	107.759	108.477
15.400	105.669	105.960	106.253	106.547	106.843	107.141	107.441	107.774	108.371
15.600	105.767	106.059	106.352	106.647	106.944	107.243	107.544	107.846	108.317
15.800	105.865	106.158	106.452	106.748	107.045	107.345	107.646	107.950	108.319
16.000	105.964	106.257	106.552	106.848	107.147	107.447	107.750	108.055	108.373
16.200	106.062	106.356	106.652	106.949	107.249	107.550	107.854	108.160	108.468
16.400	106.162	106.456	106.753	107.051	107.351	107.654	107.958	108.265	108.575
16.600	106.261	106.557	106.854	107.153	107.454	107.757	108.063	108.371	108.682
16.800	106.361	106.657	106.955	107.255	107.557	107.862	108.169	108.478	108.790
17.000	106.461	106.758	107.057	107.358	107.661	107.967	108.274	108.585	108.898
17.200	106.562	106.860	107.160	107.462	107.766	108.072	108.381	108.693	109.007
17.400	106.663	106.962	107.263	107.565	107.870	108.178	108.488	108.801	109.117
17.600	106.765	107.064	107.366	107.670	107.976	108.284	108.596	108.910	109.227
17.800	106.867	107.167	107.470	107.774	108.082	108.391	108.704	109.019	109.338
18.000	106.969	107.271	107.574	107.880	108.188	108.499	108.813	109.129	109.450
18.200	107.072	107.374	107.679	107.986	108.295	108.607	108.922	109.240	109.563
18.400	107.176	107.479	107.784	108.092	108.402	108.716	109.032	109.352	109.676
18.600	107.279	107.583	107.890	108.199	108.510	108.825	109.142	109.465	109.790
18.800	107.384	107.689	107.996	108.306	108.619	108.935	109.254	109.578	109.903
19.000	107.488	107.794	108.103	108.414	108.728	109.045	109.367	109.691	110.018
19.200	107.593	107.901	108.210	108.523	108.838	109.157	109.480	109.805	110.133
19.400	107.699	108.007	108.318	108.632	108.948	109.269	109.593	109.920	110.248
19.600	107.805	108.115	108.427	108.742	109.060	109.382	109.707	110.035	110.364
19.800	107.912	108.223	108.536	108.852	109.172	109.496	109.822	110.150	110.480
20.000	108.019	108.331	108.645	108.963	109.285	109.610	109.937	110.266	110.597
20.200	108.127	108.440	108.756	109.075	109.398	109.724	110.052	110.382	110.714
20.400	108.235	108.550	108.866	109.188	109.512	109.839	110.168	110.499	110.832
20.600	108.344	108.660	108.978	109.301	109.627	109.954	110.284	110.616	110.950
20.800	108.454	108.770	109.091	109.415	109.741	110.070	110.401	110.733	111.068
21.000	108.564	108.882	109.204	109.529	109.857	110.186	110.518	110.851	111.187
21.200	108.675	108.994	109.318	109.644	109.972	110.303	110.635	110.970	111.306
21.400	108.786	109.108	109.432	109.759	110.089	110.420	110.753	111.089	111.426
21.600	108.898	109.222	109.547	109.875	110.205	110.537	110.872	111.208	111.546
21.800	109.011	109.336	109.662	109.991	110.322	110.655	110.990	111.328	111.667
22.000	109.125	109.451	109.778	110.108	110.440	110.774	111.110	111.448	111.788
22.200	109.239	109.566	109.894	110.225	110.558	110.893	111.229	111.568	111.909
22.400	109.354	109.681	110.011	110.342	110.676	111.012	111.350	111.689	112.031
22.600	109.469	109.797	110.128	110.460	110.795	111.132	111.470	111.811	112.154
22.800	109.585	109.914	110.245	110.579	110.914	111.252	111.591	111.933	112.276
23.000	109.701	110.031	110.363	110.697	111.034	111.372	111.713	112.055	112.400

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	109.817	110.148	110.481	110.817	111.154	111.493	111.835	112.178	112.523
23.400	109.934	110.266	110.600	110.936	111.274	111.615	111.957	112.301	112.647
23.600	110.052	110.385	110.719	111.056	111.395	111.736	112.079	112.425	112.771
23.800	110.170	110.503	110.839	111.177	111.517	111.859	112.203	112.548	112.896
24.000	110.288	110.622	110.959	111.298	111.639	111.981	112.326	112.673	113.021
24.200	110.407	110.742	111.080	111.419	111.761	112.104	112.450	112.798	113.147
24.400	110.526	110.862	111.201	111.541	111.884	112.228	112.574	112.923	113.273
24.600	110.645	110.983	111.322	111.663	112.007	112.352	112.699	113.048	113.399
24.800	110.765	111.104	111.444	111.786	112.130	112.476	112.824	113.174	113.526
25.000	110.886	111.225	111.566	111.909	112.254	112.601	112.950	113.300	113.653
25.200	111.007	111.347	111.688	112.032	112.378	112.726	113.076	113.427	113.780
25.400	111.128	111.469	111.811	112.156	112.503	112.851	113.202	113.554	113.908
25.600	111.250	111.591	111.935	112.280	112.628	112.977	113.328	113.682	114.036
25.800	111.372	111.714	112.059	112.405	112.753	113.104	113.456	113.809	114.165
26.000	111.494	111.838	112.183	112.530	112.879	113.230	113.583	113.937	114.294
26.200	111.617	111.961	112.308	112.656	113.005	113.357	113.711	114.066	114.423
26.400	111.741	112.086	112.433	112.781	113.132	113.484	113.839	114.195	114.552
26.600	111.864	112.210	112.558	112.908	113.259	113.612	113.967	114.324	114.682
26.800	111.989	112.335	112.684	113.034	113.386	113.740	114.096	114.453	114.812
27.000	112.113	112.461	112.810	113.161	113.514	113.869	114.225	114.583	114.942
27.200	112.238	112.586	112.936	113.288	113.642	113.997	114.354	114.713	115.073
27.400	112.363	112.713	113.063	113.416	113.770	114.126	114.484	114.843	115.204
27.600	112.489	112.839	113.191	113.544	113.899	114.256	114.614	114.974	115.335
27.800	112.615	112.966	113.318	113.672	114.028	114.385	114.744	115.105	115.439
28.000	112.742	113.093	113.446	113.801	114.157	114.515	114.875	115.236	115.509
28.200	112.869	113.221	113.574	113.930	114.287	114.646	115.006	115.351	115.537
28.400	112.996	113.349	113.703	114.059	114.417	114.776	115.137	115.425	115.529
28.600	113.123	113.477	113.832	114.189	114.547	114.907	115.266	115.448	115.508
28.800	113.251	113.605	113.961	114.319	114.678	115.038	115.340	115.434	115.487
29.000	113.379	113.734	114.091	114.449	114.809	115.170	115.357	115.413	115.465
29.200	113.508	113.864	114.221	114.580	114.940	115.255	115.339	115.392	115.444
29.400	113.637	113.993	114.351	114.710	115.071	115.264	115.318	115.370	115.422
29.600	113.785	114.123	114.482	114.842	115.163	115.245	115.297	115.349	115.401
29.800	113.936	114.253	114.612	114.973	115.171	115.223	115.275	115.327	115.379
30.000	114.087	114.393	114.743	115.075	115.150	115.202	115.254	115.306	115.358
30.200	114.238	114.545	114.875	115.077	115.129	115.180	115.232	115.284	115.336
30.400	114.390	114.697	114.972	115.055	115.107	115.159	115.211	115.263	115.315
30.600	114.542	114.847	115.004	115.034	115.086	115.138	115.190	115.242	115.294
30.800	114.694	114.960	115.004	115.013	115.065	115.116	115.168	115.220	115.272
31.000	114.839	115.008	115.004	115.004	115.043	115.095	115.147	115.199	115.251
31.200	114.946	115.004	115.004	115.004	115.022	115.074	115.126	115.178	115.229
31.400	115.002	115.004	115.004	115.004	115.004	115.052	115.104	115.156	115.208
31.600	115.004	115.004	115.004	115.004	115.004	115.031	115.083	115.135	115.187
31.800	115.004	115.004	115.004	115.004	115.004	115.010	115.062	115.114	115.165
32.000	115.004	115.004	115.004	115.004	115.004	115.004	115.040	115.092	115.144

----- transv. metac. height m -----

trim draught	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
0.000	436.696	841.842	6338.917	630.024	323.672	218.735	165.757	133.775	112.378
0.200	340.661	529.737	562.462	453.544	273.449	195.633	152.526	125.200	106.379
0.400	276.243	314.490	303.529	282.272	234.706	175.896	140.650	117.307	100.803
0.600	213.732	216.382	210.633	201.039	187.052	159.306	130.206	110.136	95.672
0.800	166.742	165.909	162.405	156.881	149.501	139.337	121.009	103.704	90.943
1.000	136.011	135.073	132.727	129.141	124.423	118.568	110.476	97.908	86.567
1.200	115.054	114.229	112.526	110.012	106.797	102.845	98.079	91.515	82.543
1.400	99.842	99.159	97.844	95.998	93.647	90.802	87.540	83.703	78.171
1.600	88.299	87.716	86.679	85.259	83.483	81.358	78.959	76.272	72.977
1.800	79.233	78.715	77.893	76.758	75.370	73.764	71.951	69.883	67.561
2.000	71.932	71.458	70.777	69.864	68.757	67.520	66.098	64.496	62.706
2.200	65.965	65.510	64.913	64.171	63.282	62.302	61.191	59.901	58.452
2.400	60.998	60.560	60.047	59.414	58.694	57.887	56.989	55.945	54.756
2.600	56.800	56.402	55.937	55.385	54.798	54.120	53.365	52.517	51.513
2.800	53.183	52.867	52.417	51.956	51.446	50.875	50.231	49.511	48.661
3.000	50.028	49.810	49.387	48.986	48.541	48.047	47.497	46.853	46.137
3.200	47.273	47.109	46.779	46.394	46.003	45.568	45.082	44.508	43.904
3.400	44.871	44.718	44.484	44.107	43.767	43.379	42.932	42.433	41.898
3.600	42.771	42.583	42.425	42.118	41.780	41.428	41.015	40.578	40.107
3.800	40.910	40.705	40.571	40.344	40.002	39.675	39.302	38.913	38.498
4.000	39.245	39.054	38.887	38.722	38.426	38.096	37.765	37.409	37.043
4.200	37.747	37.574	37.375	37.250	37.011	36.671	36.374	36.053	35.767
4.400	36.394	36.233	36.045	35.886	35.699	35.394	35.110	34.868	34.559
4.600	35.173	35.012	34.844	34.641	34.494	34.262	34.002	33.740	33.462
4.800	34.062	33.900	33.739	33.543	33.378	33.212	32.946	32.707	32.456
5.000	33.047	32.890	32.728	32.539	32.340	32.244	32.029	31.759	31.531
5.200	32.114	31.967	31.793	31.620	31.467	31.318	31.145	30.889	30.678
5.400	31.257	31.113	30.940	30.807	30.629	30.445	30.323	30.128	29.892
5.600	30.476	30.321	30.192	30.020	29.857	29.681	29.552	29.393	29.169
5.800	29.751	29.621	29.465	29.299	29.141	28.978	28.823	28.705	28.529
6.000	29.110	28.945	28.793	28.635	28.476	28.326	28.168	28.056	27.914
6.200	28.479	28.323	28.165	28.019	27.863	27.720	27.573	27.443	27.334
6.400	27.894	27.744	27.587	27.446	27.300	27.154	27.018	26.878	26.784
6.600	27.381	27.205	27.057	26.909	26.773	26.631	26.502	26.372	26.264
6.800	26.878	26.722	26.558	26.413	26.283	26.149	26.018	25.898	25.775
7.000	26.409	26.259	26.097	25.957	25.823	25.698	25.570	25.456	25.341
7.200	25.971	25.826	25.681	25.528	25.394	25.277	25.156	25.039	24.933
7.400	25.566	25.421	25.277	25.148	25.002	24.880	24.767	24.653	24.552
7.600	25.192	25.041	24.905	24.771	24.649	24.509	24.403	24.296	24.193
7.800	24.839	24.687	24.554	24.420	24.302	24.167	24.060	23.959	23.858
8.000	24.506	24.363	24.225	24.098	23.977	23.864	23.737	23.643	23.548
8.200	24.199	24.057	23.914	23.792	23.669	23.562	23.457	23.346	23.256
8.400	23.913	23.767	23.632	23.505	23.388	23.280	23.179	23.065	22.982
8.600	23.643	23.497	23.365	23.235	23.123	23.011	22.915	22.820	22.725
8.800	23.390	23.248	23.113	22.984	22.873	22.764	22.669	22.579	22.495
9.000	23.154	23.011	22.875	22.755	22.638	22.535	22.435	22.349	22.267
9.200	22.923	22.789	22.656	22.535	22.417	22.316	22.217	22.136	22.058
9.400	22.699	22.584	22.452	22.328	22.216	22.112	22.018	21.933	21.858
9.600	22.483	22.387	22.257	22.134	22.024	21.917	21.828	21.742	21.671
9.800	22.274	22.193	22.076	21.956	21.844	21.739	21.651	21.568	21.494
10.000	22.071	22.006	21.907	21.786	21.673	21.575	21.482	21.402	21.325
10.200	21.876	21.824	21.741	21.626	21.516	21.419	21.324	21.246	21.172
10.400	21.689	21.648	21.580	21.479	21.370	21.271	21.182	21.100	21.029
10.600	21.509	21.477	21.423	21.339	21.232	21.132	21.045	20.961	20.861
10.800	21.336	21.313	21.271	21.201	21.103	21.006	20.917	20.816	20.729
11.000	21.048	21.155	21.123	21.067	20.983	20.885	20.796	20.685	20.578
11.200	20.855	21.003	20.979	20.936	20.867	20.772	20.653	20.558	20.250

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	20.663	20.856	20.841	20.808	20.752	20.639	20.548	20.421	19.603
11.600	20.498	20.572	20.708	20.684	20.612	20.539	20.402	19.814	18.990
11.800	20.309	20.400	20.579	20.537	20.474	20.412	20.115	19.109	18.366
12.000	20.213	20.222	20.455	20.392	20.363	20.285	19.290	18.400	17.761
12.200	18.825	20.054	20.079	20.244	20.255	19.564	18.453	17.706	17.184
12.400	17.189	19.891	19.950	20.105	19.977	18.448	17.612	17.041	16.662
12.600	15.575	18.737	19.818	19.862	18.456	17.354	16.797	16.421	16.138
12.800	14.075	15.679	19.692	18.355	16.874	16.335	16.018	15.788	15.591
13.000	12.568	12.718	19.614	15.331	15.346	15.310	15.251	15.181	15.093
13.200	11.093	9.801	7.937	12.339	13.719	14.282	14.501	14.587	14.608
13.400	9.657	7.929	7.953	9.494	12.286	13.345	13.749	14.006	14.132
13.600	8.272	7.945	7.969	7.995	10.825	12.259	13.066	13.461	13.663
13.800	7.937	7.961	7.985	8.012	9.413	11.298	12.357	12.893	13.248
14.000	7.953	7.977	8.002	8.029	8.269	10.342	11.656	12.339	12.795
14.200	7.969	7.994	8.019	8.046	8.075	9.426	10.834	11.790	12.321
14.400	7.986	8.011	8.036	8.064	8.093	8.505	10.157	11.245	11.877
14.600	8.003	8.028	8.054	8.082	8.112	8.143	9.467	10.602	11.434
14.800	8.020	8.045	8.072	8.100	8.131	8.162	8.767	10.070	10.994
15.000	8.037	8.063	8.090	8.119	8.150	8.182	8.361	9.522	10.556
15.200	8.055	8.081	8.109	8.139	8.170	8.203	8.237	8.971	10.031
15.400	8.073	8.099	8.128	8.158	8.190	8.223	8.259	8.553	9.586
15.600	8.091	8.118	8.147	8.178	8.210	8.244	8.280	8.326	9.137
15.800	8.110	8.138	8.167	8.198	8.231	8.266	8.302	8.341	8.712
16.000	8.129	8.157	8.187	8.219	8.252	8.288	8.325	8.364	8.527
16.200	8.148	8.177	8.208	8.240	8.274	8.310	8.348	8.388	8.430
16.400	8.168	8.197	8.228	8.261	8.296	8.333	8.371	8.412	8.455
16.600	8.188	8.218	8.250	8.283	8.318	8.356	8.395	8.436	8.480
16.800	8.208	8.239	8.271	8.305	8.341	8.379	8.419	8.461	8.506
17.000	8.229	8.260	8.293	8.328	8.365	8.403	8.444	8.487	8.532
17.200	8.250	8.282	8.315	8.351	8.388	8.428	8.469	8.513	8.559
17.400	8.272	8.304	8.338	8.374	8.412	8.452	8.495	8.539	8.586
17.600	8.293	8.326	8.361	8.398	8.437	8.478	8.521	8.566	8.614
17.800	8.316	8.349	8.385	8.422	8.462	8.504	8.547	8.593	8.642
18.000	8.338	8.373	8.409	8.447	8.487	8.530	8.574	8.621	8.671
18.200	8.361	8.396	8.433	8.472	8.513	8.556	8.602	8.650	8.700
18.400	8.385	8.420	8.458	8.498	8.540	8.584	8.630	8.679	8.730
18.600	8.409	8.445	8.483	8.524	8.566	8.611	8.658	8.708	8.761
18.800	8.433	8.470	8.509	8.550	8.594	8.639	8.688	8.738	8.791
19.000	8.458	8.495	8.535	8.577	8.621	8.668	8.717	8.769	8.823
19.200	8.483	8.521	8.562	8.605	8.650	8.697	8.747	8.800	8.855
19.400	8.508	8.548	8.589	8.632	8.678	8.727	8.778	8.831	8.887
19.600	8.534	8.574	8.616	8.661	8.708	8.757	8.809	8.863	8.920
19.800	8.561	8.602	8.644	8.690	8.738	8.788	8.841	8.896	8.953
20.000	8.588	8.629	8.673	8.719	8.768	8.819	8.873	8.928	8.987
20.200	8.615	8.657	8.702	8.749	8.799	8.851	8.905	8.962	9.021
20.400	8.643	8.686	8.731	8.780	8.830	8.883	8.938	8.996	9.056
20.600	8.671	8.715	8.762	8.811	8.862	8.916	8.972	9.030	9.091
20.800	8.700	8.745	8.792	8.842	8.894	8.949	9.006	9.065	9.127
21.000	8.729	8.775	8.823	8.874	8.927	8.982	9.040	9.100	9.163
21.200	8.759	8.806	8.855	8.907	8.960	9.017	9.075	9.136	9.200
21.400	8.789	8.837	8.887	8.939	8.994	9.051	9.111	9.173	9.237
21.600	8.820	8.869	8.920	8.973	9.028	9.086	9.147	9.209	9.275
21.800	8.851	8.901	8.953	9.007	9.063	9.122	9.183	9.247	9.313
22.000	8.883	8.933	8.986	9.041	9.098	9.158	9.220	9.285	9.352
22.200	8.915	8.967	9.020	9.076	9.134	9.195	9.258	9.323	9.391
22.400	8.948	9.000	9.054	9.111	9.170	9.232	9.296	9.362	9.431
22.600	8.981	9.034	9.089	9.147	9.207	9.269	9.334	9.402	9.471
22.800	9.015	9.069	9.125	9.183	9.244	9.307	9.373	9.442	9.512
23.000	9.049	9.104	9.161	9.220	9.282	9.346	9.413	9.482	9.554

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	9.084	9.140	9.197	9.257	9.320	9.385	9.453	9.523	9.596
23.400	9.119	9.176	9.234	9.295	9.359	9.425	9.493	9.565	9.638
23.600	9.155	9.212	9.272	9.334	9.398	9.465	9.535	9.607	9.681
23.800	9.191	9.249	9.310	9.373	9.438	9.506	9.576	9.649	9.725
24.000	9.228	9.287	9.348	9.412	9.478	9.547	9.618	9.692	9.769
24.200	9.265	9.325	9.387	9.452	9.519	9.589	9.661	9.736	9.813
24.400	9.302	9.363	9.426	9.492	9.560	9.631	9.704	9.780	9.858
24.600	9.341	9.402	9.466	9.533	9.602	9.674	9.748	9.824	9.904
24.800	9.379	9.442	9.507	9.574	9.644	9.717	9.792	9.869	9.949
25.000	9.418	9.482	9.548	9.616	9.687	9.761	9.836	9.915	9.996
25.200	9.458	9.522	9.589	9.658	9.730	9.805	9.881	9.961	10.043
25.400	9.498	9.563	9.631	9.701	9.774	9.849	9.927	10.007	10.090
25.600	9.539	9.605	9.674	9.745	9.818	9.894	9.973	10.054	10.138
25.800	9.580	9.647	9.716	9.789	9.863	9.940	10.019	10.101	10.186
26.000	9.622	9.689	9.760	9.833	9.908	9.986	10.066	10.149	10.235
26.200	9.664	9.732	9.804	9.878	9.954	10.033	10.114	10.197	10.284
26.400	9.706	9.776	9.848	9.923	10.000	10.080	10.162	10.246	10.333
26.600	9.749	9.820	9.893	9.969	10.047	10.127	10.210	10.295	10.383
26.800	9.793	9.864	9.938	10.015	10.094	10.175	10.259	10.345	10.434
27.000	9.837	9.909	9.984	10.061	10.141	10.223	10.308	10.395	10.485
27.200	9.881	9.955	10.030	10.108	10.189	10.272	10.358	10.446	10.536
27.400	9.926	10.000	10.077	10.156	10.237	10.321	10.408	10.497	10.588
27.600	9.972	10.047	10.124	10.204	10.286	10.371	10.458	10.548	10.633
27.800	10.018	10.093	10.172	10.252	10.336	10.421	10.510	10.600	10.497
28.000	10.064	10.141	10.220	10.301	10.385	10.472	10.561	10.653	10.187
28.200	10.111	10.188	10.268	10.351	10.436	10.523	10.613	10.545	9.791
28.400	10.158	10.236	10.317	10.400	10.486	10.575	10.666	10.153	9.497
28.600	10.205	10.285	10.367	10.451	10.537	10.627	10.610	9.670	9.468
28.800	10.254	10.334	10.416	10.501	10.589	10.680	10.096	9.484	9.471
29.000	10.302	10.383	10.467	10.553	10.641	10.701	9.577	9.482	9.474
29.200	10.351	10.433	10.517	10.604	10.694	9.983	9.494	9.486	9.478
29.400	10.339	10.483	10.569	10.657	10.747	9.530	9.498	9.489	9.481
29.600	10.126	10.534	10.621	10.709	9.829	9.510	9.501	9.493	9.484
29.800	10.173	10.586	10.673	10.763	9.523	9.514	9.505	9.496	9.488
30.000	10.220	10.317	10.726	9.742	9.527	9.518	9.509	9.500	9.491
30.200	10.268	10.366	10.647	9.540	9.531	9.521	9.512	9.503	9.495
30.400	10.317	10.415	9.718	9.544	9.535	9.525	9.516	9.507	9.498
30.600	10.366	10.113	9.238	9.548	9.539	9.529	9.520	9.511	9.502
30.800	10.415	9.401	9.238	9.552	9.543	9.533	9.524	9.514	9.505
31.000	9.949	9.240	9.238	9.238	9.547	9.537	9.527	9.518	9.509
31.200	9.424	9.238	9.238	9.238	9.551	9.541	9.531	9.522	9.513
31.400	9.253	9.238	9.238	9.238	9.298	9.545	9.535	9.526	9.517
31.600	9.238	9.238	9.238	9.238	9.238	9.549	9.539	9.530	9.520
31.800	9.238	9.238	9.238	9.238	9.238	9.530	9.543	9.534	9.524
32.000	9.238	9.238	9.238	9.238	9.238	9.238	9.547	9.537	9.528

----- block coefficient -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
0.200	0.7501	0.5680	0.5595	0.6578	0.8930	1.1722	1.4698	1.7799	2.0992
0.400	0.5983	0.5566	0.5735	0.6099	0.6811	0.8002	0.9400	1.0904	1.2465
0.600	0.5791	0.5726	0.5842	0.6063	0.6417	0.6992	0.7809	0.8746	0.9737
0.800	0.5849	0.5849	0.5933	0.6088	0.6319	0.6654	0.7145	0.7769	0.8461
1.000	0.5944	0.5949	0.6013	0.6129	0.6297	0.6534	0.6848	0.7265	0.7765
1.200	0.6031	0.6036	0.6085	0.6176	0.6307	0.6485	0.6714	0.6997	0.7360
1.400	0.6110	0.6113	0.6152	0.6224	0.6330	0.6468	0.6644	0.6855	0.7120
1.600	0.6182	0.6182	0.6214	0.6273	0.6359	0.6470	0.6607	0.6776	0.6981
1.800	0.6248	0.6246	0.6271	0.6320	0.6391	0.6481	0.6590	0.6730	0.6898
2.000	0.6309	0.6306	0.6324	0.6365	0.6424	0.6497	0.6588	0.6703	0.6846
2.200	0.6367	0.6361	0.6376	0.6409	0.6457	0.6517	0.6594	0.6691	0.6815
2.400	0.6422	0.6413	0.6424	0.6450	0.6489	0.6540	0.6606	0.6689	0.6798
2.600	0.6476	0.6463	0.6470	0.6490	0.6520	0.6564	0.6622	0.6696	0.6791
2.800	0.6529	0.6511	0.6513	0.6527	0.6552	0.6589	0.6640	0.6708	0.6791
3.000	0.6578	0.6559	0.6554	0.6564	0.6583	0.6615	0.6662	0.6723	0.6796
3.200	0.6623	0.6605	0.6595	0.6599	0.6615	0.6643	0.6685	0.6740	0.6805
3.400	0.6667	0.6648	0.6634	0.6633	0.6646	0.6671	0.6710	0.6759	0.6817
3.600	0.6709	0.6688	0.6673	0.6668	0.6677	0.6701	0.6735	0.6778	0.6831
3.800	0.6749	0.6726	0.6711	0.6703	0.6709	0.6730	0.6760	0.6799	0.6846
4.000	0.6787	0.6763	0.6747	0.6739	0.6742	0.6759	0.6786	0.6820	0.6862
4.200	0.6824	0.6798	0.6782	0.6774	0.6775	0.6788	0.6812	0.6842	0.6880
4.400	0.6860	0.6833	0.6816	0.6808	0.6809	0.6817	0.6837	0.6864	0.6899
4.600	0.6896	0.6867	0.6849	0.6842	0.6842	0.6847	0.6863	0.6887	0.6918
4.800	0.6931	0.6901	0.6883	0.6875	0.6874	0.6877	0.6889	0.6910	0.6938
5.000	0.6965	0.6935	0.6917	0.6907	0.6904	0.6907	0.6916	0.6933	0.6958
5.200	0.6999	0.6969	0.6950	0.6939	0.6935	0.6937	0.6943	0.6957	0.6978
5.400	0.7033	0.7003	0.6983	0.6970	0.6965	0.6965	0.6970	0.6981	0.6999
5.600	0.7067	0.7037	0.7015	0.7001	0.6994	0.6993	0.6997	0.7005	0.7020
5.800	0.7101	0.7070	0.7048	0.7032	0.7023	0.7020	0.7023	0.7030	0.7041
6.000	0.7135	0.7103	0.7080	0.7063	0.7052	0.7047	0.7048	0.7054	0.7063
6.200	0.7168	0.7136	0.7111	0.7093	0.7080	0.7074	0.7074	0.7078	0.7085
6.400	0.7202	0.7169	0.7142	0.7122	0.7109	0.7100	0.7098	0.7101	0.7107
6.600	0.7235	0.7200	0.7173	0.7151	0.7136	0.7127	0.7123	0.7124	0.7129
6.800	0.7269	0.7232	0.7203	0.7180	0.7164	0.7153	0.7147	0.7147	0.7150
7.000	0.7301	0.7264	0.7233	0.7209	0.7191	0.7179	0.7172	0.7169	0.7171
7.200	0.7333	0.7295	0.7263	0.7238	0.7218	0.7205	0.7196	0.7192	0.7192
7.400	0.7366	0.7326	0.7293	0.7266	0.7246	0.7230	0.7220	0.7214	0.7213
7.600	0.7398	0.7357	0.7323	0.7294	0.7272	0.7256	0.7244	0.7237	0.7233
7.800	0.7429	0.7387	0.7352	0.7323	0.7299	0.7281	0.7268	0.7259	0.7254
8.000	0.7461	0.7417	0.7381	0.7351	0.7326	0.7306	0.7291	0.7281	0.7275
8.200	0.7492	0.7448	0.7410	0.7378	0.7352	0.7331	0.7315	0.7303	0.7295
8.400	0.7523	0.7477	0.7438	0.7406	0.7378	0.7356	0.7338	0.7325	0.7316
8.600	0.7554	0.7507	0.7467	0.7433	0.7404	0.7381	0.7362	0.7347	0.7336
8.800	0.7585	0.7537	0.7496	0.7460	0.7430	0.7405	0.7385	0.7369	0.7356
9.000	0.7616	0.7566	0.7524	0.7487	0.7456	0.7430	0.7408	0.7391	0.7376
9.200	0.7646	0.7596	0.7552	0.7514	0.7482	0.7454	0.7431	0.7412	0.7397
9.400	0.7676	0.7625	0.7580	0.7541	0.7507	0.7479	0.7454	0.7434	0.7417
9.600	0.7705	0.7654	0.7608	0.7567	0.7533	0.7502	0.7477	0.7455	0.7438
9.800	0.7733	0.7683	0.7636	0.7594	0.7558	0.7526	0.7499	0.7477	0.7458
10.000	0.7760	0.7711	0.7663	0.7620	0.7583	0.7550	0.7522	0.7498	0.7479
10.200	0.7787	0.7738	0.7691	0.7647	0.7608	0.7574	0.7544	0.7520	0.7499
10.400	0.7813	0.7765	0.7717	0.7673	0.7633	0.7597	0.7567	0.7541	0.7520
10.600	0.7839	0.7790	0.7744	0.7699	0.7657	0.7621	0.7590	0.7563	0.7540
10.800	0.7863	0.7816	0.7769	0.7725	0.7682	0.7645	0.7612	0.7584	0.7561
11.000	0.7887	0.7840	0.7794	0.7750	0.7707	0.7668	0.7635	0.7606	0.7579
11.200	0.7910	0.7864	0.7819	0.7774	0.7731	0.7692	0.7657	0.7627	0.7596

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	0.7932	0.7888	0.7842	0.7798	0.7756	0.7716	0.7680	0.7647	0.7607
11.600	0.7953	0.7910	0.7865	0.7822	0.7779	0.7739	0.7702	0.7662	0.7613
11.800	0.7973	0.7931	0.7888	0.7845	0.7803	0.7762	0.7721	0.7671	0.7614
12.000	0.7992	0.7951	0.7910	0.7867	0.7825	0.7785	0.7734	0.7673	0.7610
12.200	0.8002	0.7971	0.7930	0.7889	0.7848	0.7801	0.7738	0.7670	0.7602
12.400	0.7996	0.7990	0.7950	0.7910	0.7868	0.7807	0.7735	0.7661	0.7589
12.600	0.7974	0.8004	0.7969	0.7930	0.7875	0.7802	0.7725	0.7647	0.7572
12.800	0.7938	0.7990	0.7988	0.7942	0.7867	0.7787	0.7707	0.7628	0.7551
13.000	0.7887	0.7946	0.8006	0.7925	0.7844	0.7763	0.7682	0.7603	0.7526
13.200	0.7822	0.7873	0.7897	0.7878	0.7807	0.7729	0.7651	0.7573	0.7497
13.400	0.7745	0.7775	0.7791	0.7803	0.7755	0.7687	0.7614	0.7539	0.7465
13.600	0.7656	0.7673	0.7689	0.7705	0.7691	0.7636	0.7570	0.7500	0.7429
13.800	0.7558	0.7574	0.7589	0.7605	0.7613	0.7577	0.7521	0.7457	0.7390
14.000	0.7462	0.7477	0.7493	0.7509	0.7525	0.7510	0.7466	0.7409	0.7347
14.200	0.7368	0.7384	0.7399	0.7415	0.7431	0.7435	0.7405	0.7357	0.7301
14.400	0.7278	0.7293	0.7308	0.7324	0.7340	0.7354	0.7339	0.7301	0.7252
14.600	0.7189	0.7204	0.7220	0.7235	0.7251	0.7267	0.7268	0.7242	0.7200
14.800	0.7104	0.7118	0.7134	0.7149	0.7165	0.7180	0.7192	0.7178	0.7145
15.000	0.7020	0.7035	0.7050	0.7065	0.7081	0.7097	0.7112	0.7110	0.7087
15.200	0.6939	0.6954	0.6968	0.6984	0.6999	0.7015	0.7031	0.7040	0.7026
15.400	0.6860	0.6874	0.6889	0.6904	0.6920	0.6935	0.6951	0.6966	0.6962
15.600	0.6783	0.6797	0.6812	0.6827	0.6842	0.6858	0.6874	0.6890	0.6896
15.800	0.6708	0.6722	0.6737	0.6752	0.6767	0.6782	0.6798	0.6814	0.6827
16.000	0.6635	0.6649	0.6664	0.6678	0.6694	0.6709	0.6725	0.6740	0.6756
16.200	0.6563	0.6578	0.6592	0.6607	0.6622	0.6637	0.6653	0.6669	0.6685
16.400	0.6494	0.6508	0.6523	0.6537	0.6552	0.6568	0.6583	0.6599	0.6615
16.600	0.6426	0.6440	0.6455	0.6469	0.6484	0.6500	0.6515	0.6531	0.6547
16.800	0.6360	0.6374	0.6389	0.6403	0.6418	0.6433	0.6449	0.6464	0.6480
17.000	0.6296	0.6310	0.6324	0.6339	0.6353	0.6368	0.6384	0.6399	0.6415
17.200	0.6233	0.6247	0.6261	0.6276	0.6290	0.6305	0.6321	0.6336	0.6352
17.400	0.6171	0.6185	0.6200	0.6214	0.6229	0.6244	0.6259	0.6274	0.6290
17.600	0.6111	0.6125	0.6139	0.6154	0.6168	0.6183	0.6199	0.6214	0.6230
17.800	0.6053	0.6067	0.6081	0.6095	0.6110	0.6125	0.6140	0.6155	0.6171
18.000	0.5996	0.6009	0.6023	0.6038	0.6052	0.6067	0.6082	0.6098	0.6114
18.200	0.5940	0.5953	0.5968	0.5982	0.5996	0.6011	0.6026	0.6042	0.6057
18.400	0.5885	0.5899	0.5913	0.5927	0.5942	0.5956	0.5971	0.5987	0.6003
18.600	0.5832	0.5845	0.5859	0.5874	0.5888	0.5903	0.5918	0.5933	0.5949
18.800	0.5779	0.5793	0.5807	0.5821	0.5836	0.5850	0.5865	0.5881	0.5897
19.000	0.5728	0.5742	0.5756	0.5770	0.5784	0.5799	0.5814	0.5830	0.5845
19.200	0.5678	0.5692	0.5706	0.5720	0.5734	0.5749	0.5764	0.5780	0.5795
19.400	0.5630	0.5643	0.5657	0.5671	0.5685	0.5700	0.5715	0.5731	0.5746
19.600	0.5582	0.5595	0.5609	0.5623	0.5638	0.5652	0.5667	0.5683	0.5698
19.800	0.5535	0.5549	0.5562	0.5576	0.5591	0.5606	0.5620	0.5636	0.5651
20.000	0.5489	0.5503	0.5516	0.5531	0.5545	0.5560	0.5575	0.5590	0.5605
20.200	0.5444	0.5458	0.5472	0.5486	0.5500	0.5515	0.5530	0.5545	0.5560
20.400	0.5400	0.5414	0.5428	0.5442	0.5456	0.5471	0.5486	0.5501	0.5516
20.600	0.5357	0.5371	0.5385	0.5399	0.5413	0.5428	0.5443	0.5458	0.5473
20.800	0.5315	0.5329	0.5343	0.5357	0.5371	0.5386	0.5400	0.5415	0.5431
21.000	0.5274	0.5288	0.5301	0.5315	0.5330	0.5344	0.5359	0.5374	0.5389
21.200	0.5234	0.5247	0.5261	0.5275	0.5289	0.5304	0.5319	0.5334	0.5349
21.400	0.5194	0.5208	0.5221	0.5235	0.5250	0.5264	0.5279	0.5294	0.5309
21.600	0.5155	0.5169	0.5183	0.5197	0.5211	0.5225	0.5240	0.5255	0.5270
21.800	0.5117	0.5131	0.5144	0.5158	0.5173	0.5187	0.5202	0.5217	0.5232
22.000	0.5080	0.5093	0.5107	0.5121	0.5135	0.5150	0.5164	0.5179	0.5194
22.200	0.5043	0.5057	0.5071	0.5085	0.5099	0.5113	0.5128	0.5143	0.5158
22.400	0.5008	0.5021	0.5035	0.5049	0.5063	0.5077	0.5092	0.5107	0.5122
22.600	0.4972	0.4986	0.5000	0.5013	0.5028	0.5042	0.5056	0.5071	0.5086
22.800	0.4938	0.4951	0.4965	0.4979	0.4993	0.5007	0.5022	0.5037	0.5052
23.000	0.4904	0.4917	0.4931	0.4945	0.4959	0.4973	0.4988	0.5003	0.5018

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	0.4871	0.4884	0.4898	0.4912	0.4926	0.4940	0.4955	0.4969	0.4984
23.400	0.4838	0.4852	0.4865	0.4879	0.4893	0.4907	0.4922	0.4937	0.4952
23.600	0.4806	0.4820	0.4833	0.4847	0.4861	0.4875	0.4890	0.4905	0.4919
23.800	0.4775	0.4788	0.4802	0.4816	0.4830	0.4844	0.4858	0.4873	0.4888
24.000	0.4744	0.4757	0.4771	0.4785	0.4799	0.4813	0.4827	0.4842	0.4857
24.200	0.4714	0.4727	0.4741	0.4754	0.4768	0.4783	0.4797	0.4812	0.4827
24.400	0.4684	0.4697	0.4711	0.4725	0.4739	0.4753	0.4767	0.4782	0.4797
24.600	0.4655	0.4668	0.4682	0.4695	0.4709	0.4724	0.4738	0.4753	0.4767
24.800	0.4626	0.4639	0.4653	0.4667	0.4681	0.4695	0.4709	0.4724	0.4739
25.000	0.4598	0.4611	0.4625	0.4638	0.4652	0.4667	0.4681	0.4695	0.4710
25.200	0.4570	0.4583	0.4597	0.4611	0.4625	0.4639	0.4653	0.4668	0.4682
25.400	0.4543	0.4556	0.4570	0.4583	0.4597	0.4612	0.4626	0.4640	0.4655
25.600	0.4516	0.4530	0.4543	0.4557	0.4571	0.4585	0.4599	0.4614	0.4628
25.800	0.4490	0.4503	0.4517	0.4530	0.4544	0.4558	0.4573	0.4587	0.4602
26.000	0.4464	0.4477	0.4491	0.4505	0.4518	0.4533	0.4547	0.4561	0.4576
26.200	0.4439	0.4452	0.4465	0.4479	0.4493	0.4507	0.4521	0.4536	0.4551
26.400	0.4414	0.4427	0.4441	0.4454	0.4468	0.4482	0.4496	0.4511	0.4526
26.600	0.4389	0.4402	0.4416	0.4430	0.4443	0.4458	0.4472	0.4486	0.4501
26.800	0.4365	0.4378	0.4392	0.4405	0.4419	0.4433	0.4448	0.4462	0.4477
27.000	0.4341	0.4355	0.4368	0.4382	0.4396	0.4410	0.4424	0.4438	0.4453
27.200	0.4318	0.4331	0.4345	0.4358	0.4372	0.4386	0.4400	0.4415	0.4430
27.400	0.4295	0.4308	0.4322	0.4335	0.4349	0.4363	0.4377	0.4392	0.4406
27.600	0.4273	0.4286	0.4299	0.4313	0.4327	0.4341	0.4355	0.4369	0.4384
27.800	0.4250	0.4264	0.4277	0.4291	0.4304	0.4318	0.4333	0.4347	0.4360
28.000	0.4228	0.4242	0.4255	0.4269	0.4283	0.4297	0.4311	0.4325	0.4335
28.200	0.4207	0.4220	0.4234	0.4247	0.4261	0.4275	0.4289	0.4303	0.4308
28.400	0.4186	0.4199	0.4212	0.4226	0.4240	0.4254	0.4268	0.4278	0.4279
28.600	0.4165	0.4178	0.4192	0.4205	0.4219	0.4233	0.4247	0.4252	0.4250
28.800	0.4145	0.4158	0.4171	0.4185	0.4198	0.4212	0.4223	0.4223	0.4221
29.000	0.4124	0.4138	0.4151	0.4165	0.4178	0.4192	0.4197	0.4195	0.4193
29.200	0.4105	0.4118	0.4131	0.4145	0.4158	0.4170	0.4169	0.4167	0.4165
29.400	0.4085	0.4098	0.4112	0.4125	0.4139	0.4144	0.4142	0.4140	0.4138
29.600	0.4065	0.4079	0.4092	0.4106	0.4117	0.4117	0.4115	0.4113	0.4111
29.800	0.4045	0.4060	0.4073	0.4087	0.4092	0.4090	0.4088	0.4086	0.4084
30.000	0.4026	0.4041	0.4055	0.4066	0.4065	0.4063	0.4061	0.4059	0.4057
30.200	0.4007	0.4022	0.4036	0.4041	0.4039	0.4037	0.4035	0.4033	0.4031
30.400	0.3988	0.4003	0.4016	0.4015	0.4013	0.4011	0.4010	0.4008	0.4006
30.600	0.3969	0.3984	0.3991	0.3990	0.3988	0.3986	0.3984	0.3982	0.3980
30.800	0.3951	0.3963	0.3965	0.3965	0.3963	0.3961	0.3959	0.3957	0.3955
31.000	0.3933	0.3940	0.3939	0.3939	0.3938	0.3936	0.3934	0.3932	0.3930
31.200	0.3912	0.3914	0.3914	0.3914	0.3914	0.3912	0.3910	0.3908	0.3906
31.400	0.3889	0.3889	0.3889	0.3889	0.3889	0.3888	0.3886	0.3884	0.3882
31.600	0.3865	0.3865	0.3865	0.3865	0.3865	0.3864	0.3862	0.3860	0.3858
31.800	0.3840	0.3840	0.3840	0.3840	0.3840	0.3840	0.3838	0.3836	0.3835
32.000	0.3816	0.3816	0.3816	0.3816	0.3816	0.3816	0.3815	0.3813	0.3811

----- waterline area m2 -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	2726.7	2673.2	4949.6	2952.5	2968.2	3015.8	3065.3	3112.2	3155.7
0.200	3682.6	4492.2	5284.3	4761.9	3898.6	3648.0	3546.1	3500.9	3468.8
0.400	4626.7	5428.2	5464.7	5379.0	4830.5	4280.3	4026.4	3889.2	3779.2
0.600	5321.1	5616.6	5611.1	5568.2	5416.8	4912.0	4506.4	4264.6	4098.4
0.800	5695.0	5752.2	5735.3	5700.2	5607.4	5433.0	4985.4	4636.9	4418.6
1.000	5854.6	5863.7	5844.5	5812.4	5752.9	5633.4	5404.7	5014.5	4738.3
1.200	5969.5	5963.8	5944.4	5907.6	5865.4	5769.9	5630.3	5369.0	5060.0
1.400	6067.1	6049.3	6032.8	5999.7	5958.0	5880.2	5754.2	5623.9	5367.0
1.600	6153.9	6139.1	6112.2	6081.3	6037.3	5967.9	5853.9	5758.2	5621.3
1.800	6233.4	6215.4	6184.6	6156.8	6111.4	6039.0	5955.7	5863.6	5775.8
2.000	6302.3	6286.4	6259.7	6223.4	6171.7	6105.7	6045.2	5959.8	5894.7
2.200	6388.5	6352.4	6324.0	6283.9	6223.5	6171.9	6119.6	6052.5	5998.5
2.400	6471.9	6410.5	6388.5	6340.1	6276.2	6232.1	6191.0	6147.3	6092.3
2.600	6548.2	6487.6	6447.2	6383.8	6334.7	6292.7	6257.8	6231.3	6170.5
2.800	6614.5	6566.1	6485.2	6437.6	6392.1	6348.0	6327.4	6310.3	6243.4
3.000	6658.4	6638.8	6542.5	6486.3	6445.5	6413.6	6399.3	6367.2	6311.3
3.200	6703.3	6686.9	6608.7	6533.3	6504.4	6478.6	6460.2	6423.5	6378.2
3.400	6756.7	6718.7	6674.2	6594.0	6557.4	6546.1	6519.8	6474.1	6430.7
3.600	6798.5	6747.2	6729.0	6666.9	6613.7	6612.1	6571.2	6524.0	6484.2
3.800	6847.4	6789.5	6769.2	6739.4	6688.9	6658.0	6620.6	6579.0	6530.5
4.000	6886.2	6836.1	6806.4	6800.8	6764.6	6706.4	6669.6	6626.9	6580.4
4.200	6929.1	6888.8	6850.6	6860.3	6836.6	6760.5	6712.0	6674.7	6644.2
4.400	6989.0	6937.3	6903.6	6905.5	6889.8	6815.1	6759.8	6732.8	6689.0
4.600	7039.8	6982.9	6973.7	6946.3	6929.7	6886.9	6823.5	6772.5	6735.4
4.800	7091.7	7050.2	7026.2	6991.9	6968.2	6943.0	6867.4	6820.0	6777.9
5.000	7143.1	7114.5	7078.5	7040.1	6994.8	6989.5	6943.5	6868.3	6814.9
5.200	7188.3	7173.3	7135.0	7083.8	7050.9	7028.4	6992.3	6910.7	6862.1
5.400	7263.2	7233.7	7184.6	7136.9	7096.6	7050.5	7031.0	6981.7	6907.5
5.600	7327.2	7276.8	7242.1	7190.0	7138.5	7091.8	7070.2	7029.4	6948.2
5.800	7390.9	7343.4	7288.9	7238.3	7177.5	7135.4	7097.5	7069.7	7013.1
6.000	7450.5	7395.0	7330.9	7279.7	7226.8	7175.8	7128.0	7106.1	7061.1
6.200	7494.2	7440.7	7378.6	7326.2	7274.3	7213.3	7169.4	7136.4	7100.2
6.400	7556.1	7487.9	7430.3	7363.8	7313.0	7259.3	7208.0	7159.0	7136.8
6.600	7612.7	7529.5	7472.1	7408.6	7358.6	7305.8	7244.1	7198.3	7167.9
6.800	7662.2	7593.6	7519.5	7461.0	7395.7	7343.4	7286.5	7235.6	7185.0
7.000	7703.5	7643.7	7558.1	7502.0	7434.2	7386.3	7332.2	7270.1	7221.8
7.200	7760.3	7688.9	7617.3	7547.0	7486.6	7422.7	7369.1	7308.5	7257.5
7.400	7813.6	7731.8	7669.1	7594.2	7528.2	7457.2	7409.2	7353.3	7290.1
7.600	7863.9	7775.8	7710.8	7635.4	7577.9	7507.0	7444.6	7389.2	7324.3
7.800	7910.5	7834.0	7755.8	7689.5	7616.7	7549.0	7477.1	7425.9	7367.9
8.000	7962.2	7878.4	7794.5	7730.5	7651.6	7595.0	7522.2	7460.6	7401.7
8.200	8016.3	7930.0	7848.3	7773.8	7703.9	7634.1	7572.7	7491.6	7433.7
8.400	8065.0	7967.1	7895.5	7809.9	7747.3	7668.5	7605.7	7530.3	7464.1
8.600	8117.6	8027.8	7941.5	7855.9	7786.2	7712.9	7644.6	7579.2	7492.7
8.800	8167.5	8072.6	7982.5	7906.7	7823.5	7757.4	7678.6	7608.7	7537.5
9.000	8207.9	8119.6	8030.9	7947.7	7854.6	7791.4	7712.3	7640.5	7583.8
9.200	8239.1	8172.4	8081.0	7992.7	7911.0	7830.2	7757.4	7674.8	7619.7
9.400	8265.6	8216.8	8124.7	8024.9	7950.8	7862.5	7787.3	7706.4	7657.4
9.600	8288.7	8250.8	8170.2	8081.6	7992.5	7904.8	7823.6	7759.3	7692.7
9.800	8309.1	8278.1	8217.2	8121.9	8028.0	7945.3	7857.9	7799.7	7724.9
10.000	8327.0	8301.3	8255.8	8163.8	8070.8	7978.7	7896.2	7836.0	7770.8
10.200	8343.0	8321.3	8284.1	8210.9	8114.5	8021.7	7948.0	7872.5	7813.8
10.400	8357.5	8338.9	8307.2	8250.5	8148.7	8052.9	7986.1	7905.2	7846.6
10.600	8370.4	8353.9	8326.8	8280.5	8187.8	8109.5	8029.0	7954.8	7880.8
10.800	8381.8	8367.3	8343.3	8301.0	8236.7	8150.1	8065.1	7994.0	7903.7
11.000	8355.2	8379.1	8356.9	8319.3	8276.5	8192.2	8110.1	8030.4	7837.9
11.200	8354.3	8389.3	8367.5	8339.8	8306.6	8238.4	8151.3	8017.6	7691.5

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	8354.2	8397.4	8375.2	8358.0	8331.4	8275.7	8188.5	7920.3	7426.9
11.600	8356.6	8358.9	8386.9	8374.4	8347.7	8309.5	8148.6	7671.1	7173.4
11.800	8351.7	8354.3	8399.0	8384.1	8361.3	8330.4	7974.4	7352.8	6900.5
12.000	8320.1	8349.6	8410.1	8391.9	8377.1	8236.8	7571.0	7028.7	6638.1
12.200	7495.8	8348.6	8350.9	8397.0	8391.3	7797.3	7149.4	6705.0	6377.4
12.400	6582.6	8346.6	8356.2	8401.6	8140.4	7208.0	6715.7	6379.5	6112.5
12.600	5668.3	7577.5	8359.1	8364.9	7263.3	6613.1	6278.2	6052.1	5848.0
12.800	4768.1	5762.5	8361.5	7307.1	6360.3	6024.7	5840.4	5710.0	5582.6
13.000	3859.2	3949.2	8372.5	5492.1	5457.3	5427.5	5398.9	5365.8	5315.4
13.200	2951.6	2134.8	751.5	3645.2	4506.1	4823.5	4955.7	5018.7	5040.0
13.400	2048.9	748.0	755.2	1863.1	3607.3	4233.1	4507.2	4669.6	4758.9
13.600	1155.6	751.7	758.8	766.1	2697.5	3588.6	4069.5	4324.1	4474.0
13.800	748.2	755.3	762.5	769.8	1795.7	2985.4	3623.2	3970.7	4196.4
14.000	751.9	759.0	766.2	773.5	975.8	2380.9	3175.9	3617.3	3907.8
14.200	755.5	762.7	769.9	777.2	784.7	1783.7	2682.1	3262.7	3612.3
14.400	759.2	766.3	773.5	780.9	788.5	1186.9	2236.0	2907.2	3320.5
14.600	762.8	770.0	777.2	784.7	792.4	800.3	1786.1	2513.5	3027.5
14.800	766.5	773.6	780.9	788.5	796.3	804.3	1333.6	2154.9	2733.4
15.000	770.1	777.3	784.7	792.4	800.2	808.3	970.5	1794.9	2438.3
15.200	773.7	781.0	788.5	796.3	804.2	812.4	820.9	1433.9	2105.4
15.400	777.4	784.8	792.4	800.2	808.2	816.5	825.1	1113.5	1807.1
15.600	781.1	788.6	796.3	804.1	812.3	820.7	829.4	855.8	1507.7
15.800	784.9	792.4	800.2	808.1	816.3	824.9	833.7	842.9	1215.6
16.000	788.7	796.3	804.0	812.2	820.5	829.2	838.1	847.4	997.2
16.200	792.5	800.1	808.0	816.2	824.7	833.5	842.6	852.0	861.7
16.400	796.3	804.0	812.1	820.3	829.0	837.9	847.1	856.6	866.5
16.600	800.1	808.0	816.1	824.6	833.2	842.3	851.6	861.3	871.4
16.800	804.0	812.0	820.2	828.8	837.6	846.7	856.3	866.1	876.3
17.000	807.9	816.1	824.4	833.0	842.0	851.3	860.9	870.9	881.2
17.200	812.0	820.1	828.6	837.4	846.5	855.9	865.7	875.8	886.2
17.400	816.0	824.3	832.9	841.8	851.0	860.6	870.5	880.7	891.3
17.600	820.1	828.5	837.2	846.2	855.6	865.3	875.3	885.7	896.4
17.800	824.2	832.7	841.6	850.7	860.2	870.1	880.3	890.8	901.5
18.000	828.4	837.0	846.0	855.3	864.9	874.9	885.2	895.9	906.5
18.200	832.6	841.4	850.5	859.9	869.7	879.8	890.3	901.0	911.6
18.400	836.8	845.8	855.0	864.6	874.5	884.8	895.4	906.0	916.7
18.600	841.2	850.2	859.6	869.4	879.4	889.8	900.5	911.1	921.9
18.800	845.6	854.8	864.3	874.1	884.3	894.9	905.6	916.2	927.0
19.000	850.0	859.4	869.0	879.0	889.3	900.0	910.6	921.3	932.3
19.200	854.5	864.0	873.8	883.9	894.4	905.1	915.7	926.5	937.5
19.400	859.1	868.7	878.6	888.9	899.5	910.2	920.8	931.7	942.7
19.600	863.7	873.5	883.5	893.9	904.7	915.2	926.0	936.9	947.9
19.800	868.4	878.3	888.5	899.0	909.7	920.3	931.2	942.1	953.1
20.000	873.2	883.2	893.5	904.2	914.8	925.5	936.4	947.3	958.4
20.200	878.0	888.1	898.6	909.3	919.9	930.7	941.6	952.6	963.7
20.400	882.8	893.1	903.7	914.3	925.1	935.9	946.8	957.8	969.1
20.600	887.8	898.2	908.9	919.4	930.2	941.1	952.0	963.2	974.4
20.800	892.7	903.3	913.9	924.6	935.4	946.3	957.3	968.5	979.8
21.000	897.8	908.5	919.0	929.8	940.6	951.5	962.6	973.8	985.2
21.200	902.9	913.6	924.2	935.0	945.8	956.8	967.9	979.2	990.6
21.400	908.1	918.6	929.3	940.1	951.0	962.1	973.3	984.6	996.0
21.600	913.2	923.8	934.5	945.4	956.3	967.4	978.6	990.0	1001.5
21.800	918.3	928.9	939.7	950.6	961.6	972.7	984.0	995.4	1006.9
22.000	923.4	934.1	944.9	955.8	966.9	978.1	989.4	1000.8	1012.4
22.200	928.5	939.3	950.1	961.1	972.2	983.5	994.8	1006.3	1017.9
22.400	933.7	944.5	955.3	966.4	977.6	988.8	1000.3	1011.8	1023.5
22.600	938.9	949.7	960.6	971.7	982.9	994.3	1005.7	1017.3	1029.0
22.800	944.1	954.9	965.9	977.1	988.3	999.7	1011.2	1022.8	1034.6
23.000	949.3	960.2	971.3	982.4	993.7	1005.1	1016.7	1028.4	1040.1

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	954.5	965.5	976.6	987.8	999.1	1010.6	1022.2	1033.9	1045.8
23.400	959.8	970.8	981.9	993.2	1004.6	1016.1	1027.7	1039.5	1051.4
23.600	965.1	976.1	987.3	998.6	1010.1	1021.6	1033.3	1045.1	1057.0
23.800	970.4	981.5	992.7	1004.1	1015.5	1027.1	1038.9	1050.7	1062.7
24.000	975.7	986.8	998.1	1009.5	1021.0	1032.7	1044.4	1056.3	1068.3
24.200	981.0	992.2	1003.6	1015.0	1026.6	1038.3	1050.0	1062.0	1073.9
24.400	986.4	997.7	1009.0	1020.5	1032.1	1043.8	1055.7	1067.6	1079.6
24.600	991.8	1003.1	1014.4	1026.0	1037.7	1049.4	1061.3	1073.2	1085.2
24.800	997.2	1008.5	1020.0	1031.6	1043.2	1055.0	1066.9	1078.8	1090.9
25.000	1002.6	1013.9	1025.5	1037.1	1048.8	1060.7	1072.5	1084.5	1096.5
25.200	1008.0	1019.4	1031.0	1042.7	1054.4	1066.3	1078.1	1090.1	1102.1
25.400	1013.5	1025.0	1036.6	1048.2	1060.0	1071.9	1083.8	1095.7	1107.8
25.600	1019.0	1030.5	1042.1	1053.8	1065.6	1077.5	1089.4	1101.4	1113.4
25.800	1024.5	1036.0	1047.7	1059.4	1071.2	1083.1	1095.0	1107.0	1119.0
26.000	1030.0	1041.6	1053.2	1065.0	1076.8	1088.7	1100.6	1112.6	1124.6
26.200	1035.5	1047.1	1058.9	1070.6	1082.4	1094.3	1106.2	1118.2	1130.2
26.400	1041.1	1052.7	1064.4	1076.2	1088.0	1099.9	1111.8	1123.8	1135.8
26.600	1046.6	1058.3	1070.0	1081.8	1093.6	1105.5	1117.4	1129.4	1141.4
26.800	1052.2	1063.9	1075.6	1087.4	1099.2	1111.1	1123.0	1135.0	1146.9
27.000	1057.8	1069.4	1081.2	1093.0	1104.8	1116.7	1128.6	1140.5	1152.5
27.200	1063.3	1075.0	1086.8	1098.5	1110.4	1122.3	1134.2	1146.1	1158.1
27.400	1068.9	1080.6	1092.3	1104.1	1116.0	1127.8	1139.7	1151.7	1163.6
27.600	1074.4	1086.2	1097.9	1109.7	1121.5	1133.4	1145.3	1157.2	1155.4
27.800	1080.0	1091.7	1103.4	1115.3	1127.1	1138.9	1150.8	1162.7	930.4
28.000	1085.6	1097.3	1109.0	1120.8	1132.6	1144.5	1156.4	1168.3	648.5
28.200	1091.1	1102.8	1114.6	1126.4	1138.2	1150.0	1161.9	985.7	346.4
28.400	1096.7	1108.4	1120.1	1131.9	1143.7	1155.6	1167.4	643.3	144.9
28.600	1102.2	1113.9	1125.7	1137.4	1149.2	1161.1	1075.8	281.0	110.4
28.800	1107.7	1119.5	1131.2	1142.9	1154.8	1166.6	635.6	117.4	110.4
29.000	1113.3	1125.0	1136.7	1148.5	1160.3	1153.7	213.1	110.4	110.4
29.200	1118.8	1130.5	1142.2	1154.0	1165.8	621.6	110.4	110.4	110.4
29.400	1103.2	1136.0	1147.8	1159.5	1171.3	152.7	110.4	110.4	110.4
29.600	1019.4	1141.5	1153.3	1165.1	617.3	110.4	110.4	110.4	110.4
29.800	1024.9	1147.1	1158.8	1170.6	111.1	110.4	110.4	110.4	110.4
30.000	1030.4	1042.2	1164.3	643.7	110.4	110.4	110.4	110.4	110.4
30.200	1036.0	1047.7	1135.1	110.4	110.4	110.4	110.4	110.4	110.4
30.400	1041.5	1053.2	637.5	110.4	110.4	110.4	110.4	110.4	110.4
30.600	1047.0	953.0	0.0	110.4	110.4	110.4	110.4	110.4	110.4
30.800	1052.5	512.0	0.0	110.4	110.4	110.4	110.4	110.4	110.4
31.000	880.4	59.6	0.0	0.0	110.4	110.4	110.4	110.4	110.4
31.200	525.3	0.0	0.0	0.0	110.4	110.4	110.4	110.4	110.4
31.400	151.3	0.0	0.0	0.0	21.2	110.4	110.4	110.4	110.4
31.600	0.0	0.0	0.0	0.0	0.0	110.4	110.4	110.4	110.4
31.800	0.0	0.0	0.0	0.0	0.0	102.5	110.4	110.4	110.4
32.000	0.0	0.0	0.0	0.0	0.0	0.0	110.4	110.4	110.4

----- moment to change trim tm/cm -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	58.1	56.7	352.7	86.0	87.1	90.3	93.5	96.4	99.2
0.200	128.6	229.6	408.1	294.3	171.6	143.8	133.1	128.2	123.8
0.400	242.5	418.5	438.7	412.6	301.2	216.5	183.4	166.7	152.0
0.600	368.0	461.5	465.3	456.9	411.0	311.4	245.6	210.2	186.6
0.800	462.5	489.4	489.1	483.3	452.1	409.4	321.0	260.4	226.7
1.000	504.7	512.6	511.0	505.3	490.3	451.1	398.4	320.3	272.6
1.200	531.7	534.3	532.1	524.6	515.7	482.2	444.0	385.3	325.3
1.400	554.8	553.3	551.4	544.6	535.3	510.4	470.0	437.8	382.4
1.600	575.9	574.8	569.5	562.8	552.4	532.1	493.1	466.9	435.2
1.800	595.9	593.4	586.5	580.5	568.7	548.0	519.3	491.3	469.6
2.000	613.7	611.3	605.4	596.5	581.9	562.7	544.6	515.3	498.0
2.200	638.8	628.4	621.8	611.0	592.3	578.0	564.5	540.2	524.6
2.400	663.3	643.6	638.6	624.2	604.5	592.4	582.1	568.0	549.3
2.600	685.3	666.0	654.0	633.4	618.9	607.5	599.1	593.1	570.5
2.800	703.8	689.9	662.0	646.6	633.4	621.6	617.7	615.5	591.5
3.000	714.4	711.3	676.9	658.8	647.2	639.3	637.9	630.0	611.8
3.200	725.7	723.2	696.6	670.9	663.3	657.4	654.8	644.5	631.6
3.400	740.7	729.4	715.7	687.7	677.8	677.2	671.0	657.6	645.2
3.600	751.3	734.0	731.0	711.1	693.9	696.9	684.8	670.8	659.2
3.800	764.7	745.0	741.4	733.7	717.2	709.3	698.2	685.9	671.3
4.000	774.4	758.0	750.1	752.6	742.6	722.9	711.8	698.9	684.8
4.200	786.5	773.6	762.6	771.4	765.9	738.5	723.4	712.1	700.2
4.400	804.9	788.2	778.9	784.2	782.0	755.6	737.0	726.0	712.5
4.600	820.5	802.4	802.4	795.6	793.5	779.4	753.3	737.0	725.5
4.800	837.0	824.5	819.3	809.3	803.9	795.8	766.0	750.8	737.4
5.000	853.7	846.5	836.8	824.2	809.4	808.2	792.3	765.0	747.7
5.200	868.7	866.6	855.1	837.7	824.1	819.2	807.8	777.4	761.6
5.400	895.4	888.0	871.2	851.9	838.4	823.0	819.5	802.0	774.9
5.600	918.4	902.1	887.3	869.3	851.4	835.6	831.0	817.4	786.9
5.800	942.2	922.3	902.8	885.1	863.7	849.3	837.4	829.9	809.0
6.000	960.2	940.0	916.7	898.5	879.9	862.0	845.5	840.7	825.0
6.200	975.5	955.7	933.0	914.0	895.6	873.9	858.5	848.6	837.3
6.400	998.3	972.3	951.0	926.4	908.2	889.0	870.7	853.5	848.4
6.600	1016.9	986.9	965.4	941.8	923.5	904.5	882.1	865.8	857.0
6.800	1035.2	1008.7	982.2	960.3	935.8	916.7	896.0	877.6	859.5
7.000	1050.3	1027.0	995.6	974.4	948.9	931.2	911.3	888.6	871.0
7.200	1072.3	1043.7	1015.6	990.4	967.5	943.2	923.3	901.0	882.3
7.400	1092.9	1059.5	1034.9	1005.4	981.9	954.8	936.8	916.0	892.5
7.600	1112.5	1076.2	1050.2	1020.2	997.8	972.5	948.6	927.7	903.4
7.800	1130.8	1099.1	1067.0	1040.5	1011.5	987.2	959.4	939.9	917.9
8.000	1151.8	1116.2	1081.5	1055.5	1023.9	1001.7	975.3	951.3	928.7
8.200	1173.9	1136.9	1102.6	1071.8	1043.5	1015.5	991.4	961.5	938.8
8.400	1193.9	1151.3	1121.0	1085.1	1059.6	1027.7	1002.6	974.8	948.1
8.600	1216.1	1176.5	1139.2	1103.0	1074.0	1044.1	1016.4	990.1	957.1
8.800	1237.2	1194.7	1155.4	1123.0	1087.9	1060.6	1028.3	999.6	971.3
9.000	1254.0	1214.2	1175.1	1139.0	1099.2	1072.9	1040.1	1009.8	988.2
9.200	1266.7	1236.8	1195.8	1157.0	1121.8	1087.3	1056.7	1022.2	1001.0
9.400	1277.4	1255.5	1213.8	1169.3	1137.1	1099.1	1066.6	1033.7	1014.8
9.600	1286.5	1269.6	1232.9	1193.0	1153.5	1115.3	1079.9	1054.2	1027.7
9.800	1294.5	1280.6	1252.9	1209.3	1167.3	1130.5	1092.9	1069.5	1039.6
10.000	1301.5	1289.8	1269.0	1226.7	1184.6	1142.6	1108.0	1083.4	1057.3
10.200	1307.7	1297.6	1280.4	1246.7	1202.1	1160.2	1129.0	1097.5	1073.9
10.400	1313.3	1304.3	1289.5	1263.1	1215.2	1172.6	1144.2	1110.2	1086.3
10.600	1318.1	1310.0	1297.0	1275.1	1231.4	1196.8	1161.8	1130.5	1100.9
10.800	1322.3	1314.9	1303.1	1282.3	1252.7	1213.9	1176.6	1147.1	1110.5
11.000	1309.1	1319.1	1307.8	1288.8	1270.0	1231.8	1195.6	1162.5	1092.2
11.200	1307.8	1322.5	1311.0	1297.3	1282.7	1252.1	1214.5	1160.5	1057.2

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	1308.3	1325.0	1312.6	1304.8	1293.1	1269.7	1230.4	1132.2	996.4
11.600	1308.6	1306.1	1317.0	1311.5	1301.1	1284.4	1220.2	1072.9	946.2
11.800	1308.4	1303.9	1321.9	1316.6	1308.7	1295.4	1172.2	1001.2	897.1
12.000	1292.6	1303.7	1326.5	1321.8	1315.6	1262.5	1069.7	939.1	858.7
12.200	965.1	1305.3	1307.2	1326.8	1321.7	1135.3	979.2	887.7	827.1
12.400	685.6	1306.1	1309.8	1330.3	1238.9	998.5	902.5	845.1	799.0
12.600	478.4	995.7	1311.5	1313.8	1013.1	891.4	842.0	810.8	777.8
12.800	333.9	497.0	1313.2	1017.8	855.9	814.6	795.5	781.7	763.9
13.000	239.7	247.2	1315.8	754.0	762.6	762.8	762.1	759.8	752.8
13.200	186.7	166.1	159.0	693.8	710.4	732.6	740.7	744.7	744.4
13.400	164.9	158.8	159.2	651.9	702.9	720.5	729.3	736.0	739.0
13.600	161.6	159.0	159.3	159.7	702.7	711.7	726.3	732.7	736.9
13.800	158.8	159.2	159.5	159.8	651.4	714.4	728.6	733.4	737.9
14.000	159.0	159.3	159.6	160.0	360.6	706.0	732.7	737.1	741.1
14.200	159.1	159.5	159.8	160.1	160.5	657.2	723.5	741.6	745.5
14.400	159.3	159.6	159.9	160.3	160.7	486.8	709.9	744.3	750.6
14.600	159.5	159.8	160.1	160.4	160.8	161.2	665.2	733.0	754.5
14.800	159.6	159.9	160.3	160.6	161.0	161.4	550.3	715.1	755.1
15.000	159.8	160.1	160.4	160.8	161.2	161.6	332.1	674.5	749.7
15.200	159.9	160.2	160.6	160.9	161.3	161.7	162.2	588.0	721.6
15.400	160.1	160.4	160.7	161.1	161.5	161.9	162.4	433.2	684.5
15.600	160.2	160.6	160.9	161.3	161.7	162.1	162.6	186.3	614.7
15.800	160.4	160.7	161.1	161.5	161.9	162.3	162.8	163.2	490.0
16.000	160.6	160.9	161.3	161.6	162.1	162.5	163.0	163.4	325.4
16.200	160.7	161.1	161.4	161.8	162.2	162.7	163.1	163.6	164.2
16.400	160.9	161.2	161.6	162.0	162.4	162.9	163.3	163.9	164.4
16.600	161.1	161.4	161.8	162.2	162.6	163.1	163.6	164.1	164.6
16.800	161.2	161.6	162.0	162.4	162.8	163.3	163.8	164.3	164.8
17.000	161.4	161.8	162.1	162.6	163.0	163.5	164.0	164.5	165.0
17.200	161.6	161.9	162.3	162.8	163.2	163.7	164.2	164.7	165.3
17.400	161.7	162.1	162.5	163.0	163.4	163.9	164.4	164.9	165.5
17.600	161.9	162.3	162.7	163.1	163.6	164.1	164.6	165.1	165.7
17.800	162.1	162.5	162.9	163.3	163.8	164.3	164.8	165.4	165.9
18.000	162.3	162.7	163.1	163.6	164.0	164.5	165.0	165.6	166.1
18.200	162.5	162.9	163.3	163.8	164.2	164.7	165.3	165.8	166.3
18.400	162.7	163.1	163.5	164.0	164.4	165.0	165.5	166.0	166.5
18.600	162.9	163.3	163.7	164.2	164.7	165.2	165.7	166.2	166.8
18.800	163.0	163.5	163.9	164.4	164.9	165.4	165.9	166.4	167.0
19.000	163.2	163.7	164.1	164.6	165.1	165.6	166.1	166.7	167.2
19.200	163.4	163.9	164.3	164.8	165.3	165.8	166.3	166.9	167.4
19.400	163.6	164.1	164.5	165.0	165.5	166.0	166.6	167.1	167.6
19.600	163.9	164.3	164.8	165.3	165.8	166.3	166.8	167.3	167.8
19.800	164.1	164.5	165.0	165.5	166.0	166.5	167.0	167.5	168.0
20.000	164.3	164.7	165.2	165.7	166.2	166.7	167.2	167.7	168.2
20.200	164.5	164.9	165.4	165.9	166.4	166.9	167.4	167.9	168.4
20.400	164.7	165.2	165.6	166.1	166.6	167.1	167.6	168.1	168.6
20.600	164.9	165.4	165.9	166.3	166.8	167.3	167.8	168.3	168.8
20.800	165.1	165.6	166.1	166.5	167.0	167.5	168.0	168.5	169.0
21.000	165.3	165.8	166.3	166.8	167.2	167.7	168.2	168.7	169.2
21.200	165.6	166.0	166.5	167.0	167.4	167.9	168.4	168.9	169.4
21.400	165.8	166.2	166.7	167.2	167.6	168.1	168.6	169.1	169.6
21.600	166.0	166.4	166.9	167.4	167.8	168.3	168.8	169.3	169.8
21.800	166.2	166.7	167.1	167.6	168.0	168.5	169.0	169.5	170.0
22.000	166.4	166.9	167.3	167.8	168.2	168.7	169.2	169.7	170.2
22.200	166.6	167.1	167.5	168.0	168.4	168.9	169.4	169.9	170.4
22.400	166.8	167.3	167.7	168.2	168.6	169.1	169.6	170.1	170.6
22.600	167.0	167.5	167.9	168.4	168.8	169.3	169.8	170.3	170.8
22.800	167.3	167.7	168.1	168.6	169.0	169.5	170.0	170.5	171.0
23.000	167.5	167.9	168.3	168.8	169.2	169.7	170.2	170.7	171.1

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	167.7	168.1	168.5	169.0	169.4	169.9	170.4	170.8	171.3
23.400	167.9	168.3	168.7	169.2	169.6	170.1	170.6	171.0	171.5
23.600	168.1	168.5	168.9	169.4	169.8	170.3	170.8	171.2	171.7
23.800	168.3	168.7	169.1	169.6	170.0	170.5	170.9	171.4	171.9
24.000	168.5	168.9	169.3	169.8	170.2	170.7	171.1	171.6	172.1
24.200	168.7	169.1	169.5	170.0	170.4	170.9	171.3	171.8	172.3
24.400	168.9	169.3	169.7	170.2	170.6	171.0	171.5	172.0	172.4
24.600	169.1	169.5	169.9	170.3	170.8	171.2	171.7	172.2	172.6
24.800	169.3	169.7	170.1	170.5	171.0	171.4	171.9	172.3	172.8
25.000	169.5	169.9	170.3	170.7	171.2	171.6	172.1	172.5	173.0
25.200	169.7	170.1	170.5	170.9	171.4	171.8	172.2	172.7	173.2
25.400	169.8	170.3	170.7	171.1	171.5	172.0	172.4	172.9	173.3
25.600	170.0	170.5	170.9	171.3	171.7	172.2	172.6	173.1	173.5
25.800	170.2	170.6	171.1	171.5	171.9	172.3	172.8	173.2	173.7
26.000	170.4	170.8	171.2	171.7	172.1	172.5	173.0	173.4	173.9
26.200	170.6	171.0	171.4	171.8	172.3	172.7	173.1	173.6	174.0
26.400	170.8	171.2	171.6	172.0	172.5	172.9	173.3	173.8	174.2
26.600	171.0	171.4	171.8	172.2	172.6	173.1	173.5	173.9	174.4
26.800	171.2	171.6	172.0	172.4	172.8	173.2	173.7	174.1	174.6
27.000	171.4	171.8	172.2	172.6	173.0	173.4	173.8	174.3	174.7
27.200	171.6	171.9	172.3	172.8	173.2	173.6	174.0	174.4	174.9
27.400	171.7	172.1	172.5	172.9	173.3	173.8	174.2	174.6	175.1
27.600	171.9	172.3	172.7	173.1	173.5	173.9	174.4	174.8	174.4
27.800	172.1	172.5	172.9	173.3	173.7	174.1	174.5	175.0	162.0
28.000	172.3	172.7	173.1	173.5	173.9	174.3	174.7	175.1	145.2
28.200	172.5	172.8	173.2	173.6	174.0	174.4	174.9	165.2	113.1
28.400	172.6	173.0	173.4	173.8	174.2	174.6	175.0	145.4	37.7
28.600	172.8	173.2	173.6	174.0	174.4	174.8	170.6	100.4	0.0
28.800	173.0	173.4	173.8	174.1	174.5	174.9	146.0	9.3	0.0
29.000	173.2	173.6	173.9	174.3	174.7	174.4	79.5	0.0	0.0
29.200	173.4	173.7	174.1	174.5	174.9	147.4	0.0	0.0	0.0
29.400	142.4	173.9	174.3	174.7	175.0	45.8	0.0	0.0	0.0
29.600	4.5	174.1	174.4	174.8	152.4	0.0	0.0	0.0	0.0
29.800	4.5	174.2	174.6	175.0	1.0	0.0	0.0	0.0	0.0
30.000	4.5	4.6	174.8	161.2	0.0	0.0	0.0	0.0	0.0
30.200	4.5	4.6	174.9	0.0	0.0	0.0	0.0	0.0	0.0
30.400	4.6	4.6	165.4	0.0	0.0	0.0	0.0	0.0	0.0
30.600	4.6	3.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0
30.800	4.6	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0
31.000	3.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
31.200	1.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
31.400	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
31.600	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
31.800	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
32.000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

	immersion/cm t/cm								
trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	27.9	27.4	50.7	30.3	30.4	30.9	31.4	31.9	32.3
0.200	37.7	46.0	54.2	48.8	40.0	37.4	36.3	35.9	35.6
0.400	47.4	55.6	56.0	55.1	49.5	43.9	41.3	39.9	38.7
0.600	54.5	57.6	57.5	57.1	55.5	50.3	46.2	43.7	42.0
0.800	58.4	59.0	58.8	58.4	57.5	55.7	51.1	47.5	45.3
1.000	60.0	60.1	59.9	59.6	59.0	57.7	55.4	51.4	48.6
1.200	61.2	61.1	60.9	60.6	60.1	59.1	57.7	55.0	51.9
1.400	62.2	62.0	61.8	61.5	61.1	60.3	59.0	57.6	55.0
1.600	63.1	62.9	62.7	62.3	61.9	61.2	60.0	59.0	57.6
1.800	63.9	63.7	63.4	63.1	62.6	61.9	61.0	60.1	59.2
2.000	64.6	64.4	64.2	63.8	63.3	62.6	62.0	61.1	60.4
2.200	65.5	65.1	64.8	64.4	63.8	63.3	62.7	62.0	61.5
2.400	66.3	65.7	65.5	65.0	64.3	63.9	63.5	63.0	62.4
2.600	67.1	66.5	66.1	65.4	64.9	64.5	64.1	63.9	63.2
2.800	67.8	67.3	66.5	66.0	65.5	65.1	64.9	64.7	64.0
3.000	68.2	68.0	67.1	66.5	66.1	65.7	65.6	65.3	64.7
3.200	68.7	68.5	67.7	67.0	66.7	66.4	66.2	65.8	65.4
3.400	69.3	68.9	68.4	67.6	67.2	67.1	66.8	66.4	65.9
3.600	69.7	69.2	69.0	68.3	67.8	67.8	67.4	66.9	66.5
3.800	70.2	69.6	69.4	69.1	68.6	68.2	67.9	67.4	66.9
4.000	70.6	70.1	69.8	69.7	69.3	68.7	68.4	67.9	67.4
4.200	71.0	70.6	70.2	70.3	70.1	69.3	68.8	68.4	68.1
4.400	71.6	71.1	70.8	70.8	70.6	69.9	69.3	69.0	68.6
4.600	72.2	71.6	71.5	71.2	71.0	70.6	69.9	69.4	69.0
4.800	72.7	72.3	72.0	71.7	71.4	71.2	70.4	69.9	69.5
5.000	73.2	72.9	72.6	72.2	71.7	71.6	71.2	70.4	69.9
5.200	73.7	73.5	73.1	72.6	72.3	72.0	71.7	70.8	70.3
5.400	74.4	74.1	73.6	73.2	72.7	72.3	72.1	71.6	70.8
5.600	75.1	74.6	74.2	73.7	73.2	72.7	72.5	72.1	71.2
5.800	75.8	75.3	74.7	74.2	73.6	73.1	72.7	72.5	71.9
6.000	76.4	75.8	75.1	74.6	74.1	73.6	73.1	72.8	72.4
6.200	76.8	76.3	75.6	75.1	74.6	73.9	73.5	73.1	72.8
6.400	77.4	76.8	76.2	75.5	75.0	74.4	73.9	73.4	73.2
6.600	78.0	77.2	76.6	75.9	75.4	74.9	74.3	73.8	73.5
6.800	78.5	77.8	77.1	76.5	75.8	75.3	74.7	74.2	73.6
7.000	79.0	78.3	77.5	76.9	76.2	75.7	75.2	74.5	74.0
7.200	79.5	78.8	78.1	77.4	76.7	76.1	75.5	74.9	74.4
7.400	80.1	79.3	78.6	77.8	77.2	76.4	75.9	75.4	74.7
7.600	80.6	79.7	79.0	78.3	77.7	76.9	76.3	75.7	75.1
7.800	81.1	80.3	79.5	78.8	78.1	77.4	76.6	76.1	75.5
8.000	81.6	80.8	79.9	79.2	78.4	77.8	77.1	76.5	75.9
8.200	82.2	81.3	80.4	79.7	79.0	78.2	77.6	76.8	76.2
8.400	82.7	81.7	80.9	80.1	79.4	78.6	78.0	77.2	76.5
8.600	83.2	82.3	81.4	80.5	79.8	79.1	78.4	77.7	76.8
8.800	83.7	82.7	81.8	81.0	80.2	79.5	78.7	78.0	77.3
9.000	84.1	83.2	82.3	81.5	80.5	79.9	79.1	78.3	77.7
9.200	84.5	83.8	82.8	81.9	81.1	80.3	79.5	78.7	78.1
9.400	84.7	84.2	83.3	82.3	81.5	80.6	79.8	79.0	78.5
9.600	85.0	84.6	83.7	82.8	81.9	81.0	80.2	79.5	78.9
9.800	85.2	84.9	84.2	83.2	82.3	81.4	80.5	79.9	79.2
10.000	85.4	85.1	84.6	83.7	82.7	81.8	80.9	80.3	79.7
10.200	85.5	85.3	84.9	84.2	83.2	82.2	81.5	80.7	80.1
10.400	85.7	85.5	85.1	84.6	83.5	82.5	81.9	81.0	80.4
10.600	85.8	85.6	85.3	84.9	83.9	83.1	82.3	81.5	80.8
10.800	85.9	85.8	85.5	85.1	84.4	83.5	82.7	81.9	81.0
11.000	85.6	85.9	85.7	85.3	84.8	84.0	83.1	82.3	80.3
11.200	85.6	86.0	85.8	85.5	85.1	84.4	83.6	82.2	78.8

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	85.6	86.1	85.8	85.7	85.4	84.8	83.9	81.2	76.1
11.600	85.7	85.7	86.0	85.8	85.6	85.2	83.5	78.6	73.5
11.800	85.6	85.6	86.1	85.9	85.7	85.4	81.7	75.4	70.7
12.000	85.3	85.6	86.2	86.0	85.9	84.4	77.6	72.0	68.0
12.200	76.8	85.6	85.6	86.1	86.0	79.9	73.3	68.7	65.4
12.400	67.5	85.6	85.7	86.1	83.4	73.9	68.8	65.4	62.7
12.600	58.1	77.7	85.7	85.7	74.4	67.8	64.4	62.0	59.9
12.800	48.9	59.1	85.7	74.9	65.2	61.8	59.9	58.5	57.2
13.000	39.6	40.5	85.8	56.3	55.9	55.6	55.3	55.0	54.5
13.200	30.3	21.9	7.7	37.4	46.2	49.4	50.8	51.4	51.7
13.400	21.0	7.7	7.7	19.1	37.0	43.4	46.2	47.9	48.8
13.600	11.8	7.7	7.8	7.9	27.6	36.8	41.7	44.3	45.9
13.800	7.7	7.7	7.8	7.9	18.4	30.6	37.1	40.7	43.0
14.000	7.7	7.8	7.9	7.9	10.0	24.4	32.6	37.1	40.1
14.200	7.7	7.8	7.9	8.0	8.0	18.3	27.5	33.4	37.0
14.400	7.8	7.9	7.9	8.0	8.1	12.2	22.9	29.8	34.0
14.600	7.8	7.9	8.0	8.0	8.1	8.2	18.3	25.8	31.0
14.800	7.9	7.9	8.0	8.1	8.2	8.2	13.7	22.1	28.0
15.000	7.9	8.0	8.0	8.1	8.2	8.3	9.9	18.4	25.0
15.200	7.9	8.0	8.1	8.2	8.2	8.3	8.4	14.7	21.6
15.400	8.0	8.0	8.1	8.2	8.3	8.4	8.5	11.4	18.5
15.600	8.0	8.1	8.2	8.2	8.3	8.4	8.5	8.8	15.5
15.800	8.0	8.1	8.2	8.3	8.4	8.5	8.5	8.6	12.5
16.000	8.1	8.2	8.2	8.3	8.4	8.5	8.6	8.7	10.2
16.200	8.1	8.2	8.3	8.4	8.5	8.5	8.6	8.7	8.8
16.400	8.2	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9
16.600	8.2	8.3	8.4	8.5	8.5	8.6	8.7	8.8	8.9
16.800	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	9.0
17.000	8.3	8.4	8.5	8.5	8.6	8.7	8.8	8.9	9.0
17.200	8.3	8.4	8.5	8.6	8.7	8.8	8.9	9.0	9.1
17.400	8.4	8.4	8.5	8.6	8.7	8.8	8.9	9.0	9.1
17.600	8.4	8.5	8.6	8.7	8.8	8.9	9.0	9.1	9.2
17.800	8.4	8.5	8.6	8.7	8.8	8.9	9.0	9.1	9.2
18.000	8.5	8.6	8.7	8.8	8.9	9.0	9.1	9.2	9.3
18.200	8.5	8.6	8.7	8.8	8.9	9.0	9.1	9.2	9.3
18.400	8.6	8.7	8.8	8.9	9.0	9.1	9.2	9.3	9.4
18.600	8.6	8.7	8.8	8.9	9.0	9.1	9.2	9.3	9.4
18.800	8.7	8.8	8.9	9.0	9.1	9.2	9.3	9.4	9.5
19.000	8.7	8.8	8.9	9.0	9.1	9.2	9.3	9.4	9.6
19.200	8.8	8.9	9.0	9.1	9.2	9.3	9.4	9.5	9.6
19.400	8.8	8.9	9.0	9.1	9.2	9.3	9.4	9.6	9.7
19.600	8.9	9.0	9.1	9.2	9.3	9.4	9.5	9.6	9.7
19.800	8.9	9.0	9.1	9.2	9.3	9.4	9.5	9.7	9.8
20.000	9.0	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8
20.200	9.0	9.1	9.2	9.3	9.4	9.5	9.7	9.8	9.9
20.400	9.0	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9
20.600	9.1	9.2	9.3	9.4	9.5	9.6	9.8	9.9	10.0
20.800	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10.0
21.000	9.2	9.3	9.4	9.5	9.6	9.8	9.9	10.0	10.1
21.200	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10.0	10.2
21.400	9.3	9.4	9.5	9.6	9.7	9.9	10.0	10.1	10.2
21.600	9.4	9.5	9.6	9.7	9.8	9.9	10.0	10.1	10.3
21.800	9.4	9.5	9.6	9.7	9.9	10.0	10.1	10.2	10.3
22.000	9.5	9.6	9.7	9.8	9.9	10.0	10.1	10.3	10.4
22.200	9.5	9.6	9.7	9.9	10.0	10.1	10.2	10.3	10.4
22.400	9.6	9.7	9.8	9.9	10.0	10.1	10.3	10.4	10.5
22.600	9.6	9.7	9.8	10.0	10.1	10.2	10.3	10.4	10.5
22.800	9.7	9.8	9.9	10.0	10.1	10.2	10.4	10.5	10.6
23.000	9.7	9.8	10.0	10.1	10.2	10.3	10.4	10.5	10.7

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	9.8	9.9	10.0	10.1	10.2	10.4	10.5	10.6	10.7
23.400	9.8	10.0	10.1	10.2	10.3	10.4	10.5	10.7	10.8
23.600	9.9	10.0	10.1	10.2	10.4	10.5	10.6	10.7	10.8
23.800	9.9	10.1	10.2	10.3	10.4	10.5	10.6	10.8	10.9
24.000	10.0	10.1	10.2	10.3	10.5	10.6	10.7	10.8	11.0
24.200	10.1	10.2	10.3	10.4	10.5	10.6	10.8	10.9	11.0
24.400	10.1	10.2	10.3	10.5	10.6	10.7	10.8	10.9	11.1
24.600	10.2	10.3	10.4	10.5	10.6	10.8	10.9	11.0	11.1
24.800	10.2	10.3	10.5	10.6	10.7	10.8	10.9	11.1	11.2
25.000	10.3	10.4	10.5	10.6	10.8	10.9	11.0	11.1	11.2
25.200	10.3	10.4	10.6	10.7	10.8	10.9	11.1	11.2	11.3
25.400	10.4	10.5	10.6	10.7	10.9	11.0	11.1	11.2	11.4
25.600	10.4	10.6	10.7	10.8	10.9	11.0	11.2	11.3	11.4
25.800	10.5	10.6	10.7	10.9	11.0	11.1	11.2	11.3	11.5
26.000	10.6	10.7	10.8	10.9	11.0	11.2	11.3	11.4	11.5
26.200	10.6	10.7	10.9	11.0	11.1	11.2	11.3	11.5	11.6
26.400	10.7	10.8	10.9	11.0	11.2	11.3	11.4	11.5	11.6
26.600	10.7	10.8	11.0	11.1	11.2	11.3	11.5	11.6	11.7
26.800	10.8	10.9	11.0	11.1	11.3	11.4	11.5	11.6	11.8
27.000	10.8	11.0	11.1	11.2	11.3	11.4	11.6	11.7	11.8
27.200	10.9	11.0	11.1	11.3	11.4	11.5	11.6	11.7	11.9
27.400	11.0	11.1	11.2	11.3	11.4	11.6	11.7	11.8	11.9
27.600	11.0	11.1	11.3	11.4	11.5	11.6	11.7	11.9	11.8
27.800	11.1	11.2	11.3	11.4	11.6	11.7	11.8	11.9	9.5
28.000	11.1	11.2	11.4	11.5	11.6	11.7	11.9	12.0	6.6
28.200	11.2	11.3	11.4	11.5	11.7	11.8	11.9	10.1	3.6
28.400	11.2	11.4	11.5	11.6	11.7	11.8	12.0	6.6	1.5
28.600	11.3	11.4	11.5	11.7	11.8	11.9	11.0	2.9	1.1
28.800	11.4	11.5	11.6	11.7	11.8	12.0	6.5	1.2	1.1
29.000	11.4	11.5	11.7	11.8	11.9	11.8	2.2	1.1	1.1
29.200	11.5	11.6	11.7	11.8	11.9	6.4	1.1	1.1	1.1
29.400	11.3	11.6	11.8	11.9	12.0	1.6	1.1	1.1	1.1
29.600	10.4	11.7	11.8	11.9	6.3	1.1	1.1	1.1	1.1
29.800	10.5	11.8	11.9	12.0	1.1	1.1	1.1	1.1	1.1
30.000	10.6	10.7	11.9	6.6	1.1	1.1	1.1	1.1	1.1
30.200	10.6	10.7	11.6	1.1	1.1	1.1	1.1	1.1	1.1
30.400	10.7	10.8	6.5	1.1	1.1	1.1	1.1	1.1	1.1
30.600	10.7	9.8	0.0	1.1	1.1	1.1	1.1	1.1	1.1
30.800	10.8	5.2	0.0	1.1	1.1	1.1	1.1	1.1	1.1
31.000	9.0	0.6	0.0	0.0	1.1	1.1	1.1	1.1	1.1
31.200	5.4	0.0	0.0	0.0	1.1	1.1	1.1	1.1	1.1
31.400	1.6	0.0	0.0	0.0	0.2	1.1	1.1	1.1	1.1
31.600	0.0	0.0	0.0	0.0	0.0	1.1	1.1	1.1	1.1
31.800	0.0	0.0	0.0	0.0	0.0	1.1	1.1	1.1	1.1
32.000	0.0	0.0	0.0	0.0	0.0	0.0	1.1	1.1	1.1

----- volume moulded m3 -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	744.9	350.9	0.0	461.2	959.6	1483.8	2029.3	2593.3	3173.6
0.200	1368.3	1036.2	1020.6	1200.0	1629.2	2138.5	2681.4	3247.2	3829.5
0.400	2182.8	2030.7	2092.4	2225.4	2485.0	2919.5	3429.6	3978.6	4547.9
0.600	3169.5	3133.7	3197.2	3318.3	3512.2	3826.9	4273.7	4786.5	5329.1
0.800	4268.3	4267.9	4329.3	4442.3	4611.4	4855.9	5213.6	5669.3	6173.9
1.000	5421.7	5426.8	5484.4	5590.9	5744.3	5960.5	6246.2	6626.6	7082.9
1.200	6601.3	6607.2	6660.9	6760.6	6903.6	7098.0	7349.0	7658.3	8056.0
1.400	7802.5	7806.2	7856.1	7948.6	8083.4	8259.9	8484.5	8754.6	9092.8
1.600	9022.0	9022.7	9068.4	9154.5	9280.6	9442.8	9642.6	9889.8	10187.9
1.800	10258.4	10255.9	10295.6	10376.0	10493.6	10641.3	10820.8	11049.2	11325.8
2.000	11509.8	11503.6	11537.9	11611.9	11719.6	11853.0	12018.0	12228.6	12489.6
2.200	12776.1	12765.4	12794.1	12860.5	12956.7	13078.3	13232.1	13427.6	13676.5
2.400	14059.8	14039.6	14063.0	14120.6	14204.5	14316.3	14460.7	14644.4	14883.1
2.600	15359.8	15326.7	15344.5	15391.0	15462.9	15566.5	15703.7	15879.9	16106.6
2.800	16674.1	16630.0	16635.3	16670.9	16733.4	16828.7	16959.3	17132.2	17345.3
3.000	17999.6	17948.5	17936.1	17961.1	18015.1	18103.0	18229.6	18397.6	18597.8
3.200	19333.0	19279.1	19248.8	19260.8	19308.1	19389.6	19513.5	19674.3	19864.0
3.400	20676.8	20618.4	20575.2	20571.4	20612.8	20689.6	20809.2	20961.8	21142.3
3.600	22030.4	21962.1	21913.6	21895.3	21927.1	22003.4	22116.2	22258.8	22431.1
3.800	23393.1	23313.5	23261.6	23234.6	23255.1	23328.4	23433.0	23566.7	23730.1
4.000	24764.1	24674.0	24616.9	24586.5	24597.4	24662.3	24759.8	24884.9	25038.2
4.200	26143.3	26044.2	25980.8	25950.9	25956.1	26006.9	26095.8	26212.4	26358.0
4.400	27532.8	27425.0	27354.2	27324.7	27327.0	27361.8	27440.2	27550.4	27688.8
4.600	28933.6	28815.4	28739.3	28708.2	28707.4	28730.4	28795.4	28898.6	29028.6
4.800	30344.3	30216.6	30136.9	30100.2	30094.9	30111.0	30162.3	30255.0	30377.4
5.000	31766.0	31631.1	31545.6	31501.2	31489.2	31503.2	31541.7	31621.4	31734.3
5.200	33197.5	33057.4	32964.5	32911.8	32892.6	32902.4	32933.4	32996.8	33099.3
5.400	34640.3	34496.4	34394.6	34332.2	34305.0	34308.8	34333.8	34383.6	34474.0
5.600	36097.5	35945.4	35835.2	35762.2	35726.5	35721.0	35741.3	35782.8	35857.3
5.800	37566.8	37404.2	37286.5	37203.0	37155.9	37141.2	37156.3	37190.8	37250.7
6.000	39048.2	38876.2	38746.3	38652.7	38593.4	38570.3	38576.0	38605.6	38656.1
6.200	40540.4	40357.7	40214.1	40111.4	40041.5	40007.0	40003.4	40028.1	40069.9
6.400	42043.4	41848.7	41693.1	41578.4	41498.3	41451.1	41439.4	41454.8	41490.4
6.600	43559.1	43348.3	43181.5	43052.9	42963.4	42905.5	42882.5	42887.9	42918.6
6.800	45084.4	44857.6	44678.8	44537.8	44436.9	44368.8	44332.5	44329.1	44351.1
7.000	46619.1	46379.3	46184.6	46032.3	45917.2	45839.7	45792.3	45777.2	45789.0
7.200	48162.4	47910.2	47700.1	47535.1	47407.5	47318.9	47260.3	47231.7	47234.5
7.400	49717.7	49450.4	49226.7	49047.4	48907.4	48804.6	48735.7	48695.5	48686.6
7.600	51282.9	50998.5	50762.7	50567.8	50415.3	50298.6	50219.0	50167.5	50144.5
7.800	52858.1	52557.4	52307.4	52098.5	51933.1	51802.4	51708.7	51646.3	51611.2
8.000	54441.9	54126.6	53860.2	53638.9	53457.8	53314.8	53205.6	53132.7	53085.6
8.200	56037.8	55705.4	55421.9	55187.4	54990.9	54835.7	54712.5	54625.3	54566.3
8.400	57643.5	57292.8	56994.6	56744.2	56534.2	56363.6	56228.2	56123.9	56053.7
8.600	59259.1	58890.1	58576.2	58307.9	58085.1	57898.5	57751.2	57633.0	57546.5
8.800	60885.4	60498.6	60166.8	59882.2	59644.2	59443.4	59281.0	59149.6	59045.7
9.000	62521.0	62116.2	61765.0	61465.5	61210.0	60996.1	60816.7	60672.2	60555.4
9.200	64164.0	63743.4	63374.4	63057.4	62784.5	62556.2	62361.4	62201.6	62073.5
9.400	65813.0	65380.5	64992.7	64656.9	64368.7	64123.2	63913.8	63737.3	63598.6
9.600	67466.9	67025.5	66619.8	66265.4	65960.7	65696.5	65472.3	65281.6	65131.3
9.800	69125.1	68676.7	68256.6	67883.9	67560.6	67279.4	67038.6	66835.3	66670.4
10.000	70787.1	70332.9	69902.1	69510.4	69167.1	68869.4	68611.1	68396.3	68216.5
10.200	72452.5	71993.5	71554.3	71145.4	70783.7	70467.3	70193.4	69965.0	69772.6
10.400	74120.9	73657.8	73211.6	72789.4	72407.6	72072.5	71784.8	71540.3	71336.1
10.600	75792.0	75325.4	74873.3	74440.7	74039.2	73686.6	73384.0	73123.2	72906.6
10.800	77465.6	76995.9	76538.6	76097.0	75679.9	75310.6	74991.2	74716.1	74483.0
11.000	79140.0	78668.9	78206.9	77757.3	77329.4	76942.7	76605.2	76315.6	76049.8
11.200	80808.6	80344.1	79877.7	79421.7	78986.0	78583.2	78228.9	77921.5	77599.9

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	82477.9	82021.1	81550.3	81089.9	80648.1	80232.3	79860.6	79513.0	79104.5
11.600	84147.4	83694.5	83224.9	82761.6	82314.4	81888.9	81496.3	81068.0	80556.4
11.800	85817.0	85364.0	84902.1	84436.3	83984.1	83551.0	83107.8	82561.5	81955.4
12.000	87482.9	87032.7	86581.5	86112.3	85656.3	85209.2	84652.5	83990.6	83300.4
12.200	89053.0	88701.2	88249.7	87789.9	87331.5	86810.2	86114.5	85354.9	84594.3
12.400	90442.4	90369.0	89919.1	89468.5	88988.7	88298.7	87491.2	86654.7	85835.2
12.600	91650.2	91986.6	91589.2	91144.3	90512.0	89668.5	88780.0	87889.9	87024.0
12.800	92675.9	93287.0	93259.8	92728.3	91856.2	90920.0	89982.2	89057.6	88159.3
13.000	93521.4	94224.6	94930.7	93974.5	93020.4	92052.9	91096.5	90156.9	89241.6
13.200	94185.4	94799.1	95080.4	94857.5	93996.1	93065.5	92122.3	91187.2	90269.9
13.400	94668.7	95035.9	95230.7	95377.7	94790.6	93958.5	93059.3	92148.0	91242.7
13.600	94972.5	95185.6	95381.8	95580.8	95403.7	94730.7	93907.8	93039.1	92158.9
13.800	95141.2	95336.0	95533.6	95734.1	95836.2	95376.3	94667.8	93861.2	93019.0
14.000	95290.9	95487.1	95686.2	95888.0	96090.0	95901.0	95338.4	94612.3	93822.6
14.200	95441.3	95639.0	95839.5	96042.8	96249.0	96306.5	95913.2	95292.8	94567.8
14.400	95592.5	95791.6	95993.4	96198.2	96406.0	96591.3	96395.9	95902.0	95254.3
14.600	95744.4	95944.9	96148.2	96354.4	96563.7	96776.1	96789.5	96439.1	95882.9
14.800	95897.0	96098.9	96303.6	96511.4	96722.2	96936.2	97092.7	96897.8	96452.5
15.000	96050.3	96253.6	96459.8	96669.1	96881.5	97097.1	97311.3	97285.8	96963.3
15.200	96204.3	96409.1	96616.8	96827.6	97041.6	97258.9	97479.6	97601.8	97411.5
15.400	96359.1	96565.3	96774.5	96986.9	97202.5	97421.4	97644.0	97847.0	97796.8
15.600	96514.6	96722.3	96933.0	97146.9	97364.2	97584.8	97809.1	98037.3	98122.4
15.800	96670.8	96880.0	97092.3	97307.8	97526.7	97749.1	97975.2	98205.2	98388.3
16.000	96827.8	97038.5	97252.4	97469.5	97690.1	97914.3	98142.2	98374.1	98603.2
16.200	96985.6	97197.8	97413.2	97632.0	97854.3	98080.3	98310.1	98544.0	98782.1
16.400	97144.0	97357.9	97574.9	97795.4	98019.4	98247.2	98478.9	98714.8	98955.0
16.600	97303.3	97518.7	97737.4	97959.6	98185.4	98415.0	98648.7	98886.6	99128.9
16.800	97463.4	97680.3	97900.7	98124.6	98352.3	98583.8	98819.4	99059.4	99303.8
17.000	97624.2	97842.8	98064.9	98290.6	98520.0	98753.5	98991.2	99233.2	99479.9
17.200	97785.8	98006.1	98229.9	98457.4	98688.7	98924.2	99163.9	99408.1	99657.0
17.400	97948.3	98170.3	98395.8	98625.1	98858.4	99095.8	99337.6	99584.0	99835.2
17.600	98111.5	98335.2	98562.6	98793.8	99029.0	99268.5	99512.4	99761.0	100014.5
17.800	98275.6	98501.1	98730.2	98963.3	99200.5	99442.1	99688.2	99939.1	100195.7
18.000	98440.6	98667.8	98898.8	99133.8	99373.1	99616.8	99865.1	100118.2	100377.9
18.200	98606.4	98835.4	99068.3	99305.4	99546.7	99792.5	100043.0	100298.9	100561.2
18.400	98773.1	99003.9	99238.8	99477.8	99721.2	99969.2	100222.1	100481.1	100745.4
18.600	98940.7	99173.4	99410.2	99651.3	99896.8	100147.1	100402.2	100664.3	100930.7
18.800	99109.1	99343.8	99582.6	99825.7	100073.5	100326.0	100584.3	100848.5	101116.9
19.000	99278.5	99515.1	99756.0	100001.2	100251.2	100506.0	100767.5	101033.7	101304.2
19.200	99448.8	99687.4	99930.3	100177.8	100430.0	100687.6	100951.6	101219.8	101492.5
19.400	99620.1	99860.7	100105.8	100355.4	100609.9	100870.7	101136.7	101407.0	101681.8
19.600	99792.3	100035.0	100282.2	100534.1	100791.0	101054.7	101322.8	101595.3	101872.2
19.800	99965.6	100210.3	100459.7	100713.9	100974.0	101239.7	101509.9	101784.5	102063.5
20.000	100139.8	100386.6	100638.3	100894.8	101157.9	101425.8	101698.1	101974.8	102255.9
20.200	100315.0	100564.1	100817.9	101077.3	101342.9	101612.9	101887.2	102166.0	102449.4
20.400	100491.2	100742.5	100998.7	101261.2	101528.8	101800.9	102077.4	102358.4	102643.8
20.600	100668.5	100922.1	101180.7	101446.1	101715.9	101990.0	102268.6	102551.7	102839.4
20.800	100846.8	101102.7	101364.5	101632.0	101903.8	102180.1	102460.9	102746.1	103035.9
21.000	101026.3	101284.4	101549.4	101818.9	102092.9	102371.3	102654.1	102941.6	103233.6
21.200	101206.8	101467.9	101735.2	102006.8	102282.9	102563.4	102848.4	103138.0	103432.3
21.400	101388.4	101652.7	101922.0	102195.8	102473.9	102756.6	103043.8	103335.6	103632.0
21.600	101571.3	101838.4	102109.8	102385.7	102666.0	102950.8	103240.2	103534.2	103832.8
21.800	101756.0	102025.2	102298.7	102576.7	102859.1	103146.1	103437.7	103733.8	104034.7
22.000	101941.7	102212.9	102488.6	102768.7	103053.3	103342.4	103636.2	103934.6	104237.6
22.200	102128.4	102401.7	102679.5	102961.7	103248.5	103539.8	103835.8	104136.4	104441.7
22.400	102316.1	102591.5	102871.4	103155.8	103444.7	103738.2	104036.4	104339.2	104646.8
22.600	102504.8	102782.3	103064.4	103350.9	103642.0	103937.7	104238.1	104543.2	104853.0
22.800	102694.5	102974.2	103258.4	103547.1	103840.4	104138.3	104440.9	104748.2	105060.3
23.000	102885.3	103167.1	103453.4	103744.3	104039.8	104339.9	104644.7	104954.3	105268.7

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	103077.0	103361.0	103649.5	103942.5	104240.2	104542.6	104849.7	105161.5	105478.2
23.400	103269.8	103555.9	103846.6	104141.8	104441.7	104746.4	105055.7	105369.8	105688.8
23.600	103463.7	103751.9	104044.7	104342.2	104644.3	104951.2	105262.8	105579.2	105900.4
23.800	103658.5	103949.0	104244.0	104543.6	104848.0	105157.1	105471.0	105789.7	106113.2
24.000	103854.4	104147.0	104444.3	104746.1	105052.8	105364.1	105680.3	106001.3	106327.2
24.200	104051.4	104346.2	104645.6	104949.7	105258.6	105572.2	105890.6	106214.0	106542.2
24.400	104249.4	104546.4	104848.0	105154.4	105465.5	105781.4	106102.1	106427.8	106758.3
24.600	104448.4	104747.6	105051.5	105360.1	105673.5	105991.7	106314.7	106642.7	106975.5
24.800	104648.5	104950.0	105256.1	105567.0	105882.6	106203.1	106528.4	106858.7	107193.8
25.000	104849.7	105153.4	105461.7	105774.9	106092.8	106415.6	106743.2	107075.8	107413.3
25.200	105052.0	105357.8	105668.5	105983.9	106304.1	106629.2	106959.1	107294.0	107633.8
25.400	105255.3	105563.4	105876.3	106194.0	106516.5	106843.9	107176.1	107513.3	107855.4
25.600	105459.6	105770.0	106085.2	106405.1	106730.0	107059.7	107394.3	107733.8	108078.2
25.800	105665.1	105977.7	106295.2	106617.4	106944.5	107276.5	107613.5	107955.3	108302.0
26.000	105871.6	106186.6	106506.3	106830.8	107160.2	107494.5	107833.8	108177.9	108527.0
26.200	106079.3	106396.4	106718.4	107045.3	107377.0	107713.6	108055.2	108401.6	108753.0
26.400	106288.0	106607.4	106931.7	107260.9	107594.9	107933.8	108277.7	108626.5	108980.2
26.600	106497.8	106819.5	107146.1	107477.5	107813.9	108155.1	108501.3	108852.4	109208.4
26.800	106708.6	107032.7	107361.5	107695.3	108034.0	108377.5	108726.0	109079.4	109437.8
27.000	106920.6	107246.9	107578.1	107914.2	108255.1	108601.0	108951.8	109307.5	109668.2
27.200	107133.7	107462.3	107795.8	108134.1	108477.4	108825.6	109178.7	109536.7	109899.7
27.400	107347.8	107678.8	108014.5	108355.2	108700.8	109051.3	109406.7	109767.0	110132.3
27.600	107563.1	107896.3	108234.4	108577.3	108925.2	109278.0	109635.8	109998.4	110365.2
27.800	107779.4	108114.9	108455.3	108800.6	109150.8	109505.9	109865.9	110230.9	110563.3
28.000	107996.9	108334.6	108677.3	109024.9	109377.4	109734.8	110097.2	110464.4	110716.7
28.200	108215.4	108555.4	108900.4	109250.3	109605.1	109964.9	110329.5	110676.5	110811.2
28.400	108435.0	108777.4	109124.7	109476.8	109833.9	110196.0	110562.9	110833.7	110852.8
28.600	108655.7	109000.4	109350.0	109704.4	110063.8	110428.2	110785.4	110917.5	110875.9
28.800	108877.5	109224.5	109576.3	109933.1	110294.8	110661.5	110948.8	110950.8	110898.0
29.000	109100.3	109449.6	109803.8	110162.9	110526.9	110895.1	111022.9	110973.7	110920.1
29.200	109324.3	109675.9	110032.3	110393.7	110760.0	111061.6	111049.5	110995.8	110942.1
29.400	109549.2	109903.2	110262.0	110625.7	110994.3	111124.0	111071.5	111017.9	110964.2
29.600	109756.5	110131.6	110492.7	110858.7	111169.5	111147.3	111093.6	111040.0	110986.3
29.800	109961.6	110361.1	110724.5	111092.8	111223.0	111169.4	111115.7	111062.0	111008.4
30.000	110167.8	110582.2	110957.4	111274.8	111245.1	111191.5	111137.8	111084.1	111030.5
30.200	110375.1	110791.8	111190.0	111320.8	111267.2	111213.5	111159.9	111106.2	111052.5
30.400	110583.4	111002.4	111364.3	111342.9	111289.3	111235.6	111182.0	111128.3	111074.6
30.600	110792.9	111207.7	111396.6	111365.0	111311.3	111257.7	111204.0	111150.4	111096.7
30.800	111003.4	111349.8	111396.6	111387.1	111333.4	111279.8	111226.1	111172.5	111118.8
31.000	111199.4	111401.6	111396.6	111396.6	111355.5	111301.8	111248.2	111194.5	111140.9
31.200	111333.3	111396.6	111396.6	111396.6	111377.6	111323.9	111270.3	111216.6	111163.0
31.400	111396.1	111396.6	111396.6	111396.6	111396.6	111346.0	111292.3	111238.7	111185.0
31.600	111396.6	111396.6	111396.6	111396.6	111396.6	111368.1	111314.4	111260.8	111207.1
31.800	111396.6	111396.6	111396.6	111396.6	111396.6	111390.1	111336.5	111282.8	111229.2
32.000	111396.6	111396.6	111396.6	111396.6	111396.6	111396.6	111358.6	111304.9	111251.3

----- total volume m3 -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	794.5	398.1	88.1	513.2	1012.9	1538.5	2085.4	2650.8	3232.3
0.200	1435.6	1116.8	1116.2	1284.9	1699.5	2204.8	2746.5	3311.9	3894.5
0.400	2267.6	2129.9	2191.9	2322.9	2572.4	2997.6	3503.7	4050.8	4619.2
0.600	3267.9	3237.1	3300.1	3420.1	3611.1	3916.8	4356.8	4866.1	5406.6
0.800	4374.3	4374.5	4435.2	4547.3	4714.4	4956.0	5305.9	5756.3	6257.9
1.000	5531.4	5536.3	5593.1	5698.8	5850.9	6064.9	6347.0	6721.2	7173.2
1.200	6713.9	6719.3	6772.3	6871.1	7013.1	7205.7	7454.9	7760.0	8152.8
1.400	7917.8	7920.8	7970.1	8061.8	8195.6	8370.7	8593.5	8861.7	9195.8
1.600	9139.8	9139.9	9184.9	9270.1	9395.2	9556.4	9754.5	10000.2	10296.2
1.800	10378.7	10375.5	10414.4	10494.0	10610.8	10757.4	10935.5	11162.5	11437.6
2.000	11632.3	11625.6	11659.1	11732.2	11839.1	11971.5	12135.5	12344.6	12604.2
2.200	12901.0	12889.6	12917.6	12983.1	13078.6	13199.2	13352.0	13546.2	13793.8
2.400	14187.2	14166.0	14188.8	14245.6	14328.6	14439.5	14583.0	14765.6	15002.9
2.600	15489.3	15455.6	15472.6	15518.3	15589.3	15692.0	15828.3	16003.4	16228.9
2.800	16806.0	16761.1	16765.5	16800.4	16862.0	16956.4	17086.2	17258.1	17470.1
3.000	18133.6	18081.8	18068.6	18092.8	18145.9	18232.9	18358.6	18525.8	18724.9
3.200	19469.3	19414.6	19383.6	19394.5	19441.1	19521.8	19644.7	19804.7	19993.5
3.400	20815.4	20756.0	20712.1	20707.4	20747.9	20823.9	20942.7	21094.3	21273.9
3.600	22171.2	22101.9	22052.6	22033.5	22064.3	22139.8	22251.8	22393.5	22565.0
3.800	23536.1	23455.7	23402.8	23374.9	23394.6	23466.9	23570.8	23703.5	23866.0
4.000	24909.3	24818.3	24760.2	24729.0	24739.0	24802.9	24899.7	25023.9	25176.3
4.200	26290.5	26190.7	26126.4	26095.4	26099.8	26149.8	26237.7	26353.5	26498.3
4.400	27682.4	27573.5	27501.8	27471.3	27472.9	27506.8	27584.2	27693.6	27831.2
4.600	29085.3	28965.9	28889.1	28857.1	28855.2	28877.4	28941.6	29043.8	29173.0
4.800	30498.2	30369.5	30288.7	30251.1	30244.9	30260.1	30310.5	30402.3	30523.9
5.000	31922.1	31786.2	31699.4	31654.3	31641.3	31654.3	31692.1	31770.9	31882.8
5.200	33355.5	33214.6	33120.7	33066.9	33046.8	33055.6	33085.8	33148.4	33249.9
5.400	34800.7	34655.8	34553.0	34489.3	34461.3	34464.2	34488.2	34537.2	34626.7
5.600	36260.1	36106.7	35995.6	35921.5	35884.9	35878.4	35897.7	35938.4	36012.0
5.800	37731.6	37567.8	37449.0	37364.5	37316.2	37300.8	37315.0	37348.3	37407.5
6.000	39215.1	39042.0	38910.8	38816.3	38755.9	38731.9	38736.6	38765.2	38814.9
6.200	40709.3	40525.6	40380.8	40277.0	40206.2	40170.4	40166.1	40189.8	40230.6
6.400	42214.7	42018.8	41862.0	41746.1	41665.0	41616.8	41604.1	41618.6	41653.2
6.600	43732.6	43520.3	43352.4	43222.7	43132.2	43073.3	43049.1	43053.8	43083.5
6.800	45260.1	45032.0	44851.9	44709.9	44607.7	44538.6	44501.3	44497.0	44518.1
7.000	46796.8	46555.9	46359.8	46206.4	46090.1	46011.7	45963.2	45947.0	45957.9
7.200	48342.5	48088.9	47877.5	47711.3	47582.5	47492.8	47433.3	47403.6	47405.4
7.400	49900.0	49631.2	49406.4	49225.6	49084.6	48980.4	48910.7	48869.5	48859.4
7.600	51467.4	51181.4	50944.3	50748.0	50594.5	50476.8	50396.0	50343.5	50319.4
7.800	53044.7	52742.7	52491.2	52281.1	52114.3	51982.6	51887.6	51824.3	51788.3
8.000	54630.8	54314.0	54046.0	53823.6	53641.0	53497.1	53386.7	53312.7	53264.8
8.200	56229.0	55895.0	55610.0	55374.3	55176.5	55020.0	54895.7	54807.3	54747.4
8.400	57836.9	57484.4	57184.9	56933.0	56721.9	56549.9	56413.4	56308.1	56236.9
8.600	59454.8	59084.2	58768.7	58498.9	58274.9	58087.0	57938.5	57819.3	57731.7
8.800	61083.4	60694.8	60361.3	60075.5	59836.0	59634.1	59470.3	59337.9	59233.0
9.000	62721.0	62314.6	61961.8	61660.9	61403.7	61188.8	61008.1	60862.7	60744.8
9.200	64366.0	63944.1	63573.5	63255.0	62980.6	62751.0	62555.0	62394.1	62265.0
9.400	66016.8	65583.4	65194.0	64856.4	64567.0	64319.9	64109.5	63931.7	63792.1
9.600	67672.5	67230.5	66823.2	66467.3	66161.1	65895.5	65670.2	65478.3	65326.8
9.800	69332.4	68883.5	68462.3	68087.9	67763.1	67480.7	67238.5	67034.1	66867.7
10.000	70996.2	70541.5	70109.9	69716.6	69371.8	69072.8	68813.1	68597.1	68416.1
10.200	72663.2	72203.8	71764.0	71353.9	70990.6	70672.8	70397.6	70167.8	69974.3
10.400	74333.3	73869.9	73423.2	73000.1	72616.8	72279.9	71991.1	71745.0	71539.8
10.600	76006.1	75539.1	75086.6	74653.3	74250.4	73896.4	73592.4	73330.2	73112.4
10.800	77681.2	77211.3	76753.7	76311.6	75893.5	75522.6	75201.5	74925.2	74690.8
11.000	79358.1	78886.0	78423.7	77973.8	77545.1	77156.8	76817.8	76526.8	76262.0
11.200	81028.3	80562.8	80096.3	79639.9	79203.6	78799.4	78443.7	78135.7	77817.7

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	82699.1	82241.5	81770.6	81309.8	80867.5	80450.8	80077.5	79731.5	79329.8
11.600	84370.2	83917.3	83446.8	82983.2	82535.6	82109.4	81716.5	81294.4	80789.1
11.800	86041.4	85588.4	85125.6	84659.5	84207.0	83773.3	83334.5	82796.2	82195.2
12.000	87711.3	87258.8	86806.7	86337.3	85880.9	85435.5	84889.5	84234.1	83548.0
12.200	89298.6	88928.9	88477.3	88016.4	87557.8	87047.1	86361.4	85607.1	84849.4
12.400	90706.0	90598.3	90148.2	89696.7	89221.7	88548.0	87748.2	86915.5	86097.6
12.600	91931.2	92234.8	91819.9	91374.8	90763.3	89930.3	89046.8	88159.0	87293.7
12.800	92974.3	93569.1	93492.0	92981.5	92125.1	91194.1	90258.6	89334.8	88436.3
13.000	93837.2	94540.5	95164.5	94261.6	93307.0	92339.3	91382.4	90442.1	89525.8
13.200	94518.2	95148.6	95454.2	95178.4	94301.0	93364.0	92417.6	91480.3	90561.1
13.400	95018.8	95409.4	95604.9	95732.2	95112.9	94269.1	93364.0	92448.9	91540.6
13.600	95338.9	95559.4	95756.3	95956.0	95743.4	95053.8	94221.7	93347.7	92463.6
13.800	95514.7	95710.2	95908.5	96109.6	96192.8	95711.2	94990.8	94177.4	93330.3
14.000	95664.7	95861.7	96061.4	96264.0	96464.2	96247.8	95670.7	94936.0	94140.4
14.200	95815.5	96013.9	96215.0	96419.1	96626.0	96664.7	96254.9	95623.8	94892.1
14.400	95967.0	96166.8	96369.4	96574.9	96783.3	96961.0	96746.7	96240.1	95585.1
14.600	96119.3	96320.5	96524.5	96731.4	96941.4	97154.5	97149.1	96785.3	96219.9
14.800	96272.3	96474.8	96680.3	96888.7	97100.2	97315.0	97461.0	97251.5	96795.5
15.000	96425.9	96629.9	96836.8	97046.8	97259.9	97476.2	97688.6	97646.9	97313.0
15.200	96580.3	96785.8	96994.2	97205.6	97420.3	97638.3	97859.8	97970.0	97767.8
15.400	96735.4	96942.4	97152.3	97365.3	97581.6	97801.2	98024.5	98222.2	98159.2
15.600	96891.3	97099.7	97311.1	97525.7	97743.6	97965.0	98190.0	98418.8	98490.9
15.800	97047.9	97257.8	97470.8	97687.0	97906.5	98129.6	98356.4	98587.1	98762.7
16.000	97205.2	97416.6	97631.2	97849.0	98070.2	98295.1	98523.8	98756.4	98983.5
16.200	97363.3	97576.2	97792.4	98011.9	98234.8	98461.5	98692.0	98926.6	99165.4
16.400	97522.2	97736.7	97954.4	98175.6	98400.3	98628.8	98861.2	99097.7	99338.6
16.600	97681.8	97897.9	98117.2	98340.1	98566.6	98797.0	99031.3	99269.9	99512.9
16.800	97842.2	98059.9	98280.9	98505.5	98733.9	98966.1	99202.4	99443.1	99688.2
17.000	98003.4	98222.7	98445.4	98671.8	98902.0	99136.2	99374.5	99617.3	99864.7
17.200	98165.4	98386.3	98610.8	98839.0	99071.1	99307.2	99547.6	99792.5	100042.1
17.400	98328.2	98550.8	98777.1	99007.1	99241.0	99479.2	99721.7	99968.8	100220.7
17.600	98491.8	98716.2	98944.2	99176.1	99412.0	99652.2	99896.8	100146.1	100400.4
17.800	98656.3	98882.4	99112.2	99346.0	99583.9	99826.2	100073.0	100324.6	100582.0
18.000	98821.6	99049.5	99281.2	99516.9	99756.8	100001.2	100250.2	100504.1	100764.7
18.200	98987.7	99217.4	99451.0	99688.7	99930.8	100177.3	100428.6	100685.3	100948.3
18.400	99154.8	99386.3	99621.8	99861.6	100105.7	100354.4	100608.0	100867.9	101132.9
18.600	99322.7	99556.1	99793.6	100035.4	100281.7	100532.6	100788.6	101051.4	101318.5
18.800	99491.5	99726.9	99966.4	100210.2	100458.7	100711.9	100971.1	101236.0	101505.2
19.000	99661.3	99898.6	100140.1	100386.1	100636.8	100892.3	101154.6	101421.5	101692.8
19.200	99831.9	100071.2	100314.9	100563.0	100815.9	101074.4	101339.1	101608.1	101881.5
19.400	100003.6	100244.9	100490.6	100741.0	100996.2	101257.8	101524.6	101795.7	102071.2
19.600	100176.2	100419.5	100667.4	100920.0	101177.8	101442.2	101711.0	101984.3	102261.9
19.800	100349.7	100595.2	100845.3	101100.2	101361.1	101627.6	101898.6	102173.9	102453.7
20.000	100524.3	100771.9	101024.2	101281.5	101545.5	101814.1	102087.1	102364.5	102646.4
20.200	100699.9	100949.7	101204.3	101464.5	101730.8	102001.5	102276.6	102556.2	102840.3
20.400	100876.5	101128.5	101385.4	101648.7	101917.1	102190.0	102467.2	102748.9	103035.1
20.600	101054.1	101308.4	101567.9	101834.0	102104.5	102379.4	102658.8	102942.6	103231.0
20.800	101232.8	101489.4	101752.1	102020.3	102292.9	102569.9	102851.4	103137.4	103428.0
21.000	101412.6	101671.5	101937.3	102207.6	102482.3	102761.4	103045.0	103333.2	103626.0
21.200	101593.5	101855.5	102123.5	102395.9	102672.7	102954.0	103239.7	103530.1	103825.1
21.400	101775.5	102040.6	102310.7	102585.2	102864.1	103147.5	103435.5	103728.0	104025.2
21.600	101958.9	102226.7	102498.9	102775.5	103056.6	103342.1	103632.3	103927.0	104226.4
21.800	102144.0	102413.9	102688.2	102966.9	103250.1	103537.8	103830.1	104127.0	104428.6
22.000	102330.0	102602.0	102878.4	103159.3	103444.6	103734.5	104029.0	104328.1	104632.0
22.200	102517.1	102791.2	103069.7	103352.7	103640.2	103932.3	104229.0	104530.3	104836.4
22.400	102705.2	102981.4	103262.0	103547.1	103836.8	104131.1	104430.0	104733.6	105041.9
22.600	102894.3	103172.5	103455.3	103742.6	104034.5	104330.9	104632.1	104937.9	105248.5
22.800	103084.4	103364.8	103649.7	103939.1	104233.2	104531.9	104835.2	105143.3	105456.1
23.000	103275.5	103558.0	103845.1	104136.7	104433.0	104733.9	105039.4	105349.8	105664.9

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	103467.6	103752.3	104041.5	104335.4	104633.8	104936.9	105244.8	105557.4	105874.8
23.400	103660.8	103947.6	104239.0	104535.1	104835.7	105141.1	105451.2	105766.0	106085.7
23.600	103855.0	104144.0	104437.6	104735.8	105038.7	105346.3	105658.7	105975.8	106297.8
23.800	104050.3	104341.4	104637.2	104937.6	105242.8	105552.6	105867.2	106186.7	106511.0
24.000	104246.5	104539.9	104837.9	105140.5	105447.9	105760.0	106076.9	106398.6	106725.3
24.200	104443.9	104739.4	105039.6	105344.5	105654.1	105968.5	106287.7	106611.7	106940.7
24.400	104642.2	104940.0	105242.4	105549.5	105861.4	106178.0	106499.5	106825.9	107157.2
24.600	104841.7	105141.6	105446.3	105755.6	106069.8	106388.7	106712.5	107041.2	107374.8
24.800	105042.2	105344.4	105651.2	105962.8	106279.3	106600.5	106926.6	107257.6	107593.5
25.000	105243.8	105548.1	105857.2	106171.1	106489.8	106813.4	107141.8	107475.1	107813.3
25.200	105446.4	105753.0	106064.4	106380.5	106701.5	107027.3	107358.1	107693.7	108034.2
25.400	105650.0	105958.9	106272.5	106591.0	106914.3	107242.4	107575.4	107913.4	108256.2
25.600	105854.8	106165.9	106481.8	106802.6	107128.1	107458.6	107793.9	108134.2	108479.4
25.800	106060.6	106374.0	106692.2	107015.2	107343.1	107675.9	108013.5	108356.1	108703.6
26.000	106267.5	106583.2	106903.7	107229.0	107559.2	107894.2	108234.2	108579.1	108929.0
26.200	106475.6	106793.5	107116.2	107443.9	107776.4	108113.7	108456.0	108803.2	109155.4
26.400	106684.7	107004.9	107329.9	107659.8	107994.6	108334.3	108678.9	109028.4	109382.9
26.600	106894.8	107217.3	107544.7	107876.9	108214.0	108556.0	108902.9	109254.7	109611.5
26.800	107106.1	107430.9	107760.5	108095.0	108434.4	108778.7	109128.0	109482.1	109841.3
27.000	107318.5	107645.5	107977.4	108314.3	108656.0	109002.6	109354.2	109710.6	110072.1
27.200	107531.9	107861.3	108195.5	108534.6	108878.6	109227.6	109581.5	109940.2	110304.0
27.400	107746.4	108078.1	108414.6	108756.0	109102.4	109453.6	109809.8	110170.9	110537.0
27.600	107962.1	108296.0	108634.9	108978.6	109327.2	109680.8	110039.3	110402.7	110770.8
27.800	108178.8	108515.0	108856.2	109202.2	109553.1	109909.0	110269.8	110635.5	110973.6
28.000	108396.6	108735.1	109078.6	109426.9	109780.2	110138.3	110501.5	110869.5	111132.6
28.200	108615.5	108956.3	109302.1	109652.7	110008.3	110368.8	110734.2	111085.5	111232.1
28.400	108835.5	109178.6	109526.7	109879.6	110237.5	110600.3	110968.0	111249.3	111277.1
28.600	109056.6	109402.0	109752.4	110107.6	110467.8	110832.8	111199.4	111340.9	111300.9
28.800	109278.8	109626.5	109979.1	110336.6	110699.1	111066.5	111363.6	111375.8	111323.0
29.000	109502.0	109852.0	110207.0	110566.8	110931.6	111301.1	111446.1	111398.7	111345.0
29.200	109726.4	110078.7	110435.9	110798.1	111165.1	111477.8	111474.5	111420.8	111367.1
29.400	109951.6	110306.4	110665.9	111030.4	111399.7	111547.1	111496.5	111442.9	111389.2
29.600	110159.3	110535.2	110897.0	111263.8	111583.2	111572.3	111518.6	111465.0	111411.3
29.800	110364.8	110765.1	111129.2	111498.2	111647.7	111594.3	111540.7	111487.0	111433.4
30.000	110571.4	110986.5	111362.5	111690.6	111670.1	111616.4	111562.8	111509.1	111455.5
30.200	110779.0	111196.5	111596.3	111745.8	111692.2	111638.5	111584.9	111531.2	111477.5
30.400	110987.8	111407.6	111780.1	111767.9	111714.3	111660.6	111606.9	111553.3	111499.6
30.600	111197.6	111615.3	111821.6	111790.0	111736.3	111682.7	111629.0	111575.4	111521.7
30.800	111408.5	111765.5	111821.6	111812.1	111758.4	111704.8	111651.1	111597.4	111543.8
31.000	111608.3	111826.3	111821.6	111821.6	111780.5	111726.8	111673.2	111619.5	111565.9
31.200	111748.8	111821.6	111821.6	111821.6	111802.6	111748.9	111695.3	111641.6	111587.9
31.400	111819.0	111821.6	111821.6	111821.6	111821.6	111771.0	111717.3	111663.7	111610.0
31.600	111821.6	111821.6	111821.6	111821.6	111821.6	111793.1	111739.4	111685.8	111632.1
31.800	111821.6	111821.6	111821.6	111821.6	111821.6	111815.1	111761.5	111707.8	111654.2
32.000	111821.6	111821.6	111821.6	111821.6	111821.6	111821.6	111783.6	111729.9	111676.3

----- long. centre of flotation m -----

trim draught	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
0.000	78.10	80.34	115.20	145.08	147.02	148.07	148.82	149.42	149.92
0.200	88.18	100.59	114.38	123.69	136.30	140.94	143.49	145.17	146.17
0.400	97.93	110.79	114.09	117.28	125.79	133.92	138.23	140.97	142.43
0.600	105.06	111.55	113.78	115.99	119.53	126.97	133.01	136.61	138.87
0.800	108.58	111.51	113.52	115.41	118.02	121.36	127.83	132.24	135.35
1.000	109.47	111.45	113.28	114.98	116.88	119.60	123.08	128.00	131.86
1.200	109.54	111.36	113.02	114.66	116.22	118.55	120.54	124.11	128.42
1.400	109.54	111.31	112.79	114.29	115.77	117.46	119.30	121.40	125.22
1.600	109.51	111.08	112.57	113.98	115.43	116.56	118.43	120.19	122.65
1.800	109.45	110.92	112.38	113.68	114.90	115.89	117.48	119.36	121.31
2.000	109.42	110.76	112.10	113.44	114.47	115.43	116.71	118.65	120.62
2.200	109.06	110.60	111.89	113.10	113.92	114.97	116.21	117.98	120.12
2.400	108.69	110.49	111.65	112.65	113.59	114.60	115.82	117.46	119.45
2.600	108.38	110.10	111.45	112.21	113.20	114.28	115.50	117.17	118.80
2.800	108.17	109.65	111.10	111.84	112.84	114.07	115.29	116.76	118.19
3.000	108.22	109.26	110.54	111.56	112.59	113.75	115.21	116.41	117.62
3.200	108.23	108.81	109.93	111.32	112.32	113.56	115.10	116.04	117.07
3.400	108.09	108.62	109.42	110.96	112.15	113.51	114.70	115.74	116.72
3.600	107.74	108.49	109.07	110.47	112.01	113.39	114.39	115.43	116.34
3.800	107.37	108.27	108.96	110.02	111.89	113.16	114.09	115.05	116.05
4.000	107.02	108.01	108.95	109.78	111.59	112.89	113.79	114.74	115.70
4.200	106.78	107.72	108.86	109.89	111.08	112.53	113.56	114.43	115.41
4.400	106.34	107.53	108.71	109.91	110.74	112.14	113.25	114.21	115.11
4.600	106.07	107.41	108.68	109.94	110.56	111.52	112.98	113.98	114.78
4.800	105.81	107.05	108.55	109.69	110.40	111.21	112.70	113.64	114.49
5.000	105.61	106.96	108.48	109.40	110.38	111.04	111.98	113.28	114.25
5.200	105.51	106.83	108.09	109.16	110.21	110.83	111.62	112.99	113.88
5.400	105.23	106.74	107.79	109.03	109.91	110.84	111.38	112.31	113.53
5.600	105.08	106.52	107.62	108.65	109.66	110.58	111.13	111.93	113.22
5.800	104.98	106.25	107.32	108.33	109.43	110.28	111.04	111.63	112.59
6.000	104.80	105.90	107.08	108.07	109.06	110.01	110.89	111.38	112.17
6.200	104.56	105.62	106.75	107.75	108.71	109.76	110.58	111.21	111.86
6.400	104.08	105.31	106.37	107.52	108.46	109.41	110.30	111.13	111.57
6.600	103.71	105.06	106.11	107.21	108.12	109.04	110.04	110.82	111.36
6.800	103.37	104.56	105.76	106.79	107.88	108.77	109.70	110.53	111.31
7.000	103.12	104.20	105.52	106.51	107.61	108.44	109.31	110.26	111.00
7.200	102.68	103.88	105.06	106.16	107.16	108.17	109.03	109.93	110.69
7.400	102.27	103.58	104.64	105.83	106.84	107.92	108.69	109.52	110.40
7.600	101.89	103.27	104.34	105.52	106.45	107.47	108.40	109.21	110.08
7.800	101.55	102.77	104.00	105.04	106.14	107.11	108.14	108.88	109.64
8.000	101.14	102.44	103.72	104.72	105.88	106.74	107.72	108.56	109.29
8.200	100.69	102.01	103.24	104.36	105.39	106.40	107.26	108.27	108.94
8.400	100.31	101.75	102.85	104.08	105.02	106.10	106.96	107.87	108.55
8.600	99.87	101.20	102.46	103.68	104.68	105.68	106.58	107.36	108.22
8.800	99.45	100.83	102.12	103.21	104.35	105.25	106.25	107.02	107.86
9.000	99.15	100.43	101.68	102.85	104.09	104.93	105.89	106.56	107.45
9.200	98.94	99.94	101.22	102.43	103.52	104.54	105.38	106.29	107.16
9.400	98.78	99.56	100.83	102.16	103.13	104.21	104.98	106.05	106.85
9.600	98.66	99.29	100.41	101.58	102.71	103.74	104.59	105.56	106.56
9.800	98.56	99.10	99.96	101.19	102.35	103.24	104.32	105.21	106.31
10.000	98.48	98.95	99.60	100.77	101.87	102.81	103.99	104.91	105.89
10.200	98.41	98.82	99.35	100.27	101.37	102.43	103.51	104.61	105.50
10.400	98.36	98.71	99.16	99.84	100.92	102.19	103.19	104.35	105.24
10.600	98.32	98.62	99.00	99.52	100.51	101.65	102.80	103.88	104.86
10.800	98.28	98.53	98.85	99.23	100.06	101.29	102.50	103.48	104.46
11.000	98.67	98.46	98.72	99.02	99.72	100.92	102.09	103.12	103.02
11.200	98.75	98.40	98.59	98.90	99.49	100.49	101.62	102.21	100.70

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	98.79	98.33	98.45	98.81	99.32	100.07	101.29	100.71	97.61
11.600	98.82	98.79	98.39	98.74	99.15	99.78	100.14	97.26	94.43
11.800	98.84	98.76	98.37	98.65	99.02	99.56	97.47	93.55	91.40
12.000	99.22	98.77	98.36	98.58	98.94	98.19	92.90	89.86	88.29
12.200	108.61	98.80	98.88	98.56	98.87	93.61	88.48	86.19	85.23
12.400	118.79	98.82	98.91	98.57	96.20	87.90	84.22	82.66	82.45
12.600	129.03	107.73	98.96	99.08	87.72	82.39	80.22	79.39	79.67
12.800	139.06	128.27	99.01	88.80	79.70	77.30	76.46	76.34	76.90
13.000	148.88	148.11	99.07	73.46	72.50	72.63	72.99	73.53	74.37
13.200	158.18	165.61	165.44	63.89	67.35	68.64	69.91	71.00	72.12
13.400	166.18	165.30	165.58	77.30	64.20	65.57	67.33	68.78	70.13
13.600	170.02	165.43	165.71	165.98	66.07	64.61	65.44	66.98	68.43
13.800	165.30	165.57	165.84	166.12	80.51	65.25	64.37	65.69	67.09
14.000	165.43	165.71	165.98	166.25	133.97	69.87	64.55	64.97	66.11
14.200	165.56	165.84	166.11	166.39	166.67	82.31	67.69	65.07	65.63
14.400	165.70	165.97	166.24	166.52	166.80	114.35	72.99	66.27	65.69
14.600	165.83	166.10	166.38	166.66	166.94	167.23	83.76	70.06	66.51
14.800	165.96	166.24	166.51	166.79	167.08	167.37	105.55	75.64	68.32
15.000	166.09	166.37	166.65	166.93	167.22	167.52	141.68	85.08	71.48
15.200	166.23	166.50	166.78	167.07	167.36	167.66	167.96	101.49	78.00
15.400	166.36	166.64	166.92	167.21	167.50	167.80	168.11	127.41	86.37
15.600	166.49	166.77	167.06	167.34	167.64	167.95	168.26	165.17	99.57
15.800	166.63	166.91	167.19	167.49	167.78	168.09	168.41	168.73	120.44
16.000	166.76	167.04	167.33	167.63	167.93	168.24	168.56	168.88	145.82
16.200	166.90	167.18	167.47	167.77	168.08	168.39	168.71	169.04	169.37
16.400	167.03	167.32	167.61	167.91	168.22	168.54	168.86	169.19	169.53
16.600	167.17	167.46	167.75	168.06	168.37	168.69	169.01	169.35	169.69
16.800	167.30	167.60	167.89	168.20	168.52	168.84	169.17	169.51	169.85
17.000	167.44	167.74	168.04	168.35	168.67	168.99	169.32	169.66	170.01
17.200	167.58	167.88	168.19	168.50	168.82	169.15	169.48	169.82	170.17
17.400	167.72	168.02	168.33	168.65	168.97	169.30	169.64	169.98	170.33
17.600	167.86	168.17	168.48	168.80	169.12	169.46	169.80	170.14	170.50
17.800	168.01	168.31	168.63	168.95	169.28	169.61	169.96	170.31	170.65
18.000	168.15	168.46	168.78	169.10	169.43	169.77	170.12	170.47	170.81
18.200	168.29	168.61	168.93	169.26	169.59	169.93	170.28	170.63	170.96
18.400	168.44	168.76	169.08	169.41	169.75	170.09	170.44	170.78	171.11
18.600	168.59	168.91	169.23	169.57	169.90	170.25	170.60	170.94	171.27
18.800	168.74	169.06	169.39	169.72	170.06	170.41	170.76	171.09	171.42
19.000	168.89	169.21	169.54	169.88	170.22	170.57	170.91	171.24	171.57
19.200	169.04	169.36	169.70	170.04	170.38	170.73	171.06	171.39	171.72
19.400	169.19	169.52	169.86	170.20	170.55	170.89	171.22	171.55	171.87
19.600	169.34	169.67	170.01	170.36	170.71	171.04	171.37	171.70	172.02
19.800	169.50	169.83	170.17	170.52	170.86	171.19	171.52	171.85	172.16
20.000	169.65	169.99	170.33	170.68	171.01	171.34	171.67	171.99	172.31
20.200	169.81	170.15	170.49	170.84	171.17	171.50	171.82	172.14	172.46
20.400	169.96	170.30	170.65	170.99	171.32	171.65	171.97	172.29	172.60
20.600	170.12	170.46	170.81	171.14	171.47	171.80	172.12	172.43	172.75
20.800	170.28	170.62	170.96	171.29	171.62	171.94	172.26	172.58	172.89
21.000	170.44	170.79	171.12	171.45	171.77	172.09	172.41	172.72	173.03
21.200	170.60	170.94	171.27	171.60	171.92	172.24	172.55	172.86	173.17
21.400	170.76	171.09	171.42	171.75	172.07	172.38	172.70	173.01	173.31
21.600	170.92	171.24	171.57	171.90	172.21	172.53	172.84	173.15	173.45
21.800	171.07	171.40	171.72	172.04	172.36	172.67	172.98	173.29	173.59
22.000	171.22	171.55	171.87	172.19	172.50	172.82	173.12	173.43	173.73
22.200	171.37	171.70	172.02	172.33	172.65	172.96	173.26	173.57	173.87
22.400	171.52	171.85	172.16	172.48	172.79	173.10	173.40	173.70	174.00
22.600	171.67	171.99	172.31	172.62	172.93	173.24	173.54	173.84	174.14
22.800	171.82	172.14	172.46	172.77	173.08	173.38	173.68	173.98	174.27
23.000	171.97	172.29	172.60	172.91	173.22	173.52	173.82	174.11	174.40

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	172.12	172.43	172.74	173.05	173.36	173.66	173.95	174.25	174.54
23.400	172.26	172.58	172.89	173.19	173.50	173.79	174.09	174.38	174.67
23.600	172.41	172.72	173.03	173.33	173.63	173.93	174.22	174.51	174.80
23.800	172.55	172.86	173.17	173.47	173.77	174.07	174.36	174.65	174.93
24.000	172.70	173.00	173.31	173.61	173.91	174.20	174.49	174.78	175.06
24.200	172.84	173.15	173.45	173.75	174.04	174.34	174.62	174.91	175.19
24.400	172.98	173.29	173.59	173.88	174.18	174.47	174.76	175.04	175.32
24.600	173.12	173.42	173.72	174.02	174.31	174.60	174.89	175.17	175.44
24.800	173.26	173.56	173.86	174.15	174.44	174.73	175.02	175.29	175.57
25.000	173.40	173.70	174.00	174.29	174.58	174.86	175.14	175.42	175.69
25.200	173.54	173.84	174.13	174.42	174.71	174.99	175.27	175.55	175.82
25.400	173.67	173.97	174.27	174.55	174.84	175.12	175.40	175.67	175.94
25.600	173.81	174.11	174.40	174.69	174.97	175.25	175.52	175.79	176.06
25.800	173.95	174.24	174.53	174.82	175.10	175.37	175.65	175.92	176.18
26.000	174.08	174.37	174.66	174.95	175.23	175.50	175.77	176.04	176.30
26.200	174.22	174.51	174.79	175.07	175.35	175.62	175.89	176.16	176.42
26.400	174.35	174.64	174.92	175.20	175.48	175.75	176.02	176.28	176.54
26.600	174.48	174.77	175.05	175.33	175.60	175.87	176.14	176.40	176.65
26.800	174.62	174.90	175.18	175.45	175.73	175.99	176.26	176.52	176.77
27.000	174.75	175.03	175.31	175.58	175.85	176.11	176.38	176.63	176.89
27.200	174.88	175.16	175.43	175.70	175.97	176.23	176.49	176.75	177.00
27.400	175.01	175.28	175.56	175.83	176.09	176.35	176.61	176.86	177.11
27.600	175.13	175.41	175.68	175.95	176.21	176.47	176.73	176.98	176.81
27.800	175.26	175.53	175.80	176.07	176.33	176.59	176.84	177.09	169.53
28.000	175.39	175.66	175.93	176.19	176.45	176.70	176.96	177.20	156.91
28.200	175.51	175.78	176.05	176.31	176.57	176.82	177.07	171.54	127.15
28.400	175.63	175.90	176.17	176.43	176.68	176.93	177.18	156.87	48.13
28.600	175.76	176.02	176.29	176.54	176.80	177.05	174.61	113.92	7.20
28.800	175.88	176.15	176.40	176.66	176.91	177.16	156.92	17.36	7.20
29.000	176.00	176.26	176.52	176.78	177.03	176.83	91.85	7.20	7.20
29.200	176.12	176.38	176.64	176.89	177.14	157.18	7.20	7.20	7.20
29.400	179.56	176.50	176.75	177.00	177.25	55.91	7.20	7.20	7.20
29.600	194.68	176.62	176.87	177.12	159.39	7.20	7.20	7.20	7.20
29.800	194.71	176.73	176.98	177.23	8.27	7.20	7.20	7.20	7.20
30.000	194.74	194.82	177.09	164.21	7.20	7.20	7.20	7.20	7.20
30.200	194.77	194.85	177.02	7.20	7.20	7.20	7.20	7.20	7.20
30.400	194.81	194.88	166.39	7.20	7.20	7.20	7.20	7.20	7.20
30.600	194.84	196.19	0.00	7.20	7.20	7.20	7.20	7.20	7.20
30.800	194.87	201.76	0.00	7.20	7.20	7.20	7.20	7.20	7.20
31.000	197.22	210.65	0.00	0.00	7.20	7.20	7.20	7.20	7.20
31.200	201.98	0.00	0.00	0.00	7.20	7.20	7.20	7.20	7.20
31.400	208.45	0.00	0.00	0.00	3.32	7.20	7.20	7.20	7.20
31.600	0.00	0.00	0.00	0.00	0.00	7.20	7.20	7.20	7.20
31.800	0.00	0.00	0.00	0.00	0.00	6.86	7.20	7.20	7.20
32.000	0.00	0.00	0.00	0.00	0.00	0.00	7.20	7.20	7.20

----- vertical center of buoy m -----

trim draught	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
0.000	0.208	0.094	-0.009	0.111	0.241	0.375	0.513	0.654	0.796
0.200	0.275	0.160	0.092	0.174	0.304	0.439	0.577	0.717	0.860
0.400	0.344	0.235	0.195	0.250	0.370	0.504	0.642	0.782	0.923
0.600	0.417	0.324	0.297	0.339	0.439	0.570	0.708	0.848	0.986
0.800	0.497	0.421	0.400	0.435	0.519	0.638	0.774	0.913	1.051
1.000	0.583	0.520	0.504	0.534	0.606	0.712	0.842	0.978	1.115
1.200	0.676	0.621	0.608	0.634	0.697	0.793	0.912	1.044	1.181
1.400	0.772	0.724	0.712	0.735	0.792	0.879	0.987	1.112	1.248
1.600	0.870	0.827	0.816	0.837	0.889	0.968	1.067	1.184	1.316
1.800	0.970	0.930	0.920	0.940	0.987	1.059	1.151	1.261	1.387
2.000	1.071	1.034	1.025	1.043	1.086	1.152	1.238	1.342	1.462
2.200	1.173	1.139	1.130	1.146	1.186	1.246	1.327	1.426	1.541
2.400	1.277	1.243	1.235	1.249	1.286	1.342	1.418	1.513	1.624
2.600	1.381	1.348	1.340	1.353	1.386	1.439	1.511	1.602	1.709
2.800	1.486	1.454	1.445	1.456	1.487	1.537	1.606	1.693	1.796
3.000	1.591	1.560	1.550	1.560	1.589	1.636	1.702	1.786	1.885
3.200	1.696	1.666	1.655	1.664	1.691	1.736	1.800	1.880	1.974
3.400	1.801	1.772	1.760	1.768	1.793	1.837	1.898	1.975	2.065
3.600	1.906	1.878	1.866	1.872	1.896	1.938	1.997	2.070	2.157
3.800	2.012	1.983	1.972	1.977	2.000	2.040	2.096	2.167	2.251
4.000	2.117	2.089	2.078	2.082	2.104	2.143	2.196	2.264	2.345
4.200	2.223	2.195	2.183	2.188	2.208	2.245	2.297	2.362	2.440
4.400	2.329	2.301	2.289	2.293	2.313	2.348	2.397	2.460	2.536
4.600	2.436	2.408	2.395	2.399	2.418	2.451	2.498	2.559	2.632
4.800	2.543	2.514	2.502	2.505	2.523	2.555	2.600	2.658	2.729
5.000	2.650	2.621	2.609	2.611	2.628	2.658	2.702	2.758	2.826
5.200	2.757	2.729	2.716	2.717	2.733	2.762	2.804	2.858	2.924
5.400	2.865	2.836	2.823	2.824	2.838	2.866	2.907	2.959	3.023
5.600	2.973	2.944	2.930	2.930	2.944	2.970	3.009	3.059	3.121
5.800	3.081	3.052	3.037	3.037	3.049	3.074	3.112	3.160	3.220
6.000	3.190	3.160	3.145	3.143	3.155	3.179	3.215	3.262	3.320
6.200	3.298	3.269	3.253	3.250	3.261	3.283	3.318	3.363	3.419
6.400	3.407	3.377	3.360	3.357	3.366	3.388	3.421	3.465	3.519
6.600	3.517	3.486	3.468	3.464	3.472	3.492	3.524	3.566	3.619
6.800	3.626	3.594	3.576	3.571	3.578	3.597	3.628	3.668	3.719
7.000	3.736	3.704	3.684	3.678	3.684	3.702	3.731	3.770	3.820
7.200	3.846	3.813	3.793	3.786	3.791	3.807	3.835	3.873	3.920
7.400	3.956	3.922	3.901	3.893	3.897	3.913	3.939	3.975	4.021
7.600	4.066	4.031	4.010	4.001	4.004	4.018	4.043	4.078	4.122
7.800	4.177	4.141	4.119	4.109	4.110	4.123	4.147	4.180	4.223
8.000	4.287	4.251	4.227	4.217	4.217	4.229	4.251	4.283	4.324
8.200	4.398	4.361	4.336	4.325	4.324	4.335	4.355	4.386	4.426
8.400	4.509	4.471	4.446	4.433	4.431	4.440	4.460	4.489	4.527
8.600	4.620	4.581	4.555	4.541	4.538	4.546	4.564	4.592	4.629
8.800	4.732	4.692	4.664	4.649	4.645	4.652	4.669	4.695	4.730
9.000	4.843	4.802	4.774	4.757	4.752	4.758	4.774	4.799	4.832
9.200	4.955	4.913	4.883	4.866	4.860	4.864	4.879	4.902	4.934
9.400	5.066	5.024	4.993	4.974	4.967	4.970	4.983	5.005	5.037
9.600	5.177	5.135	5.103	5.083	5.075	5.077	5.088	5.109	5.139
9.800	5.288	5.246	5.213	5.192	5.182	5.183	5.193	5.213	5.242
10.000	5.399	5.356	5.323	5.301	5.290	5.289	5.298	5.317	5.344
10.200	5.509	5.467	5.433	5.410	5.398	5.396	5.404	5.421	5.447
10.400	5.620	5.577	5.543	5.519	5.506	5.502	5.509	5.525	5.550
10.600	5.730	5.687	5.653	5.629	5.614	5.609	5.615	5.630	5.654
10.800	5.839	5.797	5.763	5.738	5.722	5.716	5.720	5.734	5.757
11.000	5.948	5.906	5.872	5.847	5.830	5.823	5.826	5.839	5.859
11.200	6.057	6.015	5.981	5.955	5.938	5.930	5.932	5.944	5.959

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	6.165	6.124	6.090	6.064	6.046	6.037	6.038	6.047	6.055
11.600	6.273	6.233	6.199	6.172	6.154	6.145	6.144	6.147	6.146
11.800	6.380	6.341	6.307	6.281	6.262	6.252	6.248	6.241	6.233
12.000	6.488	6.448	6.416	6.389	6.370	6.359	6.346	6.330	6.315
12.200	6.588	6.555	6.523	6.497	6.478	6.462	6.438	6.414	6.394
12.400	6.676	6.662	6.630	6.604	6.584	6.556	6.524	6.494	6.468
12.600	6.752	6.766	6.737	6.712	6.682	6.642	6.603	6.568	6.540
12.800	6.817	6.851	6.844	6.814	6.767	6.720	6.677	6.639	6.607
13.000	6.870	6.911	6.950	6.895	6.841	6.790	6.745	6.706	6.672
13.200	6.911	6.949	6.969	6.953	6.902	6.853	6.808	6.768	6.734
13.400	6.943	6.966	6.978	6.988	6.953	6.909	6.866	6.826	6.792
13.600	6.963	6.976	6.989	7.003	6.994	6.958	6.919	6.881	6.847
13.800	6.974	6.986	6.999	7.014	7.023	7.000	6.967	6.932	6.900
14.000	6.984	6.997	7.010	7.026	7.042	7.035	7.010	6.979	6.949
14.200	6.994	7.007	7.022	7.038	7.055	7.063	7.048	7.023	6.996
14.400	7.005	7.019	7.033	7.050	7.068	7.085	7.081	7.063	7.040
14.600	7.016	7.030	7.045	7.062	7.081	7.100	7.109	7.099	7.081
14.800	7.028	7.042	7.058	7.075	7.094	7.114	7.132	7.131	7.119
15.000	7.039	7.054	7.070	7.088	7.108	7.128	7.151	7.159	7.154
15.200	7.051	7.067	7.083	7.102	7.122	7.143	7.166	7.184	7.186
15.400	7.064	7.079	7.097	7.116	7.136	7.158	7.182	7.205	7.215
15.600	7.076	7.093	7.111	7.130	7.151	7.173	7.197	7.223	7.242
15.800	7.089	7.106	7.125	7.144	7.166	7.189	7.214	7.240	7.265
16.000	7.103	7.120	7.139	7.159	7.181	7.205	7.230	7.257	7.286
16.200	7.117	7.134	7.154	7.175	7.197	7.221	7.247	7.275	7.304
16.400	7.131	7.149	7.169	7.190	7.213	7.238	7.265	7.293	7.323
16.600	7.145	7.164	7.184	7.206	7.230	7.255	7.282	7.311	7.342
16.800	7.160	7.179	7.200	7.223	7.247	7.273	7.301	7.330	7.361
17.000	7.175	7.195	7.216	7.240	7.264	7.291	7.319	7.349	7.381
17.200	7.191	7.211	7.233	7.257	7.282	7.309	7.338	7.369	7.401
17.400	7.206	7.227	7.250	7.274	7.300	7.328	7.357	7.389	7.422
17.600	7.223	7.244	7.267	7.292	7.319	7.347	7.377	7.409	7.443
17.800	7.239	7.261	7.285	7.310	7.338	7.367	7.397	7.430	7.465
18.000	7.256	7.279	7.303	7.329	7.357	7.386	7.418	7.451	7.487
18.200	7.273	7.297	7.321	7.348	7.377	7.407	7.439	7.473	7.510
18.400	7.291	7.315	7.340	7.368	7.397	7.428	7.460	7.495	7.533
18.600	7.309	7.333	7.360	7.387	7.417	7.449	7.482	7.518	7.556
18.800	7.327	7.352	7.379	7.408	7.438	7.470	7.505	7.541	7.580
19.000	7.346	7.372	7.399	7.428	7.459	7.492	7.527	7.565	7.604
19.200	7.365	7.392	7.419	7.449	7.481	7.515	7.551	7.589	7.629
19.400	7.385	7.412	7.440	7.471	7.503	7.538	7.574	7.613	7.654
19.600	7.405	7.432	7.461	7.493	7.526	7.561	7.599	7.638	7.679
19.800	7.425	7.453	7.483	7.515	7.549	7.585	7.623	7.663	7.705
20.000	7.446	7.474	7.505	7.537	7.572	7.609	7.648	7.689	7.731
20.200	7.467	7.496	7.527	7.561	7.596	7.634	7.673	7.715	7.758
20.400	7.488	7.518	7.550	7.584	7.621	7.659	7.699	7.741	7.785
20.600	7.510	7.541	7.573	7.609	7.646	7.684	7.725	7.768	7.813
20.800	7.532	7.564	7.597	7.633	7.671	7.710	7.752	7.795	7.841
21.000	7.555	7.587	7.622	7.658	7.696	7.737	7.779	7.823	7.869
21.200	7.578	7.611	7.646	7.683	7.722	7.763	7.806	7.851	7.898
21.400	7.602	7.636	7.671	7.709	7.749	7.790	7.834	7.880	7.927
21.600	7.626	7.660	7.697	7.735	7.776	7.818	7.862	7.909	7.957
21.800	7.650	7.686	7.723	7.762	7.803	7.846	7.891	7.938	7.987
22.000	7.675	7.711	7.749	7.789	7.831	7.874	7.920	7.968	8.018
22.200	7.701	7.737	7.776	7.816	7.859	7.903	7.950	7.998	8.049
22.400	7.726	7.764	7.803	7.844	7.887	7.932	7.980	8.029	8.080
22.600	7.753	7.791	7.831	7.872	7.916	7.962	8.010	8.060	8.112
22.800	7.779	7.818	7.858	7.901	7.946	7.992	8.041	8.091	8.144
23.000	7.806	7.846	7.887	7.930	7.975	8.023	8.072	8.123	8.177

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	7.834	7.874	7.916	7.960	8.006	8.054	8.104	8.156	8.210
23.400	7.861	7.902	7.945	7.990	8.036	8.085	8.136	8.189	8.244
23.600	7.890	7.931	7.975	8.020	8.067	8.117	8.168	8.222	8.278
23.800	7.918	7.960	8.005	8.051	8.099	8.149	8.201	8.256	8.312
24.000	7.947	7.990	8.035	8.082	8.131	8.182	8.235	8.290	8.347
24.200	7.977	8.020	8.066	8.114	8.163	8.215	8.269	8.325	8.383
24.400	8.007	8.051	8.097	8.146	8.196	8.249	8.303	8.360	8.419
24.600	8.037	8.082	8.129	8.178	8.229	8.283	8.338	8.395	8.455
24.800	8.068	8.113	8.161	8.211	8.263	8.317	8.373	8.431	8.492
25.000	8.099	8.145	8.194	8.244	8.297	8.352	8.409	8.468	8.529
25.200	8.130	8.178	8.227	8.278	8.332	8.387	8.445	8.505	8.567
25.400	8.162	8.210	8.260	8.312	8.367	8.423	8.481	8.542	8.605
25.600	8.195	8.244	8.294	8.347	8.402	8.459	8.518	8.580	8.643
25.800	8.228	8.277	8.329	8.382	8.438	8.496	8.556	8.618	8.682
26.000	8.261	8.311	8.363	8.418	8.474	8.533	8.594	8.657	8.722
26.200	8.295	8.346	8.399	8.454	8.511	8.570	8.632	8.696	8.762
26.400	8.329	8.380	8.434	8.490	8.548	8.608	8.671	8.735	8.802
26.600	8.363	8.416	8.470	8.527	8.586	8.647	8.710	8.775	8.843
26.800	8.398	8.452	8.507	8.564	8.624	8.686	8.750	8.816	8.884
27.000	8.434	8.488	8.544	8.602	8.662	8.725	8.790	8.857	8.926
27.200	8.470	8.524	8.581	8.640	8.701	8.765	8.830	8.898	8.968
27.400	8.506	8.561	8.619	8.679	8.741	8.805	8.871	8.940	9.010
27.600	8.543	8.599	8.657	8.718	8.781	8.846	8.913	8.982	9.053
27.800	8.580	8.637	8.696	8.757	8.821	8.887	8.954	9.025	9.092
28.000	8.617	8.675	8.735	8.797	8.862	8.928	8.997	9.068	9.121
28.200	8.655	8.714	8.775	8.838	8.903	8.970	9.040	9.109	9.139
28.400	8.694	8.753	8.815	8.878	8.944	9.012	9.083	9.139	9.146
28.600	8.733	8.793	8.855	8.920	8.986	9.055	9.126	9.155	9.150
28.800	8.772	8.833	8.896	8.961	9.029	9.098	9.157	9.161	9.153
29.000	8.812	8.874	8.937	9.003	9.072	9.142	9.172	9.165	9.156
29.200	8.852	8.914	8.979	9.046	9.115	9.176	9.177	9.168	9.160
29.400	8.893	8.956	9.021	9.089	9.159	9.189	9.180	9.172	9.163
29.600	8.930	8.998	9.064	9.132	9.194	9.193	9.184	9.175	9.167
29.800	8.967	9.040	9.107	9.176	9.206	9.197	9.188	9.179	9.170
30.000	9.005	9.081	9.151	9.212	9.210	9.201	9.191	9.182	9.174
30.200	9.043	9.120	9.194	9.224	9.214	9.204	9.195	9.186	9.177
30.400	9.081	9.159	9.229	9.228	9.218	9.208	9.199	9.190	9.181
30.600	9.120	9.198	9.238	9.232	9.222	9.212	9.203	9.193	9.184
30.800	9.159	9.226	9.238	9.236	9.226	9.216	9.207	9.197	9.188
31.000	9.197	9.238	9.238	9.238	9.230	9.220	9.210	9.201	9.192
31.200	9.223	9.238	9.238	9.238	9.234	9.224	9.214	9.205	9.196
31.400	9.236	9.238	9.238	9.238	9.238	9.228	9.218	9.209	9.199
31.600	9.238	9.238	9.238	9.238	9.238	9.232	9.222	9.213	9.203
31.800	9.238	9.238	9.238	9.238	9.238	9.237	9.226	9.217	9.207
32.000	9.238	9.238	9.238	9.238	9.238	9.238	9.231	9.221	9.211

----- wetted surface area m2 -----									
trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	2823	2726	4964	2983	3029	3100	3135	3214	3290
0.200	3796	4566	5346	4819	3935	3753	3663	3633	3660
0.400	4795	5582	5585	5462	4909	4386	4162	4055	4061
0.600	5564	5821	5783	5702	5550	5058	4695	4491	4407
0.800	5983	6008	5962	5900	5780	5618	5201	4932	4779
1.000	6199	6176	6129	6071	5989	5865	5683	5370	5116
1.200	6371	6335	6289	6229	6164	6056	5976	5761	5519
1.400	6530	6484	6442	6386	6323	6248	6169	6086	5873
1.600	6682	6640	6588	6535	6471	6421	6348	6274	6168
1.800	6830	6785	6731	6681	6631	6589	6519	6446	6379
2.000	6970	6928	6878	6821	6779	6742	6682	6608	6557
2.200	7129	7068	7017	6964	6938	6891	6833	6768	6731
2.400	7291	7201	7157	7117	7080	7034	6981	6932	6889
2.600	7453	7353	7292	7267	7223	7177	7129	7089	7032
2.800	7613	7511	7434	7412	7366	7311	7279	7248	7175
3.000	7770	7671	7593	7547	7502	7457	7431	7384	7316
3.200	7931	7840	7755	7680	7640	7601	7572	7519	7459
3.400	8085	8003	7920	7823	7774	7751	7710	7650	7592
3.600	8232	8173	8079	7979	7910	7898	7841	7780	7726
3.800	8376	8333	8229	8139	8070	8024	7971	7915	7854
4.000	8522	8471	8394	8300	8227	8153	8101	8044	7985
4.200	8658	8609	8552	8469	8387	8287	8225	8173	8128
4.400	8807	8744	8693	8632	8537	8423	8354	8312	8250
4.600	8946	8873	8847	8801	8680	8583	8499	8429	8373
4.800	9084	9026	8984	8930	8833	8736	8620	8554	8494
5.000	9219	9176	9118	9062	8997	8881	8779	8680	8609
5.200	9351	9321	9262	9190	9134	9027	8921	8801	8735
5.400	9509	9464	9395	9322	9260	9188	9056	8955	8859
5.600	9662	9588	9535	9458	9382	9311	9200	9096	8980
5.800	9809	9742	9661	9586	9499	9436	9355	9229	9128
6.000	9952	9873	9780	9708	9632	9557	9487	9373	9269
6.200	10074	9998	9913	9835	9761	9674	9610	9523	9400
6.400	10214	10125	10045	9954	9881	9804	9731	9661	9544
6.600	10349	10243	10167	10081	10008	9933	9847	9783	9691
6.800	10478	10389	10295	10215	10127	10053	9975	9903	9832
7.000	10598	10517	10423	10338	10249	10179	10104	10018	9953
7.200	10731	10643	10552	10465	10384	10298	10224	10143	10073
7.400	10861	10766	10683	10592	10508	10413	10348	10272	10188
7.600	10988	10888	10806	10713	10640	10550	10467	10392	10311
7.800	11110	11023	10931	10847	10759	10675	10580	10514	10440
8.000	11243	11145	11047	10969	10872	10804	10715	10634	10562
8.200	11375	11273	11182	11095	11008	10924	10849	10748	10684
8.400	11503	11390	11308	11213	11133	11037	10965	10879	10809
8.600	11632	11532	11433	11338	11255	11167	11088	11015	10927
8.800	11758	11656	11553	11468	11375	11294	11203	11134	11059
9.000	11872	11782	11684	11589	11488	11413	11326	11262	11187
9.200	11979	11913	11814	11712	11625	11536	11458	11378	11304
9.400	12080	12033	11939	11826	11746	11651	11581	11491	11423
9.600	12179	12143	12063	11967	11870	11782	11707	11624	11540
9.800	12276	12246	12188	12089	11988	11911	11824	11746	11650
10.000	12371	12346	12303	12213	12119	12036	11942	11866	11781
10.200	12465	12443	12409	12342	12249	12157	12073	11984	11905
10.400	12557	12538	12510	12462	12376	12269	12191	12094	12021
10.600	12649	12632	12609	12574	12499	12409	12314	12227	12148
10.800	12740	12725	12706	12681	12625	12529	12429	12352	12275
11.000	12866	12817	12802	12784	12741	12652	12557	12476	12508
11.200	12962	12909	12898	12882	12847	12777	12691	12650	12835

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	13057	13000	12995	12977	12948	12903	12815	12893	13227
11.600	13148	13135	13087	13070	13052	13017	13014	13345	13671
11.800	13247	13235	13178	13168	13157	13131	13375	13837	14064
12.000	13411	13338	13288	13270	13258	13354	13942	14282	14531
12.200	14500	13440	13436	13376	13359	13907	14474	14774	14949
12.400	15533	13547	13540	13485	13693	14607	15057	15260	15354
12.600	16556	14603	13644	13635	14713	15338	15609	15744	15765
12.800	17558	16552	13748	14814	15747	16028	16162	16209	16210
13.000	18510	18464	13995	16701	16724	16701	16705	16664	16604
13.200	19502	20377	21793	18606	17761	17401	17196	17083	16991
13.400	20456	21816	21825	20513	18711	18067	17745	17525	17360
13.600	21431	21849	21857	21866	19691	18804	18273	17967	17741
13.800	21872	21881	21890	21898	20654	19465	18797	18406	18146
14.000	21904	21913	21922	21930	21690	20119	19314	18830	18499
14.200	21936	21945	21954	21962	21970	20742	19823	19264	18860
14.400	21968	21977	21986	21994	22002	21418	20344	19679	19205
14.600	22001	22009	22018	22026	22034	22042	20823	20080	19567
14.800	22033	22041	22050	22058	22066	22074	21326	20501	19943
15.000	22065	22073	22082	22090	22098	22106	21886	20901	20275
15.200	22097	22105	22114	22122	22130	22138	22146	21297	20673
15.400	22129	22137	22146	22154	22162	22170	22178	21731	20978
15.600	22161	22169	22177	22186	22194	22202	22210	22201	21344
15.800	22193	22201	22209	22218	22226	22234	22242	22251	21690
16.000	22225	22233	22241	22249	22258	22266	22274	22283	22070
16.200	22257	22265	22273	22281	22290	22298	22306	22315	22324
16.400	22289	22297	22305	22313	22321	22330	22338	22347	22356
16.600	22321	22329	22337	22345	22353	22362	22370	22379	22388
16.800	22352	22361	22369	22377	22385	22394	22402	22411	22420
17.000	22384	22393	22401	22409	22417	22426	22435	22444	22453
17.200	22416	22424	22433	22441	22449	22458	22467	22476	22485
17.400	22448	22456	22465	22473	22481	22490	22499	22508	22518
17.600	22480	22488	22497	22505	22514	22522	22531	22541	22550
17.800	22512	22520	22528	22537	22546	22555	22564	22573	22584
18.000	22544	22552	22561	22569	22578	22587	22596	22606	22617
18.200	22576	22584	22593	22601	22610	22619	22629	22639	22651
18.400	22608	22616	22625	22633	22642	22652	22661	22672	22684
18.600	22640	22648	22657	22666	22675	22684	22694	22706	22717
18.800	22672	22680	22689	22698	22707	22716	22727	22739	22751
19.000	22704	22712	22721	22730	22739	22749	22761	22772	22784
19.200	22736	22744	22753	22762	22772	22782	22794	22806	22818
19.400	22768	22777	22786	22795	22804	22816	22827	22839	22851
19.600	22800	22809	22818	22827	22837	22849	22861	22873	22885
19.800	22832	22841	22850	22860	22870	22882	22894	22906	22918
20.000	22864	22873	22883	22892	22904	22916	22928	22940	22952
20.200	22897	22906	22915	22925	22937	22949	22961	22973	22985
20.400	22929	22938	22948	22959	22971	22983	22995	23007	23019
20.600	22961	22971	22980	22992	23004	23016	23028	23040	23052
20.800	22994	23003	23014	23026	23038	23050	23062	23074	23086
21.000	23026	23036	23047	23059	23071	23083	23095	23107	23120
21.200	23058	23069	23081	23093	23105	23117	23129	23141	23153
21.400	23091	23102	23114	23126	23138	23150	23162	23175	23187
21.600	23124	23136	23148	23160	23172	23184	23196	23208	23220
21.800	23157	23169	23181	23193	23205	23217	23230	23242	23254
22.000	23191	23203	23215	23227	23239	23251	23263	23275	23288
22.200	23224	23236	23248	23260	23272	23285	23297	23309	23321
22.400	23258	23270	23282	23294	23306	23318	23330	23343	23355
22.600	23291	23303	23315	23327	23339	23352	23364	23376	23389
22.800	23325	23337	23349	23361	23373	23385	23397	23410	23422
23.000	23358	23370	23382	23394	23407	23419	23431	23443	23456

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	23391	23404	23416	23428	23440	23452	23465	23477	23489
23.400	23425	23437	23449	23462	23474	23486	23498	23511	23523
23.600	23459	23471	23483	23495	23507	23520	23532	23544	23557
23.800	23492	23504	23517	23529	23541	23553	23566	23578	23591
24.000	23526	23538	23550	23562	23575	23587	23600	23612	23624
24.200	23559	23571	23584	23596	23608	23621	23633	23646	23658
24.400	23593	23605	23617	23630	23642	23654	23667	23679	23692
24.600	23626	23639	23651	23663	23676	23688	23700	23713	23725
24.800	23660	23672	23685	23697	23709	23722	23734	23746	23759
25.000	23693	23706	23718	23731	23743	23755	23768	23780	23793
25.200	23727	23740	23752	23764	23777	23789	23802	23814	23827
25.400	23761	23773	23785	23798	23810	23823	23835	23848	23860
25.600	23794	23807	23819	23832	23844	23857	23869	23881	23894
25.800	23828	23840	23853	23865	23878	23890	23903	23915	23928
26.000	23862	23874	23887	23899	23911	23924	23936	23949	23962
26.200	23895	23908	23920	23933	23945	23958	23970	23983	23995
26.400	23929	23941	23954	23966	23979	23992	24004	24017	24029
26.600	23963	23975	23988	24000	24013	24025	24038	24050	24063
26.800	23996	24009	24021	24034	24046	24059	24072	24084	24097
27.000	24030	24042	24055	24067	24080	24093	24105	24118	24131
27.200	24064	24076	24089	24101	24114	24126	24139	24152	24164
27.400	24097	24110	24122	24135	24148	24160	24173	24186	24198
27.600	24131	24143	24156	24169	24181	24194	24207	24219	24250
27.800	24165	24178	24190	24202	24215	24228	24241	24253	24509
28.000	24198	24211	24224	24236	24249	24262	24274	24287	24816
28.200	24232	24245	24257	24270	24283	24295	24308	24515	25140
28.400	24266	24278	24291	24304	24316	24329	24342	24878	25357
28.600	24300	24312	24325	24338	24350	24363	24475	25260	25405
28.800	24333	24346	24359	24371	24384	24397	24945	25440	25417
29.000	24367	24380	24392	24405	24418	24448	25382	25459	25429
29.200	24401	24413	24426	24439	24452	25013	25502	25472	25442
29.400	24455	24447	24460	24473	24485	25501	25514	25484	25454
29.600	24568	24481	24494	24506	25075	25556	25526	25496	25466
29.800	24590	24515	24528	24540	25594	25568	25538	25508	25478
30.000	24611	24654	24561	25091	25610	25580	25551	25521	25491
30.200	24633	24675	24629	25652	25623	25593	25563	25533	25503
30.400	24654	24697	25262	25665	25635	25605	25575	25545	25515
30.600	24676	24822	25805	25677	25647	25617	25587	25557	25528
30.800	24697	25277	25805	25689	25659	25630	25600	25570	25540
31.000	24892	25729	25805	25805	25672	25642	25612	25582	25552
31.200	25261	25805	25805	25805	25684	25654	25624	25594	25565
31.400	25638	25805	25805	25805	25783	25666	25637	25607	25577
31.600	25805	25805	25805	25805	25805	25679	25649	25619	25589
31.800	25805	25805	25805	25805	25805	25698	25661	25631	25601
32.000	25805	25805	25805	25805	25805	25674	25643	25614	

----- waterplane coefficient -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	0.2892	0.2747	0.4730	0.3058	0.3161	0.3243	0.3312	0.3372	0.3426
0.200	0.3941	0.4757	0.5767	0.5044	0.4181	0.3935	0.3838	0.3798	0.3769
0.400	0.4982	0.5908	0.5971	0.5868	0.5203	0.4628	0.4365	0.4223	0.4108
0.600	0.5776	0.6139	0.6134	0.6087	0.5915	0.5321	0.4890	0.4634	0.4458
0.800	0.6215	0.6290	0.6272	0.6233	0.6127	0.5919	0.5416	0.5042	0.4809
1.000	0.6403	0.6413	0.6392	0.6357	0.6288	0.6157	0.5887	0.5456	0.5160
1.200	0.6529	0.6524	0.6503	0.6463	0.6416	0.6309	0.6156	0.5850	0.5512
1.400	0.6637	0.6618	0.6600	0.6563	0.6518	0.6429	0.6291	0.6144	0.5850
1.600	0.6733	0.6717	0.6688	0.6654	0.6604	0.6528	0.6403	0.6296	0.6138
1.800	0.6821	0.6801	0.6766	0.6737	0.6687	0.6608	0.6514	0.6413	0.6319
2.000	0.6897	0.6879	0.6850	0.6810	0.6753	0.6680	0.6613	0.6519	0.6449
2.200	0.6990	0.6952	0.6921	0.6876	0.6810	0.6754	0.6696	0.6622	0.6563
2.400	0.7083	0.7016	0.6991	0.6938	0.6869	0.6819	0.6774	0.6725	0.6665
2.600	0.7166	0.7099	0.7056	0.6987	0.6932	0.6887	0.6848	0.6819	0.6751
2.800	0.7239	0.7186	0.7098	0.7045	0.6996	0.6948	0.6924	0.6905	0.6832
3.000	0.7288	0.7265	0.7160	0.7099	0.7054	0.7019	0.7004	0.6968	0.6905
3.200	0.7336	0.7319	0.7233	0.7149	0.7119	0.7091	0.7069	0.7030	0.6980
3.400	0.7395	0.7354	0.7304	0.7217	0.7178	0.7164	0.7135	0.7086	0.7038
3.600	0.7442	0.7384	0.7365	0.7297	0.7238	0.7237	0.7193	0.7140	0.7097
3.800	0.7495	0.7431	0.7410	0.7376	0.7321	0.7288	0.7246	0.7201	0.7148
4.000	0.7538	0.7483	0.7449	0.7444	0.7405	0.7340	0.7300	0.7254	0.7202
4.200	0.7583	0.7540	0.7499	0.7510	0.7483	0.7400	0.7348	0.7306	0.7272
4.400	0.7650	0.7594	0.7557	0.7558	0.7542	0.7460	0.7398	0.7370	0.7322
4.600	0.7706	0.7647	0.7633	0.7604	0.7586	0.7538	0.7469	0.7414	0.7373
4.800	0.7762	0.7717	0.7692	0.7654	0.7627	0.7601	0.7518	0.7465	0.7420
5.000	0.7820	0.7788	0.7750	0.7706	0.7657	0.7652	0.7600	0.7518	0.7461
5.200	0.7870	0.7852	0.7810	0.7755	0.7719	0.7693	0.7655	0.7565	0.7511
5.400	0.7950	0.7919	0.7865	0.7814	0.7768	0.7718	0.7698	0.7643	0.7562
5.600	0.8021	0.7967	0.7927	0.7870	0.7815	0.7764	0.7739	0.7696	0.7607
5.800	0.8091	0.8038	0.7980	0.7924	0.7858	0.7811	0.7770	0.7740	0.7677
6.000	0.8157	0.8095	0.8026	0.7968	0.7911	0.7856	0.7804	0.7778	0.7730
6.200	0.8202	0.8145	0.8076	0.8021	0.7963	0.7898	0.7848	0.7812	0.7773
6.400	0.8271	0.8198	0.8134	0.8062	0.8007	0.7946	0.7891	0.7838	0.7812
6.600	0.8335	0.8244	0.8180	0.8109	0.8056	0.7998	0.7931	0.7880	0.7847
6.800	0.8389	0.8312	0.8232	0.8168	0.8097	0.8040	0.7976	0.7922	0.7866
7.000	0.8434	0.8368	0.8275	0.8214	0.8137	0.8086	0.8027	0.7960	0.7906
7.200	0.8494	0.8417	0.8338	0.8263	0.8195	0.8127	0.8068	0.8000	0.7946
7.400	0.8554	0.8465	0.8396	0.8315	0.8242	0.8164	0.8111	0.8050	0.7982
7.600	0.8609	0.8510	0.8441	0.8357	0.8296	0.8218	0.8151	0.8090	0.8017
7.800	0.8661	0.8576	0.8491	0.8418	0.8339	0.8265	0.8187	0.8130	0.8066
8.000	0.8715	0.8625	0.8535	0.8464	0.8378	0.8315	0.8235	0.8168	0.8104
8.200	0.8776	0.8682	0.8591	0.8511	0.8434	0.8358	0.8291	0.8203	0.8138
8.400	0.8830	0.8724	0.8644	0.8551	0.8482	0.8397	0.8326	0.8243	0.8171
8.600	0.8886	0.8789	0.8695	0.8599	0.8524	0.8443	0.8370	0.8298	0.8204
8.800	0.8942	0.8839	0.8740	0.8656	0.8566	0.8493	0.8404	0.8330	0.8251
9.000	0.8987	0.8890	0.8791	0.8700	0.8603	0.8529	0.8442	0.8365	0.8303
9.200	0.9022	0.8948	0.8847	0.8751	0.8660	0.8573	0.8492	0.8404	0.8343
9.400	0.9052	0.8997	0.8895	0.8787	0.8706	0.8609	0.8526	0.8439	0.8384
9.600	0.9077	0.9035	0.8943	0.8848	0.8751	0.8653	0.8566	0.8495	0.8423
9.800	0.9100	0.9065	0.8996	0.8893	0.8790	0.8698	0.8604	0.8540	0.8459
10.000	0.9120	0.9091	0.9040	0.8939	0.8835	0.8735	0.8643	0.8579	0.8507
10.200	0.9137	0.9113	0.9072	0.8989	0.8884	0.8783	0.8702	0.8620	0.8555
10.400	0.9153	0.9133	0.9097	0.9034	0.8920	0.8818	0.8744	0.8656	0.8590
10.600	0.9167	0.9149	0.9119	0.9067	0.8963	0.8878	0.8791	0.8709	0.8629
10.800	0.9180	0.9164	0.9137	0.9090	0.9018	0.8924	0.8831	0.8753	0.8655
11.000	0.9143	0.9177	0.9152	0.9111	0.9063	0.8970	0.8878	0.8792	0.8556
11.200	0.9150	0.9188	0.9164	0.9133	0.9096	0.9020	0.8925	0.8779	0.8392

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	0.9150	0.9197	0.9171	0.9154	0.9124	0.9061	0.8965	0.8673	0.8102
11.600	0.9153	0.9155	0.9186	0.9172	0.9142	0.9099	0.8922	0.8363	0.7822
11.800	0.9147	0.9150	0.9199	0.9183	0.9157	0.9123	0.8689	0.8014	0.7525
12.000	0.9024	0.9145	0.9163	0.9191	0.9175	0.8997	0.8244	0.7659	0.7235
12.200	0.8119	0.9144	0.9146	0.9197	0.9190	0.8480	0.7783	0.7303	0.6951
12.400	0.7120	0.9142	0.9152	0.9202	0.8831	0.7833	0.7309	0.6947	0.6659
12.600	0.6118	0.8120	0.9156	0.9162	0.7864	0.7182	0.6830	0.6590	0.6372
12.800	0.5132	0.6131	0.9158	0.7823	0.6875	0.6537	0.6350	0.6216	0.6081
13.000	0.4137	0.4145	0.0671	0.5834	0.5886	0.5883	0.5867	0.5839	0.5788
13.200	0.3142	0.2158	0.0822	0.3848	0.4844	0.5222	0.5381	0.5459	0.5488
13.400	0.2154	0.0818	0.0826	0.1859	0.3860	0.4575	0.4890	0.5077	0.5180
13.600	0.1180	0.0822	0.0830	0.0838	0.2863	0.3869	0.4411	0.4699	0.4868
13.800	0.0818	0.0826	0.0834	0.0842	0.1876	0.3209	0.3922	0.4312	0.4564
14.000	0.0822	0.0830	0.0838	0.0846	0.0978	0.2547	0.3432	0.3925	0.4248
14.200	0.0826	0.0834	0.0842	0.0850	0.0858	0.1892	0.2891	0.3536	0.3925
14.400	0.0830	0.0838	0.0846	0.0854	0.0863	0.1239	0.2403	0.3147	0.3605
14.600	0.0834	0.0842	0.0850	0.0858	0.0867	0.0875	0.1910	0.2708	0.3284
14.800	0.0838	0.0846	0.0854	0.0863	0.0871	0.0880	0.1414	0.2323	0.2963
15.000	0.0842	0.0850	0.0858	0.0867	0.0875	0.0884	0.1017	0.1929	0.2639
15.200	0.0846	0.0854	0.0863	0.0871	0.0880	0.0889	0.0898	0.1533	0.2275
15.400	0.0850	0.0858	0.0867	0.0875	0.0884	0.0893	0.0903	0.1182	0.1948
15.600	0.0854	0.0863	0.0871	0.0880	0.0889	0.0898	0.0907	0.0917	0.1620
15.800	0.0859	0.0867	0.0875	0.0884	0.0893	0.0902	0.0912	0.0922	0.1300
16.000	0.0863	0.0871	0.0880	0.0888	0.0898	0.0907	0.0917	0.0927	0.1061
16.200	0.0867	0.0875	0.0884	0.0893	0.0902	0.0912	0.0922	0.0932	0.0943
16.400	0.0871	0.0879	0.0888	0.0897	0.0907	0.0917	0.0927	0.0937	0.0948
16.600	0.0875	0.0884	0.0893	0.0902	0.0912	0.0921	0.0932	0.0942	0.0953
16.800	0.0879	0.0888	0.0897	0.0907	0.0916	0.0926	0.0937	0.0948	0.0959
17.000	0.0884	0.0893	0.0902	0.0911	0.0921	0.0931	0.0942	0.0953	0.0964
17.200	0.0888	0.0897	0.0906	0.0916	0.0926	0.0936	0.0947	0.0958	0.0970
17.400	0.0893	0.0902	0.0911	0.0921	0.0931	0.0941	0.0952	0.0964	0.0975
17.600	0.0897	0.0906	0.0916	0.0926	0.0936	0.0947	0.0958	0.0969	0.0981
17.800	0.0902	0.0911	0.0921	0.0931	0.0941	0.0952	0.0963	0.0975	0.0986
18.000	0.0906	0.0916	0.0925	0.0936	0.0946	0.0957	0.0968	0.0980	0.0992
18.200	0.0911	0.0920	0.0930	0.0941	0.0951	0.0963	0.0974	0.0986	0.0997
18.400	0.0915	0.0925	0.0935	0.0946	0.0957	0.0968	0.0980	0.0991	0.1003
18.600	0.0920	0.0930	0.0940	0.0951	0.0962	0.0973	0.0985	0.0997	0.1009
18.800	0.0925	0.0935	0.0946	0.0956	0.0968	0.0979	0.0991	0.1002	0.1014
19.000	0.0930	0.0940	0.0951	0.0962	0.0973	0.0985	0.0996	0.1008	0.1020
19.200	0.0935	0.0945	0.0956	0.0967	0.0978	0.0990	0.1002	0.1014	0.1026
19.400	0.0940	0.0950	0.0961	0.0973	0.0984	0.0996	0.1007	0.1019	0.1031
19.600	0.0945	0.0956	0.0967	0.0978	0.0990	0.1001	0.1013	0.1025	0.1037
19.800	0.0950	0.0961	0.0972	0.0984	0.0995	0.1007	0.1019	0.1031	0.1043
20.000	0.0955	0.0966	0.0978	0.0989	0.1001	0.1013	0.1025	0.1037	0.1049
20.200	0.0961	0.0972	0.0983	0.0995	0.1006	0.1018	0.1030	0.1042	0.1055
20.400	0.0966	0.0977	0.0989	0.1000	0.1012	0.1024	0.1036	0.1048	0.1060
20.600	0.0971	0.0983	0.0994	0.1006	0.1018	0.1030	0.1042	0.1054	0.1066
20.800	0.0977	0.0988	0.1000	0.1012	0.1023	0.1035	0.1047	0.1060	0.1072
21.000	0.0982	0.0994	0.1006	0.1017	0.1029	0.1041	0.1053	0.1066	0.1078
21.200	0.0988	0.1000	0.1011	0.1023	0.1035	0.1047	0.1059	0.1071	0.1084
21.400	0.0993	0.1005	0.1017	0.1029	0.1041	0.1053	0.1065	0.1077	0.1090
21.600	0.0999	0.1011	0.1023	0.1034	0.1046	0.1059	0.1071	0.1083	0.1096
21.800	0.1005	0.1016	0.1028	0.1040	0.1052	0.1064	0.1077	0.1089	0.1102
22.000	0.1010	0.1022	0.1034	0.1046	0.1058	0.1070	0.1083	0.1095	0.1108
22.200	0.1016	0.1028	0.1040	0.1052	0.1064	0.1076	0.1089	0.1101	0.1114
22.400	0.1022	0.1033	0.1045	0.1057	0.1070	0.1082	0.1095	0.1107	0.1120
22.600	0.1027	0.1039	0.1051	0.1063	0.1076	0.1088	0.1101	0.1113	0.1126
22.800	0.1033	0.1045	0.1057	0.1069	0.1081	0.1094	0.1107	0.1119	0.1132
23.000	0.1039	0.1051	0.1063	0.1075	0.1087	0.1100	0.1113	0.1125	0.1138

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	0.1044	0.1056	0.1069	0.1081	0.1093	0.1106	0.1119	0.1131	0.1144
23.400	0.1050	0.1062	0.1074	0.1087	0.1099	0.1112	0.1125	0.1138	0.1151
23.600	0.1056	0.1068	0.1080	0.1093	0.1105	0.1118	0.1131	0.1144	0.1157
23.800	0.1062	0.1074	0.1086	0.1099	0.1111	0.1124	0.1137	0.1150	0.1163
24.000	0.1068	0.1080	0.1092	0.1105	0.1117	0.1130	0.1143	0.1156	0.1169
24.200	0.1073	0.1086	0.1098	0.1111	0.1123	0.1136	0.1149	0.1162	0.1175
24.400	0.1079	0.1092	0.1104	0.1117	0.1130	0.1142	0.1155	0.1168	0.1182
24.600	0.1085	0.1098	0.1110	0.1123	0.1136	0.1148	0.1162	0.1175	0.1188
24.800	0.1091	0.1104	0.1116	0.1129	0.1142	0.1155	0.1168	0.1181	0.1194
25.000	0.1097	0.1110	0.1122	0.1135	0.1148	0.1161	0.1174	0.1187	0.1200
25.200	0.1103	0.1116	0.1128	0.1141	0.1154	0.1167	0.1180	0.1193	0.1206
25.400	0.1109	0.1122	0.1134	0.1147	0.1160	0.1173	0.1186	0.1199	0.1212
25.600	0.1115	0.1128	0.1140	0.1153	0.1166	0.1179	0.1192	0.1205	0.1219
25.800	0.1121	0.1134	0.1147	0.1159	0.1172	0.1185	0.1198	0.1212	0.1225
26.000	0.1127	0.1140	0.1153	0.1166	0.1179	0.1192	0.1205	0.1218	0.1231
26.200	0.1133	0.1146	0.1159	0.1172	0.1185	0.1198	0.1211	0.1224	0.1237
26.400	0.1139	0.1152	0.1165	0.1178	0.1191	0.1204	0.1217	0.1230	0.1243
26.600	0.1145	0.1158	0.1171	0.1184	0.1197	0.1210	0.1223	0.1236	0.1249
26.800	0.1151	0.1164	0.1177	0.1190	0.1203	0.1216	0.1229	0.1242	0.1255
27.000	0.1158	0.1170	0.1183	0.1196	0.1209	0.1222	0.1235	0.1248	0.1261
27.200	0.1164	0.1177	0.1189	0.1202	0.1215	0.1228	0.1241	0.1254	0.1268
27.400	0.1170	0.1183	0.1195	0.1208	0.1221	0.1234	0.1247	0.1261	0.1274
27.600	0.1176	0.1189	0.1202	0.1215	0.1228	0.1241	0.1254	0.1267	0.1283
27.800	0.1182	0.1195	0.1208	0.1221	0.1234	0.1247	0.1260	0.1273	0.1289
28.000	0.1188	0.1201	0.1214	0.1227	0.1240	0.1253	0.1266	0.1279	0.1295
28.200	0.1194	0.1207	0.1220	0.1233	0.1246	0.1259	0.1272	0.1285	0.1301
28.400	0.1200	0.1213	0.1226	0.1239	0.1252	0.1265	0.1278	0.1291	0.1307
28.600	0.1206	0.1219	0.1232	0.1245	0.1258	0.1271	0.1284	0.1297	0.1313
28.800	0.1212	0.1225	0.1238	0.1251	0.1264	0.1277	0.1290	0.1303	0.1319
29.000	0.1218	0.1231	0.1244	0.1257	0.1270	0.1283	0.1296	0.1309	0.1325
29.200	0.1225	0.1237	0.1250	0.1263	0.1276	0.1289	0.1302	0.1315	0.1331
29.400	0.1231	0.1243	0.1256	0.1269	0.1282	0.1295	0.1308	0.1321	0.1337
29.600	0.1237	0.1249	0.1262	0.1275	0.1288	0.1301	0.1314	0.1327	0.1343
29.800	0.1243	0.1256	0.1268	0.1281	0.1294	0.1307	0.1320	0.1333	0.1349
30.000	0.1249	0.1261	0.1274	0.1287	0.1300	0.1313	0.1326	0.1339	0.1355
30.200	0.1255	0.1267	0.1280	0.1293	0.1306	0.1319	0.1332	0.1345	0.1361
30.400	0.1261	0.1273	0.1286	0.1299	0.1312	0.1325	0.1338	0.1351	0.1367
30.600	0.1267	0.1279	0.1292	0.1305	0.1318	0.1331	0.1344	0.1357	0.1373
30.800	0.1273	0.1285	0.1298	0.1311	0.1324	0.1337	0.1350	0.1363	0.1379
31.000	0.1279	0.1291	0.1304	0.1317	0.1330	0.1343	0.1356	0.1369	0.1385
31.200	0.1285	0.1297	0.1310	0.1323	0.1336	0.1349	0.1362	0.1375	0.1391
31.400	0.1291	0.1303	0.1316	0.1329	0.1342	0.1355	0.1368	0.1381	0.1397
31.600	0.1297	0.1309	0.1322	0.1335	0.1348	0.1361	0.1374	0.1387	0.1403
31.800	0.1303	0.1315	0.1328	0.1341	0.1354	0.1367	0.1380	0.1393	0.1409
32.000	0.1309	0.1321	0.1334	0.1347	0.1360	0.1373	0.1386	0.1399	0.1415

----- prismatic coefficient -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	*****	*****	323146	*****	*****	*****	*****	*****	*****
0.200	0.7911	0.5990	0.5900	0.6937	0.9418	1.2363	1.5502	1.8772	2.2139
0.400	0.6249	0.5814	0.5990	0.6371	0.7115	0.8358	0.9819	1.1391	1.3021
0.600	0.6009	0.5941	0.6062	0.6291	0.6659	0.7255	0.8102	0.9075	1.0103
0.800	0.6038	0.6037	0.6124	0.6284	0.6523	0.6869	0.7375	0.8019	0.8733
1.000	0.6110	0.6116	0.6180	0.6300	0.6473	0.6717	0.7039	0.7468	0.7982
1.200	0.6178	0.6183	0.6234	0.6327	0.6461	0.6643	0.6878	0.7167	0.7539
1.400	0.6241	0.6244	0.6284	0.6358	0.6465	0.6607	0.6786	0.7002	0.7273
1.600	0.6299	0.6299	0.6331	0.6391	0.6479	0.6593	0.6732	0.6905	0.7113
1.800	0.6353	0.6352	0.6377	0.6426	0.6499	0.6591	0.6702	0.6843	0.7015
2.000	0.6405	0.6402	0.6421	0.6462	0.6522	0.6596	0.6688	0.6805	0.6950
2.200	0.6454	0.6449	0.6463	0.6497	0.6546	0.6607	0.6685	0.6783	0.6909
2.400	0.6504	0.6494	0.6505	0.6532	0.6570	0.6622	0.6689	0.6774	0.6884
2.600	0.6552	0.6538	0.6545	0.6565	0.6596	0.6640	0.6699	0.6774	0.6870
2.800	0.6599	0.6582	0.6584	0.6598	0.6622	0.6660	0.6712	0.6780	0.6865
3.000	0.6644	0.6625	0.6620	0.6630	0.6650	0.6682	0.6729	0.6791	0.6865
3.200	0.6686	0.6667	0.6657	0.6661	0.6677	0.6705	0.6748	0.6804	0.6870
3.400	0.6726	0.6707	0.6693	0.6692	0.6705	0.6730	0.6769	0.6819	0.6878
3.600	0.6765	0.6744	0.6729	0.6724	0.6733	0.6757	0.6791	0.6835	0.6888
3.800	0.6802	0.6779	0.6764	0.6756	0.6762	0.6784	0.6814	0.6853	0.6900
4.000	0.6838	0.6814	0.6798	0.6789	0.6792	0.6810	0.6837	0.6872	0.6914
4.200	0.6873	0.6847	0.6830	0.6822	0.6824	0.6837	0.6861	0.6891	0.6929
4.400	0.6907	0.6880	0.6862	0.6855	0.6855	0.6864	0.6884	0.6911	0.6946
4.600	0.6941	0.6912	0.6894	0.6887	0.6887	0.6892	0.6908	0.6932	0.6964
4.800	0.6974	0.6945	0.6926	0.6918	0.6917	0.6920	0.6932	0.6953	0.6982
5.000	0.7007	0.6977	0.6958	0.6949	0.6946	0.6949	0.6957	0.6975	0.7000
5.200	0.7039	0.7010	0.6990	0.6979	0.6975	0.6977	0.6983	0.6997	0.7019
5.400	0.7072	0.7042	0.7022	0.7009	0.7003	0.7004	0.7009	0.7019	0.7038
5.600	0.7105	0.7075	0.7053	0.7039	0.7032	0.7031	0.7035	0.7043	0.7057
5.800	0.7138	0.7107	0.7084	0.7068	0.7060	0.7057	0.7060	0.7066	0.7078
6.000	0.7171	0.7139	0.7115	0.7098	0.7087	0.7083	0.7084	0.7089	0.7099
6.200	0.7203	0.7171	0.7145	0.7127	0.7115	0.7108	0.7108	0.7112	0.7120
6.400	0.7236	0.7202	0.7175	0.7156	0.7142	0.7134	0.7132	0.7134	0.7141
6.600	0.7268	0.7233	0.7205	0.7184	0.7169	0.7159	0.7155	0.7156	0.7162
6.800	0.7301	0.7264	0.7235	0.7212	0.7196	0.7185	0.7179	0.7178	0.7182
7.000	0.7333	0.7295	0.7264	0.7240	0.7222	0.7210	0.7203	0.7200	0.7202
7.200	0.7364	0.7325	0.7293	0.7268	0.7249	0.7235	0.7226	0.7222	0.7222
7.400	0.7396	0.7356	0.7322	0.7296	0.7275	0.7260	0.7249	0.7243	0.7242
7.600	0.7427	0.7386	0.7351	0.7323	0.7301	0.7284	0.7273	0.7265	0.7262
7.800	0.7458	0.7415	0.7380	0.7351	0.7327	0.7309	0.7296	0.7287	0.7282
8.000	0.7489	0.7445	0.7409	0.7378	0.7353	0.7334	0.7319	0.7309	0.7302
8.200	0.7519	0.7475	0.7437	0.7405	0.7379	0.7358	0.7342	0.7330	0.7322
8.400	0.7550	0.7504	0.7465	0.7432	0.7405	0.7382	0.7365	0.7351	0.7342
8.600	0.7581	0.7533	0.7493	0.7459	0.7430	0.7406	0.7388	0.7373	0.7361
8.800	0.7611	0.7563	0.7521	0.7486	0.7456	0.7431	0.7410	0.7394	0.7381
9.000	0.7641	0.7592	0.7549	0.7512	0.7481	0.7455	0.7433	0.7415	0.7401
9.200	0.7671	0.7621	0.7577	0.7539	0.7506	0.7479	0.7455	0.7436	0.7421
9.400	0.7700	0.7650	0.7604	0.7565	0.7531	0.7502	0.7478	0.7457	0.7441
9.600	0.7729	0.7678	0.7632	0.7591	0.7556	0.7526	0.7500	0.7478	0.7461
9.800	0.7757	0.7706	0.7659	0.7617	0.7581	0.7549	0.7522	0.7500	0.7481
10.000	0.7784	0.7734	0.7686	0.7643	0.7606	0.7573	0.7544	0.7521	0.7501
10.200	0.7810	0.7761	0.7713	0.7669	0.7630	0.7596	0.7567	0.7542	0.7521
10.400	0.7836	0.7787	0.7740	0.7695	0.7655	0.7619	0.7589	0.7563	0.7541
10.600	0.7861	0.7813	0.7766	0.7721	0.7679	0.7643	0.7611	0.7584	0.7562
10.800	0.7885	0.7838	0.7791	0.7746	0.7704	0.7666	0.7633	0.7605	0.7582
11.000	0.7909	0.7862	0.7816	0.7771	0.7728	0.7689	0.7656	0.7627	0.7600
11.200	0.7931	0.7885	0.7840	0.7795	0.7752	0.7713	0.7678	0.7648	0.7616

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	0.7953	0.7908	0.7863	0.7819	0.7776	0.7736	0.7700	0.7667	0.7627
11.600	0.7973	0.7930	0.7886	0.7842	0.7800	0.7759	0.7722	0.7681	0.7633
11.800	0.7993	0.7951	0.7908	0.7865	0.7823	0.7782	0.7741	0.7690	0.7634
12.000	0.8012	0.7971	0.7930	0.7887	0.7845	0.7804	0.7753	0.7692	0.7629
12.200	0.8022	0.7990	0.7950	0.7908	0.7867	0.7820	0.7757	0.7689	0.7620
12.400	0.8015	0.8009	0.7969	0.7929	0.7887	0.7825	0.7754	0.7680	0.7607
12.600	0.7993	0.8023	0.7988	0.7949	0.7894	0.7820	0.7743	0.7665	0.7590
12.800	0.7956	0.8009	0.8006	0.7961	0.7886	0.7805	0.7725	0.7646	0.7568
13.000	0.7905	0.7964	0.8024	0.7943	0.7863	0.7781	0.7700	0.7621	0.7543
13.200	0.7961	0.8013	0.8037	0.8018	0.7945	0.7866	0.7787	0.7708	0.7630
13.400	0.8002	0.8033	0.8049	0.8062	0.8012	0.7942	0.7866	0.7789	0.7712
13.600	0.8028	0.8046	0.8062	0.8079	0.8064	0.8007	0.7938	0.7864	0.7790
13.800	0.8042	0.8058	0.8075	0.8092	0.8101	0.8062	0.8002	0.7934	0.7862
14.000	0.8054	0.8071	0.8088	0.8105	0.8122	0.8106	0.8059	0.7997	0.7930
14.200	0.8067	0.8084	0.8101	0.8118	0.8135	0.8140	0.8107	0.8055	0.7993
14.400	0.8080	0.8097	0.8114	0.8131	0.8149	0.8164	0.8148	0.8106	0.8051
14.600	0.8093	0.8110	0.8127	0.8144	0.8162	0.8180	0.8181	0.8152	0.8105
14.800	0.8106	0.8123	0.8140	0.8158	0.8175	0.8194	0.8207	0.8190	0.8153
15.000	0.8119	0.8136	0.8153	0.8171	0.8189	0.8207	0.8225	0.8223	0.8196
15.200	0.8132	0.8149	0.8167	0.8184	0.8202	0.8221	0.8240	0.8250	0.8234
15.400	0.8145	0.8162	0.8180	0.8198	0.8216	0.8235	0.8253	0.8271	0.8266
15.600	0.8158	0.8175	0.8193	0.8211	0.8230	0.8248	0.8267	0.8287	0.8294
15.800	0.8171	0.8189	0.8207	0.8225	0.8243	0.8262	0.8281	0.8301	0.8316
16.000	0.8184	0.8202	0.8220	0.8239	0.8257	0.8276	0.8296	0.8315	0.8334
16.200	0.8198	0.8216	0.8234	0.8252	0.8271	0.8290	0.8310	0.8329	0.8350
16.400	0.8211	0.8229	0.8248	0.8266	0.8285	0.8304	0.8324	0.8344	0.8364
16.600	0.8225	0.8243	0.8261	0.8280	0.8299	0.8319	0.8338	0.8358	0.8379
16.800	0.8238	0.8256	0.8275	0.8294	0.8313	0.8333	0.8353	0.8373	0.8394
17.000	0.8252	0.8270	0.8289	0.8308	0.8327	0.8347	0.8367	0.8388	0.8409
17.200	0.8265	0.8284	0.8303	0.8322	0.8342	0.8362	0.8382	0.8403	0.8424
17.400	0.8279	0.8298	0.8317	0.8336	0.8356	0.8376	0.8397	0.8417	0.8439
17.600	0.8293	0.8312	0.8331	0.8351	0.8370	0.8391	0.8411	0.8432	0.8454
17.800	0.8307	0.8326	0.8345	0.8365	0.8385	0.8405	0.8426	0.8447	0.8469
18.000	0.8321	0.8340	0.8359	0.8379	0.8400	0.8420	0.8441	0.8463	0.8484
18.200	0.8335	0.8354	0.8374	0.8394	0.8414	0.8435	0.8456	0.8478	0.8500
18.400	0.8349	0.8368	0.8388	0.8408	0.8429	0.8450	0.8471	0.8493	0.8516
18.600	0.8363	0.8383	0.8403	0.8423	0.8444	0.8465	0.8487	0.8509	0.8531
18.800	0.8377	0.8397	0.8417	0.8438	0.8459	0.8480	0.8502	0.8524	0.8547
19.000	0.8392	0.8412	0.8432	0.8453	0.8474	0.8495	0.8517	0.8540	0.8563
19.200	0.8406	0.8426	0.8447	0.8468	0.8489	0.8511	0.8533	0.8556	0.8579
19.400	0.8420	0.8441	0.8461	0.8483	0.8504	0.8526	0.8549	0.8571	0.8595
19.600	0.8435	0.8455	0.8476	0.8498	0.8519	0.8542	0.8564	0.8587	0.8611
19.800	0.8450	0.8470	0.8491	0.8513	0.8535	0.8557	0.8580	0.8603	0.8627
20.000	0.8464	0.8485	0.8506	0.8528	0.8550	0.8573	0.8596	0.8619	0.8643
20.200	0.8479	0.8500	0.8522	0.8544	0.8566	0.8589	0.8612	0.8636	0.8660
20.400	0.8494	0.8515	0.8537	0.8559	0.8582	0.8605	0.8628	0.8652	0.8676
20.600	0.8509	0.8530	0.8552	0.8575	0.8598	0.8621	0.8644	0.8668	0.8693
20.800	0.8524	0.8546	0.8568	0.8590	0.8613	0.8637	0.8661	0.8685	0.8709
21.000	0.8539	0.8561	0.8583	0.8606	0.8629	0.8653	0.8677	0.8701	0.8726
21.200	0.8555	0.8577	0.8599	0.8622	0.8646	0.8669	0.8693	0.8718	0.8743
21.400	0.8570	0.8592	0.8615	0.8638	0.8662	0.8686	0.8710	0.8734	0.8760
21.600	0.8585	0.8608	0.8631	0.8654	0.8678	0.8702	0.8726	0.8751	0.8777
21.800	0.8601	0.8624	0.8647	0.8670	0.8694	0.8718	0.8743	0.8768	0.8794
22.000	0.8617	0.8640	0.8663	0.8687	0.8711	0.8735	0.8760	0.8785	0.8811
22.200	0.8632	0.8656	0.8679	0.8703	0.8727	0.8752	0.8777	0.8802	0.8828
22.400	0.8648	0.8672	0.8695	0.8719	0.8744	0.8769	0.8794	0.8819	0.8845
22.600	0.8664	0.8688	0.8712	0.8736	0.8760	0.8785	0.8811	0.8837	0.8863
22.800	0.8680	0.8704	0.8728	0.8752	0.8777	0.8802	0.8828	0.8854	0.8880
23.000	0.8696	0.8720	0.8744	0.8769	0.8794	0.8819	0.8845	0.8871	0.8898

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	0.8713	0.8737	0.8761	0.8786	0.8811	0.8837	0.8862	0.8889	0.8916
23.400	0.8729	0.8753	0.8778	0.8803	0.8828	0.8854	0.8880	0.8906	0.8933
23.600	0.8745	0.8770	0.8794	0.8820	0.8845	0.8871	0.8897	0.8924	0.8951
23.800	0.8762	0.8786	0.8811	0.8837	0.8862	0.8888	0.8915	0.8942	0.8969
24.000	0.8778	0.8803	0.8828	0.8854	0.8880	0.8906	0.8933	0.8960	0.8987
24.200	0.8795	0.8820	0.8845	0.8871	0.8897	0.8924	0.8950	0.8978	0.9006
24.400	0.8812	0.8837	0.8862	0.8888	0.8915	0.8941	0.8968	0.8996	0.9024
24.600	0.8829	0.8854	0.8880	0.8906	0.8932	0.8959	0.8986	0.9014	0.9042
24.800	0.8845	0.8871	0.8897	0.8923	0.8950	0.8977	0.9004	0.9032	0.9061
25.000	0.8862	0.8888	0.8914	0.8941	0.8968	0.8995	0.9023	0.9051	0.9079
25.200	0.8880	0.8905	0.8932	0.8958	0.8985	0.9013	0.9041	0.9069	0.9098
25.400	0.8897	0.8923	0.8949	0.8976	0.9003	0.9031	0.9059	0.9088	0.9117
25.600	0.8914	0.8940	0.8967	0.8994	0.9021	0.9049	0.9078	0.9106	0.9135
25.800	0.8931	0.8958	0.8985	0.9012	0.9040	0.9068	0.9096	0.9125	0.9154
26.000	0.8949	0.8975	0.9002	0.9030	0.9058	0.9086	0.9115	0.9144	0.9173
26.200	0.8966	0.8993	0.9020	0.9048	0.9076	0.9105	0.9133	0.9163	0.9192
26.400	0.8984	0.9011	0.9038	0.9066	0.9095	0.9123	0.9152	0.9182	0.9212
26.600	0.9002	0.9029	0.9057	0.9085	0.9113	0.9142	0.9171	0.9201	0.9231
26.800	0.9020	0.9047	0.9075	0.9103	0.9132	0.9161	0.9190	0.9220	0.9250
27.000	0.9038	0.9065	0.9093	0.9121	0.9150	0.9180	0.9209	0.9239	0.9270
27.200	0.9056	0.9083	0.9111	0.9140	0.9169	0.9199	0.9228	0.9259	0.9289
27.400	0.9074	0.9102	0.9130	0.9159	0.9188	0.9218	0.9248	0.9278	0.9309
27.600	0.9092	0.9120	0.9149	0.9178	0.9207	0.9237	0.9267	0.9298	0.9329
27.800	0.9110	0.9138	0.9167	0.9196	0.9226	0.9256	0.9286	0.9317	0.9345
28.000	0.9128	0.9157	0.9186	0.9215	0.9245	0.9275	0.9306	0.9337	0.9358
28.200	0.9147	0.9176	0.9205	0.9234	0.9264	0.9295	0.9326	0.9355	0.9366
28.400	0.9166	0.9194	0.9224	0.9254	0.9284	0.9314	0.9345	0.9368	0.9370
28.600	0.9184	0.9213	0.9243	0.9273	0.9303	0.9334	0.9364	0.9375	0.9372
28.800	0.9203	0.9232	0.9262	0.9292	0.9323	0.9354	0.9378	0.9378	0.9374
29.000	0.9222	0.9251	0.9281	0.9312	0.9342	0.9373	0.9384	0.9380	0.9376
29.200	0.9241	0.9270	0.9301	0.9331	0.9362	0.9388	0.9387	0.9382	0.9377
29.400	0.9260	0.9290	0.9320	0.9351	0.9382	0.9393	0.9388	0.9384	0.9379
29.600	0.9277	0.9309	0.9339	0.9370	0.9397	0.9395	0.9390	0.9386	0.9381
29.800	0.9295	0.9328	0.9359	0.9390	0.9401	0.9397	0.9392	0.9388	0.9383
30.000	0.9312	0.9347	0.9379	0.9406	0.9403	0.9399	0.9394	0.9389	0.9385
30.200	0.9329	0.9365	0.9398	0.9409	0.9405	0.9400	0.9396	0.9391	0.9387
30.400	0.9347	0.9383	0.9413	0.9411	0.9407	0.9402	0.9398	0.9393	0.9389
30.600	0.9365	0.9400	0.9416	0.9413	0.9409	0.9404	0.9400	0.9395	0.9390
30.800	0.9383	0.9412	0.9416	0.9415	0.9411	0.9406	0.9401	0.9397	0.9392
31.000	0.9399	0.9416	0.9416	0.9416	0.9412	0.9408	0.9403	0.9399	0.9394
31.200	0.9410	0.9416	0.9416	0.9416	0.9414	0.9410	0.9405	0.9401	0.9396
31.400	0.9416	0.9416	0.9416	0.9416	0.9416	0.9412	0.9407	0.9402	0.9398
31.600	0.9416	0.9416	0.9416	0.9416	0.9416	0.9413	0.9409	0.9404	0.9400
31.800	0.9416	0.9416	0.9416	0.9416	0.9416	0.9415	0.9411	0.9406	0.9402
32.000	0.9416	0.9416	0.9416	0.9416	0.9416	0.9416	0.9413	0.9408	0.9404

----- midship section coeff. -----

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
0.200	0.948	0.948	0.948	0.948	0.948	0.948	0.948	0.948	0.948
0.400	0.957	0.957	0.957	0.957	0.957	0.957	0.957	0.957	0.957
0.600	0.964	0.964	0.964	0.964	0.964	0.964	0.964	0.964	0.964
0.800	0.969	0.969	0.969	0.969	0.969	0.969	0.969	0.969	0.969
1.000	0.973	0.973	0.973	0.973	0.973	0.973	0.973	0.973	0.973
1.200	0.976	0.976	0.976	0.976	0.976	0.976	0.976	0.976	0.976
1.400	0.979	0.979	0.979	0.979	0.979	0.979	0.979	0.979	0.979
1.600	0.981	0.981	0.981	0.981	0.981	0.981	0.981	0.981	0.981
1.800	0.983	0.983	0.983	0.983	0.983	0.983	0.983	0.983	0.983
2.000	0.985	0.985	0.985	0.985	0.985	0.985	0.985	0.985	0.985
2.200	0.986	0.986	0.986	0.986	0.986	0.986	0.986	0.986	0.986
2.400	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988
2.600	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988	0.988
2.800	0.989	0.989	0.989	0.989	0.989	0.989	0.989	0.989	0.989
3.000	0.990	0.990	0.990	0.990	0.990	0.990	0.990	0.990	0.990
3.200	0.991	0.991	0.991	0.991	0.991	0.991	0.991	0.991	0.991
3.400	0.991	0.991	0.991	0.991	0.991	0.991	0.991	0.991	0.991
3.600	0.992	0.992	0.992	0.992	0.992	0.992	0.992	0.992	0.992
3.800	0.992	0.992	0.992	0.992	0.992	0.992	0.992	0.992	0.992
4.000	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993
4.200	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993
4.400	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993
4.600	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993	0.993
4.800	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994
5.000	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994
5.200	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994
5.400	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994	0.994
5.600	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
5.800	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
6.000	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
6.200	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
6.400	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
6.600	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995	0.995
6.800	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
7.000	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
7.200	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
7.400	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
7.600	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
7.800	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
8.000	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
8.200	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
8.400	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996	0.996
8.600	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
8.800	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
9.000	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
9.200	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
9.400	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
9.600	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
9.800	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
10.000	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
10.200	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
10.400	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
10.600	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
10.800	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
11.000	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
11.200	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
11.400	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
11.600	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
11.800	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997	0.997
12.000	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
12.200	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
12.400	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
12.600	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
12.800	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
13.000	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998	0.998
13.200	0.983	0.983	0.983	0.983	0.983	0.983	0.983	0.983	0.983
13.400	0.968	0.968	0.968	0.968	0.968	0.968	0.968	0.968	0.968
13.600	0.954	0.954	0.954	0.954	0.954	0.954	0.954	0.954	0.954
13.800	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940	0.940
14.000	0.926	0.926	0.926	0.926	0.926	0.926	0.926	0.926	0.926
14.200	0.913	0.913	0.913	0.913	0.913	0.913	0.913	0.913	0.913
14.400	0.901	0.901	0.901	0.901	0.901	0.901	0.901	0.901	0.901
14.600	0.888	0.888	0.888	0.888	0.888	0.888	0.888	0.888	0.888
14.800	0.876	0.876	0.876	0.876	0.876	0.876	0.876	0.876	0.876
15.000	0.865	0.865	0.865	0.865	0.865	0.865	0.865	0.865	0.865
15.200	0.853	0.853	0.853	0.853	0.853	0.853	0.853	0.853	0.853
15.400	0.842	0.842	0.842	0.842	0.842	0.842	0.842	0.842	0.842
15.600	0.831	0.831	0.831	0.831	0.831	0.831	0.831	0.831	0.831
15.800	0.821	0.821	0.821	0.821	0.821	0.821	0.821	0.821	0.821
16.000	0.811	0.811	0.811	0.811	0.811	0.811	0.811	0.811	0.811
16.200	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801	0.801
16.400	0.791	0.791	0.791	0.791	0.791	0.791	0.791	0.791	0.791
16.600	0.781	0.781	0.781	0.781	0.781	0.781	0.781	0.781	0.781
16.800	0.772	0.772	0.772	0.772	0.772	0.772	0.772	0.772	0.772
17.000	0.763	0.763	0.763	0.763	0.763	0.763	0.763	0.763	0.763
17.200	0.754	0.754	0.754	0.754	0.754	0.754	0.754	0.754	0.754
17.400	0.745	0.745	0.745	0.745	0.745	0.745	0.745	0.745	0.745
17.600	0.737	0.737	0.737	0.737	0.737	0.737	0.737	0.737	0.737
17.800	0.729	0.729	0.729	0.729	0.729	0.729	0.729	0.729	0.729
18.000	0.721	0.721	0.721	0.721	0.721	0.721	0.721	0.721	0.721
18.200	0.713	0.713	0.713	0.713	0.713	0.713	0.713	0.713	0.713
18.400	0.705	0.705	0.705	0.705	0.705	0.705	0.705	0.705	0.705
18.600	0.697	0.697	0.697	0.697	0.697	0.697	0.697	0.697	0.697
18.800	0.690	0.690	0.690	0.690	0.690	0.690	0.690	0.690	0.690
19.000	0.683	0.683	0.683	0.683	0.683	0.683	0.683	0.683	0.683
19.200	0.676	0.676	0.676	0.676	0.676	0.676	0.676	0.676	0.676
19.400	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669	0.669
19.600	0.662	0.662	0.662	0.662	0.662	0.662	0.662	0.662	0.662
19.800	0.655	0.655	0.655	0.655	0.655	0.655	0.655	0.655	0.655
20.000	0.649	0.649	0.649	0.649	0.649	0.649	0.649	0.649	0.649
20.200	0.642	0.642	0.642	0.642	0.642	0.642	0.642	0.642	0.642
20.400	0.636	0.636	0.636	0.636	0.636	0.636	0.636	0.636	0.636
20.600	0.630	0.630	0.630	0.630	0.630	0.630	0.630	0.630	0.630
20.800	0.624	0.624	0.624	0.624	0.624	0.624	0.624	0.624	0.624
21.000	0.618	0.618	0.618	0.618	0.618	0.618	0.618	0.618	0.618
21.200	0.612	0.612	0.612	0.612	0.612	0.612	0.612	0.612	0.612
21.400	0.606	0.606	0.606	0.606	0.606	0.606	0.606	0.606	0.606
21.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600	0.600
21.800	0.595	0.595	0.595	0.595	0.595	0.595	0.595	0.595	0.595
22.000	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590	0.590
22.200	0.584	0.584	0.584	0.584	0.584	0.584	0.584	0.584	0.584
22.400	0.579	0.579	0.579	0.579	0.579	0.579	0.579	0.579	0.579
22.600	0.574	0.574	0.574	0.574	0.574	0.574	0.574	0.574	0.574
22.800	0.569	0.569	0.569	0.569	0.569	0.569	0.569	0.569	0.569
23.000	0.564	0.564	0.564	0.564	0.564	0.564	0.564	0.564	0.564

trim	-2.000	-1.000	0.000	1.000	2.000	3.000	4.000	5.000	6.000
draught									
23.200	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559	0.559
23.400	0.554	0.554	0.554	0.554	0.554	0.554	0.554	0.554	0.554
23.600	0.550	0.550	0.550	0.550	0.550	0.550	0.550	0.550	0.550
23.800	0.545	0.545	0.545	0.545	0.545	0.545	0.545	0.545	0.545
24.000	0.540	0.540	0.540	0.540	0.540	0.540	0.540	0.540	0.540
24.200	0.536	0.536	0.536	0.536	0.536	0.536	0.536	0.536	0.536
24.400	0.532	0.532	0.532	0.532	0.532	0.532	0.532	0.532	0.532
24.600	0.527	0.527	0.527	0.527	0.527	0.527	0.527	0.527	0.527
24.800	0.523	0.523	0.523	0.523	0.523	0.523	0.523	0.523	0.523
25.000	0.519	0.519	0.519	0.519	0.519	0.519	0.519	0.519	0.519
25.200	0.515	0.515	0.515	0.515	0.515	0.515	0.515	0.515	0.515
25.400	0.511	0.511	0.511	0.511	0.511	0.511	0.511	0.511	0.511
25.600	0.507	0.507	0.507	0.507	0.507	0.507	0.507	0.507	0.507
25.800	0.503	0.503	0.503	0.503	0.503	0.503	0.503	0.503	0.503
26.000	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499	0.499
26.200	0.495	0.495	0.495	0.495	0.495	0.495	0.495	0.495	0.495
26.400	0.491	0.491	0.491	0.491	0.491	0.491	0.491	0.491	0.491
26.600	0.488	0.488	0.488	0.488	0.488	0.488	0.488	0.488	0.488
26.800	0.484	0.484	0.484	0.484	0.484	0.484	0.484	0.484	0.484
27.000	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480	0.480
27.200	0.477	0.477	0.477	0.477	0.477	0.477	0.477	0.477	0.477
27.400	0.473	0.473	0.473	0.473	0.473	0.473	0.473	0.473	0.473
27.600	0.470	0.470	0.470	0.470	0.470	0.470	0.470	0.470	0.470
27.800	0.467	0.467	0.467	0.467	0.467	0.467	0.467	0.467	0.467
28.000	0.463	0.463	0.463	0.463	0.463	0.463	0.463	0.463	0.463
28.200	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460	0.460
28.400	0.457	0.457	0.457	0.457	0.457	0.457	0.457	0.457	0.457
28.600	0.453	0.453	0.453	0.453	0.453	0.453	0.453	0.453	0.453
28.800	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450	0.450
29.000	0.447	0.447	0.447	0.447	0.447	0.447	0.447	0.447	0.447
29.200	0.444	0.444	0.444	0.444	0.444	0.444	0.444	0.444	0.444
29.400	0.441	0.441	0.441	0.441	0.441	0.441	0.441	0.441	0.441
29.600	0.438	0.438	0.438	0.438	0.438	0.438	0.438	0.438	0.438
29.800	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435	0.435
30.000	0.432	0.432	0.432	0.432	0.432	0.432	0.432	0.432	0.432
30.200	0.429	0.429	0.429	0.429	0.429	0.429	0.429	0.429	0.429
30.400	0.427	0.427	0.427	0.427	0.427	0.427	0.427	0.427	0.427
30.600	0.424	0.424	0.424	0.424	0.424	0.424	0.424	0.424	0.424
30.800	0.421	0.421	0.421	0.421	0.421	0.421	0.421	0.421	0.421
31.000	0.418	0.418	0.418	0.418	0.418	0.418	0.418	0.418	0.418
31.200	0.416	0.416	0.416	0.416	0.416	0.416	0.416	0.416	0.416
31.400	0.413	0.413	0.413	0.413	0.413	0.413	0.413	0.413	0.413
31.600	0.410	0.410	0.410	0.410	0.410	0.410	0.410	0.410	0.410
31.800	0.408	0.408	0.408	0.408	0.408	0.408	0.408	0.408	0.408
32.000	0.405	0.405	0.405	0.405	0.405	0.405	0.405	0.405	0.405

5.2 Cross curves

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: -2.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	0.000	13.226	15.588	16.757	17.198	17.113	16.253	14.811	12.968
0.100	0.000	13.038	15.465	16.651	17.104	17.044	16.205	14.794	12.990
0.200	0.000	12.806	15.293	16.506	16.989	16.969	16.167	14.793	13.028
0.300	0.000	12.541	15.099	16.343	16.862	16.890	16.127	14.792	13.068
0.400	0.000	12.242	14.888	16.165	16.719	16.801	16.081	14.790	13.112
0.500	-0.000	11.903	14.652	15.967	16.560	16.702	16.030	14.790	13.164
0.600	-0.000	11.523	14.388	15.749	16.383	16.593	15.977	14.794	13.228
0.700	-0.000	11.109	14.098	15.513	16.194	16.477	15.924	14.803	13.304
0.800	-0.000	10.675	13.790	15.264	15.998	16.358	15.873	14.818	13.388
0.900	-0.000	10.232	13.473	15.008	15.798	16.240	15.823	14.837	13.479
1.000	-0.000	9.789	13.154	14.749	15.599	16.124	15.778	14.859	13.573
1.100	-0.000	9.353	12.840	14.494	15.403	16.010	15.735	14.884	13.666
1.200	-0.000	8.928	12.532	14.245	15.211	15.899	15.696	14.911	13.750
1.300	-0.000	8.520	12.233	14.002	15.024	15.791	15.658	14.940	13.811
1.400	-0.000	8.128	11.942	13.766	14.841	15.686	15.621	14.969	13.854
1.500	-0.000	7.754	11.658	13.538	14.661	15.585	15.587	14.998	13.881
1.600	-0.000	7.400	11.384	13.317	14.482	15.486	15.553	15.027	13.896
1.700	-0.000	7.066	11.119	13.103	14.307	15.391	15.520	15.051	13.900
1.800	-0.000	6.753	10.863	12.896	14.136	15.298	15.489	15.062	13.896
1.900	-0.000	6.460	10.617	12.694	13.969	15.208	15.459	15.061	13.885
2.000	-0.000	6.186	10.380	12.500	13.807	15.120	15.429	15.050	13.869
2.100	-0.000	5.933	10.150	12.311	13.651	15.035	15.401	15.031	13.849
2.200	-0.000	5.699	9.928	12.128	13.499	14.952	15.373	15.005	13.826
2.300	-0.000	5.484	9.712	11.951	13.352	14.871	15.346	14.973	13.801
2.400	-0.001	5.286	9.502	11.779	13.209	14.791	15.318	14.935	13.775
2.500	-0.001	5.102	9.298	11.612	13.070	14.713	15.285	14.894	13.747
2.600	-0.001	4.932	9.100	11.451	12.936	14.637	15.246	14.848	13.719
2.700	-0.001	4.775	8.907	11.294	12.806	14.562	15.202	14.800	13.691
2.800	-0.001	4.628	8.719	11.141	12.680	14.489	15.152	14.748	13.662
2.900	-0.001	4.489	8.536	10.992	12.558	14.418	15.097	14.695	13.633
3.000	-0.001	4.359	8.357	10.847	12.439	14.348	15.038	14.640	13.604
3.100	-0.001	4.238	8.183	10.706	12.323	14.279	14.976	14.584	13.576
3.200	-0.001	4.124	8.012	10.568	12.210	14.210	14.911	14.526	13.547
3.300	-0.001	4.017	7.846	10.434	12.100	14.141	14.842	14.467	13.519
3.400	-0.001	3.916	7.684	10.302	11.992	14.073	14.770	14.407	13.491
3.500	-0.001	3.822	7.527	10.174	11.887	14.006	14.696	14.346	13.464
3.600	-0.001	3.733	7.374	10.048	11.784	13.939	14.621	14.285	13.437
3.700	-0.001	3.649	7.226	9.925	11.684	13.869	14.544	14.224	13.410
3.800	-0.001	3.569	7.082	9.804	11.585	13.796	14.466	14.162	13.384
3.900	-0.001	3.494	6.944	9.686	11.489	13.722	14.387	14.100	13.359
4.000	-0.001	3.422	6.812	9.570	11.396	13.645	14.308	14.038	13.333
4.100	-0.001	3.355	6.685	9.457	11.303	13.567	14.229	13.974	13.308
4.200	-0.001	3.291	6.564	9.345	11.213	13.487	14.148	13.912	13.282
4.300	-0.001	3.231	6.449	9.236	11.125	13.406	14.067	13.849	13.257
4.400	-0.001	3.174	6.339	9.128	11.038	13.323	13.985	13.788	13.230
4.500	-0.001	3.119	6.233	9.022	10.953	13.239	13.903	13.726	13.203
4.600	-0.001	3.067	6.132	8.918	10.870	13.154	13.818	13.665	13.175
4.700	-0.001	3.018	6.035	8.816	10.788	13.068	13.733	13.604	13.147
4.800	-0.001	2.971	5.942	8.714	10.708	12.981	13.646	13.542	13.120
4.900	-0.001	2.925	5.854	8.614	10.628	12.892	13.558	13.480	13.092
5.000	-0.001	2.882	5.769	8.516	10.550	12.803	13.470	13.419	13.063
5.100	-0.001	2.841	5.688	8.419	10.474	12.714	13.382	13.357	13.034
5.200	-0.001	2.802	5.610	8.323	10.398	12.623	13.294	13.295	13.004
5.300	-0.001	2.764	5.536	8.229	10.323	12.532	13.207	13.233	12.974

5.400 -0.001 2.728 5.464 8.137 10.249 12.440 13.119 13.170 12.942

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.693	5.395	8.046	10.175	12.347	13.032	13.106	12.908
5.600	-0.001	2.660	5.329	7.957	10.101	12.254	12.944	13.041	12.872
5.700	-0.001	2.628	5.265	7.870	10.027	12.160	12.857	12.975	12.834
5.800	-0.001	2.597	5.203	7.785	9.952	12.065	12.769	12.908	12.793
5.900	-0.001	2.568	5.144	7.702	9.875	11.969	12.682	12.841	12.751
6.000	-0.001	2.539	5.087	7.623	9.796	11.873	12.595	12.775	12.706
6.100	-0.001	2.512	5.032	7.545	9.715	11.776	12.508	12.710	12.657
6.200	-0.000	2.486	4.979	7.470	9.632	11.678	12.420	12.645	12.604
6.300	-0.000	2.460	4.928	7.397	9.548	11.580	12.333	12.580	12.550
6.400	-0.000	2.436	4.879	7.325	9.462	11.481	12.246	12.516	12.493
6.500	-0.000	2.412	4.831	7.256	9.374	11.382	12.158	12.452	12.435
6.600	-0.000	2.389	4.786	7.189	9.286	11.282	12.071	12.389	12.375
6.700	-0.000	2.367	4.741	7.124	9.195	11.181	11.983	12.326	12.314
6.800	-0.000	2.346	4.699	7.060	9.104	11.081	11.895	12.263	12.252
6.900	-0.000	2.325	4.658	6.998	9.011	10.979	11.808	12.200	12.188
7.000	-0.000	2.305	4.618	6.938	8.918	10.878	11.720	12.138	12.125
7.100	-0.000	2.286	4.580	6.879	8.823	10.775	11.632	12.074	12.061
7.200	-0.000	2.268	4.543	6.821	8.728	10.673	11.544	12.010	11.997
7.300	-0.000	2.250	4.507	6.763	8.631	10.569	11.456	11.944	11.933
7.400	-0.000	2.232	4.473	6.706	8.533	10.464	11.367	11.876	11.868
7.500	-0.000	2.216	4.439	6.649	8.434	10.359	11.278	11.807	11.802
7.600	-0.000	2.199	4.407	6.591	8.335	10.254	11.189	11.737	11.735
7.700	-0.000	2.184	4.375	6.533	8.235	10.148	11.100	11.664	11.667
7.800	-0.000	2.169	4.345	6.474	8.134	10.042	11.010	11.590	11.600
7.900	-0.000	2.154	4.315	6.415	8.034	9.935	10.920	11.513	11.531
8.000	-0.000	2.140	4.286	6.354	7.933	9.828	10.828	11.435	11.462
8.100	-0.000	2.126	4.258	6.293	7.833	9.721	10.737	11.354	11.392
8.200	-0.000	2.113	4.230	6.230	7.732	9.612	10.644	11.272	11.322
8.300	-0.000	2.100	4.204	6.167	7.631	9.503	10.552	11.186	11.251
8.400	-0.000	2.088	4.177	6.103	7.530	9.394	10.460	11.100	11.179
8.500	-0.000	2.076	4.151	6.037	7.428	9.284	10.368	11.012	11.106
8.600	-0.000	2.064	4.126	5.971	7.327	9.173	10.275	10.925	11.033
8.700	-0.000	2.053	4.101	5.903	7.226	9.061	10.181	10.836	10.959
8.800	-0.000	2.042	4.076	5.834	7.125	8.949	10.087	10.747	10.885
8.900	-0.000	2.031	4.050	5.764	7.023	8.836	9.992	10.658	10.811
9.000	-0.000	2.020	4.024	5.693	6.922	8.722	9.896	10.567	10.736
9.100	-0.000	2.010	3.998	5.621	6.821	8.608	9.799	10.476	10.660
9.200	-0.000	2.000	3.970	5.548	6.719	8.494	9.701	10.384	10.584
9.300	-0.000	1.990	3.941	5.474	6.618	8.379	9.601	10.292	10.508
9.400	-0.000	1.980	3.910	5.399	6.517	8.264	9.499	10.199	10.431
9.500	-0.000	1.970	3.878	5.323	6.417	8.148	9.396	10.105	10.353
9.600	-0.000	1.960	3.844	5.246	6.316	8.032	9.292	10.011	10.275
9.700	-0.000	1.950	3.808	5.168	6.215	7.917	9.186	9.917	10.197
9.800	-0.000	1.941	3.769	5.089	6.115	7.801	9.078	9.822	10.118
9.900	-0.000	1.931	3.728	5.009	6.014	7.686	8.969	9.727	10.039
10.000	-0.000	1.922	3.685	4.928	5.914	7.570	8.859	9.631	9.960
10.100	-0.000	1.913	3.639	4.847	5.814	7.456	8.747	9.535	9.880
10.200	-0.000	1.903	3.591	4.764	5.713	7.342	8.635	9.438	9.800
10.300	-0.000	1.894	3.540	4.681	5.613	7.228	8.523	9.341	9.720
10.400	-0.000	1.885	3.487	4.596	5.513	7.115	8.411	9.243	9.640
10.500	-0.000	1.875	3.432	4.511	5.413	7.003	8.298	9.145	9.559
10.600	-0.000	1.865	3.374	4.425	5.313	6.892	8.185	9.047	9.478
10.700	-0.000	1.853	3.313	4.338	5.212	6.781	8.072	8.948	9.396
10.800	-0.000	1.840	3.251	4.250	5.112	6.672	7.959	8.849	9.315
10.900	-0.000	1.825	3.185	4.162	5.011	6.563	7.846	8.749	9.234
11.000	-0.000	1.807	3.118	4.072	4.911	6.455	7.733	8.650	9.153
11.100	-0.000	1.786	3.049	3.982	4.811	6.349	7.621	8.550	9.074
11.200	-0.000	1.763	2.978	3.891	4.711	6.244	7.510	8.452	8.996
11.300	-0.000	1.736	2.904	3.800	4.611	6.139	7.398	8.353	8.919
11.400	-0.000	1.704	2.828	3.707	4.510	6.033	7.287	8.255	8.842

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.669	2.749	3.613	4.410	5.927	7.176	8.158	8.766
11.600	-0.000	1.629	2.669	3.518	4.309	5.820	7.065	8.061	8.691
11.700	-0.000	1.584	2.585	3.422	4.208	5.711	6.956	7.965	8.617
11.800	-0.000	1.534	2.500	3.325	4.107	5.601	6.847	7.871	8.544
11.900	-0.000	1.478	2.412	3.227	4.006	5.489	6.739	7.777	8.472
12.000	-0.000	1.416	2.321	3.127	3.904	5.375	6.632	7.684	8.400
12.100	-0.000	1.351	2.231	3.030	3.806	5.267	6.530	7.595	8.334
12.200	-0.000	1.287	2.146	2.940	3.716	5.169	6.440	7.515	8.275
12.300	-0.000	1.225	2.067	2.858	3.635	5.082	6.360	7.444	8.223
12.400	-0.000	1.166	1.995	2.783	3.562	5.005	6.289	7.383	8.178
12.500	-0.000	1.110	1.928	2.716	3.497	4.938	6.228	7.330	8.136
12.600	-0.000	1.058	1.869	2.656	3.439	4.880	6.176	7.285	8.100
12.700	-0.000	1.010	1.815	2.603	3.389	4.830	6.132	7.246	8.068
12.800	-0.000	0.967	1.767	2.556	3.345	4.788	6.095	7.215	8.040
12.900	-0.000	0.928	1.725	2.516	3.308	4.753	6.066	7.189	8.016
13.000	-0.000	0.894	1.689	2.482	3.276	4.725	6.042	7.168	7.997
13.100	-0.000	0.863	1.658	2.453	3.250	4.703	6.024	7.151	7.982
13.200	-0.000	0.837	1.632	2.429	3.228	4.686	6.011	7.138	7.970
13.300	-0.000	0.815	1.611	2.409	3.211	4.674	6.002	7.129	7.961
13.400	-0.000	0.797	1.594	2.394	3.198	4.666	5.995	7.122	7.956
13.500	-0.000	0.783	1.581	2.382	3.188	4.660	5.991	7.117	7.952
13.600	-0.000	0.772	1.571	2.374	3.180	4.656	5.988	7.113	7.950
13.700	-0.000	0.765	1.564	2.368	3.175	4.654	5.986	7.111	7.948
13.800	-0.000	0.759	1.558	2.363	3.171	4.651	5.984	7.109	7.946
13.900	-0.000	0.753	1.553	2.358	3.167	4.649	5.982	7.107	7.945
14.000	-0.000	0.748	1.548	2.353	3.162	4.646	5.980	7.104	7.943
14.100	-0.000	0.742	1.542	2.348	3.158	4.644	5.978	7.102	7.941
14.200	-0.000	0.737	1.537	2.343	3.154	4.641	5.977	7.100	7.940
14.300	-0.000	0.732	1.532	2.339	3.150	4.639	5.975	7.098	7.938
14.400	-0.000	0.727	1.526	2.334	3.146	4.637	5.973	7.096	7.936
14.500	-0.000	0.722	1.521	2.329	3.142	4.634	5.971	7.093	7.935
14.600	-0.000	0.718	1.516	2.325	3.138	4.632	5.969	7.091	7.933
14.700	-0.000	0.714	1.511	2.320	3.134	4.630	5.967	7.089	7.931
14.800	-0.000	0.711	1.506	2.315	3.130	4.627	5.965	7.087	7.929
14.900	-0.000	0.708	1.501	2.311	3.126	4.625	5.963	7.085	7.928
15.000	-0.000	0.706	1.496	2.306	3.122	4.623	5.962	7.082	7.926
15.100	-0.000	0.704	1.491	2.302	3.118	4.621	5.960	7.080	7.924
15.200	-0.000	0.703	1.486	2.297	3.115	4.619	5.958	7.078	7.923
15.300	-0.000	0.703	1.481	2.293	3.111	4.617	5.956	7.076	7.921
15.400	-0.000	0.704	1.477	2.288	3.107	4.615	5.954	7.074	7.919
15.500	-0.000	0.705	1.472	2.284	3.103	4.613	5.952	7.071	7.918
15.600	-0.000	0.705	1.468	2.280	3.100	4.611	5.951	7.069	7.916
15.700	-0.000	0.706	1.463	2.275	3.096	4.609	5.949	7.067	7.914
15.800	-0.000	0.707	1.459	2.271	3.092	4.607	5.947	7.065	7.913
15.900	-0.000	0.708	1.455	2.267	3.089	4.605	5.945	7.062	7.911
16.000	-0.000	0.709	1.452	2.263	3.085	4.603	5.943	7.060	7.909
16.100	-0.000	0.710	1.448	2.259	3.082	4.601	5.941	7.058	7.908
16.200	-0.000	0.710	1.445	2.255	3.078	4.599	5.940	7.056	7.906
16.300	-0.000	0.711	1.442	2.251	3.075	4.598	5.938	7.053	7.904
16.400	-0.000	0.712	1.439	2.247	3.071	4.596	5.936	7.051	7.903
16.500	-0.000	0.713	1.437	2.243	3.068	4.594	5.934	7.049	7.901
16.600	-0.000	0.714	1.435	2.239	3.065	4.592	5.933	7.047	7.899
16.700	-0.000	0.715	1.434	2.236	3.061	4.591	5.931	7.044	7.898
16.800	-0.000	0.716	1.433	2.232	3.058	4.589	5.929	7.042	7.896
16.900	-0.000	0.717	1.433	2.229	3.055	4.588	5.927	7.040	7.895
17.000	-0.000	0.717	1.433	2.225	3.052	4.586	5.926	7.038	7.893
17.100	-0.000	0.718	1.434	2.222	3.049	4.585	5.924	7.035	7.891
17.200	-0.000	0.719	1.436	2.219	3.046	4.583	5.922	7.033	7.890
17.300	-0.000	0.720	1.437	2.216	3.043	4.582	5.921	7.031	7.888
17.400	-0.000	0.721	1.439	2.213	3.040	4.580	5.919	7.029	7.887

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.722	1.441	2.210	3.037	4.579	5.917	7.026	7.885
17.600	-0.000	0.723	1.443	2.208	3.034	4.578	5.916	7.024	7.884
17.700	-0.000	0.724	1.445	2.205	3.032	4.576	5.914	7.022	7.882
17.800	-0.000	0.725	1.447	2.203	3.029	4.575	5.912	7.020	7.881
17.900	-0.000	0.726	1.449	2.201	3.026	4.574	5.911	7.018	7.879
18.000	-0.000	0.727	1.451	2.199	3.024	4.572	5.909	7.015	7.878
18.100	-0.000	0.728	1.453	2.198	3.022	4.571	5.907	7.013	7.876
18.200	-0.000	0.729	1.455	2.197	3.019	4.570	5.906	7.011	7.875
18.300	-0.000	0.730	1.457	2.196	3.017	4.569	5.904	7.009	7.873
18.400	-0.000	0.731	1.459	2.195	3.015	4.567	5.902	7.007	7.872
18.500	-0.000	0.732	1.461	2.195	3.013	4.566	5.901	7.005	7.870
18.600	-0.000	0.733	1.463	2.196	3.011	4.565	5.899	7.003	7.869
18.700	-0.000	0.734	1.465	2.196	3.009	4.564	5.898	7.000	7.867
18.800	-0.000	0.735	1.467	2.197	3.007	4.563	5.896	6.998	7.866
18.900	-0.000	0.736	1.469	2.199	3.006	4.562	5.894	6.996	7.864
19.000	-0.000	0.737	1.472	2.201	3.004	4.561	5.893	6.994	7.863
19.100	-0.000	0.738	1.474	2.203	3.003	4.560	5.891	6.992	7.862
19.200	-0.000	0.740	1.476	2.206	3.001	4.559	5.890	6.990	7.860
19.300	-0.000	0.741	1.478	2.209	3.000	4.558	5.888	6.988	7.859
19.400	-0.000	0.742	1.481	2.213	2.999	4.557	5.887	6.986	7.857
19.500	-0.000	0.743	1.483	2.216	2.998	4.556	5.885	6.984	7.856
19.600	-0.000	0.744	1.485	2.219	2.998	4.555	5.883	6.982	7.854
19.700	-0.000	0.745	1.487	2.223	2.997	4.554	5.882	6.980	7.853
19.800	-0.000	0.746	1.490	2.226	2.997	4.553	5.880	6.978	7.851
19.900	-0.000	0.748	1.492	2.230	2.997	4.553	5.879	6.977	7.850
20.000	-0.000	0.749	1.494	2.233	2.997	4.552	5.877	6.975	7.848
20.100	-0.000	0.750	1.497	2.237	2.998	4.551	5.876	6.973	7.847
20.200	-0.000	0.751	1.499	2.241	2.998	4.550	5.874	6.971	7.846
20.300	-0.000	0.752	1.502	2.244	2.999	4.549	5.873	6.969	7.844
20.400	-0.000	0.754	1.504	2.248	3.001	4.549	5.871	6.968	7.843
20.500	-0.000	0.755	1.506	2.252	3.002	4.548	5.870	6.966	7.842
20.600	-0.000	0.756	1.509	2.255	3.004	4.548	5.868	6.964	7.840
20.700	-0.000	0.757	1.511	2.259	3.007	4.547	5.867	6.962	7.839
20.800	-0.000	0.759	1.514	2.263	3.009	4.546	5.865	6.961	7.839
20.900	-0.000	0.760	1.516	2.267	3.013	4.546	5.864	6.959	7.838
21.000	-0.000	0.761	1.519	2.270	3.016	4.545	5.862	6.957	7.836
21.100	-0.000	0.762	1.522	2.274	3.020	4.545	5.861	6.955	7.835
21.200	-0.000	0.764	1.524	2.278	3.024	4.544	5.859	6.954	7.834
21.300	-0.000	0.765	1.527	2.282	3.029	4.544	5.858	6.952	7.833
21.400	-0.000	0.766	1.529	2.286	3.034	4.543	5.857	6.950	7.832
21.500	-0.000	0.768	1.532	2.290	3.039	4.543	5.855	6.949	7.830
21.600	-0.000	0.769	1.535	2.294	3.044	4.543	5.854	6.947	7.829
21.700	-0.000	0.770	1.538	2.298	3.050	4.542	5.852	6.945	7.828
21.800	-0.000	0.772	1.541	2.302	3.055	4.542	5.851	6.944	7.827
21.900	-0.000	0.773	1.543	2.307	3.061	4.542	5.850	6.942	7.826
22.000	-0.000	0.775	1.546	2.311	3.066	4.542	5.848	6.941	7.825
22.100	-0.000	0.776	1.549	2.315	3.072	4.542	5.847	6.940	7.824
22.200	-0.000	0.777	1.552	2.319	3.078	4.542	5.845	6.939	7.823
22.300	-0.000	0.779	1.554	2.324	3.083	4.542	5.844	6.937	7.822
22.400	-0.000	0.780	1.557	2.328	3.089	4.542	5.843	6.936	7.821
22.500	-0.000	0.782	1.560	2.333	3.094	4.542	5.841	6.934	7.820
22.600	-0.000	0.783	1.563	2.336	3.099	4.542	5.840	6.933	7.820
22.700	-0.000	0.785	1.566	2.341	3.105	4.543	5.838	6.932	7.819
22.800	-0.000	0.786	1.568	2.345	3.110	4.543	5.837	6.930	7.818
22.900	-0.000	0.788	1.572	2.350	3.115	4.543	5.837	6.929	7.817
23.000	-0.000	0.789	1.575	2.354	3.120	4.545	5.835	6.928	7.816
23.100	-0.000	0.791	1.578	2.359	3.125	4.546	5.834	6.927	7.815
23.200	-0.000	0.792	1.581	2.363	3.130	4.546	5.833	6.926	7.815
23.300	-0.000	0.793	1.584	2.368	3.134	4.547	5.831	6.924	7.814
23.400	-0.000	0.795	1.587	2.372	3.139	4.548	5.830	6.923	7.813

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.797	1.590	2.377	3.143	4.549	5.829	6.922	7.812
23.600	-0.000	0.798	1.593	2.382	3.147	4.551	5.828	6.921	7.812
23.700	-0.000	0.800	1.596	2.386	3.152	4.552	5.827	6.920	7.811
23.800	-0.000	0.801	1.600	2.391	3.156	4.554	5.826	6.919	7.810
23.900	-0.000	0.803	1.603	2.396	3.160	4.556	5.825	6.919	7.810
24.000	-0.000	0.805	1.606	2.400	3.164	4.558	5.825	6.918	7.809
24.100	-0.000	0.806	1.609	2.405	3.170	4.560	5.824	6.917	7.808
24.200	-0.000	0.808	1.612	2.410	3.172	4.562	5.823	6.916	7.808
24.300	-0.000	0.809	1.616	2.414	3.176	4.567	5.823	6.915	7.807
24.400	-0.000	0.811	1.619	2.419	3.180	4.567	5.822	6.915	7.807
24.500	-0.000	0.813	1.622	2.423	3.182	4.570	5.822	6.914	7.806
24.600	-0.000	0.814	1.626	2.428	3.187	4.573	5.821	6.914	7.806
24.700	-0.000	0.816	1.629	2.432	3.191	4.576	5.821	6.913	7.806
24.800	-0.000	0.818	1.632	2.434	3.195	4.579	5.822	6.912	7.805
24.900	-0.000	0.820	1.636	2.440	3.198	4.583	5.822	6.913	7.805
25.000	-0.000	0.821	1.639	2.443	3.202	4.586	5.823	6.913	7.805
25.100	-0.000	0.823	1.643	2.447	3.205	4.589	5.823	6.912	7.804
25.200	-0.000	0.824	1.646	2.450	3.209	4.592	5.823	6.912	7.804
25.300	-0.000	0.826	1.649	2.453	3.212	4.595	5.824	6.912	7.804
25.400	-0.000	0.828	1.653	2.457	3.215	4.598	5.825	6.912	7.804
25.500	-0.000	0.830	1.656	2.460	3.218	4.601	5.826	6.911	7.804
25.600	-0.000	0.832	1.660	2.463	3.219	4.604	5.827	6.911	7.804
25.700	-0.000	0.834	1.664	2.466	3.222	4.606	5.828	6.912	7.804
25.800	-0.000	0.835	1.667	2.469	3.227	4.609	5.829	6.912	7.804
25.900	-0.000	0.837	1.671	2.472	3.230	4.611	5.830	6.912	7.804
26.000	-0.000	0.839	1.674	2.475	3.232	4.611	5.832	6.912	7.804

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: -1.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	0.000	13.990	16.196	17.163	17.445	17.229	16.227	14.672	12.718
0.100	0.000	13.799	16.084	17.050	17.357	17.155	16.175	14.647	12.727
0.200	0.000	13.533	15.888	16.891	17.236	17.072	16.134	14.645	12.764
0.300	0.000	13.217	15.654	16.704	17.090	16.977	16.087	14.643	12.810
0.400	-0.000	12.838	15.378	16.482	16.913	16.865	16.032	14.644	12.870
0.500	-0.000	12.390	15.058	16.227	16.709	16.739	15.974	14.653	12.948
0.600	-0.000	11.895	14.703	15.950	16.487	16.605	15.916	14.669	13.043
0.700	-0.000	11.376	14.330	15.661	16.260	16.469	15.860	14.692	13.149
0.800	-0.000	10.858	13.955	15.372	16.032	16.338	15.808	14.720	13.261
0.900	-0.000	10.353	13.587	15.088	15.810	16.211	15.761	14.750	13.374
1.000	-0.000	9.865	13.230	14.809	15.594	16.087	15.716	14.781	13.486
1.100	-0.000	9.401	12.886	14.536	15.385	15.968	15.675	14.814	13.596
1.200	-0.000	8.960	12.556	14.269	15.184	15.852	15.636	14.847	13.686
1.300	-0.000	8.541	12.239	14.012	14.990	15.741	15.598	14.879	13.751
1.400	-0.000	8.142	11.934	13.764	14.802	15.634	15.563	14.911	13.796
1.500	-0.000	7.763	11.642	13.526	14.621	15.531	15.528	14.943	13.826
1.600	-0.001	7.403	11.362	13.297	14.446	15.431	15.495	14.971	13.842
1.700	-0.001	7.062	11.093	13.077	14.275	15.334	15.463	14.991	13.848
1.800	-0.001	6.742	10.836	12.864	14.105	15.241	15.432	15.001	13.845
1.900	-0.001	6.444	10.588	12.659	13.938	15.150	15.402	15.001	13.836
2.000	-0.001	6.169	10.350	12.461	13.775	15.062	15.373	14.991	13.822
2.100	-0.001	5.915	10.120	12.271	13.615	14.976	15.345	14.973	13.805
2.200	-0.001	5.682	9.899	12.088	13.461	14.893	15.317	14.948	13.784
2.300	-0.001	5.469	9.684	11.911	13.311	14.812	15.289	14.916	13.762
2.400	-0.001	5.271	9.476	11.739	13.167	14.733	15.258	14.880	13.738
2.500	-0.001	5.089	9.274	11.573	13.027	14.656	15.224	14.840	13.713
2.600	-0.001	4.919	9.076	11.412	12.892	14.581	15.185	14.796	13.688
2.700	-0.001	4.762	8.885	11.255	12.762	14.506	15.141	14.749	13.662
2.800	-0.001	4.615	8.698	11.103	12.635	14.433	15.093	14.700	13.636
2.900	-0.001	4.479	8.516	10.955	12.513	14.362	15.040	14.649	13.609
3.000	-0.001	4.351	8.339	10.811	12.394	14.292	14.983	14.596	13.583
3.100	-0.001	4.230	8.165	10.670	12.279	14.224	14.922	14.541	13.557
3.200	-0.001	4.116	7.995	10.533	12.167	14.157	14.858	14.485	13.531
3.300	-0.001	4.009	7.829	10.399	12.057	14.090	14.792	14.428	13.505
3.400	-0.001	3.909	7.667	10.268	11.950	14.024	14.722	14.370	13.479
3.500	-0.001	3.814	7.509	10.140	11.845	13.959	14.651	14.312	13.454
3.600	-0.001	3.725	7.356	10.016	11.743	13.892	14.577	14.253	13.429
3.700	-0.001	3.641	7.208	9.893	11.644	13.824	14.501	14.193	13.404
3.800	-0.001	3.562	7.065	9.774	11.546	13.753	14.424	14.133	13.380
3.900	-0.001	3.487	6.928	9.657	11.451	13.681	14.347	14.074	13.356
4.000	-0.001	3.415	6.797	9.542	11.358	13.606	14.268	14.013	13.332
4.100	-0.001	3.348	6.671	9.429	11.266	13.529	14.190	13.952	13.309
4.200	-0.001	3.284	6.550	9.319	11.177	13.450	14.111	13.891	13.286
4.300	-0.001	3.223	6.435	9.210	11.089	13.371	14.032	13.830	13.262
4.400	-0.001	3.166	6.325	9.103	11.003	13.289	13.952	13.769	13.237
4.500	-0.001	3.112	6.220	8.999	10.919	13.207	13.872	13.709	13.211
4.600	-0.001	3.060	6.119	8.895	10.836	13.123	13.791	13.649	13.184
4.700	-0.001	3.011	6.023	8.794	10.755	13.039	13.709	13.589	13.156
4.800	-0.001	2.964	5.930	8.694	10.675	12.953	13.625	13.529	13.128
4.900	-0.001	2.919	5.841	8.595	10.596	12.866	13.541	13.469	13.100
5.000	-0.001	2.876	5.756	8.497	10.518	12.778	13.455	13.409	13.072
5.100	-0.001	2.835	5.675	8.401	10.442	12.690	13.369	13.349	13.043
5.200	-0.001	2.795	5.597	8.305	10.367	12.601	13.282	13.289	13.014
5.300	-0.001	2.758	5.523	8.212	10.293	12.511	13.196	13.229	12.984
5.400	-0.001	2.722	5.451	8.119	10.220	12.420	13.110	13.169	12.954

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.687	5.382	8.028	10.147	12.329	13.024	13.108	12.922
5.600	-0.001	2.654	5.316	7.939	10.075	12.237	12.938	13.046	12.889
5.700	-0.001	2.622	5.252	7.852	10.004	12.144	12.852	12.983	12.853
5.800	-0.001	2.591	5.190	7.766	9.931	12.051	12.766	12.919	12.815
5.900	-0.001	2.562	5.131	7.684	9.856	11.957	12.680	12.854	12.776
6.000	-0.001	2.533	5.074	7.603	9.779	11.862	12.594	12.789	12.734
6.100	-0.001	2.505	5.019	7.525	9.700	11.766	12.508	12.725	12.690
6.200	-0.000	2.479	4.966	7.450	9.619	11.670	12.422	12.661	12.642
6.300	-0.000	2.453	4.915	7.377	9.536	11.574	12.336	12.598	12.592
6.400	-0.000	2.429	4.866	7.306	9.452	11.477	12.250	12.535	12.538
6.500	-0.000	2.405	4.818	7.237	9.366	11.379	12.165	12.472	12.482
6.600	-0.000	2.382	4.772	7.171	9.279	11.281	12.079	12.410	12.425
6.700	-0.000	2.360	4.728	7.106	9.190	11.183	11.993	12.348	12.366
6.800	-0.000	2.339	4.686	7.043	9.100	11.084	11.907	12.287	12.306
6.900	-0.000	2.318	4.645	6.982	9.009	10.984	11.822	12.225	12.244
7.000	-0.000	2.299	4.605	6.923	8.917	10.885	11.736	12.164	12.181
7.100	-0.000	2.279	4.567	6.865	8.823	10.784	11.650	12.102	12.118
7.200	-0.000	2.261	4.530	6.808	8.729	10.683	11.563	12.040	12.054
7.300	-0.000	2.243	4.494	6.753	8.634	10.582	11.477	11.977	11.991
7.400	-0.000	2.226	4.460	6.698	8.538	10.481	11.390	11.913	11.927
7.500	-0.000	2.209	4.426	6.643	8.441	10.378	11.304	11.847	11.863
7.600	-0.000	2.193	4.394	6.588	8.343	10.275	11.216	11.780	11.798
7.700	-0.000	2.177	4.363	6.532	8.245	10.171	11.129	11.711	11.732
7.800	-0.000	2.162	4.333	6.475	8.146	10.066	11.041	11.641	11.666
7.900	-0.000	2.147	4.304	6.417	8.048	9.961	10.953	11.568	11.598
8.000	-0.000	2.133	4.275	6.358	7.949	9.856	10.864	11.494	11.530
8.100	-0.000	2.120	4.247	6.298	7.850	9.751	10.775	11.417	11.462
8.200	-0.000	2.107	4.220	6.237	7.751	9.645	10.685	11.339	11.392
8.300	-0.000	2.094	4.194	6.176	7.652	9.539	10.595	11.258	11.323
8.400	-0.000	2.081	4.168	6.113	7.553	9.432	10.504	11.175	11.252
8.500	-0.000	2.070	4.143	6.049	7.454	9.324	10.413	11.090	11.181
8.600	-0.000	2.058	4.119	5.984	7.355	9.215	10.322	11.004	11.109
8.700	-0.000	2.047	4.094	5.919	7.255	9.106	10.231	10.918	11.037
8.800	-0.000	2.036	4.071	5.852	7.156	8.996	10.139	10.831	10.964
8.900	-0.000	2.025	4.047	5.784	7.057	8.885	10.047	10.743	10.891
9.000	-0.000	2.015	4.024	5.715	6.957	8.774	9.954	10.655	10.817
9.100	-0.000	2.005	4.000	5.645	6.858	8.662	9.861	10.565	10.742
9.200	-0.000	1.995	3.975	5.573	6.758	8.550	9.766	10.475	10.667
9.300	-0.000	1.985	3.948	5.501	6.659	8.437	9.670	10.384	10.591
9.400	-0.000	1.976	3.921	5.428	6.560	8.323	9.572	10.293	10.515
9.500	-0.000	1.966	3.891	5.354	6.461	8.209	9.474	10.201	10.439
9.600	-0.000	1.957	3.859	5.279	6.361	8.095	9.373	10.108	10.362
9.700	-0.000	1.948	3.826	5.203	6.262	7.981	9.271	10.015	10.285
9.800	-0.000	1.939	3.789	5.125	6.163	7.866	9.168	9.922	10.207
9.900	-0.000	1.930	3.751	5.047	6.065	7.752	9.063	9.827	10.129
10.000	-0.000	1.921	3.710	4.969	5.966	7.638	8.956	9.733	10.051
10.100	-0.000	1.912	3.666	4.889	5.867	7.524	8.848	9.638	9.972
10.200	-0.000	1.903	3.620	4.808	5.768	7.411	8.739	9.542	9.893
10.300	-0.000	1.895	3.571	4.726	5.669	7.299	8.628	9.446	9.813
10.400	-0.000	1.886	3.520	4.644	5.571	7.187	8.517	9.349	9.733
10.500	-0.000	1.878	3.467	4.560	5.472	7.075	8.406	9.253	9.652
10.600	-0.000	1.869	3.411	4.476	5.373	6.965	8.295	9.155	9.572
10.700	-0.000	1.860	3.353	4.391	5.274	6.855	8.184	9.057	9.490
10.800	-0.000	1.850	3.292	4.305	5.175	6.747	8.072	8.959	9.409
10.900	-0.000	1.838	3.230	4.218	5.075	6.639	7.960	8.861	9.327
11.000	-0.000	1.825	3.165	4.130	4.976	6.531	7.848	8.762	9.246
11.100	-0.000	1.808	3.097	4.041	4.877	6.425	7.736	8.662	9.165
11.200	-0.000	1.789	3.027	3.952	4.777	6.320	7.624	8.563	9.085
11.300	-0.000	1.767	2.956	3.861	4.678	6.216	7.513	8.463	9.005
11.400	-0.000	1.740	2.882	3.770	4.579	6.113	7.402	8.365	8.927

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.709	2.806	3.678	4.480	6.011	7.292	8.267	8.850
11.600	-0.000	1.673	2.728	3.585	4.380	5.907	7.182	8.169	8.774
11.700	-0.000	1.632	2.647	3.491	4.281	5.803	7.072	8.073	8.698
11.800	-0.000	1.586	2.564	3.395	4.181	5.698	6.963	7.977	8.624
11.900	-0.000	1.534	2.479	3.299	4.081	5.591	6.855	7.881	8.550
12.000	-0.000	1.476	2.391	3.201	3.980	5.482	6.748	7.787	8.477
12.100	-0.000	1.413	2.300	3.102	3.880	5.371	6.642	7.694	8.406
12.200	-0.000	1.343	2.206	3.002	3.779	5.260	6.536	7.601	8.335
12.300	-0.000	1.268	2.111	2.902	3.679	5.149	6.433	7.509	8.267
12.400	-0.000	1.195	2.021	2.809	3.588	5.051	6.340	7.427	8.207
12.500	-0.000	1.126	1.940	2.727	3.509	4.966	6.261	7.358	8.156
12.600	-0.000	1.062	1.868	2.656	3.440	4.895	6.196	7.300	8.112
12.700	-0.000	1.005	1.805	2.594	3.381	4.835	6.142	7.253	8.073
12.800	-0.000	0.954	1.751	2.542	3.333	4.787	6.099	7.216	8.042
12.900	-0.000	0.910	1.706	2.498	3.292	4.749	6.066	7.186	8.016
13.000	-0.000	0.873	1.668	2.463	3.260	4.720	6.042	7.164	7.996
13.100	-0.000	0.842	1.637	2.434	3.234	4.699	6.024	7.148	7.981
13.200	-0.000	0.816	1.613	2.412	3.215	4.685	6.013	7.137	7.971
13.300	-0.000	0.797	1.595	2.396	3.201	4.676	6.006	7.129	7.965
13.400	-0.000	0.783	1.582	2.385	3.191	4.671	6.002	7.124	7.962
13.500	-0.000	0.774	1.574	2.377	3.184	4.667	5.999	7.121	7.960
13.600	-0.000	0.768	1.568	2.372	3.180	4.665	5.997	7.119	7.958
13.700	-0.000	0.762	1.563	2.367	3.176	4.663	5.996	7.117	7.956
13.800	-0.000	0.756	1.557	2.363	3.172	4.660	5.994	7.115	7.955
13.900	-0.000	0.750	1.552	2.358	3.168	4.658	5.992	7.112	7.953
14.000	-0.000	0.745	1.547	2.353	3.164	4.655	5.990	7.110	7.951
14.100	-0.000	0.739	1.541	2.348	3.159	4.653	5.988	7.108	7.950
14.200	-0.000	0.734	1.536	2.343	3.155	4.651	5.986	7.106	7.948
14.300	-0.000	0.729	1.531	2.339	3.151	4.648	5.984	7.104	7.946
14.400	-0.000	0.724	1.525	2.334	3.147	4.646	5.982	7.101	7.945
14.500	-0.000	0.719	1.520	2.329	3.143	4.644	5.980	7.099	7.943
14.600	-0.000	0.715	1.515	2.325	3.139	4.641	5.978	7.097	7.941
14.700	-0.000	0.711	1.510	2.320	3.135	4.639	5.977	7.095	7.940
14.800	-0.000	0.708	1.505	2.316	3.132	4.637	5.975	7.092	7.938
14.900	-0.000	0.706	1.500	2.311	3.128	4.635	5.973	7.090	7.936
15.000	-0.000	0.704	1.495	2.306	3.124	4.633	5.971	7.088	7.935
15.100	-0.000	0.703	1.490	2.302	3.120	4.631	5.969	7.086	7.933
15.200	-0.000	0.703	1.485	2.298	3.116	4.629	5.967	7.084	7.931
15.300	-0.000	0.704	1.480	2.293	3.112	4.627	5.965	7.081	7.930
15.400	-0.000	0.705	1.476	2.289	3.109	4.625	5.963	7.079	7.928
15.500	-0.000	0.706	1.471	2.284	3.105	4.623	5.962	7.077	7.926
15.600	-0.000	0.707	1.467	2.280	3.101	4.621	5.960	7.075	7.925
15.700	-0.000	0.707	1.463	2.276	3.098	4.619	5.958	7.072	7.923
15.800	-0.000	0.708	1.458	2.272	3.094	4.617	5.956	7.070	7.921
15.900	-0.000	0.709	1.455	2.268	3.091	4.615	5.954	7.068	7.920
16.000	-0.000	0.710	1.451	2.264	3.087	4.613	5.953	7.065	7.918
16.100	-0.000	0.711	1.447	2.260	3.084	4.612	5.951	7.063	7.916
16.200	-0.000	0.712	1.444	2.256	3.080	4.610	5.949	7.061	7.915
16.300	-0.000	0.712	1.441	2.252	3.077	4.608	5.947	7.059	7.913
16.400	-0.000	0.713	1.439	2.248	3.073	4.607	5.946	7.056	7.911
16.500	-0.000	0.714	1.437	2.244	3.070	4.605	5.944	7.054	7.910
16.600	-0.000	0.715	1.435	2.240	3.067	4.604	5.942	7.052	7.908
16.700	-0.000	0.716	1.434	2.237	3.064	4.602	5.940	7.050	7.907
16.800	-0.000	0.717	1.434	2.233	3.060	4.600	5.939	7.047	7.905
16.900	-0.000	0.718	1.434	2.230	3.057	4.599	5.937	7.045	7.903
17.000	-0.000	0.719	1.435	2.227	3.054	4.597	5.935	7.043	7.902
17.100	-0.000	0.720	1.436	2.224	3.051	4.596	5.933	7.041	7.900
17.200	-0.000	0.721	1.438	2.220	3.048	4.595	5.932	7.038	7.899
17.300	-0.000	0.722	1.440	2.217	3.046	4.593	5.930	7.036	7.897
17.400	-0.000	0.723	1.442	2.215	3.043	4.592	5.928	7.034	7.896

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.724	1.444	2.212	3.040	4.590	5.927	7.032	7.894
17.600	-0.000	0.724	1.446	2.210	3.037	4.589	5.925	7.029	7.893
17.700	-0.000	0.725	1.448	2.207	3.035	4.588	5.923	7.027	7.891
17.800	-0.000	0.726	1.450	2.205	3.032	4.587	5.922	7.025	7.890
17.900	-0.000	0.727	1.452	2.203	3.030	4.585	5.920	7.023	7.888
18.000	-0.000	0.728	1.454	2.201	3.027	4.584	5.918	7.021	7.887
18.100	-0.000	0.729	1.456	2.200	3.025	4.583	5.917	7.018	7.885
18.200	-0.000	0.731	1.458	2.199	3.023	4.582	5.915	7.016	7.884
18.300	-0.000	0.732	1.460	2.198	3.021	4.581	5.914	7.014	7.882
18.400	-0.000	0.733	1.462	2.198	3.018	4.579	5.912	7.012	7.881
18.500	-0.000	0.734	1.464	2.198	3.017	4.578	5.910	7.010	7.879
18.600	-0.000	0.735	1.466	2.198	3.015	4.577	5.909	7.008	7.878
18.700	-0.000	0.736	1.468	2.199	3.013	4.576	5.907	7.006	7.876
18.800	-0.000	0.737	1.471	2.201	3.011	4.575	5.905	7.003	7.875
18.900	-0.000	0.738	1.473	2.203	3.010	4.574	5.904	7.001	7.873
19.000	-0.000	0.739	1.475	2.205	3.008	4.573	5.902	6.999	7.872
19.100	-0.000	0.740	1.477	2.208	3.007	4.572	5.901	6.997	7.870
19.200	-0.000	0.741	1.479	2.211	3.006	4.571	5.899	6.995	7.869
19.300	-0.000	0.742	1.482	2.214	3.005	4.570	5.897	6.993	7.867
19.400	-0.000	0.744	1.484	2.218	3.004	4.569	5.896	6.991	7.866
19.500	-0.000	0.745	1.486	2.221	3.003	4.569	5.894	6.989	7.864
19.600	-0.000	0.746	1.489	2.225	3.003	4.568	5.893	6.988	7.863
19.700	-0.000	0.747	1.491	2.228	3.002	4.567	5.891	6.986	7.861
19.800	-0.000	0.748	1.493	2.232	3.002	4.566	5.890	6.984	7.860
19.900	-0.000	0.749	1.496	2.235	3.002	4.565	5.888	6.982	7.858
20.000	-0.000	0.751	1.498	2.239	3.002	4.564	5.886	6.980	7.857
20.100	-0.000	0.752	1.500	2.242	3.003	4.564	5.885	6.978	7.856
20.200	-0.000	0.753	1.503	2.246	3.004	4.563	5.883	6.976	7.854
20.300	-0.000	0.754	1.505	2.250	3.005	4.562	5.882	6.975	7.853
20.400	-0.000	0.755	1.508	2.254	3.006	4.562	5.880	6.973	7.851
20.500	-0.000	0.757	1.510	2.257	3.008	4.561	5.879	6.971	7.851
20.600	-0.000	0.758	1.513	2.261	3.010	4.560	5.877	6.969	7.850
20.700	-0.000	0.759	1.515	2.265	3.013	4.560	5.876	6.968	7.849
20.800	-0.000	0.760	1.518	2.269	3.016	4.559	5.874	6.966	7.847
20.900	-0.000	0.762	1.520	2.273	3.019	4.559	5.873	6.964	7.846
21.000	-0.000	0.763	1.523	2.276	3.023	4.558	5.871	6.962	7.845
21.100	-0.000	0.764	1.526	2.280	3.027	4.558	5.870	6.960	7.844
21.200	-0.000	0.766	1.528	2.284	3.032	4.557	5.868	6.959	7.843
21.300	-0.000	0.767	1.531	2.288	3.037	4.557	5.867	6.957	7.841
21.400	-0.000	0.769	1.534	2.292	3.042	4.556	5.865	6.955	7.840
21.500	-0.000	0.770	1.537	2.296	3.047	4.556	5.864	6.953	7.839
21.600	-0.000	0.771	1.539	2.301	3.053	4.556	5.862	6.952	7.838
21.700	-0.000	0.772	1.542	2.305	3.058	4.555	5.861	6.950	7.837
21.800	-0.000	0.774	1.545	2.309	3.064	4.555	5.860	6.950	7.836
21.900	-0.000	0.775	1.547	2.313	3.069	4.555	5.858	6.948	7.835
22.000	-0.000	0.777	1.550	2.318	3.076	4.555	5.857	6.947	7.834
22.100	-0.000	0.778	1.553	2.322	3.081	4.555	5.855	6.945	7.833
22.200	-0.000	0.780	1.556	2.327	3.087	4.555	5.854	6.944	7.832
22.300	-0.000	0.781	1.559	2.330	3.092	4.555	5.852	6.942	7.831
22.400	-0.000	0.782	1.562	2.334	3.098	4.555	5.851	6.941	7.830
22.500	-0.000	0.784	1.565	2.339	3.104	4.555	5.849	6.939	7.829
22.600	-0.000	0.785	1.568	2.343	3.109	4.555	5.849	6.938	7.828
22.700	-0.000	0.787	1.571	2.348	3.114	4.556	5.847	6.937	7.827
22.800	-0.000	0.788	1.574	2.352	3.120	4.556	5.846	6.936	7.826
22.900	-0.000	0.790	1.576	2.357	3.125	4.557	5.844	6.934	7.825
23.000	-0.000	0.791	1.580	2.362	3.130	4.557	5.843	6.933	7.825
23.100	-0.000	0.793	1.583	2.366	3.136	4.558	5.842	6.932	7.824
23.200	-0.000	0.794	1.586	2.370	3.142	4.558	5.840	6.931	7.823
23.300	-0.000	0.796	1.589	2.375	3.147	4.559	5.839	6.930	7.822
23.400	-0.000	0.797	1.592	2.378	3.152	4.560	5.838	6.928	7.821

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.799	1.595	2.384	3.157	4.561	5.837	6.927	7.821
23.600	-0.000	0.801	1.598	2.389	3.161	4.563	5.836	6.926	7.820
23.700	-0.000	0.802	1.600	2.394	3.166	4.564	5.835	6.925	7.819
23.800	-0.000	0.804	1.605	2.397	3.170	4.566	5.834	6.925	7.818
23.900	-0.000	0.806	1.608	2.403	3.174	4.568	5.833	6.924	7.818
24.000	-0.000	0.807	1.611	2.408	3.178	4.568	5.832	6.923	7.817
24.100	-0.000	0.808	1.614	2.413	3.182	4.572	5.831	6.922	7.816
24.200	-0.000	0.810	1.617	2.418	3.186	4.572	5.831	6.921	7.816
24.300	-0.000	0.812	1.621	2.423	3.190	4.577	5.830	6.921	7.815
24.400	-0.000	0.814	1.624	2.426	3.194	4.579	5.830	6.920	7.815
24.500	-0.000	0.815	1.628	2.432	3.197	4.582	5.830	6.919	7.814
24.600	-0.000	0.817	1.631	2.437	3.201	4.585	5.830	6.919	7.814
24.700	-0.000	0.819	1.634	2.442	3.205	4.588	5.830	6.919	7.813
24.800	-0.000	0.821	1.638	2.444	3.208	4.591	5.830	6.918	7.813
24.900	-0.000	0.822	1.641	2.450	3.212	4.594	5.830	6.917	7.813
25.000	-0.000	0.824	1.645	2.455	3.215	4.597	5.830	6.917	7.812
25.100	-0.000	0.826	1.648	2.459	3.218	4.600	5.830	6.917	7.812
25.200	-0.000	0.827	1.652	2.463	3.221	4.603	5.828	6.917	7.812
25.300	-0.000	0.829	1.655	2.466	3.225	4.606	5.832	6.916	7.812
25.400	-0.000	0.831	1.659	2.469	3.228	4.609	5.832	6.916	7.811
25.500	-0.000	0.833	1.662	2.472	3.230	4.611	5.833	6.916	7.811
25.600	-0.000	0.835	1.666	2.476	3.233	4.614	5.834	6.916	7.811
25.700	-0.000	0.836	1.669	2.479	3.236	4.616	5.835	6.916	7.811
25.800	-0.000	0.838	1.673	2.482	3.239	4.619	5.837	6.916	7.811
25.900	-0.000	0.840	1.677	2.484	3.241	4.618	5.838	6.917	7.811
26.000	-0.000	0.842	1.680	2.487	3.244	4.623	5.840	6.917	7.811

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: -1.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	0.000	14.999	16.945	17.588	17.765	17.358	16.202	14.510	12.422
0.100	0.000	14.784	16.819	17.483	17.680	17.283	16.149	14.478	12.420
0.200	0.000	14.442	16.571	17.298	17.531	17.187	16.102	14.476	12.463
0.300	-0.000	13.976	16.234	17.037	17.324	17.057	16.041	14.482	12.538
0.400	-0.000	13.387	15.816	16.714	17.067	16.901	15.974	14.502	12.648
0.500	-0.000	12.744	15.361	16.366	16.792	16.738	15.910	14.533	12.779
0.600	-0.000	12.105	14.904	16.019	16.521	16.579	15.851	14.570	12.917
0.700	-0.000	11.494	14.456	15.685	16.260	16.430	15.796	14.609	13.053
0.800	-0.000	10.915	14.027	15.367	16.010	16.289	15.746	14.647	13.185
0.900	-0.000	10.371	13.619	15.064	15.772	16.155	15.700	14.685	13.312
1.000	-0.001	9.865	13.235	14.775	15.546	16.028	15.657	14.722	13.433
1.100	-0.001	9.391	12.872	14.500	15.330	15.905	15.615	14.760	13.540
1.200	-0.001	8.946	12.527	14.237	15.123	15.789	15.576	14.796	13.623
1.300	-0.001	8.526	12.199	13.980	14.925	15.677	15.539	14.831	13.687
1.400	-0.001	8.127	11.889	13.728	14.735	15.569	15.503	14.865	13.733
1.500	-0.001	7.748	11.594	13.484	14.552	15.466	15.469	14.896	13.763
1.600	-0.001	7.388	11.314	13.250	14.376	15.365	15.436	14.919	13.782
1.700	-0.001	7.045	11.046	13.026	14.206	15.268	15.405	14.933	13.790
1.800	-0.001	6.721	10.789	12.811	14.042	15.174	15.374	14.939	13.791
1.900	-0.001	6.420	10.543	12.606	13.881	15.084	15.345	14.936	13.785
2.000	-0.001	6.144	10.306	12.408	13.722	14.996	15.316	14.925	13.775
2.100	-0.001	5.892	10.077	12.218	13.564	14.910	15.288	14.907	13.760
2.200	-0.001	5.661	9.857	12.035	13.410	14.827	15.259	14.883	13.743
2.300	-0.001	5.448	9.644	11.859	13.260	14.747	15.229	14.853	13.723
2.400	-0.001	5.252	9.438	11.689	13.115	14.668	15.195	14.819	13.703
2.500	-0.001	5.071	9.238	11.524	12.976	14.591	15.158	14.781	13.681
2.600	-0.001	4.903	9.044	11.365	12.841	14.516	15.118	14.739	13.658
2.700	-0.001	4.747	8.855	11.209	12.711	14.444	15.073	14.694	13.635
2.800	-0.001	4.601	8.670	11.058	12.585	14.372	15.026	14.647	13.611
2.900	-0.001	4.465	8.489	10.911	12.463	14.301	14.974	14.598	13.588
3.000	-0.001	4.338	8.313	10.769	12.345	14.232	14.919	14.547	13.564
3.100	-0.001	4.219	8.141	10.629	12.230	14.164	14.860	14.495	13.541
3.200	-0.001	4.107	7.973	10.493	12.119	14.098	14.798	14.441	13.517
3.300	-0.001	4.001	7.808	10.361	12.010	14.033	14.734	14.386	13.493
3.400	-0.001	3.900	7.647	10.231	11.904	13.970	14.667	14.331	13.470
3.500	-0.001	3.806	7.490	10.104	11.801	13.906	14.598	14.275	13.446
3.600	-0.001	3.717	7.337	9.980	11.700	13.840	14.526	14.218	13.423
3.700	-0.001	3.633	7.188	9.859	11.601	13.774	14.453	14.160	13.400
3.800	-0.001	3.554	7.046	9.740	11.505	13.705	14.378	14.102	13.377
3.900	-0.001	3.479	6.910	9.625	11.410	13.634	14.302	14.044	13.355
4.000	-0.001	3.408	6.780	9.511	11.318	13.561	14.225	13.986	13.333
4.100	-0.001	3.341	6.655	9.400	11.228	13.486	14.148	13.927	13.311
4.200	-0.001	3.277	6.536	9.290	11.139	13.409	14.071	13.868	13.289
4.300	-0.001	3.217	6.421	9.182	11.052	13.331	13.993	13.809	13.266
4.400	-0.001	3.159	6.311	9.077	10.967	13.251	13.916	13.749	13.242
4.500	-0.001	3.104	6.207	8.973	10.883	13.170	13.837	13.690	13.217
4.600	-0.001	3.053	6.106	8.870	10.801	13.088	13.759	13.631	13.191
4.700	-0.001	3.004	6.010	8.770	10.720	13.005	13.679	13.572	13.164
4.800	-0.001	2.957	5.918	8.670	10.640	12.921	13.599	13.514	13.137
4.900	-0.001	2.912	5.829	8.572	10.562	12.835	13.518	13.456	13.108
5.000	-0.001	2.869	5.744	8.476	10.485	12.749	13.435	13.398	13.080
5.100	-0.001	2.829	5.663	8.380	10.410	12.662	13.351	13.340	13.052
5.200	-0.001	2.789	5.585	8.286	10.335	12.574	13.267	13.281	13.023
5.300	-0.001	2.751	5.509	8.192	10.262	12.486	13.182	13.223	12.993
5.400	-0.001	2.715	5.438	8.100	10.189	12.396	13.097	13.164	12.963

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.680	5.369	8.009	10.117	12.306	13.012	13.105	12.932
5.600	-0.001	2.647	5.302	7.920	10.046	12.215	12.928	13.047	12.901
5.700	-0.001	2.615	5.239	7.833	9.975	12.124	12.843	12.987	12.867
5.800	-0.001	2.584	5.177	7.747	9.904	12.032	12.759	12.926	12.832
5.900	-0.001	2.555	5.118	7.664	9.832	11.939	12.674	12.864	12.795
6.000	-0.001	2.526	5.061	7.584	9.757	11.846	12.589	12.802	12.756
6.100	-0.001	2.499	5.006	7.505	9.680	11.752	12.505	12.739	12.715
6.200	-0.000	2.472	4.953	7.430	9.601	11.658	12.421	12.676	12.672
6.300	-0.000	2.447	4.902	7.357	9.520	11.563	12.336	12.614	12.626
6.400	-0.000	2.422	4.853	7.286	9.437	11.468	12.252	12.553	12.576
6.500	-0.000	2.398	4.805	7.218	9.353	11.373	12.168	12.491	12.524
6.600	-0.000	2.376	4.760	7.151	9.267	11.276	12.084	12.430	12.469
6.700	-0.000	2.354	4.716	7.087	9.180	11.180	12.001	12.370	12.412
6.800	-0.000	2.333	4.673	7.025	9.092	11.083	11.917	12.310	12.354
6.900	-0.000	2.312	4.632	6.964	9.002	10.985	11.832	12.249	12.294
7.000	-0.000	2.292	4.592	6.906	8.911	10.887	11.748	12.189	12.234
7.100	-0.000	2.273	4.554	6.848	8.819	10.789	11.664	12.128	12.172
7.200	-0.000	2.254	4.517	6.793	8.726	10.690	11.580	12.067	12.109
7.300	-0.000	2.236	4.481	6.739	8.632	10.591	11.495	12.007	12.047
7.400	-0.000	2.219	4.447	6.686	8.538	10.491	11.410	11.945	11.984
7.500	-0.000	2.203	4.414	6.633	8.442	10.391	11.325	11.882	11.920
7.600	-0.000	2.187	4.381	6.580	8.346	10.290	11.240	11.818	11.857
7.700	-0.000	2.171	4.350	6.526	8.250	10.189	11.154	11.752	11.792
7.800	-0.000	2.156	4.320	6.470	8.153	10.086	11.068	11.685	11.727
7.900	-0.000	2.141	4.291	6.414	8.056	9.983	10.982	11.616	11.661
8.000	-0.000	2.127	4.263	6.357	7.959	9.880	10.896	11.546	11.594
8.100	-0.000	2.113	4.236	6.298	7.862	9.776	10.809	11.473	11.527
8.200	-0.000	2.100	4.209	6.239	7.765	9.673	10.721	11.398	11.459
8.300	-0.000	2.087	4.183	6.179	7.667	9.568	10.633	11.322	11.390
8.400	-0.000	2.075	4.158	6.118	7.570	9.463	10.544	11.243	11.321
8.500	-0.000	2.063	4.134	6.055	7.473	9.358	10.455	11.163	11.251
8.600	-0.000	2.052	4.110	5.992	7.376	9.252	10.366	11.079	11.181
8.700	-0.000	2.041	4.086	5.928	7.279	9.145	10.277	10.995	11.110
8.800	-0.000	2.030	4.063	5.863	7.181	9.037	10.187	10.909	11.038
8.900	-0.000	2.019	4.041	5.797	7.084	8.929	10.097	10.823	10.966
9.000	-0.000	2.009	4.019	5.730	6.987	8.820	10.007	10.737	10.893
9.100	-0.000	1.999	3.997	5.662	6.889	8.710	9.916	10.650	10.820
9.200	-0.000	1.990	3.974	5.593	6.792	8.600	9.825	10.562	10.746
9.300	-0.000	1.980	3.951	5.522	6.694	8.489	9.732	10.473	10.672
9.400	-0.000	1.971	3.926	5.451	6.597	8.377	9.639	10.383	10.597
9.500	-0.000	1.962	3.899	5.379	6.499	8.266	9.544	10.293	10.522
9.600	-0.000	1.953	3.870	5.306	6.402	8.153	9.447	10.202	10.446
9.700	-0.000	1.944	3.838	5.231	6.305	8.040	9.349	10.110	10.370
9.800	-0.000	1.936	3.803	5.156	6.207	7.928	9.250	10.018	10.293
9.900	-0.000	1.927	3.767	5.080	6.110	7.815	9.149	9.925	10.216
10.000	-0.000	1.919	3.728	5.003	6.013	7.702	9.047	9.831	10.138
10.100	-0.000	1.911	3.686	4.925	5.916	7.589	8.943	9.737	10.060
10.200	-0.000	1.902	3.642	4.846	5.818	7.478	8.837	9.643	9.981
10.300	-0.000	1.894	3.596	4.766	5.721	7.366	8.730	9.548	9.902
10.400	-0.000	1.886	3.547	4.686	5.623	7.255	8.621	9.452	9.823
10.500	-0.000	1.878	3.495	4.604	5.526	7.145	8.512	9.357	9.743
10.600	-0.000	1.870	3.442	4.522	5.428	7.036	8.402	9.260	9.662
10.700	-0.000	1.862	3.386	4.438	5.331	6.927	8.292	9.164	9.582
10.800	-0.000	1.854	3.328	4.354	5.233	6.819	8.182	9.066	9.501
10.900	-0.000	1.845	3.267	4.269	5.135	6.712	8.071	8.969	9.420
11.000	-0.000	1.835	3.204	4.183	5.037	6.605	7.961	8.871	9.338
11.100	-0.000	1.823	3.139	4.096	4.939	6.500	7.850	8.773	9.257
11.200	-0.000	1.808	3.072	4.008	4.841	6.396	7.739	8.674	9.175
11.300	-0.000	1.790	3.002	3.919	4.743	6.292	7.628	8.575	9.095
11.400	-0.000	1.767	2.930	3.829	4.644	6.189	7.517	8.475	9.015

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.740	2.856	3.738	4.546	6.088	7.406	8.375	8.935
11.600	-0.000	1.708	2.780	3.646	4.448	5.988	7.297	8.277	8.857
11.700	-0.000	1.670	2.702	3.554	4.349	5.887	7.188	8.180	8.780
11.800	-0.000	1.627	2.622	3.460	4.251	5.786	7.079	8.083	8.704
11.900	-0.000	1.579	2.539	3.366	4.153	5.684	6.971	7.986	8.629
12.000	-0.000	1.526	2.454	3.270	4.054	5.580	6.864	7.891	8.555
12.100	-0.000	1.467	2.366	3.174	3.955	5.475	6.757	7.797	8.482
12.200	-0.000	1.402	2.275	3.076	3.855	5.368	6.652	7.703	8.410
12.300	-0.000	1.331	2.182	2.976	3.756	5.258	6.547	7.611	8.339
12.400	-0.000	1.254	2.085	2.876	3.656	5.149	6.443	7.518	8.270
12.500	-0.000	1.170	1.986	2.774	3.556	5.039	6.339	7.424	8.202
12.600	-0.000	1.085	1.889	2.677	3.462	4.937	6.242	7.339	8.140
12.700	-0.000	1.010	1.808	2.597	3.386	4.855	6.166	7.272	8.088
12.800	-0.000	0.945	1.740	2.532	3.325	4.793	6.109	7.221	8.047
12.900	-0.000	0.892	1.687	2.481	3.278	4.747	6.068	7.185	8.016
13.000	-0.000	0.851	1.646	2.443	3.244	4.716	6.041	7.161	7.995
13.100	-0.000	0.819	1.617	2.416	3.219	4.697	6.025	7.145	7.981
13.200	-0.000	0.798	1.597	2.398	3.204	4.686	6.016	7.136	7.974
13.300	-0.000	0.785	1.585	2.388	3.194	4.681	6.013	7.132	7.971
13.400	-0.000	0.777	1.578	2.382	3.190	4.679	6.011	7.129	7.969
13.500	-0.000	0.771	1.573	2.377	3.185	4.676	6.009	7.127	7.968
13.600	-0.000	0.765	1.567	2.372	3.181	4.674	6.007	7.125	7.966
13.700	-0.000	0.759	1.562	2.367	3.177	4.672	6.005	7.123	7.964
13.800	-0.000	0.754	1.556	2.363	3.173	4.669	6.003	7.120	7.962
13.900	-0.000	0.748	1.551	2.358	3.169	4.667	6.001	7.118	7.961
14.000	-0.000	0.742	1.545	2.353	3.165	4.664	5.999	7.116	7.959
14.100	-0.000	0.737	1.540	2.348	3.161	4.662	5.997	7.114	7.957
14.200	-0.000	0.731	1.535	2.344	3.157	4.660	5.995	7.112	7.956
14.300	-0.000	0.726	1.530	2.339	3.153	4.658	5.993	7.109	7.954
14.400	-0.000	0.721	1.524	2.334	3.149	4.655	5.991	7.107	7.952
14.500	-0.000	0.717	1.519	2.330	3.145	4.653	5.990	7.105	7.951
14.600	-0.000	0.712	1.514	2.325	3.141	4.651	5.988	7.103	7.949
14.700	-0.000	0.709	1.509	2.321	3.137	4.649	5.986	7.100	7.947
14.800	-0.000	0.706	1.504	2.316	3.133	4.647	5.984	7.098	7.946
14.900	-0.000	0.704	1.499	2.311	3.129	4.645	5.982	7.096	7.944
15.000	-0.000	0.704	1.494	2.307	3.125	4.643	5.980	7.094	7.942
15.100	-0.000	0.704	1.489	2.303	3.122	4.641	5.978	7.091	7.941
15.200	-0.000	0.705	1.484	2.298	3.118	4.639	5.977	7.089	7.939
15.300	-0.000	0.705	1.480	2.294	3.114	4.637	5.975	7.087	7.937
15.400	-0.000	0.706	1.475	2.289	3.111	4.635	5.973	7.085	7.936
15.500	-0.000	0.707	1.471	2.285	3.107	4.633	5.971	7.082	7.934
15.600	-0.000	0.708	1.467	2.281	3.103	4.631	5.969	7.080	7.932
15.700	-0.000	0.709	1.462	2.277	3.100	4.629	5.967	7.078	7.931
15.800	-0.000	0.709	1.458	2.273	3.096	4.628	5.966	7.075	7.929
15.900	-0.000	0.710	1.455	2.269	3.093	4.626	5.964	7.073	7.927
16.000	-0.000	0.711	1.451	2.265	3.089	4.624	5.962	7.071	7.926
16.100	-0.000	0.712	1.447	2.261	3.086	4.622	5.960	7.069	7.924
16.200	-0.000	0.713	1.444	2.257	3.082	4.621	5.959	7.066	7.922
16.300	-0.000	0.714	1.442	2.253	3.079	4.619	5.957	7.064	7.921
16.400	-0.000	0.715	1.439	2.249	3.076	4.618	5.955	7.062	7.919
16.500	-0.000	0.716	1.437	2.246	3.073	4.616	5.953	7.059	7.918
16.600	-0.000	0.716	1.436	2.242	3.069	4.615	5.951	7.057	7.916
16.700	-0.000	0.717	1.435	2.238	3.066	4.613	5.950	7.055	7.914
16.800	-0.000	0.718	1.435	2.235	3.063	4.612	5.948	7.053	7.913
16.900	-0.000	0.719	1.436	2.232	3.060	4.610	5.946	7.050	7.911
17.000	-0.000	0.720	1.437	2.229	3.057	4.609	5.945	7.048	7.910
17.100	-0.000	0.721	1.439	2.225	3.054	4.607	5.943	7.046	7.908
17.200	-0.000	0.722	1.441	2.222	3.051	4.606	5.941	7.043	7.907
17.300	-0.000	0.723	1.443	2.220	3.049	4.605	5.940	7.041	7.905
17.400	-0.000	0.724	1.445	2.217	3.046	4.603	5.938	7.039	7.904

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.725	1.447	2.214	3.043	4.602	5.936	7.037	7.902
17.600	-0.000	0.726	1.449	2.212	3.041	4.601	5.935	7.034	7.901
17.700	-0.000	0.727	1.451	2.210	3.038	4.599	5.933	7.032	7.899
17.800	-0.000	0.728	1.453	2.208	3.036	4.598	5.931	7.030	7.897
17.900	-0.000	0.729	1.455	2.206	3.033	4.597	5.929	7.028	7.896
18.000	-0.000	0.730	1.457	2.204	3.031	4.596	5.928	7.026	7.894
18.100	-0.000	0.731	1.459	2.203	3.029	4.595	5.926	7.023	7.893
18.200	-0.000	0.732	1.461	2.202	3.027	4.594	5.925	7.021	7.891
18.300	-0.000	0.733	1.463	2.201	3.025	4.593	5.923	7.019	7.890
18.400	-0.000	0.734	1.465	2.201	3.023	4.591	5.921	7.017	7.889
18.500	-0.000	0.735	1.467	2.201	3.021	4.590	5.920	7.015	7.887
18.600	-0.000	0.736	1.469	2.201	3.019	4.589	5.918	7.013	7.885
18.700	-0.000	0.737	1.472	2.203	3.017	4.588	5.916	7.011	7.884
18.800	-0.000	0.738	1.474	2.204	3.016	4.587	5.915	7.009	7.882
18.900	-0.000	0.740	1.476	2.207	3.014	4.586	5.913	7.007	7.881
19.000	-0.000	0.741	1.478	2.210	3.013	4.585	5.911	7.005	7.879
19.100	-0.000	0.742	1.481	2.213	3.012	4.584	5.910	7.003	7.878
19.200	-0.000	0.743	1.483	2.216	3.011	4.584	5.908	7.001	7.876
19.300	-0.000	0.744	1.485	2.220	3.010	4.583	5.907	6.999	7.875
19.400	-0.000	0.745	1.487	2.223	3.009	4.582	5.905	6.997	7.873
19.500	-0.000	0.746	1.490	2.226	3.008	4.581	5.903	6.995	7.872
19.600	-0.000	0.748	1.492	2.230	3.008	4.580	5.902	6.993	7.870
19.700	-0.000	0.749	1.494	2.234	3.008	4.579	5.900	6.991	7.869
19.800	-0.000	0.750	1.497	2.237	3.008	4.579	5.899	6.989	7.867
19.900	-0.000	0.751	1.499	2.241	3.008	4.578	5.897	6.987	7.866
20.000	-0.000	0.752	1.502	2.244	3.008	4.577	5.895	6.985	7.864
20.100	-0.000	0.754	1.504	2.248	3.009	4.576	5.894	6.984	7.864
20.200	-0.000	0.755	1.507	2.252	3.010	4.576	5.892	6.982	7.863
20.300	-0.000	0.756	1.509	2.255	3.011	4.575	5.891	6.980	7.861
20.400	-0.000	0.757	1.512	2.259	3.012	4.574	5.889	6.978	7.860
20.500	-0.000	0.759	1.514	2.263	3.014	4.574	5.888	6.976	7.859
20.600	-0.000	0.760	1.517	2.267	3.016	4.573	5.886	6.974	7.857
20.700	-0.000	0.761	1.519	2.271	3.019	4.573	5.885	6.973	7.856
20.800	-0.000	0.762	1.522	2.275	3.022	4.572	5.883	6.971	7.855
20.900	-0.000	0.764	1.524	2.279	3.026	4.571	5.882	6.969	7.854
21.000	-0.000	0.765	1.527	2.282	3.030	4.571	5.880	6.967	7.853
21.100	-0.000	0.767	1.530	2.286	3.035	4.571	5.879	6.965	7.851
21.200	-0.000	0.768	1.532	2.290	3.040	4.570	5.877	6.964	7.850
21.300	-0.000	0.769	1.535	2.295	3.045	4.570	5.876	6.962	7.849
21.400	-0.000	0.770	1.538	2.299	3.050	4.569	5.874	6.960	7.848
21.500	-0.000	0.772	1.541	2.303	3.056	4.569	5.873	6.960	7.847
21.600	-0.000	0.773	1.543	2.307	3.061	4.569	5.871	6.958	7.846
21.700	-0.000	0.775	1.546	2.311	3.068	4.568	5.869	6.956	7.845
21.800	-0.000	0.776	1.549	2.316	3.073	4.568	5.868	6.955	7.843
21.900	-0.000	0.777	1.552	2.320	3.079	4.568	5.866	6.953	7.842
22.000	-0.000	0.779	1.555	2.324	3.084	4.568	5.865	6.952	7.841
22.100	-0.000	0.780	1.558	2.328	3.090	4.568	5.863	6.950	7.840
22.200	-0.000	0.782	1.560	2.333	3.096	4.567	5.862	6.949	7.839
22.300	-0.000	0.783	1.563	2.337	3.101	4.567	5.861	6.947	7.838
22.400	-0.000	0.785	1.566	2.341	3.107	4.568	5.860	6.946	7.837
22.500	-0.000	0.786	1.569	2.346	3.113	4.568	5.858	6.945	7.836
22.600	-0.000	0.788	1.572	2.350	3.119	4.568	5.857	6.943	7.835
22.700	-0.000	0.789	1.575	2.355	3.123	4.568	5.855	6.942	7.834
22.800	-0.000	0.791	1.578	2.359	3.129	4.569	5.854	6.941	7.833
22.900	-0.000	0.792	1.581	2.364	3.136	4.569	5.852	6.939	7.832
23.000	-0.000	0.794	1.584	2.368	3.142	4.569	5.851	6.938	7.832
23.100	-0.000	0.795	1.587	2.373	3.147	4.570	5.849	6.937	7.831
23.200	-0.000	0.797	1.591	2.378	3.153	4.570	5.848	6.936	7.830
23.300	-0.000	0.798	1.594	2.382	3.159	4.571	5.847	6.935	7.829
23.400	-0.000	0.800	1.597	2.387	3.164	4.572	5.846	6.934	7.828

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.801	1.600	2.392	3.169	4.573	5.845	6.933	7.827
23.600	-0.000	0.803	1.603	2.397	3.174	4.574	5.843	6.932	7.827
23.700	-0.000	0.805	1.606	2.401	3.179	4.578	5.842	6.931	7.826
23.800	-0.000	0.806	1.610	2.406	3.184	4.578	5.841	6.930	7.825
23.900	-0.000	0.808	1.613	2.411	3.188	4.579	5.841	6.929	7.824
24.000	-0.000	0.810	1.615	2.416	3.192	4.581	5.840	6.928	7.824
24.100	-0.000	0.811	1.620	2.421	3.196	4.584	5.839	6.927	7.823
24.200	-0.000	0.813	1.623	2.426	3.200	4.586	5.840	6.927	7.822
24.300	-0.000	0.815	1.625	2.431	3.204	4.588	5.839	6.926	7.822
24.400	-0.000	0.816	1.630	2.436	3.207	4.591	5.838	6.926	7.821
24.500	-0.000	0.818	1.633	2.441	3.211	4.594	5.838	6.925	7.821
24.600	-0.000	0.820	1.636	2.446	3.212	4.597	5.838	6.924	7.820
24.700	-0.000	0.822	1.640	2.451	3.218	4.600	5.838	6.924	7.820
24.800	-0.000	0.823	1.643	2.455	3.221	4.603	5.838	6.923	7.819
24.900	-0.000	0.824	1.647	2.460	3.225	4.606	5.838	6.923	7.819
25.000	-0.000	0.827	1.649	2.465	3.228	4.609	5.838	6.922	7.819
25.100	-0.000	0.829	1.654	2.469	3.229	4.611	5.838	6.922	7.818
25.200	-0.000	0.830	1.657	2.474	3.234	4.614	5.838	6.921	7.818
25.300	-0.000	0.832	1.661	2.478	3.237	4.617	5.839	6.921	7.818
25.400	-0.000	0.834	1.664	2.482	3.240	4.619	5.840	6.921	7.817
25.500	-0.000	0.836	1.668	2.485	3.242	4.622	5.841	6.921	7.817
25.600	-0.000	0.837	1.672	2.488	3.245	4.624	5.842	6.921	7.817
25.700	-0.000	0.839	1.675	2.491	3.248	4.626	5.843	6.921	7.817
25.800	-0.000	0.841	1.679	2.494	3.250	4.626	5.844	6.921	7.817
25.900	-0.000	0.843	1.682	2.496	3.252	4.631	5.846	6.921	7.817
26.000	-0.000	0.845	1.686	2.499	3.254	4.633	5.848	6.922	7.817

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: -0.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	0.000	16.518	17.816	18.213	18.237	17.550	16.187	14.301	12.022
0.100	0.000	16.207	17.622	18.075	18.115	17.471	16.137	14.276	12.024
0.200	-0.000	15.460	17.124	17.691	17.812	17.292	16.063	14.301	12.152
0.300	-0.000	14.532	16.506	17.213	17.435	17.073	15.981	14.352	12.341
0.400	-0.000	13.642	15.912	16.761	17.080	16.868	15.908	14.406	12.528
0.500	-0.001	12.836	15.367	16.346	16.755	16.682	15.842	14.459	12.703
0.600	-0.001	12.107	14.866	15.963	16.456	16.514	15.783	14.507	12.865
0.700	-0.001	11.449	14.404	15.608	16.180	16.357	15.730	14.553	13.016
0.800	-0.001	10.853	13.973	15.277	15.923	16.211	15.682	14.597	13.157
0.900	-0.001	10.308	13.564	14.967	15.681	16.074	15.637	14.639	13.289
1.000	-0.001	9.803	13.168	14.675	15.452	15.945	15.594	14.680	13.404
1.100	-0.001	9.332	12.796	14.398	15.234	15.823	15.553	14.720	13.499
1.200	-0.001	8.891	12.448	14.135	15.027	15.706	15.514	14.758	13.574
1.300	-0.001	8.476	12.121	13.883	14.829	15.595	15.477	14.793	13.630
1.400	-0.001	8.084	11.813	13.643	14.640	15.488	15.442	14.825	13.672
1.500	-0.001	7.711	11.520	13.408	14.458	15.386	15.408	14.850	13.701
1.600	-0.001	7.354	11.243	13.177	14.283	15.287	15.376	14.867	13.720
1.700	-0.001	7.014	10.978	12.954	14.114	15.191	15.345	14.876	13.730
1.800	-0.001	6.690	10.725	12.741	13.951	15.098	15.314	14.876	13.734
1.900	-0.001	6.390	10.482	12.537	13.794	15.008	15.285	14.870	13.732
2.000	-0.001	6.115	10.248	12.341	13.641	14.921	15.257	14.856	13.726
2.100	-0.001	5.864	10.022	12.154	13.492	14.837	15.228	14.837	13.716
2.200	-0.001	5.634	9.804	11.973	13.344	14.754	15.197	14.813	13.702
2.300	-0.001	5.423	9.594	11.798	13.198	14.674	15.164	14.785	13.687
2.400	-0.001	5.229	9.390	11.630	13.056	14.596	15.128	14.752	13.669
2.500	-0.001	5.049	9.193	11.467	12.918	14.520	15.089	14.717	13.650
2.600	-0.001	4.882	9.002	11.309	12.784	14.446	15.047	14.677	13.631
2.700	-0.001	4.728	8.815	11.156	12.655	14.374	15.002	14.635	13.610
2.800	-0.001	4.584	8.633	11.007	12.530	14.303	14.953	14.590	13.590
2.900	-0.001	4.449	8.455	10.862	12.409	14.235	14.902	14.544	13.569
3.000	-0.001	4.322	8.281	10.721	12.292	14.167	14.848	14.495	13.548
3.100	-0.001	4.204	8.111	10.583	12.178	14.101	14.791	14.445	13.526
3.200	-0.001	4.093	7.944	10.449	12.067	14.036	14.731	14.394	13.505
3.300	-0.001	3.989	7.782	10.318	11.960	13.973	14.669	14.342	13.483
3.400	-0.001	3.891	7.623	10.190	11.855	13.910	14.605	14.289	13.462
3.500	-0.001	3.797	7.468	10.064	11.754	13.847	14.538	14.235	13.441
3.600	-0.001	3.709	7.315	9.942	11.654	13.783	14.469	14.180	13.419
3.700	-0.001	3.625	7.168	9.822	11.557	13.718	14.399	14.125	13.398
3.800	-0.001	3.546	7.026	9.705	11.461	13.651	14.327	14.069	13.377
3.900	-0.001	3.471	6.891	9.590	11.368	13.582	14.254	14.013	13.356
4.000	-0.001	3.401	6.761	9.478	11.277	13.511	14.179	13.957	13.335
4.100	-0.001	3.334	6.637	9.368	11.188	13.438	14.104	13.900	13.314
4.200	-0.001	3.270	6.519	9.259	11.100	13.363	14.028	13.843	13.292
4.300	-0.001	3.210	6.406	9.153	11.014	13.287	13.952	13.786	13.269
4.400	-0.001	3.153	6.297	9.048	10.929	13.209	13.876	13.728	13.246
4.500	-0.001	3.098	6.193	8.945	10.846	13.129	13.799	13.670	13.222
4.600	-0.001	3.046	6.093	8.843	10.765	13.049	13.722	13.612	13.198
4.700	-0.001	2.996	5.997	8.743	10.684	12.967	13.645	13.554	13.172
4.800	-0.001	2.950	5.905	8.645	10.605	12.884	13.567	13.497	13.144
4.900	-0.001	2.905	5.817	8.548	10.528	12.800	13.489	13.441	13.116
5.000	-0.001	2.863	5.732	8.452	10.451	12.716	13.410	13.384	13.088
5.100	-0.001	2.822	5.650	8.357	10.376	12.630	13.329	13.328	13.059
5.200	-0.001	2.782	5.572	8.263	10.301	12.543	13.247	13.271	13.030
5.300	-0.001	2.745	5.497	8.171	10.228	12.456	13.164	13.215	13.000
5.400	-0.001	2.709	5.425	8.079	10.156	12.368	13.081	13.158	12.970

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.674	5.355	7.989	10.085	12.279	12.998	13.101	12.940
5.600	-0.001	2.641	5.289	7.900	10.014	12.190	12.915	13.044	12.908
5.700	-0.001	2.608	5.225	7.812	9.944	12.100	12.831	12.987	12.877
5.800	-0.001	2.578	5.163	7.727	9.873	12.009	12.748	12.930	12.844
5.900	-0.001	2.548	5.104	7.644	9.801	11.918	12.665	12.871	12.810
6.000	-0.001	2.519	5.047	7.564	9.728	11.826	12.582	12.812	12.774
6.100	-0.001	2.492	4.992	7.486	9.653	11.734	12.500	12.751	12.735
6.200	-0.000	2.465	4.939	7.411	9.577	11.642	12.417	12.691	12.694
6.300	-0.000	2.440	4.888	7.337	9.498	11.549	12.334	12.630	12.652
6.400	-0.000	2.416	4.839	7.266	9.417	11.455	12.252	12.569	12.607
6.500	-0.000	2.392	4.792	7.198	9.335	11.362	12.169	12.509	12.558
6.600	-0.000	2.369	4.747	7.132	9.251	11.267	12.087	12.450	12.507
6.700	-0.000	2.347	4.703	7.068	9.165	11.172	12.005	12.390	12.453
6.800	-0.000	2.326	4.660	7.006	9.078	11.077	11.923	12.331	12.397
6.900	-0.000	2.305	4.619	6.946	8.990	10.982	11.840	12.271	12.339
7.000	-0.000	2.286	4.580	6.887	8.901	10.885	11.758	12.212	12.282
7.100	-0.000	2.267	4.541	6.831	8.811	10.789	11.676	12.152	12.222
7.200	-0.000	2.248	4.504	6.776	8.719	10.692	11.593	12.092	12.162
7.300	-0.000	2.230	4.469	6.722	8.626	10.595	11.510	12.033	12.100
7.400	-0.000	2.213	4.435	6.671	8.533	10.497	11.427	11.973	12.037
7.500	-0.000	2.196	4.401	6.620	8.439	10.399	11.343	11.912	11.974
7.600	-0.000	2.180	4.369	6.568	8.344	10.300	11.260	11.851	11.911
7.700	-0.000	2.165	4.338	6.515	8.249	10.201	11.176	11.788	11.848
7.800	-0.000	2.150	4.308	6.462	8.154	10.101	11.092	11.724	11.784
7.900	-0.000	2.135	4.279	6.407	8.059	10.001	11.008	11.658	11.720
8.000	-0.000	2.121	4.251	6.351	7.964	9.899	10.923	11.591	11.655
8.100	-0.000	2.107	4.224	6.294	7.869	9.798	10.838	11.522	11.588
8.200	-0.000	2.094	4.197	6.236	7.773	9.695	10.753	11.451	11.521
8.300	-0.000	2.081	4.172	6.177	7.678	9.593	10.668	11.378	11.454
8.400	-0.000	2.069	4.147	6.118	7.582	9.490	10.581	11.303	11.386
8.500	-0.000	2.057	4.123	6.057	7.487	9.386	10.494	11.226	11.317
8.600	-0.000	2.046	4.100	5.995	7.391	9.282	10.407	11.147	11.248
8.700	-0.000	2.035	4.077	5.933	7.296	9.178	10.319	11.067	11.178
8.800	-0.000	2.024	4.054	5.869	7.201	9.072	10.232	10.984	11.108
8.900	-0.000	2.013	4.032	5.805	7.105	8.966	10.144	10.899	11.037
9.000	-0.000	2.003	4.011	5.739	7.010	8.859	10.055	10.814	10.966
9.100	-0.000	1.994	3.990	5.673	6.915	8.752	9.967	10.729	10.894
9.200	-0.000	1.984	3.969	5.606	6.819	8.644	9.878	10.643	10.822
9.300	-0.000	1.975	3.949	5.538	6.724	8.536	9.788	10.556	10.749
9.400	-0.000	1.966	3.926	5.468	6.628	8.427	9.698	10.469	10.676
9.500	-0.000	1.957	3.902	5.398	6.533	8.317	9.607	10.380	10.602
9.600	-0.000	1.949	3.874	5.327	6.437	8.207	9.515	10.291	10.527
9.700	-0.000	1.940	3.844	5.255	6.342	8.096	9.421	10.200	10.451
9.800	-0.000	1.932	3.812	5.181	6.246	7.985	9.325	10.110	10.375
9.900	-0.000	1.924	3.777	5.107	6.151	7.873	9.229	10.018	10.299
10.000	-0.000	1.916	3.739	5.032	6.055	7.762	9.130	9.926	10.222
10.100	-0.000	1.908	3.700	4.956	5.959	7.651	9.030	9.833	10.144
10.200	-0.000	1.900	3.658	4.879	5.863	7.540	8.929	9.740	10.066
10.300	-0.000	1.892	3.613	4.801	5.767	7.430	8.826	9.646	9.988
10.400	-0.000	1.885	3.566	4.722	5.671	7.321	8.721	9.552	9.909
10.500	-0.000	1.877	3.517	4.643	5.575	7.212	8.614	9.457	9.830
10.600	-0.000	1.869	3.466	4.562	5.479	7.103	8.506	9.362	9.751
10.700	-0.000	1.862	3.412	4.481	5.383	6.995	8.398	9.267	9.671
10.800	-0.000	1.854	3.356	4.398	5.287	6.889	8.289	9.171	9.591
10.900	-0.000	1.847	3.298	4.315	5.190	6.782	8.180	9.074	9.510
11.000	-0.000	1.839	3.237	4.231	5.094	6.677	8.070	8.977	9.429
11.100	-0.000	1.831	3.175	4.145	4.997	6.573	7.961	8.880	9.348
11.200	-0.000	1.820	3.110	4.059	4.901	6.469	7.851	8.782	9.266
11.300	-0.000	1.805	3.042	3.972	4.804	6.366	7.741	8.683	9.184
11.400	-0.000	1.786	2.973	3.883	4.707	6.265	7.631	8.585	9.103

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.762	2.901	3.794	4.610	6.164	7.521	8.486	9.022
11.600	-0.000	1.733	2.827	3.704	4.513	6.064	7.412	8.386	8.943
11.700	-0.000	1.698	2.751	3.613	4.415	5.965	7.302	8.287	8.863
11.800	-0.000	1.659	2.673	3.521	4.318	5.867	7.193	8.189	8.786
11.900	-0.000	1.614	2.592	3.428	4.221	5.769	7.086	8.092	8.709
12.000	-0.000	1.564	2.509	3.334	4.124	5.670	6.979	7.995	8.634
12.100	-0.000	1.509	2.424	3.240	4.026	5.570	6.872	7.900	8.560
12.200	-0.000	1.449	2.336	3.144	3.928	5.468	6.766	7.805	8.487
12.300	-0.000	1.383	2.246	3.047	3.830	5.365	6.662	7.711	8.415
12.400	-0.000	1.311	2.153	2.949	3.732	5.258	6.558	7.619	8.344
12.500	-0.000	1.233	2.057	2.849	3.634	5.151	6.455	7.527	8.274
12.600	-0.000	1.148	1.958	2.749	3.535	5.043	6.352	7.434	8.206
12.700	-0.000	1.056	1.856	2.646	3.436	4.934	6.249	7.341	8.139
12.800	-0.000	0.958	1.753	2.545	3.340	4.827	6.147	7.250	8.072
12.900	-0.000	0.879	1.674	2.471	3.270	4.753	6.077	7.189	8.022
13.000	-0.000	0.827	1.625	2.425	3.228	4.713	6.041	7.157	7.994
13.100	-0.000	0.799	1.598	2.401	3.207	4.697	6.027	7.144	7.983
13.200	-0.000	0.787	1.588	2.392	3.199	4.693	6.024	7.140	7.980
13.300	-0.000	0.781	1.583	2.387	3.195	4.690	6.022	7.137	7.978
13.400	-0.000	0.775	1.577	2.382	3.191	4.688	6.020	7.135	7.976
13.500	-0.000	0.769	1.572	2.377	3.186	4.685	6.018	7.133	7.975
13.600	-0.000	0.763	1.566	2.372	3.182	4.683	6.016	7.131	7.973
13.700	-0.000	0.758	1.561	2.368	3.178	4.681	6.014	7.129	7.971
13.800	-0.000	0.752	1.555	2.363	3.174	4.678	6.012	7.126	7.970
13.900	-0.000	0.746	1.550	2.358	3.170	4.676	6.010	7.124	7.968
14.000	-0.000	0.740	1.545	2.353	3.166	4.674	6.008	7.122	7.966
14.100	-0.000	0.735	1.540	2.349	3.162	4.671	6.006	7.120	7.965
14.200	-0.000	0.730	1.534	2.344	3.158	4.669	6.005	7.117	7.963
14.300	-0.000	0.724	1.529	2.339	3.154	4.667	6.003	7.115	7.961
14.400	-0.000	0.719	1.524	2.335	3.150	4.665	6.001	7.113	7.959
14.500	-0.000	0.715	1.519	2.330	3.146	4.663	5.999	7.111	7.958
14.600	-0.000	0.711	1.514	2.326	3.142	4.661	5.997	7.108	7.956
14.700	-0.000	0.707	1.509	2.321	3.139	4.659	5.995	7.106	7.954
14.800	-0.000	0.705	1.504	2.317	3.135	4.656	5.993	7.104	7.953
14.900	-0.000	0.704	1.499	2.312	3.131	4.654	5.991	7.102	7.951
15.000	-0.000	0.704	1.494	2.308	3.127	4.653	5.990	7.099	7.949
15.100	-0.000	0.705	1.489	2.303	3.123	4.651	5.988	7.097	7.948
15.200	-0.000	0.706	1.484	2.299	3.120	4.649	5.986	7.095	7.946
15.300	-0.000	0.707	1.480	2.295	3.116	4.647	5.984	7.092	7.944
15.400	-0.000	0.707	1.475	2.290	3.112	4.645	5.982	7.090	7.943
15.500	-0.000	0.708	1.471	2.286	3.109	4.643	5.980	7.088	7.941
15.600	-0.000	0.709	1.467	2.282	3.105	4.641	5.979	7.085	7.939
15.700	-0.000	0.710	1.463	2.278	3.102	4.640	5.977	7.083	7.938
15.800	-0.000	0.711	1.459	2.274	3.098	4.638	5.975	7.081	7.936
15.900	-0.000	0.712	1.455	2.270	3.095	4.636	5.973	7.079	7.934
16.000	-0.000	0.712	1.452	2.266	3.091	4.635	5.971	7.076	7.933
16.100	-0.000	0.713	1.448	2.262	3.088	4.633	5.970	7.074	7.931
16.200	-0.000	0.714	1.445	2.258	3.085	4.632	5.968	7.072	7.930
16.300	-0.000	0.715	1.443	2.254	3.082	4.630	5.966	7.069	7.928
16.400	-0.000	0.716	1.440	2.251	3.078	4.628	5.964	7.067	7.926
16.500	-0.000	0.717	1.438	2.247	3.075	4.627	5.963	7.065	7.925
16.600	-0.000	0.718	1.437	2.244	3.072	4.625	5.961	7.062	7.923
16.700	-0.000	0.719	1.437	2.240	3.069	4.624	5.959	7.060	7.922
16.800	-0.000	0.720	1.437	2.237	3.066	4.623	5.957	7.058	7.920
16.900	-0.000	0.721	1.438	2.234	3.063	4.621	5.956	7.056	7.919
17.000	-0.000	0.722	1.440	2.231	3.060	4.620	5.954	7.053	7.917
17.100	-0.000	0.723	1.442	2.228	3.057	4.619	5.952	7.051	7.915
17.200	-0.000	0.724	1.444	2.225	3.055	4.617	5.951	7.049	7.914
17.300	-0.000	0.724	1.446	2.222	3.052	4.616	5.949	7.046	7.912
17.400	-0.000	0.725	1.448	2.220	3.049	4.615	5.947	7.044	7.911

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.726	1.450	2.217	3.047	4.613	5.946	7.042	7.909
17.600	-0.000	0.727	1.452	2.215	3.044	4.612	5.944	7.040	7.908
17.700	-0.000	0.728	1.454	2.213	3.042	4.611	5.942	7.037	7.906
17.800	-0.000	0.729	1.456	2.211	3.039	4.610	5.940	7.035	7.905
17.900	-0.000	0.731	1.458	2.209	3.037	4.609	5.939	7.033	7.903
18.000	-0.000	0.732	1.460	2.208	3.035	4.608	5.937	7.031	7.902
18.100	-0.000	0.733	1.462	2.206	3.033	4.606	5.935	7.029	7.900
18.200	-0.000	0.734	1.464	2.205	3.031	4.605	5.934	7.027	7.899
18.300	-0.000	0.735	1.466	2.205	3.029	4.604	5.932	7.024	7.897
18.400	-0.000	0.736	1.468	2.204	3.027	4.603	5.931	7.022	7.895
18.500	-0.000	0.737	1.471	2.205	3.025	4.602	5.929	7.020	7.894
18.600	-0.000	0.738	1.473	2.205	3.024	4.601	5.927	7.018	7.892
18.700	-0.000	0.739	1.475	2.207	3.022	4.600	5.926	7.016	7.891
18.800	-0.000	0.740	1.477	2.209	3.021	4.599	5.924	7.014	7.889
18.900	-0.000	0.741	1.479	2.211	3.019	4.598	5.922	7.012	7.887
19.000	-0.000	0.742	1.482	2.214	3.018	4.598	5.921	7.010	7.886
19.100	-0.000	0.744	1.484	2.218	3.017	4.597	5.919	7.008	7.884
19.200	-0.000	0.745	1.486	2.221	3.016	4.596	5.917	7.006	7.883
19.300	-0.000	0.746	1.489	2.225	3.015	4.595	5.916	7.004	7.881
19.400	-0.000	0.747	1.491	2.228	3.015	4.594	5.914	7.002	7.880
19.500	-0.000	0.748	1.493	2.232	3.014	4.593	5.913	7.000	7.878
19.600	-0.000	0.749	1.496	2.235	3.014	4.593	5.911	6.998	7.877
19.700	-0.000	0.751	1.498	2.239	3.014	4.592	5.909	6.996	7.875
19.800	-0.000	0.752	1.501	2.243	3.014	4.591	5.908	6.995	7.875
19.900	-0.000	0.753	1.503	2.246	3.014	4.590	5.906	6.993	7.874
20.000	-0.000	0.754	1.505	2.250	3.015	4.590	5.905	6.991	7.872
20.100	-0.000	0.756	1.508	2.254	3.015	4.589	5.903	6.989	7.871
20.200	-0.000	0.757	1.510	2.258	3.016	4.588	5.902	6.987	7.869
20.300	-0.000	0.758	1.513	2.261	3.018	4.588	5.900	6.985	7.868
20.400	-0.000	0.759	1.515	2.265	3.019	4.587	5.898	6.983	7.867
20.500	-0.000	0.761	1.518	2.269	3.021	4.586	5.897	6.981	7.865
20.600	-0.000	0.762	1.521	2.273	3.023	4.586	5.895	6.979	7.864
20.700	-0.000	0.763	1.523	2.277	3.026	4.585	5.894	6.978	7.863
20.800	-0.000	0.765	1.526	2.281	3.029	4.585	5.892	6.976	7.862
20.900	-0.000	0.766	1.529	2.285	3.033	4.584	5.891	6.974	7.860
21.000	-0.000	0.767	1.532	2.289	3.038	4.584	5.889	6.972	7.859
21.100	-0.000	0.769	1.534	2.293	3.043	4.583	5.887	6.970	7.858
21.200	-0.000	0.770	1.537	2.297	3.048	4.583	5.886	6.970	7.857
21.300	-0.000	0.771	1.540	2.301	3.053	4.583	5.884	6.968	7.856
21.400	-0.000	0.773	1.542	2.306	3.059	4.582	5.883	6.967	7.854
21.500	-0.000	0.774	1.545	2.310	3.065	4.582	5.881	6.965	7.853
21.600	-0.000	0.775	1.548	2.314	3.071	4.582	5.879	6.963	7.852
21.700	-0.000	0.777	1.551	2.317	3.076	4.581	5.878	6.962	7.851
21.800	-0.000	0.778	1.553	2.322	3.082	4.581	5.876	6.960	7.850
21.900	-0.000	0.780	1.556	2.326	3.088	4.581	5.875	6.959	7.849
22.000	-0.000	0.781	1.559	2.331	3.093	4.581	5.874	6.957	7.848
22.100	-0.000	0.783	1.562	2.335	3.099	4.581	5.872	6.956	7.847
22.200	-0.000	0.784	1.564	2.339	3.104	4.581	5.871	6.954	7.846
22.300	-0.000	0.786	1.568	2.344	3.110	4.581	5.869	6.953	7.844
22.400	-0.000	0.787	1.571	2.348	3.116	4.581	5.868	6.951	7.843
22.500	-0.000	0.789	1.574	2.353	3.122	4.581	5.866	6.950	7.842
22.600	-0.000	0.790	1.578	2.357	3.128	4.581	5.864	6.949	7.841
22.700	-0.000	0.792	1.580	2.362	3.134	4.581	5.863	6.947	7.840
22.800	-0.000	0.793	1.583	2.366	3.140	4.581	5.862	6.946	7.839
22.900	-0.000	0.795	1.586	2.371	3.146	4.581	5.860	6.945	7.838
23.000	-0.000	0.796	1.589	2.376	3.152	4.582	5.859	6.944	7.837
23.100	-0.000	0.798	1.592	2.380	3.158	4.582	5.857	6.942	7.837
23.200	-0.000	0.799	1.596	2.385	3.164	4.583	5.856	6.941	7.836
23.300	-0.000	0.801	1.599	2.390	3.170	4.583	5.855	6.940	7.835
23.400	-0.000	0.803	1.602	2.395	3.175	4.586	5.853	6.939	7.834

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.804	1.605	2.399	3.181	4.585	5.852	6.938	7.833
23.600	-0.000	0.806	1.608	2.404	3.186	4.586	5.851	6.937	7.832
23.700	-0.000	0.807	1.612	2.409	3.191	4.588	5.850	6.936	7.831
23.800	-0.000	0.809	1.615	2.414	3.196	4.589	5.849	6.935	7.831
23.900	-0.000	0.811	1.618	2.419	3.201	4.591	5.849	6.934	7.830
24.000	-0.000	0.812	1.622	2.422	3.205	4.593	5.849	6.934	7.829
24.100	-0.000	0.814	1.625	2.429	3.210	4.593	5.848	6.933	7.829
24.200	-0.000	0.816	1.628	2.434	3.214	4.598	5.847	6.932	7.828
24.300	-0.000	0.817	1.632	2.439	3.217	4.600	5.847	6.931	7.827
24.400	-0.000	0.819	1.635	2.444	3.221	4.603	5.846	6.931	7.827
24.500	-0.000	0.821	1.638	2.449	3.224	4.606	5.846	6.930	7.826
24.600	-0.000	0.823	1.642	2.454	3.228	4.609	5.845	6.929	7.825
24.700	-0.000	0.824	1.645	2.459	3.231	4.611	5.845	6.929	7.825
24.800	-0.000	0.826	1.649	2.464	3.234	4.614	5.845	6.928	7.825
24.900	-0.000	0.828	1.652	2.470	3.237	4.617	5.845	6.927	7.824
25.000	-0.000	0.830	1.656	2.474	3.240	4.620	5.845	6.927	7.824
25.100	-0.000	0.831	1.659	2.479	3.243	4.622	5.846	6.927	7.823
25.200	-0.000	0.833	1.663	2.484	3.246	4.622	5.846	6.926	7.823
25.300	-0.000	0.835	1.667	2.488	3.249	4.627	5.847	6.926	7.823
25.400	-0.000	0.837	1.670	2.492	3.252	4.629	5.847	6.926	7.822
25.500	-0.000	0.839	1.674	2.496	3.254	4.632	5.848	6.926	7.822
25.600	-0.000	0.840	1.677	2.500	3.256	4.634	5.849	6.926	7.822
25.700	-0.000	0.842	1.681	2.503	3.259	4.636	5.851	6.926	7.822
25.800	-0.000	0.844	1.685	2.506	3.261	4.638	5.852	6.926	7.822
25.900	-0.000	0.846	1.689	2.508	3.263	4.640	5.854	6.926	7.822
26.000	-0.000	0.848	1.692	2.511	3.265	4.642	5.856	6.926	7.823

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 0.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	0.000	0.000	-0.000	-0.000	-0.000	-0.000	0.000	-0.000	-0.000
0.100	-0.001	16.657	17.887	18.248	18.230	17.497	16.083	14.146	11.818
0.200	-0.001	15.337	16.999	17.572	17.698	17.192	15.979	14.235	12.109
0.300	-0.001	14.295	16.285	17.027	17.274	16.949	15.895	14.307	12.342
0.400	-0.001	13.411	15.673	16.558	16.908	16.743	15.823	14.369	12.544
0.500	-0.001	12.637	15.132	16.142	16.583	16.560	15.761	14.424	12.723
0.600	-0.001	11.921	14.641	15.766	16.288	16.394	15.706	14.474	12.885
0.700	-0.001	11.282	14.189	15.420	16.016	16.241	15.657	14.522	13.035
0.800	-0.001	10.704	13.771	15.099	15.765	16.098	15.610	14.567	13.173
0.900	-0.001	10.176	13.382	14.797	15.529	15.964	15.566	14.611	13.291
1.000	-0.001	9.687	13.018	14.512	15.306	15.837	15.525	14.652	13.389
1.100	-0.001	9.231	12.668	14.241	15.095	15.717	15.485	14.691	13.469
1.200	-0.001	8.802	12.329	13.985	14.894	15.604	15.448	14.728	13.533
1.300	-0.001	8.398	12.010	13.741	14.701	15.495	15.411	14.760	13.581
1.400	-0.001	8.015	11.709	13.508	14.515	15.391	15.377	14.785	13.618
1.500	-0.001	7.650	11.423	13.286	14.337	15.291	15.344	14.803	13.644
1.600	-0.001	7.302	11.151	13.072	14.166	15.194	15.312	14.813	13.662
1.700	-0.001	6.968	10.892	12.861	14.001	15.101	15.281	14.817	13.674
1.800	-0.001	6.649	10.644	12.654	13.843	15.011	15.252	14.813	13.679
1.900	-0.001	6.352	10.407	12.454	13.689	14.923	15.223	14.803	13.680
2.000	-0.001	6.081	10.177	12.262	13.541	14.838	15.194	14.787	13.677
2.100	-0.001	5.833	9.956	12.077	13.398	14.754	15.163	14.767	13.671
2.200	-0.001	5.605	9.742	11.900	13.259	14.674	15.130	14.742	13.662
2.300	-0.001	5.395	9.535	11.729	13.122	14.595	15.095	14.714	13.651
2.400	-0.001	5.202	9.334	11.563	12.986	14.518	15.057	14.683	13.638
2.500	-0.001	5.025	9.140	11.402	12.852	14.443	15.016	14.648	13.622
2.600	-0.001	4.860	8.951	11.246	12.721	14.370	14.972	14.611	13.606
2.700	-0.001	4.706	8.767	11.095	12.594	14.299	14.926	14.572	13.588
2.800	-0.001	4.563	8.588	10.949	12.470	14.230	14.877	14.530	13.570
2.900	-0.001	4.430	8.413	10.806	12.350	14.163	14.825	14.486	13.552
3.000	-0.001	4.305	8.243	10.668	12.235	14.098	14.772	14.440	13.533
3.100	-0.001	4.188	8.076	10.532	12.123	14.034	14.716	14.393	13.514
3.200	-0.001	4.078	7.912	10.400	12.013	13.971	14.658	14.344	13.495
3.300	-0.001	3.974	7.752	10.271	11.907	13.909	14.598	14.295	13.476
3.400	-0.001	3.877	7.595	10.145	11.804	13.847	14.536	14.244	13.456
3.500	-0.001	3.785	7.442	10.022	11.704	13.784	14.472	14.192	13.437
3.600	-0.001	3.699	7.292	9.901	11.606	13.722	14.407	14.140	13.417
3.700	-0.001	3.617	7.146	9.783	11.510	13.658	14.339	14.087	13.398
3.800	-0.001	3.538	7.005	9.668	11.416	13.593	14.271	14.034	13.379
3.900	-0.001	3.464	6.871	9.554	11.324	13.525	14.201	13.980	13.359
4.000	-0.001	3.393	6.742	9.443	11.234	13.456	14.129	13.925	13.338
4.100	-0.001	3.326	6.619	9.334	11.146	13.385	14.056	13.870	13.317
4.200	-0.001	3.263	6.501	9.226	11.059	13.312	13.983	13.815	13.295
4.300	-0.001	3.203	6.388	9.121	10.974	13.238	13.908	13.760	13.273
4.400	-0.001	3.146	6.281	9.017	10.890	13.162	13.833	13.704	13.250
4.500	-0.001	3.092	6.178	8.915	10.808	13.084	13.758	13.648	13.227
4.600	-0.001	3.040	6.078	8.814	10.727	13.005	13.683	13.592	13.202
4.700	-0.001	2.990	5.983	8.715	10.647	12.925	13.608	13.535	13.177
4.800	-0.001	2.943	5.891	8.617	10.569	12.844	13.532	13.479	13.151
4.900	-0.001	2.898	5.803	8.521	10.491	12.761	13.456	13.424	13.124
5.000	-0.001	2.855	5.718	8.425	10.415	12.678	13.379	13.369	13.095
5.100	-0.001	2.814	5.637	8.331	10.340	12.593	13.301	13.314	13.066
5.200	-0.001	2.775	5.559	8.238	10.266	12.508	13.222	13.260	13.036
5.300	-0.001	2.738	5.484	8.147	10.194	12.422	13.142	13.205	13.007
5.400	-0.001	2.702	5.412	8.056	10.122	12.336	13.062	13.150	12.977

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.667	5.342	7.967	10.051	12.248	12.980	13.095	12.946
5.600	-0.001	2.634	5.275	7.878	9.980	12.160	12.899	13.040	12.915
5.700	-0.001	2.602	5.210	7.791	9.910	12.072	12.817	12.985	12.885
5.800	-0.001	2.571	5.149	7.707	9.839	11.982	12.735	12.930	12.853
5.900	-0.001	2.541	5.090	7.624	9.768	11.893	12.654	12.874	12.821
6.000	-0.001	2.512	5.033	7.544	9.695	11.803	12.573	12.818	12.786
6.100	-0.001	2.485	4.979	7.467	9.622	11.712	12.492	12.761	12.750
6.200	-0.001	2.459	4.926	7.391	9.547	11.621	12.411	12.703	12.712
6.300	-0.000	2.433	4.875	7.318	9.470	11.530	12.330	12.644	12.672
6.400	-0.000	2.409	4.826	7.248	9.391	11.439	12.249	12.585	12.630
6.500	-0.000	2.385	4.779	7.179	9.311	11.347	12.168	12.526	12.586
6.600	-0.000	2.363	4.734	7.113	9.229	11.254	12.087	12.468	12.538
6.700	-0.000	2.341	4.690	7.048	9.146	11.161	12.007	12.409	12.488
6.800	-0.000	2.319	4.648	6.986	9.060	11.068	11.926	12.350	12.436
6.900	-0.000	2.299	4.607	6.927	8.974	10.974	11.846	12.292	12.381
7.000	-0.000	2.279	4.567	6.869	8.886	10.880	11.765	12.233	12.325
7.100	-0.000	2.260	4.529	6.813	8.797	10.785	11.684	12.175	12.268
7.200	-0.000	2.242	4.493	6.758	8.707	10.690	11.603	12.116	12.209
7.300	-0.000	2.224	4.457	6.705	8.616	10.595	11.522	12.057	12.149
7.400	-0.000	2.207	4.423	6.654	8.524	10.499	11.441	11.998	12.088
7.500	-0.000	2.190	4.389	6.603	8.431	10.402	11.359	11.939	12.026
7.600	-0.000	2.174	4.357	6.552	8.338	10.306	11.277	11.879	11.963
7.700	-0.000	2.159	4.326	6.500	8.245	10.209	11.195	11.819	11.901
7.800	-0.000	2.144	4.296	6.448	8.151	10.111	11.113	11.757	11.838
7.900	-0.000	2.129	4.267	6.395	8.057	10.013	11.030	11.694	11.774
8.000	-0.000	2.115	4.239	6.341	7.964	9.914	10.948	11.630	11.710
8.100	-0.000	2.102	4.212	6.285	7.870	9.814	10.865	11.564	11.646
8.200	-0.000	2.088	4.186	6.229	7.777	9.714	10.782	11.497	11.580
8.300	-0.000	2.076	4.160	6.171	7.683	9.613	10.698	11.427	11.514
8.400	-0.000	2.063	4.136	6.113	7.589	9.511	10.614	11.356	11.447
8.500	-0.000	2.051	4.112	6.054	7.496	9.410	10.529	11.283	11.380
8.600	-0.000	2.040	4.089	5.993	7.402	9.308	10.444	11.208	11.312
8.700	-0.000	2.029	4.066	5.932	7.308	9.205	10.359	11.131	11.244
8.800	-0.000	2.018	4.044	5.870	7.215	9.102	10.273	11.052	11.175
8.900	-0.000	2.008	4.023	5.807	7.121	8.998	10.187	10.971	11.105
9.000	-0.000	1.997	4.002	5.744	7.028	8.894	10.101	10.888	11.035
9.100	-0.000	1.988	3.982	5.679	6.935	8.789	10.014	10.804	10.965
9.200	-0.000	1.978	3.962	5.613	6.841	8.683	9.927	10.719	10.893
9.300	-0.000	1.969	3.942	5.547	6.748	8.577	9.840	10.634	10.821
9.400	-0.000	1.960	3.921	5.480	6.654	8.470	9.752	10.549	10.749
9.500	-0.000	1.952	3.899	5.411	6.561	8.363	9.664	10.462	10.676
9.600	-0.000	1.943	3.873	5.342	6.467	8.255	9.575	10.375	10.603
9.700	-0.000	1.935	3.845	5.272	6.374	8.146	9.486	10.286	10.529
9.800	-0.000	1.927	3.814	5.201	6.280	8.037	9.394	10.197	10.454
9.900	-0.000	1.919	3.781	5.129	6.186	7.928	9.301	10.107	10.378
10.000	-0.000	1.912	3.745	5.055	6.092	7.818	9.207	10.017	10.302
10.100	-0.000	1.904	3.707	4.981	5.998	7.709	9.111	9.925	10.226
10.200	-0.000	1.897	3.667	4.906	5.904	7.600	9.013	9.834	10.149
10.300	-0.000	1.890	3.625	4.830	5.809	7.491	8.914	9.741	10.071
10.400	-0.000	1.882	3.580	4.754	5.715	7.383	8.813	9.648	9.993
10.500	-0.000	1.875	3.533	4.676	5.620	7.275	8.710	9.555	9.915
10.600	-0.000	1.868	3.484	4.597	5.526	7.168	8.606	9.461	9.836
10.700	-0.000	1.861	3.432	4.517	5.431	7.061	8.500	9.366	9.757
10.800	-0.000	1.854	3.378	4.437	5.336	6.956	8.393	9.271	9.677
10.900	-0.000	1.847	3.322	4.355	5.241	6.851	8.285	9.176	9.597
11.000	-0.000	1.840	3.264	4.273	5.146	6.746	8.177	9.080	9.516
11.100	-0.000	1.833	3.203	4.189	5.051	6.643	8.068	8.983	9.436
11.200	-0.000	1.825	3.141	4.105	4.956	6.540	7.960	8.886	9.354
11.300	-0.000	1.813	3.076	4.019	4.861	6.439	7.851	8.789	9.273
11.400	-0.000	1.797	3.009	3.933	4.766	6.338	7.742	8.691	9.191

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.775	2.940	3.845	4.670	6.238	7.633	8.593	9.110
11.600	-0.000	1.748	2.868	3.757	4.574	6.139	7.525	8.494	9.029
11.700	-0.000	1.717	2.794	3.668	4.478	6.041	7.416	8.396	8.950
11.800	-0.000	1.681	2.718	3.577	4.382	5.944	7.308	8.296	8.870
11.900	-0.000	1.639	2.639	3.486	4.286	5.848	7.200	8.198	8.792
12.000	-0.000	1.593	2.558	3.394	4.190	5.752	7.092	8.099	8.715
12.100	-0.000	1.541	2.475	3.301	4.094	5.656	6.986	8.003	8.640
12.200	-0.000	1.485	2.390	3.207	3.998	5.559	6.880	7.907	8.565
12.300	-0.000	1.423	2.303	3.113	3.902	5.461	6.776	7.812	8.492
12.400	-0.000	1.355	2.213	3.017	3.805	5.361	6.672	7.719	8.420
12.500	-0.000	1.282	2.120	2.920	3.709	5.259	6.569	7.626	8.349
12.600	-0.000	1.203	2.025	2.822	3.612	5.154	6.467	7.534	8.279
12.700	-0.000	1.117	1.928	2.723	3.515	5.048	6.366	7.442	8.211
12.800	-0.000	1.025	1.827	2.622	3.417	4.942	6.264	7.350	8.144
12.900	-0.000	0.925	1.723	2.521	3.321	4.835	6.162	7.257	8.077
13.000	-0.000	0.817	1.617	2.419	3.225	4.726	6.056	7.166	8.004
13.100	-0.000	0.792	1.593	2.397	3.204	4.704	6.035	7.148	7.988
13.200	-0.000	0.786	1.587	2.392	3.200	4.701	6.033	7.146	7.986
13.300	-0.000	0.780	1.582	2.387	3.196	4.699	6.031	7.143	7.985
13.400	-0.000	0.774	1.576	2.382	3.192	4.696	6.029	7.141	7.983
13.500	-0.000	0.768	1.571	2.377	3.188	4.694	6.027	7.139	7.981
13.600	-0.000	0.762	1.565	2.373	3.184	4.692	6.025	7.137	7.980
13.700	-0.000	0.756	1.560	2.368	3.180	4.690	6.023	7.134	7.978
13.800	-0.000	0.750	1.555	2.363	3.175	4.687	6.021	7.132	7.976
13.900	-0.000	0.745	1.549	2.358	3.171	4.685	6.020	7.130	7.975
14.000	-0.000	0.739	1.544	2.354	3.167	4.683	6.018	7.128	7.973
14.100	-0.000	0.734	1.539	2.349	3.164	4.681	6.016	7.125	7.971
14.200	-0.000	0.729	1.534	2.344	3.160	4.678	6.014	7.123	7.969
14.300	-0.000	0.724	1.529	2.340	3.156	4.676	6.012	7.121	7.968
14.400	-0.000	0.719	1.523	2.335	3.152	4.674	6.010	7.119	7.966
14.500	-0.000	0.715	1.518	2.331	3.148	4.672	6.008	7.116	7.964
14.600	-0.000	0.711	1.513	2.326	3.144	4.670	6.006	7.114	7.963
14.700	-0.000	0.708	1.508	2.322	3.140	4.668	6.004	7.112	7.961
14.800	-0.000	0.705	1.504	2.317	3.136	4.666	6.003	7.109	7.959
14.900	-0.000	0.705	1.499	2.313	3.133	4.664	6.001	7.107	7.958
15.000	-0.000	0.705	1.494	2.308	3.129	4.662	5.999	7.105	7.956
15.100	-0.000	0.706	1.489	2.304	3.125	4.660	5.997	7.102	7.954
15.200	-0.000	0.707	1.485	2.300	3.122	4.659	5.995	7.100	7.953
15.300	-0.000	0.708	1.480	2.296	3.118	4.657	5.993	7.098	7.951
15.400	-0.000	0.709	1.476	2.291	3.115	4.655	5.992	7.096	7.949
15.500	-0.000	0.709	1.472	2.287	3.111	4.653	5.990	7.093	7.948
15.600	-0.000	0.710	1.468	2.283	3.107	4.652	5.988	7.091	7.946
15.700	-0.000	0.711	1.464	2.279	3.104	4.650	5.986	7.089	7.944
15.800	-0.000	0.712	1.460	2.275	3.101	4.648	5.984	7.086	7.943
15.900	-0.000	0.713	1.457	2.271	3.097	4.647	5.983	7.084	7.941
16.000	-0.000	0.714	1.453	2.267	3.094	4.645	5.981	7.082	7.939
16.100	-0.000	0.715	1.450	2.264	3.091	4.644	5.979	7.079	7.938
16.200	-0.000	0.716	1.447	2.260	3.087	4.642	5.977	7.077	7.936
16.300	-0.000	0.716	1.445	2.256	3.084	4.641	5.976	7.075	7.935
16.400	-0.000	0.717	1.442	2.253	3.081	4.639	5.974	7.072	7.933
16.500	-0.000	0.718	1.441	2.249	3.078	4.638	5.972	7.070	7.932
16.600	-0.000	0.719	1.440	2.246	3.075	4.636	5.970	7.068	7.930
16.700	-0.000	0.720	1.439	2.243	3.072	4.635	5.969	7.065	7.928
16.800	-0.000	0.721	1.439	2.239	3.069	4.634	5.967	7.063	7.927
16.900	-0.000	0.722	1.441	2.236	3.066	4.632	5.965	7.061	7.925
17.000	-0.000	0.723	1.443	2.233	3.063	4.631	5.963	7.058	7.924
17.100	-0.000	0.724	1.445	2.231	3.061	4.630	5.962	7.056	7.922
17.200	-0.000	0.725	1.447	2.228	3.058	4.628	5.960	7.054	7.921
17.300	-0.000	0.726	1.449	2.225	3.055	4.627	5.958	7.052	7.919
17.400	-0.000	0.727	1.451	2.223	3.053	4.626	5.957	7.049	7.917

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.728	1.453	2.221	3.050	4.625	5.955	7.047	7.916
17.600	-0.000	0.729	1.455	2.218	3.048	4.624	5.953	7.045	7.914
17.700	-0.000	0.730	1.457	2.216	3.046	4.622	5.951	7.043	7.913
17.800	-0.000	0.731	1.459	2.215	3.043	4.621	5.950	7.040	7.911
17.900	-0.000	0.732	1.461	2.213	3.041	4.620	5.948	7.038	7.910
18.000	-0.000	0.733	1.463	2.212	3.039	4.619	5.946	7.036	7.908
18.100	-0.000	0.734	1.465	2.211	3.037	4.618	5.945	7.034	7.906
18.200	-0.000	0.735	1.467	2.210	3.035	4.617	5.943	7.032	7.905
18.300	-0.000	0.736	1.470	2.209	3.033	4.616	5.941	7.030	7.903
18.400	-0.000	0.737	1.472	2.209	3.032	4.615	5.940	7.028	7.901
18.500	-0.000	0.739	1.474	2.209	3.030	4.614	5.938	7.026	7.900
18.600	-0.000	0.740	1.476	2.210	3.029	4.613	5.937	7.024	7.898
18.700	-0.000	0.741	1.478	2.211	3.027	4.612	5.935	7.021	7.897
18.800	-0.000	0.742	1.481	2.213	3.026	4.611	5.933	7.019	7.895
18.900	-0.000	0.743	1.483	2.216	3.025	4.610	5.932	7.017	7.893
19.000	-0.000	0.744	1.485	2.220	3.024	4.610	5.930	7.015	7.892
19.100	-0.000	0.745	1.488	2.223	3.023	4.609	5.928	7.014	7.890
19.200	-0.000	0.747	1.490	2.227	3.022	4.608	5.927	7.012	7.889
19.300	-0.000	0.748	1.492	2.230	3.022	4.607	5.925	7.010	7.887
19.400	-0.000	0.749	1.495	2.234	3.021	4.606	5.923	7.008	7.886
19.500	-0.000	0.750	1.497	2.237	3.021	4.606	5.922	7.006	7.885
19.600	-0.000	0.751	1.499	2.241	3.021	4.605	5.920	7.004	7.884
19.700	-0.000	0.753	1.502	2.245	3.021	4.604	5.919	7.002	7.882
19.800	-0.000	0.754	1.504	2.248	3.021	4.603	5.917	7.000	7.881
19.900	-0.000	0.755	1.507	2.252	3.021	4.603	5.915	6.998	7.880
20.000	-0.000	0.756	1.509	2.256	3.022	4.602	5.914	6.996	7.878
20.100	-0.000	0.757	1.512	2.260	3.023	4.601	5.912	6.994	7.877
20.200	-0.000	0.759	1.514	2.263	3.024	4.601	5.911	6.992	7.876
20.300	-0.000	0.760	1.517	2.267	3.025	4.600	5.909	6.990	7.874
20.400	-0.000	0.761	1.519	2.271	3.027	4.600	5.907	6.988	7.873
20.500	-0.000	0.763	1.522	2.275	3.029	4.599	5.906	6.986	7.872
20.600	-0.000	0.764	1.525	2.279	3.031	4.598	5.904	6.985	7.870
20.700	-0.000	0.765	1.527	2.283	3.034	4.598	5.902	6.983	7.869
20.800	-0.000	0.767	1.530	2.287	3.037	4.597	5.901	6.981	7.868
20.900	-0.000	0.768	1.533	2.291	3.041	4.597	5.899	6.980	7.866
21.000	-0.000	0.769	1.536	2.296	3.046	4.597	5.898	6.979	7.865
21.100	-0.000	0.771	1.539	2.300	3.051	4.596	5.896	6.977	7.864
21.200	-0.000	0.772	1.541	2.304	3.057	4.596	5.894	6.975	7.863
21.300	-0.000	0.773	1.544	2.308	3.063	4.595	5.893	6.973	7.861
21.400	-0.000	0.775	1.547	2.312	3.068	4.595	5.891	6.972	7.860
21.500	-0.000	0.776	1.549	2.317	3.074	4.595	5.889	6.970	7.859
21.600	-0.000	0.777	1.552	2.320	3.080	4.594	5.888	6.969	7.858
21.700	-0.000	0.779	1.555	2.325	3.085	4.594	5.887	6.967	7.857
21.800	-0.000	0.781	1.558	2.329	3.091	4.595	5.885	6.965	7.856
21.900	-0.000	0.782	1.561	2.333	3.098	4.594	5.884	6.964	7.854
22.000	-0.000	0.784	1.564	2.338	3.102	4.594	5.882	6.962	7.853
22.100	-0.000	0.785	1.567	2.342	3.108	4.594	5.880	6.961	7.852
22.200	-0.000	0.786	1.570	2.347	3.114	4.594	5.879	6.960	7.851
22.300	-0.000	0.788	1.573	2.351	3.120	4.594	5.877	6.958	7.850
22.400	-0.000	0.789	1.576	2.356	3.126	4.593	5.876	6.957	7.849
22.500	-0.000	0.791	1.580	2.360	3.132	4.593	5.874	6.955	7.848
22.600	-0.000	0.793	1.581	2.365	3.138	4.593	5.872	6.954	7.847
22.700	-0.000	0.794	1.585	2.369	3.144	4.593	5.871	6.953	7.846
22.800	-0.000	0.796	1.588	2.374	3.150	4.593	5.869	6.951	7.845
22.900	-0.000	0.797	1.591	2.379	3.156	4.594	5.868	6.950	7.844
23.000	-0.000	0.799	1.594	2.383	3.163	4.594	5.867	6.949	7.843
23.100	-0.000	0.800	1.597	2.388	3.169	4.595	5.865	6.948	7.842
23.200	-0.000	0.802	1.601	2.393	3.174	4.595	5.864	6.947	7.841
23.300	-0.000	0.803	1.604	2.398	3.180	4.593	5.863	6.945	7.840
23.400	-0.000	0.805	1.607	2.402	3.186	4.596	5.861	6.944	7.839

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.807	1.610	2.407	3.192	4.597	5.860	6.943	7.838
23.600	-0.000	0.808	1.614	2.412	3.197	4.598	5.860	6.942	7.837
23.700	-0.000	0.810	1.617	2.417	3.203	4.600	5.859	6.942	7.836
23.800	-0.000	0.812	1.620	2.422	3.208	4.601	5.858	6.941	7.835
23.900	-0.000	0.813	1.624	2.427	3.213	4.603	5.857	6.940	7.835
24.000	-0.000	0.815	1.627	2.432	3.218	4.605	5.853	6.939	7.834
24.100	-0.000	0.817	1.630	2.437	3.222	4.607	5.856	6.938	7.833
24.200	-0.000	0.818	1.634	2.442	3.226	4.610	5.855	6.937	7.832
24.300	-0.000	0.820	1.637	2.447	3.231	4.612	5.854	6.936	7.832
24.400	-0.000	0.822	1.641	2.452	3.234	4.615	5.854	6.936	7.831
24.500	-0.000	0.824	1.644	2.457	3.238	4.617	5.853	6.935	7.831
24.600	-0.000	0.825	1.647	2.463	3.241	4.620	5.853	6.934	7.830
24.700	-0.000	0.827	1.651	2.468	3.244	4.623	5.853	6.934	7.829
24.800	-0.000	0.829	1.654	2.471	3.247	4.626	5.850	6.933	7.829
24.900	-0.000	0.831	1.658	2.478	3.250	4.628	5.853	6.932	7.828
25.000	-0.000	0.833	1.662	2.483	3.253	4.631	5.853	6.932	7.828
25.100	-0.000	0.834	1.665	2.488	3.253	4.633	5.853	6.931	7.828
25.200	-0.000	0.836	1.669	2.493	3.258	4.635	5.854	6.931	7.827
25.300	-0.000	0.838	1.672	2.497	3.261	4.637	5.854	6.931	7.827
25.400	-0.000	0.840	1.676	2.502	3.263	4.639	5.855	6.931	7.827
25.500	-0.000	0.842	1.680	2.506	3.265	4.641	5.856	6.931	7.827
25.600	-0.000	0.843	1.683	2.510	3.268	4.643	5.857	6.931	7.827
25.700	-0.000	0.845	1.687	2.513	3.270	4.645	5.858	6.931	7.827
25.800	-0.000	0.847	1.691	2.517	3.271	4.647	5.860	6.931	7.827
25.900	-0.000	0.849	1.695	2.519	3.273	4.649	5.862	6.931	7.827
26.000	-0.000	0.851	1.698	2.522	3.275	4.651	5.864	6.932	7.827

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 0.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	13.201	14.563	15.162	15.414	15.312	14.674	13.768	12.873
0.100	-0.001	14.187	15.992	16.734	16.991	16.698	15.701	14.197	12.348
0.200	-0.001	14.004	15.963	16.751	17.036	16.773	15.778	14.260	12.380
0.300	-0.001	13.411	15.556	16.445	16.803	16.653	15.747	14.315	12.522
0.400	-0.001	12.776	15.103	16.098	16.533	16.505	15.703	14.367	12.676
0.500	-0.001	12.160	14.659	15.757	16.266	16.356	15.658	14.417	12.826
0.600	-0.001	11.576	14.241	15.430	16.010	16.212	15.612	14.464	12.969
0.700	-0.001	11.017	13.847	15.121	15.767	16.075	15.568	14.510	13.098
0.800	-0.001	10.483	13.475	14.828	15.537	15.945	15.526	14.554	13.210
0.900	-0.001	9.986	13.123	14.550	15.318	15.820	15.486	14.595	13.304
1.000	-0.001	9.524	12.790	14.286	15.111	15.702	15.447	14.634	13.383
1.100	-0.001	9.090	12.472	14.035	14.912	15.589	15.410	14.670	13.447
1.200	-0.001	8.681	12.168	13.796	14.721	15.480	15.375	14.701	13.499
1.300	-0.001	8.292	11.869	13.566	14.538	15.377	15.340	14.725	13.540
1.400	-0.001	7.922	11.580	13.347	14.361	15.277	15.307	14.742	13.571
1.500	-0.001	7.568	11.304	13.136	14.192	15.181	15.275	14.753	13.595
1.600	-0.001	7.230	11.042	12.933	14.028	15.088	15.244	14.757	13.612
1.700	-0.001	6.907	10.790	12.737	13.871	14.998	15.214	14.754	13.623
1.800	-0.001	6.598	10.549	12.546	13.719	14.911	15.185	14.747	13.629
1.900	-0.001	6.308	10.317	12.357	13.572	14.826	15.156	14.734	13.632
2.000	-0.001	6.041	10.093	12.171	13.430	14.743	15.125	14.717	13.632
2.100	-0.001	5.797	9.878	11.991	13.292	14.663	15.092	14.696	13.629
2.200	-0.001	5.572	9.669	11.817	13.159	14.584	15.057	14.671	13.623
2.300	-0.001	5.365	9.467	11.649	13.029	14.507	15.020	14.643	13.616
2.400	-0.001	5.174	9.270	11.488	12.902	14.432	14.980	14.612	13.606
2.500	-0.001	4.998	9.079	11.330	12.775	14.360	14.938	14.579	13.596
2.600	-0.001	4.834	8.893	11.178	12.650	14.289	14.893	14.543	13.583
2.700	-0.001	4.682	8.712	11.029	12.527	14.220	14.846	14.506	13.568
2.800	-0.001	4.541	8.537	10.885	12.406	14.154	14.797	14.466	13.553
2.900	-0.001	4.409	8.365	10.746	12.289	14.089	14.746	14.426	13.537
3.000	-0.001	4.285	8.198	10.610	12.175	14.026	14.693	14.383	13.521
3.100	-0.001	4.169	8.034	10.477	12.064	13.964	14.638	14.339	13.504
3.200	-0.001	4.061	7.874	10.348	11.957	13.903	14.581	14.293	13.487
3.300	-0.001	3.959	7.717	10.221	11.853	13.842	14.523	14.245	13.470
3.400	-0.001	3.863	7.563	10.097	11.751	13.781	14.463	14.197	13.453
3.500	-0.001	3.772	7.413	9.976	11.652	13.719	14.402	14.148	13.435
3.600	-0.001	3.686	7.265	9.858	11.555	13.657	14.340	14.099	13.418
3.700	-0.001	3.605	7.122	9.741	11.461	13.594	14.276	14.048	13.400
3.800	-0.001	3.529	6.983	9.628	11.368	13.530	14.210	13.996	13.381
3.900	-0.001	3.456	6.850	9.516	11.278	13.464	14.143	13.944	13.362
4.000	-0.001	3.386	6.722	9.406	11.189	13.397	14.075	13.892	13.341
4.100	-0.001	3.319	6.599	9.298	11.102	13.328	14.005	13.839	13.320
4.200	-0.001	3.256	6.482	9.192	11.016	13.257	13.934	13.785	13.299
4.300	-0.001	3.196	6.370	9.087	10.932	13.184	13.861	13.732	13.276
4.400	-0.001	3.139	6.263	8.984	10.849	13.110	13.788	13.678	13.253
4.500	-0.001	3.085	6.160	8.883	10.768	13.034	13.715	13.624	13.230
4.600	-0.001	3.033	6.062	8.783	10.688	12.957	13.641	13.570	13.206
4.700	-0.001	2.984	5.967	8.685	10.608	12.879	13.567	13.516	13.181
4.800	-0.001	2.937	5.876	8.587	10.530	12.799	13.493	13.461	13.156
4.900	-0.001	2.892	5.788	8.491	10.454	12.718	13.419	13.407	13.129
5.000	-0.001	2.849	5.704	8.397	10.378	12.636	13.344	13.353	13.101
5.100	-0.001	2.807	5.623	8.304	10.304	12.553	13.269	13.300	13.072
5.200	-0.001	2.768	5.545	8.211	10.230	12.469	13.193	13.247	13.042
5.300	-0.001	2.730	5.470	8.120	10.158	12.385	13.116	13.194	13.012
5.400	-0.001	2.694	5.398	8.031	10.086	12.300	13.038	13.141	12.982

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.660	5.328	7.942	10.015	12.214	12.959	13.088	12.952
5.600	-0.001	2.626	5.262	7.855	9.945	12.127	12.880	13.035	12.922
5.700	-0.001	2.595	5.198	7.769	9.874	12.040	12.800	12.982	12.891
5.800	-0.001	2.564	5.136	7.686	9.803	11.953	12.720	12.929	12.860
5.900	-0.001	2.534	5.076	7.604	9.732	11.865	12.640	12.875	12.828
6.000	-0.001	2.506	5.019	7.524	9.660	11.776	12.561	12.822	12.795
6.100	-0.001	2.478	4.965	7.447	9.587	11.687	12.481	12.768	12.761
6.200	-0.001	2.452	4.913	7.372	9.512	11.598	12.402	12.712	12.725
6.300	-0.000	2.427	4.862	7.299	9.437	11.508	12.323	12.657	12.687
6.400	-0.000	2.402	4.813	7.229	9.360	11.418	12.244	12.600	12.648
6.500	-0.000	2.379	4.766	7.160	9.282	11.328	12.165	12.542	12.607
6.600	-0.000	2.356	4.721	7.094	9.202	11.237	12.086	12.485	12.564
6.700	-0.000	2.334	4.677	7.030	9.120	11.146	12.007	12.427	12.518
6.800	-0.000	2.313	4.635	6.968	9.037	11.055	11.927	12.369	12.469
6.900	-0.000	2.293	4.594	6.908	8.953	10.963	11.848	12.311	12.418
7.000	-0.000	2.273	4.555	6.850	8.867	10.870	11.769	12.254	12.364
7.100	-0.000	2.254	4.517	6.794	8.779	10.777	11.690	12.196	12.308
7.200	-0.000	2.236	4.481	6.740	8.691	10.684	11.611	12.138	12.251
7.300	-0.000	2.218	4.445	6.687	8.601	10.591	11.531	12.080	12.193
7.400	-0.000	2.201	4.411	6.635	8.511	10.497	11.452	12.022	12.135
7.500	-0.000	2.184	4.378	6.584	8.420	10.402	11.372	11.964	12.075
7.600	-0.000	2.168	4.346	6.533	8.328	10.307	11.292	11.905	12.013
7.700	-0.000	2.153	4.315	6.482	8.236	10.212	11.212	11.846	11.951
7.800	-0.000	2.138	4.285	6.431	8.144	10.116	11.131	11.787	11.888
7.900	-0.000	2.123	4.256	6.379	8.052	10.020	11.050	11.726	11.826
8.000	-0.000	2.109	4.228	6.326	7.959	9.923	10.969	11.664	11.763
8.100	-0.000	2.096	4.200	6.272	7.867	9.826	10.888	11.601	11.699
8.200	-0.000	2.083	4.174	6.217	7.775	9.727	10.807	11.537	11.635
8.300	-0.000	2.070	4.149	6.161	7.683	9.629	10.725	11.471	11.571
8.400	-0.000	2.058	4.124	6.104	7.591	9.529	10.643	11.403	11.506
8.500	-0.000	2.046	4.100	6.046	7.500	9.429	10.561	11.333	11.440
8.600	-0.000	2.034	4.077	5.987	7.408	9.329	10.478	11.261	11.373
8.700	-0.000	2.023	4.055	5.927	7.316	9.228	10.395	11.188	11.306
8.800	-0.000	2.012	4.034	5.867	7.224	9.127	10.311	11.112	11.238
8.900	-0.000	2.002	4.013	5.805	7.132	9.025	10.227	11.035	11.169
9.000	-0.000	1.992	3.992	5.743	7.041	8.923	10.142	10.956	11.100
9.100	-0.000	1.982	3.972	5.680	6.949	8.820	10.058	10.874	11.031
9.200	-0.000	1.973	3.953	5.616	6.858	8.717	9.973	10.791	10.960
9.300	-0.000	1.964	3.933	5.551	6.766	8.613	9.888	10.708	10.890
9.400	-0.000	1.955	3.912	5.486	6.675	8.509	9.803	10.624	10.818
9.500	-0.000	1.946	3.890	5.419	6.583	8.403	9.717	10.539	10.747
9.600	-0.000	1.938	3.866	5.352	6.491	8.298	9.630	10.453	10.674
9.700	-0.000	1.930	3.840	5.284	6.400	8.192	9.544	10.367	10.602
9.800	-0.000	1.922	3.811	5.214	6.308	8.085	9.456	10.280	10.528
9.900	-0.000	1.915	3.780	5.144	6.216	7.978	9.367	10.192	10.454
10.000	-0.000	1.907	3.746	5.073	6.124	7.870	9.277	10.103	10.379
10.100	-0.000	1.900	3.710	5.001	6.031	7.763	9.185	10.014	10.304
10.200	-0.000	1.893	3.671	4.928	5.939	7.655	9.091	9.923	10.228
10.300	-0.000	1.886	3.630	4.854	5.846	7.548	8.995	9.832	10.151
10.400	-0.000	1.879	3.587	4.779	5.753	7.442	8.898	9.740	10.074
10.500	-0.000	1.872	3.542	4.703	5.660	7.335	8.799	9.648	9.996
10.600	-0.000	1.865	3.495	4.626	5.567	7.230	8.699	9.555	9.918
10.700	-0.000	1.859	3.445	4.549	5.474	7.124	8.597	9.462	9.839
10.800	-0.000	1.852	3.394	4.470	5.381	7.020	8.493	9.368	9.761
10.900	-0.000	1.846	3.340	4.390	5.288	6.916	8.387	9.274	9.681
11.000	-0.000	1.839	3.284	4.309	5.195	6.813	8.280	9.179	9.601
11.100	-0.000	1.832	3.226	4.227	5.101	6.711	8.173	9.084	9.521
11.200	-0.000	1.824	3.165	4.145	5.008	6.610	8.066	8.988	9.441
11.300	-0.000	1.814	3.103	4.061	4.914	6.509	7.958	8.892	9.360
11.400	-0.000	1.799	3.038	3.977	4.820	6.409	7.850	8.795	9.279

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.780	2.971	3.891	4.726	6.311	7.742	8.697	9.198
11.600	-0.000	1.755	2.902	3.805	4.632	6.213	7.634	8.600	9.117
11.700	-0.000	1.727	2.830	3.717	4.538	6.116	7.527	8.502	9.037
11.800	-0.000	1.693	2.756	3.629	4.444	6.020	7.419	8.403	8.957
11.900	-0.000	1.655	2.680	3.540	4.349	5.925	7.312	8.304	8.878
12.000	-0.000	1.612	2.602	3.450	4.254	5.830	7.205	8.206	8.800
12.100	-0.000	1.563	2.521	3.358	4.160	5.737	7.099	8.107	8.723
12.200	-0.000	1.510	2.438	3.266	4.065	5.643	6.993	8.010	8.646
12.300	-0.000	1.452	2.353	3.173	3.970	5.549	6.888	7.913	8.571
12.400	-0.000	1.388	2.266	3.080	3.875	5.454	6.784	7.818	8.497
12.500	-0.000	1.319	2.177	2.985	3.780	5.358	6.682	7.725	8.425
12.600	-0.000	1.244	2.085	2.890	3.686	5.260	6.580	7.632	8.354
12.700	-0.000	1.164	1.991	2.794	3.590	5.159	6.479	7.540	8.284
12.800	-0.000	1.078	1.895	2.696	3.495	5.055	6.378	7.449	8.215
12.900	-0.000	0.988	1.798	2.599	3.401	4.952	6.279	7.358	8.148
13.000	-0.000	0.906	1.710	2.513	3.317	4.855	6.185	7.273	8.088
13.100	-0.000	0.838	1.640	2.444	3.251	4.775	6.106	7.205	8.037
13.200	-0.000	0.793	1.596	2.401	3.210	4.722	6.054	7.161	8.001
13.300	-0.000	0.778	1.581	2.387	3.197	4.708	6.040	7.149	7.991
13.400	-0.000	0.772	1.576	2.382	3.193	4.705	6.038	7.147	7.989
13.500	-0.000	0.766	1.570	2.378	3.189	4.703	6.036	7.145	7.988
13.600	-0.000	0.760	1.565	2.373	3.185	4.701	6.035	7.143	7.986
13.700	-0.000	0.755	1.559	2.368	3.181	4.698	6.033	7.140	7.984
13.800	-0.000	0.749	1.554	2.363	3.177	4.696	6.031	7.138	7.983
13.900	-0.000	0.744	1.549	2.359	3.173	4.694	6.029	7.136	7.981
14.000	-0.000	0.738	1.544	2.354	3.169	4.692	6.027	7.134	7.979
14.100	-0.000	0.733	1.538	2.349	3.165	4.690	6.025	7.131	7.977
14.200	-0.000	0.728	1.533	2.345	3.161	4.688	6.023	7.129	7.976
14.300	-0.000	0.724	1.528	2.340	3.157	4.686	6.021	7.127	7.974
14.400	-0.000	0.720	1.523	2.336	3.153	4.684	6.019	7.124	7.972
14.500	-0.000	0.716	1.518	2.331	3.150	4.682	6.017	7.122	7.971
14.600	-0.000	0.713	1.513	2.327	3.146	4.680	6.016	7.120	7.969
14.700	-0.000	0.710	1.508	2.322	3.142	4.678	6.014	7.117	7.967
14.800	-0.000	0.708	1.504	2.318	3.138	4.676	6.012	7.115	7.966
14.900	-0.000	0.707	1.499	2.314	3.135	4.674	6.010	7.113	7.964
15.000	-0.000	0.707	1.494	2.309	3.131	4.672	6.008	7.110	7.962
15.100	-0.000	0.707	1.490	2.305	3.127	4.670	6.006	7.108	7.961
15.200	-0.000	0.708	1.485	2.301	3.124	4.669	6.005	7.106	7.959
15.300	-0.000	0.709	1.481	2.297	3.120	4.667	6.003	7.103	7.957
15.400	-0.000	0.710	1.477	2.293	3.117	4.665	6.001	7.101	7.956
15.500	-0.000	0.711	1.473	2.289	3.113	4.664	5.999	7.099	7.954
15.600	-0.000	0.712	1.469	2.285	3.110	4.662	5.997	7.096	7.952
15.700	-0.000	0.713	1.465	2.281	3.106	4.660	5.995	7.094	7.951
15.800	-0.000	0.713	1.462	2.277	3.103	4.659	5.994	7.092	7.949
15.900	-0.000	0.714	1.458	2.273	3.100	4.657	5.992	7.089	7.947
16.000	-0.000	0.715	1.455	2.269	3.097	4.656	5.990	7.087	7.946
16.100	-0.000	0.716	1.452	2.265	3.093	4.654	5.988	7.085	7.944
16.200	-0.000	0.717	1.450	2.262	3.090	4.653	5.987	7.082	7.943
16.300	-0.000	0.718	1.448	2.258	3.087	4.651	5.985	7.080	7.941
16.400	-0.000	0.719	1.446	2.255	3.084	4.650	5.983	7.078	7.939
16.500	-0.000	0.720	1.444	2.252	3.081	4.649	5.981	7.075	7.938
16.600	-0.000	0.721	1.443	2.248	3.078	4.647	5.980	7.073	7.936
16.700	-0.000	0.722	1.443	2.245	3.075	4.646	5.978	7.071	7.935
16.800	-0.000	0.723	1.443	2.242	3.072	4.644	5.976	7.068	7.933
16.900	-0.000	0.724	1.444	2.239	3.070	4.643	5.974	7.066	7.931
17.000	-0.000	0.725	1.446	2.236	3.067	4.642	5.973	7.064	7.930
17.100	-0.000	0.726	1.448	2.234	3.064	4.641	5.971	7.061	7.928
17.200	-0.000	0.727	1.450	2.231	3.062	4.639	5.969	7.059	7.927
17.300	-0.000	0.728	1.452	2.229	3.059	4.638	5.968	7.057	7.925
17.400	-0.000	0.729	1.454	2.226	3.057	4.637	5.966	7.055	7.923

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.730	1.456	2.224	3.054	4.636	5.964	7.052	7.922
17.600	-0.000	0.731	1.458	2.222	3.052	4.635	5.962	7.050	7.920
17.700	-0.000	0.732	1.460	2.221	3.050	4.634	5.961	7.048	7.918
17.800	-0.000	0.733	1.462	2.219	3.048	4.632	5.959	7.046	7.917
17.900	-0.000	0.734	1.464	2.218	3.046	4.631	5.957	7.044	7.915
18.000	-0.000	0.735	1.466	2.217	3.044	4.630	5.956	7.042	7.913
18.100	-0.000	0.736	1.469	2.216	3.042	4.629	5.954	7.039	7.912
18.200	-0.000	0.737	1.471	2.215	3.040	4.628	5.952	7.037	7.910
18.300	-0.000	0.738	1.473	2.215	3.038	4.627	5.951	7.035	7.909
18.400	-0.000	0.739	1.475	2.215	3.037	4.626	5.949	7.033	7.907
18.500	-0.000	0.740	1.477	2.215	3.035	4.625	5.947	7.031	7.905
18.600	-0.000	0.741	1.480	2.216	3.034	4.625	5.946	7.029	7.904
18.700	-0.000	0.743	1.482	2.218	3.033	4.624	5.944	7.027	7.902
18.800	-0.000	0.744	1.484	2.219	3.032	4.623	5.942	7.025	7.900
18.900	-0.000	0.745	1.487	2.222	3.031	4.622	5.941	7.023	7.899
19.000	-0.000	0.746	1.489	2.225	3.030	4.621	5.939	7.021	7.897
19.100	-0.000	0.747	1.491	2.229	3.029	4.620	5.937	7.019	7.896
19.200	-0.000	0.748	1.494	2.232	3.028	4.620	5.936	7.017	7.895
19.300	-0.000	0.750	1.496	2.236	3.028	4.619	5.934	7.015	7.894
19.400	-0.000	0.751	1.498	2.240	3.028	4.618	5.932	7.013	7.892
19.500	-0.000	0.752	1.501	2.243	3.028	4.617	5.931	7.011	7.891
19.600	-0.000	0.753	1.503	2.247	3.028	4.617	5.929	7.009	7.889
19.700	-0.000	0.754	1.506	2.251	3.028	4.616	5.928	7.007	7.888
19.800	-0.000	0.756	1.508	2.254	3.028	4.615	5.926	7.005	7.887
19.900	-0.000	0.757	1.511	2.258	3.029	4.615	5.924	7.003	7.885
20.000	-0.000	0.758	1.513	2.262	3.030	4.614	5.923	7.001	7.884
20.100	-0.000	0.759	1.516	2.266	3.031	4.614	5.921	6.999	7.882
20.200	-0.000	0.761	1.518	2.270	3.032	4.613	5.919	6.997	7.881
20.300	-0.000	0.762	1.521	2.273	3.034	4.612	5.918	6.995	7.880
20.400	-0.000	0.764	1.524	2.277	3.036	4.612	5.916	6.994	7.878
20.500	-0.000	0.765	1.527	2.281	3.038	4.611	5.914	6.992	7.877
20.600	-0.000	0.766	1.529	2.285	3.040	4.611	5.913	6.991	7.876
20.700	-0.000	0.767	1.532	2.290	3.043	4.610	5.911	6.989	7.874
20.800	-0.000	0.769	1.535	2.294	3.047	4.610	5.910	6.987	7.873
20.900	-0.000	0.770	1.537	2.298	3.051	4.610	5.908	6.986	7.872
21.000	-0.000	0.772	1.540	2.302	3.056	4.609	5.906	6.984	7.870
21.100	-0.000	0.773	1.543	2.306	3.061	4.609	5.904	6.982	7.869
21.200	-0.000	0.774	1.545	2.311	3.066	4.608	5.903	6.980	7.868
21.300	-0.000	0.776	1.548	2.314	3.072	4.608	5.901	6.979	7.867
21.400	-0.000	0.777	1.551	2.319	3.077	4.608	5.900	6.977	7.865
21.500	-0.000	0.779	1.554	2.322	3.083	4.608	5.899	6.976	7.864
21.600	-0.000	0.780	1.557	2.327	3.089	4.608	5.897	6.974	7.863
21.700	-0.000	0.781	1.560	2.332	3.094	4.607	5.895	6.972	7.862
21.800	-0.000	0.783	1.563	2.336	3.100	4.607	5.893	6.971	7.861
21.900	-0.000	0.784	1.566	2.340	3.106	4.607	5.892	6.969	7.859
22.000	-0.000	0.786	1.569	2.345	3.112	4.607	5.890	6.968	7.858
22.100	-0.000	0.787	1.572	2.349	3.118	4.606	5.888	6.966	7.857
22.200	-0.000	0.789	1.575	2.354	3.124	4.606	5.887	6.965	7.856
22.300	-0.000	0.790	1.578	2.358	3.130	4.606	5.885	6.964	7.855
22.400	-0.000	0.792	1.581	2.363	3.136	4.606	5.884	6.962	7.854
22.500	-0.000	0.793	1.584	2.367	3.142	4.606	5.882	6.961	7.852
22.600	-0.000	0.795	1.586	2.372	3.148	4.606	5.880	6.959	7.851
22.700	-0.000	0.797	1.590	2.377	3.154	4.606	5.879	6.958	7.850
22.800	-0.000	0.798	1.592	2.381	3.160	4.604	5.877	6.957	7.849
22.900	-0.000	0.799	1.596	2.386	3.166	4.604	5.876	6.956	7.848
23.000	-0.000	0.801	1.599	2.391	3.172	4.606	5.874	6.954	7.847
23.100	-0.000	0.803	1.603	2.396	3.179	4.607	5.873	6.953	7.846
23.200	-0.000	0.805	1.606	2.399	3.185	4.607	5.872	6.952	7.845
23.300	-0.000	0.806	1.609	2.404	3.191	4.608	5.872	6.951	7.844
23.400	-0.000	0.808	1.612	2.410	3.196	4.608	5.870	6.950	7.843

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.809	1.616	2.415	3.202	4.609	5.869	6.950	7.842
23.600	-0.000	0.811	1.619	2.420	3.208	4.610	5.865	6.948	7.841
23.700	-0.000	0.813	1.622	2.425	3.213	4.612	5.867	6.947	7.840
23.800	-0.000	0.814	1.626	2.430	3.218	4.613	5.866	6.946	7.840
23.900	-0.000	0.816	1.629	2.435	3.224	4.612	5.865	6.945	7.839
24.000	-0.000	0.818	1.632	2.440	3.229	4.617	5.864	6.944	7.838
24.100	-0.000	0.820	1.636	2.445	3.233	4.619	5.863	6.943	7.837
24.200	-0.000	0.821	1.639	2.450	3.238	4.621	5.863	6.942	7.836
24.300	-0.000	0.823	1.643	2.456	3.242	4.624	5.862	6.942	7.836
24.400	-0.000	0.825	1.646	2.461	3.246	4.626	5.861	6.941	7.835
24.500	-0.000	0.827	1.650	2.466	3.250	4.629	5.861	6.940	7.834
24.600	-0.000	0.828	1.653	2.470	3.254	4.632	5.858	6.939	7.834
24.700	-0.000	0.830	1.657	2.476	3.257	4.634	5.860	6.939	7.833
24.800	-0.000	0.832	1.660	2.481	3.260	4.636	5.860	6.938	7.833
24.900	-0.000	0.834	1.664	2.485	3.262	4.639	5.860	6.937	7.832
25.000	-0.000	0.835	1.667	2.492	3.265	4.641	5.860	6.937	7.832
25.100	-0.000	0.837	1.671	2.496	3.267	4.643	5.861	6.936	7.831
25.200	-0.000	0.839	1.675	2.501	3.270	4.645	5.861	6.936	7.831
25.300	-0.000	0.841	1.678	2.506	3.272	4.647	5.862	6.936	7.831
25.400	-0.000	0.843	1.682	2.510	3.274	4.649	5.863	6.936	7.831
25.500	-0.000	0.845	1.686	2.515	3.276	4.651	5.864	6.936	7.831
25.600	-0.000	0.847	1.690	2.519	3.278	4.653	5.865	6.936	7.831
25.700	-0.000	0.848	1.693	2.522	3.280	4.654	5.866	6.936	7.831
25.800	-0.000	0.850	1.697	2.526	3.282	4.656	5.868	6.936	7.831
25.900	-0.000	0.852	1.701	2.529	3.283	4.658	5.870	6.936	7.831
26.000	-0.000	0.854	1.705	2.532	3.284	4.659	5.872	6.937	7.831

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 1.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	11.703	13.286	14.038	14.425	14.601	14.307	13.750	13.223
0.100	-0.001	12.191	14.063	14.982	15.478	15.660	15.122	14.148	12.953
0.200	-0.001	12.351	14.398	15.405	15.927	16.058	15.420	14.277	12.832
0.300	-0.001	12.273	14.465	15.518	16.046	16.175	15.520	14.336	12.827
0.400	-0.001	11.948	14.289	15.398	15.961	16.143	15.532	14.382	12.898
0.500	-0.001	11.533	14.023	15.194	15.805	16.062	15.516	14.426	12.991
0.600	-0.001	11.089	13.725	14.963	15.623	15.963	15.489	14.469	13.085
0.700	-0.001	10.637	13.418	14.720	15.432	15.856	15.458	14.510	13.174
0.800	-0.001	10.190	13.112	14.478	15.240	15.749	15.425	14.549	13.253
0.900	-0.001	9.746	12.810	14.242	15.050	15.642	15.391	14.586	13.323
1.000	-0.001	9.320	12.516	14.013	14.865	15.537	15.358	14.620	13.382
1.100	-0.001	8.914	12.231	13.790	14.686	15.435	15.325	14.647	13.432
1.200	-0.001	8.529	11.955	13.575	14.512	15.336	15.292	14.669	13.473
1.300	-0.001	8.160	11.687	13.365	14.345	15.240	15.261	14.684	13.506
1.400	-0.001	7.807	11.425	13.162	14.182	15.146	15.230	14.694	13.532
1.500	-0.001	7.468	11.165	12.966	14.026	15.056	15.200	14.697	13.553
1.600	-0.001	7.143	10.914	12.776	13.874	14.968	15.171	14.696	13.568
1.700	-0.001	6.832	10.672	12.591	13.726	14.883	15.142	14.690	13.579
1.800	-0.001	6.535	10.439	12.412	13.583	14.799	15.112	14.679	13.586
1.900	-0.001	6.256	10.215	12.237	13.444	14.718	15.081	14.664	13.591
2.000	-0.001	5.996	9.998	12.065	13.309	14.638	15.049	14.645	13.592
2.100	-0.001	5.756	9.788	11.894	13.177	14.561	15.014	14.623	13.591
2.200	-0.001	5.535	9.586	11.726	13.049	14.486	14.978	14.598	13.588
2.300	-0.001	5.332	9.389	11.562	12.925	14.413	14.939	14.570	13.584
2.400	-0.001	5.143	9.198	11.404	12.804	14.341	14.898	14.540	13.577
2.500	-0.001	4.969	9.011	11.251	12.685	14.272	14.855	14.508	13.570
2.600	-0.001	4.807	8.829	11.103	12.568	14.204	14.810	14.474	13.561
2.700	-0.001	4.657	8.652	10.959	12.452	14.139	14.763	14.439	13.550
2.800	-0.001	4.517	8.480	10.818	12.337	14.075	14.714	14.401	13.538
2.900	-0.001	4.386	8.312	10.681	12.223	14.012	14.664	14.363	13.525
3.000	-0.001	4.264	8.148	10.548	12.112	13.951	14.611	14.323	13.511
3.100	-0.001	4.150	7.988	10.418	12.003	13.891	14.557	14.282	13.496
3.200	-0.001	4.043	7.831	10.291	11.898	13.831	14.502	14.239	13.482
3.300	-0.001	3.942	7.677	10.168	11.795	13.772	14.446	14.195	13.466
3.400	-0.001	3.847	7.527	10.047	11.696	13.712	14.388	14.149	13.451
3.500	-0.001	3.758	7.380	9.928	11.598	13.651	14.329	14.102	13.435
3.600	-0.001	3.673	7.236	9.811	11.503	13.589	14.269	14.054	13.419
3.700	-0.001	3.592	7.095	9.697	11.410	13.527	14.208	14.006	13.402
3.800	-0.001	3.517	6.958	9.585	11.319	13.463	14.145	13.957	13.384
3.900	-0.001	3.445	6.827	9.474	11.229	13.399	14.081	13.906	13.365
4.000	-0.001	3.377	6.700	9.366	11.142	13.333	14.016	13.856	13.345
4.100	-0.001	3.311	6.579	9.259	11.056	13.266	13.949	13.805	13.324
4.200	-0.001	3.249	6.463	9.154	10.971	13.197	13.881	13.754	13.302
4.300	-0.001	3.189	6.351	9.051	10.888	13.127	13.811	13.703	13.280
4.400	-0.001	3.132	6.244	8.949	10.806	13.054	13.741	13.651	13.256
4.500	-0.001	3.078	6.142	8.849	10.726	12.980	13.669	13.599	13.233
4.600	-0.001	3.026	6.043	8.749	10.646	12.904	13.597	13.547	13.209
4.700	-0.001	2.977	5.949	8.652	10.568	12.827	13.525	13.494	13.184
4.800	-0.001	2.930	5.859	8.555	10.491	12.750	13.452	13.442	13.158
4.900	-0.001	2.885	5.772	8.460	10.415	12.670	13.379	13.389	13.132
5.000	-0.001	2.842	5.689	8.367	10.340	12.590	13.306	13.337	13.105
5.100	-0.001	2.801	5.608	8.274	10.266	12.509	13.233	13.285	13.076
5.200	-0.001	2.761	5.530	8.183	10.193	12.426	13.159	13.233	13.047
5.300	-0.001	2.723	5.455	8.093	10.121	12.344	13.085	13.182	13.017
5.400	-0.001	2.687	5.384	8.004	10.050	12.260	13.010	13.131	12.988

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.652	5.315	7.916	9.979	12.176	12.934	13.080	12.958
5.600	-0.001	2.619	5.248	7.830	9.908	12.091	12.858	13.029	12.927
5.700	-0.001	2.587	5.184	7.746	9.837	12.006	12.780	12.978	12.896
5.800	-0.001	2.557	5.123	7.663	9.766	11.920	12.703	12.926	12.865
5.900	-0.001	2.527	5.064	7.582	9.694	11.833	12.625	12.875	12.834
6.000	-0.001	2.499	5.007	7.504	9.622	11.746	12.547	12.824	12.801
6.100	-0.001	2.472	4.952	7.427	9.549	11.659	12.469	12.772	12.769
6.200	-0.001	2.445	4.899	7.352	9.475	11.571	12.391	12.720	12.735
6.300	-0.000	2.420	4.849	7.280	9.401	11.483	12.314	12.666	12.699
6.400	-0.000	2.396	4.800	7.210	9.325	11.395	12.236	12.612	12.662
6.500	-0.000	2.372	4.754	7.142	9.248	11.306	12.159	12.556	12.624
6.600	-0.000	2.350	4.709	7.076	9.170	11.217	12.081	12.500	12.584
6.700	-0.000	2.328	4.665	7.012	9.090	11.128	12.004	12.443	12.541
6.800	-0.000	2.307	4.623	6.950	9.009	11.038	11.926	12.387	12.496
6.900	-0.000	2.287	4.582	6.890	8.926	10.948	11.849	12.330	12.448
7.000	-0.000	2.267	4.543	6.832	8.842	10.857	11.771	12.273	12.397
7.100	-0.000	2.248	4.505	6.776	8.757	10.766	11.694	12.216	12.344
7.200	-0.000	2.230	4.469	6.722	8.670	10.675	11.616	12.159	12.290
7.300	-0.000	2.212	4.434	6.668	8.583	10.583	11.538	12.101	12.234
7.400	-0.000	2.195	4.399	6.616	8.494	10.491	11.460	12.044	12.177
7.500	-0.000	2.178	4.366	6.564	8.405	10.398	11.382	11.986	12.118
7.600	-0.000	2.162	4.334	6.513	8.315	10.305	11.304	11.929	12.059
7.700	-0.000	2.147	4.303	6.462	8.224	10.212	11.225	11.871	11.998
7.800	-0.000	2.132	4.273	6.410	8.134	10.118	11.146	11.812	11.936
7.900	-0.000	2.118	4.244	6.359	8.042	10.023	11.068	11.754	11.874
8.000	-0.000	2.104	4.216	6.307	7.951	9.928	10.988	11.694	11.812
8.100	-0.000	2.090	4.189	6.254	7.861	9.833	10.909	11.634	11.750
8.200	-0.000	2.077	4.163	6.200	7.770	9.737	10.829	11.572	11.688
8.300	-0.000	2.065	4.138	6.146	7.679	9.640	10.749	11.508	11.624
8.400	-0.000	2.052	4.113	6.090	7.589	9.542	10.669	11.443	11.560
8.500	-0.000	2.040	4.089	6.034	7.499	9.444	10.589	11.377	11.496
8.600	-0.000	2.029	4.067	5.977	7.409	9.346	10.508	11.308	11.430
8.700	-0.000	2.018	4.044	5.918	7.319	9.247	10.428	11.238	11.364
8.800	-0.000	2.007	4.023	5.859	7.229	9.147	10.346	11.166	11.297
8.900	-0.000	1.997	4.002	5.799	7.139	9.048	10.265	11.092	11.230
9.000	-0.000	1.987	3.982	5.738	7.049	8.948	10.182	11.016	11.162
9.100	-0.000	1.977	3.962	5.677	6.959	8.847	10.099	10.938	11.093
9.200	-0.000	1.967	3.942	5.614	6.869	8.746	10.016	10.858	11.024
9.300	-0.000	1.958	3.922	5.551	6.779	8.644	9.933	10.777	10.954
9.400	-0.000	1.949	3.901	5.487	6.690	8.542	9.850	10.694	10.884
9.500	-0.000	1.941	3.878	5.422	6.600	8.439	9.767	10.611	10.813
9.600	-0.000	1.933	3.855	5.356	6.510	8.336	9.682	10.527	10.742
9.700	-0.000	1.925	3.829	5.290	6.420	8.232	9.597	10.443	10.670
9.800	-0.000	1.917	3.802	5.223	6.330	8.127	9.512	10.357	10.598
9.900	-0.000	1.909	3.772	5.154	6.240	8.023	9.426	10.271	10.525
10.000	-0.000	1.902	3.741	5.085	6.150	7.917	9.339	10.184	10.452
10.100	-0.000	1.895	3.706	5.015	6.059	7.812	9.251	10.097	10.377
10.200	-0.000	1.888	3.670	4.944	5.969	7.707	9.161	10.008	10.303
10.300	-0.000	1.881	3.631	4.872	5.878	7.602	9.069	9.919	10.227
10.400	-0.000	1.875	3.589	4.799	5.787	7.497	8.976	9.829	10.151
10.500	-0.000	1.868	3.546	4.725	5.696	7.392	8.881	9.738	10.074
10.600	-0.000	1.862	3.500	4.650	5.605	7.288	8.784	9.646	9.997
10.700	-0.000	1.856	3.453	4.574	5.513	7.185	8.686	9.554	9.919
10.800	-0.000	1.850	3.403	4.497	5.422	7.082	8.585	9.462	9.841
10.900	-0.000	1.843	3.351	4.419	5.330	6.979	8.484	9.369	9.762
11.000	-0.000	1.837	3.298	4.340	5.239	6.877	8.380	9.275	9.684
11.100	-0.000	1.830	3.242	4.261	5.147	6.777	8.275	9.181	9.604
11.200	-0.000	1.820	3.184	4.180	5.055	6.676	8.169	9.086	9.525
11.300	-0.000	1.809	3.123	4.098	4.963	6.577	8.062	8.991	9.445
11.400	-0.000	1.794	3.060	4.015	4.871	6.479	7.955	8.895	9.365

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.777	2.996	3.932	4.778	6.381	7.848	8.799	9.285
11.600	-0.000	1.755	2.928	3.847	4.686	6.284	7.741	8.702	9.205
11.700	-0.000	1.728	2.859	3.762	4.593	6.189	7.635	8.605	9.124
11.800	-0.000	1.697	2.788	3.676	4.501	6.094	7.528	8.507	9.044
11.900	-0.000	1.661	2.714	3.589	4.408	6.000	7.422	8.409	8.964
12.000	-0.000	1.621	2.638	3.500	4.315	5.906	7.316	8.311	8.885
12.100	-0.000	1.576	2.560	3.411	4.222	5.814	7.210	8.212	8.807
12.200	-0.000	1.525	2.480	3.322	4.129	5.722	7.105	8.114	8.729
12.300	-0.000	1.470	2.397	3.231	4.036	5.632	7.001	8.016	8.653
12.400	-0.000	1.410	2.313	3.139	3.943	5.540	6.897	7.919	8.577
12.500	-0.000	1.344	2.226	3.047	3.849	5.448	6.793	7.823	8.502
12.600	-0.000	1.274	2.138	2.954	3.756	5.355	6.691	7.729	8.430
12.700	-0.000	1.199	2.048	2.860	3.663	5.261	6.590	7.636	8.359
12.800	-0.000	1.124	1.960	2.770	3.574	5.166	6.491	7.546	8.290
12.900	-0.000	1.053	1.877	2.685	3.490	5.072	6.397	7.461	8.224
13.000	-0.000	0.985	1.801	2.607	3.413	4.982	6.309	7.382	8.165
13.100	-0.000	0.922	1.732	2.537	3.344	4.900	6.230	7.310	8.113
13.200	-0.000	0.866	1.672	2.477	3.285	4.829	6.161	7.249	8.070
13.300	-0.000	0.819	1.623	2.430	3.240	4.772	6.105	7.201	8.034
13.400	-0.000	0.784	1.589	2.396	3.208	4.733	6.066	7.168	8.008
13.500	-0.000	0.766	1.570	2.379	3.191	4.713	6.047	7.152	7.994
13.600	-0.000	0.759	1.564	2.373	3.186	4.709	6.044	7.149	7.992
13.700	-0.000	0.754	1.559	2.368	3.182	4.707	6.042	7.146	7.990
13.800	-0.000	0.748	1.554	2.364	3.178	4.705	6.040	7.144	7.988
13.900	-0.000	0.743	1.548	2.359	3.174	4.703	6.038	7.142	7.987
14.000	-0.000	0.738	1.543	2.355	3.171	4.701	6.036	7.139	7.985
14.100	-0.000	0.733	1.538	2.350	3.167	4.699	6.034	7.137	7.983
14.200	-0.000	0.729	1.533	2.345	3.163	4.697	6.032	7.135	7.982
14.300	-0.000	0.725	1.528	2.341	3.159	4.695	6.030	7.132	7.980
14.400	-0.000	0.721	1.523	2.336	3.155	4.693	6.028	7.130	7.978
14.500	-0.000	0.718	1.518	2.332	3.151	4.691	6.027	7.128	7.976
14.600	-0.000	0.716	1.513	2.328	3.148	4.689	6.025	7.125	7.975
14.700	-0.000	0.714	1.509	2.323	3.144	4.687	6.023	7.123	7.973
14.800	-0.000	0.712	1.504	2.319	3.140	4.686	6.021	7.121	7.971
14.900	-0.000	0.711	1.499	2.315	3.137	4.684	6.019	7.118	7.970
15.000	-0.000	0.710	1.495	2.310	3.133	4.682	6.017	7.116	7.968
15.100	-0.000	0.710	1.491	2.306	3.130	4.680	6.015	7.114	7.966
15.200	-0.000	0.710	1.486	2.302	3.126	4.679	6.014	7.111	7.965
15.300	-0.000	0.711	1.482	2.298	3.123	4.677	6.012	7.109	7.963
15.400	-0.000	0.711	1.478	2.294	3.119	4.675	6.010	7.106	7.961
15.500	-0.000	0.712	1.474	2.290	3.116	4.674	6.008	7.104	7.960
15.600	-0.000	0.713	1.471	2.286	3.112	4.672	6.006	7.102	7.958
15.700	-0.000	0.714	1.467	2.282	3.109	4.671	6.005	7.099	7.956
15.800	-0.000	0.715	1.464	2.278	3.106	4.669	6.003	7.097	7.955
15.900	-0.000	0.716	1.461	2.275	3.103	4.668	6.001	7.095	7.953
16.000	-0.000	0.717	1.458	2.271	3.099	4.666	5.999	7.092	7.952
16.100	-0.000	0.717	1.456	2.268	3.096	4.665	5.998	7.090	7.950
16.200	-0.000	0.718	1.453	2.264	3.093	4.663	5.996	7.088	7.948
16.300	-0.000	0.719	1.451	2.261	3.090	4.662	5.994	7.085	7.947
16.400	-0.000	0.720	1.450	2.257	3.087	4.661	5.992	7.083	7.945
16.500	-0.000	0.721	1.449	2.254	3.084	4.659	5.991	7.081	7.943
16.600	-0.000	0.722	1.448	2.251	3.081	4.658	5.989	7.078	7.942
16.700	-0.000	0.723	1.448	2.248	3.079	4.657	5.987	7.076	7.940
16.800	-0.000	0.724	1.448	2.245	3.076	4.655	5.985	7.074	7.939
16.900	-0.000	0.725	1.449	2.242	3.073	4.654	5.984	7.071	7.937
17.000	-0.000	0.726	1.450	2.240	3.071	4.653	5.982	7.069	7.935
17.100	-0.000	0.727	1.451	2.237	3.068	4.652	5.980	7.067	7.934
17.200	-0.000	0.728	1.453	2.235	3.066	4.650	5.979	7.064	7.932
17.300	-0.000	0.729	1.455	2.233	3.063	4.649	5.977	7.062	7.930
17.400	-0.000	0.730	1.457	2.230	3.061	4.648	5.975	7.060	7.929

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.731	1.459	2.229	3.059	4.647	5.973	7.058	7.927
17.600	-0.000	0.732	1.461	2.227	3.056	4.646	5.972	7.056	7.925
17.700	-0.000	0.733	1.463	2.225	3.054	4.645	5.970	7.053	7.924
17.800	-0.000	0.734	1.466	2.224	3.052	4.644	5.968	7.051	7.922
17.900	-0.000	0.735	1.468	2.223	3.050	4.643	5.967	7.049	7.920
18.000	-0.000	0.736	1.470	2.222	3.049	4.642	5.965	7.047	7.919
18.100	-0.000	0.738	1.472	2.221	3.047	4.641	5.963	7.045	7.917
18.200	-0.000	0.739	1.474	2.221	3.045	4.640	5.961	7.043	7.915
18.300	-0.000	0.740	1.476	2.221	3.044	4.639	5.960	7.041	7.914
18.400	-0.000	0.741	1.479	2.221	3.042	4.638	5.958	7.039	7.912
18.500	-0.000	0.742	1.481	2.222	3.041	4.637	5.956	7.037	7.910
18.600	-0.000	0.743	1.483	2.223	3.040	4.636	5.955	7.035	7.909
18.700	-0.000	0.744	1.486	2.225	3.039	4.635	5.953	7.033	7.907
18.800	-0.000	0.746	1.488	2.227	3.038	4.635	5.951	7.031	7.907
18.900	-0.000	0.747	1.490	2.229	3.037	4.634	5.950	7.029	7.905
19.000	-0.000	0.748	1.493	2.232	3.036	4.633	5.948	7.027	7.904
19.100	-0.000	0.749	1.495	2.235	3.036	4.632	5.946	7.025	7.902
19.200	-0.000	0.750	1.497	2.238	3.035	4.631	5.945	7.023	7.900
19.300	-0.000	0.751	1.500	2.242	3.035	4.631	5.943	7.020	7.899
19.400	-0.000	0.753	1.502	2.245	3.035	4.630	5.941	7.018	7.897
19.500	-0.000	0.754	1.505	2.249	3.035	4.629	5.940	7.016	7.896
19.600	-0.000	0.755	1.507	2.253	3.035	4.629	5.938	7.014	7.895
19.700	-0.000	0.756	1.510	2.257	3.036	4.628	5.936	7.012	7.893
19.800	-0.000	0.758	1.512	2.260	3.036	4.627	5.935	7.010	7.892
19.900	-0.000	0.759	1.515	2.264	3.037	4.627	5.933	7.008	7.890
20.000	-0.000	0.760	1.517	2.268	3.038	4.626	5.931	7.006	7.889
20.100	-0.000	0.761	1.520	2.272	3.039	4.625	5.930	7.004	7.887
20.200	-0.000	0.763	1.523	2.276	3.041	4.625	5.928	7.003	7.886
20.300	-0.000	0.764	1.526	2.280	3.043	4.624	5.926	7.002	7.885
20.400	-0.000	0.766	1.528	2.284	3.045	4.624	5.925	7.000	7.883
20.500	-0.000	0.767	1.531	2.288	3.047	4.623	5.923	6.998	7.882
20.600	-0.000	0.768	1.533	2.292	3.050	4.623	5.921	6.996	7.881
20.700	-0.000	0.770	1.536	2.296	3.054	4.623	5.920	6.995	7.879
20.800	-0.000	0.771	1.539	2.300	3.057	4.622	5.918	6.993	7.878
20.900	-0.000	0.772	1.541	2.305	3.061	4.622	5.916	6.991	7.877
21.000	-0.000	0.774	1.544	2.308	3.066	4.621	5.914	6.989	7.875
21.100	-0.000	0.775	1.547	2.313	3.070	4.621	5.913	6.988	7.874
21.200	-0.000	0.777	1.550	2.317	3.075	4.621	5.912	6.986	7.873
21.300	-0.000	0.778	1.553	2.320	3.081	4.621	5.910	6.984	7.871
21.400	-0.000	0.779	1.556	2.326	3.086	4.621	5.908	6.983	7.870
21.500	-0.000	0.781	1.558	2.330	3.092	4.620	5.907	6.981	7.869
21.600	-0.000	0.782	1.562	2.334	3.098	4.620	5.905	6.980	7.867
21.700	-0.000	0.784	1.565	2.339	3.104	4.620	5.903	6.978	7.866
21.800	-0.000	0.785	1.567	2.343	3.109	4.619	5.902	6.976	7.865
21.900	-0.000	0.787	1.570	2.348	3.115	4.619	5.900	6.975	7.864
22.000	-0.000	0.788	1.573	2.352	3.121	4.619	5.898	6.973	7.862
22.100	-0.000	0.790	1.576	2.357	3.127	4.619	5.896	6.972	7.861
22.200	-0.000	0.791	1.580	2.361	3.133	4.618	5.895	6.971	7.860
22.300	-0.000	0.793	1.583	2.366	3.139	4.618	5.893	6.969	7.859
22.400	-0.000	0.794	1.586	2.370	3.146	4.618	5.892	6.968	7.858
22.500	-0.000	0.796	1.589	2.375	3.152	4.618	5.890	6.966	7.857
22.600	-0.000	0.798	1.592	2.380	3.158	4.618	5.888	6.965	7.855
22.700	-0.000	0.799	1.595	2.384	3.164	4.618	5.887	6.964	7.854
22.800	-0.000	0.801	1.598	2.389	3.170	4.618	5.885	6.963	7.853
22.900	-0.000	0.802	1.601	2.394	3.176	4.619	5.884	6.961	7.852
23.000	-0.000	0.804	1.605	2.399	3.181	4.619	5.884	6.960	7.851
23.100	-0.000	0.806	1.608	2.404	3.188	4.619	5.882	6.960	7.850
23.200	-0.000	0.807	1.611	2.408	3.194	4.619	5.881	6.958	7.849
23.300	-0.000	0.809	1.614	2.413	3.200	4.620	5.880	6.957	7.848
23.400	-0.000	0.811	1.618	2.418	3.206	4.621	5.878	6.956	7.847

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.812	1.621	2.423	3.212	4.621	5.877	6.955	7.846
23.600	-0.000	0.814	1.624	2.428	3.218	4.620	5.876	6.954	7.845
23.700	-0.000	0.816	1.628	2.433	3.223	4.623	5.875	6.953	7.844
23.800	-0.000	0.817	1.631	2.438	3.229	4.625	5.874	6.951	7.843
23.900	-0.000	0.819	1.633	2.443	3.234	4.626	5.870	6.950	7.842
24.000	-0.000	0.821	1.638	2.449	3.239	4.628	5.872	6.949	7.842
24.100	-0.000	0.822	1.641	2.454	3.244	4.630	5.871	6.948	7.841
24.200	-0.000	0.824	1.645	2.459	3.248	4.633	5.870	6.948	7.840
24.300	-0.000	0.826	1.648	2.464	3.253	4.635	5.870	6.947	7.839
24.400	-0.000	0.828	1.652	2.469	3.257	4.638	5.869	6.946	7.838
24.500	-0.000	0.829	1.655	2.475	3.261	4.640	5.869	6.945	7.838
24.600	-0.000	0.831	1.659	2.480	3.265	4.643	5.868	6.944	7.837
24.700	-0.000	0.833	1.663	2.485	3.269	4.645	5.868	6.944	7.837
24.800	-0.000	0.835	1.666	2.490	3.272	4.647	5.868	6.943	7.836
24.900	-0.000	0.837	1.670	2.495	3.274	4.649	5.868	6.942	7.836
25.000	-0.000	0.838	1.673	2.500	3.277	4.651	5.868	6.942	7.835
25.100	-0.000	0.840	1.677	2.504	3.279	4.653	5.868	6.942	7.835
25.200	-0.000	0.842	1.681	2.509	3.281	4.655	5.869	6.941	7.835
25.300	-0.000	0.844	1.685	2.514	3.283	4.657	5.869	6.941	7.834
25.400	-0.000	0.846	1.688	2.518	3.285	4.658	5.870	6.941	7.834
25.500	-0.000	0.848	1.692	2.522	3.287	4.660	5.871	6.941	7.834
25.600	-0.000	0.850	1.696	2.526	3.288	4.662	5.873	6.941	7.834
25.700	-0.000	0.852	1.700	2.530	3.290	4.663	5.874	6.941	7.834
25.800	-0.000	0.853	1.703	2.533	3.291	4.665	5.876	6.941	7.834
25.900	-0.000	0.855	1.707	2.537	3.292	4.666	5.878	6.941	7.834
26.000	-0.000	0.857	1.711	2.540	3.293	4.667	5.880	6.942	7.835

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 1.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	10.766	12.516	13.374	13.852	14.196	14.094	13.743	13.378
0.100	-0.002	11.047	12.995	13.969	14.527	14.954	14.721	14.086	13.256
0.200	-0.001	11.154	13.253	14.313	14.925	15.362	15.046	14.250	13.172
0.300	-0.001	11.154	13.380	14.496	15.141	15.575	15.215	14.339	13.133
0.400	-0.001	11.058	13.414	14.577	15.243	15.676	15.301	14.394	13.126
0.500	-0.001	10.833	13.324	14.536	15.228	15.688	15.333	14.437	13.152
0.600	-0.001	10.524	13.148	14.414	15.139	15.648	15.334	14.476	13.197
0.700	-0.001	10.178	12.932	14.256	15.017	15.584	15.322	14.512	13.247
0.800	-0.001	9.815	12.697	14.079	14.878	15.509	15.302	14.545	13.296
0.900	-0.001	9.447	12.453	13.894	14.730	15.427	15.279	14.573	13.343
1.000	-0.001	9.076	12.205	13.705	14.578	15.342	15.254	14.597	13.385
1.100	-0.001	8.708	11.957	13.515	14.427	15.255	15.227	14.615	13.422
1.200	-0.001	8.349	11.712	13.327	14.277	15.169	15.200	14.628	13.454
1.300	-0.001	8.004	11.470	13.141	14.130	15.083	15.173	14.635	13.480
1.400	-0.001	7.671	11.233	12.958	13.985	14.998	15.146	14.639	13.501
1.500	-0.001	7.350	11.000	12.779	13.843	14.914	15.118	14.637	13.519
1.600	-0.001	7.041	10.768	12.603	13.704	14.833	15.091	14.631	13.532
1.700	-0.001	6.744	10.539	12.431	13.568	14.753	15.062	14.622	13.543
1.800	-0.001	6.462	10.317	12.263	13.435	14.675	15.031	14.608	13.550
1.900	-0.001	6.195	10.101	12.099	13.305	14.599	14.999	14.592	13.555
2.000	-0.001	5.944	9.892	11.939	13.178	14.524	14.965	14.572	13.558
2.100	-0.001	5.711	9.690	11.781	13.054	14.452	14.929	14.550	13.559
2.200	-0.001	5.494	9.493	11.624	12.932	14.381	14.892	14.525	13.558
2.300	-0.001	5.294	9.303	11.468	12.814	14.312	14.852	14.498	13.556
2.400	-0.001	5.109	9.117	11.315	12.699	14.246	14.811	14.469	13.552
2.500	-0.001	4.938	8.937	11.167	12.586	14.180	14.768	14.438	13.547
2.600	-0.001	4.779	8.760	11.023	12.475	14.117	14.723	14.405	13.541
2.700	-0.001	4.631	8.587	10.883	12.366	14.054	14.676	14.371	13.533
2.800	-0.001	4.493	8.419	10.746	12.259	13.993	14.628	14.336	13.525
2.900	-0.001	4.363	8.254	10.613	12.151	13.934	14.578	14.300	13.515
3.000	-0.001	4.243	8.094	10.483	12.045	13.875	14.527	14.262	13.503
3.100	-0.001	4.130	7.937	10.356	11.940	13.816	14.475	14.223	13.491
3.200	-0.001	4.024	7.784	10.232	11.837	13.757	14.421	14.183	13.478
3.300	-0.001	3.924	7.634	10.111	11.736	13.698	14.367	14.141	13.465
3.400	-0.001	3.830	7.487	9.992	11.638	13.639	14.311	14.098	13.451
3.500	-0.001	3.742	7.343	9.876	11.542	13.579	14.254	14.054	13.437
3.600	-0.001	3.658	7.202	9.762	11.449	13.519	14.196	14.008	13.421
3.700	-0.001	3.579	7.065	9.650	11.357	13.457	14.137	13.962	13.405
3.800	-0.001	3.504	6.931	9.539	11.267	13.394	14.077	13.915	13.387
3.900	-0.001	3.432	6.802	9.430	11.179	13.331	14.016	13.867	13.368
4.000	-0.001	3.365	6.677	9.323	11.093	13.266	13.953	13.819	13.348
4.100	-0.001	3.301	6.557	9.218	11.008	13.200	13.889	13.770	13.327
4.200	-0.001	3.240	6.442	9.115	10.924	13.133	13.824	13.721	13.305
4.300	-0.001	3.181	6.331	9.012	10.842	13.064	13.757	13.671	13.283
4.400	-0.001	3.125	6.224	8.911	10.761	12.993	13.689	13.622	13.260
4.500	-0.001	3.071	6.123	8.812	10.681	12.921	13.620	13.572	13.236
4.600	-0.001	3.019	6.025	8.714	10.603	12.847	13.550	13.522	13.211
4.700	-0.001	2.970	5.931	8.617	10.526	12.772	13.480	13.472	13.186
4.800	-0.001	2.923	5.841	8.521	10.449	12.696	13.409	13.421	13.160
4.900	-0.001	2.878	5.754	8.427	10.374	12.619	13.337	13.371	13.134
5.000	-0.001	2.835	5.672	8.334	10.300	12.540	13.266	13.320	13.107
5.100	-0.001	2.794	5.592	8.243	10.227	12.461	13.195	13.270	13.079
5.200	-0.001	2.754	5.515	8.153	10.155	12.380	13.123	13.219	13.052
5.300	-0.001	2.716	5.441	8.063	10.083	12.299	13.051	13.169	13.023
5.400	-0.001	2.680	5.369	7.975	10.012	12.217	12.979	13.120	12.993

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.645	5.300	7.889	9.941	12.135	12.906	13.071	12.962
5.600	-0.001	2.612	5.234	7.804	9.870	12.052	12.832	13.022	12.932
5.700	-0.001	2.580	5.171	7.721	9.799	11.968	12.757	12.972	12.901
5.800	-0.001	2.550	5.110	7.639	9.727	11.884	12.682	12.923	12.870
5.900	-0.001	2.520	5.051	7.560	9.655	11.799	12.607	12.874	12.839
6.000	-0.001	2.492	4.994	7.482	9.583	11.714	12.531	12.824	12.806
6.100	-0.001	2.465	4.939	7.406	9.510	11.628	12.455	12.775	12.774
6.200	-0.001	2.439	4.887	7.332	9.436	11.542	12.379	12.724	12.741
6.300	-0.000	2.414	4.836	7.261	9.362	11.455	12.303	12.673	12.708
6.400	-0.000	2.389	4.787	7.191	9.287	11.369	12.227	12.621	12.674
6.500	-0.000	2.366	4.741	7.123	9.211	11.281	12.151	12.568	12.637
6.600	-0.000	2.343	4.696	7.058	9.133	11.194	12.075	12.514	12.599
6.700	-0.000	2.322	4.653	6.994	9.055	11.106	11.999	12.458	12.558
6.800	-0.000	2.301	4.611	6.933	8.976	11.018	11.923	12.403	12.516
6.900	-0.000	2.280	4.570	6.873	8.895	10.929	11.847	12.347	12.472
7.000	-0.000	2.261	4.531	6.815	8.813	10.841	11.771	12.291	12.425
7.100	-0.000	2.242	4.494	6.759	8.730	10.751	11.695	12.234	12.376
7.200	-0.000	2.224	4.457	6.703	8.646	10.662	11.619	12.178	12.324
7.300	-0.000	2.206	4.422	6.649	8.560	10.572	11.543	12.122	12.270
7.400	-0.000	2.189	4.388	6.596	8.473	10.482	11.466	12.065	12.214
7.500	-0.000	2.172	4.355	6.544	8.386	10.391	11.390	12.008	12.158
7.600	-0.000	2.157	4.323	6.492	8.297	10.299	11.313	11.951	12.100
7.700	-0.000	2.141	4.292	6.440	8.208	10.207	11.236	11.894	12.042
7.800	-0.000	2.127	4.262	6.388	8.119	10.115	11.159	11.836	11.982
7.900	-0.000	2.112	4.234	6.337	8.030	10.022	11.082	11.779	11.921
8.000	-0.000	2.098	4.206	6.285	7.940	9.929	11.005	11.721	11.860
8.100	-0.000	2.085	4.179	6.233	7.851	9.835	10.927	11.662	11.799
8.200	-0.000	2.072	4.152	6.180	7.761	9.741	10.849	11.602	11.737
8.300	-0.000	2.059	4.127	6.127	7.672	9.647	10.771	11.541	11.675
8.400	-0.000	2.047	4.103	6.073	7.583	9.551	10.693	11.479	11.612
8.500	-0.000	2.035	4.079	6.018	7.494	9.455	10.615	11.415	11.548
8.600	-0.000	2.024	4.056	5.962	7.406	9.359	10.536	11.349	11.483
8.700	-0.000	2.013	4.034	5.905	7.317	9.262	10.457	11.282	11.418
8.800	-0.000	2.002	4.012	5.847	7.229	9.164	10.378	11.213	11.353
8.900	-0.000	1.992	3.992	5.789	7.141	9.066	10.299	11.142	11.286
9.000	-0.000	1.981	3.971	5.729	7.053	8.968	10.219	11.070	11.219
9.100	-0.000	1.972	3.951	5.669	6.964	8.869	10.138	10.996	11.152
9.200	-0.000	1.962	3.930	5.608	6.876	8.770	10.057	10.919	11.084
9.300	-0.000	1.953	3.909	5.546	6.788	8.671	9.976	10.841	11.015
9.400	-0.000	1.944	3.888	5.484	6.700	8.571	9.894	10.761	10.946
9.500	-0.000	1.936	3.865	5.420	6.612	8.470	9.813	10.679	10.876
9.600	-0.000	1.928	3.841	5.356	6.524	8.369	9.731	10.597	10.806
9.700	-0.000	1.920	3.816	5.291	6.435	8.267	9.648	10.514	10.735
9.800	-0.000	1.912	3.789	5.226	6.347	8.165	9.564	10.430	10.664
9.900	-0.000	1.904	3.760	5.159	6.259	8.063	9.480	10.346	10.592
10.000	-0.000	1.897	3.730	5.092	6.170	7.960	9.396	10.261	10.520
10.100	-0.000	1.890	3.697	5.024	6.082	7.857	9.310	10.175	10.447
10.200	-0.000	1.883	3.662	4.955	5.993	7.754	9.224	10.088	10.373
10.300	-0.000	1.877	3.625	4.885	5.904	7.651	9.136	10.000	10.299
10.400	-0.000	1.870	3.586	4.814	5.815	7.548	9.047	9.912	10.224
10.500	-0.000	1.864	3.544	4.741	5.726	7.446	8.955	9.823	10.149
10.600	-0.000	1.858	3.501	4.668	5.637	7.343	8.862	9.733	10.072
10.700	-0.000	1.852	3.455	4.594	5.547	7.241	8.767	9.643	9.996
10.800	-0.000	1.846	3.407	4.519	5.458	7.140	8.671	9.551	9.919
10.900	-0.000	1.840	3.357	4.443	5.368	7.039	8.573	9.459	9.841
11.000	-0.000	1.833	3.305	4.366	5.278	6.939	8.473	9.367	9.763
11.100	-0.000	1.824	3.251	4.288	5.188	6.840	8.371	9.274	9.685
11.200	-0.000	1.814	3.195	4.209	5.098	6.741	8.268	9.180	9.607
11.300	-0.000	1.801	3.137	4.130	5.007	6.643	8.163	9.086	9.528
11.400	-0.000	1.786	3.076	4.049	4.917	6.546	8.057	8.991	9.449

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.768	3.013	3.967	4.827	6.449	7.951	8.896	9.370
11.600	-0.000	1.747	2.948	3.885	4.736	6.354	7.845	8.801	9.290
11.700	-0.000	1.722	2.881	3.801	4.645	6.259	7.739	8.704	9.211
11.800	-0.000	1.693	2.812	3.717	4.554	6.166	7.634	8.608	9.131
11.900	-0.000	1.660	2.741	3.632	4.463	6.073	7.528	8.510	9.051
12.000	-0.000	1.622	2.667	3.546	4.372	5.981	7.423	8.413	8.971
12.100	-0.000	1.579	2.592	3.459	4.281	5.890	7.319	8.315	8.892
12.200	-0.000	1.531	2.514	3.372	4.190	5.799	7.214	8.217	8.814
12.300	-0.000	1.478	2.435	3.283	4.099	5.710	7.111	8.119	8.736
12.400	-0.000	1.421	2.353	3.194	4.007	5.621	7.007	8.021	8.659
12.500	-0.000	1.360	2.270	3.105	3.916	5.533	6.905	7.924	8.583
12.600	-0.000	1.296	2.187	3.016	3.826	5.445	6.803	7.828	8.509
12.700	-0.000	1.233	2.106	2.930	3.740	5.358	6.703	7.736	8.437
12.800	-0.000	1.170	2.028	2.848	3.657	5.272	6.607	7.647	8.369
12.900	-0.000	1.110	1.954	2.770	3.579	5.187	6.516	7.565	8.305
13.000	-0.000	1.051	1.884	2.697	3.505	5.104	6.430	7.488	8.246
13.100	-0.000	0.994	1.818	2.628	3.437	5.024	6.351	7.416	8.191
13.200	-0.000	0.940	1.757	2.566	3.375	4.949	6.278	7.351	8.143
13.300	-0.000	0.889	1.701	2.510	3.319	4.882	6.214	7.293	8.101
13.400	-0.000	0.844	1.653	2.461	3.272	4.825	6.158	7.243	8.066
13.500	-0.000	0.806	1.613	2.422	3.234	4.778	6.112	7.204	8.036
13.600	-0.000	0.777	1.582	2.392	3.206	4.743	6.077	7.174	8.014
13.700	-0.000	0.757	1.563	2.373	3.188	4.722	6.057	7.157	8.000
13.800	-0.000	0.748	1.553	2.364	3.180	4.714	6.049	7.150	7.994
13.900	-0.000	0.743	1.548	2.360	3.176	4.712	6.047	7.147	7.992
14.000	-0.000	0.738	1.543	2.355	3.172	4.710	6.045	7.145	7.990
14.100	-0.000	0.734	1.538	2.351	3.168	4.708	6.043	7.143	7.989
14.200	-0.000	0.730	1.533	2.346	3.165	4.706	6.041	7.140	7.987
14.300	-0.000	0.727	1.528	2.342	3.161	4.704	6.040	7.138	7.985
14.400	-0.000	0.724	1.523	2.337	3.157	4.702	6.038	7.136	7.984
14.500	-0.000	0.722	1.518	2.333	3.153	4.701	6.036	7.133	7.982
14.600	-0.000	0.720	1.514	2.329	3.150	4.699	6.034	7.131	7.980
14.700	-0.000	0.718	1.509	2.324	3.146	4.697	6.032	7.128	7.978
14.800	-0.000	0.717	1.504	2.320	3.142	4.695	6.030	7.126	7.977
14.900	-0.000	0.715	1.500	2.316	3.139	4.693	6.028	7.124	7.975
15.000	-0.000	0.714	1.496	2.312	3.135	4.692	6.027	7.121	7.973
15.100	-0.000	0.714	1.492	2.307	3.132	4.690	6.025	7.119	7.972
15.200	-0.000	0.713	1.488	2.303	3.128	4.689	6.023	7.117	7.970
15.300	-0.000	0.713	1.484	2.299	3.125	4.687	6.021	7.114	7.968
15.400	-0.000	0.713	1.480	2.295	3.122	4.685	6.019	7.112	7.967
15.500	-0.000	0.714	1.476	2.292	3.118	4.684	6.017	7.110	7.965
15.600	-0.000	0.714	1.473	2.288	3.115	4.682	6.016	7.107	7.964
15.700	-0.000	0.715	1.470	2.284	3.112	4.681	6.014	7.105	7.962
15.800	-0.000	0.716	1.467	2.280	3.109	4.679	6.012	7.102	7.960
15.900	-0.000	0.717	1.464	2.277	3.105	4.678	6.010	7.100	7.959
16.000	-0.000	0.718	1.462	2.273	3.102	4.676	6.009	7.098	7.957
16.100	-0.000	0.719	1.459	2.270	3.099	4.675	6.007	7.095	7.955
16.200	-0.000	0.720	1.458	2.266	3.096	4.674	6.005	7.093	7.954
16.300	-0.000	0.721	1.456	2.263	3.093	4.672	6.003	7.091	7.952
16.400	-0.000	0.722	1.455	2.260	3.090	4.671	6.002	7.088	7.950
16.500	-0.000	0.723	1.454	2.257	3.088	4.670	6.000	7.086	7.949
16.600	-0.000	0.724	1.454	2.254	3.085	4.668	5.998	7.084	7.947
16.700	-0.000	0.725	1.454	2.251	3.082	4.667	5.996	7.081	7.946
16.800	-0.000	0.726	1.454	2.248	3.080	4.666	5.995	7.079	7.944
16.900	-0.000	0.727	1.454	2.246	3.077	4.665	5.993	7.077	7.942
17.000	-0.000	0.728	1.455	2.243	3.074	4.664	5.991	7.074	7.940
17.100	-0.000	0.729	1.456	2.241	3.072	4.662	5.989	7.072	7.939
17.200	-0.000	0.730	1.457	2.239	3.070	4.661	5.988	7.070	7.937
17.300	-0.000	0.731	1.459	2.237	3.067	4.660	5.986	7.068	7.935
17.400	-0.000	0.732	1.461	2.235	3.065	4.659	5.984	7.065	7.934

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.733	1.463	2.233	3.063	4.658	5.982	7.063	7.932
17.600	-0.000	0.734	1.465	2.232	3.061	4.657	5.981	7.061	7.930
17.700	-0.000	0.735	1.467	2.231	3.059	4.656	5.979	7.059	7.928
17.800	-0.000	0.736	1.469	2.229	3.057	4.655	5.977	7.057	7.927
17.900	-0.000	0.737	1.471	2.229	3.055	4.654	5.976	7.055	7.925
18.000	-0.000	0.738	1.473	2.228	3.054	4.653	5.974	7.053	7.923
18.100	-0.000	0.739	1.476	2.228	3.052	4.652	5.972	7.050	7.922
18.200	-0.000	0.740	1.478	2.228	3.051	4.651	5.970	7.048	7.920
18.300	-0.000	0.742	1.480	2.228	3.049	4.650	5.969	7.046	7.918
18.400	-0.000	0.743	1.482	2.229	3.048	4.649	5.967	7.044	7.917
18.500	-0.000	0.744	1.485	2.230	3.047	4.648	5.965	7.042	7.916
18.600	-0.000	0.745	1.487	2.231	3.046	4.647	5.964	7.040	7.915
18.700	-0.000	0.746	1.489	2.232	3.045	4.647	5.962	7.038	7.913
18.800	-0.000	0.747	1.492	2.234	3.044	4.646	5.960	7.036	7.911
18.900	-0.000	0.749	1.494	2.237	3.043	4.645	5.959	7.034	7.910
19.000	-0.000	0.750	1.496	2.239	3.043	4.644	5.957	7.032	7.908
19.100	-0.000	0.751	1.499	2.242	3.043	4.644	5.955	7.030	7.907
19.200	-0.000	0.752	1.501	2.245	3.042	4.643	5.954	7.028	7.905
19.300	-0.000	0.753	1.504	2.248	3.042	4.642	5.952	7.026	7.904
19.400	-0.000	0.755	1.506	2.251	3.042	4.641	5.950	7.024	7.902
19.500	-0.000	0.756	1.509	2.255	3.043	4.641	5.949	7.022	7.901
19.600	-0.000	0.757	1.511	2.259	3.043	4.640	5.947	7.020	7.899
19.700	-0.000	0.758	1.514	2.263	3.044	4.639	5.945	7.018	7.898
19.800	-0.000	0.760	1.516	2.266	3.045	4.639	5.944	7.016	7.896
19.900	-0.000	0.761	1.519	2.270	3.046	4.638	5.942	7.014	7.895
20.000	-0.000	0.762	1.522	2.274	3.047	4.638	5.940	7.013	7.894
20.100	-0.000	0.764	1.524	2.278	3.049	4.637	5.939	7.011	7.892
20.200	-0.000	0.765	1.527	2.283	3.050	4.637	5.937	7.009	7.891
20.300	-0.000	0.766	1.530	2.287	3.053	4.636	5.935	7.007	7.889
20.400	-0.000	0.768	1.532	2.291	3.055	4.636	5.933	7.005	7.888
20.500	-0.000	0.769	1.535	2.295	3.058	4.635	5.931	7.004	7.886
20.600	-0.000	0.770	1.538	2.299	3.061	4.635	5.930	7.002	7.885
20.700	-0.000	0.772	1.540	2.303	3.064	4.634	5.928	7.000	7.884
20.800	-0.000	0.773	1.543	2.307	3.068	4.634	5.926	6.998	7.882
20.900	-0.000	0.775	1.546	2.311	3.072	4.634	5.926	6.997	7.881
21.000	-0.000	0.776	1.549	2.315	3.076	4.634	5.924	6.995	7.879
21.100	-0.000	0.777	1.552	2.320	3.081	4.634	5.922	6.993	7.878
21.200	-0.000	0.779	1.555	2.324	3.085	4.633	5.920	6.992	7.877
21.300	-0.000	0.780	1.558	2.328	3.090	4.633	5.918	6.990	7.875
21.400	-0.000	0.782	1.560	2.333	3.096	4.632	5.917	6.988	7.874
21.500	-0.000	0.783	1.563	2.337	3.101	4.632	5.915	6.987	7.873
21.600	-0.000	0.785	1.566	2.341	3.107	4.632	5.913	6.985	7.871
21.700	-0.000	0.786	1.569	2.346	3.113	4.632	5.911	6.984	7.870
21.800	-0.000	0.788	1.572	2.350	3.119	4.631	5.910	6.982	7.869
21.900	-0.000	0.789	1.575	2.355	3.125	4.631	5.908	6.981	7.868
22.000	-0.000	0.791	1.578	2.358	3.131	4.631	5.906	6.979	7.866
22.100	-0.000	0.792	1.581	2.364	3.137	4.631	5.904	6.978	7.865
22.200	-0.000	0.794	1.585	2.369	3.143	4.631	5.903	6.976	7.864
22.300	-0.000	0.795	1.588	2.373	3.150	4.631	5.901	6.975	7.863
22.400	-0.000	0.797	1.591	2.378	3.156	4.631	5.899	6.973	7.861
22.500	-0.000	0.799	1.593	2.383	3.162	4.631	5.898	6.972	7.860
22.600	-0.000	0.800	1.597	2.387	3.168	4.631	5.896	6.971	7.859
22.700	-0.000	0.801	1.600	2.392	3.174	4.631	5.893	6.969	7.858
22.800	-0.000	0.803	1.604	2.397	3.180	4.631	5.895	6.969	7.857
22.900	-0.000	0.805	1.606	2.402	3.186	4.631	5.893	6.968	7.856
23.000	-0.000	0.807	1.610	2.407	3.192	4.631	5.892	6.966	7.854
23.100	-0.000	0.808	1.613	2.412	3.198	4.629	5.890	6.965	7.853
23.200	-0.000	0.810	1.617	2.417	3.204	4.632	5.889	6.964	7.852
23.300	-0.000	0.812	1.620	2.421	3.210	4.632	5.888	6.962	7.851
23.400	-0.000	0.813	1.623	2.426	3.216	4.633	5.886	6.961	7.850

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.815	1.627	2.431	3.221	4.633	5.885	6.960	7.849
23.600	-0.000	0.817	1.630	2.437	3.227	4.634	5.884	6.959	7.848
23.700	-0.000	0.818	1.633	2.442	3.233	4.635	5.883	6.958	7.847
23.800	-0.000	0.820	1.637	2.447	3.238	4.636	5.882	6.957	7.846
23.900	-0.000	0.822	1.640	2.452	3.243	4.638	5.881	6.956	7.845
24.000	-0.000	0.824	1.644	2.457	3.248	4.640	5.880	6.955	7.845
24.100	-0.000	0.825	1.647	2.462	3.253	4.639	5.879	6.954	7.844
24.200	-0.000	0.827	1.651	2.468	3.258	4.644	5.878	6.953	7.843
24.300	-0.000	0.829	1.654	2.473	3.263	4.644	5.875	6.952	7.842
24.400	-0.000	0.831	1.658	2.478	3.267	4.649	5.877	6.951	7.841
24.500	-0.000	0.832	1.661	2.483	3.271	4.651	5.876	6.950	7.841
24.600	-0.000	0.834	1.665	2.488	3.275	4.653	5.876	6.950	7.840
24.700	-0.000	0.836	1.669	2.493	3.279	4.655	5.875	6.949	7.840
24.800	-0.000	0.838	1.672	2.498	3.282	4.657	5.875	6.948	7.839
24.900	-0.000	0.840	1.676	2.503	3.285	4.659	5.875	6.948	7.839
25.000	-0.000	0.842	1.680	2.507	3.288	4.661	5.875	6.947	7.838
25.100	-0.000	0.843	1.683	2.512	3.290	4.663	5.876	6.947	7.838
25.200	-0.000	0.845	1.687	2.516	3.292	4.664	5.876	6.946	7.838
25.300	-0.000	0.847	1.691	2.521	3.294	4.666	5.877	6.946	7.838
25.400	-0.000	0.849	1.695	2.525	3.295	4.667	5.878	6.946	7.838
25.500	-0.000	0.851	1.698	2.529	3.297	4.669	5.879	6.946	7.837
25.600	-0.000	0.853	1.702	2.533	3.298	4.670	5.881	6.946	7.837
25.700	-0.000	0.855	1.706	2.537	3.299	4.672	5.882	6.946	7.837
25.800	-0.000	0.857	1.710	2.540	3.300	4.673	5.884	6.947	7.837
25.900	-0.000	0.859	1.714	2.543	3.301	4.674	5.886	6.949	7.838
26.000	-0.000	0.861	1.718	2.546	3.302	4.675	5.889	6.949	7.838

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 2.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	10.057	11.937	12.880	13.429	13.899	13.922	13.732	13.453
0.100	-0.002	10.228	12.260	13.291	13.901	14.442	14.426	14.029	13.387
0.200	-0.001	10.293	12.455	13.557	14.214	14.804	14.731	14.204	13.333
0.300	-0.001	10.288	12.563	13.720	14.414	15.034	14.918	14.311	13.299
0.400	-0.001	10.227	12.614	13.816	14.536	15.172	15.035	14.382	13.283
0.500	-0.001	10.113	12.616	13.863	14.606	15.254	15.108	14.432	13.279
0.600	-0.001	9.930	12.553	13.847	14.616	15.285	15.147	14.471	13.289
0.700	-0.001	9.682	12.424	13.769	14.567	15.271	15.160	14.501	13.310
0.800	-0.001	9.399	12.256	13.655	14.483	15.232	15.159	14.525	13.337
0.900	-0.001	9.096	12.067	13.519	14.380	15.178	15.149	14.544	13.366
1.000	-0.001	8.783	11.866	13.371	14.267	15.116	15.134	14.559	13.393
1.100	-0.001	8.464	11.658	13.217	14.146	15.048	15.116	14.569	13.419
1.200	-0.001	8.144	11.445	13.059	14.022	14.977	15.096	14.576	13.442
1.300	-0.001	7.826	11.232	12.898	13.898	14.904	15.074	14.578	13.462
1.400	-0.001	7.515	11.018	12.737	13.772	14.830	15.051	14.576	13.479
1.500	-0.001	7.215	10.807	12.576	13.647	14.757	15.027	14.571	13.493
1.600	-0.001	6.924	10.597	12.417	13.522	14.684	15.000	14.562	13.505
1.700	-0.001	6.645	10.389	12.259	13.399	14.612	14.972	14.551	13.514
1.800	-0.001	6.378	10.181	12.103	13.277	14.541	14.941	14.536	13.522
1.900	-0.001	6.124	9.977	11.950	13.157	14.472	14.908	14.518	13.527
2.000	-0.001	5.884	9.777	11.799	13.039	14.404	14.873	14.498	13.531
2.100	-0.001	5.659	9.583	11.652	12.923	14.337	14.837	14.476	13.533
2.200	-0.001	5.449	9.394	11.507	12.809	14.272	14.799	14.451	13.534
2.300	-0.001	5.254	9.209	11.363	12.698	14.209	14.760	14.425	13.533
2.400	-0.001	5.073	9.030	11.219	12.589	14.147	14.719	14.397	13.531
2.500	-0.001	4.905	8.855	11.078	12.482	14.086	14.676	14.368	13.528
2.600	-0.001	4.749	8.684	10.938	12.377	14.026	14.632	14.337	13.524
2.700	-0.001	4.603	8.517	10.803	12.274	13.968	14.586	14.304	13.519
2.800	-0.001	4.467	8.354	10.670	12.172	13.910	14.539	14.271	13.513
2.900	-0.001	4.340	8.193	10.541	12.072	13.852	14.491	14.236	13.505
3.000	-0.001	4.221	8.037	10.415	11.972	13.795	14.441	14.200	13.497
3.100	-0.001	4.109	7.883	10.291	11.872	13.738	14.391	14.163	13.487
3.200	-0.001	4.005	7.734	10.170	11.773	13.680	14.339	14.125	13.476
3.300	-0.001	3.906	7.587	10.051	11.675	13.622	14.286	14.086	13.465
3.400	-0.001	3.813	7.443	9.935	11.578	13.564	14.232	14.045	13.452
3.500	-0.001	3.725	7.302	9.821	11.484	13.505	14.178	14.003	13.438
3.600	-0.001	3.643	7.165	9.709	11.392	13.445	14.122	13.960	13.423
3.700	-0.001	3.564	7.031	9.599	11.302	13.384	14.064	13.916	13.407
3.800	-0.001	3.490	6.901	9.490	11.213	13.323	14.006	13.871	13.390
3.900	-0.001	3.420	6.774	9.384	11.127	13.260	13.947	13.826	13.371
4.000	-0.001	3.353	6.651	9.278	11.041	13.196	13.887	13.780	13.351
4.100	-0.001	3.289	6.533	9.174	10.957	13.131	13.826	13.733	13.330
4.200	-0.001	3.228	6.418	9.072	10.875	13.065	13.763	13.686	13.308
4.300	-0.001	3.171	6.309	8.971	10.794	12.997	13.699	13.639	13.286
4.400	-0.001	3.116	6.203	8.871	10.714	12.928	13.634	13.592	13.262
4.500	-0.001	3.063	6.102	8.773	10.635	12.858	13.568	13.544	13.238
4.600	-0.001	3.012	6.005	8.676	10.558	12.786	13.500	13.496	13.213
4.700	-0.001	2.963	5.912	8.580	10.482	12.713	13.432	13.448	13.187
4.800	-0.001	2.916	5.822	8.486	10.406	12.639	13.363	13.400	13.161
4.900	-0.001	2.871	5.736	8.393	10.332	12.563	13.294	13.351	13.135
5.000	-0.001	2.828	5.654	8.301	10.259	12.487	13.225	13.303	13.109
5.100	-0.001	2.787	5.574	8.210	10.187	12.410	13.155	13.254	13.082
5.200	-0.001	2.747	5.498	8.121	10.115	12.331	13.085	13.205	13.055
5.300	-0.001	2.710	5.425	8.033	10.044	12.252	13.015	13.157	13.026
5.400	-0.001	2.674	5.355	7.946	9.972	12.172	12.945	13.109	12.997

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.639	5.286	7.861	9.901	12.091	12.874	13.061	12.967
5.600	-0.001	2.605	5.220	7.777	9.830	12.009	12.803	13.014	12.936
5.700	-0.001	2.573	5.157	7.695	9.759	11.927	12.731	12.967	12.905
5.800	-0.001	2.542	5.096	7.614	9.687	11.845	12.659	12.919	12.874
5.900	-0.001	2.513	5.038	7.536	9.615	11.762	12.586	12.872	12.843
6.000	-0.001	2.485	4.981	7.459	9.543	11.678	12.512	12.824	12.811
6.100	-0.001	2.458	4.927	7.384	9.470	11.594	12.438	12.776	12.780
6.200	-0.001	2.432	4.875	7.312	9.396	11.510	12.364	12.727	12.748
6.300	-0.000	2.407	4.824	7.241	9.322	11.425	12.290	12.678	12.715
6.400	-0.000	2.383	4.776	7.172	9.247	11.339	12.215	12.628	12.681
6.500	-0.000	2.360	4.729	7.105	9.171	11.254	12.141	12.577	12.647
6.600	-0.000	2.337	4.684	7.040	9.095	11.168	12.067	12.525	12.610
6.700	-0.000	2.315	4.640	6.976	9.017	11.082	11.992	12.471	12.572
6.800	-0.000	2.294	4.599	6.915	8.939	10.995	11.918	12.417	12.532
6.900	-0.000	2.274	4.559	6.856	8.860	10.908	11.843	12.362	12.491
7.000	-0.000	2.255	4.520	6.798	8.780	10.821	11.769	12.307	12.447
7.100	-0.000	2.236	4.482	6.741	8.699	10.734	11.694	12.252	12.401
7.200	-0.000	2.218	4.446	6.685	8.616	10.646	11.620	12.196	12.352
7.300	-0.000	2.200	4.411	6.630	8.533	10.558	11.545	12.140	12.301
7.400	-0.000	2.183	4.377	6.576	8.448	10.469	11.470	12.084	12.248
7.500	-0.000	2.167	4.344	6.523	8.362	10.380	11.395	12.028	12.193
7.600	-0.000	2.151	4.312	6.470	8.276	10.290	11.320	11.972	12.138
7.700	-0.000	2.136	4.281	6.418	8.189	10.200	11.245	11.915	12.082
7.800	-0.000	2.121	4.252	6.366	8.101	10.109	11.170	11.859	12.024
7.900	-0.000	2.107	4.223	6.314	8.014	10.018	11.094	11.802	11.965
8.000	-0.000	2.093	4.195	6.262	7.926	9.926	11.019	11.744	11.905
8.100	-0.000	2.080	4.168	6.210	7.838	9.834	10.943	11.687	11.844
8.200	-0.000	2.067	4.142	6.158	7.750	9.742	10.867	11.629	11.783
8.300	-0.000	2.054	4.117	6.105	7.662	9.649	10.791	11.570	11.721
8.400	-0.000	2.042	4.092	6.052	7.574	9.556	10.715	11.509	11.659
8.500	-0.000	2.030	4.069	5.998	7.486	9.462	10.638	11.448	11.597
8.600	-0.000	2.019	4.046	5.943	7.399	9.368	10.561	11.385	11.533
8.700	-0.000	2.008	4.024	5.888	7.312	9.273	10.485	11.320	11.470
8.800	-0.000	1.997	4.002	5.832	7.225	9.177	10.407	11.254	11.405
8.900	-0.000	1.987	3.981	5.775	7.138	9.081	10.330	11.187	11.340
9.000	-0.000	1.977	3.960	5.717	7.052	8.984	10.252	11.117	11.274
9.100	-0.000	1.967	3.939	5.658	6.965	8.888	10.174	11.046	11.207
9.200	-0.000	1.957	3.917	5.598	6.879	8.791	10.096	10.973	11.140
9.300	-0.000	1.948	3.896	5.538	6.792	8.693	10.016	10.898	11.072
9.400	-0.000	1.939	3.873	5.477	6.706	8.595	9.937	10.821	11.004
9.500	-0.000	1.931	3.850	5.415	6.619	8.497	9.856	10.742	10.935
9.600	-0.000	1.923	3.825	5.352	6.532	8.398	9.776	10.662	10.866
9.700	-0.000	1.915	3.800	5.289	6.446	8.298	9.695	10.580	10.796
9.800	-0.000	1.907	3.773	5.225	6.359	8.199	9.613	10.498	10.725
9.900	-0.000	1.900	3.745	5.160	6.273	8.098	9.531	10.415	10.655
10.000	-0.000	1.892	3.715	5.094	6.186	7.998	9.448	10.332	10.584
10.100	-0.000	1.885	3.683	5.027	6.099	7.897	9.365	10.248	10.512
10.200	-0.000	1.879	3.650	4.960	6.013	7.796	9.281	10.163	10.440
10.300	-0.000	1.872	3.615	4.892	5.926	7.696	9.196	10.077	10.367
10.400	-0.000	1.866	3.577	4.822	5.839	7.595	9.110	9.991	10.293
10.500	-0.000	1.860	3.538	4.752	5.752	7.495	9.022	9.903	10.219
10.600	-0.000	1.854	3.496	4.681	5.664	7.394	8.933	9.815	10.144
10.700	-0.000	1.848	3.452	4.609	5.577	7.295	8.842	9.726	10.069
10.800	-0.000	1.841	3.406	4.536	5.489	7.195	8.749	9.637	9.993
10.900	-0.000	1.834	3.358	4.462	5.401	7.096	8.654	9.546	9.916
11.000	-0.000	1.826	3.307	4.387	5.313	6.998	8.558	9.455	9.840
11.100	-0.000	1.816	3.255	4.311	5.225	6.900	8.460	9.363	9.763
11.200	-0.000	1.804	3.200	4.234	5.136	6.803	8.360	9.271	9.685
11.300	-0.000	1.791	3.144	4.156	5.048	6.706	8.258	9.178	9.608
11.400	-0.000	1.775	3.085	4.077	4.959	6.610	8.155	9.084	9.530

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.757	3.024	3.998	4.871	6.515	8.051	8.990	9.452
11.600	-0.000	1.735	2.962	3.917	4.782	6.421	7.946	8.895	9.373
11.700	-0.000	1.711	2.897	3.836	4.693	6.328	7.841	8.800	9.294
11.800	-0.000	1.682	2.830	3.754	4.604	6.236	7.736	8.704	9.215
11.900	-0.000	1.650	2.761	3.671	4.515	6.144	7.631	8.608	9.136
12.000	-0.000	1.614	2.690	3.587	4.426	6.053	7.527	8.511	9.057
12.100	-0.000	1.573	2.617	3.503	4.337	5.963	7.423	8.414	8.977
12.200	-0.000	1.528	2.543	3.417	4.247	5.874	7.320	8.317	8.898
12.300	-0.000	1.478	2.466	3.331	4.158	5.786	7.217	8.219	8.820
12.400	-0.000	1.425	2.389	3.246	4.070	5.699	7.115	8.121	8.742
12.500	-0.000	1.371	2.312	3.162	3.983	5.615	7.014	8.025	8.666
12.600	-0.000	1.317	2.238	3.080	3.900	5.533	6.916	7.931	8.592
12.700	-0.000	1.264	2.165	3.002	3.819	5.452	6.819	7.840	8.521
12.800	-0.000	1.211	2.095	2.925	3.741	5.373	6.726	7.753	8.453
12.900	-0.000	1.158	2.027	2.852	3.666	5.295	6.636	7.671	8.388
13.000	-0.000	1.106	1.960	2.782	3.595	5.218	6.551	7.593	8.328
13.100	-0.000	1.054	1.897	2.715	3.527	5.143	6.471	7.520	8.272
13.200	-0.000	1.004	1.838	2.653	3.465	5.070	6.396	7.454	8.221
13.300	-0.000	0.956	1.781	2.594	3.406	5.000	6.328	7.393	8.174
13.400	-0.000	0.910	1.729	2.541	3.353	4.935	6.265	7.337	8.132
13.500	-0.000	0.868	1.682	2.493	3.306	4.878	6.210	7.287	8.096
13.600	-0.000	0.830	1.640	2.451	3.265	4.828	6.161	7.243	8.065
13.700	-0.000	0.797	1.604	2.416	3.231	4.786	6.120	7.208	8.040
13.800	-0.000	0.770	1.576	2.388	3.204	4.754	6.089	7.180	8.019
13.900	-0.000	0.752	1.556	2.368	3.186	4.732	6.067	7.162	8.005
14.000	-0.000	0.741	1.544	2.357	3.175	4.721	6.056	7.152	7.997
14.100	-0.000	0.736	1.538	2.351	3.170	4.717	6.052	7.148	7.994
14.200	-0.000	0.733	1.533	2.347	3.167	4.715	6.050	7.146	7.992
14.300	-0.000	0.730	1.528	2.343	3.163	4.714	6.049	7.144	7.991
14.400	-0.000	0.728	1.523	2.338	3.159	4.712	6.047	7.141	7.989
14.500	-0.000	0.726	1.519	2.334	3.155	4.710	6.045	7.139	7.987
14.600	-0.000	0.724	1.514	2.330	3.152	4.708	6.043	7.136	7.985
14.700	-0.000	0.723	1.510	2.325	3.148	4.707	6.041	7.134	7.984
14.800	-0.000	0.721	1.505	2.321	3.145	4.705	6.039	7.132	7.982
14.900	-0.000	0.720	1.501	2.317	3.141	4.703	6.037	7.129	7.980
15.000	-0.000	0.719	1.497	2.313	3.138	4.702	6.036	7.127	7.979
15.100	-0.000	0.718	1.493	2.309	3.134	4.700	6.034	7.124	7.977
15.200	-0.000	0.717	1.489	2.305	3.131	4.698	6.032	7.122	7.975
15.300	-0.000	0.717	1.485	2.301	3.128	4.697	6.030	7.120	7.974
15.400	-0.000	0.716	1.482	2.297	3.124	4.695	6.028	7.117	7.972
15.500	-0.000	0.716	1.479	2.293	3.121	4.694	6.027	7.115	7.970
15.600	-0.000	0.717	1.476	2.290	3.118	4.692	6.025	7.113	7.969
15.700	-0.000	0.717	1.473	2.286	3.115	4.691	6.023	7.110	7.967
15.800	-0.000	0.718	1.470	2.283	3.111	4.689	6.021	7.108	7.965
15.900	-0.000	0.719	1.468	2.279	3.108	4.688	6.019	7.105	7.964
16.000	-0.000	0.719	1.466	2.276	3.105	4.687	6.018	7.103	7.962
16.100	-0.000	0.720	1.464	2.272	3.102	4.685	6.016	7.101	7.960
16.200	-0.000	0.721	1.462	2.269	3.100	4.684	6.014	7.098	7.959
16.300	-0.000	0.722	1.461	2.266	3.097	4.683	6.012	7.096	7.957
16.400	-0.000	0.723	1.461	2.263	3.094	4.681	6.011	7.094	7.955
16.500	-0.000	0.724	1.460	2.260	3.091	4.680	6.009	7.091	7.954
16.600	-0.000	0.725	1.460	2.257	3.089	4.679	6.007	7.089	7.952
16.700	-0.000	0.726	1.460	2.255	3.086	4.678	6.005	7.087	7.950
16.800	-0.000	0.727	1.460	2.252	3.083	4.677	6.004	7.084	7.948
16.900	-0.000	0.728	1.460	2.250	3.081	4.675	6.002	7.082	7.947
17.000	-0.000	0.729	1.461	2.247	3.079	4.674	6.000	7.080	7.945
17.100	-0.000	0.730	1.462	2.245	3.076	4.673	5.998	7.078	7.943
17.200	-0.000	0.731	1.462	2.243	3.074	4.672	5.997	7.075	7.941
17.300	-0.000	0.732	1.464	2.242	3.072	4.671	5.995	7.073	7.940
17.400	-0.000	0.733	1.465	2.240	3.070	4.670	5.993	7.071	7.938

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.735	1.467	2.239	3.068	4.669	5.991	7.069	7.936
17.600	-0.000	0.736	1.468	2.237	3.066	4.668	5.990	7.067	7.934
17.700	-0.000	0.737	1.470	2.236	3.064	4.667	5.988	7.065	7.933
17.800	-0.000	0.738	1.472	2.236	3.062	4.666	5.986	7.062	7.931
17.900	-0.000	0.739	1.475	2.235	3.061	4.665	5.985	7.060	7.929
18.000	-0.000	0.740	1.477	2.235	3.059	4.664	5.983	7.058	7.928
18.100	-0.000	0.741	1.479	2.235	3.058	4.663	5.981	7.056	7.926
18.200	-0.000	0.742	1.481	2.235	3.056	4.662	5.979	7.054	7.926
18.300	-0.000	0.743	1.484	2.235	3.055	4.661	5.978	7.052	7.924
18.400	-0.000	0.745	1.486	2.236	3.054	4.660	5.976	7.050	7.922
18.500	-0.000	0.746	1.488	2.238	3.053	4.659	5.974	7.048	7.921
18.600	-0.000	0.747	1.491	2.239	3.052	4.659	5.973	7.046	7.919
18.700	-0.000	0.748	1.493	2.241	3.052	4.658	5.971	7.044	7.917
18.800	-0.000	0.749	1.495	2.242	3.051	4.657	5.969	7.041	7.916
18.900	-0.000	0.750	1.498	2.245	3.050	4.656	5.968	7.039	7.914
19.000	-0.000	0.752	1.500	2.247	3.050	4.656	5.966	7.037	7.913
19.100	-0.000	0.753	1.503	2.249	3.050	4.655	5.964	7.035	7.911
19.200	-0.000	0.754	1.505	2.252	3.050	4.654	5.962	7.033	7.910
19.300	-0.000	0.755	1.508	2.255	3.050	4.653	5.961	7.031	7.908
19.400	-0.000	0.757	1.510	2.258	3.051	4.653	5.959	7.029	7.907
19.500	-0.000	0.758	1.513	2.262	3.051	4.652	5.957	7.027	7.905
19.600	-0.000	0.759	1.515	2.265	3.052	4.651	5.956	7.025	7.904
19.700	-0.000	0.761	1.518	2.269	3.053	4.651	5.954	7.024	7.902
19.800	-0.000	0.762	1.521	2.273	3.054	4.650	5.952	7.022	7.901
19.900	-0.000	0.763	1.523	2.277	3.055	4.650	5.950	7.020	7.899
20.000	-0.000	0.764	1.526	2.281	3.057	4.649	5.949	7.018	7.898
20.100	-0.000	0.766	1.529	2.285	3.059	4.649	5.947	7.016	7.896
20.200	-0.000	0.767	1.531	2.289	3.061	4.648	5.945	7.015	7.895
20.300	-0.000	0.769	1.534	2.293	3.064	4.648	5.943	7.013	7.893
20.400	-0.000	0.770	1.537	2.297	3.066	4.647	5.941	7.011	7.892
20.500	-0.000	0.771	1.539	2.301	3.069	4.647	5.940	7.009	7.890
20.600	-0.000	0.773	1.542	2.305	3.072	4.646	5.939	7.007	7.889
20.700	-0.000	0.774	1.545	2.309	3.076	4.647	5.937	7.006	7.888
20.800	-0.000	0.776	1.548	2.314	3.079	4.646	5.935	7.004	7.886
20.900	-0.000	0.777	1.551	2.317	3.083	4.646	5.933	7.002	7.885
21.000	-0.000	0.778	1.554	2.322	3.087	4.645	5.932	7.001	7.883
21.100	-0.000	0.780	1.556	2.327	3.092	4.645	5.930	6.999	7.882
21.200	-0.000	0.781	1.559	2.331	3.096	4.645	5.928	6.997	7.881
21.300	-0.000	0.783	1.562	2.335	3.101	4.644	5.926	6.996	7.879
21.400	-0.000	0.784	1.565	2.340	3.106	4.644	5.925	6.994	7.878
21.500	-0.000	0.786	1.568	2.344	3.112	4.644	5.923	6.992	7.876
21.600	-0.000	0.787	1.571	2.349	3.117	4.643	5.921	6.991	7.875
21.700	-0.000	0.789	1.573	2.352	3.123	4.643	5.919	6.989	7.874
21.800	-0.000	0.790	1.577	2.358	3.129	4.643	5.917	6.988	7.872
21.900	-0.000	0.792	1.580	2.362	3.135	4.643	5.916	6.986	7.871
22.000	-0.000	0.793	1.583	2.367	3.141	4.643	5.914	6.985	7.870
22.100	-0.000	0.795	1.587	2.372	3.147	4.643	5.912	6.983	7.868
22.200	-0.000	0.796	1.590	2.376	3.154	4.642	5.911	6.982	7.867
22.300	-0.000	0.797	1.593	2.381	3.160	4.642	5.909	6.981	7.866
22.400	-0.000	0.800	1.596	2.386	3.166	4.642	5.910	6.979	7.865
22.500	-0.000	0.801	1.599	2.391	3.172	4.642	5.907	6.978	7.863
22.600	-0.000	0.803	1.602	2.395	3.178	4.642	5.906	6.977	7.862
22.700	-0.000	0.804	1.606	2.402	3.184	4.643	5.904	6.976	7.861
22.800	-0.000	0.806	1.609	2.405	3.190	4.643	5.903	6.974	7.860
22.900	-0.000	0.807	1.612	2.410	3.196	4.643	5.901	6.973	7.859
23.000	-0.000	0.809	1.615	2.415	3.202	4.643	5.900	6.972	7.858
23.100	-0.000	0.811	1.619	2.418	3.208	4.643	5.898	6.970	7.856
23.200	-0.000	0.813	1.622	2.425	3.214	4.644	5.897	6.969	7.855
23.300	-0.000	0.814	1.625	2.430	3.219	4.644	5.896	6.968	7.854
23.400	-0.000	0.816	1.629	2.435	3.225	4.644	5.894	6.967	7.853

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.818	1.632	2.440	3.231	4.645	5.893	6.965	7.852
23.600	-0.000	0.819	1.636	2.445	3.236	4.643	5.892	6.964	7.851
23.700	-0.000	0.821	1.639	2.450	3.242	4.644	5.891	6.963	7.850
23.800	-0.000	0.823	1.643	2.455	3.247	4.648	5.890	6.962	7.849
23.900	-0.000	0.825	1.646	2.461	3.250	4.649	5.889	6.961	7.848
24.000	-0.000	0.826	1.648	2.466	3.257	4.651	5.888	6.960	7.847
24.100	-0.000	0.828	1.653	2.471	3.262	4.651	5.887	6.959	7.846
24.200	-0.000	0.830	1.657	2.476	3.267	4.655	5.886	6.958	7.846
24.300	-0.000	0.832	1.660	2.480	3.272	4.658	5.885	6.957	7.845
24.400	-0.000	0.834	1.664	2.486	3.276	4.660	5.884	6.956	7.844
24.500	-0.000	0.835	1.667	2.491	3.278	4.662	5.884	6.956	7.844
24.600	-0.000	0.837	1.671	2.496	3.284	4.664	5.883	6.955	7.843
24.700	-0.000	0.839	1.675	2.501	3.288	4.665	5.883	6.954	7.843
24.800	-0.000	0.841	1.678	2.506	3.291	4.667	5.883	6.954	7.842
24.900	-0.000	0.843	1.682	2.510	3.295	4.669	5.883	6.953	7.842
25.000	-0.000	0.845	1.686	2.515	3.298	4.670	5.883	6.952	7.841
25.100	-0.000	0.847	1.690	2.519	3.300	4.672	5.883	6.952	7.841
25.200	-0.000	0.848	1.693	2.523	3.302	4.673	5.884	6.952	7.841
25.300	-0.000	0.850	1.697	2.527	3.304	4.675	5.885	6.952	7.841
25.400	-0.000	0.852	1.701	2.531	3.305	4.676	5.886	6.951	7.840
25.500	-0.000	0.854	1.705	2.535	3.306	4.677	5.887	6.951	7.840
25.600	-0.000	0.856	1.709	2.539	3.307	4.679	5.889	6.953	7.840
25.700	-0.000	0.858	1.712	2.542	3.308	4.680	5.891	6.953	7.840
25.800	-0.000	0.860	1.716	2.546	3.309	4.681	5.893	6.954	7.841
25.900	-0.000	0.862	1.720	2.549	3.309	4.682	5.895	6.954	7.841
26.000	-0.000	0.864	1.724	2.552	3.310	4.683	5.897	6.954	7.841

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 2.50 M

UNIT: m

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INITIAL	HEELING ANGLE (DEGREES)								
DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	9.473	11.462	12.477	13.084	13.657	13.775	13.716	13.491
0.100	-0.002	9.578	11.693	12.779	13.435	14.067	14.190	13.978	13.454
0.200	-0.001	9.613	11.841	12.988	13.685	14.364	14.468	14.150	13.418
0.300	-0.001	9.600	11.930	13.126	13.857	14.574	14.655	14.264	13.392
0.400	-0.001	9.546	11.976	13.214	13.972	14.719	14.782	14.340	13.376
0.500	-0.001	9.457	11.987	13.267	14.049	14.817	14.871	14.393	13.367
0.600	-0.001	9.334	11.967	13.290	14.096	14.884	14.933	14.431	13.364
0.700	-0.001	9.171	11.909	13.279	14.109	14.921	14.975	14.460	13.366
0.800	-0.001	8.962	11.806	13.224	14.081	14.924	14.996	14.481	13.376
0.900	-0.001	8.720	11.669	13.136	14.022	14.902	15.002	14.495	13.390
1.000	-0.001	8.459	11.511	13.027	13.944	14.865	15.000	14.505	13.406
1.100	-0.001	8.186	11.340	12.905	13.854	14.818	14.991	14.511	13.422
1.200	-0.001	7.907	11.160	12.775	13.756	14.765	14.979	14.513	13.437
1.300	-0.001	7.625	10.974	12.640	13.653	14.708	14.964	14.512	13.452
1.400	-0.001	7.343	10.786	12.501	13.547	14.649	14.945	14.507	13.465
1.500	-0.001	7.065	10.596	12.360	13.439	14.588	14.923	14.500	13.476
1.600	-0.001	6.794	10.406	12.218	13.330	14.525	14.898	14.489	13.486
1.700	-0.001	6.533	10.217	12.075	13.220	14.463	14.871	14.476	13.494
1.800	-0.001	6.283	10.028	11.933	13.110	14.401	14.841	14.461	13.501
1.900	-0.001	6.044	9.839	11.792	13.001	14.339	14.809	14.443	13.506
2.000	-0.001	5.817	9.652	11.652	12.893	14.279	14.775	14.423	13.510
2.100	-0.001	5.602	9.468	11.515	12.787	14.219	14.739	14.401	13.513
2.200	-0.001	5.400	9.287	11.379	12.681	14.160	14.701	14.378	13.514
2.300	-0.001	5.210	9.110	11.245	12.577	14.102	14.662	14.353	13.515
2.400	-0.001	5.034	8.938	11.113	12.475	14.045	14.622	14.326	13.515
2.500	-0.001	4.870	8.769	10.981	12.374	13.988	14.581	14.298	13.513
2.600	-0.001	4.717	8.604	10.849	12.275	13.933	14.538	14.268	13.511
2.700	-0.001	4.574	8.442	10.719	12.177	13.878	14.494	14.238	13.508
2.800	-0.001	4.441	8.284	10.590	12.081	13.823	14.448	14.206	13.503
2.900	-0.001	4.316	8.129	10.465	11.985	13.768	14.402	14.172	13.498
3.000	-0.001	4.199	7.976	10.342	11.891	13.713	14.354	14.138	13.492
3.100	-0.001	4.089	7.826	10.222	11.797	13.657	14.305	14.103	13.485
3.200	-0.001	3.985	7.680	10.104	11.703	13.600	14.256	14.066	13.476
3.300	-0.001	3.887	7.536	9.988	11.609	13.543	14.205	14.029	13.466
3.400	-0.001	3.795	7.396	9.874	11.516	13.486	14.153	13.990	13.453
3.500	-0.001	3.708	7.258	9.762	11.424	13.428	14.099	13.951	13.440
3.600	-0.001	3.626	7.125	9.653	11.334	13.369	14.046	13.911	13.425
3.700	-0.001	3.549	6.994	9.545	11.245	13.309	13.990	13.870	13.409
3.800	-0.001	3.475	6.866	9.438	11.158	13.248	13.934	13.827	13.391
3.900	-0.001	3.406	6.742	9.333	11.072	13.186	13.877	13.784	13.372
4.000	-0.001	3.339	6.622	9.229	10.988	13.123	13.818	13.740	13.353
4.100	-0.001	3.277	6.506	9.127	10.905	13.059	13.759	13.696	13.333
4.200	-0.001	3.216	6.393	9.026	10.824	12.994	13.699	13.651	13.311
4.300	-0.001	3.159	6.285	8.927	10.744	12.927	13.638	13.606	13.288
4.400	-0.001	3.104	6.180	8.828	10.665	12.860	13.575	13.561	13.264
4.500	-0.001	3.052	6.080	8.731	10.588	12.791	13.512	13.516	13.239
4.600	-0.001	3.002	5.984	8.636	10.511	12.722	13.448	13.470	13.214
4.700	-0.001	2.954	5.892	8.541	10.436	12.651	13.382	13.424	13.188
4.800	-0.001	2.908	5.803	8.448	10.362	12.579	13.316	13.378	13.163
4.900	-0.001	2.863	5.718	8.357	10.289	12.505	13.249	13.332	13.137
5.000	-0.001	2.821	5.636	8.266	10.217	12.431	13.181	13.285	13.111
5.100	-0.001	2.780	5.557	8.176	10.145	12.355	13.113	13.238	13.084
5.200	-0.001	2.740	5.481	8.088	10.073	12.279	13.045	13.192	13.056
5.300	-0.001	2.703	5.408	8.001	10.002	12.201	12.977	13.145	13.028
5.400	-0.001	2.667	5.338	7.916	9.931	12.123	12.909	13.098	13.000

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.632	5.271	7.832	9.860	12.044	12.841	13.052	12.970
5.600	-0.001	2.599	5.206	7.749	9.788	11.965	12.772	13.006	12.940
5.700	-0.001	2.567	5.144	7.668	9.717	11.884	12.702	12.960	12.909
5.800	-0.001	2.536	5.083	7.589	9.645	11.803	12.633	12.915	12.878
5.900	-0.001	2.507	5.025	7.511	9.573	11.722	12.562	12.869	12.847
6.000	-0.001	2.479	4.968	7.436	9.501	11.640	12.491	12.823	12.816
6.100	-0.001	2.452	4.915	7.362	9.428	11.558	12.419	12.776	12.785
6.200	-0.001	2.426	4.863	7.290	9.354	11.475	12.347	12.729	12.752
6.300	-0.000	2.401	4.812	7.220	9.280	11.391	12.275	12.681	12.720
6.400	-0.000	2.377	4.764	7.152	9.205	11.308	12.202	12.633	12.687
6.500	-0.000	2.353	4.717	7.086	9.130	11.224	12.129	12.583	12.653
6.600	-0.000	2.331	4.672	7.021	9.054	11.139	12.057	12.533	12.619
6.700	-0.000	2.309	4.629	6.959	8.977	11.055	11.984	12.482	12.582
6.800	-0.000	2.288	4.587	6.898	8.900	10.970	11.911	12.430	12.545
6.900	-0.000	2.268	4.547	6.838	8.822	10.884	11.838	12.376	12.505
7.000	-0.000	2.249	4.508	6.780	8.743	10.799	11.765	12.322	12.464
7.100	-0.000	2.230	4.471	6.723	8.663	10.713	11.692	12.268	12.421
7.200	-0.000	2.212	4.435	6.666	8.582	10.626	11.619	12.213	12.375
7.300	-0.000	2.195	4.400	6.611	8.501	10.540	11.546	12.158	12.327
7.400	-0.000	2.178	4.366	6.556	8.418	10.453	11.472	12.103	12.278
7.500	-0.000	2.162	4.333	6.502	8.335	10.365	11.399	12.047	12.226
7.600	-0.000	2.146	4.301	6.449	8.251	10.277	11.326	11.992	12.173
7.700	-0.000	2.131	4.271	6.395	8.166	10.189	11.252	11.936	12.118
7.800	-0.000	2.116	4.241	6.343	8.080	10.100	11.179	11.880	12.062
7.900	-0.000	2.102	4.212	6.290	7.994	10.010	11.105	11.823	12.005
8.000	-0.000	2.088	4.185	6.238	7.907	9.920	11.031	11.766	11.947
8.100	-0.000	2.074	4.158	6.186	7.821	9.830	10.957	11.709	11.887
8.200	-0.000	2.062	4.132	6.133	7.734	9.739	10.883	11.652	11.826
8.300	-0.000	2.049	4.107	6.081	7.648	9.648	10.809	11.594	11.765
8.400	-0.000	2.037	4.083	6.028	7.562	9.557	10.734	11.536	11.704
8.500	-0.000	2.025	4.059	5.975	7.475	9.465	10.660	11.477	11.643
8.600	-0.000	2.014	4.036	5.921	7.389	9.373	10.585	11.416	11.580
8.700	-0.000	2.003	4.014	5.867	7.304	9.280	10.510	11.354	11.517
8.800	-0.000	1.992	3.992	5.812	7.218	9.186	10.435	11.290	11.454
8.900	-0.000	1.982	3.970	5.756	7.133	9.092	10.359	11.225	11.390
9.000	-0.000	1.972	3.948	5.700	7.047	8.998	10.283	11.159	11.325
9.100	-0.000	1.962	3.926	5.643	6.962	8.903	10.207	11.090	11.259
9.200	-0.000	1.953	3.904	5.584	6.877	8.807	10.131	11.020	11.193
9.300	-0.000	1.944	3.881	5.525	6.792	8.712	10.054	10.948	11.126
9.400	-0.000	1.935	3.858	5.466	6.707	8.616	9.976	10.875	11.059
9.500	-0.000	1.926	3.833	5.405	6.622	8.519	9.897	10.799	10.991
9.600	-0.000	1.918	3.808	5.344	6.537	8.423	9.818	10.722	10.923
9.700	-0.000	1.910	3.782	5.282	6.452	8.326	9.739	10.643	10.854
9.800	-0.000	1.902	3.755	5.219	6.367	8.228	9.659	10.562	10.784
9.900	-0.000	1.895	3.727	5.156	6.282	8.130	9.578	10.481	10.715
10.000	-0.000	1.888	3.697	5.091	6.197	8.032	9.497	10.399	10.644
10.100	-0.000	1.881	3.666	5.026	6.112	7.933	9.415	10.316	10.574
10.200	-0.000	1.874	3.634	4.960	6.027	7.835	9.333	10.233	10.502
10.300	-0.000	1.868	3.599	4.894	5.942	7.737	9.250	10.149	10.430
10.400	-0.000	1.861	3.563	4.826	5.857	7.638	9.166	10.064	10.358
10.500	-0.000	1.855	3.525	4.758	5.772	7.540	9.082	9.979	10.285
10.600	-0.000	1.849	3.486	4.688	5.687	7.442	8.996	9.892	10.211
10.700	-0.000	1.842	3.443	4.618	5.601	7.344	8.908	9.805	10.137
10.800	-0.000	1.835	3.399	4.547	5.515	7.247	8.819	9.717	10.063
10.900	-0.000	1.827	3.352	4.475	5.430	7.150	8.728	9.628	9.988
11.000	-0.000	1.817	3.304	4.401	5.343	7.053	8.635	9.539	9.913
11.100	-0.000	1.806	3.253	4.328	5.257	6.957	8.541	9.448	9.837
11.200	-0.000	1.793	3.200	4.253	5.171	6.862	8.444	9.357	9.761
11.300	-0.000	1.778	3.145	4.177	5.084	6.767	8.346	9.265	9.685
11.400	-0.000	1.761	3.088	4.100	4.997	6.673	8.247	9.173	9.608

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.742	3.029	4.023	4.911	6.579	8.145	9.080	9.531
11.600	-0.000	1.720	2.969	3.944	4.824	6.487	8.042	8.986	9.454
11.700	-0.000	1.695	2.906	3.865	4.737	6.395	7.939	8.892	9.376
11.800	-0.000	1.666	2.842	3.785	4.650	6.304	7.835	8.797	9.298
11.900	-0.000	1.635	2.775	3.705	4.563	6.214	7.731	8.702	9.220
12.000	-0.000	1.599	2.707	3.623	4.475	6.124	7.628	8.606	9.141
12.100	-0.000	1.560	2.636	3.541	4.388	6.036	7.525	8.510	9.062
12.200	-0.000	1.518	2.565	3.459	4.302	5.948	7.422	8.413	8.983
12.300	-0.000	1.473	2.494	3.377	4.216	5.863	7.321	8.317	8.905
12.400	-0.000	1.428	2.424	3.298	4.133	5.779	7.221	8.222	8.827
12.500	-0.000	1.382	2.355	3.221	4.052	5.698	7.124	8.128	8.752
12.600	-0.000	1.336	2.288	3.145	3.974	5.619	7.028	8.036	8.678
12.700	-0.000	1.290	2.222	3.072	3.897	5.542	6.935	7.947	8.607
12.800	-0.000	1.243	2.157	3.000	3.823	5.468	6.844	7.861	8.539
12.900	-0.000	1.197	2.093	2.931	3.752	5.395	6.757	7.779	8.474
13.000	-0.000	1.151	2.032	2.864	3.683	5.324	6.672	7.700	8.413
13.100	-0.000	1.105	1.972	2.799	3.616	5.253	6.592	7.626	8.355
13.200	-0.000	1.060	1.913	2.737	3.553	5.184	6.515	7.557	8.301
13.300	-0.000	1.015	1.857	2.678	3.493	5.116	6.443	7.492	8.251
13.400	-0.000	0.971	1.805	2.623	3.438	5.050	6.377	7.433	8.205
13.500	-0.000	0.929	1.755	2.571	3.386	4.988	6.316	7.379	8.164
13.600	-0.000	0.890	1.709	2.524	3.339	4.931	6.260	7.329	8.127
13.700	-0.000	0.853	1.667	2.481	3.297	4.879	6.211	7.284	8.094
13.800	-0.000	0.820	1.630	2.443	3.260	4.834	6.167	7.245	8.067
13.900	-0.000	0.791	1.597	2.411	3.229	4.796	6.130	7.213	8.043
14.000	-0.000	0.768	1.571	2.384	3.203	4.766	6.100	7.187	8.024
14.100	-0.000	0.750	1.550	2.365	3.184	4.743	6.078	7.167	8.010
14.200	-0.000	0.740	1.537	2.351	3.172	4.729	6.064	7.155	8.000
14.300	-0.000	0.734	1.528	2.344	3.165	4.723	6.058	7.149	7.996
14.400	-0.000	0.732	1.524	2.339	3.161	4.721	6.056	7.147	7.994
14.500	-0.000	0.730	1.519	2.335	3.158	4.719	6.054	7.144	7.992
14.600	-0.000	0.729	1.515	2.331	3.154	4.718	6.052	7.142	7.990
14.700	-0.000	0.727	1.511	2.327	3.151	4.716	6.050	7.140	7.989
14.800	-0.000	0.726	1.506	2.322	3.147	4.714	6.048	7.137	7.987
14.900	-0.000	0.725	1.502	2.318	3.144	4.713	6.046	7.135	7.985
15.000	-0.000	0.723	1.498	2.314	3.140	4.711	6.045	7.132	7.984
15.100	-0.000	0.722	1.495	2.311	3.137	4.710	6.043	7.130	7.982
15.200	-0.000	0.722	1.491	2.307	3.134	4.708	6.041	7.128	7.980
15.300	-0.000	0.721	1.488	2.303	3.130	4.707	6.039	7.125	7.979
15.400	-0.000	0.720	1.485	2.299	3.127	4.705	6.037	7.123	7.977
15.500	-0.000	0.720	1.482	2.296	3.124	4.704	6.036	7.120	7.975
15.600	-0.000	0.720	1.479	2.292	3.121	4.702	6.034	7.118	7.974
15.700	-0.000	0.720	1.476	2.288	3.118	4.701	6.032	7.115	7.972
15.800	-0.000	0.720	1.474	2.285	3.115	4.700	6.030	7.113	7.970
15.900	-0.000	0.721	1.472	2.282	3.112	4.698	6.028	7.111	7.969
16.000	-0.000	0.721	1.470	2.278	3.109	4.697	6.027	7.108	7.967
16.100	-0.000	0.722	1.469	2.275	3.106	4.696	6.025	7.106	7.965
16.200	-0.000	0.723	1.468	2.272	3.103	4.694	6.023	7.104	7.964
16.300	-0.000	0.724	1.467	2.269	3.100	4.693	6.021	7.101	7.962
16.400	-0.000	0.725	1.467	2.266	3.098	4.692	6.020	7.099	7.960
16.500	-0.000	0.726	1.466	2.264	3.095	4.691	6.018	7.097	7.958
16.600	-0.000	0.727	1.466	2.261	3.092	4.689	6.016	7.094	7.956
16.700	-0.000	0.728	1.466	2.258	3.090	4.688	6.014	7.092	7.955
16.800	-0.000	0.729	1.466	2.256	3.088	4.687	6.013	7.090	7.953
16.900	-0.000	0.730	1.466	2.254	3.085	4.686	6.011	7.088	7.951
17.000	-0.000	0.731	1.467	2.252	3.083	4.685	6.009	7.085	7.949
17.100	-0.000	0.732	1.467	2.250	3.081	4.684	6.007	7.083	7.947
17.200	-0.000	0.733	1.468	2.248	3.079	4.683	6.006	7.081	7.946
17.300	-0.000	0.734	1.469	2.247	3.077	4.682	6.004	7.079	7.944
17.400	-0.000	0.735	1.470	2.245	3.075	4.681	6.002	7.077	7.942

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.736	1.471	2.244	3.073	4.679	6.000	7.074	7.940
17.600	-0.000	0.737	1.473	2.243	3.071	4.678	5.999	7.072	7.939
17.700	-0.000	0.738	1.475	2.243	3.069	4.678	5.997	7.070	7.937
17.800	-0.000	0.740	1.476	2.242	3.068	4.677	5.995	7.068	7.935
17.900	-0.000	0.741	1.478	2.242	3.066	4.676	5.993	7.066	7.935
18.000	-0.000	0.742	1.481	2.242	3.065	4.675	5.992	7.064	7.933
18.100	-0.000	0.743	1.483	2.242	3.064	4.674	5.990	7.062	7.931
18.200	-0.000	0.744	1.485	2.243	3.063	4.673	5.988	7.060	7.930
18.300	-0.000	0.745	1.487	2.244	3.061	4.672	5.987	7.058	7.928
18.400	-0.000	0.746	1.490	2.245	3.061	4.671	5.985	7.055	7.926
18.500	-0.000	0.748	1.492	2.246	3.060	4.670	5.983	7.053	7.925
18.600	-0.000	0.749	1.495	2.247	3.059	4.670	5.981	7.051	7.923
18.700	-0.000	0.750	1.497	2.249	3.059	4.669	5.980	7.049	7.922
18.800	-0.000	0.751	1.499	2.251	3.058	4.668	5.978	7.047	7.920
18.900	-0.000	0.752	1.502	2.253	3.058	4.667	5.976	7.045	7.918
19.000	-0.000	0.754	1.504	2.255	3.058	4.667	5.975	7.043	7.917
19.100	-0.000	0.755	1.507	2.258	3.058	4.666	5.973	7.040	7.915
19.200	-0.000	0.756	1.509	2.260	3.058	4.665	5.971	7.038	7.914
19.300	-0.000	0.757	1.512	2.263	3.058	4.665	5.969	7.036	7.912
19.400	-0.000	0.759	1.514	2.266	3.059	4.664	5.968	7.036	7.911
19.500	-0.000	0.760	1.517	2.269	3.060	4.663	5.966	7.034	7.909
19.600	-0.000	0.761	1.520	2.272	3.061	4.663	5.964	7.032	7.908
19.700	-0.000	0.763	1.522	2.277	3.062	4.662	5.962	7.030	7.906
19.800	-0.000	0.764	1.525	2.280	3.063	4.662	5.960	7.028	7.905
19.900	-0.000	0.765	1.528	2.284	3.065	4.661	5.959	7.026	7.903
20.000	-0.000	0.767	1.531	2.288	3.068	4.661	5.957	7.024	7.902
20.100	-0.000	0.768	1.533	2.292	3.070	4.660	5.955	7.022	7.900
20.200	-0.000	0.769	1.536	2.297	3.072	4.660	5.953	7.020	7.899
20.300	-0.000	0.771	1.538	2.299	3.074	4.659	5.952	7.018	7.897
20.400	-0.000	0.772	1.541	2.304	3.077	4.660	5.951	7.017	7.896
20.500	-0.000	0.774	1.545	2.308	3.080	4.659	5.949	7.015	7.894
20.600	-0.000	0.775	1.547	2.312	3.084	4.658	5.947	7.013	7.893
20.700	-0.000	0.776	1.551	2.317	3.087	4.658	5.945	7.011	7.891
20.800	-0.000	0.778	1.553	2.321	3.091	4.657	5.943	7.010	7.890
20.900	-0.000	0.779	1.555	2.325	3.093	4.657	5.941	7.008	7.888
21.000	-0.000	0.781	1.558	2.330	3.099	4.657	5.940	7.006	7.887
21.100	-0.000	0.782	1.561	2.334	3.103	4.656	5.938	7.005	7.885
21.200	-0.000	0.784	1.564	2.338	3.108	4.656	5.936	7.003	7.884
21.300	-0.000	0.785	1.567	2.343	3.113	4.656	5.934	7.001	7.882
21.400	-0.000	0.787	1.570	2.347	3.118	4.655	5.932	7.000	7.881
21.500	-0.000	0.788	1.573	2.352	3.121	4.655	5.931	6.998	7.880
21.600	-0.000	0.790	1.576	2.356	3.128	4.655	5.929	6.996	7.878
21.700	-0.000	0.791	1.579	2.360	3.134	4.654	5.927	6.995	7.877
21.800	-0.000	0.793	1.582	2.365	3.140	4.654	5.925	6.993	7.876
21.900	-0.000	0.794	1.585	2.370	3.146	4.654	5.924	6.992	7.874
22.000	-0.000	0.796	1.589	2.375	3.152	4.654	5.922	6.991	7.873
22.100	-0.000	0.797	1.592	2.379	3.158	4.654	5.920	6.989	7.871
22.200	-0.000	0.799	1.595	2.384	3.164	4.654	5.920	6.988	7.870
22.300	-0.000	0.801	1.597	2.389	3.170	4.654	5.918	6.987	7.869
22.400	-0.000	0.802	1.601	2.394	3.176	4.654	5.917	6.986	7.868
22.500	-0.000	0.804	1.605	2.399	3.182	4.654	5.915	6.984	7.866
22.600	-0.000	0.805	1.608	2.403	3.188	4.654	5.913	6.983	7.865
22.700	-0.000	0.807	1.611	2.408	3.194	4.654	5.912	6.981	7.864
22.800	-0.000	0.809	1.614	2.413	3.200	4.654	5.910	6.980	7.863
22.900	-0.000	0.810	1.618	2.418	3.206	4.654	5.909	6.978	7.861
23.000	-0.000	0.812	1.621	2.422	3.212	4.655	5.908	6.977	7.860
23.100	-0.000	0.814	1.624	2.428	3.215	4.655	5.906	6.976	7.859
23.200	-0.000	0.815	1.628	2.432	3.223	4.653	5.905	6.975	7.858
23.300	-0.000	0.817	1.631	2.438	3.229	4.656	5.904	6.973	7.857
23.400	-0.000	0.819	1.635	2.443	3.234	4.656	5.902	6.972	7.856

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.821	1.638	2.447	3.240	4.657	5.901	6.971	7.855
23.600	-0.000	0.822	1.641	2.454	3.245	4.658	5.900	6.970	7.854
23.700	-0.000	0.824	1.645	2.459	3.250	4.658	5.899	6.969	7.853
23.800	-0.000	0.826	1.648	2.464	3.255	4.659	5.897	6.967	7.852
23.900	-0.000	0.828	1.652	2.469	3.261	4.661	5.896	6.966	7.851
24.000	-0.000	0.829	1.655	2.475	3.266	4.662	5.895	6.965	7.850
24.100	-0.000	0.831	1.659	2.480	3.270	4.664	5.894	6.964	7.849
24.200	-0.000	0.833	1.663	2.485	3.275	4.666	5.893	6.963	7.849
24.300	-0.000	0.835	1.666	2.490	3.280	4.668	5.893	6.963	7.848
24.400	-0.000	0.837	1.670	2.495	3.284	4.670	5.892	6.962	7.847
24.500	-0.000	0.838	1.674	2.499	3.288	4.672	5.891	6.961	7.847
24.600	-0.000	0.840	1.677	2.504	3.292	4.674	5.891	6.960	7.846
24.700	-0.000	0.842	1.681	2.509	3.296	4.675	5.891	6.960	7.846
24.800	-0.000	0.844	1.685	2.513	3.299	4.677	5.890	6.959	7.845
24.900	-0.000	0.846	1.688	2.518	3.303	4.678	5.890	6.958	7.845
25.000	-0.000	0.848	1.692	2.522	3.306	4.680	5.891	6.958	7.844
25.100	-0.000	0.850	1.696	2.526	3.308	4.681	5.891	6.958	7.844
25.200	-0.000	0.852	1.700	2.530	3.311	4.682	5.892	6.957	7.844
25.300	-0.000	0.854	1.704	2.534	3.313	4.683	5.893	6.960	7.844
25.400	-0.000	0.855	1.707	2.537	3.314	4.685	5.894	6.959	7.844
25.500	-0.000	0.857	1.711	2.541	3.315	4.686	5.895	6.958	7.844
25.600	-0.000	0.859	1.715	2.544	3.316	4.687	5.897	6.959	7.844
25.700	-0.000	0.861	1.719	2.548	3.317	4.688	5.899	6.959	7.844
25.800	-0.000	0.863	1.723	2.550	3.317	4.688	5.901	6.959	7.844
25.900	-0.000	0.865	1.727	2.553	3.317	4.689	5.903	6.959	7.844
26.000	-0.000	0.867	1.730	2.556	3.317	4.690	5.906	6.960	7.845

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 3.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	8.970	11.055	12.132	12.790	13.451	13.650	13.698	13.513
0.100	-0.002	9.033	11.227	12.364	13.062	13.774	13.987	13.927	13.492
0.200	-0.001	9.048	11.342	12.531	13.265	14.021	14.239	14.085	13.469
0.300	-0.001	9.026	11.414	12.647	13.413	14.205	14.421	14.194	13.450
0.400	-0.001	8.974	11.451	12.724	13.517	14.341	14.552	14.270	13.436
0.500	-0.001	8.896	11.461	12.774	13.590	14.440	14.648	14.324	13.426
0.600	-0.001	8.795	11.449	12.802	13.640	14.513	14.719	14.362	13.420
0.700	-0.001	8.670	11.415	12.810	13.670	14.566	14.773	14.391	13.415
0.800	-0.001	8.516	11.356	12.795	13.679	14.601	14.814	14.412	13.414
0.900	-0.001	8.332	11.267	12.751	13.662	14.613	14.840	14.427	13.417
1.000	-0.001	8.119	11.150	12.680	13.619	14.604	14.852	14.436	13.423
1.100	-0.001	7.889	11.014	12.590	13.558	14.579	14.855	14.441	13.431
1.200	-0.001	7.648	10.864	12.486	13.484	14.545	14.851	14.441	13.440
1.300	-0.001	7.400	10.706	12.374	13.402	14.505	14.840	14.438	13.449
1.400	-0.001	7.149	10.541	12.256	13.314	14.459	14.825	14.432	13.458
1.500	-0.001	6.899	10.372	12.134	13.223	14.411	14.806	14.423	13.467
1.600	-0.001	6.651	10.202	12.009	13.129	14.360	14.784	14.412	13.474
1.700	-0.001	6.411	10.030	11.882	13.034	14.309	14.759	14.399	13.481
1.800	-0.001	6.178	9.859	11.755	12.937	14.256	14.731	14.384	13.486
1.900	-0.001	5.955	9.688	11.627	12.840	14.203	14.701	14.367	13.491
2.000	-0.001	5.742	9.516	11.499	12.743	14.150	14.668	14.348	13.495
2.100	-0.001	5.538	9.344	11.371	12.645	14.098	14.634	14.327	13.498
2.200	-0.001	5.346	9.174	11.246	12.549	14.045	14.598	14.304	13.501
2.300	-0.001	5.164	9.006	11.121	12.453	13.993	14.561	14.281	13.502
2.400	-0.001	4.993	8.841	10.997	12.357	13.941	14.522	14.255	13.503
2.500	-0.001	4.833	8.678	10.874	12.263	13.889	14.483	14.228	13.503
2.600	-0.001	4.684	8.519	10.751	12.169	13.838	14.442	14.200	13.501
2.700	-0.001	4.544	8.362	10.629	12.076	13.786	14.400	14.171	13.500
2.800	-0.001	4.413	8.209	10.507	11.985	13.733	14.356	14.140	13.497
2.900	-0.001	4.290	8.059	10.386	11.894	13.680	14.312	14.108	13.493
3.000	-0.001	4.175	7.911	10.267	11.804	13.627	14.266	14.075	13.489
3.100	-0.001	4.066	7.766	10.150	11.715	13.573	14.219	14.041	13.483
3.200	-0.001	3.964	7.623	10.035	11.626	13.517	14.171	14.007	13.476
3.300	-0.001	3.868	7.483	9.922	11.537	13.462	14.122	13.971	13.466
3.400	-0.001	3.777	7.346	9.811	11.449	13.405	14.072	13.935	13.455
3.500	-0.001	3.691	7.212	9.702	11.360	13.348	14.020	13.898	13.441
3.600	-0.001	3.610	7.081	9.594	11.272	13.290	13.968	13.860	13.426
3.700	-0.001	3.533	6.953	9.487	11.185	13.230	13.914	13.821	13.410
3.800	-0.001	3.460	6.828	9.382	11.099	13.170	13.860	13.781	13.392
3.900	-0.001	3.391	6.707	9.279	11.015	13.110	13.805	13.741	13.374
4.000	-0.001	3.325	6.590	9.178	10.932	13.048	13.748	13.700	13.354
4.100	-0.001	3.263	6.476	9.077	10.851	12.984	13.691	13.658	13.333
4.200	-0.001	3.203	6.365	8.978	10.771	12.920	13.633	13.615	13.311
4.300	-0.001	3.146	6.259	8.880	10.692	12.855	13.574	13.573	13.289
4.400	-0.001	3.092	6.156	8.783	10.615	12.789	13.514	13.530	13.265
4.500	-0.001	3.040	6.057	8.688	10.538	12.722	13.454	13.487	13.240
4.600	-0.001	2.991	5.962	8.594	10.463	12.654	13.392	13.443	13.215
4.700	-0.001	2.944	5.871	8.501	10.390	12.585	13.330	13.399	13.190
4.800	-0.001	2.899	5.783	8.410	10.317	12.515	13.266	13.356	13.165
4.900	-0.001	2.855	5.698	8.319	10.244	12.444	13.202	13.311	13.138
5.000	-0.001	2.813	5.617	8.230	10.173	12.371	13.137	13.267	13.112
5.100	-0.001	2.772	5.539	8.141	10.101	12.298	13.071	13.222	13.085
5.200	-0.001	2.733	5.463	8.054	10.030	12.223	13.005	13.178	13.058
5.300	-0.001	2.696	5.391	7.969	9.959	12.148	12.938	13.133	13.030
5.400	-0.001	2.660	5.322	7.885	9.888	12.072	12.872	13.088	13.002

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.626	5.255	7.802	9.817	11.995	12.805	13.043	12.973
5.600	-0.001	2.593	5.191	7.721	9.745	11.917	12.739	12.999	12.943
5.700	-0.001	2.561	5.129	7.641	9.674	11.838	12.671	12.954	12.913
5.800	-0.001	2.531	5.069	7.563	9.602	11.759	12.604	12.910	12.882
5.900	-0.001	2.501	5.012	7.486	9.530	11.680	12.536	12.866	12.852
6.000	-0.001	2.473	4.956	7.412	9.457	11.599	12.467	12.821	12.820
6.100	-0.001	2.445	4.902	7.339	9.384	11.518	12.398	12.775	12.789
6.200	-0.001	2.419	4.850	7.268	9.311	11.437	12.328	12.729	12.757
6.300	-0.000	2.394	4.800	7.198	9.237	11.356	12.258	12.683	12.724
6.400	-0.000	2.370	4.752	7.131	9.163	11.274	12.187	12.636	12.691
6.500	-0.000	2.347	4.706	7.066	9.088	11.191	12.116	12.588	12.658
6.600	-0.000	2.325	4.661	7.002	9.012	11.108	12.045	12.539	12.624
6.700	-0.000	2.303	4.618	6.940	8.936	11.025	11.973	12.490	12.589
6.800	-0.000	2.283	4.576	6.879	8.859	10.942	11.902	12.439	12.553
6.900	-0.000	2.263	4.536	6.820	8.782	10.858	11.830	12.388	12.516
7.000	-0.000	2.243	4.497	6.761	8.704	10.774	11.759	12.335	12.477
7.100	-0.000	2.225	4.459	6.704	8.625	10.689	11.688	12.282	12.436
7.200	-0.000	2.207	4.423	6.647	8.545	10.605	11.616	12.228	12.394
7.300	-0.000	2.189	4.389	6.591	8.465	10.519	11.544	12.174	12.349
7.400	-0.000	2.172	4.355	6.536	8.384	10.434	11.473	12.119	12.303
7.500	-0.000	2.156	4.322	6.481	8.303	10.348	11.401	12.065	12.254
7.600	-0.000	2.141	4.291	6.427	8.221	10.261	11.329	12.010	12.203
7.700	-0.000	2.125	4.260	6.373	8.138	10.175	11.257	11.955	12.150
7.800	-0.000	2.111	4.231	6.320	8.055	10.087	11.185	11.899	12.096
7.900	-0.000	2.097	4.202	6.266	7.971	10.000	11.113	11.843	12.040
8.000	-0.000	2.083	4.175	6.214	7.886	9.911	11.041	11.787	11.984
8.100	-0.000	2.069	4.148	6.161	7.801	9.823	10.969	11.730	11.926
8.200	-0.000	2.056	4.122	6.108	7.716	9.734	10.897	11.674	11.867
8.300	-0.000	2.044	4.097	6.056	7.631	9.644	10.824	11.616	11.807
8.400	-0.000	2.032	4.073	6.003	7.546	9.555	10.752	11.559	11.746
8.500	-0.000	2.020	4.049	5.950	7.462	9.465	10.679	11.501	11.685
8.600	-0.000	2.009	4.027	5.897	7.377	9.374	10.606	11.443	11.624
8.700	-0.000	1.998	4.004	5.843	7.292	9.283	10.533	11.383	11.562
8.800	-0.000	1.988	3.981	5.789	7.208	9.192	10.460	11.321	11.499
8.900	-0.000	1.977	3.959	5.735	7.123	9.100	10.386	11.259	11.436
9.000	-0.000	1.967	3.936	5.679	7.039	9.007	10.312	11.195	11.372
9.100	-0.000	1.958	3.913	5.623	6.955	8.914	10.238	11.129	11.308
9.200	-0.000	1.948	3.890	5.567	6.871	8.821	10.163	11.062	11.242
9.300	-0.000	1.939	3.866	5.509	6.788	8.727	10.088	10.993	11.177
9.400	-0.000	1.930	3.842	5.451	6.704	8.633	10.012	10.922	11.111
9.500	-0.000	1.922	3.816	5.392	6.621	8.538	9.935	10.850	11.044
9.600	-0.000	1.914	3.791	5.332	6.537	8.444	9.858	10.775	10.977
9.700	-0.000	1.906	3.764	5.272	6.454	8.349	9.780	10.700	10.909
9.800	-0.000	1.898	3.736	5.210	6.371	8.254	9.701	10.622	10.840
9.900	-0.000	1.891	3.708	5.148	6.287	8.158	9.622	10.542	10.771
10.000	-0.000	1.883	3.678	5.085	6.204	8.062	9.543	10.462	10.701
10.100	-0.000	1.876	3.647	5.021	6.121	7.966	9.463	10.381	10.631
10.200	-0.000	1.870	3.615	4.957	6.038	7.870	9.382	10.299	10.560
10.300	-0.000	1.863	3.581	4.891	5.954	7.773	9.300	10.216	10.490
10.400	-0.000	1.857	3.546	4.825	5.871	7.677	9.218	10.133	10.419
10.500	-0.000	1.850	3.509	4.758	5.788	7.581	9.136	10.049	10.347
10.600	-0.000	1.843	3.470	4.690	5.704	7.485	9.052	9.964	10.275
10.700	-0.000	1.835	3.429	4.622	5.621	7.389	8.967	9.879	10.202
10.800	-0.000	1.827	3.387	4.552	5.537	7.294	8.881	9.792	10.129
10.900	-0.000	1.817	3.342	4.482	5.453	7.199	8.794	9.705	10.056
11.000	-0.000	1.806	3.295	4.411	5.369	7.105	8.705	9.617	9.982
11.100	-0.000	1.794	3.245	4.339	5.285	7.011	8.614	9.528	9.908
11.200	-0.000	1.779	3.194	4.266	5.200	6.917	8.521	9.439	9.834
11.300	-0.000	1.763	3.141	4.193	5.116	6.825	8.426	9.348	9.759
11.400	-0.000	1.745	3.086	4.118	5.031	6.732	8.330	9.257	9.683

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.725	3.029	4.043	4.947	6.641	8.232	9.166	9.607
11.600	-0.000	1.701	2.970	3.966	4.862	6.550	8.133	9.073	9.531
11.700	-0.000	1.675	2.909	3.890	4.777	6.459	8.032	8.980	9.455
11.800	-0.000	1.647	2.847	3.812	4.692	6.370	7.930	8.886	9.377
11.900	-0.000	1.615	2.783	3.733	4.607	6.281	7.827	8.792	9.300
12.000	-0.000	1.580	2.717	3.655	4.522	6.194	7.725	8.697	9.222
12.100	-0.000	1.543	2.651	3.577	4.438	6.107	7.624	8.602	9.144
12.200	-0.000	1.506	2.586	3.500	4.356	6.023	7.523	8.508	9.067
12.300	-0.000	1.467	2.522	3.424	4.276	5.940	7.424	8.414	8.990
12.400	-0.000	1.429	2.458	3.351	4.198	5.860	7.327	8.322	8.913
12.500	-0.000	1.390	2.396	3.278	4.121	5.782	7.232	8.231	8.839
12.600	-0.000	1.350	2.334	3.208	4.047	5.705	7.139	8.141	8.766
12.700	-0.000	1.310	2.273	3.139	3.974	5.631	7.049	8.054	8.695
12.800	-0.000	1.269	2.213	3.071	3.903	5.559	6.960	7.969	8.627
12.900	-0.000	1.228	2.154	3.005	3.834	5.489	6.875	7.887	8.562
13.000	-0.000	1.188	2.097	2.941	3.767	5.422	6.792	7.808	8.499
13.100	-0.000	1.147	2.040	2.879	3.703	5.355	6.712	7.733	8.440
13.200	-0.000	1.106	1.984	2.818	3.640	5.289	6.635	7.662	8.384
13.300	-0.000	1.065	1.930	2.760	3.581	5.225	6.561	7.595	8.332
13.400	-0.000	1.025	1.878	2.704	3.524	5.161	6.491	7.532	8.284
13.500	-0.000	0.985	1.828	2.650	3.469	5.099	6.426	7.474	8.238
13.600	-0.000	0.947	1.780	2.600	3.419	5.040	6.366	7.420	8.196
13.700	-0.000	0.910	1.735	2.554	3.372	4.983	6.310	7.371	8.158
13.800	-0.000	0.875	1.694	2.511	3.329	4.931	6.260	7.326	8.124
13.900	-0.000	0.843	1.655	2.472	3.291	4.884	6.214	7.285	8.094
14.000	-0.000	0.814	1.621	2.437	3.256	4.842	6.174	7.249	8.069
14.100	-0.000	0.789	1.591	2.407	3.227	4.807	6.140	7.218	8.047
14.200	-0.000	0.769	1.566	2.381	3.203	4.777	6.111	7.192	8.029
14.300	-0.000	0.753	1.546	2.361	3.184	4.755	6.089	7.173	8.014
14.400	-0.000	0.742	1.531	2.347	3.170	4.740	6.073	7.159	8.004
14.500	-0.000	0.736	1.521	2.337	3.161	4.731	6.064	7.151	7.998
14.600	-0.000	0.733	1.516	2.332	3.156	4.727	6.061	7.147	7.995
14.700	-0.000	0.732	1.512	2.328	3.153	4.726	6.059	7.145	7.993
14.800	-0.000	0.730	1.508	2.324	3.150	4.724	6.057	7.143	7.992
14.900	-0.000	0.729	1.504	2.320	3.146	4.723	6.055	7.140	7.990
15.000	-0.000	0.728	1.500	2.316	3.143	4.721	6.054	7.138	7.988
15.100	-0.000	0.727	1.497	2.312	3.140	4.720	6.052	7.135	7.987
15.200	-0.000	0.726	1.494	2.309	3.136	4.718	6.050	7.133	7.985
15.300	-0.000	0.725	1.491	2.305	3.133	4.717	6.048	7.130	7.983
15.400	-0.000	0.725	1.488	2.301	3.130	4.715	6.046	7.128	7.982
15.500	-0.000	0.724	1.485	2.298	3.127	4.714	6.045	7.126	7.980
15.600	-0.000	0.724	1.483	2.294	3.124	4.712	6.043	7.123	7.978
15.700	-0.000	0.723	1.481	2.291	3.121	4.711	6.041	7.121	7.976
15.800	-0.000	0.723	1.479	2.288	3.118	4.710	6.039	7.118	7.975
15.900	-0.000	0.723	1.477	2.284	3.115	4.708	6.037	7.116	7.973
16.000	-0.000	0.724	1.476	2.281	3.112	4.707	6.036	7.114	7.971
16.100	-0.000	0.724	1.475	2.278	3.109	4.706	6.034	7.111	7.969
16.200	-0.000	0.725	1.474	2.275	3.107	4.705	6.032	7.109	7.968
16.300	-0.000	0.726	1.473	2.273	3.104	4.703	6.030	7.107	7.966
16.400	-0.000	0.727	1.473	2.270	3.102	4.702	6.028	7.104	7.964
16.500	-0.000	0.728	1.473	2.267	3.099	4.701	6.027	7.102	7.962
16.600	-0.000	0.729	1.472	2.265	3.097	4.700	6.025	7.100	7.960
16.700	-0.000	0.730	1.472	2.263	3.094	4.699	6.023	7.098	7.959
16.800	-0.000	0.731	1.472	2.261	3.092	4.697	6.021	7.095	7.957
16.900	-0.000	0.732	1.473	2.259	3.090	4.696	6.020	7.093	7.955
17.000	-0.000	0.733	1.473	2.257	3.088	4.695	6.018	7.091	7.953
17.100	-0.000	0.734	1.473	2.255	3.086	4.694	6.016	7.089	7.951
17.200	-0.000	0.735	1.474	2.254	3.084	4.693	6.014	7.087	7.950
17.300	-0.000	0.736	1.475	2.252	3.082	4.692	6.013	7.084	7.948
17.400	-0.000	0.737	1.476	2.251	3.080	4.691	6.011	7.082	7.946

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.738	1.477	2.251	3.078	4.690	6.009	7.080	7.944
17.600	-0.000	0.739	1.478	2.250	3.076	4.689	6.007	7.078	7.944
17.700	-0.000	0.740	1.480	2.249	3.075	4.688	6.006	7.076	7.942
17.800	-0.000	0.741	1.481	2.249	3.074	4.687	6.004	7.074	7.940
17.900	-0.000	0.743	1.483	2.249	3.072	4.686	6.002	7.072	7.939
18.000	-0.000	0.744	1.485	2.250	3.071	4.686	6.000	7.069	7.937
18.100	-0.000	0.745	1.487	2.250	3.070	4.685	5.999	7.067	7.935
18.200	-0.000	0.746	1.489	2.251	3.069	4.684	5.997	7.065	7.934
18.300	-0.000	0.747	1.491	2.252	3.068	4.683	5.995	7.063	7.932
18.400	-0.000	0.748	1.494	2.253	3.067	4.682	5.993	7.061	7.930
18.500	-0.000	0.750	1.496	2.255	3.067	4.681	5.992	7.058	7.929
18.600	-0.000	0.751	1.499	2.256	3.066	4.681	5.990	7.056	7.927
18.700	-0.000	0.752	1.501	2.258	3.066	4.680	5.988	7.054	7.926
18.800	-0.000	0.753	1.503	2.260	3.066	4.679	5.986	7.052	7.924
18.900	-0.000	0.755	1.506	2.262	3.066	4.678	5.985	7.050	7.922
19.000	-0.000	0.756	1.508	2.264	3.066	4.678	5.983	7.048	7.921
19.100	-0.000	0.757	1.511	2.266	3.066	4.677	5.981	7.046	7.919
19.200	-0.000	0.758	1.514	2.269	3.067	4.676	5.980	7.045	7.918
19.300	-0.000	0.760	1.516	2.271	3.067	4.676	5.978	7.043	7.916
19.400	-0.000	0.761	1.519	2.274	3.068	4.675	5.976	7.041	7.914
19.500	-0.000	0.762	1.521	2.278	3.069	4.675	5.974	7.039	7.913
19.600	-0.000	0.764	1.524	2.281	3.071	4.674	5.972	7.037	7.911
19.700	-0.000	0.765	1.527	2.284	3.072	4.673	5.970	7.035	7.910
19.800	-0.000	0.766	1.529	2.287	3.074	4.673	5.969	7.033	7.908
19.900	-0.000	0.768	1.532	2.291	3.076	4.672	5.967	7.031	7.907
20.000	-0.000	0.769	1.534	2.295	3.078	4.672	5.966	7.030	7.905
20.100	-0.000	0.770	1.538	2.298	3.081	4.671	5.964	7.028	7.904
20.200	-0.000	0.772	1.540	2.302	3.083	4.672	5.962	7.026	7.902
20.300	-0.000	0.773	1.543	2.307	3.086	4.671	5.960	7.024	7.900
20.400	-0.000	0.775	1.546	2.310	3.089	4.670	5.958	7.022	7.899
20.500	-0.000	0.776	1.549	2.315	3.092	4.670	5.957	7.020	7.897
20.600	-0.000	0.777	1.552	2.319	3.095	4.670	5.955	7.019	7.896
20.700	-0.000	0.779	1.554	2.324	3.098	4.669	5.953	7.017	7.894
20.800	-0.000	0.780	1.557	2.328	3.103	4.669	5.951	7.015	7.893
20.900	-0.000	0.782	1.560	2.332	3.107	4.668	5.949	7.014	7.891
21.000	-0.000	0.783	1.563	2.337	3.111	4.668	5.947	7.012	7.890
21.100	-0.000	0.785	1.566	2.341	3.115	4.667	5.946	7.010	7.888
21.200	-0.000	0.786	1.569	2.346	3.120	4.667	5.944	7.009	7.887
21.300	-0.000	0.788	1.572	2.350	3.125	4.667	5.942	7.007	7.885
21.400	-0.000	0.789	1.575	2.355	3.130	4.666	5.940	7.005	7.884
21.500	-0.000	0.791	1.578	2.359	3.135	4.666	5.938	7.004	7.883
21.600	-0.000	0.792	1.581	2.364	3.140	4.666	5.936	7.002	7.881
21.700	-0.000	0.794	1.584	2.369	3.145	4.666	5.935	7.001	7.880
21.800	-0.000	0.795	1.588	2.372	3.151	4.666	5.933	6.999	7.878
21.900	-0.000	0.797	1.591	2.378	3.157	4.666	5.933	6.998	7.877
22.000	-0.000	0.799	1.594	2.383	3.162	4.666	5.931	6.997	7.876
22.100	-0.000	0.800	1.597	2.387	3.168	4.665	5.929	6.996	7.874
22.200	-0.000	0.802	1.600	2.391	3.174	4.665	5.928	6.994	7.873
22.300	-0.000	0.803	1.603	2.397	3.180	4.665	5.926	6.993	7.872
22.400	-0.000	0.805	1.607	2.402	3.186	4.665	5.924	6.991	7.870
22.500	-0.000	0.807	1.610	2.407	3.192	4.665	5.923	6.990	7.869
22.600	-0.000	0.808	1.613	2.412	3.198	4.665	5.921	6.988	7.868
22.700	-0.000	0.810	1.617	2.417	3.204	4.666	5.917	6.987	7.866
22.800	-0.000	0.812	1.620	2.421	3.208	4.666	5.918	6.985	7.865
22.900	-0.000	0.813	1.623	2.427	3.216	4.666	5.917	6.984	7.864
23.000	-0.000	0.815	1.625	2.432	3.221	4.666	5.915	6.983	7.863
23.100	-0.000	0.817	1.630	2.437	3.227	4.666	5.914	6.981	7.862
23.200	-0.000	0.818	1.633	2.442	3.232	4.667	5.913	6.980	7.860
23.300	-0.000	0.820	1.637	2.447	3.238	4.667	5.911	6.979	7.859
23.400	-0.000	0.822	1.640	2.452	3.243	4.668	5.910	6.977	7.858

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.824	1.644	2.457	3.248	4.668	5.909	6.976	7.857
23.600	-0.000	0.825	1.647	2.462	3.254	4.666	5.907	6.975	7.856
23.700	-0.000	0.827	1.651	2.468	3.259	4.670	5.906	6.974	7.855
23.800	-0.000	0.829	1.654	2.473	3.264	4.671	5.905	6.973	7.854
23.900	-0.000	0.831	1.658	2.478	3.269	4.672	5.904	6.972	7.854
24.000	-0.000	0.832	1.662	2.483	3.273	4.673	5.903	6.971	7.853
24.100	-0.000	0.834	1.665	2.488	3.278	4.675	5.902	6.970	7.852
24.200	-0.000	0.836	1.669	2.493	3.283	4.677	5.901	6.969	7.851
24.300	-0.000	0.838	1.672	2.498	3.287	4.679	5.900	6.968	7.851
24.400	-0.000	0.840	1.676	2.503	3.291	4.680	5.900	6.967	7.850
24.500	-0.000	0.842	1.680	2.507	3.296	4.682	5.899	6.966	7.850
24.600	-0.000	0.843	1.683	2.512	3.299	4.683	5.898	6.966	7.849
24.700	-0.000	0.845	1.687	2.516	3.303	4.685	5.898	6.965	7.849
24.800	-0.000	0.847	1.691	2.520	3.307	4.686	5.898	6.964	7.848
24.900	-0.000	0.849	1.695	2.525	3.310	4.687	5.898	6.964	7.848
25.000	-0.000	0.851	1.699	2.529	3.313	4.689	5.898	6.963	7.847
25.100	-0.000	0.853	1.702	2.532	3.315	4.690	5.899	6.965	7.847
25.200	-0.000	0.855	1.706	2.536	3.318	4.691	5.899	6.964	7.847
25.300	-0.000	0.857	1.710	2.540	3.320	4.692	5.901	6.964	7.847
25.400	-0.000	0.859	1.714	2.543	3.321	4.693	5.902	6.964	7.847
25.500	-0.000	0.861	1.718	2.546	3.323	4.693	5.903	6.964	7.847
25.600	-0.000	0.863	1.722	2.549	3.324	4.694	5.905	6.964	7.847
25.700	-0.000	0.865	1.725	2.552	3.324	4.695	5.907	6.964	7.847
25.800	-0.000	0.867	1.729	2.555	3.324	4.696	5.910	6.965	7.848
25.900	-0.000	0.869	1.732	2.557	3.324	4.696	5.912	6.966	7.848
26.000	-0.000	0.871	1.736	2.559	3.323	4.696	5.914	6.966	7.849

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 3.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	8.525	10.697	11.829	12.531	13.270	13.541	13.671	13.526
0.100	-0.002	8.560	10.829	12.012	12.750	13.534	13.820	13.870	13.516
0.200	-0.002	8.560	10.919	12.147	12.917	13.742	14.038	14.011	13.502
0.300	-0.001	8.532	10.975	12.245	13.044	13.903	14.208	14.113	13.489
0.400	-0.001	8.481	11.004	12.312	13.137	14.028	14.338	14.186	13.478
0.500	-0.001	8.411	11.012	12.358	13.204	14.123	14.439	14.239	13.469
0.600	-0.001	8.321	11.000	12.385	13.252	14.196	14.516	14.278	13.461
0.700	-0.001	8.213	10.973	12.396	13.285	14.252	14.576	14.306	13.454
0.800	-0.001	8.086	10.929	12.392	13.304	14.295	14.624	14.327	13.448
0.900	-0.001	7.940	10.868	12.372	13.308	14.325	14.663	14.343	13.444
1.000	-0.001	7.771	10.787	12.333	13.295	14.341	14.690	14.354	13.444
1.100	-0.001	7.582	10.684	12.273	13.262	14.339	14.704	14.360	13.446
1.200	-0.001	7.376	10.563	12.195	13.212	14.324	14.708	14.361	13.449
1.300	-0.001	7.160	10.430	12.105	13.150	14.300	14.703	14.358	13.454
1.400	-0.001	6.940	10.288	12.007	13.080	14.268	14.693	14.352	13.459
1.500	-0.001	6.718	10.141	11.903	13.005	14.233	14.678	14.344	13.464
1.600	-0.001	6.496	9.990	11.795	12.926	14.194	14.659	14.333	13.470
1.700	-0.001	6.278	9.836	11.684	12.844	14.152	14.637	14.321	13.474
1.800	-0.001	6.065	9.680	11.570	12.760	14.109	14.612	14.306	13.479
1.900	-0.001	5.858	9.524	11.455	12.675	14.065	14.585	14.290	13.483
2.000	-0.001	5.659	9.368	11.340	12.588	14.020	14.556	14.272	13.487
2.100	-0.001	5.469	9.210	11.224	12.500	13.974	14.525	14.252	13.490
2.200	-0.001	5.286	9.052	11.107	12.412	13.928	14.492	14.231	13.492
2.300	-0.001	5.113	8.895	10.991	12.324	13.882	14.457	14.208	13.494
2.400	-0.001	4.949	8.738	10.876	12.236	13.835	14.421	14.184	13.495
2.500	-0.001	4.794	8.583	10.760	12.148	13.787	14.383	14.158	13.496
2.600	-0.001	4.648	8.430	10.646	12.060	13.739	14.344	14.131	13.496
2.700	-0.001	4.512	8.279	10.531	11.973	13.690	14.304	14.103	13.495
2.800	-0.001	4.384	8.131	10.416	11.886	13.640	14.263	14.074	13.494
2.900	-0.001	4.263	7.985	10.301	11.799	13.589	14.220	14.043	13.491
3.000	-0.001	4.150	7.842	10.187	11.714	13.538	14.176	14.012	13.487
3.100	-0.001	4.043	7.701	10.074	11.629	13.485	14.131	13.980	13.482
3.200	-0.001	3.943	7.563	9.963	11.545	13.432	14.085	13.947	13.475
3.300	-0.001	3.848	7.426	9.853	11.460	13.377	14.038	13.913	13.466
3.400	-0.001	3.758	7.292	9.744	11.375	13.322	13.989	13.879	13.455
3.500	-0.001	3.673	7.161	9.637	11.291	13.265	13.939	13.844	13.442
3.600	-0.001	3.592	7.033	9.531	11.207	13.208	13.889	13.808	13.427
3.700	-0.001	3.516	6.909	9.427	11.123	13.150	13.837	13.772	13.411
3.800	-0.001	3.444	6.787	9.324	11.039	13.090	13.785	13.735	13.393
3.900	-0.001	3.375	6.669	9.223	10.956	13.030	13.731	13.697	13.374
4.000	-0.001	3.310	6.554	9.122	10.875	12.969	13.677	13.658	13.354
4.100	-0.001	3.248	6.443	9.024	10.795	12.908	13.621	13.619	13.333
4.200	-0.001	3.189	6.335	8.927	10.716	12.845	13.566	13.579	13.311
4.300	-0.001	3.133	6.231	8.831	10.639	12.781	13.509	13.539	13.288
4.400	-0.001	3.079	6.130	8.736	10.563	12.717	13.451	13.498	13.265
4.500	-0.001	3.028	6.033	8.643	10.488	12.651	13.393	13.458	13.241
4.600	-0.001	2.980	5.939	8.550	10.414	12.585	13.335	13.416	13.217
4.700	-0.001	2.933	5.849	8.459	10.342	12.518	13.275	13.375	13.192
4.800	-0.001	2.888	5.762	8.369	10.269	12.449	13.214	13.333	13.166
4.900	-0.001	2.845	5.678	8.280	10.198	12.380	13.153	13.291	13.139
5.000	-0.001	2.804	5.598	8.192	10.127	12.309	13.090	13.249	13.113
5.100	-0.001	2.765	5.520	8.105	10.056	12.238	13.027	13.206	13.086
5.200	-0.001	2.726	5.446	8.020	9.985	12.165	12.963	13.164	13.058
5.300	-0.001	2.689	5.374	7.935	9.914	12.092	12.898	13.121	13.031
5.400	-0.001	2.653	5.305	7.853	9.843	12.017	12.834	13.078	13.002

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.619	5.239	7.771	9.772	11.942	12.769	13.035	12.974
5.600	-0.001	2.586	5.175	7.691	9.700	11.866	12.704	12.992	12.946
5.700	-0.001	2.555	5.114	7.613	9.629	11.790	12.639	12.948	12.917
5.800	-0.001	2.524	5.055	7.536	9.557	11.713	12.573	12.905	12.886
5.900	-0.001	2.495	4.998	7.461	9.485	11.635	12.508	12.861	12.855
6.000	-0.001	2.467	4.943	7.387	9.413	11.556	12.441	12.818	12.824
6.100	-0.001	2.440	4.890	7.315	9.340	11.477	12.374	12.773	12.792
6.200	-0.001	2.414	4.838	7.245	9.267	11.397	12.306	12.729	12.760
6.300	-0.000	2.388	4.789	7.177	9.193	11.317	12.238	12.683	12.728
6.400	-0.000	2.364	4.741	7.110	9.119	11.237	12.169	12.637	12.695
6.500	-0.000	2.341	4.695	7.045	9.044	11.156	12.100	12.591	12.662
6.600	-0.000	2.319	4.650	6.982	8.969	11.075	12.031	12.544	12.628
6.700	-0.000	2.297	4.607	6.920	8.894	10.994	11.961	12.496	12.594
6.800	-0.000	2.277	4.566	6.860	8.817	10.912	11.891	12.447	12.559
6.900	-0.000	2.257	4.525	6.800	8.740	10.830	11.822	12.397	12.524
7.000	-0.000	2.238	4.487	6.741	8.663	10.747	11.752	12.347	12.487
7.100	-0.000	2.219	4.449	6.683	8.585	10.664	11.682	12.295	12.448
7.200	-0.000	2.201	4.413	6.626	8.506	10.580	11.612	12.242	12.408
7.300	-0.000	2.184	4.378	6.570	8.427	10.496	11.542	12.189	12.367
7.400	-0.000	2.167	4.344	6.514	8.348	10.412	11.472	12.135	12.323
7.500	-0.000	2.151	4.312	6.459	8.268	10.328	11.401	12.081	12.276
7.600	-0.000	2.135	4.281	6.404	8.188	10.243	11.331	12.026	12.228
7.700	-0.000	2.120	4.250	6.350	8.107	10.158	11.261	11.972	12.177
7.800	-0.000	2.106	4.221	6.296	8.025	10.072	11.191	11.917	12.125
7.900	-0.000	2.092	4.193	6.243	7.944	9.986	11.120	11.862	12.072
8.000	-0.000	2.078	4.165	6.189	7.861	9.900	11.050	11.806	12.017
8.100	-0.000	2.065	4.138	6.136	7.778	9.813	10.980	11.750	11.961
8.200	-0.000	2.052	4.113	6.083	7.695	9.725	10.909	11.694	11.904
8.300	-0.000	2.039	4.088	6.030	7.612	9.638	10.838	11.637	11.846
8.400	-0.000	2.027	4.063	5.977	7.528	9.550	10.767	11.580	11.786
8.500	-0.000	2.016	4.040	5.924	7.445	9.461	10.697	11.523	11.725
8.600	-0.000	2.004	4.016	5.871	7.361	9.372	10.626	11.466	11.664
8.700	-0.000	1.994	3.993	5.818	7.278	9.283	10.554	11.407	11.603
8.800	-0.000	1.983	3.970	5.764	7.195	9.194	10.483	11.348	11.542
8.900	-0.000	1.973	3.946	5.710	7.111	9.104	10.411	11.288	11.480
9.000	-0.000	1.963	3.923	5.656	7.028	9.013	10.339	11.226	11.417
9.100	-0.000	1.953	3.899	5.601	6.946	8.923	10.266	11.162	11.354
9.200	-0.000	1.944	3.875	5.546	6.863	8.831	10.193	11.098	11.290
9.300	-0.000	1.935	3.850	5.490	6.780	8.739	10.119	11.031	11.225
9.400	-0.000	1.926	3.825	5.433	6.698	8.647	10.045	10.963	11.159
9.500	-0.000	1.918	3.799	5.375	6.616	8.555	9.970	10.894	11.093
9.600	-0.000	1.910	3.773	5.317	6.534	8.462	9.895	10.823	11.027
9.700	-0.000	1.902	3.745	5.258	6.452	8.369	9.818	10.750	10.960
9.800	-0.000	1.894	3.717	5.198	6.371	8.276	9.741	10.675	10.892
9.900	-0.000	1.887	3.688	5.137	6.289	8.182	9.664	10.599	10.824
10.000	-0.000	1.879	3.658	5.075	6.207	8.089	9.585	10.521	10.755
10.100	-0.000	1.872	3.626	5.012	6.126	7.995	9.507	10.441	10.685
10.200	-0.000	1.866	3.594	4.949	6.044	7.901	9.427	10.360	10.616
10.300	-0.000	1.859	3.560	4.885	5.962	7.807	9.347	10.279	10.546
10.400	-0.000	1.852	3.525	4.820	5.881	7.712	9.267	10.197	10.476
10.500	-0.000	1.844	3.489	4.754	5.799	7.618	9.185	10.115	10.406
10.600	-0.000	1.836	3.450	4.688	5.717	7.525	9.103	10.031	10.335
10.700	-0.000	1.827	3.411	4.621	5.636	7.431	9.021	9.947	10.264
10.800	-0.000	1.817	3.369	4.554	5.554	7.338	8.937	9.863	10.193
10.900	-0.000	1.806	3.325	4.485	5.472	7.245	8.853	9.777	10.121
11.000	-0.000	1.794	3.280	4.416	5.390	7.153	8.767	9.690	10.048
11.100	-0.000	1.780	3.233	4.346	5.308	7.061	8.679	9.603	9.976
11.200	-0.000	1.764	3.183	4.275	5.226	6.970	8.589	9.515	9.902
11.300	-0.000	1.746	3.132	4.203	5.143	6.879	8.498	9.426	9.828
11.400	-0.000	1.727	3.078	4.131	5.061	6.789	8.406	9.337	9.754

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.705	3.023	4.057	4.978	6.699	8.312	9.246	9.679
11.600	-0.000	1.680	2.966	3.984	4.895	6.610	8.216	9.155	9.604
11.700	-0.000	1.653	2.907	3.909	4.813	6.521	8.118	9.063	9.529
11.800	-0.000	1.624	2.847	3.834	4.730	6.434	8.019	8.971	9.453
11.900	-0.000	1.593	2.787	3.759	4.648	6.347	7.920	8.878	9.377
12.000	-0.000	1.561	2.726	3.685	4.567	6.263	7.820	8.785	9.300
12.100	-0.000	1.528	2.666	3.612	4.488	6.179	7.721	8.693	9.224
12.200	-0.000	1.495	2.607	3.541	4.411	6.098	7.623	8.602	9.149
12.300	-0.000	1.461	2.548	3.470	4.335	6.019	7.527	8.511	9.074
12.400	-0.000	1.427	2.490	3.402	4.261	5.941	7.432	8.421	9.000
12.500	-0.000	1.393	2.433	3.334	4.188	5.865	7.340	8.332	8.927
12.600	-0.000	1.359	2.376	3.267	4.117	5.791	7.249	8.245	8.854
12.700	-0.000	1.324	2.320	3.202	4.048	5.719	7.161	8.160	8.784
12.800	-0.000	1.289	2.264	3.138	3.980	5.649	7.074	8.077	8.716
12.900	-0.000	1.253	2.210	3.075	3.913	5.580	6.990	7.995	8.650
13.000	-0.000	1.216	2.155	3.014	3.848	5.514	6.908	7.917	8.587
13.100	-0.000	1.180	2.102	2.954	3.785	5.449	6.829	7.841	8.527
13.200	-0.000	1.144	2.049	2.895	3.724	5.387	6.752	7.769	8.469
13.300	-0.000	1.107	1.997	2.838	3.665	5.325	6.679	7.700	8.414
13.400	-0.000	1.071	1.947	2.783	3.608	5.265	6.608	7.634	8.363
13.500	-0.000	1.034	1.897	2.729	3.553	5.205	6.540	7.573	8.315
13.600	-0.000	0.998	1.850	2.678	3.501	5.147	6.476	7.515	8.271
13.700	-0.000	0.963	1.804	2.629	3.452	5.090	6.416	7.461	8.229
13.800	-0.000	0.929	1.760	2.583	3.405	5.035	6.360	7.411	8.190
13.900	-0.000	0.897	1.719	2.540	3.362	4.983	6.308	7.366	8.155
14.000	-0.000	0.866	1.681	2.500	3.322	4.935	6.262	7.324	8.124
14.100	-0.000	0.838	1.646	2.464	3.286	4.891	6.220	7.286	8.095
14.200	-0.000	0.813	1.614	2.432	3.254	4.852	6.182	7.252	8.071
14.300	-0.000	0.790	1.586	2.403	3.226	4.818	6.149	7.223	8.050
14.400	-0.000	0.772	1.562	2.379	3.202	4.790	6.122	7.198	8.033
14.500	-0.000	0.757	1.542	2.359	3.183	4.767	6.099	7.178	8.019
14.600	-0.000	0.746	1.527	2.344	3.169	4.751	6.083	7.164	8.008
14.700	-0.000	0.739	1.517	2.333	3.159	4.740	6.072	7.154	8.001
14.800	-0.000	0.735	1.510	2.326	3.152	4.734	6.066	7.148	7.996
14.900	-0.000	0.734	1.506	2.322	3.149	4.732	6.064	7.146	7.995
15.000	-0.000	0.733	1.503	2.318	3.146	4.731	6.062	7.143	7.993
15.100	-0.000	0.732	1.500	2.314	3.142	4.729	6.061	7.141	7.991
15.200	-0.000	0.731	1.497	2.311	3.139	4.728	6.059	7.138	7.989
15.300	-0.000	0.730	1.494	2.307	3.136	4.727	6.057	7.136	7.988
15.400	-0.000	0.729	1.492	2.304	3.133	4.725	6.055	7.133	7.986
15.500	-0.000	0.729	1.489	2.300	3.130	4.724	6.053	7.131	7.984
15.600	-0.000	0.728	1.487	2.297	3.127	4.722	6.052	7.129	7.982
15.700	-0.000	0.727	1.486	2.294	3.124	4.721	6.050	7.126	7.981
15.800	-0.000	0.727	1.484	2.290	3.121	4.720	6.048	7.124	7.979
15.900	-0.000	0.727	1.483	2.287	3.119	4.719	6.046	7.122	7.977
16.000	-0.000	0.727	1.482	2.284	3.116	4.717	6.044	7.119	7.975
16.100	-0.000	0.727	1.481	2.282	3.113	4.716	6.043	7.117	7.973
16.200	-0.000	0.727	1.480	2.279	3.111	4.715	6.041	7.115	7.971
16.300	-0.000	0.728	1.480	2.276	3.108	4.714	6.039	7.112	7.970
16.400	-0.000	0.729	1.479	2.274	3.106	4.712	6.037	7.110	7.968
16.500	-0.000	0.729	1.479	2.272	3.103	4.711	6.035	7.108	7.966
16.600	-0.000	0.730	1.479	2.269	3.101	4.710	6.034	7.105	7.964
16.700	-0.000	0.731	1.479	2.267	3.099	4.709	6.032	7.103	7.962
16.800	-0.000	0.732	1.479	2.265	3.097	4.708	6.030	7.101	7.960
16.900	-0.000	0.733	1.479	2.264	3.094	4.707	6.028	7.099	7.959
17.000	-0.000	0.734	1.479	2.262	3.092	4.706	6.027	7.097	7.957
17.100	-0.000	0.735	1.480	2.261	3.090	4.705	6.025	7.094	7.955
17.200	-0.000	0.737	1.480	2.260	3.089	4.704	6.023	7.092	7.953
17.300	-0.000	0.738	1.481	2.259	3.087	4.703	6.021	7.090	7.953
17.400	-0.000	0.739	1.482	2.258	3.085	4.702	6.019	7.088	7.951

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.740	1.483	2.257	3.084	4.701	6.018	7.086	7.949
17.600	-0.000	0.741	1.484	2.257	3.082	4.700	6.016	7.084	7.947
17.700	-0.000	0.742	1.485	2.257	3.081	4.699	6.014	7.081	7.946
17.800	-0.000	0.743	1.486	2.257	3.080	4.698	6.012	7.079	7.944
17.900	-0.000	0.744	1.488	2.257	3.078	4.697	6.011	7.077	7.942
18.000	-0.000	0.746	1.490	2.258	3.077	4.696	6.009	7.075	7.941
18.100	-0.000	0.747	1.491	2.259	3.077	4.696	6.007	7.072	7.939
18.200	-0.000	0.748	1.493	2.260	3.076	4.695	6.005	7.070	7.937
18.300	-0.000	0.749	1.496	2.261	3.075	4.694	6.004	7.068	7.936
18.400	-0.000	0.750	1.498	2.262	3.075	4.693	6.002	7.066	7.934
18.500	-0.000	0.752	1.500	2.264	3.074	4.692	6.000	7.064	7.932
18.600	-0.000	0.753	1.503	2.265	3.074	4.692	5.998	7.062	7.931
18.700	-0.000	0.754	1.505	2.267	3.074	4.691	5.997	7.060	7.929
18.800	-0.000	0.755	1.508	2.269	3.074	4.690	5.995	7.057	7.928
18.900	-0.000	0.757	1.510	2.271	3.074	4.689	5.993	7.057	7.926
19.000	-0.000	0.758	1.513	2.273	3.074	4.689	5.991	7.055	7.924
19.100	-0.000	0.759	1.516	2.275	3.075	4.688	5.989	7.053	7.923
19.200	-0.000	0.760	1.518	2.278	3.076	4.687	5.988	7.050	7.921
19.300	-0.000	0.762	1.520	2.281	3.077	4.687	5.986	7.048	7.919
19.400	-0.000	0.763	1.523	2.283	3.078	4.686	5.984	7.046	7.918
19.500	-0.000	0.764	1.526	2.286	3.079	4.686	5.982	7.045	7.916
19.600	-0.000	0.766	1.528	2.289	3.081	4.685	5.980	7.043	7.915
19.700	-0.000	0.767	1.531	2.292	3.083	4.684	5.979	7.041	7.913
19.800	-0.000	0.769	1.534	2.295	3.085	4.684	5.977	7.039	7.911
19.900	-0.000	0.770	1.537	2.299	3.087	4.684	5.976	7.037	7.910
20.000	-0.000	0.771	1.539	2.302	3.089	4.684	5.974	7.035	7.908
20.100	-0.000	0.773	1.542	2.306	3.092	4.683	5.972	7.033	7.907
20.200	-0.000	0.774	1.545	2.310	3.095	4.683	5.970	7.031	7.905
20.300	-0.000	0.775	1.548	2.314	3.097	4.682	5.968	7.030	7.904
20.400	-0.000	0.777	1.551	2.318	3.099	4.682	5.966	7.028	7.902
20.500	-0.000	0.778	1.554	2.322	3.103	4.681	5.964	7.026	7.900
20.600	-0.000	0.780	1.556	2.327	3.107	4.681	5.962	7.024	7.899
20.700	-0.000	0.781	1.559	2.331	3.111	4.680	5.961	7.023	7.897
20.800	-0.000	0.783	1.562	2.335	3.115	4.680	5.959	7.021	7.896
20.900	-0.000	0.784	1.565	2.340	3.119	4.679	5.957	7.019	7.894
21.000	-0.000	0.786	1.568	2.344	3.123	4.679	5.955	7.018	7.893
21.100	-0.000	0.787	1.571	2.349	3.128	4.679	5.953	7.016	7.891
21.200	-0.000	0.789	1.574	2.353	3.131	4.678	5.951	7.014	7.890
21.300	-0.000	0.790	1.577	2.358	3.137	4.678	5.949	7.013	7.888
21.400	-0.000	0.792	1.580	2.362	3.142	4.678	5.948	7.011	7.887
21.500	-0.000	0.793	1.583	2.367	3.147	4.678	5.946	7.010	7.885
21.600	-0.000	0.795	1.587	2.372	3.152	4.677	5.945	7.008	7.884
21.700	-0.000	0.796	1.590	2.376	3.157	4.677	5.944	7.007	7.882
21.800	-0.000	0.798	1.593	2.381	3.163	4.677	5.942	7.006	7.881
21.900	-0.000	0.800	1.596	2.386	3.168	4.677	5.940	7.003	7.880
22.000	-0.000	0.801	1.599	2.391	3.174	4.677	5.939	7.003	7.878
22.100	-0.000	0.803	1.602	2.395	3.179	4.674	5.937	7.001	7.877
22.200	-0.000	0.804	1.606	2.400	3.185	4.677	5.935	7.000	7.875
22.300	-0.000	0.806	1.609	2.405	3.191	4.677	5.934	6.998	7.874
22.400	-0.000	0.808	1.612	2.410	3.196	4.677	5.932	6.997	7.873
22.500	-0.000	0.809	1.616	2.415	3.202	4.677	5.931	6.995	7.871
22.600	-0.000	0.811	1.618	2.420	3.208	4.677	5.929	6.994	7.870
22.700	-0.000	0.813	1.622	2.425	3.214	4.677	5.925	6.992	7.869
22.800	-0.000	0.814	1.626	2.430	3.219	4.677	5.926	6.991	7.868
22.900	-0.000	0.815	1.629	2.435	3.225	4.677	5.925	6.989	7.866
23.000	-0.000	0.818	1.632	2.440	3.231	4.677	5.923	6.988	7.865
23.100	-0.000	0.820	1.636	2.445	3.236	4.678	5.922	6.987	7.864
23.200	-0.000	0.821	1.639	2.450	3.241	4.678	5.920	6.985	7.863
23.300	-0.000	0.823	1.643	2.454	3.245	4.678	5.919	6.984	7.862
23.400	-0.000	0.825	1.646	2.461	3.252	4.679	5.918	6.983	7.861

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.827	1.650	2.466	3.257	4.679	5.916	6.982	7.860
23.600	-0.000	0.828	1.653	2.471	3.262	4.680	5.915	6.981	7.859
23.700	-0.000	0.830	1.657	2.476	3.267	4.681	5.914	6.979	7.858
23.800	-0.000	0.832	1.660	2.482	3.272	4.679	5.913	6.978	7.857
23.900	-0.000	0.834	1.664	2.487	3.276	4.683	5.912	6.977	7.857
24.000	-0.000	0.836	1.668	2.492	3.281	4.684	5.911	6.976	7.856
24.100	-0.000	0.837	1.671	2.496	3.285	4.686	5.910	6.975	7.855
24.200	-0.000	0.839	1.675	2.501	3.290	4.687	5.909	6.974	7.854
24.300	-0.000	0.841	1.679	2.506	3.294	4.689	5.908	6.974	7.854
24.400	-0.000	0.843	1.682	2.510	3.298	4.690	5.907	6.973	7.853
24.500	-0.000	0.845	1.686	2.515	3.302	4.691	5.907	6.972	7.853
24.600	-0.000	0.847	1.690	2.519	3.306	4.693	5.906	6.971	7.852
24.700	-0.000	0.849	1.694	2.523	3.309	4.694	5.906	6.971	7.852
24.800	-0.000	0.850	1.697	2.527	3.313	4.695	5.906	6.972	7.851
24.900	-0.000	0.852	1.701	2.531	3.316	4.696	5.906	6.971	7.851
25.000	-0.000	0.854	1.705	2.535	3.319	4.697	5.906	6.971	7.851
25.100	-0.000	0.856	1.709	2.538	3.321	4.698	5.906	6.970	7.851
25.200	-0.000	0.858	1.713	2.542	3.323	4.699	5.907	6.970	7.851
25.300	-0.000	0.860	1.717	2.545	3.325	4.700	5.909	6.970	7.851
25.400	-0.000	0.862	1.721	2.548	3.327	4.700	5.910	6.970	7.851
25.500	-0.000	0.864	1.724	2.551	3.328	4.701	5.912	6.970	7.851
25.600	-0.000	0.866	1.728	2.553	3.329	4.701	5.914	6.970	7.851
25.700	-0.000	0.868	1.731	2.556	3.330	4.702	5.916	6.971	7.851
25.800	-0.000	0.870	1.735	2.558	3.330	4.702	5.918	6.971	7.852
25.900	-0.000	0.872	1.738	2.560	3.329	4.703	5.920	6.972	7.852
26.000	-0.000	0.874	1.741	2.562	3.328	4.703	5.922	6.973	7.853

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 4.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	8.127	10.376	11.557	12.299	13.110	13.446	13.631	13.534
0.100	-0.002	8.143	10.479	11.704	12.478	13.330	13.681	13.806	13.532
0.200	-0.002	8.132	10.549	11.815	12.619	13.508	13.872	13.934	13.526
0.300	-0.001	8.099	10.592	11.898	12.728	13.650	14.025	14.027	13.518
0.400	-0.001	8.049	10.614	11.956	12.811	13.764	14.148	14.097	13.510
0.500	-0.001	7.982	10.618	11.997	12.872	13.853	14.246	14.149	13.501
0.600	-0.001	7.900	10.606	12.021	12.917	13.924	14.324	14.187	13.493
0.700	-0.001	7.802	10.581	12.032	12.949	13.980	14.387	14.215	13.486
0.800	-0.001	7.691	10.543	12.031	12.970	14.024	14.439	14.236	13.479
0.900	-0.001	7.565	10.493	12.019	12.981	14.060	14.480	14.252	13.472
1.000	-0.001	7.424	10.431	11.995	12.981	14.085	14.513	14.263	13.467
1.100	-0.001	7.268	10.353	11.958	12.968	14.101	14.537	14.271	13.464
1.200	-0.001	7.096	10.260	11.904	12.941	14.104	14.550	14.275	13.463
1.300	-0.001	6.913	10.152	11.837	12.900	14.096	14.555	14.274	13.464
1.400	-0.001	6.722	10.033	11.759	12.848	14.079	14.551	14.270	13.466
1.500	-0.001	6.526	9.906	11.673	12.789	14.056	14.542	14.263	13.468
1.600	-0.001	6.330	9.774	11.580	12.723	14.028	14.528	14.253	13.471
1.700	-0.001	6.134	9.637	11.484	12.654	13.996	14.511	14.241	13.475
1.800	-0.001	5.942	9.497	11.384	12.582	13.962	14.490	14.227	13.478
1.900	-0.001	5.753	9.355	11.281	12.507	13.926	14.466	14.212	13.481
2.000	-0.001	5.569	9.211	11.177	12.431	13.888	14.441	14.195	13.484
2.100	-0.001	5.392	9.067	11.072	12.353	13.849	14.413	14.177	13.487
2.200	-0.001	5.221	8.922	10.965	12.273	13.810	14.383	14.157	13.489
2.300	-0.001	5.057	8.775	10.858	12.193	13.769	14.351	14.135	13.491
2.400	-0.001	4.901	8.628	10.750	12.111	13.726	14.317	14.112	13.493
2.500	-0.001	4.752	8.481	10.642	12.030	13.683	14.282	14.088	13.494
2.600	-0.001	4.610	8.336	10.534	11.948	13.638	14.245	14.062	13.494
2.700	-0.001	4.477	8.191	10.426	11.865	13.591	14.207	14.035	13.494
2.800	-0.001	4.352	8.048	10.318	11.783	13.544	14.168	14.007	13.493
2.900	-0.001	4.234	7.907	10.210	11.701	13.495	14.127	13.979	13.491
3.000	-0.001	4.124	7.769	10.102	11.620	13.446	14.086	13.949	13.487
3.100	-0.001	4.019	7.632	9.994	11.539	13.395	14.042	13.919	13.481
3.200	-0.001	3.920	7.497	9.886	11.458	13.342	13.998	13.888	13.474
3.300	-0.001	3.826	7.365	9.779	11.377	13.289	13.952	13.856	13.465
3.400	-0.001	3.737	7.235	9.673	11.297	13.235	13.905	13.823	13.454
3.500	-0.001	3.654	7.108	9.569	11.216	13.180	13.858	13.790	13.441
3.600	-0.001	3.574	6.984	9.466	11.136	13.124	13.809	13.757	13.427
3.700	-0.001	3.499	6.862	9.364	11.055	13.066	13.759	13.723	13.411
3.800	-0.001	3.427	6.744	9.263	10.975	13.008	13.708	13.688	13.393
3.900	-0.001	3.359	6.629	9.164	10.895	12.949	13.657	13.653	13.374
4.000	-0.001	3.295	6.517	9.065	10.816	12.890	13.604	13.617	13.354
4.100	-0.001	3.233	6.408	8.969	10.738	12.829	13.551	13.580	13.333
4.200	-0.001	3.175	6.303	8.874	10.661	12.768	13.498	13.543	13.311
4.300	-0.001	3.119	6.201	8.780	10.585	12.706	13.443	13.506	13.288
4.400	-0.001	3.066	6.102	8.687	10.511	12.643	13.388	13.467	13.266
4.500	-0.001	3.016	6.006	8.595	10.437	12.579	13.332	13.429	13.242
4.600	-0.001	2.968	5.914	8.505	10.364	12.514	13.276	13.390	13.218
4.700	-0.001	2.921	5.825	8.415	10.292	12.448	13.218	13.350	13.192
4.800	-0.001	2.877	5.740	8.326	10.221	12.381	13.160	13.310	13.167
4.900	-0.001	2.835	5.657	8.239	10.150	12.313	13.101	13.271	13.141
5.000	-0.001	2.794	5.578	8.153	10.079	12.245	13.042	13.230	13.114
5.100	-0.001	2.755	5.501	8.067	10.008	12.175	12.981	13.190	13.087
5.200	-0.001	2.718	5.428	7.984	9.937	12.105	12.919	13.150	13.059
5.300	-0.001	2.682	5.357	7.901	9.867	12.033	12.857	13.109	13.031
5.400	-0.001	2.646	5.289	7.820	9.796	11.961	12.795	13.068	13.004

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.613	5.223	7.740	9.725	11.888	12.732	13.027	12.976
5.600	-0.001	2.580	5.159	7.661	9.654	11.814	12.669	12.985	12.948
5.700	-0.001	2.549	5.098	7.584	9.583	11.739	12.605	12.943	12.919
5.800	-0.001	2.518	5.040	7.509	9.511	11.663	12.541	12.900	12.889
5.900	-0.001	2.489	4.983	7.435	9.439	11.587	12.477	12.857	12.858
6.000	-0.001	2.461	4.929	7.362	9.367	11.511	12.413	12.814	12.827
6.100	-0.001	2.434	4.876	7.291	9.294	11.433	12.348	12.771	12.795
6.200	-0.001	2.408	4.826	7.222	9.221	11.355	12.283	12.727	12.763
6.300	-0.000	2.383	4.777	7.155	9.148	11.277	12.217	12.683	12.731
6.400	-0.000	2.359	4.729	7.089	9.074	11.198	12.150	12.638	12.698
6.500	-0.000	2.335	4.683	7.024	9.000	11.119	12.083	12.593	12.665
6.600	-0.000	2.313	4.639	6.961	8.925	11.040	12.015	12.547	12.631
6.700	-0.000	2.292	4.596	6.899	8.850	10.960	11.947	12.500	12.597
6.800	-0.000	2.271	4.555	6.839	8.774	10.879	11.879	12.453	12.564
6.900	-0.000	2.251	4.515	6.779	8.698	10.799	11.811	12.405	12.529
7.000	-0.000	2.232	4.476	6.720	8.621	10.717	11.743	12.355	12.494
7.100	-0.000	2.214	4.439	6.662	8.544	10.636	11.674	12.305	12.458
7.200	-0.000	2.196	4.403	6.605	8.466	10.554	11.606	12.254	12.419
7.300	-0.000	2.179	4.368	6.548	8.388	10.471	11.538	12.202	12.379
7.400	-0.000	2.162	4.334	6.492	8.310	10.388	11.469	12.149	12.338
7.500	-0.000	2.146	4.302	6.437	8.231	10.305	11.401	12.096	12.294
7.600	-0.000	2.130	4.270	6.381	8.152	10.222	11.332	12.042	12.248
7.700	-0.000	2.115	4.240	6.327	8.072	10.138	11.263	11.988	12.200
7.800	-0.000	2.101	4.211	6.273	7.993	10.054	11.195	11.933	12.151
7.900	-0.000	2.087	4.183	6.219	7.913	9.969	11.126	11.878	12.099
8.000	-0.000	2.073	4.156	6.165	7.832	9.885	11.057	11.824	12.046
8.100	-0.000	2.060	4.129	6.111	7.751	9.800	10.989	11.768	11.992
8.200	-0.000	2.047	4.103	6.058	7.670	9.714	10.920	11.713	11.936
8.300	-0.000	2.035	4.078	6.004	7.589	9.628	10.851	11.657	11.880
8.400	-0.000	2.023	4.054	5.951	7.507	9.542	10.782	11.600	11.822
8.500	-0.000	2.011	4.030	5.898	7.425	9.455	10.713	11.543	11.763
8.600	-0.000	2.000	4.006	5.845	7.343	9.368	10.644	11.486	11.703
8.700	-0.000	1.989	3.982	5.791	7.261	9.281	10.574	11.429	11.642
8.800	-0.000	1.979	3.958	5.738	7.178	9.193	10.504	11.371	11.581
8.900	-0.000	1.969	3.933	5.684	7.096	9.105	10.434	11.312	11.520
9.000	-0.000	1.959	3.909	5.631	7.015	9.016	10.364	11.253	11.459
9.100	-0.000	1.949	3.884	5.577	6.933	8.928	10.292	11.191	11.396
9.200	-0.000	1.940	3.859	5.522	6.851	8.838	10.221	11.129	11.333
9.300	-0.000	1.931	3.834	5.467	6.770	8.749	10.148	11.065	11.269
9.400	-0.000	1.922	3.808	5.411	6.689	8.659	10.076	11.000	11.205
9.500	-0.000	1.914	3.781	5.355	6.608	8.568	10.002	10.933	11.139
9.600	-0.000	1.905	3.754	5.298	6.528	8.477	9.928	10.864	11.074
9.700	-0.000	1.898	3.726	5.240	6.447	8.386	9.854	10.794	11.007
9.800	-0.000	1.890	3.697	5.181	6.367	8.295	9.778	10.722	10.940
9.900	-0.000	1.883	3.667	5.121	6.287	8.203	9.702	10.649	10.872
10.000	-0.000	1.875	3.637	5.061	6.207	8.112	9.625	10.574	10.805
10.100	-0.000	1.868	3.605	5.000	6.127	8.020	9.548	10.497	10.737
10.200	-0.000	1.861	3.572	4.938	6.047	7.928	9.470	10.418	10.668
10.300	-0.000	1.854	3.538	4.875	5.967	7.837	9.391	10.338	10.599
10.400	-0.000	1.846	3.503	4.811	5.887	7.745	9.312	10.257	10.531
10.500	-0.000	1.837	3.466	4.747	5.807	7.653	9.232	10.176	10.462
10.600	-0.000	1.828	3.428	4.682	5.727	7.561	9.151	10.094	10.392
10.700	-0.000	1.817	3.389	4.616	5.647	7.470	9.070	10.011	10.322
10.800	-0.000	1.806	3.347	4.550	5.567	7.378	8.988	9.928	10.252
10.900	-0.000	1.794	3.305	4.483	5.487	7.288	8.905	9.844	10.181
11.000	-0.000	1.780	3.260	4.416	5.407	7.198	8.822	9.759	10.110
11.100	-0.000	1.764	3.214	4.348	5.327	7.108	8.737	9.673	10.038
11.200	-0.000	1.747	3.167	4.279	5.246	7.018	8.651	9.586	9.966
11.300	-0.000	1.727	3.117	4.209	5.166	6.930	8.563	9.499	9.893
11.400	-0.000	1.706	3.066	4.138	5.086	6.841	8.473	9.411	9.821

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.683	3.012	4.067	5.005	6.754	8.382	9.322	9.747
11.600	-0.000	1.657	2.957	3.996	4.925	6.667	8.290	9.233	9.674
11.700	-0.000	1.630	2.902	3.925	4.845	6.581	8.196	9.143	9.601
11.800	-0.000	1.602	2.846	3.854	4.767	6.497	8.102	9.052	9.527
11.900	-0.000	1.573	2.790	3.783	4.689	6.413	8.007	8.962	9.453
12.000	-0.000	1.543	2.734	3.715	4.613	6.332	7.912	8.872	9.379
12.100	-0.000	1.514	2.679	3.647	4.538	6.252	7.817	8.782	9.305
12.200	-0.000	1.484	2.625	3.580	4.465	6.174	7.722	8.694	9.232
12.300	-0.000	1.454	2.572	3.515	4.393	6.097	7.628	8.605	9.159
12.400	-0.000	1.424	2.519	3.450	4.323	6.022	7.536	8.518	9.087
12.500	-0.000	1.394	2.466	3.386	4.254	5.948	7.446	8.432	9.015
12.600	-0.000	1.363	2.414	3.324	4.186	5.876	7.357	8.347	8.945
12.700	-0.000	1.333	2.362	3.262	4.119	5.805	7.270	8.264	8.875
12.800	-0.000	1.302	2.310	3.201	4.053	5.737	7.185	8.182	8.807
12.900	-0.000	1.271	2.259	3.141	3.989	5.669	7.103	8.102	8.741
13.000	-0.000	1.239	2.208	3.082	3.926	5.604	7.022	8.024	8.678
13.100	-0.000	1.206	2.158	3.024	3.865	5.540	6.944	7.949	8.616
13.200	-0.000	1.174	2.108	2.967	3.805	5.479	6.868	7.875	8.556
13.300	-0.000	1.141	2.059	2.912	3.747	5.419	6.794	7.805	8.500
13.400	-0.000	1.108	2.010	2.858	3.690	5.361	6.723	7.738	8.446
13.500	-0.000	1.076	1.962	2.805	3.636	5.304	6.654	7.674	8.395
13.600	-0.000	1.043	1.916	2.754	3.583	5.247	6.588	7.613	8.348
13.700	-0.000	1.011	1.871	2.705	3.532	5.192	6.525	7.556	8.303
13.800	-0.000	0.979	1.827	2.657	3.484	5.138	6.466	7.503	8.262
13.900	-0.000	0.947	1.784	2.612	3.438	5.085	6.410	7.452	8.223
14.000	-0.000	0.917	1.744	2.569	3.394	5.034	6.358	7.406	8.186
14.100	-0.000	0.888	1.706	2.529	3.354	4.985	6.309	7.363	8.153
14.200	-0.000	0.861	1.671	2.492	3.317	4.940	6.265	7.324	8.124
14.300	-0.000	0.836	1.638	2.458	3.283	4.899	6.226	7.288	8.097
14.400	-0.000	0.813	1.608	2.427	3.253	4.862	6.190	7.256	8.074
14.500	-0.000	0.793	1.582	2.400	3.226	4.830	6.159	7.228	8.054
14.600	-0.000	0.775	1.559	2.376	3.203	4.802	6.132	7.204	8.037
14.700	-0.000	0.761	1.540	2.357	3.184	4.779	6.110	7.184	8.023
14.800	-0.000	0.750	1.525	2.341	3.169	4.762	6.093	7.168	8.011
14.900	-0.000	0.743	1.514	2.330	3.158	4.750	6.081	7.157	8.003
15.000	-0.000	0.739	1.507	2.321	3.150	4.743	6.073	7.150	7.998
15.100	-0.000	0.737	1.503	2.316	3.145	4.739	6.069	7.146	7.995
15.200	-0.000	0.736	1.500	2.313	3.142	4.738	6.067	7.144	7.994
15.300	-0.000	0.735	1.498	2.309	3.139	4.736	6.066	7.141	7.992
15.400	-0.000	0.734	1.496	2.306	3.136	4.735	6.064	7.139	7.990
15.500	-0.000	0.733	1.494	2.303	3.133	4.734	6.062	7.136	7.988
15.600	-0.000	0.733	1.492	2.300	3.131	4.732	6.060	7.134	7.986
15.700	-0.000	0.732	1.491	2.297	3.128	4.731	6.058	7.132	7.984
15.800	-0.000	0.731	1.490	2.294	3.125	4.730	6.057	7.129	7.982
15.900	-0.000	0.731	1.489	2.291	3.122	4.729	6.055	7.127	7.981
16.000	-0.000	0.731	1.488	2.288	3.120	4.727	6.053	7.125	7.979
16.100	-0.000	0.731	1.487	2.285	3.117	4.726	6.051	7.122	7.977
16.200	-0.000	0.731	1.487	2.283	3.115	4.725	6.049	7.120	7.975
16.300	-0.000	0.731	1.486	2.280	3.112	4.724	6.048	7.118	7.973
16.400	-0.000	0.731	1.486	2.278	3.110	4.723	6.046	7.115	7.971
16.500	-0.000	0.732	1.486	2.276	3.108	4.722	6.044	7.113	7.969
16.600	-0.000	0.732	1.485	2.274	3.106	4.721	6.042	7.111	7.968
16.700	-0.000	0.733	1.485	2.272	3.103	4.719	6.040	7.109	7.966
16.800	-0.000	0.734	1.486	2.271	3.101	4.718	6.039	7.106	7.964
16.900	-0.000	0.735	1.486	2.269	3.099	4.717	6.037	7.104	7.962
17.000	-0.000	0.736	1.486	2.268	3.098	4.716	6.035	7.102	7.962
17.100	-0.000	0.737	1.486	2.267	3.096	4.715	6.033	7.100	7.960
17.200	-0.000	0.738	1.487	2.266	3.094	4.714	6.031	7.098	7.958
17.300	-0.000	0.740	1.488	2.265	3.092	4.713	6.030	7.096	7.956
17.400	-0.000	0.741	1.488	2.265	3.091	4.712	6.028	7.093	7.954

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.742	1.489	2.265	3.090	4.711	6.026	7.091	7.953
17.600	-0.000	0.743	1.490	2.265	3.088	4.711	6.024	7.089	7.951
17.700	-0.000	0.744	1.491	2.265	3.087	4.710	6.023	7.087	7.949
17.800	-0.000	0.745	1.492	2.265	3.086	4.709	6.021	7.084	7.948
17.900	-0.000	0.746	1.494	2.266	3.085	4.708	6.019	7.082	7.946
18.000	-0.000	0.748	1.495	2.267	3.084	4.707	6.017	7.080	7.944
18.100	-0.000	0.749	1.497	2.267	3.084	4.706	6.015	7.078	7.943
18.200	-0.000	0.750	1.498	2.269	3.083	4.705	6.014	7.076	7.941
18.300	-0.000	0.751	1.500	2.270	3.082	4.705	6.012	7.073	7.939
18.400	-0.000	0.752	1.502	2.271	3.082	4.704	6.010	7.071	7.938
18.500	-0.000	0.754	1.505	2.273	3.082	4.703	6.008	7.069	7.936
18.600	-0.000	0.755	1.507	2.274	3.082	4.702	6.007	7.068	7.934
18.700	-0.000	0.756	1.509	2.276	3.082	4.702	6.005	7.066	7.933
18.800	-0.000	0.757	1.512	2.278	3.082	4.701	6.003	7.064	7.931
18.900	-0.000	0.759	1.515	2.280	3.083	4.700	6.001	7.062	7.929
19.000	-0.000	0.760	1.517	2.283	3.084	4.700	5.999	7.060	7.928
19.100	-0.000	0.761	1.520	2.285	3.084	4.699	5.997	7.058	7.926
19.200	-0.000	0.763	1.522	2.287	3.085	4.698	5.995	7.056	7.924
19.300	-0.000	0.764	1.525	2.290	3.087	4.698	5.993	7.054	7.923
19.400	-0.000	0.765	1.528	2.292	3.089	4.697	5.992	7.052	7.921
19.500	-0.000	0.767	1.530	2.295	3.090	4.697	5.991	7.050	7.919
19.600	-0.000	0.768	1.533	2.298	3.092	4.696	5.989	7.048	7.918
19.700	-0.000	0.769	1.536	2.301	3.094	4.696	5.987	7.046	7.916
19.800	-0.000	0.771	1.539	2.304	3.096	4.696	5.985	7.044	7.914
19.900	-0.000	0.772	1.541	2.307	3.098	4.695	5.983	7.043	7.913
20.000	-0.000	0.774	1.544	2.311	3.102	4.694	5.981	7.041	7.911
20.100	-0.000	0.775	1.547	2.313	3.103	4.694	5.979	7.039	7.910
20.200	-0.000	0.776	1.550	2.317	3.106	4.693	5.977	7.037	7.908
20.300	-0.000	0.778	1.553	2.321	3.110	4.693	5.976	7.035	7.906
20.400	-0.000	0.779	1.556	2.326	3.113	4.692	5.974	7.034	7.905
20.500	-0.000	0.781	1.558	2.330	3.115	4.692	5.972	7.032	7.903
20.600	-0.000	0.782	1.561	2.334	3.120	4.691	5.970	7.030	7.902
20.700	-0.000	0.784	1.564	2.338	3.124	4.691	5.968	7.028	7.900
20.800	-0.000	0.785	1.567	2.342	3.126	4.691	5.966	7.027	7.898
20.900	-0.000	0.787	1.570	2.347	3.132	4.690	5.964	7.025	7.897
21.000	-0.000	0.788	1.573	2.352	3.136	4.692	5.962	7.023	7.895
21.100	-0.000	0.790	1.576	2.356	3.141	4.690	5.960	7.022	7.894
21.200	-0.000	0.791	1.579	2.361	3.145	4.690	5.959	7.020	7.892
21.300	-0.000	0.793	1.583	2.366	3.150	4.689	5.960	7.019	7.891
21.400	-0.000	0.794	1.586	2.369	3.154	4.689	5.956	7.018	7.889
21.500	-0.000	0.795	1.589	2.375	3.159	4.689	5.955	7.016	7.888
21.600	-0.000	0.797	1.592	2.380	3.163	4.689	5.953	7.015	7.886
21.700	-0.000	0.799	1.595	2.384	3.169	4.688	5.951	7.013	7.885
21.800	-0.000	0.801	1.598	2.389	3.175	4.688	5.949	7.011	7.884
21.900	-0.000	0.802	1.601	2.394	3.180	4.685	5.948	7.010	7.882
22.000	-0.000	0.804	1.604	2.399	3.185	4.688	5.946	7.008	7.881
22.100	-0.000	0.806	1.608	2.404	3.190	4.688	5.944	7.007	7.879
22.200	-0.000	0.807	1.611	2.409	3.196	4.688	5.943	7.005	7.878
22.300	-0.000	0.809	1.615	2.413	3.201	4.688	5.941	7.004	7.877
22.400	-0.000	0.811	1.618	2.418	3.207	4.688	5.940	7.002	7.875
22.500	-0.000	0.812	1.621	2.423	3.212	4.688	5.938	7.001	7.874
22.600	-0.000	0.814	1.623	2.428	3.218	4.688	5.934	6.999	7.873
22.700	-0.000	0.816	1.628	2.434	3.223	4.688	5.932	6.998	7.872
22.800	-0.000	0.817	1.631	2.439	3.229	4.688	5.933	6.996	7.870
22.900	-0.000	0.819	1.634	2.444	3.234	4.688	5.932	6.995	7.869
23.000	-0.000	0.821	1.638	2.449	3.240	4.688	5.931	6.994	7.868
23.100	-0.000	0.822	1.642	2.454	3.245	4.689	5.929	6.992	7.867
23.200	-0.000	0.824	1.645	2.459	3.250	4.689	5.928	6.991	7.866
23.300	-0.000	0.826	1.649	2.465	3.255	4.689	5.926	6.990	7.865
23.400	-0.000	0.828	1.652	2.470	3.260	4.690	5.925	6.988	7.864

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.830	1.656	2.475	3.265	4.691	5.924	6.987	7.863
23.600	-0.000	0.831	1.659	2.480	3.270	4.691	5.923	6.986	7.862
23.700	-0.000	0.833	1.663	2.485	3.275	4.692	5.921	6.985	7.861
23.800	-0.000	0.835	1.667	2.490	3.279	4.693	5.920	6.984	7.860
23.900	-0.000	0.837	1.670	2.495	3.284	4.694	5.919	6.983	7.860
24.000	-0.000	0.839	1.674	2.500	3.288	4.695	5.918	6.982	7.859
24.100	-0.000	0.840	1.678	2.504	3.292	4.696	5.917	6.981	7.858
24.200	-0.000	0.842	1.681	2.509	3.296	4.697	5.916	6.980	7.858
24.300	-0.000	0.844	1.685	2.513	3.300	4.698	5.916	6.979	7.857
24.400	-0.000	0.846	1.689	2.518	3.304	4.700	5.915	6.978	7.857
24.500	-0.000	0.848	1.693	2.522	3.308	4.701	5.914	6.980	7.856
24.600	-0.000	0.850	1.696	2.526	3.311	4.702	5.914	6.979	7.856
24.700	-0.000	0.852	1.700	2.530	3.315	4.703	5.913	6.978	7.855
24.800	-0.000	0.854	1.704	2.533	3.318	4.704	5.913	6.977	7.855
24.900	-0.000	0.856	1.708	2.537	3.321	4.705	5.913	6.977	7.855
25.000	-0.000	0.858	1.712	2.540	3.323	4.705	5.914	6.976	7.855
25.100	-0.000	0.860	1.716	2.544	3.326	4.706	5.914	6.976	7.855
25.200	-0.000	0.862	1.720	2.547	3.328	4.707	5.915	6.976	7.855
25.300	-0.000	0.864	1.723	2.549	3.330	4.707	5.917	6.976	7.855
25.400	-0.000	0.866	1.727	2.552	3.331	4.708	5.918	6.976	7.855
25.500	-0.000	0.868	1.730	2.555	3.332	4.708	5.920	6.976	7.855
25.600	-0.000	0.870	1.734	2.557	3.333	4.708	5.922	6.976	7.855
25.700	-0.000	0.872	1.737	2.559	3.334	4.708	5.924	6.977	7.856
25.800	-0.000	0.874	1.740	2.561	3.334	4.709	5.926	6.977	7.856
25.900	-0.000	0.876	1.743	2.562	3.333	4.709	5.928	6.978	7.857
26.000	-0.000	0.878	1.745	2.563	3.333	4.709	5.931	6.979	7.858

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 4.50 M

UNIT: m

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INITIAL	HEELING ANGLE (DEGREES)								
DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	7.765	10.084	11.308	12.088	12.965	13.362	13.581	13.541
0.100	-0.002	7.767	10.164	11.428	12.238	13.152	13.564	13.736	13.545
0.200	-0.002	7.748	10.218	11.521	12.358	13.307	13.731	13.853	13.543
0.300	-0.001	7.713	10.251	11.591	12.452	13.433	13.870	13.940	13.539
0.400	-0.001	7.662	10.266	11.643	12.525	13.536	13.984	14.006	13.533
0.500	-0.001	7.597	10.266	11.678	12.581	13.620	14.078	14.056	13.526
0.600	-0.001	7.519	10.254	11.699	12.623	13.687	14.155	14.094	13.519
0.700	-0.001	7.429	10.229	11.709	12.654	13.741	14.217	14.122	13.512
0.800	-0.001	7.328	10.195	11.709	12.675	13.786	14.268	14.143	13.504
0.900	-0.001	7.216	10.151	11.700	12.688	13.822	14.309	14.158	13.497
1.000	-0.001	7.093	10.097	11.682	12.692	13.852	14.342	14.169	13.490
1.100	-0.001	6.960	10.034	11.656	12.689	13.874	14.369	14.178	13.485
1.200	-0.001	6.817	9.961	11.620	12.677	13.890	14.389	14.183	13.481
1.300	-0.001	6.662	9.875	11.572	12.653	13.896	14.402	14.185	13.478
1.400	-0.001	6.499	9.778	11.513	12.619	13.893	14.406	14.184	13.477
1.500	-0.001	6.329	9.671	11.444	12.575	13.882	14.404	14.179	13.477
1.600	-0.001	6.157	9.557	11.367	12.524	13.865	14.396	14.171	13.478
1.700	-0.001	5.983	9.436	11.284	12.467	13.843	14.383	14.161	13.480
1.800	-0.001	5.811	9.311	11.197	12.405	13.817	14.367	14.148	13.482
1.900	-0.001	5.640	9.182	11.106	12.341	13.788	14.347	14.134	13.484
2.000	-0.001	5.472	9.051	11.012	12.273	13.757	14.325	14.118	13.486
2.100	-0.001	5.308	8.918	10.916	12.203	13.724	14.300	14.100	13.489
2.200	-0.001	5.149	8.783	10.819	12.131	13.689	14.273	14.081	13.491
2.300	-0.001	4.995	8.647	10.720	12.058	13.653	14.243	14.061	13.492
2.400	-0.001	4.847	8.510	10.620	11.983	13.615	14.212	14.039	13.494
2.500	-0.001	4.705	8.373	10.519	11.908	13.575	14.180	14.016	13.495
2.600	-0.001	4.570	8.235	10.417	11.832	13.533	14.145	13.992	13.496
2.700	-0.001	4.441	8.098	10.316	11.755	13.489	14.109	13.967	13.496
2.800	-0.001	4.319	7.962	10.215	11.678	13.445	14.072	13.941	13.494
2.900	-0.001	4.204	7.826	10.113	11.601	13.398	14.034	13.915	13.491
3.000	-0.001	4.095	7.691	10.010	11.523	13.350	13.994	13.887	13.487
3.100	-0.001	3.992	7.559	9.907	11.445	13.301	13.952	13.858	13.480
3.200	-0.001	3.895	7.429	9.805	11.368	13.250	13.910	13.829	13.472
3.300	-0.001	3.803	7.301	9.702	11.291	13.199	13.866	13.799	13.463
3.400	-0.001	3.716	7.175	9.599	11.214	13.146	13.821	13.768	13.452
3.500	-0.001	3.633	7.052	9.497	11.137	13.092	13.775	13.737	13.440
3.600	-0.001	3.555	6.931	9.397	11.060	13.037	13.728	13.706	13.425
3.700	-0.001	3.480	6.813	9.298	10.983	12.981	13.680	13.674	13.409
3.800	-0.001	3.410	6.698	9.199	10.907	12.924	13.631	13.642	13.392
3.900	-0.001	3.343	6.586	9.102	10.831	12.867	13.582	13.609	13.373
4.000	-0.001	3.279	6.477	9.007	10.755	12.809	13.532	13.575	13.352
4.100	-0.001	3.218	6.371	8.912	10.679	12.750	13.481	13.542	13.331
4.200	-0.001	3.160	6.268	8.819	10.604	12.690	13.429	13.507	13.310
4.300	-0.001	3.105	6.169	8.727	10.530	12.629	13.377	13.472	13.289
4.400	-0.001	3.053	6.072	8.636	10.457	12.567	13.324	13.436	13.266
4.500	-0.001	3.003	5.979	8.546	10.384	12.505	13.270	13.400	13.242
4.600	-0.001	2.955	5.888	8.457	10.313	12.441	13.216	13.363	13.218
4.700	-0.001	2.909	5.801	8.369	10.241	12.377	13.161	13.326	13.193
4.800	-0.001	2.866	5.717	8.282	10.170	12.312	13.105	13.288	13.167
4.900	-0.001	2.824	5.635	8.197	10.100	12.246	13.048	13.251	13.141
5.000	-0.001	2.784	5.557	8.112	10.029	12.179	12.991	13.213	13.114
5.100	-0.001	2.745	5.482	8.029	9.958	12.111	12.933	13.174	13.087
5.200	-0.001	2.708	5.409	7.947	9.888	12.042	12.874	13.136	13.060
5.300	-0.001	2.673	5.339	7.866	9.818	11.972	12.815	13.097	13.033
5.400	-0.001	2.639	5.272	7.786	9.747	11.902	12.754	13.058	13.006

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.605	5.207	7.708	9.677	11.830	12.694	13.018	12.978
5.600	-0.001	2.573	5.144	7.631	9.606	11.758	12.632	12.977	12.949
5.700	-0.001	2.542	5.083	7.555	9.535	11.685	12.570	12.936	12.920
5.800	-0.001	2.512	5.025	7.481	9.463	11.612	12.508	12.895	12.891
5.900	-0.001	2.483	4.969	7.408	9.392	11.538	12.446	12.853	12.861
6.000	-0.001	2.455	4.915	7.337	9.320	11.463	12.383	12.810	12.830
6.100	-0.001	2.428	4.863	7.267	9.247	11.387	12.320	12.768	12.798
6.200	-0.001	2.403	4.813	7.199	9.175	11.311	12.257	12.725	12.766
6.300	-0.000	2.378	4.764	7.132	9.101	11.235	12.193	12.682	12.733
6.400	-0.000	2.354	4.717	7.067	9.028	11.158	12.128	12.638	12.701
6.500	-0.000	2.330	4.672	7.002	8.954	11.080	12.063	12.594	12.668
6.600	-0.000	2.308	4.628	6.939	8.880	11.002	11.998	12.549	12.635
6.700	-0.000	2.286	4.585	6.877	8.805	10.924	11.932	12.503	12.602
6.800	-0.000	2.266	4.544	6.816	8.730	10.845	11.865	12.457	12.568
6.900	-0.000	2.246	4.505	6.757	8.654	10.765	11.799	12.410	12.534
7.000	-0.000	2.227	4.466	6.698	8.578	10.686	11.732	12.362	12.499
7.100	-0.000	2.208	4.429	6.639	8.501	10.606	11.666	12.313	12.463
7.200	-0.000	2.191	4.393	6.582	8.425	10.525	11.599	12.264	12.427
7.300	-0.000	2.173	4.358	6.525	8.347	10.444	11.532	12.213	12.388
7.400	-0.000	2.157	4.325	6.469	8.270	10.363	11.465	12.161	12.349
7.500	-0.000	2.141	4.292	6.413	8.192	10.281	11.398	12.109	12.307
7.600	-0.000	2.125	4.261	6.358	8.114	10.199	11.331	12.056	12.264
7.700	-0.000	2.110	4.231	6.303	8.036	10.116	11.264	12.002	12.219
7.800	-0.000	2.096	4.201	6.248	7.957	10.034	11.197	11.948	12.172
7.900	-0.000	2.082	4.173	6.194	7.879	9.951	11.130	11.894	12.123
8.000	-0.000	2.068	4.146	6.140	7.800	9.867	11.063	11.839	12.071
8.100	-0.000	2.055	4.120	6.086	7.721	9.784	10.996	11.785	12.019
8.200	-0.000	2.043	4.094	6.032	7.642	9.700	10.929	11.730	11.965
8.300	-0.000	2.030	4.069	5.978	7.562	9.616	10.862	11.674	11.910
8.400	-0.000	2.018	4.044	5.925	7.482	9.531	10.795	11.618	11.854
8.500	-0.000	2.007	4.019	5.871	7.402	9.446	10.728	11.562	11.797
8.600	-0.000	1.996	3.994	5.818	7.321	9.361	10.660	11.505	11.739
8.700	-0.000	1.985	3.969	5.765	7.240	9.275	10.592	11.449	11.680
8.800	-0.000	1.974	3.945	5.711	7.159	9.189	10.524	11.391	11.619
8.900	-0.000	1.964	3.920	5.658	7.079	9.103	10.455	11.333	11.558
9.000	-0.000	1.954	3.895	5.604	6.998	9.016	10.386	11.275	11.497
9.100	-0.000	1.945	3.869	5.551	6.918	8.929	10.316	11.216	11.435
9.200	-0.000	1.936	3.843	5.497	6.837	8.842	10.246	11.156	11.373
9.300	-0.000	1.927	3.817	5.442	6.758	8.755	10.175	11.094	11.310
9.400	-0.000	1.918	3.790	5.387	6.678	8.667	10.104	11.031	11.247
9.500	-0.000	1.910	3.763	5.331	6.598	8.579	10.032	10.966	11.182
9.600	-0.000	1.902	3.735	5.275	6.519	8.490	9.959	10.900	11.117
9.700	-0.000	1.894	3.706	5.218	6.439	8.401	9.886	10.833	11.051
9.800	-0.000	1.886	3.676	5.161	6.360	8.311	9.812	10.763	10.985
9.900	-0.000	1.879	3.646	5.102	6.281	8.222	9.737	10.692	10.919
10.000	-0.000	1.872	3.615	5.043	6.203	8.132	9.662	10.620	10.852
10.100	-0.000	1.864	3.583	4.983	6.124	8.042	9.586	10.546	10.785
10.200	-0.000	1.856	3.549	4.923	6.046	7.953	9.509	10.470	10.718
10.300	-0.000	1.848	3.515	4.861	5.967	7.863	9.432	10.393	10.651
10.400	-0.000	1.839	3.479	4.799	5.889	7.773	9.354	10.314	10.583
10.500	-0.000	1.829	3.442	4.736	5.811	7.684	9.275	10.233	10.514
10.600	-0.000	1.818	3.404	4.672	5.732	7.594	9.195	10.153	10.445
10.700	-0.000	1.807	3.364	4.608	5.654	7.505	9.115	10.071	10.376
10.800	-0.000	1.794	3.323	4.543	5.576	7.416	9.035	9.989	10.307
10.900	-0.000	1.779	3.281	4.478	5.497	7.327	8.953	9.906	10.237
11.000	-0.000	1.764	3.237	4.412	5.419	7.239	8.871	9.822	10.167
11.100	-0.000	1.747	3.192	4.345	5.341	7.151	8.789	9.738	10.097
11.200	-0.000	1.728	3.146	4.278	5.263	7.064	8.705	9.653	10.026
11.300	-0.000	1.707	3.097	4.210	5.185	6.978	8.620	9.567	9.957
11.400	-0.000	1.684	3.048	4.142	5.107	6.892	8.534	9.480	9.885

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.660	2.997	4.074	5.029	6.806	8.446	9.393	9.813
11.600	-0.000	1.635	2.946	4.006	4.952	6.722	8.358	9.305	9.741
11.700	-0.000	1.608	2.894	3.939	4.877	6.640	8.268	9.217	9.669
11.800	-0.000	1.582	2.843	3.872	4.802	6.559	8.179	9.129	9.597
11.900	-0.000	1.555	2.792	3.807	4.729	6.479	8.089	9.042	9.526
12.000	-0.000	1.528	2.742	3.743	4.658	6.401	7.998	8.955	9.454
12.100	-0.000	1.500	2.692	3.680	4.587	6.324	7.908	8.868	9.383
12.200	-0.000	1.473	2.642	3.618	4.518	6.248	7.818	8.782	9.312
12.300	-0.000	1.446	2.592	3.556	4.450	6.174	7.727	8.697	9.241
12.400	-0.000	1.419	2.543	3.496	4.382	6.101	7.638	8.612	9.171
12.500	-0.000	1.392	2.495	3.436	4.316	6.029	7.549	8.529	9.102
12.600	-0.000	1.365	2.447	3.376	4.251	5.959	7.463	8.446	9.033
12.700	-0.000	1.337	2.398	3.318	4.187	5.890	7.377	8.365	8.965
12.800	-0.000	1.310	2.350	3.260	4.124	5.823	7.294	8.285	8.898
12.900	-0.000	1.283	2.303	3.203	4.062	5.757	7.213	8.206	8.832
13.000	-0.000	1.255	2.255	3.146	4.001	5.693	7.133	8.130	8.768
13.100	-0.000	1.226	2.208	3.090	3.941	5.630	7.056	8.055	8.706
13.200	-0.000	1.198	2.161	3.036	3.883	5.568	6.980	7.982	8.646
13.300	-0.000	1.168	2.114	2.982	3.826	5.509	6.907	7.911	8.588
13.400	-0.000	1.139	2.068	2.929	3.770	5.451	6.836	7.842	8.532
13.500	-0.000	1.109	2.022	2.877	3.716	5.396	6.767	7.777	8.480
13.600	-0.000	1.080	1.977	2.827	3.663	5.341	6.700	7.714	8.429
13.700	-0.000	1.050	1.933	2.778	3.612	5.287	6.636	7.654	8.381
13.800	-0.000	1.021	1.890	2.730	3.563	5.235	6.574	7.597	8.336
13.900	-0.000	0.992	1.848	2.685	3.515	5.183	6.515	7.544	8.294
14.000	-0.000	0.964	1.808	2.640	3.470	5.133	6.459	7.493	8.255
14.100	-0.000	0.936	1.768	2.598	3.427	5.083	6.407	7.446	8.218
14.200	-0.000	0.908	1.731	2.558	3.386	5.036	6.358	7.402	8.184
14.300	-0.000	0.882	1.695	2.520	3.348	4.990	6.312	7.361	8.153
14.400	-0.000	0.858	1.662	2.485	3.313	4.947	6.270	7.324	8.124
14.500	-0.000	0.835	1.632	2.453	3.281	4.908	6.232	7.290	8.099
14.600	-0.000	0.814	1.605	2.424	3.252	4.873	6.198	7.260	8.076
14.700	-0.000	0.796	1.580	2.398	3.226	4.842	6.168	7.233	8.056
14.800	-0.000	0.780	1.559	2.375	3.204	4.815	6.142	7.209	8.040
14.900	-0.000	0.766	1.540	2.356	3.185	4.792	6.120	7.189	8.026
15.000	-0.000	0.756	1.526	2.340	3.169	4.774	6.103	7.173	8.014
15.100	-0.000	0.748	1.515	2.328	3.157	4.761	6.089	7.161	8.006
15.200	-0.000	0.743	1.508	2.319	3.149	4.752	6.080	7.153	8.000
15.300	-0.000	0.740	1.503	2.312	3.143	4.747	6.075	7.147	7.996
15.400	-0.000	0.739	1.501	2.309	3.140	4.745	6.072	7.144	7.993
15.500	-0.000	0.738	1.499	2.306	3.137	4.744	6.071	7.142	7.991
15.600	-0.000	0.737	1.498	2.303	3.134	4.742	6.069	7.139	7.990
15.700	-0.000	0.737	1.497	2.300	3.132	4.741	6.067	7.137	7.988
15.800	-0.000	0.736	1.496	2.297	3.129	4.740	6.065	7.135	7.986
15.900	-0.000	0.736	1.495	2.294	3.126	4.739	6.063	7.132	7.984
16.000	-0.000	0.735	1.494	2.292	3.124	4.738	6.061	7.130	7.982
16.100	-0.000	0.735	1.494	2.289	3.121	4.736	6.060	7.128	7.980
16.200	-0.000	0.735	1.493	2.287	3.119	4.735	6.058	7.126	7.978
16.300	-0.000	0.735	1.493	2.285	3.117	4.734	6.056	7.123	7.976
16.400	-0.000	0.735	1.493	2.283	3.115	4.733	6.054	7.121	7.975
16.500	-0.000	0.735	1.492	2.281	3.112	4.732	6.052	7.119	7.973
16.600	-0.000	0.735	1.492	2.279	3.110	4.731	6.051	7.117	7.971
16.700	-0.000	0.736	1.492	2.278	3.108	4.730	6.049	7.114	7.970
16.800	-0.000	0.736	1.492	2.277	3.106	4.729	6.047	7.112	7.968
16.900	-0.000	0.737	1.493	2.275	3.105	4.728	6.045	7.110	7.967
17.000	-0.000	0.738	1.493	2.274	3.103	4.727	6.043	7.108	7.965
17.100	-0.000	0.739	1.493	2.274	3.101	4.726	6.042	7.105	7.963
17.200	-0.000	0.740	1.494	2.273	3.100	4.725	6.040	7.103	7.961
17.300	-0.000	0.741	1.494	2.273	3.098	4.724	6.038	7.101	7.960
17.400	-0.000	0.743	1.495	2.273	3.097	4.723	6.036	7.099	7.958

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.744	1.496	2.273	3.096	4.722	6.034	7.096	7.956
17.600	-0.000	0.745	1.497	2.273	3.095	4.721	6.033	7.094	7.954
17.700	-0.000	0.746	1.498	2.273	3.094	4.720	6.031	7.092	7.953
17.800	-0.000	0.747	1.499	2.274	3.093	4.719	6.029	7.090	7.951
17.900	-0.000	0.748	1.500	2.275	3.092	4.719	6.027	7.087	7.949
18.000	-0.000	0.750	1.501	2.275	3.091	4.718	6.025	7.085	7.948
18.100	-0.000	0.751	1.503	2.277	3.091	4.717	6.024	7.083	7.946
18.200	-0.000	0.752	1.504	2.278	3.090	4.716	6.022	7.081	7.944
18.300	-0.000	0.753	1.506	2.279	3.090	4.715	6.020	7.080	7.942
18.400	-0.000	0.755	1.508	2.280	3.090	4.715	6.018	7.078	7.941
18.500	-0.000	0.756	1.510	2.282	3.090	4.714	6.016	7.076	7.939
18.600	-0.000	0.757	1.512	2.284	3.091	4.713	6.014	7.074	7.937
18.700	-0.000	0.758	1.514	2.286	3.091	4.713	6.012	7.072	7.936
18.800	-0.000	0.760	1.516	2.288	3.091	4.712	6.011	7.069	7.934
18.900	-0.000	0.761	1.519	2.290	3.092	4.711	6.009	7.067	7.932
19.000	-0.000	0.762	1.521	2.292	3.093	4.711	6.007	7.066	7.931
19.100	-0.000	0.764	1.524	2.294	3.095	4.710	6.005	7.064	7.929
19.200	-0.000	0.765	1.527	2.297	3.096	4.709	6.004	7.062	7.927
19.300	-0.000	0.766	1.529	2.299	3.098	4.709	6.002	7.060	7.926
19.400	-0.000	0.768	1.532	2.301	3.099	4.709	6.000	7.058	7.924
19.500	-0.000	0.769	1.535	2.304	3.101	4.708	5.998	7.056	7.922
19.600	-0.000	0.770	1.537	2.307	3.103	4.708	5.996	7.054	7.921
19.700	-0.000	0.772	1.540	2.310	3.105	4.707	5.994	7.052	7.919
19.800	-0.000	0.773	1.543	2.313	3.107	4.706	5.992	7.050	7.917
19.900	-0.000	0.774	1.546	2.316	3.110	4.706	5.990	7.048	7.916
20.000	-0.000	0.776	1.549	2.319	3.113	4.705	5.988	7.046	7.914
20.100	-0.000	0.777	1.552	2.323	3.116	4.705	5.986	7.045	7.912
20.200	-0.000	0.779	1.555	2.327	3.117	4.704	5.985	7.043	7.911
20.300	-0.000	0.780	1.558	2.330	3.121	4.704	5.983	7.041	7.909
20.400	-0.000	0.782	1.560	2.334	3.126	4.703	5.981	7.039	7.908
20.500	-0.000	0.783	1.563	2.338	3.129	4.703	5.979	7.038	7.906
20.600	-0.000	0.785	1.566	2.342	3.133	4.702	5.977	7.036	7.904
20.700	-0.000	0.786	1.569	2.346	3.137	4.702	5.975	7.034	7.903
20.800	-0.000	0.788	1.572	2.351	3.141	4.704	5.973	7.032	7.901
20.900	-0.000	0.789	1.575	2.354	3.145	4.701	5.971	7.031	7.900
21.000	-0.000	0.791	1.579	2.360	3.149	4.701	5.969	7.029	7.898
21.100	-0.000	0.792	1.582	2.364	3.152	4.701	5.969	7.028	7.897
21.200	-0.000	0.794	1.585	2.369	3.158	4.700	5.967	7.027	7.895
21.300	-0.000	0.795	1.588	2.374	3.163	4.700	5.965	7.025	7.894
21.400	-0.000	0.797	1.591	2.378	3.166	4.700	5.964	7.023	7.892
21.500	-0.000	0.799	1.594	2.382	3.172	4.700	5.962	7.022	7.891
21.600	-0.000	0.800	1.597	2.388	3.177	4.699	5.960	7.020	7.889
21.700	-0.000	0.802	1.599	2.393	3.182	4.699	5.958	7.018	7.888
21.800	-0.000	0.803	1.604	2.397	3.187	4.699	5.957	7.017	7.886
21.900	-0.000	0.805	1.606	2.402	3.192	4.699	5.955	7.015	7.885
22.000	-0.000	0.807	1.610	2.407	3.197	4.699	5.953	7.014	7.883
22.100	-0.000	0.808	1.614	2.412	3.202	4.699	5.952	7.012	7.882
22.200	-0.000	0.810	1.617	2.417	3.207	4.701	5.950	7.010	7.881
22.300	-0.000	0.812	1.620	2.422	3.212	4.698	5.948	7.009	7.879
22.400	-0.000	0.813	1.624	2.427	3.217	4.700	5.947	7.007	7.878
22.500	-0.000	0.815	1.627	2.430	3.223	4.698	5.945	7.006	7.877
22.600	-0.000	0.817	1.630	2.437	3.228	4.699	5.944	7.005	7.875
22.700	-0.000	0.819	1.634	2.442	3.233	4.699	5.942	7.003	7.874
22.800	-0.000	0.820	1.637	2.447	3.238	4.699	5.941	7.002	7.873
22.900	-0.000	0.822	1.641	2.453	3.243	4.699	5.939	7.000	7.872
23.000	-0.000	0.824	1.644	2.458	3.249	4.699	5.938	6.999	7.871
23.100	-0.000	0.825	1.648	2.463	3.254	4.700	5.936	6.998	7.870
23.200	-0.000	0.827	1.651	2.468	3.259	4.700	5.935	6.996	7.869
23.300	-0.000	0.829	1.655	2.474	3.264	4.700	5.934	6.995	7.868
23.400	-0.000	0.831	1.658	2.479	3.268	4.701	5.932	6.994	7.867

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.833	1.662	2.484	3.273	4.701	5.931	6.993	7.866
23.600	-0.000	0.834	1.666	2.489	3.278	4.702	5.930	6.992	7.865
23.700	-0.000	0.836	1.669	2.494	3.282	4.703	5.929	6.991	7.865
23.800	-0.000	0.838	1.673	2.498	3.287	4.704	5.928	6.990	7.864
23.900	-0.000	0.840	1.677	2.503	3.291	4.704	5.927	6.989	7.863
24.000	-0.000	0.842	1.680	2.508	3.295	4.705	5.926	6.988	7.862
24.100	-0.000	0.844	1.684	2.512	3.299	4.706	5.925	6.987	7.862
24.200	-0.000	0.846	1.688	2.516	3.303	4.707	5.924	6.986	7.861
24.300	-0.000	0.847	1.691	2.521	3.306	4.708	5.923	6.987	7.861
24.400	-0.000	0.849	1.695	2.525	3.310	4.709	5.922	6.986	7.860
24.500	-0.000	0.851	1.699	2.529	3.313	4.710	5.922	6.985	7.860
24.600	-0.000	0.853	1.703	2.532	3.316	4.711	5.922	6.984	7.859
24.700	-0.000	0.855	1.707	2.536	3.319	4.711	5.921	6.984	7.859
24.800	-0.000	0.857	1.711	2.539	3.322	4.712	5.921	6.983	7.859
24.900	-0.000	0.859	1.715	2.542	3.325	4.713	5.921	6.982	7.859
25.000	-0.000	0.861	1.718	2.545	3.327	4.713	5.922	6.982	7.859
25.100	-0.000	0.863	1.722	2.548	3.329	4.714	5.922	6.982	7.859
25.200	-0.000	0.865	1.726	2.551	3.331	4.714	5.924	6.982	7.859
25.300	-0.000	0.867	1.730	2.553	3.333	4.714	5.925	6.982	7.859
25.400	-0.000	0.869	1.733	2.556	3.334	4.715	5.927	6.982	7.859
25.500	-0.000	0.871	1.736	2.558	3.335	4.715	5.929	6.982	7.860
25.600	-0.000	0.873	1.739	2.559	3.336	4.715	5.931	6.983	7.860
25.700	-0.000	0.875	1.742	2.561	3.336	4.715	5.934	6.983	7.861
25.800	-0.000	0.877	1.744	2.562	3.336	4.714	5.935	6.984	7.862
25.900	-0.000	0.879	1.746	2.563	3.335	4.714	5.937	6.985	7.862
26.000	-0.000	0.881	1.748	2.564	3.335	4.714	5.939	6.986	7.864

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 5.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	7.434	9.814	11.078	11.894	12.834	13.286	13.528	13.548
0.100	-0.002	7.426	9.876	11.179	12.021	12.995	13.462	13.663	13.556
0.200	-0.002	7.402	9.918	11.257	12.124	13.131	13.611	13.768	13.558
0.300	-0.001	7.363	9.942	11.317	12.207	13.244	13.737	13.851	13.556
0.400	-0.001	7.311	9.951	11.361	12.272	13.338	13.842	13.914	13.553
0.500	-0.001	7.248	9.948	11.391	12.322	13.415	13.931	13.963	13.547
0.600	-0.001	7.173	9.934	11.410	12.361	13.479	14.004	14.000	13.541
0.700	-0.001	7.089	9.911	11.419	12.391	13.532	14.064	14.028	13.534
0.800	-0.001	6.996	9.879	11.419	12.412	13.576	14.112	14.049	13.527
0.900	-0.001	6.895	9.839	11.413	12.426	13.613	14.152	14.064	13.520
1.000	-0.001	6.786	9.792	11.399	12.433	13.644	14.185	14.076	13.513
1.100	-0.001	6.670	9.738	11.378	12.434	13.669	14.212	14.084	13.506
1.200	-0.001	6.546	9.676	11.350	12.428	13.689	14.234	14.090	13.501
1.300	-0.001	6.414	9.606	11.316	12.415	13.703	14.251	14.094	13.496
1.400	-0.001	6.276	9.527	11.272	12.395	13.711	14.262	14.095	13.493
1.500	-0.001	6.130	9.438	11.218	12.365	13.711	14.266	14.092	13.491
1.600	-0.001	5.980	9.341	11.156	12.327	13.704	14.264	14.087	13.490
1.700	-0.001	5.827	9.235	11.086	12.281	13.691	14.256	14.078	13.490
1.800	-0.001	5.673	9.123	11.011	12.230	13.674	14.244	14.068	13.491
1.900	-0.001	5.519	9.007	10.931	12.174	13.652	14.228	14.055	13.492
2.000	-0.001	5.367	8.887	10.847	12.115	13.627	14.209	14.040	13.493
2.100	-0.001	5.216	8.765	10.760	12.053	13.599	14.187	14.024	13.495
2.200	-0.001	5.070	8.640	10.670	11.988	13.569	14.162	14.006	13.496
2.300	-0.001	4.927	8.513	10.579	11.921	13.536	14.136	13.987	13.498
2.400	-0.001	4.788	8.386	10.486	11.853	13.501	14.107	13.967	13.499
2.500	-0.001	4.654	8.258	10.392	11.784	13.464	14.077	13.946	13.500
2.600	-0.001	4.525	8.128	10.298	11.713	13.426	14.045	13.924	13.500
2.700	-0.001	4.401	7.998	10.202	11.642	13.385	14.011	13.900	13.499
2.800	-0.001	4.283	7.868	10.106	11.569	13.342	13.976	13.876	13.496
2.900	-0.001	4.171	7.739	10.009	11.496	13.297	13.939	13.851	13.492
3.000	-0.001	4.065	7.610	9.912	11.423	13.252	13.901	13.825	13.486
3.100	-0.001	3.964	7.482	9.815	11.349	13.204	13.862	13.798	13.479
3.200	-0.001	3.869	7.356	9.717	11.275	13.155	13.821	13.771	13.471
3.300	-0.001	3.778	7.233	9.619	11.201	13.105	13.779	13.743	13.461
3.400	-0.001	3.693	7.111	9.521	11.128	13.054	13.736	13.714	13.450
3.500	-0.001	3.612	6.992	9.423	11.054	13.002	13.692	13.686	13.436
3.600	-0.001	3.535	6.876	9.325	10.981	12.949	13.647	13.656	13.422
3.700	-0.001	3.462	6.762	9.229	10.908	12.895	13.602	13.627	13.406
3.800	-0.001	3.392	6.650	9.134	10.836	12.840	13.555	13.597	13.389
3.900	-0.001	3.326	6.541	9.039	10.763	12.783	13.507	13.566	13.371
4.000	-0.001	3.263	6.436	8.946	10.690	12.727	13.459	13.535	13.351
4.100	-0.001	3.203	6.333	8.854	10.618	12.669	13.410	13.503	13.331
4.200	-0.001	3.146	6.232	8.762	10.545	12.610	13.361	13.471	13.310
4.300	-0.001	3.092	6.135	8.672	10.473	12.551	13.311	13.439	13.288
4.400	-0.001	3.040	6.041	8.583	10.402	12.491	13.260	13.406	13.265
4.500	-0.001	2.990	5.949	8.495	10.330	12.430	13.208	13.372	13.242
4.600	-0.001	2.942	5.861	8.408	10.259	12.368	13.155	13.337	13.218
4.700	-0.001	2.897	5.775	8.322	10.189	12.305	13.102	13.302	13.193
4.800	-0.001	2.854	5.692	8.237	10.118	12.241	13.049	13.267	13.167
4.900	-0.001	2.812	5.612	8.153	10.048	12.176	12.994	13.231	13.141
5.000	-0.001	2.773	5.535	8.071	9.978	12.111	12.940	13.195	13.115
5.100	-0.001	2.735	5.461	7.989	9.907	12.045	12.884	13.159	13.089
5.200	-0.001	2.698	5.390	7.909	9.837	11.978	12.828	13.122	13.062
5.300	-0.001	2.663	5.321	7.829	9.767	11.910	12.771	13.085	13.035
5.400	-0.001	2.629	5.254	7.751	9.697	11.841	12.713	13.047	13.007

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.597	5.190	7.675	9.627	11.771	12.654	13.009	12.979
5.600	-0.001	2.566	5.128	7.599	9.556	11.701	12.595	12.970	12.950
5.700	-0.001	2.535	5.068	7.525	9.485	11.630	12.535	12.930	12.921
5.800	-0.001	2.506	5.010	7.452	9.414	11.558	12.474	12.889	12.892
5.900	-0.001	2.477	4.955	7.381	9.343	11.486	12.414	12.848	12.862
6.000	-0.001	2.449	4.901	7.311	9.271	11.413	12.353	12.807	12.832
6.100	-0.001	2.423	4.849	7.242	9.199	11.339	12.291	12.765	12.800
6.200	-0.001	2.397	4.799	7.175	9.127	11.265	12.230	12.722	12.768
6.300	-0.000	2.372	4.751	7.109	9.054	11.190	12.168	12.680	12.736
6.400	-0.000	2.348	4.705	7.044	8.981	11.115	12.105	12.637	12.703
6.500	-0.000	2.325	4.660	6.980	8.907	11.039	12.042	12.593	12.671
6.600	-0.000	2.303	4.617	6.917	8.833	10.963	11.979	12.549	12.638
6.700	-0.000	2.282	4.575	6.855	8.759	10.886	11.915	12.505	12.605
6.800	-0.000	2.261	4.534	6.793	8.684	10.808	11.850	12.460	12.571
6.900	-0.000	2.241	4.494	6.733	8.609	10.730	11.786	12.414	12.537
7.000	-0.000	2.222	4.456	6.674	8.534	10.652	11.721	12.367	12.502
7.100	-0.000	2.203	4.419	6.616	8.458	10.573	11.656	12.320	12.467
7.200	-0.000	2.185	4.384	6.558	8.382	10.494	11.591	12.271	12.431
7.300	-0.000	2.168	4.349	6.501	8.306	10.415	11.526	12.222	12.395
7.400	-0.000	2.152	4.316	6.445	8.229	10.335	11.460	12.171	12.357
7.500	-0.000	2.136	4.283	6.389	8.152	10.254	11.395	12.120	12.317
7.600	-0.000	2.121	4.252	6.333	8.075	10.174	11.330	12.068	12.276
7.700	-0.000	2.106	4.222	6.278	7.998	10.093	11.264	12.015	12.233
7.800	-0.000	2.091	4.193	6.223	7.921	10.011	11.199	11.962	12.188
7.900	-0.000	2.077	4.165	6.168	7.843	9.930	11.134	11.908	12.141
8.000	-0.000	2.064	4.137	6.114	7.766	9.848	11.068	11.854	12.093
8.100	-0.000	2.051	4.111	6.060	7.688	9.766	11.003	11.800	12.042
8.200	-0.000	2.038	4.084	6.006	7.610	9.684	10.938	11.745	11.990
8.300	-0.000	2.026	4.059	5.952	7.532	9.601	10.872	11.690	11.937
8.400	-0.000	2.014	4.033	5.899	7.454	9.518	10.807	11.635	11.883
8.500	-0.000	2.003	4.008	5.845	7.375	9.435	10.741	11.579	11.828
8.600	-0.000	1.992	3.982	5.791	7.296	9.352	10.675	11.523	11.771
8.700	-0.000	1.981	3.957	5.738	7.217	9.268	10.608	11.467	11.713
8.800	-0.000	1.971	3.931	5.684	7.138	9.184	10.541	11.410	11.654
8.900	-0.000	1.960	3.905	5.631	7.058	9.099	10.474	11.353	11.594
9.000	-0.000	1.951	3.879	5.578	6.979	9.014	10.406	11.295	11.533
9.100	-0.000	1.941	3.853	5.524	6.900	8.929	10.338	11.237	11.472
9.200	-0.000	1.932	3.826	5.470	6.821	8.844	10.269	11.178	11.410
9.300	-0.000	1.923	3.799	5.416	6.742	8.758	10.199	11.119	11.348
9.400	-0.000	1.915	3.772	5.361	6.664	8.672	10.129	11.057	11.285
9.500	-0.000	1.906	3.744	5.306	6.585	8.586	10.059	10.995	11.222
9.600	-0.000	1.898	3.715	5.250	6.507	8.499	9.987	10.931	11.158
9.700	-0.000	1.890	3.686	5.194	6.429	8.412	9.915	10.866	11.093
9.800	-0.000	1.883	3.655	5.137	6.351	8.325	9.843	10.799	11.028
9.900	-0.000	1.875	3.624	5.080	6.273	8.237	9.770	10.731	10.963
10.000	-0.000	1.867	3.592	5.022	6.196	8.150	9.696	10.661	10.897
10.100	-0.000	1.859	3.559	4.963	6.118	8.062	9.621	10.589	10.831
10.200	-0.000	1.850	3.525	4.904	6.041	7.974	9.545	10.516	10.765
10.300	-0.000	1.840	3.490	4.844	5.964	7.887	9.469	10.441	10.698
10.400	-0.000	1.830	3.454	4.783	5.888	7.799	9.392	10.365	10.630
10.500	-0.000	1.819	3.417	4.721	5.811	7.712	9.315	10.287	10.563
10.600	-0.000	1.807	3.378	4.659	5.734	7.625	9.236	10.207	10.495
10.700	-0.000	1.794	3.338	4.596	5.658	7.537	9.158	10.127	10.427
10.800	-0.000	1.780	3.298	4.533	5.581	7.451	9.078	10.046	10.359
10.900	-0.000	1.764	3.255	4.469	5.505	7.364	8.998	9.964	10.292
11.000	-0.000	1.746	3.212	4.404	5.428	7.278	8.917	9.881	10.223
11.100	-0.000	1.727	3.167	4.339	5.352	7.192	8.836	9.798	10.154
11.200	-0.000	1.707	3.122	4.274	5.276	7.107	8.753	9.714	10.085
11.300	-0.000	1.685	3.075	4.208	5.200	7.023	8.671	9.630	10.015
11.400	-0.000	1.662	3.028	4.143	5.126	6.940	8.588	9.545	9.946

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.638	2.981	4.079	5.052	6.858	8.504	9.459	9.876
11.600	-0.000	1.613	2.933	4.015	4.979	6.777	8.420	9.374	9.806
11.700	-0.000	1.588	2.886	3.952	4.907	6.697	8.335	9.288	9.736
11.800	-0.000	1.563	2.840	3.891	4.837	6.620	8.250	9.203	9.666
11.900	-0.000	1.538	2.794	3.830	4.768	6.543	8.165	9.119	9.596
12.000	-0.000	1.513	2.748	3.770	4.701	6.468	8.079	9.035	9.527
12.100	-0.000	1.488	2.702	3.711	4.634	6.394	7.994	8.951	9.458
12.200	-0.000	1.463	2.656	3.653	4.568	6.321	7.908	8.868	9.389
12.300	-0.000	1.438	2.610	3.595	4.503	6.249	7.822	8.785	9.321
12.400	-0.000	1.413	2.565	3.538	4.439	6.179	7.736	8.704	9.253
12.500	-0.000	1.388	2.520	3.481	4.376	6.109	7.650	8.623	9.185
12.600	-0.000	1.364	2.475	3.425	4.313	6.041	7.566	8.542	9.118
12.700	-0.000	1.339	2.430	3.369	4.252	5.973	7.482	8.463	9.052
12.800	-0.000	1.314	2.386	3.314	4.191	5.907	7.400	8.385	8.986
12.900	-0.000	1.290	2.341	3.260	4.131	5.843	7.320	8.308	8.922
13.000	-0.000	1.265	2.297	3.206	4.072	5.779	7.242	8.232	8.858
13.100	-0.000	1.240	2.252	3.152	4.014	5.717	7.165	8.158	8.796
13.200	-0.000	1.215	2.208	3.100	3.957	5.656	7.090	8.086	8.735
13.300	-0.000	1.190	2.164	3.048	3.901	5.597	7.017	8.015	8.676
13.400	-0.000	1.164	2.121	2.997	3.846	5.540	6.946	7.946	8.620
13.500	-0.000	1.137	2.077	2.946	3.793	5.484	6.877	7.879	8.565
13.600	-0.000	1.110	2.034	2.897	3.741	5.429	6.810	7.815	8.513
13.700	-0.000	1.084	1.992	2.848	3.690	5.377	6.745	7.754	8.463
13.800	-0.000	1.057	1.950	2.801	3.640	5.326	6.682	7.695	8.415
13.900	-0.000	1.031	1.909	2.755	3.592	5.275	6.622	7.638	8.370
14.000	-0.000	1.004	1.869	2.711	3.546	5.226	6.564	7.585	8.328
14.100	-0.000	0.978	1.830	2.668	3.501	5.177	6.508	7.534	8.288
14.200	-0.000	0.952	1.792	2.626	3.459	5.130	6.456	7.487	8.250
14.300	-0.000	0.927	1.755	2.586	3.418	5.084	6.406	7.442	8.215
14.400	-0.000	0.902	1.721	2.549	3.380	5.039	6.359	7.400	8.183
14.500	-0.000	0.879	1.688	2.513	3.344	4.996	6.316	7.361	8.153
14.600	-0.000	0.856	1.657	2.480	3.310	4.955	6.275	7.326	8.125
14.700	-0.000	0.835	1.629	2.449	3.279	4.918	6.239	7.293	8.101
14.800	-0.000	0.816	1.603	2.422	3.252	4.884	6.207	7.264	8.079
14.900	-0.000	0.799	1.580	2.396	3.227	4.854	6.178	7.238	8.059
15.000	-0.000	0.784	1.560	2.374	3.205	4.828	6.152	7.214	8.043
15.100	-0.000	0.771	1.543	2.355	3.186	4.805	6.130	7.194	8.028
15.200	-0.000	0.761	1.529	2.339	3.171	4.787	6.112	7.178	8.017
15.300	-0.000	0.753	1.519	2.327	3.158	4.773	6.098	7.165	8.008
15.400	-0.000	0.747	1.511	2.317	3.149	4.763	6.088	7.156	8.001
15.500	-0.000	0.744	1.506	2.311	3.142	4.756	6.081	7.149	7.996
15.600	-0.000	0.742	1.504	2.306	3.138	4.753	6.077	7.145	7.993
15.700	-0.000	0.741	1.503	2.303	3.136	4.751	6.075	7.143	7.991
15.800	-0.000	0.741	1.502	2.301	3.133	4.750	6.073	7.140	7.989
15.900	-0.000	0.740	1.501	2.298	3.131	4.749	6.072	7.138	7.987
16.000	-0.000	0.740	1.501	2.296	3.128	4.748	6.070	7.136	7.985
16.100	-0.000	0.739	1.500	2.294	3.126	4.747	6.068	7.133	7.983
16.200	-0.000	0.739	1.500	2.292	3.124	4.745	6.066	7.131	7.982
16.300	-0.000	0.739	1.500	2.290	3.121	4.744	6.064	7.129	7.980
16.400	-0.000	0.739	1.499	2.288	3.119	4.743	6.062	7.127	7.978
16.500	-0.000	0.739	1.499	2.287	3.117	4.742	6.061	7.124	7.977
16.600	-0.000	0.739	1.499	2.285	3.115	4.741	6.059	7.122	7.975
16.700	-0.000	0.739	1.499	2.284	3.114	4.740	6.057	7.120	7.974
16.800	-0.000	0.739	1.499	2.283	3.112	4.739	6.055	7.118	7.972
16.900	-0.000	0.740	1.500	2.282	3.110	4.738	6.053	7.115	7.970
17.000	-0.000	0.741	1.500	2.281	3.109	4.737	6.052	7.113	7.968
17.100	-0.000	0.741	1.500	2.281	3.107	4.736	6.050	7.111	7.966
17.200	-0.000	0.742	1.501	2.281	3.106	4.735	6.048	7.108	7.965
17.300	-0.000	0.743	1.501	2.280	3.105	4.734	6.046	7.106	7.963
17.400	-0.000	0.745	1.502	2.281	3.103	4.733	6.044	7.104	7.961

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.746	1.503	2.281	3.102	4.733	6.042	7.102	7.959
17.600	-0.000	0.747	1.503	2.281	3.101	4.732	6.041	7.099	7.958
17.700	-0.000	0.748	1.504	2.282	3.101	4.731	6.039	7.097	7.956
17.800	-0.000	0.749	1.505	2.283	3.100	4.730	6.037	7.095	7.954
17.900	-0.000	0.750	1.506	2.284	3.099	4.729	6.035	7.093	7.953
18.000	-0.000	0.752	1.508	2.285	3.099	4.728	6.033	7.092	7.951
18.100	-0.000	0.753	1.509	2.286	3.099	4.728	6.031	7.090	7.949
18.200	-0.000	0.754	1.510	2.287	3.099	4.727	6.029	7.087	7.947
18.300	-0.000	0.755	1.512	2.288	3.099	4.726	6.028	7.085	7.946
18.400	-0.000	0.757	1.514	2.290	3.099	4.725	6.026	7.083	7.944
18.500	-0.000	0.758	1.516	2.292	3.099	4.725	6.024	7.081	7.942
18.600	-0.000	0.759	1.517	2.294	3.100	4.724	6.022	7.079	7.940
18.700	-0.000	0.761	1.519	2.296	3.100	4.723	6.020	7.077	7.939
18.800	-0.000	0.762	1.522	2.298	3.101	4.723	6.018	7.075	7.937
18.900	-0.000	0.763	1.524	2.300	3.103	4.722	6.017	7.073	7.935
19.000	-0.000	0.765	1.527	2.302	3.104	4.721	6.015	7.071	7.934
19.100	-0.000	0.766	1.529	2.304	3.105	4.721	6.013	7.069	7.932
19.200	-0.000	0.767	1.531	2.306	3.107	4.721	6.011	7.067	7.930
19.300	-0.000	0.769	1.534	2.308	3.108	4.720	6.009	7.065	7.929
19.400	-0.000	0.770	1.537	2.311	3.110	4.719	6.007	7.063	7.927
19.500	-0.000	0.771	1.540	2.313	3.112	4.719	6.005	7.061	7.925
19.600	-0.000	0.773	1.542	2.316	3.114	4.718	6.003	7.059	7.924
19.700	-0.000	0.774	1.545	2.319	3.117	4.718	6.001	7.058	7.922
19.800	-0.000	0.776	1.548	2.321	3.119	4.717	6.000	7.056	7.920
19.900	-0.000	0.777	1.551	2.326	3.122	4.717	5.998	7.054	7.919
20.000	-0.000	0.778	1.554	2.329	3.125	4.716	5.996	7.052	7.917
20.100	-0.000	0.780	1.557	2.332	3.127	4.715	5.994	7.050	7.915
20.200	-0.000	0.781	1.560	2.336	3.132	4.715	5.992	7.048	7.914
20.300	-0.000	0.783	1.563	2.339	3.135	4.714	5.990	7.047	7.912
20.400	-0.000	0.784	1.566	2.343	3.138	4.714	5.988	7.045	7.910
20.500	-0.000	0.786	1.569	2.347	3.142	4.714	5.986	7.043	7.909
20.600	-0.000	0.787	1.572	2.351	3.146	4.713	5.984	7.042	7.907
20.700	-0.000	0.789	1.575	2.355	3.150	4.711	5.982	7.040	7.906
20.800	-0.000	0.790	1.578	2.359	3.154	4.713	5.982	7.038	7.904
20.900	-0.000	0.792	1.580	2.363	3.158	4.710	5.980	7.037	7.902
21.000	-0.000	0.793	1.584	2.368	3.162	4.712	5.975	7.036	7.901
21.100	-0.000	0.795	1.586	2.372	3.167	4.712	5.976	7.034	7.899
21.200	-0.000	0.797	1.590	2.377	3.171	4.711	5.974	7.032	7.898
21.300	-0.000	0.798	1.593	2.382	3.176	4.711	5.969	7.030	7.896
21.400	-0.000	0.800	1.596	2.386	3.180	4.708	5.971	7.029	7.895
21.500	-0.000	0.801	1.600	2.391	3.185	4.710	5.969	7.027	7.893
21.600	-0.000	0.803	1.603	2.396	3.190	4.710	5.967	7.025	7.892
21.700	-0.000	0.805	1.606	2.399	3.194	4.710	5.965	7.024	7.890
21.800	-0.000	0.806	1.609	2.406	3.199	4.710	5.964	7.022	7.889
21.900	-0.000	0.808	1.613	2.411	3.204	4.710	5.962	7.021	7.888
22.000	-0.000	0.810	1.616	2.416	3.209	4.709	5.960	7.019	7.886
22.100	-0.000	0.811	1.619	2.421	3.214	4.709	5.959	7.018	7.885
22.200	-0.000	0.813	1.623	2.426	3.216	4.709	5.957	7.016	7.883
22.300	-0.000	0.815	1.626	2.431	3.223	4.709	5.955	7.014	7.882
22.400	-0.000	0.816	1.629	2.436	3.228	4.709	5.954	7.013	7.881
22.500	-0.000	0.818	1.633	2.441	3.233	4.709	5.952	7.012	7.879
22.600	-0.000	0.820	1.636	2.446	3.238	4.709	5.948	7.010	7.878
22.700	-0.000	0.821	1.640	2.451	3.243	4.709	5.947	7.009	7.877
22.800	-0.000	0.823	1.643	2.456	3.248	4.710	5.948	7.007	7.876
22.900	-0.000	0.825	1.647	2.462	3.253	4.710	5.946	7.006	7.875
23.000	-0.000	0.827	1.650	2.467	3.257	4.710	5.945	7.005	7.874
23.100	-0.000	0.829	1.654	2.472	3.262	4.710	5.943	7.003	7.873
23.200	-0.000	0.830	1.657	2.477	3.267	4.711	5.942	7.002	7.872
23.300	-0.000	0.832	1.661	2.482	3.272	4.711	5.941	7.001	7.871
23.400	-0.000	0.834	1.665	2.487	3.276	4.712	5.939	7.000	7.870

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.836	1.668	2.492	3.281	4.712	5.938	6.998	7.870
23.600	-0.000	0.838	1.672	2.497	3.285	4.713	5.937	6.997	7.869
23.700	-0.000	0.839	1.676	2.502	3.289	4.713	5.936	6.996	7.868
23.800	-0.000	0.841	1.679	2.506	3.293	4.714	5.935	6.995	7.867
23.900	-0.000	0.843	1.683	2.511	3.297	4.715	5.934	6.994	7.866
24.000	-0.000	0.845	1.687	2.515	3.301	4.715	5.933	6.995	7.866
24.100	-0.000	0.847	1.690	2.519	3.305	4.716	5.932	6.994	7.865
24.200	-0.000	0.849	1.694	2.523	3.308	4.717	5.931	6.993	7.865
24.300	-0.000	0.851	1.698	2.527	3.312	4.717	5.931	6.992	7.864
24.400	-0.000	0.853	1.702	2.531	3.315	4.718	5.930	6.991	7.864
24.500	-0.000	0.855	1.706	2.535	3.318	4.718	5.930	6.991	7.864
24.600	-0.000	0.857	1.710	2.538	3.321	4.719	5.929	6.990	7.863
24.700	-0.000	0.859	1.714	2.541	3.323	4.719	5.929	6.989	7.863
24.800	-0.000	0.861	1.717	2.544	3.326	4.720	5.929	6.989	7.863
24.900	-0.000	0.862	1.721	2.547	3.328	4.720	5.929	6.989	7.863
25.000	-0.000	0.864	1.725	2.550	3.330	4.721	5.930	6.989	7.863
25.100	-0.000	0.866	1.729	2.552	3.332	4.721	5.931	6.988	7.863
25.200	-0.000	0.868	1.732	2.555	3.333	4.721	5.932	6.988	7.863
25.300	-0.000	0.870	1.735	2.557	3.335	4.721	5.934	6.988	7.864
25.400	-0.000	0.873	1.738	2.558	3.336	4.721	5.935	6.989	7.864
25.500	-0.000	0.875	1.741	2.560	3.336	4.721	5.938	6.989	7.865
25.600	-0.000	0.877	1.743	2.561	3.337	4.721	5.940	6.989	7.865
25.700	-0.000	0.879	1.746	2.562	3.337	4.720	5.941	6.990	7.866
25.800	-0.000	0.881	1.748	2.563	3.336	4.720	5.943	6.991	7.867
25.900	-0.000	0.883	1.749	2.563	3.336	4.719	5.944	6.992	7.868
26.000	-0.000	0.885	1.751	2.563	3.335	4.719	5.946	6.994	7.870

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 5.50 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	7.129	9.562	10.865	11.715	12.713	13.218	13.477	13.554
0.100	-0.002	7.114	9.610	10.950	11.824	12.855	13.374	13.593	13.565
0.200	-0.002	7.085	9.641	11.017	11.913	12.975	13.507	13.687	13.570
0.300	-0.001	7.043	9.659	11.068	11.986	13.076	13.622	13.761	13.571
0.400	-0.001	6.990	9.663	11.106	12.044	13.162	13.719	13.821	13.568
0.500	-0.001	6.927	9.657	11.132	12.090	13.234	13.801	13.868	13.565
0.600	-0.001	6.856	9.643	11.149	12.127	13.295	13.869	13.905	13.559
0.700	-0.001	6.778	9.621	11.158	12.156	13.346	13.925	13.933	13.554
0.800	-0.001	6.693	9.592	11.160	12.177	13.390	13.972	13.955	13.547
0.900	-0.001	6.601	9.556	11.154	12.191	13.427	14.010	13.970	13.541
1.000	-0.001	6.503	9.513	11.142	12.199	13.458	14.041	13.982	13.534
1.100	-0.001	6.400	9.465	11.126	12.203	13.485	14.068	13.991	13.527
1.200	-0.001	6.291	9.411	11.103	12.201	13.507	14.090	13.997	13.521
1.300	-0.001	6.176	9.351	11.075	12.193	13.525	14.108	14.001	13.516
1.400	-0.001	6.056	9.284	11.039	12.180	13.538	14.121	14.003	13.511
1.500	-0.001	5.931	9.209	10.997	12.160	13.545	14.129	14.003	13.508
1.600	-0.001	5.801	9.125	10.947	12.132	13.547	14.132	14.000	13.505
1.700	-0.001	5.667	9.034	10.890	12.097	13.542	14.129	13.994	13.504
1.800	-0.001	5.530	8.935	10.825	12.056	13.532	14.122	13.986	13.503
1.900	-0.001	5.392	8.830	10.755	12.009	13.517	14.110	13.975	13.503
2.000	-0.001	5.255	8.721	10.681	11.958	13.498	14.094	13.962	13.504
2.100	-0.001	5.118	8.609	10.602	11.903	13.475	14.075	13.948	13.504
2.200	-0.001	4.984	8.493	10.521	11.845	13.448	14.053	13.932	13.506
2.300	-0.001	4.852	8.376	10.437	11.785	13.419	14.029	13.915	13.507
2.400	-0.001	4.723	8.257	10.351	11.722	13.387	14.003	13.896	13.507
2.500	-0.001	4.597	8.137	10.263	11.658	13.352	13.974	13.877	13.507
2.600	-0.001	4.475	8.015	10.174	11.592	13.316	13.944	13.856	13.505
2.700	-0.001	4.358	7.892	10.084	11.525	13.277	13.912	13.835	13.502
2.800	-0.001	4.244	7.769	9.993	11.457	13.237	13.879	13.812	13.498
2.900	-0.001	4.135	7.646	9.902	11.389	13.194	13.844	13.789	13.493
3.000	-0.001	4.032	7.523	9.809	11.319	13.150	13.808	13.765	13.486
3.100	-0.001	3.933	7.402	9.717	11.250	13.105	13.770	13.740	13.478
3.200	-0.001	3.840	7.281	9.624	11.180	13.058	13.732	13.715	13.469
3.300	-0.001	3.752	7.162	9.531	11.110	13.010	13.692	13.689	13.458
3.400	-0.001	3.669	7.045	9.437	11.040	12.961	13.651	13.662	13.446
3.500	-0.001	3.589	6.930	9.344	10.970	12.911	13.610	13.635	13.433
3.600	-0.001	3.514	6.818	9.251	10.900	12.859	13.567	13.608	13.419
3.700	-0.001	3.442	6.708	9.158	10.831	12.807	13.523	13.581	13.403
3.800	-0.001	3.374	6.600	9.066	10.761	12.753	13.479	13.553	13.386
3.900	-0.001	3.309	6.495	8.974	10.692	12.699	13.433	13.524	13.369
4.000	-0.001	3.247	6.392	8.883	10.622	12.643	13.387	13.495	13.351
4.100	-0.001	3.188	6.292	8.793	10.553	12.587	13.340	13.466	13.331
4.200	-0.001	3.131	6.195	8.704	10.484	12.530	13.293	13.436	13.310
4.300	-0.001	3.077	6.100	8.616	10.414	12.472	13.244	13.406	13.287
4.400	-0.001	3.026	6.008	8.529	10.344	12.413	13.195	13.375	13.265
4.500	-0.001	2.977	5.919	8.443	10.274	12.353	13.145	13.344	13.242
4.600	-0.001	2.930	5.832	8.358	10.204	12.293	13.095	13.312	13.217
4.700	-0.001	2.885	5.748	8.273	10.134	12.231	13.044	13.279	13.192
4.800	-0.001	2.842	5.667	8.190	10.064	12.169	12.992	13.246	13.167
4.900	-0.001	2.801	5.589	8.109	9.994	12.106	12.940	13.212	13.142
5.000	-0.001	2.762	5.513	8.028	9.925	12.042	12.887	13.178	13.116
5.100	-0.001	2.724	5.440	7.948	9.855	11.977	12.834	13.143	13.090
5.200	-0.001	2.688	5.370	7.869	9.785	11.912	12.780	13.108	13.063
5.300	-0.001	2.654	5.302	7.792	9.715	11.846	12.725	13.073	13.036
5.400	-0.001	2.620	5.236	7.716	9.645	11.778	12.670	13.036	13.008

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.588	5.173	7.641	9.575	11.711	12.613	12.999	12.980
5.600	-0.001	2.557	5.111	7.567	9.505	11.642	12.556	12.961	12.951
5.700	-0.001	2.527	5.052	7.494	9.434	11.573	12.498	12.923	12.922
5.800	-0.001	2.498	4.995	7.423	9.364	11.503	12.439	12.883	12.893
5.900	-0.001	2.470	4.940	7.353	9.293	11.432	12.380	12.843	12.863
6.000	-0.001	2.443	4.887	7.284	9.221	11.361	12.321	12.803	12.833
6.100	-0.001	2.417	4.836	7.216	9.150	11.289	12.261	12.761	12.802
6.200	-0.001	2.392	4.786	7.150	9.077	11.217	12.202	12.720	12.770
6.300	-0.000	2.367	4.739	7.084	9.005	11.144	12.141	12.678	12.739
6.400	-0.000	2.343	4.693	7.019	8.932	11.070	12.081	12.635	12.706
6.500	-0.000	2.320	4.648	6.955	8.859	10.996	12.020	12.592	12.674
6.600	-0.000	2.298	4.605	6.893	8.786	10.921	11.958	12.549	12.641
6.700	-0.000	2.277	4.563	6.831	8.712	10.846	11.896	12.506	12.607
6.800	-0.000	2.256	4.523	6.769	8.638	10.770	11.834	12.461	12.573
6.900	-0.000	2.237	4.484	6.709	8.564	10.693	11.771	12.416	12.539
7.000	-0.000	2.217	4.446	6.650	8.489	10.616	11.708	12.370	12.504
7.100	-0.000	2.199	4.410	6.591	8.414	10.539	11.645	12.324	12.469
7.200	-0.000	2.181	4.374	6.533	8.339	10.461	11.581	12.277	12.434
7.300	-0.000	2.164	4.340	6.476	8.263	10.383	11.518	12.228	12.398
7.400	-0.000	2.147	4.307	6.419	8.187	10.305	11.454	12.179	12.362
7.500	-0.000	2.131	4.275	6.363	8.111	10.226	11.391	12.129	12.324
7.600	-0.000	2.116	4.244	6.307	8.035	10.147	11.327	12.078	12.284
7.700	-0.000	2.101	4.213	6.252	7.959	10.067	11.263	12.027	12.243
7.800	-0.000	2.087	4.184	6.197	7.883	9.988	11.200	11.974	12.200
7.900	-0.000	2.073	4.156	6.142	7.806	9.907	11.136	11.921	12.156
8.000	-0.000	2.060	4.128	6.088	7.730	9.827	11.073	11.867	12.110
8.100	-0.000	2.047	4.101	6.033	7.653	9.746	11.009	11.813	12.062
8.200	-0.000	2.034	4.074	5.979	7.576	9.666	10.945	11.759	12.012
8.300	-0.000	2.022	4.048	5.925	7.499	9.584	10.881	11.705	11.961
8.400	-0.000	2.010	4.021	5.872	7.422	9.503	10.817	11.650	11.908
8.500	-0.000	1.999	3.995	5.818	7.345	9.422	10.753	11.595	11.854
8.600	-0.000	1.988	3.969	5.765	7.268	9.340	10.688	11.539	11.799
8.700	-0.000	1.977	3.943	5.711	7.190	9.258	10.623	11.483	11.743
8.800	-0.000	1.967	3.917	5.658	7.113	9.176	10.557	11.427	11.685
8.900	-0.000	1.957	3.890	5.604	7.035	9.093	10.491	11.370	11.626
9.000	-0.000	1.947	3.863	5.550	6.958	9.010	10.424	11.313	11.566
9.100	-0.000	1.938	3.836	5.496	6.880	8.927	10.357	11.256	11.505
9.200	-0.000	1.928	3.809	5.442	6.802	8.843	10.290	11.198	11.444
9.300	-0.000	1.920	3.781	5.388	6.725	8.759	10.221	11.139	11.382
9.400	-0.000	1.911	3.753	5.333	6.647	8.675	10.152	11.080	11.320
9.500	-0.000	1.903	3.724	5.278	6.570	8.590	10.083	11.019	11.258
9.600	-0.000	1.895	3.694	5.223	6.493	8.506	10.013	10.957	11.195
9.700	-0.000	1.887	3.664	5.167	6.416	8.421	9.942	10.894	11.132
9.800	-0.000	1.879	3.633	5.111	6.339	8.336	9.871	10.829	11.069
9.900	-0.000	1.870	3.601	5.055	6.263	8.250	9.799	10.763	11.004
10.000	-0.000	1.862	3.569	4.998	6.186	8.165	9.726	10.696	10.939
10.100	-0.000	1.852	3.535	4.940	6.110	8.079	9.653	10.627	10.873
10.200	-0.000	1.842	3.500	4.882	6.034	7.993	9.578	10.556	10.808
10.300	-0.000	1.832	3.465	4.823	5.959	7.908	9.503	10.483	10.742
10.400	-0.000	1.820	3.428	4.764	5.883	7.822	9.428	10.410	10.676
10.500	-0.000	1.808	3.390	4.704	5.808	7.737	9.351	10.334	10.609
10.600	-0.000	1.794	3.352	4.643	5.733	7.652	9.274	10.257	10.544
10.700	-0.000	1.780	3.312	4.582	5.658	7.567	9.196	10.178	10.477
10.800	-0.000	1.764	3.271	4.520	5.584	7.482	9.118	10.099	10.410
10.900	-0.000	1.746	3.228	4.457	5.509	7.398	9.039	10.018	10.343
11.000	-0.000	1.727	3.185	4.394	5.434	7.314	8.960	9.936	10.275
11.100	-0.000	1.707	3.141	4.331	5.361	7.231	8.879	9.854	10.208
11.200	-0.000	1.686	3.097	4.268	5.287	7.148	8.799	9.771	10.140
11.300	-0.000	1.664	3.053	4.206	5.215	7.067	8.718	9.688	10.072
11.400	-0.000	1.641	3.008	4.144	5.143	6.986	8.638	9.605	10.004

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.617	2.964	4.083	5.073	6.908	8.558	9.522	9.935
11.600	-0.000	1.594	2.921	4.024	5.005	6.830	8.478	9.441	9.867
11.700	-0.000	1.570	2.878	3.965	4.938	6.754	8.398	9.358	9.800
11.800	-0.000	1.547	2.835	3.908	4.871	6.680	8.317	9.276	9.732
11.900	-0.000	1.523	2.793	3.851	4.806	6.606	8.236	9.194	9.664
12.000	-0.000	1.500	2.751	3.795	4.742	6.533	8.155	9.113	9.597
12.100	-0.000	1.476	2.709	3.739	4.678	6.462	8.073	9.032	9.530
12.200	-0.000	1.453	2.667	3.684	4.616	6.392	7.992	8.952	9.463
12.300	-0.000	1.430	2.625	3.630	4.554	6.322	7.910	8.872	9.397
12.400	-0.000	1.407	2.583	3.576	4.493	6.254	7.829	8.793	9.331
12.500	-0.000	1.384	2.541	3.523	4.432	6.186	7.747	8.714	9.266
12.600	-0.000	1.361	2.499	3.470	4.372	6.120	7.665	8.637	9.201
12.700	-0.000	1.339	2.458	3.417	4.313	6.054	7.584	8.559	9.136
12.800	-0.000	1.316	2.416	3.365	4.255	5.989	7.504	8.483	9.072
12.900	-0.000	1.294	2.375	3.313	4.197	5.926	7.425	8.408	9.009
13.000	-0.000	1.272	2.333	3.261	4.140	5.863	7.347	8.333	8.947
13.100	-0.000	1.250	2.292	3.210	4.084	5.802	7.271	8.260	8.885
13.200	-0.000	1.227	2.250	3.159	4.028	5.742	7.197	8.188	8.824
13.300	-0.000	1.205	2.209	3.109	3.974	5.684	7.125	8.117	8.765
13.400	-0.000	1.182	2.168	3.060	3.920	5.626	7.054	8.048	8.708
13.500	-0.000	1.159	2.127	3.011	3.867	5.570	6.985	7.981	8.652
13.600	-0.000	1.135	2.086	2.963	3.816	5.516	6.918	7.916	8.598
13.700	-0.000	1.111	2.045	2.915	3.765	5.463	6.852	7.853	8.546
13.800	-0.000	1.087	2.005	2.869	3.716	5.412	6.789	7.793	8.497
13.900	-0.000	1.063	1.965	2.824	3.668	5.362	6.728	7.734	8.449
14.000	-0.000	1.039	1.926	2.779	3.621	5.314	6.668	7.678	8.404
14.100	-0.000	1.015	1.888	2.736	3.576	5.266	6.611	7.625	8.362
14.200	-0.000	0.991	1.851	2.694	3.532	5.220	6.556	7.575	8.321
14.300	-0.000	0.967	1.815	2.653	3.490	5.174	6.504	7.527	8.283
14.400	-0.000	0.944	1.779	2.614	3.450	5.129	6.454	7.482	8.248
14.500	-0.000	0.921	1.746	2.577	3.412	5.086	6.406	7.439	8.215
14.600	-0.000	0.898	1.713	2.541	3.375	5.044	6.362	7.399	8.184
14.700	-0.000	0.877	1.683	2.507	3.341	5.003	6.320	7.362	8.155
14.800	-0.000	0.856	1.654	2.476	3.309	4.964	6.282	7.328	8.127
14.900	-0.000	0.837	1.628	2.447	3.279	4.928	6.247	7.297	8.103
15.000	-0.000	0.819	1.604	2.420	3.252	4.896	6.215	7.268	8.081
15.100	-0.000	0.803	1.583	2.396	3.228	4.866	6.187	7.243	8.062
15.200	-0.000	0.789	1.564	2.375	3.207	4.840	6.161	7.220	8.045
15.300	-0.000	0.777	1.548	2.356	3.188	4.818	6.140	7.200	8.031
15.400	-0.000	0.766	1.535	2.340	3.173	4.799	6.122	7.183	8.019
15.500	-0.000	0.758	1.525	2.328	3.160	4.785	6.107	7.170	8.010
15.600	-0.000	0.752	1.517	2.318	3.150	4.774	6.096	7.159	8.002
15.700	-0.000	0.748	1.512	2.311	3.143	4.766	6.087	7.152	7.997
15.800	-0.000	0.746	1.509	2.306	3.138	4.761	6.082	7.146	7.993
15.900	-0.000	0.745	1.508	2.303	3.135	4.759	6.080	7.143	7.990
16.000	-0.000	0.744	1.507	2.301	3.133	4.758	6.078	7.141	7.988
16.100	-0.000	0.744	1.507	2.299	3.130	4.757	6.076	7.139	7.987
16.200	-0.000	0.744	1.507	2.297	3.128	4.756	6.074	7.137	7.986
16.300	-0.000	0.743	1.506	2.295	3.126	4.755	6.072	7.134	7.984
16.400	-0.000	0.743	1.506	2.294	3.124	4.754	6.070	7.132	7.982
16.500	-0.000	0.743	1.506	2.293	3.123	4.753	6.069	7.130	7.980
16.600	-0.000	0.743	1.506	2.291	3.121	4.751	6.067	7.128	7.979
16.700	-0.000	0.743	1.506	2.290	3.119	4.750	6.065	7.125	7.977
16.800	-0.000	0.743	1.506	2.290	3.117	4.749	6.063	7.123	7.975
16.900	-0.000	0.743	1.507	2.289	3.116	4.749	6.061	7.120	7.973
17.000	-0.000	0.744	1.507	2.289	3.115	4.748	6.059	7.118	7.971
17.100	-0.000	0.744	1.507	2.288	3.113	4.747	6.058	7.116	7.970
17.200	-0.000	0.745	1.508	2.288	3.112	4.746	6.056	7.114	7.968
17.300	-0.000	0.746	1.508	2.289	3.111	4.745	6.054	7.111	7.966
17.400	-0.000	0.747	1.509	2.289	3.110	4.744	6.052	7.109	7.964

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.748	1.510	2.289	3.109	4.743	6.050	7.107	7.963
17.600	-0.000	0.749	1.510	2.290	3.108	4.742	6.048	7.105	7.961
17.700	-0.000	0.750	1.511	2.291	3.108	4.741	6.047	7.104	7.959
17.800	-0.000	0.751	1.512	2.292	3.107	4.741	6.045	7.101	7.957
17.900	-0.000	0.753	1.513	2.293	3.107	4.740	6.043	7.099	7.956
18.000	-0.000	0.754	1.514	2.294	3.107	4.739	6.041	7.097	7.954
18.100	-0.000	0.755	1.516	2.295	3.107	4.738	6.039	7.095	7.952
18.200	-0.000	0.756	1.517	2.297	3.107	4.738	6.037	7.093	7.950
18.300	-0.000	0.758	1.519	2.298	3.107	4.737	6.035	7.091	7.949
18.400	-0.000	0.759	1.520	2.300	3.108	4.736	6.033	7.089	7.947
18.500	-0.000	0.760	1.522	2.302	3.108	4.735	6.031	7.087	7.945
18.600	-0.000	0.762	1.524	2.304	3.110	4.735	6.029	7.085	7.943
18.700	-0.000	0.763	1.525	2.306	3.111	4.734	6.028	7.083	7.942
18.800	-0.000	0.764	1.527	2.308	3.112	4.733	6.026	7.081	7.940
18.900	-0.000	0.766	1.529	2.310	3.113	4.733	6.024	7.079	7.938
19.000	-0.000	0.767	1.532	2.311	3.114	4.733	6.022	7.077	7.937
19.100	-0.000	0.768	1.534	2.314	3.116	4.732	6.020	7.075	7.935
19.200	-0.000	0.770	1.537	2.316	3.118	4.731	6.018	7.073	7.933
19.300	-0.000	0.771	1.538	2.318	3.120	4.731	6.016	7.071	7.931
19.400	-0.000	0.772	1.542	2.320	3.121	4.730	6.014	7.069	7.930
19.500	-0.000	0.774	1.544	2.324	3.124	4.729	6.012	7.067	7.928
19.600	-0.000	0.775	1.547	2.326	3.126	4.729	6.010	7.065	7.926
19.700	-0.000	0.777	1.550	2.329	3.129	4.728	6.008	7.063	7.925
19.800	-0.000	0.778	1.553	2.332	3.132	4.728	6.006	7.061	7.923
19.900	-0.000	0.780	1.556	2.335	3.135	4.727	6.004	7.060	7.921
20.000	-0.000	0.781	1.559	2.338	3.138	4.727	6.002	7.058	7.920
20.100	-0.000	0.782	1.562	2.342	3.141	4.726	6.000	7.056	7.918
20.200	-0.000	0.784	1.565	2.345	3.145	4.726	5.998	7.054	7.916
20.300	-0.000	0.785	1.568	2.349	3.148	4.725	5.996	7.052	7.915
20.400	-0.000	0.787	1.571	2.352	3.152	4.723	5.994	7.051	7.913
20.500	-0.000	0.788	1.574	2.356	3.155	4.724	5.993	7.049	7.912
20.600	-0.000	0.790	1.577	2.360	3.159	4.724	5.992	7.048	7.910
20.700	-0.000	0.791	1.580	2.364	3.163	4.724	5.990	7.046	7.908
20.800	-0.000	0.793	1.583	2.368	3.167	4.723	5.988	7.044	7.907
20.900	-0.000	0.795	1.586	2.372	3.172	4.723	5.986	7.043	7.905
21.000	-0.000	0.796	1.589	2.376	3.176	4.723	5.985	7.041	7.904
21.100	-0.000	0.798	1.592	2.379	3.180	4.722	5.983	7.039	7.902
21.200	-0.000	0.799	1.596	2.385	3.185	4.722	5.981	7.038	7.901
21.300	-0.000	0.801	1.599	2.390	3.189	4.722	5.979	7.036	7.899
21.400	-0.000	0.803	1.602	2.395	3.193	4.721	5.974	7.034	7.898
21.500	-0.000	0.804	1.605	2.400	3.198	4.721	5.976	7.033	7.896
21.600	-0.000	0.806	1.608	2.404	3.202	4.721	5.974	7.031	7.895
21.700	-0.000	0.807	1.612	2.409	3.205	4.721	5.972	7.029	7.893
21.800	-0.000	0.809	1.615	2.414	3.211	4.720	5.970	7.028	7.892
21.900	-0.000	0.811	1.618	2.419	3.216	4.720	5.969	7.026	7.890
22.000	-0.000	0.812	1.622	2.424	3.221	4.720	5.967	7.024	7.889
22.100	-0.000	0.814	1.625	2.429	3.225	4.720	5.965	7.023	7.888
22.200	-0.000	0.816	1.628	2.434	3.230	4.720	5.964	7.021	7.886
22.300	-0.000	0.818	1.632	2.439	3.232	4.720	5.962	7.020	7.885
22.400	-0.000	0.819	1.635	2.445	3.239	4.720	5.960	7.018	7.884
22.500	-0.000	0.821	1.639	2.450	3.243	4.720	5.959	7.017	7.883
22.600	-0.000	0.823	1.642	2.455	3.248	4.720	5.957	7.015	7.882
22.700	-0.000	0.825	1.646	2.460	3.253	4.720	5.956	7.014	7.880
22.800	-0.000	0.826	1.649	2.465	3.257	4.720	5.954	7.013	7.879
22.900	-0.000	0.828	1.653	2.471	3.262	4.720	5.953	7.011	7.878
23.000	-0.000	0.830	1.656	2.476	3.266	4.721	5.951	7.010	7.877
23.100	-0.000	0.832	1.660	2.481	3.271	4.721	5.950	7.009	7.876
23.200	-0.000	0.833	1.664	2.486	3.275	4.721	5.949	7.007	7.876
23.300	-0.000	0.835	1.667	2.491	3.279	4.722	5.947	7.006	7.875
23.400	-0.000	0.837	1.671	2.496	3.284	4.722	5.946	7.005	7.874

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.839	1.675	2.501	3.288	4.722	5.945	7.004	7.873
23.600	-0.000	0.841	1.678	2.505	3.292	4.723	5.944	7.003	7.872
23.700	-0.000	0.843	1.682	2.510	3.296	4.723	5.943	7.002	7.871
23.800	-0.000	0.845	1.686	2.514	3.300	4.724	5.942	7.002	7.871
23.900	-0.000	0.846	1.689	2.518	3.303	4.724	5.941	7.001	7.870
24.000	-0.000	0.848	1.693	2.522	3.307	4.725	5.940	7.000	7.870
24.100	-0.000	0.850	1.697	2.526	3.310	4.725	5.939	7.000	7.869
24.200	-0.000	0.852	1.701	2.530	3.313	4.726	5.938	6.999	7.869
24.300	-0.000	0.854	1.705	2.533	3.316	4.726	5.938	6.998	7.868
24.400	-0.000	0.856	1.709	2.537	3.319	4.726	5.937	6.997	7.868
24.500	-0.000	0.858	1.713	2.540	3.322	4.727	5.937	6.997	7.868
24.600	-0.000	0.860	1.716	2.543	3.324	4.727	5.937	6.996	7.868
24.700	-0.000	0.862	1.720	2.546	3.327	4.727	5.937	6.996	7.867
24.800	-0.000	0.864	1.724	2.549	3.329	4.727	5.937	6.995	7.867
24.900	-0.000	0.866	1.728	2.551	3.330	4.728	5.938	6.995	7.868
25.000	-0.000	0.868	1.731	2.553	3.332	4.728	5.938	6.995	7.868
25.100	-0.000	0.870	1.734	2.555	3.333	4.728	5.939	6.995	7.868
25.200	-0.000	0.872	1.737	2.557	3.334	4.727	5.942	6.995	7.868
25.300	-0.000	0.874	1.740	2.559	3.335	4.727	5.943	6.995	7.869
25.400	-0.000	0.876	1.743	2.560	3.336	4.727	5.944	6.995	7.870
25.500	-0.000	0.878	1.745	2.561	3.336	4.727	5.946	6.996	7.870
25.600	-0.000	0.880	1.747	2.562	3.336	4.726	5.948	6.996	7.871
25.700	-0.000	0.882	1.749	2.562	3.336	4.725	5.949	6.997	7.872
25.800	-0.000	0.884	1.750	2.562	3.336	4.725	5.950	6.999	7.873
25.900	-0.000	0.886	1.751	2.562	3.335	4.724	5.952	7.000	7.875
26.000	-0.000	0.889	1.752	2.561	3.334	4.723	5.953	7.002	7.876

KN AS A FUNCTION OF DRAUGHT AND HEELING ANGLE

INITIAL TRIM: 6.00 M

UNIT: m

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INITIAL DRAUGHT	HEELING ANGLE (DEGREES)								
	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
0.000	-0.002	6.847	9.325	10.667	11.548	12.603	13.156	13.427	13.560
0.100	-0.002	6.826	9.362	10.739	11.642	12.727	13.295	13.529	13.573
0.200	-0.002	6.792	9.385	10.796	11.720	12.835	13.416	13.612	13.580
0.300	-0.001	6.747	9.396	10.840	11.785	12.927	13.520	13.679	13.583
0.400	-0.001	6.694	9.398	10.873	11.838	13.006	13.609	13.734	13.582
0.500	-0.001	6.634	9.392	10.898	11.882	13.074	13.684	13.779	13.580
0.600	-0.001	6.567	9.378	10.913	11.917	13.132	13.748	13.814	13.576
0.700	-0.001	6.494	9.357	10.922	11.945	13.181	13.800	13.842	13.571
0.800	-0.001	6.415	9.331	10.924	11.965	13.224	13.844	13.864	13.566
0.900	-0.001	6.332	9.298	10.920	11.981	13.261	13.881	13.881	13.560
1.000	-0.001	6.243	9.260	10.911	11.990	13.293	13.912	13.893	13.554
1.100	-0.001	6.150	9.216	10.896	11.994	13.320	13.937	13.902	13.547
1.200	-0.001	6.052	9.167	10.876	11.994	13.342	13.958	13.908	13.541
1.300	-0.001	5.951	9.113	10.851	11.988	13.361	13.975	13.913	13.536
1.400	-0.001	5.845	9.052	10.820	11.978	13.376	13.988	13.915	13.531
1.500	-0.001	5.735	8.986	10.785	11.963	13.388	13.997	13.916	13.527
1.600	-0.001	5.622	8.913	10.743	11.943	13.394	14.003	13.915	13.523
1.700	-0.001	5.505	8.833	10.695	11.916	13.396	14.004	13.911	13.520
1.800	-0.001	5.384	8.746	10.641	11.883	13.392	14.000	13.905	13.519
1.900	-0.001	5.262	8.652	10.580	11.844	13.383	13.992	13.897	13.518
2.000	-0.001	5.138	8.553	10.514	11.800	13.368	13.979	13.887	13.518
2.100	-0.001	5.015	8.451	10.444	11.753	13.349	13.963	13.874	13.518
2.200	-0.001	4.892	8.345	10.370	11.701	13.327	13.944	13.860	13.518
2.300	-0.001	4.771	8.235	10.293	11.647	13.300	13.923	13.845	13.517
2.400	-0.001	4.651	8.124	10.213	11.589	13.270	13.898	13.828	13.516
2.500	-0.001	4.534	8.010	10.131	11.530	13.238	13.872	13.810	13.514
2.600	-0.001	4.420	7.896	10.048	11.469	13.204	13.844	13.791	13.510
2.700	-0.001	4.309	7.781	9.963	11.407	13.168	13.814	13.771	13.506
2.800	-0.001	4.201	7.664	9.877	11.343	13.130	13.782	13.750	13.500
2.900	-0.001	4.097	7.548	9.790	11.279	13.089	13.749	13.728	13.493
3.000	-0.001	3.997	7.432	9.703	11.214	13.048	13.715	13.706	13.486
3.100	-0.001	3.902	7.316	9.615	11.149	13.004	13.680	13.683	13.477
3.200	-0.001	3.811	7.202	9.527	11.083	12.960	13.643	13.660	13.467
3.300	-0.001	3.725	7.089	9.439	11.017	12.914	13.606	13.636	13.456
3.400	-0.001	3.643	6.977	9.351	10.950	12.866	13.567	13.612	13.443
3.500	-0.001	3.566	6.866	9.262	10.883	12.818	13.528	13.587	13.430
3.600	-0.001	3.492	6.758	9.173	10.817	12.768	13.487	13.562	13.415
3.700	-0.001	3.422	6.652	9.084	10.751	12.718	13.445	13.536	13.400
3.800	-0.001	3.355	6.548	8.995	10.684	12.666	13.403	13.510	13.385
3.900	-0.001	3.291	6.447	8.906	10.618	12.613	13.359	13.484	13.367
4.000	-0.001	3.230	6.348	8.818	10.552	12.559	13.315	13.457	13.349
4.100	-0.001	3.172	6.251	8.731	10.485	12.505	13.270	13.430	13.329
4.200	-0.001	3.116	6.156	8.645	10.419	12.449	13.224	13.402	13.309
4.300	-0.001	3.063	6.064	8.559	10.351	12.392	13.178	13.374	13.287
4.400	-0.001	3.012	5.974	8.474	10.283	12.334	13.131	13.345	13.264
4.500	-0.001	2.964	5.887	8.389	10.215	12.276	13.083	13.316	13.241
4.600	-0.001	2.917	5.802	8.306	10.147	12.217	13.034	13.286	13.216
4.700	-0.001	2.873	5.720	8.224	10.078	12.157	12.985	13.256	13.192
4.800	-0.001	2.831	5.641	8.143	10.009	12.096	12.936	13.225	13.168
4.900	-0.001	2.790	5.564	8.063	9.939	12.034	12.886	13.193	13.143
5.000	-0.001	2.751	5.490	7.984	9.870	11.972	12.835	13.161	13.117
5.100	-0.001	2.714	5.418	7.906	9.801	11.909	12.783	13.128	13.091
5.200	-0.001	2.678	5.349	7.829	9.731	11.845	12.731	13.095	13.064
5.300	-0.001	2.643	5.282	7.754	9.662	11.780	12.678	13.060	13.037
5.400	-0.001	2.610	5.217	7.679	9.592	11.715	12.625	13.025	13.009

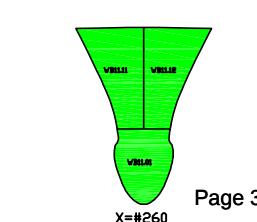
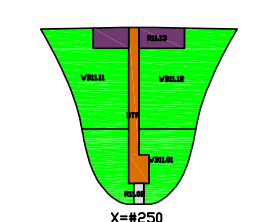
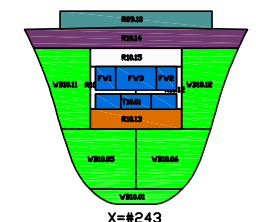
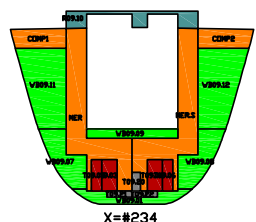
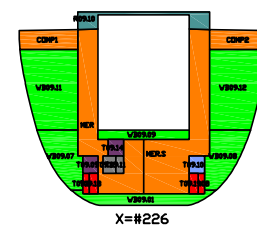
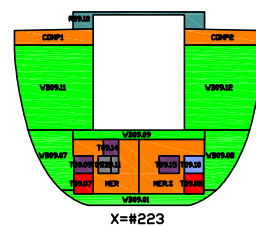
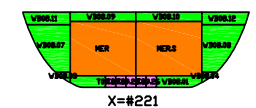
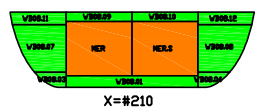
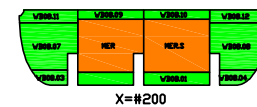
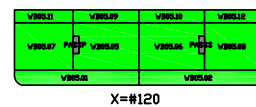
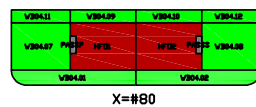
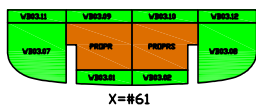
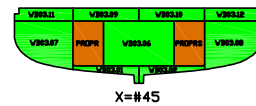
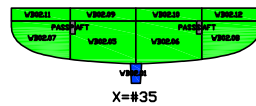
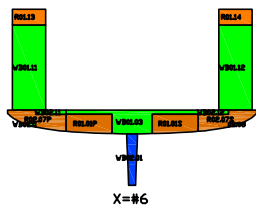
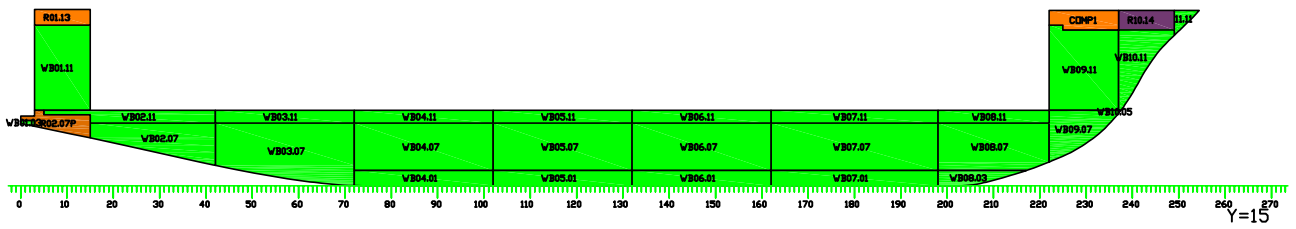
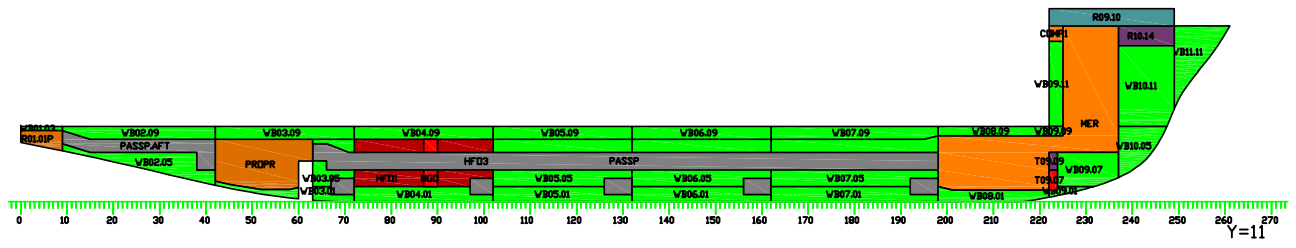
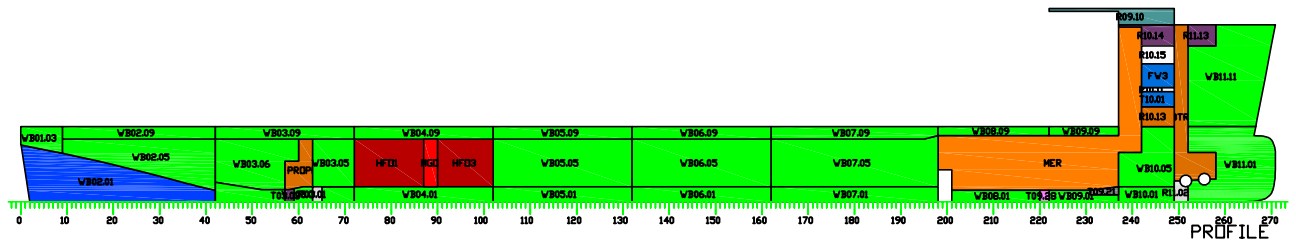
DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
5.500	-0.001	2.579	5.155	7.606	9.522	11.648	12.571	12.989	12.981
5.600	-0.001	2.548	5.094	7.534	9.452	11.581	12.516	12.952	12.952
5.700	-0.001	2.519	5.036	7.463	9.382	11.514	12.460	12.915	12.923
5.800	-0.001	2.490	4.980	7.393	9.312	11.445	12.403	12.877	12.893
5.900	-0.001	2.463	4.926	7.324	9.241	11.377	12.346	12.838	12.863
6.000	-0.001	2.436	4.873	7.257	9.170	11.307	12.289	12.798	12.833
6.100	-0.001	2.411	4.823	7.190	9.099	11.237	12.231	12.758	12.803
6.200	-0.001	2.386	4.773	7.123	9.027	11.167	12.173	12.717	12.772
6.300	-0.000	2.362	4.726	7.058	8.955	11.095	12.114	12.675	12.741
6.400	-0.000	2.338	4.680	6.994	8.883	11.023	12.055	12.633	12.709
6.500	-0.000	2.316	4.636	6.930	8.811	10.951	11.996	12.591	12.676
6.600	-0.000	2.294	4.593	6.868	8.738	10.878	11.937	12.548	12.643
6.700	-0.000	2.272	4.552	6.806	8.665	10.804	11.876	12.505	12.609
6.800	-0.000	2.252	4.512	6.744	8.591	10.730	11.816	12.462	12.575
6.900	-0.000	2.232	4.473	6.684	8.517	10.655	11.755	12.417	12.541
7.000	-0.000	2.213	4.436	6.624	8.443	10.579	11.694	12.372	12.506
7.100	-0.000	2.195	4.400	6.566	8.369	10.503	11.633	12.327	12.471
7.200	-0.000	2.177	4.365	6.508	8.294	10.427	11.571	12.280	12.436
7.300	-0.000	2.159	4.331	6.450	8.220	10.350	11.509	12.233	12.400
7.400	-0.000	2.143	4.298	6.393	8.145	10.273	11.447	12.185	12.364
7.500	-0.000	2.127	4.266	6.337	8.070	10.196	11.385	12.136	12.327
7.600	-0.000	2.111	4.235	6.281	7.995	10.118	11.323	12.087	12.289
7.700	-0.000	2.097	4.205	6.225	7.919	10.040	11.262	12.036	12.250
7.800	-0.000	2.082	4.176	6.170	7.844	9.962	11.200	11.984	12.209
7.900	-0.000	2.068	4.147	6.115	7.768	9.883	11.138	11.932	12.167
8.000	-0.000	2.055	4.119	6.061	7.693	9.804	11.076	11.879	12.123
8.100	-0.000	2.042	4.091	6.006	7.617	9.725	11.014	11.826	12.077
8.200	-0.000	2.030	4.064	5.952	7.541	9.646	10.952	11.772	12.030
8.300	-0.000	2.018	4.036	5.898	7.465	9.566	10.889	11.718	11.980
8.400	-0.000	2.006	4.009	5.845	7.389	9.486	10.827	11.663	11.929
8.500	-0.000	1.995	3.982	5.791	7.313	9.407	10.763	11.609	11.877
8.600	-0.000	1.984	3.955	5.737	7.237	9.326	10.700	11.553	11.823
8.700	-0.000	1.973	3.929	5.684	7.161	9.246	10.636	11.498	11.768
8.800	-0.000	1.963	3.902	5.630	7.085	9.166	10.571	11.442	11.712
8.900	-0.000	1.953	3.874	5.576	7.009	9.085	10.507	11.386	11.655
9.000	-0.000	1.943	3.847	5.522	6.933	9.003	10.441	11.329	11.596
9.100	-0.000	1.934	3.819	5.468	6.857	8.922	10.375	11.272	11.537
9.200	-0.000	1.925	3.791	5.414	6.781	8.840	10.308	11.215	11.476
9.300	-0.000	1.916	3.762	5.360	6.704	8.758	10.241	11.157	11.415
9.400	-0.000	1.908	3.733	5.305	6.628	8.676	10.173	11.098	11.353
9.500	-0.000	1.899	3.703	5.250	6.552	8.593	10.105	11.039	11.292
9.600	-0.000	1.891	3.673	5.195	6.476	8.510	10.036	10.979	11.230
9.700	-0.000	1.882	3.642	5.140	6.400	8.427	9.966	10.918	11.167
9.800	-0.000	1.874	3.610	5.084	6.325	8.344	9.896	10.855	11.104
9.900	-0.000	1.864	3.578	5.028	6.250	8.260	9.825	10.791	11.041
10.000	-0.000	1.855	3.544	4.971	6.175	8.177	9.753	10.726	10.977
10.100	-0.000	1.844	3.510	4.915	6.100	8.094	9.681	10.659	10.913
10.200	-0.000	1.833	3.475	4.857	6.025	8.010	9.608	10.590	10.848
10.300	-0.000	1.822	3.438	4.800	5.951	7.926	9.535	10.520	10.785
10.400	-0.000	1.809	3.401	4.741	5.877	7.843	9.460	10.449	10.720
10.500	-0.000	1.795	3.363	4.683	5.803	7.760	9.385	10.376	10.655
10.600	-0.000	1.780	3.324	4.623	5.729	7.676	9.309	10.301	10.589
10.700	-0.000	1.763	3.284	4.563	5.656	7.594	9.232	10.224	10.523
10.800	-0.000	1.746	3.243	4.503	5.583	7.511	9.155	10.146	10.457
10.900	-0.000	1.727	3.201	4.442	5.511	7.429	9.077	10.066	10.391
11.000	-0.000	1.707	3.159	4.382	5.439	7.348	8.999	9.986	10.325
11.100	-0.000	1.687	3.116	4.322	5.368	7.268	8.921	9.906	10.259
11.200	-0.000	1.665	3.074	4.262	5.298	7.188	8.843	9.826	10.192
11.300	-0.000	1.643	3.032	4.203	5.229	7.110	8.764	9.746	10.126
11.400	-0.000	1.621	2.990	4.145	5.161	7.033	8.687	9.665	10.059

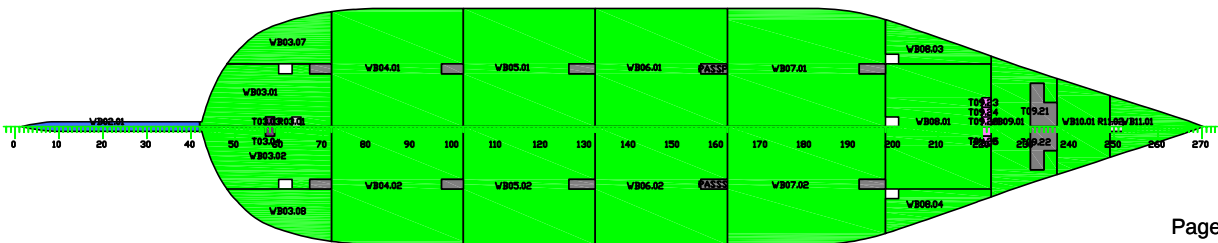
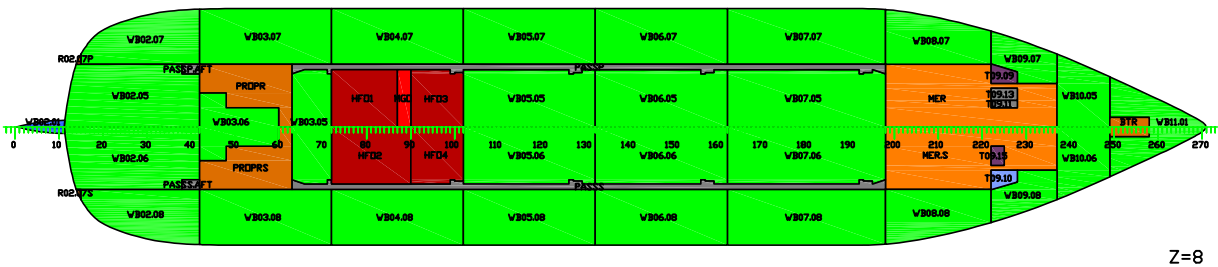
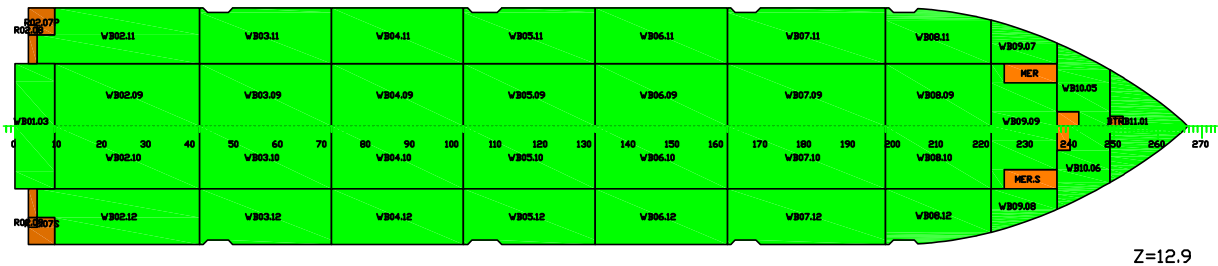
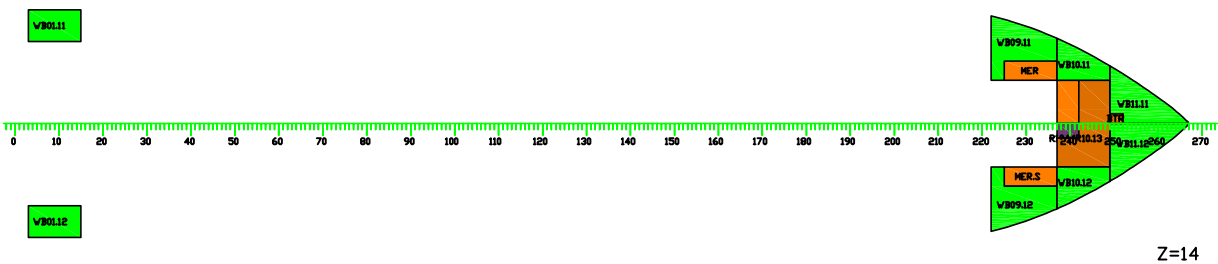
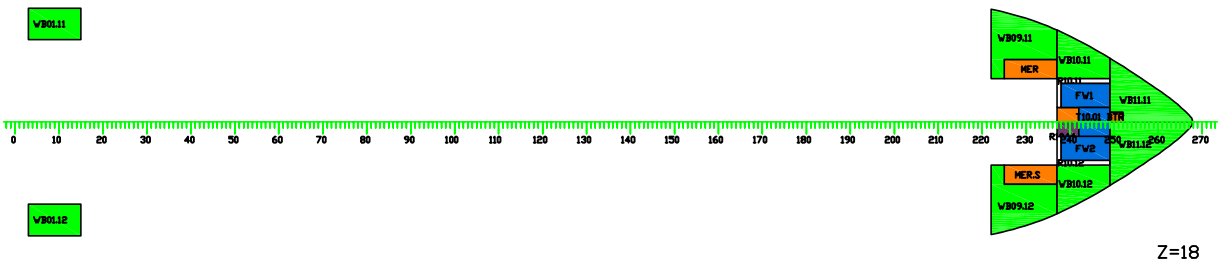
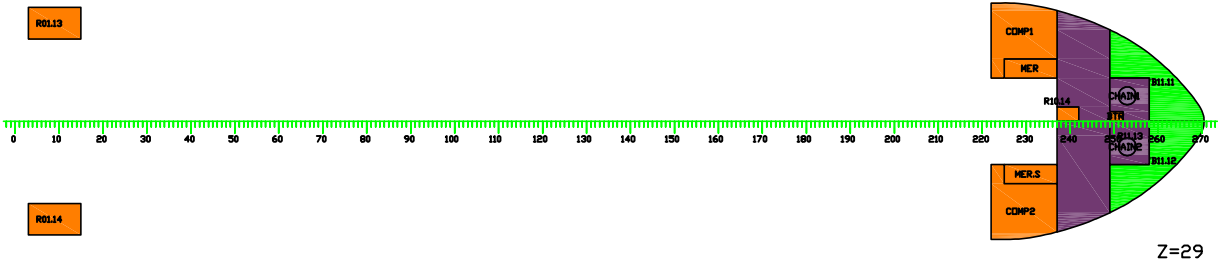
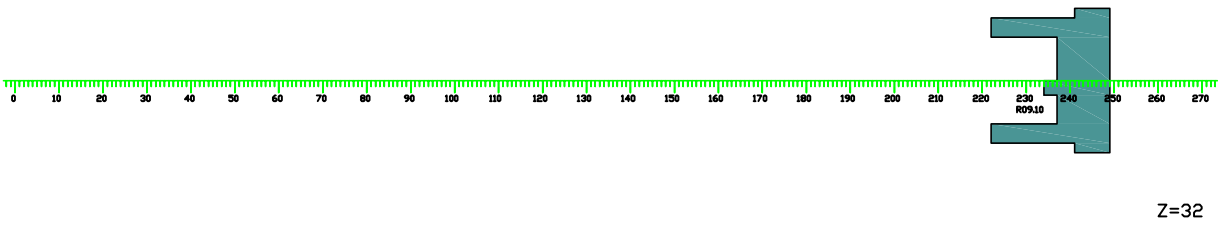
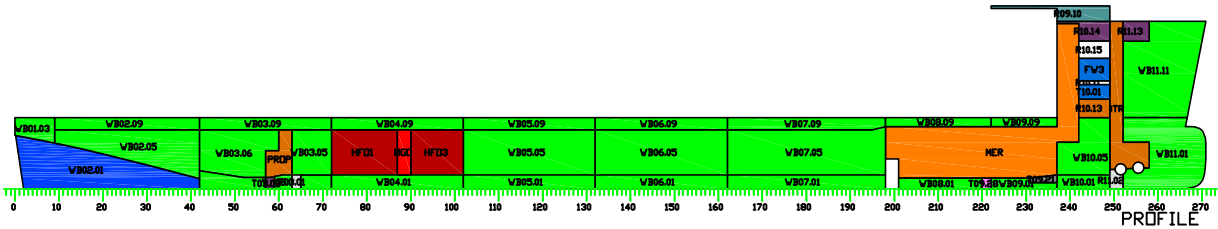
DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
11.500	-0.000	1.599	2.949	4.088	5.095	6.958	8.609	9.584	9.993
11.600	-0.000	1.577	2.909	4.032	5.030	6.883	8.532	9.504	9.927
11.700	-0.000	1.554	2.869	3.977	4.966	6.810	8.456	9.424	9.861
11.800	-0.000	1.532	2.830	3.923	4.903	6.738	8.379	9.345	9.795
11.900	-0.000	1.509	2.791	3.870	4.841	6.667	8.302	9.266	9.730
12.000	-0.000	1.487	2.752	3.817	4.780	6.597	8.225	9.188	9.665
12.100	-0.000	1.465	2.713	3.765	4.720	6.528	8.147	9.109	9.600
12.200	-0.000	1.443	2.675	3.713	4.660	6.460	8.070	9.031	9.535
12.300	-0.000	1.421	2.636	3.662	4.601	6.392	7.992	8.954	9.470
12.400	-0.000	1.400	2.597	3.611	4.543	6.326	7.914	8.877	9.407
12.500	-0.000	1.379	2.558	3.560	4.485	6.260	7.836	8.801	9.343
12.600	-0.000	1.358	2.520	3.510	4.428	6.196	7.758	8.726	9.280
12.700	-0.000	1.337	2.481	3.460	4.371	6.132	7.681	8.651	9.217
12.800	-0.000	1.317	2.442	3.411	4.315	6.069	7.603	8.576	9.155
12.900	-0.000	1.296	2.403	3.361	4.259	6.007	7.526	8.503	9.093
13.000	-0.000	1.276	2.364	3.312	4.204	5.945	7.450	8.430	9.032
13.100	-0.000	1.256	2.326	3.263	4.150	5.885	7.375	8.358	8.972
13.200	-0.000	1.236	2.287	3.215	4.096	5.826	7.301	8.287	8.912
13.300	-0.000	1.216	2.248	3.167	4.043	5.768	7.229	8.218	8.853
13.400	-0.000	1.195	2.210	3.119	3.991	5.711	7.159	8.150	8.796
13.500	-0.000	1.175	2.171	3.072	3.939	5.655	7.090	8.083	8.739
13.600	-0.000	1.154	2.132	3.025	3.888	5.601	7.023	8.018	8.684
13.700	-0.000	1.133	2.094	2.979	3.838	5.548	6.958	7.954	8.631
13.800	-0.000	1.112	2.056	2.934	3.790	5.496	6.894	7.893	8.580
13.900	-0.000	1.090	2.018	2.889	3.742	5.446	6.832	7.831	8.531
14.000	-0.000	1.068	1.980	2.845	3.695	5.398	6.772	7.774	8.484
14.100	-0.000	1.046	1.943	2.802	3.650	5.351	6.714	7.718	8.439
14.200	-0.000	1.024	1.907	2.760	3.605	5.305	6.658	7.665	8.396
14.300	-0.000	1.002	1.871	2.719	3.563	5.260	6.603	7.615	8.355
14.400	-0.000	0.980	1.837	2.680	3.521	5.215	6.551	7.566	8.316
14.500	-0.000	0.958	1.803	2.641	3.481	5.172	6.501	7.521	8.280
14.600	-0.000	0.937	1.770	2.604	3.443	5.130	6.453	7.478	8.246
14.700	-0.000	0.916	1.739	2.569	3.407	5.089	6.408	7.437	8.214
14.800	-0.000	0.896	1.709	2.535	3.372	5.049	6.365	7.399	8.184
14.900	-0.000	0.876	1.680	2.503	3.339	5.011	6.325	7.363	8.156
15.000	-0.000	0.857	1.654	2.473	3.308	4.974	6.288	7.330	8.130
15.100	-0.000	0.839	1.629	2.445	3.280	4.939	6.254	7.300	8.106
15.200	-0.000	0.823	1.607	2.420	3.254	4.907	6.223	7.272	8.085
15.300	-0.000	0.808	1.587	2.397	3.231	4.879	6.195	7.247	8.066
15.400	-0.000	0.794	1.570	2.376	3.210	4.853	6.171	7.225	8.050
15.500	-0.000	0.782	1.554	2.358	3.192	4.831	6.149	7.205	8.035
15.600	-0.000	0.772	1.542	2.343	3.176	4.812	6.131	7.188	8.023
15.700	-0.000	0.764	1.532	2.330	3.163	4.797	6.115	7.174	8.013
15.800	-0.000	0.758	1.524	2.320	3.153	4.785	6.104	7.163	8.005
15.900	-0.000	0.753	1.519	2.313	3.145	4.776	6.094	7.155	7.999
16.000	-0.000	0.750	1.515	2.307	3.139	4.770	6.088	7.149	7.994
16.100	-0.000	0.749	1.514	2.304	3.135	4.767	6.084	7.144	7.991
16.200	-0.000	0.748	1.513	2.302	3.133	4.766	6.082	7.142	7.989
16.300	-0.000	0.748	1.513	2.301	3.132	4.765	6.080	7.140	7.987
16.400	-0.000	0.748	1.513	2.300	3.130	4.764	6.078	7.137	7.985
16.500	-0.000	0.747	1.513	2.299	3.128	4.763	6.077	7.135	7.984
16.600	-0.000	0.747	1.513	2.298	3.126	4.762	6.075	7.133	7.982
16.700	-0.000	0.747	1.513	2.297	3.125	4.761	6.073	7.130	7.980
16.800	-0.000	0.747	1.513	2.297	3.123	4.760	6.071	7.128	7.978
16.900	-0.000	0.747	1.514	2.297	3.122	4.759	6.069	7.126	7.977
17.000	-0.000	0.747	1.514	2.296	3.121	4.758	6.067	7.123	7.975
17.100	-0.000	0.748	1.514	2.297	3.120	4.757	6.065	7.121	7.973
17.200	-0.000	0.748	1.515	2.297	3.119	4.756	6.063	7.119	7.971
17.300	-0.000	0.749	1.515	2.297	3.118	4.755	6.062	7.117	7.969
17.400	-0.000	0.749	1.516	2.298	3.117	4.754	6.060	7.114	7.968

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
17.500	-0.000	0.750	1.517	2.298	3.117	4.753	6.058	7.113	7.966
17.600	-0.000	0.751	1.517	2.299	3.116	4.753	6.056	7.111	7.964
17.700	-0.000	0.753	1.518	2.300	3.116	4.752	6.054	7.109	7.962
17.800	-0.000	0.754	1.519	2.301	3.115	4.751	6.052	7.107	7.961
17.900	-0.000	0.755	1.521	2.302	3.115	4.750	6.050	7.105	7.959
18.000	-0.000	0.756	1.522	2.304	3.115	4.750	6.048	7.102	7.957
18.100	-0.000	0.757	1.523	2.306	3.116	4.749	6.046	7.100	7.955
18.200	-0.000	0.759	1.523	2.307	3.116	4.748	6.044	7.098	7.954
18.300	-0.000	0.760	1.525	2.308	3.117	4.747	6.042	7.096	7.952
18.400	-0.000	0.761	1.527	2.310	3.118	4.747	6.041	7.094	7.950
18.500	-0.000	0.762	1.528	2.312	3.119	4.746	6.039	7.092	7.948
18.600	-0.000	0.764	1.530	2.313	3.120	4.745	6.037	7.090	7.947
18.700	-0.000	0.765	1.531	2.315	3.121	4.745	6.035	7.088	7.945
18.800	-0.000	0.767	1.533	2.317	3.122	4.745	6.033	7.086	7.943
18.900	-0.000	0.768	1.535	2.319	3.123	4.744	6.031	7.084	7.941
19.000	-0.000	0.769	1.537	2.321	3.125	4.743	6.029	7.082	7.940
19.100	-0.000	0.771	1.540	2.324	3.127	4.742	6.027	7.080	7.938
19.200	-0.000	0.772	1.542	2.326	3.129	4.742	6.025	7.078	7.936
19.300	-0.000	0.773	1.544	2.329	3.131	4.741	6.023	7.076	7.934
19.400	-0.000	0.775	1.547	2.331	3.133	4.741	6.021	7.074	7.933
19.500	-0.000	0.776	1.550	2.334	3.136	4.740	6.019	7.072	7.931
19.600	-0.000	0.778	1.552	2.337	3.139	4.739	6.017	7.071	7.929
19.700	-0.000	0.779	1.555	2.339	3.140	4.739	6.015	7.069	7.928
19.800	-0.000	0.781	1.558	2.342	3.144	4.738	6.013	7.067	7.926
19.900	-0.000	0.782	1.561	2.345	3.148	4.738	6.011	7.065	7.924
20.000	-0.000	0.784	1.564	2.348	3.151	4.737	6.009	7.063	7.923
20.100	-0.000	0.785	1.567	2.352	3.154	4.737	6.007	7.062	7.921
20.200	-0.000	0.787	1.571	2.355	3.158	4.736	6.005	7.060	7.919
20.300	-0.000	0.788	1.573	2.358	3.161	4.736	6.004	7.057	7.918
20.400	-0.000	0.790	1.576	2.362	3.165	4.735	6.002	7.057	7.916
20.500	-0.000	0.791	1.580	2.366	3.169	4.735	6.001	7.055	7.914
20.600	-0.000	0.793	1.582	2.369	3.173	4.735	5.999	7.053	7.913
20.700	-0.000	0.794	1.585	2.373	3.177	4.734	5.997	7.052	7.911
20.800	-0.000	0.796	1.588	2.377	3.181	4.734	5.995	7.050	7.910
20.900	-0.000	0.797	1.592	2.381	3.185	4.733	5.993	7.048	7.908
21.000	-0.000	0.798	1.594	2.386	3.189	4.733	5.991	7.046	7.907
21.100	-0.000	0.801	1.598	2.390	3.194	4.733	5.989	7.045	7.905
21.200	-0.000	0.802	1.601	2.394	3.198	4.732	5.988	7.043	7.903
21.300	-0.000	0.804	1.604	2.399	3.202	4.732	5.986	7.041	7.902
21.400	-0.000	0.805	1.608	2.404	3.207	4.732	5.981	7.039	7.900
21.500	-0.000	0.807	1.611	2.408	3.211	4.731	5.982	7.038	7.899
21.600	-0.000	0.809	1.614	2.413	3.215	4.731	5.980	7.036	7.897
21.700	-0.000	0.810	1.617	2.418	3.219	4.731	5.979	7.035	7.896
21.800	-0.000	0.812	1.621	2.423	3.224	4.731	5.976	7.033	7.895
21.900	-0.000	0.814	1.624	2.428	3.228	4.731	5.975	7.031	7.893
22.000	-0.000	0.815	1.626	2.433	3.232	4.730	5.974	7.030	7.892
22.100	-0.000	0.817	1.631	2.437	3.237	4.730	5.972	7.028	7.891
22.200	-0.000	0.819	1.634	2.443	3.241	4.730	5.970	7.027	7.889
22.300	-0.000	0.821	1.638	2.448	3.243	4.730	5.969	7.025	7.888
22.400	-0.000	0.822	1.641	2.454	3.250	4.730	5.967	7.024	7.887
22.500	-0.000	0.824	1.645	2.459	3.254	4.730	5.965	7.022	7.886
22.600	-0.000	0.826	1.648	2.464	3.258	4.730	5.964	7.021	7.885
22.700	-0.000	0.828	1.652	2.469	3.262	4.730	5.962	7.019	7.884
22.800	-0.000	0.829	1.655	2.475	3.266	4.731	5.961	7.018	7.883
22.900	-0.000	0.831	1.659	2.480	3.271	4.731	5.959	7.017	7.882
23.000	-0.000	0.833	1.663	2.485	3.275	4.731	5.958	7.015	7.881
23.100	-0.000	0.835	1.666	2.490	3.279	4.731	5.957	7.014	7.880
23.200	-0.000	0.837	1.670	2.495	3.283	4.731	5.955	7.013	7.879
23.300	-0.000	0.839	1.674	2.499	3.287	4.732	5.954	7.012	7.878
23.400	-0.000	0.840	1.677	2.504	3.291	4.732	5.953	7.011	7.877

DRAUGHT	0.0	5.0	10.0	15.0	20.0	30.0	40.0	50.0	60.0
23.500	-0.000	0.842	1.681	2.508	3.295	4.732	5.952	7.011	7.877
23.600	-0.000	0.844	1.685	2.513	3.298	4.733	5.951	7.010	7.876
23.700	-0.000	0.846	1.689	2.517	3.302	4.733	5.950	7.009	7.875
23.800	-0.000	0.848	1.692	2.521	3.305	4.733	5.949	7.008	7.875
23.900	-0.000	0.850	1.696	2.525	3.309	4.734	5.948	7.007	7.874
24.000	-0.000	0.852	1.700	2.529	3.312	4.734	5.947	7.006	7.874
24.100	-0.000	0.854	1.704	2.532	3.315	4.734	5.946	7.005	7.873
24.200	-0.000	0.856	1.708	2.536	3.318	4.735	5.946	7.004	7.873
24.300	-0.000	0.858	1.712	2.539	3.320	4.735	5.945	7.004	7.873
24.400	-0.000	0.860	1.715	2.542	3.323	4.735	5.945	7.003	7.872
24.500	-0.000	0.861	1.719	2.545	3.325	4.735	5.945	7.003	7.872
24.600	-0.000	0.863	1.723	2.548	3.327	4.735	5.945	7.002	7.872
24.700	-0.000	0.865	1.727	2.550	3.329	4.735	5.945	7.002	7.872
24.800	-0.000	0.867	1.730	2.552	3.331	4.735	5.945	7.001	7.872
24.900	-0.000	0.869	1.733	2.554	3.332	4.734	5.946	7.001	7.872
25.000	-0.000	0.872	1.737	2.556	3.333	4.734	5.948	7.001	7.873
25.100	-0.000	0.874	1.739	2.558	3.334	4.734	5.949	7.001	7.873
25.200	-0.000	0.876	1.742	2.559	3.334	4.733	5.950	7.001	7.874
25.300	-0.000	0.878	1.744	2.560	3.335	4.733	5.951	7.002	7.875
25.400	-0.000	0.880	1.746	2.561	3.335	4.732	5.953	7.002	7.875
25.500	-0.000	0.882	1.748	2.561	3.335	4.732	5.954	7.003	7.876
25.600	-0.000	0.884	1.749	2.561	3.335	4.731	5.955	7.004	7.878
25.700	-0.000	0.886	1.751	2.561	3.335	4.730	5.956	7.005	7.879
25.800	-0.000	0.888	1.751	2.560	3.334	4.729	5.957	7.007	7.880
25.900	-0.000	0.890	1.752	2.559	3.333	4.728	5.958	7.009	7.882
26.000	-0.000	0.892	1.752	2.558	3.332	4.727	5.960	7.011	7.883

5.3 Tank data





Explanation of filling colours:

	MAP
	CHA
	MMA
	FW
	HFO
	CORR
	DRY
	ACC
	VOID
	STO
	LO
	DO
	TFW
	GWA
	BIW
	SLU
	MIS
	BW
	FB

5.3.1 Tank information tables

NAME	DES	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	FRMIN #	FRMAX #	IY*RHO tm	IY/IMO tm
CAPACITY OF Ballast Water (RHO=1.025)										
WB01.03	Water Ballast 01.03	352.3	345.2	3.19	0.00	11.41	0.00	9.00	1139.0	4889
WB01.11	Water Ballast 01.11	811.4	795.2	7.20	17.83	20.35	3.00	15.00	155.9	153
WB01.12	Water Ballast 01.12	811.4	795.2	7.20-17.83	20.35		3.00	15.00	155.9	153
WB02.05	Water Ballast 02.05	1517.9	1487.6	23.01	5.35	7.78	9.00	42.00	2872.6	1403
WB02.06	Water Ballast 02.06	1519.8	1489.4	23.02	-5.35	7.78	9.00	42.00	2883.5	1405
WB02.07	Water Ballast 02.07	947.8	928.9	24.37	15.78	8.38	1.56	42.00	1437.0	622
WB02.08	Water Ballast 02.08	947.8	928.9	24.37-15.78	8.38		1.56	42.00	1437.0	622
WB02.09	Water Ballast 02.09	654.7	641.6	20.51	5.65	11.90	9.00	42.00	3257.4	3209
WB02.10	Water Ballast 02.10	654.7	641.6	20.51	-5.65	11.90	9.00	42.00	3257.4	3209
WB02.11	Water Ballast 02.11	560.3	549.1	20.84	16.51	11.95	5.00	42.00	2104.2	2015
WB02.12	Water Ballast 02.12	560.3	549.1	20.84-16.51	11.95		5.00	42.00	2104.2	2015
WB03.01	Water Ballast 03.01	480.2	470.6	47.42	5.24	1.46	42.00	72.00	2067.3	973
WB03.02	Water Ballast 03.02	487.7	478.0	47.48	-5.17	1.46	42.00	72.00	2125.5	996
WB03.05	Water Ballast 03.05	1262.9	1237.6	53.97	0.00	6.71	63.00	72.00	5953.7	6470
WB03.06	Water Ballast 03.06	956.8	937.7	39.47	0.00	6.85	42.00	60.00	1034.9	1091
WB03.07	Water Ballast 03.07	2118.9	2076.5	46.50	16.11	6.30	42.00	72.00	2051.3	1513
WB03.08	Water Ballast 03.08	2118.9	2076.5	46.50-16.11	6.30		42.00	72.00	2051.3	1513
WB03.09	Water Ballast 03.09	600.6	588.6	45.60	5.69	11.90	42.00	72.00	3017.2	2957
WB03.10	Water Ballast 03.10	600.6	588.6	45.60	-5.69	11.90	42.00	72.00	3017.2	2957
WB03.11	Water Ballast 03.11	531.0	520.4	45.66	16.41	11.90	42.00	72.00	2127.8	2064
WB03.12	Water Ballast 03.12	531.0	520.4	45.66-16.41	11.90		42.00	72.00	2127.8	2064
WB04.01	Water Ballast 04.01	1303.0	1276.9	69.62	10.53	1.31	72.00	102.00	19690.1	19111
WB04.02	Water Ballast 04.02	1303.0	1276.9	69.62-10.53	1.31		72.00	102.00	19690.1	19111
WB04.07	Water Ballast 04.07	1992.6	1952.8	69.60	16.44	6.70	72.00	102.00	2127.8	2085
WB04.08	Water Ballast 04.08	1992.6	1952.8	69.60-16.44	6.70		72.00	102.00	2127.8	2085
WB04.09	Water Ballast 04.09	600.6	588.6	69.60	5.69	11.90	72.00	102.00	3017.2	2957
WB04.10	Water Ballast 04.10	600.6	588.6	69.60	-5.69	11.90	72.00	102.00	3017.2	2957
WB04.11	Water Ballast 04.11	534.6	523.9	69.60	16.44	11.90	72.00	102.00	2127.8	2085
WB04.12	Water Ballast 04.12	534.6	523.9	69.60-16.44	11.90		72.00	102.00	2127.8	2085
WB05.01	Water Ballast 05.01	1314.4	1288.1	93.51	10.63	1.30	102.00	132.00	20130.2	19362
WB05.02	Water Ballast 05.02	1314.4	1288.1	93.51-10.63	1.30		102.00	132.00	20130.2	19362
WB05.05	Water Ballast 05.05	2149.2	2106.2	93.53	5.48	6.71	102.00	132.00	2254.9	2782
WB05.06	Water Ballast 05.06	2149.2	2106.2	93.53	-5.48	6.71	102.00	132.00	2254.9	2782
WB05.07	Water Ballast 05.07	1992.6	1952.7	93.60	16.44	6.70	102.00	132.00	2127.8	2085
WB05.08	Water Ballast 05.08	1992.6	1952.7	93.60-16.44	6.70		102.00	132.00	2127.8	2085

NAME	DES		VOLM m3	VNET m3	CGX m	CGY m	CGZ m	FRMIN #	FRMAX #	IY*RHO tm	IY/IMO tm
WB05.09	Water Ballast	05.09	600.6	588.6	93.60	5.69	11.90	102.00	132.00	3017.2	2957
WB05.10	Water Ballast	05.10	600.6	588.6	93.60	-5.69	11.90	102.00	132.00	3017.2	2957
WB05.11	Water Ballast	05.11	530.4	519.8	93.66	16.40	11.90	102.00	132.00	2127.8	2061
WB05.12	Water Ballast	05.12	530.4	519.8	93.66	-16.40	11.90	102.00	132.00	2127.8	2061
WB06.01	Water Ballast	06.01	1314.4	1288.1	117.51	10.63	1.30	132.00	162.00	20130.3	19362
WB06.02	Water Ballast	06.02	1314.4	1288.1	117.51	-10.63	1.30	132.00	162.00	20130.3	19362
WB06.05	Water Ballast	06.05	2149.2	2106.2	117.53	5.48	6.71	132.00	162.00	2254.9	2782
WB06.06	Water Ballast	06.06	2149.2	2106.2	117.53	-5.48	6.71	132.00	162.00	2254.9	2782
WB06.07	Water Ballast	06.07	1992.6	1952.7	117.60	16.44	6.70	132.00	162.00	2127.8	2085
WB06.08	Water Ballast	06.08	1992.6	1952.7	117.60	-16.44	6.70	132.00	162.00	2127.8	2085
WB06.09	Water Ballast	06.09	600.6	588.6	117.60	5.69	11.90	132.00	162.00	3017.2	2957
WB06.10	Water Ballast	06.10	600.6	588.6	117.60	-5.69	11.90	132.00	162.00	3017.2	2957
WB06.11	Water Ballast	06.11	534.6	523.9	117.60	16.44	11.90	132.00	162.00	2127.8	2085
WB06.12	Water Ballast	06.12	534.6	523.9	117.60	-16.44	11.90	132.00	162.00	2127.8	2085
WB07.01	Water Ballast	07.01	1548.7	1517.8	143.70	10.44	1.31	162.00	198.00	22953.9	22614
WB07.02	Water Ballast	07.02	1548.7	1517.8	143.70	-10.44	1.31	162.00	198.00	22953.9	22614
WB07.05	Water Ballast	07.05	2588.1	2536.4	143.95	5.48	6.72	162.00	198.00	2692.1	3018
WB07.06	Water Ballast	07.06	2588.1	2536.4	143.95	-5.48	6.72	162.00	198.00	2692.1	3018
WB07.07	Water Ballast	07.07	2379.9	2332.3	143.94	16.42	6.71	162.00	198.00	2538.6	2487
WB07.08	Water Ballast	07.08	2379.9	2332.3	143.94	-16.42	6.71	162.00	198.00	2538.6	2487
WB07.09	Water Ballast	07.09	712.5	698.3	143.84	5.69	11.91	162.00	198.00	3620.7	3488
WB07.10	Water Ballast	07.10	712.5	698.3	143.84	-5.69	11.91	162.00	198.00	3620.7	3488
WB07.11	Water Ballast	07.11	637.9	625.2	144.06	16.41	11.90	162.00	198.00	2553.4	2481
WB07.12	Water Ballast	07.12	637.9	625.2	144.06	-16.41	11.90	162.00	198.00	2553.4	2481
WB08.01	Water Ballast	08.01	850.2	833.2	167.59	-0.04	1.03	198.00	222.00	19202.7	12257
WB08.03	Water Ballast	08.03	188.8	185.0	165.55	14.11	1.49	198.00	222.00	284.5	292
WB08.04	Water Ballast	08.04	188.8	185.0	165.55	-14.11	1.49	198.00	222.00	284.5	292
WB08.07	Water Ballast	08.07	1194.2	1170.3	167.13	15.42	7.04	198.00	222.00	966.9	1081
WB08.08	Water Ballast	08.08	1194.2	1170.3	167.13	-15.42	7.04	198.00	222.00	966.9	1081
WB08.09	Water Ballast	08.09	349.4	342.5	168.00	5.69	12.20	198.00	222.00	2413.8	2366
WB08.10	Water Ballast	08.10	349.4	342.5	168.00	-5.69	12.20	198.00	222.00	2413.8	2366
WB08.11	Water Ballast	08.11	389.6	381.8	167.62	16.01	11.90	198.00	222.00	1371.9	1450
WB08.12	Water Ballast	08.12	389.6	381.8	167.62	-16.01	11.90	198.00	222.00	1371.9	1450
WB09.01	Water Ballast	09.01	436.6	427.9	182.90	-0.00	1.12	222.00	237.00	7426.9	7489
WB09.07	Water Ballast	09.07	630.5	617.9	183.26	12.72	8.16	222.00	237.00	408.3	443
WB09.08	Water Ballast	09.08	630.5	617.9	183.26	-12.72	8.16	222.00	237.00	408.3	443
WB09.09	Water Ballast	09.09	329.3	322.7	183.21	0.00	12.20	222.00	237.00	5617.5	7741
WB09.11	Water Ballast	09.11	1459.9	1430.7	182.74	14.97	20.53	222.00	237.00	885.2	1203

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NAME	DES	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	FRMIN #	FRMAX #	IY*RHO tm	IY/IMO tm
WB09.12	Water Ballast 09.12	1459.9	1430.7	182.74	-14.97	20.53	222.00	237.00	885.2	1203
WB10.01	Water Ballast 10.01	312.8	306.6	194.02	0.00	1.44	237.00	249.00	2069.7	2981
WB10.05	Water Ballast 10.05	1041.4	1020.5	194.21	5.79	8.04	237.00	249.00	1247.6	1599
WB10.06	Water Ballast 10.06	1041.4	1020.5	194.25	-5.75	8.04	237.00	249.00	1247.5	1599
WB10.11	Water Ballast 10.11	951.9	932.8	193.86	11.70	20.71	237.00	249.00	399.6	602
WB10.12	Water Ballast 10.12	951.9	932.8	193.86	-11.70	20.71	237.00	249.00	399.6	602
WB11.01	Water Ballast 11.01	1632.3	1599.7	205.41	-0.01	7.51	249.00	270.88	2483.2	2634
WB11.11	Water Ballast 11.11	1896.0	1858.1	205.72	5.21	22.57	249.00	270.84	1401.9	1629
WB11.12	Water Ballast 11.12	1954.4	1915.3	205.56	-5.08	22.49	249.00	270.84	1455.6	1705
SUBTOTAL		89803.4	88007.3	111.50	-0.00	8.62			343266.6	

CAPACITY OF Heavy Fuel Oil (RH0=0.991)

HF01	HFO Storage Tank 1	1083.3	1061.6	63.60	5.52	6.69	72.00	87.00	1106.7	1361
HF02	HFO Storage Tank 2	1300.0	1274.0	64.80	-5.52	6.69	72.00	90.00	1328.1	1633
HF03	HFO Storage Tank 3	851.3	834.2	76.75	5.43	6.73	90.00	102.00	851.8	1060
HF04	HFO Storage Tank 4	851.3	834.2	76.75	-5.43	6.73	90.00	102.00	851.8	1060
T09.03	HFO Settling Tank P	57.2	56.1	186.77	3.94	4.97	231.00	236.00	6.0	5
T09.04	HFO Settling Tank S	45.4	44.5	187.18	-3.94	5.00	232.00	236.00	4.8	4
T09.05	HFO Service Tank P	38.1	37.4	186.77	6.13	4.97	231.00	236.00	1.8	2
T09.06	HFO Service Tank S	30.2	29.6	187.18	-6.13	5.00	232.00	236.00	1.4	1
SUBTOTAL		4256.8	4171.6	74.18	-0.26	6.64			4152.2	

CAPACITY OF Diesel Oil (RH0=0.9)

T09.07	MDO Storage P	35.2	34.5	179.30	9.58	3.75	222.00	228.00	6.5	7
T09.08	MDO Storage S	35.2	34.5	179.30	-9.58	3.75	222.00	228.00	6.5	7
MGO	MGO Storage Tank	216.7	212.3	70.80	5.52	6.69	87.00	90.00	201.0	247
T09.18	MDO Service P	12.6	12.3	181.20	8.63	3.75	225.00	228.00	0.6	1
T09.19	MDO Service S	12.6	12.3	181.20	-8.63	3.75	225.00	228.00	0.6	1
SUBTOTAL		312.2	306.0	104.17	3.83	5.79			215.3	

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CAPACITY OF Lubricating Oil (RHO=0.9)										
T03.03	L.O. DRAIN TK.	5.8	5.7	46.41	0.86	1.11	57.00	59.00	0.6	1
T03.04	ST L.O. DRAIN TK.	5.8	5.7	46.41	-0.86	1.11	57.00	59.00	0.6	1
T09.09	Used LO	41.0	40.2	179.80	9.33	7.00	222.00	228.00	9.3	11
T09.14	NO.1 M/E & A/E L.O. STORAGE TANK	36.5	35.8	180.00	4.81	9.95	222.00	228.00	6.5	6
T09.15	NO.2 M/E & A/E L.O. STORAGE TANK	25.2	24.7	178.80	-5.25	7.00	222.00	225.00	7.7	8
SUBTOTAL										
		114.3	112.0	166.11	3.72	7.34			24.8	
CAPACITY OF Fresh Water (RHO=1)										
FW1	Fresh Water P	223.7	219.2	194.91	4.97	19.93	238.00	249.00	38.1	42
FW2	Fresh Water S	223.7	219.2	194.91	-4.97	19.93	238.00	249.00	38.1	42
FW3	Fresh Water C	159.7	156.5	196.40	0.00	21.76	242.00	249.00	160.1	157
T10.01	Potable Water	77.2	75.6	196.40	0.00	17.71	242.00	249.00	67.5	66
SUBTOTAL										
		684.4	670.7	195.43	0.00	20.11			303.7	
CAPACITY OF Technical Water (RHO=1)										
T09.10	Feed Water	41.0	40.2	179.80	-9.33	7.00	222.00	228.00	10.4	12
CAPACITY OF Gray Water (RHO=1)										
T09.11	Black Water	18.9	18.5	180.00	4.16	7.00	222.00	228.00	0.9	1
T09.13	Gray Water	31.5	30.9	180.00	5.91	7.00	222.00	228.00	4.2	4
SUBTOTAL										
		50.4	49.4	180.00	5.25	7.00			5.1	

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NAME	DES	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	FRMIN #	FRMAX #	IY*RHO tm	IY/IMO tm
CAPACITY OF Bilge Water (RHO=1)										
T09.20	Bilge Water Settling	10.6	10.4	188.38	-0.73	3.97	234.00	237.00	0.6	1
T09.22	Bilge Water	31.1	30.5	187.07	-3.20	1.74	231.00	237.00	135.1	86
SUBTOTAL		41.7	40.9	187.40	-2.57	2.31			135.7	
CAPACITY OF Sludge (RHO=2.38)										
T09.21	Sludge	31.1	30.5	187.07	3.20	1.74	231.00	237.00	321.5	205
CAPACITY OF Miscellaneous (RHO=1)										
T09.23	Clean F.O. Drain Tank P	5.6	5.5	176.80	4.37	1.00	220.00	222.00	0.7	1
T09.24	Dirty Oil Drain Tank P	5.6	5.5	176.80	2.62	1.00	220.00	222.00	0.7	1
T09.25	Dirty Oil Drain Tank S	5.6	5.5	176.80	-0.87	1.00	220.00	222.00	0.7	1
T09.26	Clean F.O. Drain Tank S	5.6	5.5	176.80	-2.62	1.00	220.00	222.00	0.7	1
T09.27	FO Overflow Tank	24.5	24.0	173.20	-0.88	3.75	214.00	219.00	1.8	2
T09.28	Cool F.W. Drain Tk.	5.6	5.5	176.80	0.87	1.00	220.00	222.00	0.7	1
SUBTOTAL		52.5	51.5	175.12	0.06	2.28			5.4	
CAPACITY OF Fixed FW ballast (RHO=1)										
WB02.01	Water Ballast 02.01	327.7	321.1	14.73	0.00	3.55	0.10	42.00	12.5	10
CAPACITY OF Void (RHO=0)										
R10.11	Cofferdam	139.0	136.3	193.45	5.72	20.09	237.00	249.00	0.0	0
R10.12	Cofferdam	139.3	136.5	193.45	-5.73	20.08	237.00	249.00	0.0	0
R10.15	Void	403.3	395.2	194.85	0.00	25.35	237.00	249.00	0.0	0
SUBTOTAL		681.6	668.0	194.28	-0.00	23.20			0.0	

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NAME	DES	VOLM m3	VNET m3	CGX m	CGY m	CGZ m	FRMIN #	FRMAX #	IY*RHO tm	IY/IMO tm
CAPACITY OF Dry Tank (RHO=0)										
R03.01	Dry Tank	7.3	7.1	51.20	0.88	1.30	63.00	65.00	0.0	0
R11.02	Dry Tank	13.1	12.8	200.31	-0.00	1.59	249.00	252.00	0.0	0
SUBTOTAL		20.4	19.9	146.97	0.31	1.48			0.0	
TOTAL		96417.3	94489.0	110.91	0.00	8.68			348453.2	

5.3.2 Tank capacities tables

Compartment ident: WB01.03
Compartment descr: Water Ballast 01.03
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 0.00 (0.00 m)
Fore end at frame 9.00 (7.20 m)
Lowest point 8.37 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.05	17.3	17.7	5.52	0.00	9.08	146.49
10.0	1.44	34.5	35.4	4.76	0.00	9.36	518.65
15.0	1.70	51.8	53.1	4.15	0.00	9.55	1638.48
20.0	1.93	69.0	70.8	3.79	0.00	9.71	2947.44
25.0	2.14	86.3	88.5	3.55	0.00	9.85	4337.58
30.0	2.34	103.6	106.2	3.37	0.00	9.98	5632.58
35.0	2.52	120.8	123.8	3.24	0.00	10.09	6813.31
40.0	2.70	138.1	141.5	3.13	0.00	10.20	7897.30
45.0	2.87	155.3	159.2	3.04	0.00	10.31	8911.18
50.0	3.12	172.6	176.9	3.01	0.00	10.42	1138.96
55.0	3.37	189.9	194.6	3.00	0.00	10.52	1138.96
60.0	3.62	207.1	212.3	2.98	0.00	10.64	1138.96
65.0	3.87	224.4	230.0	2.97	0.00	10.75	1138.96
70.0	3.99	241.7	247.7	3.01	0.00	10.86	7241.34
75.0	4.09	258.9	265.4	3.05	0.00	10.96	7241.34
80.0	4.20	276.2	283.1	3.09	0.00	11.06	7241.34
85.0	4.31	293.4	300.8	3.12	0.00	11.15	7241.34
90.0	4.42	310.7	318.5	3.14	0.00	11.24	7241.34
95.0	4.52	328.0	336.2	3.17	0.00	11.32	7241.34
100.0	4.63	345.2	353.9	3.19	0.00	11.41	7241.34

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB01.11
 Compartment descr: Water Ballast 01.11
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 3.00 (2.40 m)
 Fore end at frame 15.00 (12.00 m)
 Lowest point 13.00 m above BL
 Highest point 27.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.73	39.8	40.8	7.20	17.83	13.37	155.89
10.0	1.47	79.5	81.5	7.20	17.83	13.73	155.89
15.0	2.20	119.3	122.3	7.20	17.83	14.10	155.89
20.0	2.94	159.0	163.0	7.20	17.83	14.47	155.89
25.0	3.67	198.8	203.8	7.20	17.83	14.84	155.89
30.0	4.41	238.6	244.5	7.20	17.83	15.20	155.89
35.0	5.15	278.3	285.3	7.20	17.83	15.57	155.89
40.0	5.88	318.1	326.0	7.20	17.83	15.94	155.89
45.0	6.61	357.8	366.8	7.20	17.83	16.31	155.89
50.0	7.35	397.6	407.5	7.20	17.83	16.67	155.89
55.0	8.09	437.4	448.3	7.20	17.83	17.04	155.89
60.0	8.82	477.1	489.1	7.20	17.83	17.41	155.89
65.0	9.56	516.9	529.8	7.20	17.83	17.78	155.89
70.0	10.29	556.6	570.6	7.20	17.83	18.15	155.89
75.0	11.02	596.4	611.3	7.20	17.83	18.51	155.89
80.0	11.76	636.2	652.1	7.20	17.83	18.88	155.89
85.0	12.50	675.9	692.8	7.20	17.83	19.25	155.89
90.0	13.23	715.7	733.6	7.20	17.83	19.61	155.89
95.0	13.97	755.5	774.3	7.20	17.83	19.98	155.89
100.0	14.70	795.2	815.1	7.20	17.83	20.35	155.89

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB01.12
Compartment descr: Water Ballast 01.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 3.00 (2.40 m)
Fore end at frame 15.00 (12.00 m)
Lowest point 13.00 m above BL
Highest point 27.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.73	39.8	40.8	7.20-17.83	13.37	155.89	
10.0	1.47	79.5	81.5	7.20-17.83	13.73	155.89	
15.0	2.20	119.3	122.3	7.20-17.83	14.10	155.89	
20.0	2.94	159.0	163.0	7.20-17.83	14.47	155.89	
25.0	3.67	198.8	203.8	7.20-17.83	14.84	155.89	
30.0	4.41	238.6	244.5	7.20-17.83	15.20	155.89	
35.0	5.15	278.3	285.3	7.20-17.83	15.57	155.89	
40.0	5.88	318.1	326.0	7.20-17.83	15.94	155.89	
45.0	6.61	357.8	366.8	7.20-17.83	16.31	155.89	
50.0	7.35	397.6	407.5	7.20-17.83	16.67	155.89	
55.0	8.09	437.4	448.3	7.20-17.83	17.04	155.89	
60.0	8.82	477.1	489.1	7.20-17.83	17.41	155.89	
65.0	9.56	516.9	529.8	7.20-17.83	17.78	155.89	
70.0	10.29	556.6	570.6	7.20-17.83	18.15	155.89	
75.0	11.02	596.4	611.3	7.20-17.83	18.51	155.89	
80.0	11.76	636.2	652.1	7.20-17.83	18.88	155.89	
85.0	12.50	675.9	692.8	7.20-17.83	19.25	155.89	
90.0	13.23	715.7	733.6	7.20-17.83	19.61	155.89	
95.0	13.97	755.5	774.3	7.20-17.83	19.98	155.89	
100.0	14.70	795.2	815.1	7.20-17.83	20.35	155.89	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.05
Compartment descr: Water Ballast 02.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 9.00 (7.20 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 1.94 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	2.18	74.4	76.2	31.06	4.80	3.49	839.65
10.0	2.92	148.8	152.5	30.05	5.12	4.00	1220.80
15.0	3.51	223.1	228.7	29.33	5.24	4.39	1522.81
20.0	4.00	297.5	305.0	28.62	5.29	4.72	1780.51
25.0	4.43	371.9	381.2	28.03	5.32	5.01	1941.69
30.0	4.82	446.3	457.4	27.49	5.36	5.27	2176.55
35.0	5.18	520.7	533.7	26.99	5.37	5.51	2383.88
40.0	5.51	595.0	609.9	26.49	5.38	5.74	2576.58
45.0	5.82	669.4	686.1	26.03	5.39	5.94	2689.32
50.0	6.11	743.8	762.4	25.60	5.41	6.14	2872.59
55.0	6.39	818.2	838.6	25.18	5.41	6.33	3020.05
60.0	6.66	892.5	914.9	24.80	5.42	6.51	2603.55
65.0	6.93	966.9	991.1	24.45	5.40	6.68	2689.10
70.0	7.21	1041.3	1067.3	24.15	5.40	6.84	2689.10
75.0	7.48	1115.7	1143.6	23.89	5.39	7.01	2619.93
80.0	7.75	1190.1	1219.8	23.67	5.38	7.17	2619.93
85.0	8.03	1264.4	1296.0	23.47	5.37	7.32	2619.93
90.0	8.30	1338.8	1372.3	23.30	5.36	7.48	2545.82
95.0	8.58	1413.2	1448.5	23.15	5.35	7.63	2545.82
100.0	8.86	1487.6	1524.8	23.01	5.35	7.78	2545.82

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.06
Compartment descr: Water Ballast 02.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 9.00 (7.20 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 1.94 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	2.17	74.5	76.3	31.04	-4.84	3.50	862.80
10.0	2.94	148.9	152.7	30.10	-5.11	4.01	1218.77
15.0	3.50	223.4	229.0	29.29	-5.25	4.40	1599.02
20.0	4.00	297.9	305.3	28.61	-5.30	4.72	1746.73
25.0	4.44	372.4	381.7	28.05	-5.33	5.01	1939.17
30.0	4.82	446.8	458.0	27.50	-5.35	5.27	2228.65
35.0	5.18	521.3	534.3	26.99	-5.38	5.51	2321.96
40.0	5.51	595.8	610.7	26.50	-5.39	5.74	2519.72
45.0	5.82	670.2	687.0	26.04	-5.41	5.94	2694.61
50.0	6.12	744.7	763.3	25.64	-5.41	6.14	2883.54
55.0	6.39	819.2	839.7	25.20	-5.42	6.33	3048.62
60.0	6.66	893.6	916.0	24.81	-5.43	6.51	2661.55
65.0	6.93	968.1	992.3	24.46	-5.42	6.68	2742.41
70.0	7.20	1042.6	1068.7	24.16	-5.41	6.85	2616.32
75.0	7.48	1117.1	1145.0	23.90	-5.40	7.01	2616.32
80.0	7.75	1191.5	1221.3	23.68	-5.39	7.17	2616.32
85.0	8.03	1266.0	1297.6	23.48	-5.38	7.32	2616.32
90.0	8.30	1340.5	1374.0	23.31	-5.37	7.48	2545.41
95.0	8.58	1414.9	1450.3	23.16	-5.36	7.63	2545.41
100.0	8.86	1489.4	1526.6	23.02	-5.35	7.78	2545.41

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.07
Compartment descr: Water Ballast 02.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 1.56 (1.25 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 2.89 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	2.31	46.4	47.6	30.73	13.93	4.58	213.92
10.0	3.03	92.9	95.2	29.90	14.33	5.09	422.33
15.0	3.54	139.3	142.8	29.21	14.55	5.46	591.06
20.0	3.96	185.8	190.4	28.63	14.72	5.75	735.95
25.0	4.33	232.2	238.0	28.14	14.85	6.01	909.35
30.0	4.66	278.7	285.6	27.72	14.95	6.24	1046.77
35.0	4.96	325.1	333.2	27.30	15.03	6.45	1160.79
40.0	5.23	371.6	380.8	26.92	15.11	6.64	1273.28
45.0	5.49	418.0	428.4	26.59	15.18	6.82	1359.64
50.0	5.75	464.4	476.1	26.31	15.25	6.99	1437.00
55.0	5.99	510.9	523.7	26.05	15.31	7.15	1509.41
60.0	6.22	557.3	571.3	25.82	15.38	7.31	1577.17
65.0	6.45	603.8	618.9	25.62	15.43	7.45	1638.59
70.0	6.68	650.2	666.5	25.42	15.49	7.60	1697.38
75.0	6.89	696.7	714.1	25.23	15.54	7.74	1752.99
80.0	7.11	743.1	761.7	25.04	15.59	7.87	1805.45
85.0	7.31	789.6	809.3	24.87	15.64	8.00	1855.06
90.0	7.52	836.0	856.9	24.70	15.69	8.13	1901.67
95.0	7.71	882.4	904.5	24.53	15.73	8.26	1946.74
100.0	7.91	928.9	952.1	24.37	15.78	8.38	1990.33

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.08
Compartment descr: Water Ballast 02.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 1.56 (1.25 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 2.89 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	2.31	46.4	47.6	30.73-13.93	4.58	213.92	
10.0	3.03	92.9	95.2	29.90-14.33	5.09	422.33	
15.0	3.54	139.3	142.8	29.21-14.55	5.46	591.06	
20.0	3.96	185.8	190.4	28.63-14.72	5.75	735.95	
25.0	4.33	232.2	238.0	28.14-14.85	6.01	909.35	
30.0	4.66	278.7	285.6	27.72-14.95	6.24	1046.77	
35.0	4.96	325.1	333.2	27.30-15.03	6.45	1160.79	
40.0	5.23	371.6	380.8	26.92-15.11	6.64	1273.28	
45.0	5.49	418.0	428.4	26.59-15.18	6.82	1359.64	
50.0	5.75	464.4	476.1	26.31-15.25	6.99	1437.00	
55.0	5.99	510.9	523.7	26.05-15.31	7.15	1509.41	
60.0	6.22	557.3	571.3	25.82-15.38	7.31	1577.17	
65.0	6.45	603.8	618.9	25.62-15.43	7.45	1638.59	
70.0	6.68	650.2	666.5	25.42-15.49	7.60	1697.38	
75.0	6.89	696.7	714.1	25.23-15.54	7.74	1752.99	
80.0	7.11	743.1	761.7	25.04-15.59	7.87	1805.45	
85.0	7.31	789.6	809.3	24.87-15.64	8.00	1855.06	
90.0	7.52	836.0	856.9	24.70-15.69	8.13	1901.67	
95.0	7.71	882.4	904.5	24.53-15.73	8.26	1946.74	
100.0	7.91	928.9	952.1	24.37-15.78	8.38	1990.33	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.09
 Compartment descr: Water Ballast 02.09
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 9.00 (7.20 m)
 Fore end at frame 42.00 (33.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	32.1	32.9	20.60	5.61	10.86	3196.46
10.0	0.22	64.2	65.8	20.60	5.61	10.91	3196.46
15.0	0.33	96.2	98.6	20.60	5.61	10.97	3196.46
20.0	0.44	128.3	131.5	20.60	5.61	11.02	3196.46
25.0	0.55	160.4	164.4	20.60	5.61	11.08	3196.46
30.0	0.67	192.5	197.3	20.60	5.61	11.13	3196.46
35.0	0.78	224.6	230.2	20.60	5.61	11.19	3257.36
40.0	0.89	256.6	263.1	20.59	5.61	11.24	3257.36
45.0	1.00	288.7	295.9	20.59	5.61	11.30	3257.36
50.0	1.11	320.8	328.8	20.58	5.62	11.35	3257.36
55.0	1.22	352.9	361.7	20.57	5.62	11.41	3257.36
60.0	1.33	385.0	394.6	20.57	5.62	11.46	3257.36
65.0	1.44	417.0	427.5	20.57	5.62	11.52	3257.36
70.0	1.55	449.1	460.4	20.56	5.63	11.57	3318.95
75.0	1.65	481.2	493.2	20.55	5.63	11.63	3318.95
80.0	1.76	513.3	526.1	20.54	5.64	11.68	3318.95
85.0	1.87	545.4	559.0	20.53	5.64	11.74	3318.95
90.0	1.98	577.4	591.9	20.52	5.64	11.79	3318.95
95.0	2.09	609.5	624.8	20.52	5.64	11.85	3318.95
100.0	2.20	641.6	657.6	20.51	5.65	11.90	3318.95

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.10
Compartment descr: Water Ballast 02.10
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 9.00 (7.20 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	32.1	32.9	20.60	-5.61	10.86	3196.46
10.0	0.22	64.2	65.8	20.60	-5.61	10.91	3196.46
15.0	0.33	96.2	98.6	20.60	-5.61	10.97	3196.46
20.0	0.44	128.3	131.5	20.60	-5.61	11.02	3196.46
25.0	0.55	160.4	164.4	20.60	-5.61	11.08	3196.46
30.0	0.67	192.5	197.3	20.60	-5.61	11.13	3196.46
35.0	0.78	224.6	230.2	20.60	-5.61	11.19	3257.36
40.0	0.89	256.6	263.1	20.59	-5.61	11.24	3257.36
45.0	1.00	288.7	295.9	20.59	-5.61	11.30	3257.36
50.0	1.11	320.8	328.8	20.58	-5.62	11.35	3257.36
55.0	1.22	352.9	361.7	20.57	-5.62	11.41	3257.36
60.0	1.33	385.0	394.6	20.57	-5.62	11.46	3257.36
65.0	1.44	417.0	427.5	20.57	-5.62	11.52	3257.36
70.0	1.55	449.1	460.4	20.56	-5.63	11.57	3318.95
75.0	1.65	481.2	493.2	20.55	-5.63	11.63	3318.95
80.0	1.76	513.3	526.1	20.54	-5.64	11.68	3318.95
85.0	1.87	545.4	559.0	20.53	-5.64	11.74	3318.95
90.0	1.98	577.4	591.9	20.52	-5.64	11.79	3318.95
95.0	2.09	609.5	624.8	20.52	-5.64	11.85	3318.95
100.0	2.20	641.6	657.6	20.51	-5.65	11.90	3318.95

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.11
Compartment descr: Water Ballast 02.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 5.00 (4.00 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.12	27.5	28.1	21.85	16.56	10.86	1983.41
10.0	0.24	54.9	56.3	21.82	16.57	10.92	2002.30
15.0	0.36	82.4	84.4	21.80	16.58	10.98	2019.55
20.0	0.47	109.8	112.6	21.78	16.59	11.04	2036.15
25.0	0.59	137.3	140.7	21.76	16.60	11.10	2051.99
30.0	0.71	164.7	168.9	21.74	16.61	11.16	2064.38
35.0	0.82	192.2	197.0	21.72	16.61	11.21	2075.73
40.0	0.94	219.6	225.1	21.71	16.62	11.27	2086.65
45.0	1.06	247.1	253.3	21.69	16.63	11.33	2096.44
50.0	1.17	274.6	281.4	21.68	16.63	11.39	2104.16
55.0	1.29	302.0	309.6	21.67	16.64	11.45	2109.48
60.0	1.40	329.5	337.7	21.65	16.64	11.51	2111.32
65.0	1.51	356.9	365.9	21.55	16.63	11.56	2476.43
70.0	1.61	384.4	394.0	21.41	16.60	11.62	2476.43
75.0	1.71	411.8	422.1	21.28	16.58	11.68	2476.43
80.0	1.81	439.3	450.3	21.17	16.56	11.73	2476.43
85.0	1.90	466.8	478.4	21.07	16.55	11.79	2476.43
90.0	2.00	494.2	506.6	20.99	16.53	11.84	2476.43
95.0	2.10	521.7	534.7	20.91	16.52	11.89	2476.42
100.0	2.20	549.1	562.9	20.84	16.51	11.95	2476.42

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB02.12
Compartment descr: Water Ballast 02.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 5.00 (4.00 m)
Fore end at frame 42.00 (33.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.12	27.5	28.1	21.85-16.56	10.86	1983.41	
10.0	0.24	54.9	56.3	21.82-16.57	10.92	2002.30	
15.0	0.36	82.4	84.4	21.80-16.58	10.98	2019.55	
20.0	0.47	109.8	112.6	21.78-16.59	11.04	2036.15	
25.0	0.59	137.3	140.7	21.76-16.60	11.10	2051.99	
30.0	0.71	164.7	168.9	21.74-16.61	11.16	2064.38	
35.0	0.82	192.2	197.0	21.72-16.61	11.21	2075.73	
40.0	0.94	219.6	225.1	21.71-16.62	11.27	2086.65	
45.0	1.06	247.1	253.3	21.69-16.63	11.33	2096.44	
50.0	1.17	274.6	281.4	21.68-16.63	11.39	2104.16	
55.0	1.29	302.0	309.6	21.67-16.64	11.45	2109.48	
60.0	1.40	329.5	337.7	21.65-16.64	11.51	2111.32	
65.0	1.51	356.9	365.9	21.55-16.63	11.56	2476.43	
70.0	1.61	384.4	394.0	21.41-16.60	11.62	2476.43	
75.0	1.71	411.8	422.1	21.28-16.58	11.68	2476.43	
80.0	1.81	439.3	450.3	21.17-16.56	11.73	2476.43	
85.0	1.90	466.8	478.4	21.07-16.55	11.79	2476.43	
90.0	2.00	494.2	506.6	20.99-16.53	11.84	2476.43	
95.0	2.10	521.7	534.7	20.91-16.52	11.89	2476.42	
100.0	2.20	549.1	562.9	20.84-16.51	11.95	2476.42	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.01
 Compartment descr: Water Ballast 03.01
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
 Fore end at frame 72.00 (57.60 m)
 Lowest point -0.00 m above BL
 Highest point 3.69 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.24	23.5	24.1	52.00	4.70	0.13	1297.30
10.0	0.43	47.1	48.2	51.56	4.83	0.23	1472.70
15.0	0.59	70.6	72.3	51.17	4.88	0.33	1611.25
20.0	0.74	94.1	96.5	50.83	4.94	0.41	1726.85
25.0	0.88	117.6	120.6	50.54	4.99	0.49	1845.44
30.0	1.01	141.2	144.7	50.28	5.03	0.57	1951.53
35.0	1.14	164.7	168.8	50.03	5.08	0.64	2049.55
40.0	1.26	188.2	192.9	49.80	5.10	0.71	1914.98
45.0	1.38	211.7	217.0	49.57	5.11	0.78	2019.91
50.0	1.50	235.3	241.2	49.35	5.12	0.84	2067.25
55.0	1.61	258.8	265.3	49.14	5.13	0.91	2134.15
60.0	1.72	282.3	289.4	48.95	5.15	0.97	2172.69
65.0	1.82	305.9	313.5	48.77	5.16	1.03	2260.30
70.0	1.93	329.4	337.6	48.59	5.17	1.09	2325.75
75.0	2.03	352.9	361.7	48.42	5.19	1.15	2364.05
80.0	2.14	376.4	385.9	48.28	5.20	1.21	1705.85
85.0	2.28	400.0	410.0	48.16	5.20	1.27	1658.14
90.0	2.42	423.5	434.1	48.06	5.21	1.33	1508.67
95.0	2.58	447.0	458.2	47.98	5.21	1.39	1483.78
100.0	3.69	470.6	482.3	47.42	5.24	1.46	0.00

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.02
 Compartment descr: Water Ballast 03.02
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
 Fore end at frame 72.00 (57.60 m)
 Lowest point -0.00 m above BL
 Highest point 3.69 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.24	23.9	24.5	51.99	-4.59	0.13	1352.83
10.0	0.42	47.8	49.0	51.55	-4.73	0.23	1517.72
15.0	0.59	71.7	73.5	51.18	-4.79	0.32	1657.09
20.0	0.74	95.6	98.0	50.85	-4.85	0.41	1777.19
25.0	0.88	119.5	122.5	50.56	-4.91	0.49	1896.90
30.0	1.01	143.4	147.0	50.30	-4.95	0.56	2004.57
35.0	1.13	167.3	171.5	50.06	-5.00	0.64	2105.76
40.0	1.26	191.2	196.0	49.83	-5.03	0.71	1970.83
45.0	1.38	215.1	220.5	49.60	-5.04	0.77	2078.18
50.0	1.49	239.0	245.0	49.38	-5.05	0.84	2125.52
55.0	1.61	262.9	269.5	49.18	-5.06	0.90	2193.56
60.0	1.72	286.8	294.0	48.99	-5.08	0.97	2233.90
65.0	1.83	310.7	318.5	48.81	-5.09	1.03	2320.94
70.0	1.93	334.6	342.9	48.63	-5.11	1.09	2387.39
75.0	2.03	358.5	367.4	48.47	-5.12	1.15	2426.64
80.0	2.14	382.4	391.9	48.32	-5.13	1.21	1765.41
85.0	2.28	406.3	416.4	48.21	-5.14	1.27	1718.26
90.0	2.42	430.2	440.9	48.12	-5.14	1.33	1569.45
95.0	2.58	454.1	465.4	48.03	-5.15	1.39	1545.96
100.0	3.69	478.0	489.9	47.48	-5.17	1.46	0.00

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.05
Compartment descr: Water Ballast 03.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 63.00 (50.40 m)
Fore end at frame 72.00 (57.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.42	61.9	63.4	53.85	0.00	2.81	5655.60
10.0	0.84	123.8	126.9	53.85	0.00	3.02	5655.60
15.0	1.26	185.6	190.3	53.85	0.00	3.23	5655.60
20.0	1.67	247.5	253.7	53.87	0.00	3.44	7095.96
25.0	2.06	309.4	317.1	53.89	0.00	3.65	7095.96
30.0	2.45	371.3	380.6	53.90	0.00	3.85	7095.96
35.0	2.83	433.2	444.0	53.91	0.00	4.05	7095.96
40.0	3.24	495.0	507.4	53.91	0.00	4.24	5953.73
45.0	3.66	556.9	570.8	53.91	0.00	4.45	5953.73
50.0	4.07	618.8	634.3	53.91	0.00	4.65	5953.73
55.0	4.48	680.7	697.7	53.91	0.00	4.85	5190.19
60.0	4.92	742.6	761.1	53.92	0.00	5.05	5190.19
65.0	5.35	804.4	824.6	53.93	0.00	5.26	5190.19
70.0	5.78	866.3	888.0	53.93	0.00	5.47	5190.19
75.0	6.20	928.2	951.4	53.94	0.00	5.68	5656.96
80.0	6.62	990.1	1014.8	53.95	0.00	5.88	6158.36
85.0	7.03	1052.0	1078.3	53.96	0.00	6.09	6158.36
90.0	7.43	1113.8	1141.7	53.97	0.00	6.30	7241.34
95.0	7.81	1175.7	1205.1	53.97	0.00	6.51	7241.34
100.0	8.20	1237.6	1268.5	53.97	0.00	6.71	7241.34

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.06
Compartment descr: Water Ballast 03.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 60.00 (48.00 m)
Lowest point 2.10 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.92	46.9	48.1	41.64	0.00	2.64	634.97
10.0	1.40	93.8	96.1	40.40	0.00	2.96	964.63
15.0	1.84	140.7	144.2	39.86	0.00	3.21	964.63
20.0	2.27	187.5	192.2	39.59	0.00	3.45	964.63
25.0	2.71	234.4	240.3	39.42	0.00	3.68	964.63
30.0	3.15	281.3	288.3	39.31	0.00	3.90	964.63
35.0	3.59	328.2	336.4	39.24	0.00	4.13	964.63
40.0	4.03	375.1	384.5	39.18	0.00	4.35	964.63
45.0	4.47	422.0	432.5	39.13	0.00	4.57	964.63
50.0	4.90	468.8	480.6	39.10	0.00	4.79	1034.95
55.0	5.28	515.7	528.6	39.16	0.00	5.01	1034.95
60.0	5.66	562.6	576.7	39.22	0.00	5.22	1034.95
65.0	6.04	609.5	624.7	39.27	0.00	5.43	1034.95
70.0	6.42	656.4	672.8	39.31	0.00	5.64	1034.95
75.0	6.80	703.3	720.9	39.34	0.00	5.85	1034.95
80.0	7.18	750.2	768.9	39.38	0.00	6.05	1034.95
85.0	7.56	797.0	817.0	39.40	0.00	6.25	1034.95
90.0	7.94	843.9	865.0	39.43	0.00	6.45	1034.95
95.0	8.32	890.8	913.1	39.45	0.00	6.65	1034.95
100.0	8.70	937.7	961.1	39.47	0.00	6.85	1034.95

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.07
Compartment descr: Water Ballast 03.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 72.00 (57.60 m)
Lowest point -0.00 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.67	103.8	106.4	52.43	14.67	1.06	645.19
10.0	2.46	207.7	212.8	51.05	14.94	1.57	1004.10
15.0	3.09	311.5	319.3	50.07	15.11	1.98	1282.64
20.0	3.63	415.3	425.7	49.33	15.25	2.33	1471.42
25.0	4.14	519.1	532.1	48.78	15.37	2.64	1623.20
30.0	4.62	623.0	638.5	48.37	15.48	2.93	1749.66
35.0	5.08	726.8	744.9	48.04	15.57	3.20	1853.75
40.0	5.54	830.6	851.4	47.78	15.66	3.47	1937.80
45.0	5.99	934.4	957.8	47.56	15.73	3.72	2002.74
50.0	6.43	1038.3	1064.2	47.38	15.79	3.97	2051.30
55.0	6.87	1142.1	1170.6	47.23	15.84	4.22	2085.61
60.0	7.31	1245.9	1277.0	47.10	15.89	4.45	2107.64
65.0	7.75	1349.7	1383.5	46.98	15.93	4.69	2120.33
70.0	8.18	1453.6	1489.9	46.88	15.97	4.93	2126.12
75.0	8.62	1557.4	1596.3	46.80	16.00	5.16	2127.65
80.0	9.06	1661.2	1702.7	46.72	16.03	5.39	2127.84
85.0	9.49	1765.0	1809.2	46.66	16.05	5.62	2127.84
90.0	9.93	1868.9	1915.6	46.60	16.07	5.84	2127.84
95.0	10.36	1972.7	2022.0	46.55	16.09	6.07	2127.84
100.0	10.80	2076.5	2128.4	46.50	16.11	6.30	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.08
Compartment descr: Water Ballast 03.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 72.00 (57.60 m)
Lowest point -0.00 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.67	103.8	106.4	52.43-14.67	1.06	645.19	
10.0	2.46	207.7	212.8	51.05-14.94	1.57	1004.10	
15.0	3.09	311.5	319.3	50.07-15.11	1.98	1282.64	
20.0	3.63	415.3	425.7	49.33-15.25	2.33	1471.42	
25.0	4.14	519.1	532.1	48.78-15.37	2.64	1623.20	
30.0	4.62	623.0	638.5	48.37-15.48	2.93	1749.66	
35.0	5.08	726.8	744.9	48.04-15.57	3.20	1853.75	
40.0	5.54	830.6	851.4	47.78-15.66	3.47	1937.80	
45.0	5.99	934.4	957.8	47.56-15.73	3.72	2002.74	
50.0	6.43	1038.3	1064.2	47.38-15.79	3.97	2051.30	
55.0	6.87	1142.1	1170.6	47.23-15.84	4.22	2085.61	
60.0	7.31	1245.9	1277.0	47.10-15.89	4.45	2107.64	
65.0	7.75	1349.7	1383.5	46.98-15.93	4.69	2120.33	
70.0	8.18	1453.6	1489.9	46.88-15.97	4.93	2126.12	
75.0	8.62	1557.4	1596.3	46.80-16.00	5.16	2127.65	
80.0	9.06	1661.2	1702.7	46.72-16.03	5.39	2127.84	
85.0	9.49	1765.0	1809.2	46.66-16.05	5.62	2127.84	
90.0	9.93	1868.9	1915.6	46.60-16.07	5.84	2127.84	
95.0	10.36	1972.7	2022.0	46.55-16.09	6.07	2127.84	
100.0	10.80	2076.5	2128.4	46.50-16.11	6.30	2127.84	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.09
Compartment descr: Water Ballast 03.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 72.00 (57.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	45.60	5.69	10.85	3017.23
10.0	0.22	58.9	60.3	45.60	5.69	10.91	3017.23
15.0	0.33	88.3	90.5	45.60	5.69	10.96	3017.23
20.0	0.44	117.7	120.7	45.60	5.69	11.02	3017.23
25.0	0.55	147.1	150.8	45.60	5.69	11.07	3017.23
30.0	0.66	176.6	181.0	45.60	5.69	11.13	3017.23
35.0	0.77	206.0	211.2	45.60	5.69	11.18	3017.23
40.0	0.88	235.4	241.3	45.60	5.69	11.24	3017.23
45.0	0.99	264.9	271.5	45.60	5.69	11.30	3017.23
50.0	1.10	294.3	301.7	45.60	5.69	11.35	3017.23
55.0	1.21	323.7	331.8	45.60	5.69	11.40	3017.23
60.0	1.32	353.2	362.0	45.60	5.69	11.46	3017.23
65.0	1.43	382.6	392.1	45.60	5.69	11.51	3017.23
70.0	1.54	412.0	422.3	45.60	5.69	11.57	3017.23
75.0	1.65	441.4	452.5	45.60	5.69	11.63	3017.23
80.0	1.76	470.9	482.6	45.60	5.69	11.68	3017.23
85.0	1.87	500.3	512.8	45.60	5.69	11.73	3017.23
90.0	1.98	529.7	543.0	45.60	5.69	11.79	3017.23
95.0	2.09	559.2	573.1	45.60	5.69	11.84	3017.23
100.0	2.20	588.6	603.3	45.60	5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.10
Compartment descr: Water Ballast 03.10
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 72.00 (57.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	45.60	-5.69	10.85	3017.23
10.0	0.22	58.9	60.3	45.60	-5.69	10.91	3017.23
15.0	0.33	88.3	90.5	45.60	-5.69	10.96	3017.23
20.0	0.44	117.7	120.7	45.60	-5.69	11.02	3017.23
25.0	0.55	147.1	150.8	45.60	-5.69	11.07	3017.23
30.0	0.66	176.6	181.0	45.60	-5.69	11.13	3017.23
35.0	0.77	206.0	211.2	45.60	-5.69	11.18	3017.23
40.0	0.88	235.4	241.3	45.60	-5.69	11.24	3017.23
45.0	0.99	264.9	271.5	45.60	-5.69	11.30	3017.23
50.0	1.10	294.3	301.7	45.60	-5.69	11.35	3017.23
55.0	1.21	323.7	331.8	45.60	-5.69	11.40	3017.23
60.0	1.32	353.2	362.0	45.60	-5.69	11.46	3017.23
65.0	1.43	382.6	392.1	45.60	-5.69	11.51	3017.23
70.0	1.54	412.0	422.3	45.60	-5.69	11.57	3017.23
75.0	1.65	441.4	452.5	45.60	-5.69	11.63	3017.23
80.0	1.76	470.9	482.6	45.60	-5.69	11.68	3017.23
85.0	1.87	500.3	512.8	45.60	-5.69	11.73	3017.23
90.0	1.98	529.7	543.0	45.60	-5.69	11.79	3017.23
95.0	2.09	559.2	573.1	45.60	-5.69	11.84	3017.23
100.0	2.20	588.6	603.3	45.60	-5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.11
Compartment descr: Water Ballast 03.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
Fore end at frame 72.00 (57.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.0	26.7	45.60	16.44	10.85	2127.84
10.0	0.22	52.0	53.3	45.60	16.44	10.91	2127.84
15.0	0.33	78.1	80.0	45.60	16.44	10.96	2127.84
20.0	0.44	104.1	106.7	45.60	16.44	11.02	2127.84
25.0	0.55	130.1	133.3	45.60	16.44	11.07	2127.84
30.0	0.66	156.1	160.0	45.60	16.44	11.13	2127.84
35.0	0.76	182.1	186.7	45.60	16.44	11.18	2127.84
40.0	0.87	208.2	213.4	45.60	16.44	11.24	2127.84
45.0	0.98	234.2	240.0	45.60	16.44	11.29	2127.84
50.0	1.09	260.2	266.7	45.60	16.44	11.35	2127.84
55.0	1.20	286.2	293.4	45.60	16.44	11.40	2057.52
60.0	1.31	312.2	320.0	45.61	16.43	11.46	2051.21
65.0	1.42	338.2	346.7	45.62	16.43	11.51	2045.10
70.0	1.53	364.3	373.4	45.63	16.42	11.57	2044.37
75.0	1.65	390.3	400.0	45.63	16.42	11.62	2044.37
80.0	1.76	416.3	426.7	45.64	16.42	11.68	2044.37
85.0	1.87	442.3	453.4	45.65	16.41	11.73	2044.37
90.0	1.98	468.3	480.0	45.65	16.41	11.79	2044.37
95.0	2.09	494.4	506.7	45.65	16.41	11.84	2044.37
100.0	2.20	520.4	533.4	45.66	16.41	11.90	2044.37

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB03.12
 Compartment descr: Water Ballast 03.12
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 42.00 (33.60 m)
 Fore end at frame 72.00 (57.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.0	26.7	45.60-16.44	10.85	2127.84	
10.0	0.22	52.0	53.3	45.60-16.44	10.91	2127.84	
15.0	0.33	78.1	80.0	45.60-16.44	10.96	2127.84	
20.0	0.44	104.1	106.7	45.60-16.44	11.02	2127.84	
25.0	0.55	130.1	133.3	45.60-16.44	11.07	2127.84	
30.0	0.66	156.1	160.0	45.60-16.44	11.13	2127.84	
35.0	0.76	182.1	186.7	45.60-16.44	11.18	2127.84	
40.0	0.87	208.2	213.4	45.60-16.44	11.24	2127.84	
45.0	0.98	234.2	240.0	45.60-16.44	11.29	2127.84	
50.0	1.09	260.2	266.7	45.60-16.44	11.35	2127.84	
55.0	1.20	286.2	293.4	45.60-16.44	11.40	2057.40	
60.0	1.31	312.2	320.0	45.61-16.43	11.46	2051.12	
65.0	1.42	338.3	346.7	45.62-16.43	11.51	2045.23	
70.0	1.53	364.3	373.4	45.63-16.42	11.57	2044.69	
75.0	1.65	390.3	400.0	45.63-16.42	11.62	2044.69	
80.0	1.76	416.3	426.7	45.64-16.42	11.68	2044.69	
85.0	1.87	442.3	453.4	45.65-16.41	11.73	2044.69	
90.0	1.98	468.3	480.1	45.65-16.41	11.79	2044.69	
95.0	2.09	494.4	506.7	45.65-16.41	11.84	2044.69	
100.0	2.20	520.4	533.4	45.66-16.41	11.90	2044.44	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.01
Compartment descr: Water Ballast 04.01
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.14	63.8	65.4	69.91	9.84	0.07	16474.00
10.0	0.27	127.7	130.9	69.86	9.97	0.14	17253.20
15.0	0.41	191.5	196.3	69.83	10.05	0.21	17796.21
20.0	0.54	255.4	261.8	69.81	10.11	0.27	18225.44
25.0	0.67	319.2	327.2	69.80	10.17	0.34	18576.03
30.0	0.80	383.1	392.7	69.78	10.22	0.40	18872.29
35.0	0.93	446.9	458.1	69.77	10.26	0.47	19125.48
40.0	1.06	510.8	523.5	69.76	10.29	0.53	19342.70
45.0	1.18	574.6	589.0	69.76	10.32	0.60	19529.54
50.0	1.31	638.5	654.4	69.74	10.35	0.66	19690.09
55.0	1.44	702.3	719.9	69.72	10.38	0.73	19826.96
60.0	1.57	766.2	785.3	69.70	10.40	0.79	19941.83
65.0	1.70	830.0	850.8	69.69	10.42	0.86	20037.47
70.0	1.83	893.9	916.2	69.67	10.44	0.92	20112.60
75.0	1.96	957.7	981.6	69.66	10.46	0.99	20169.01
80.0	2.09	1021.5	1047.1	69.65	10.48	1.05	20213.07
85.0	2.22	1085.4	1112.5	69.64	10.49	1.12	20247.78
90.0	2.34	1149.2	1178.0	69.63	10.51	1.18	20276.71
95.0	2.47	1213.1	1243.4	69.62	10.52	1.25	20299.06
100.0	2.60	1276.9	1308.9	69.62	10.53	1.31	20318.55

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.02
Compartment descr: Water Ballast 04.02
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.14	63.8	65.4	69.91	-9.84	0.07	16474.00
10.0	0.27	127.7	130.9	69.86	-9.97	0.14	17253.20
15.0	0.41	191.5	196.3	69.83	-10.05	0.21	17796.21
20.0	0.54	255.4	261.8	69.81	-10.11	0.27	18225.44
25.0	0.67	319.2	327.2	69.80	-10.17	0.34	18576.03
30.0	0.80	383.1	392.7	69.78	-10.22	0.40	18872.29
35.0	0.93	446.9	458.1	69.77	-10.26	0.47	19125.48
40.0	1.06	510.8	523.5	69.76	-10.29	0.53	19342.70
45.0	1.18	574.6	589.0	69.76	-10.32	0.60	19529.54
50.0	1.31	638.5	654.4	69.74	-10.35	0.66	19690.09
55.0	1.44	702.3	719.9	69.72	-10.38	0.73	19826.96
60.0	1.57	766.2	785.3	69.70	-10.40	0.79	19941.83
65.0	1.70	830.0	850.8	69.69	-10.42	0.86	20037.47
70.0	1.83	893.9	916.2	69.67	-10.44	0.92	20112.60
75.0	1.96	957.7	981.6	69.66	-10.46	0.99	20169.01
80.0	2.09	1021.5	1047.1	69.65	-10.48	1.05	20213.07
85.0	2.22	1085.4	1112.5	69.64	-10.49	1.12	20247.78
90.0	2.34	1149.2	1178.0	69.63	-10.51	1.18	20276.71
95.0	2.47	1213.1	1243.4	69.62	-10.52	1.25	20299.06
100.0	2.60	1276.9	1308.9	69.62	-10.53	1.31	20318.55

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.07
 Compartment descr: Water Ballast 04.07
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
 Fore end at frame 102.00 (81.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	69.60	16.44	2.80	2127.85
10.0	0.82	195.3	200.2	69.60	16.44	3.01	2127.85
15.0	1.23	292.9	300.2	69.60	16.44	3.21	2127.85
20.0	1.64	390.6	400.3	69.60	16.44	3.42	2127.85
25.0	2.05	488.2	500.4	69.60	16.44	3.62	2127.85
30.0	2.46	585.8	600.5	69.60	16.44	3.83	2127.85
35.0	2.87	683.5	700.5	69.60	16.44	4.03	2127.85
40.0	3.28	781.1	800.6	69.60	16.44	4.24	2127.85
45.0	3.69	878.7	900.7	69.60	16.44	4.44	2127.85
50.0	4.10	976.4	1000.8	69.60	16.44	4.65	2127.85
55.0	4.51	1074.0	1100.9	69.60	16.44	4.85	2127.85
60.0	4.92	1171.7	1200.9	69.60	16.44	5.06	2127.85
65.0	5.33	1269.3	1301.0	69.60	16.44	5.26	2127.85
70.0	5.74	1366.9	1401.1	69.60	16.44	5.47	2127.85
75.0	6.15	1464.6	1501.2	69.60	16.44	5.67	2127.85
80.0	6.56	1562.2	1601.3	69.60	16.44	5.88	2127.85
85.0	6.97	1659.8	1701.3	69.60	16.44	6.08	2127.85
90.0	7.38	1757.5	1801.4	69.60	16.44	6.29	2127.85
95.0	7.79	1855.1	1901.5	69.60	16.44	6.49	2127.85
100.0	8.20	1952.8	2001.6	69.60	16.44	6.70	2127.85

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.08
Compartment descr: Water Ballast 04.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	69.60-16.44	2.80	2127.85	
10.0	0.82	195.3	200.2	69.60-16.44	3.01	2127.85	
15.0	1.23	292.9	300.2	69.60-16.44	3.21	2127.85	
20.0	1.64	390.6	400.3	69.60-16.44	3.42	2127.85	
25.0	2.05	488.2	500.4	69.60-16.44	3.62	2127.85	
30.0	2.46	585.8	600.5	69.60-16.44	3.83	2127.85	
35.0	2.87	683.5	700.5	69.60-16.44	4.03	2127.85	
40.0	3.28	781.1	800.6	69.60-16.44	4.24	2127.85	
45.0	3.69	878.7	900.7	69.60-16.44	4.44	2127.85	
50.0	4.10	976.4	1000.8	69.60-16.44	4.65	2127.85	
55.0	4.51	1074.0	1100.9	69.60-16.44	4.85	2127.85	
60.0	4.92	1171.7	1200.9	69.60-16.44	5.06	2127.85	
65.0	5.33	1269.3	1301.0	69.60-16.44	5.26	2127.85	
70.0	5.74	1366.9	1401.1	69.60-16.44	5.47	2127.85	
75.0	6.15	1464.6	1501.2	69.60-16.44	5.67	2127.85	
80.0	6.56	1562.2	1601.3	69.60-16.44	5.88	2127.85	
85.0	6.97	1659.8	1701.3	69.60-16.44	6.08	2127.85	
90.0	7.38	1757.5	1801.4	69.60-16.44	6.29	2127.85	
95.0	7.79	1855.1	1901.5	69.60-16.44	6.49	2127.85	
100.0	8.20	1952.8	2001.6	69.60-16.44	6.70	2127.85	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.09
Compartment descr: Water Ballast 04.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	69.60	5.69	10.85	3017.23
10.0	0.22	58.9	60.3	69.60	5.69	10.91	3017.23
15.0	0.33	88.3	90.5	69.60	5.69	10.96	3017.23
20.0	0.44	117.7	120.7	69.60	5.69	11.02	3017.23
25.0	0.55	147.1	150.8	69.60	5.69	11.07	3017.23
30.0	0.66	176.6	181.0	69.60	5.69	11.13	3017.23
35.0	0.77	206.0	211.2	69.60	5.69	11.19	3017.23
40.0	0.88	235.4	241.3	69.60	5.69	11.24	3017.23
45.0	0.99	264.9	271.5	69.60	5.69	11.30	3017.23
50.0	1.10	294.3	301.7	69.60	5.69	11.35	3017.23
55.0	1.21	323.7	331.8	69.60	5.69	11.40	3017.23
60.0	1.32	353.2	362.0	69.60	5.69	11.46	3017.23
65.0	1.43	382.6	392.1	69.60	5.69	11.51	3017.23
70.0	1.54	412.0	422.3	69.60	5.69	11.57	3017.23
75.0	1.65	441.4	452.5	69.60	5.69	11.63	3017.23
80.0	1.76	470.9	482.6	69.60	5.69	11.68	3017.23
85.0	1.87	500.3	512.8	69.60	5.69	11.73	3017.23
90.0	1.98	529.7	543.0	69.60	5.69	11.79	3017.23
95.0	2.09	559.2	573.1	69.60	5.69	11.85	3017.23
100.0	2.20	588.6	603.3	69.60	5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.10
 Compartment descr: Water Ballast 04.10
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
 Fore end at frame 102.00 (81.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	69.60	-5.69	10.85	3017.23
10.0	0.22	58.9	60.3	69.60	-5.69	10.91	3017.23
15.0	0.33	88.3	90.5	69.60	-5.69	10.96	3017.23
20.0	0.44	117.7	120.7	69.60	-5.69	11.02	3017.23
25.0	0.55	147.1	150.8	69.60	-5.69	11.07	3017.23
30.0	0.66	176.6	181.0	69.60	-5.69	11.13	3017.23
35.0	0.77	206.0	211.2	69.60	-5.69	11.19	3017.23
40.0	0.88	235.4	241.3	69.60	-5.69	11.24	3017.23
45.0	0.99	264.9	271.5	69.60	-5.69	11.30	3017.23
50.0	1.10	294.3	301.7	69.60	-5.69	11.35	3017.23
55.0	1.21	323.7	331.8	69.60	-5.69	11.40	3017.23
60.0	1.32	353.2	362.0	69.60	-5.69	11.46	3017.23
65.0	1.43	382.6	392.1	69.60	-5.69	11.51	3017.23
70.0	1.54	412.0	422.3	69.60	-5.69	11.57	3017.23
75.0	1.65	441.4	452.5	69.60	-5.69	11.63	3017.23
80.0	1.76	470.9	482.6	69.60	-5.69	11.68	3017.23
85.0	1.87	500.3	512.8	69.60	-5.69	11.73	3017.23
90.0	1.98	529.7	543.0	69.60	-5.69	11.79	3017.23
95.0	2.09	559.2	573.1	69.60	-5.69	11.85	3017.23
100.0	2.20	588.6	603.3	69.60	-5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.11
Compartment descr: Water Ballast 04.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.2	26.9	69.60	16.44	10.85	2127.84
10.0	0.22	52.4	53.7	69.60	16.44	10.91	2127.84
15.0	0.33	78.6	80.6	69.60	16.44	10.97	2127.84
20.0	0.44	104.8	107.4	69.60	16.44	11.02	2127.84
25.0	0.55	131.0	134.3	69.60	16.44	11.07	2127.84
30.0	0.66	157.2	161.1	69.60	16.44	11.13	2127.84
35.0	0.77	183.4	188.0	69.60	16.44	11.18	2127.84
40.0	0.88	209.6	214.8	69.60	16.44	11.24	2127.84
45.0	0.99	235.8	241.7	69.60	16.44	11.30	2127.84
50.0	1.10	262.0	268.5	69.60	16.44	11.35	2127.84
55.0	1.21	288.1	295.4	69.60	16.44	11.40	2127.84
60.0	1.32	314.3	322.2	69.60	16.44	11.46	2127.84
65.0	1.43	340.5	349.1	69.60	16.44	11.51	2127.84
70.0	1.54	366.7	375.9	69.60	16.44	11.57	2127.84
75.0	1.65	392.9	402.8	69.60	16.44	11.62	2127.84
80.0	1.76	419.1	429.6	69.60	16.44	11.68	2127.84
85.0	1.87	445.3	456.5	69.60	16.44	11.73	2127.84
90.0	1.98	471.5	483.3	69.60	16.44	11.79	2127.84
95.0	2.09	497.7	510.2	69.60	16.44	11.84	2127.84
100.0	2.20	523.9	537.0	69.60	16.44	11.90	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB04.12
Compartment descr: Water Ballast 04.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
Fore end at frame 102.00 (81.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.2	26.9	69.60-16.44	10.85	2127.84	
10.0	0.22	52.4	53.7	69.60-16.44	10.91	2127.84	
15.0	0.33	78.6	80.6	69.60-16.44	10.97	2127.84	
20.0	0.44	104.8	107.4	69.60-16.44	11.02	2127.84	
25.0	0.55	131.0	134.3	69.60-16.44	11.07	2127.84	
30.0	0.66	157.2	161.1	69.60-16.44	11.13	2127.84	
35.0	0.77	183.4	188.0	69.60-16.44	11.18	2127.84	
40.0	0.88	209.6	214.8	69.60-16.44	11.24	2127.84	
45.0	0.99	235.8	241.7	69.60-16.44	11.30	2127.84	
50.0	1.10	262.0	268.5	69.60-16.44	11.35	2127.84	
55.0	1.21	288.1	295.4	69.60-16.44	11.40	2127.84	
60.0	1.32	314.3	322.2	69.60-16.44	11.46	2127.84	
65.0	1.43	340.5	349.1	69.60-16.44	11.51	2127.84	
70.0	1.54	366.7	375.9	69.60-16.44	11.57	2127.84	
75.0	1.65	392.9	402.8	69.60-16.44	11.62	2127.84	
80.0	1.76	419.1	429.6	69.60-16.44	11.68	2127.84	
85.0	1.87	445.3	456.5	69.60-16.44	11.73	2127.84	
90.0	1.98	471.5	483.3	69.60-16.44	11.79	2127.84	
95.0	2.09	497.7	510.2	69.60-16.44	11.84	2127.84	
100.0	2.20	523.9	537.0	69.60-16.44	11.90	2127.84	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.01
 Compartment descr: Water Ballast 05.01
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
 Fore end at frame 132.00 (105.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	64.4	66.0	93.60	10.15	0.07	17673.23
10.0	0.27	128.8	132.0	93.60	10.23	0.13	18284.28
15.0	0.40	193.2	198.0	93.60	10.29	0.20	18718.10
20.0	0.53	257.6	264.1	93.60	10.34	0.27	19058.20
25.0	0.66	322.0	330.1	93.60	10.38	0.33	19330.57
30.0	0.79	386.4	396.1	93.60	10.41	0.40	19554.98
35.0	0.92	450.8	462.1	93.60	10.44	0.46	19742.34
40.0	1.05	515.2	528.1	93.60	10.47	0.53	19898.21
45.0	1.17	579.7	594.1	93.60	10.49	0.59	20026.86
50.0	1.30	644.1	660.2	93.59	10.51	0.66	20130.24
55.0	1.43	708.5	726.2	93.57	10.53	0.72	20214.55
60.0	1.56	772.9	792.2	93.56	10.55	0.79	20279.13
65.0	1.69	837.3	858.2	93.55	10.56	0.85	20325.86
70.0	1.82	901.7	924.2	93.55	10.58	0.92	20356.36
75.0	1.95	966.1	990.2	93.54	10.59	0.98	20367.89
80.0	2.08	1030.5	1056.3	93.53	10.60	1.05	20370.92
85.0	2.21	1094.9	1122.3	93.53	10.61	1.11	20370.92
90.0	2.34	1159.3	1188.3	93.52	10.62	1.18	20370.93
95.0	2.47	1223.7	1254.3	93.52	10.62	1.24	20370.93
100.0	2.60	1288.1	1320.3	93.51	10.63	1.30	20370.93

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.02
 Compartment descr: Water Ballast 05.02
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
 Fore end at frame 132.00 (105.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	64.4	66.0	93.60-10.15	0.07	17673.23	
10.0	0.27	128.8	132.0	93.60-10.23	0.13	18284.28	
15.0	0.40	193.2	198.0	93.60-10.29	0.20	18718.10	
20.0	0.53	257.6	264.1	93.60-10.34	0.27	19058.20	
25.0	0.66	322.0	330.1	93.60-10.38	0.33	19330.57	
30.0	0.79	386.4	396.1	93.60-10.41	0.40	19554.98	
35.0	0.92	450.8	462.1	93.60-10.44	0.46	19742.34	
40.0	1.05	515.2	528.1	93.60-10.47	0.53	19898.21	
45.0	1.17	579.7	594.1	93.60-10.49	0.59	20026.86	
50.0	1.30	644.1	660.2	93.59-10.51	0.66	20130.24	
55.0	1.43	708.5	726.2	93.57-10.53	0.72	20214.55	
60.0	1.56	772.9	792.2	93.56-10.55	0.79	20279.13	
65.0	1.69	837.3	858.2	93.55-10.56	0.85	20325.86	
70.0	1.82	901.7	924.2	93.55-10.58	0.92	20356.36	
75.0	1.95	966.1	990.2	93.54-10.59	0.98	20367.89	
80.0	2.08	1030.5	1056.3	93.53-10.60	1.05	20370.92	
85.0	2.21	1094.9	1122.3	93.53-10.61	1.11	20370.92	
90.0	2.34	1159.3	1188.3	93.52-10.62	1.18	20370.93	
95.0	2.47	1223.7	1254.3	93.52-10.62	1.24	20370.93	
100.0	2.60	1288.1	1320.3	93.51-10.63	1.30	20370.93	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.05
Compartment descr: Water Ballast 05.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
Fore end at frame 132.00 (105.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	105.3	107.9	93.30	5.53	2.80	2809.29
10.0	0.81	210.6	215.9	93.30	5.53	3.01	2809.29
15.0	1.22	315.9	323.8	93.30	5.53	3.21	2809.29
20.0	1.62	421.2	431.8	93.33	5.55	3.41	3003.41
25.0	2.01	526.6	539.7	93.38	5.58	3.61	3003.41
30.0	2.41	631.9	647.7	93.41	5.59	3.81	3003.41
35.0	2.80	737.2	755.6	93.43	5.60	4.01	3003.41
40.0	3.23	842.5	863.6	93.45	5.56	4.21	2254.91
45.0	3.66	947.8	971.5	93.46	5.52	4.42	2254.91
50.0	4.10	1053.1	1079.4	93.47	5.48	4.62	2254.91
55.0	4.53	1158.4	1187.4	93.48	5.45	4.83	2254.91
60.0	4.96	1263.7	1295.3	93.48	5.43	5.04	2254.91
65.0	5.40	1369.0	1403.3	93.49	5.41	5.25	2254.91
70.0	5.83	1474.4	1511.2	93.49	5.39	5.46	2254.91
75.0	6.23	1579.7	1619.2	93.50	5.41	5.67	3017.23
80.0	6.63	1685.0	1727.1	93.51	5.42	5.88	3017.23
85.0	7.02	1790.3	1835.0	93.51	5.44	6.09	3017.23
90.0	7.41	1895.6	1943.0	93.52	5.45	6.30	3017.23
95.0	7.81	2000.9	2050.9	93.52	5.46	6.50	3017.23
100.0	8.20	2106.2	2158.9	93.53	5.48	6.71	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.06
Compartment descr: Water Ballast 05.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
Fore end at frame 132.00 (105.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	105.3	107.9	93.30	-5.53	2.80	2809.29
10.0	0.81	210.6	215.9	93.30	-5.53	3.01	2809.29
15.0	1.22	315.9	323.8	93.30	-5.53	3.21	2809.29
20.0	1.62	421.2	431.8	93.33	-5.55	3.41	3003.41
25.0	2.01	526.6	539.7	93.38	-5.58	3.61	3003.41
30.0	2.41	631.9	647.7	93.41	-5.59	3.81	3003.41
35.0	2.80	737.2	755.6	93.43	-5.60	4.01	3003.41
40.0	3.23	842.5	863.6	93.45	-5.56	4.21	2254.91
45.0	3.66	947.8	971.5	93.46	-5.52	4.42	2254.91
50.0	4.10	1053.1	1079.4	93.47	-5.48	4.62	2254.91
55.0	4.53	1158.4	1187.4	93.48	-5.45	4.83	2254.91
60.0	4.96	1263.7	1295.3	93.48	-5.43	5.04	2254.91
65.0	5.40	1369.0	1403.3	93.49	-5.41	5.25	2254.91
70.0	5.83	1474.4	1511.2	93.49	-5.39	5.46	2254.91
75.0	6.23	1579.7	1619.2	93.50	-5.41	5.67	3017.23
80.0	6.63	1685.0	1727.1	93.51	-5.42	5.88	3017.23
85.0	7.02	1790.3	1835.0	93.51	-5.44	6.09	3017.23
90.0	7.41	1895.6	1943.0	93.52	-5.45	6.30	3017.23
95.0	7.81	2000.9	2050.9	93.52	-5.46	6.50	3017.23
100.0	8.20	2106.2	2158.9	93.53	-5.48	6.71	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.07
 Compartment descr: Water Ballast 05.07
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
 Fore end at frame 132.00 (105.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	93.60	16.44	2.80	2127.84
10.0	0.82	195.3	200.2	93.60	16.44	3.01	2127.84
15.0	1.23	292.9	300.2	93.60	16.44	3.21	2127.84
20.0	1.64	390.5	400.3	93.60	16.44	3.42	2127.84
25.0	2.05	488.2	500.4	93.60	16.44	3.62	2127.84
30.0	2.46	585.8	600.5	93.60	16.44	3.83	2127.84
35.0	2.87	683.5	700.5	93.60	16.44	4.03	2127.84
40.0	3.28	781.1	800.6	93.60	16.44	4.24	2127.84
45.0	3.69	878.7	900.7	93.60	16.44	4.44	2127.84
50.0	4.10	976.4	1000.8	93.60	16.44	4.65	2127.84
55.0	4.51	1074.0	1100.9	93.60	16.44	4.85	2127.84
60.0	4.92	1171.6	1200.9	93.60	16.44	5.06	2127.84
65.0	5.33	1269.3	1301.0	93.60	16.44	5.26	2127.84
70.0	5.74	1366.9	1401.1	93.60	16.44	5.47	2127.84
75.0	6.15	1464.6	1501.2	93.60	16.44	5.67	2127.84
80.0	6.56	1562.2	1601.3	93.60	16.44	5.88	2127.84
85.0	6.97	1659.8	1701.3	93.60	16.44	6.08	2127.84
90.0	7.38	1757.5	1801.4	93.60	16.44	6.29	2127.84
95.0	7.79	1855.1	1901.5	93.60	16.44	6.49	2127.84
100.0	8.20	1952.7	2001.6	93.60	16.44	6.70	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.08
 Compartment descr: Water Ballast 05.08
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
 Fore end at frame 132.00 (105.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	93.60-16.44	2.80	2127.84	
10.0	0.82	195.3	200.2	93.60-16.44	3.01	2127.84	
15.0	1.23	292.9	300.2	93.60-16.44	3.21	2127.84	
20.0	1.64	390.5	400.3	93.60-16.44	3.42	2127.84	
25.0	2.05	488.2	500.4	93.60-16.44	3.62	2127.84	
30.0	2.46	585.8	600.5	93.60-16.44	3.83	2127.84	
35.0	2.87	683.5	700.5	93.60-16.44	4.03	2127.84	
40.0	3.28	781.1	800.6	93.60-16.44	4.24	2127.84	
45.0	3.69	878.7	900.7	93.60-16.44	4.44	2127.84	
50.0	4.10	976.4	1000.8	93.60-16.44	4.65	2127.84	
55.0	4.51	1074.0	1100.9	93.60-16.44	4.85	2127.84	
60.0	4.92	1171.6	1200.9	93.60-16.44	5.06	2127.84	
65.0	5.33	1269.3	1301.0	93.60-16.44	5.26	2127.84	
70.0	5.74	1366.9	1401.1	93.60-16.44	5.47	2127.84	
75.0	6.15	1464.6	1501.2	93.60-16.44	5.67	2127.84	
80.0	6.56	1562.2	1601.3	93.60-16.44	5.88	2127.84	
85.0	6.97	1659.8	1701.3	93.60-16.44	6.08	2127.84	
90.0	7.38	1757.5	1801.4	93.60-16.44	6.29	2127.84	
95.0	7.79	1855.1	1901.5	93.60-16.44	6.49	2127.84	
100.0	8.20	1952.7	2001.6	93.60-16.44	6.70	2127.84	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.09
Compartment descr: Water Ballast 05.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
Fore end at frame 132.00 (105.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	93.60	5.69	10.85	3017.23
10.0	0.22	58.9	60.3	93.60	5.69	10.91	3017.23
15.0	0.33	88.3	90.5	93.60	5.69	10.96	3017.23
20.0	0.44	117.7	120.7	93.60	5.69	11.02	3017.23
25.0	0.55	147.1	150.8	93.60	5.69	11.07	3017.23
30.0	0.66	176.6	181.0	93.60	5.69	11.13	3017.23
35.0	0.77	206.0	211.2	93.60	5.69	11.18	3017.23
40.0	0.88	235.4	241.3	93.60	5.69	11.24	3017.23
45.0	0.99	264.9	271.5	93.60	5.69	11.30	3017.23
50.0	1.10	294.3	301.7	93.60	5.69	11.35	3017.23
55.0	1.21	323.7	331.8	93.60	5.69	11.40	3017.23
60.0	1.32	353.2	362.0	93.60	5.69	11.46	3017.23
65.0	1.43	382.6	392.1	93.60	5.69	11.51	3017.23
70.0	1.54	412.0	422.3	93.60	5.69	11.57	3017.23
75.0	1.65	441.4	452.5	93.60	5.69	11.63	3017.23
80.0	1.76	470.9	482.6	93.60	5.69	11.68	3017.23
85.0	1.87	500.3	512.8	93.60	5.69	11.73	3017.23
90.0	1.98	529.7	543.0	93.60	5.69	11.79	3017.23
95.0	2.09	559.2	573.1	93.60	5.69	11.84	3017.23
100.0	2.20	588.6	603.3	93.60	5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.10
 Compartment descr: Water Ballast 05.10
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
 Fore end at frame 132.00 (105.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	93.60	-5.69	10.85	3017.23
10.0	0.22	58.9	60.3	93.60	-5.69	10.91	3017.23
15.0	0.33	88.3	90.5	93.60	-5.69	10.96	3017.23
20.0	0.44	117.7	120.7	93.60	-5.69	11.02	3017.23
25.0	0.55	147.1	150.8	93.60	-5.69	11.07	3017.23
30.0	0.66	176.6	181.0	93.60	-5.69	11.13	3017.23
35.0	0.77	206.0	211.2	93.60	-5.69	11.18	3017.23
40.0	0.88	235.4	241.3	93.60	-5.69	11.24	3017.23
45.0	0.99	264.9	271.5	93.60	-5.69	11.30	3017.23
50.0	1.10	294.3	301.7	93.60	-5.69	11.35	3017.23
55.0	1.21	323.7	331.8	93.60	-5.69	11.40	3017.23
60.0	1.32	353.2	362.0	93.60	-5.69	11.46	3017.23
65.0	1.43	382.6	392.1	93.60	-5.69	11.51	3017.23
70.0	1.54	412.0	422.3	93.60	-5.69	11.57	3017.23
75.0	1.65	441.4	452.5	93.60	-5.69	11.63	3017.23
80.0	1.76	470.9	482.6	93.60	-5.69	11.68	3017.23
85.0	1.87	500.3	512.8	93.60	-5.69	11.73	3017.23
90.0	1.98	529.7	543.0	93.60	-5.69	11.79	3017.23
95.0	2.09	559.2	573.1	93.60	-5.69	11.84	3017.23
100.0	2.20	588.6	603.3	93.60	-5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.11
Compartment descr: Water Ballast 05.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
Fore end at frame 132.00 (105.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.0	26.6	93.60	16.44	10.85	2127.83
10.0	0.22	52.0	53.3	93.60	16.44	10.91	2127.83
15.0	0.33	78.0	79.9	93.60	16.44	10.96	2127.83
20.0	0.44	104.0	106.6	93.60	16.44	11.02	2127.83
25.0	0.55	129.9	133.2	93.60	16.44	11.07	2127.83
30.0	0.65	155.9	159.8	93.60	16.44	11.13	2127.83
35.0	0.76	181.9	186.5	93.60	16.44	11.18	2127.83
40.0	0.87	207.9	213.1	93.60	16.44	11.24	2127.83
45.0	0.98	233.9	239.8	93.60	16.44	11.29	2127.83
50.0	1.09	259.9	266.4	93.60	16.44	11.35	2127.83
55.0	1.20	285.9	293.0	93.60	16.44	11.40	2032.56
60.0	1.31	311.9	319.7	93.61	16.43	11.45	2032.56
65.0	1.42	337.9	346.3	93.62	16.42	11.51	2032.56
70.0	1.53	363.9	373.0	93.63	16.42	11.56	2032.56
75.0	1.64	389.8	399.6	93.64	16.42	11.62	2032.56
80.0	1.76	415.8	426.2	93.64	16.41	11.67	2032.55
85.0	1.87	441.8	452.9	93.65	16.41	11.73	2032.55
90.0	1.98	467.8	479.5	93.66	16.41	11.78	2032.55
95.0	2.09	493.8	506.2	93.66	16.40	11.84	2032.55
100.0	2.20	519.8	532.8	93.66	16.40	11.90	2032.22

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB05.12
Compartment descr: Water Ballast 05.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 102.00 (81.60 m)
Fore end at frame 132.00 (105.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.0	26.6	93.60-16.44	10.85	2127.83	
10.0	0.22	52.0	53.3	93.60-16.44	10.91	2127.83	
15.0	0.33	78.0	79.9	93.60-16.44	10.96	2127.83	
20.0	0.44	104.0	106.6	93.60-16.44	11.02	2127.83	
25.0	0.55	129.9	133.2	93.60-16.44	11.07	2127.83	
30.0	0.65	155.9	159.8	93.60-16.44	11.13	2127.83	
35.0	0.76	181.9	186.5	93.60-16.44	11.18	2127.83	
40.0	0.87	207.9	213.1	93.60-16.44	11.24	2127.83	
45.0	0.98	233.9	239.8	93.60-16.44	11.29	2127.83	
50.0	1.09	259.9	266.4	93.60-16.44	11.35	2127.83	
55.0	1.20	285.9	293.0	93.60-16.44	11.40	2032.56	
60.0	1.31	311.9	319.7	93.61-16.43	11.45	2032.56	
65.0	1.42	337.9	346.3	93.62-16.42	11.51	2032.56	
70.0	1.53	363.9	373.0	93.63-16.42	11.56	2032.56	
75.0	1.64	389.8	399.6	93.64-16.42	11.62	2032.56	
80.0	1.76	415.8	426.2	93.64-16.41	11.67	2032.56	
85.0	1.87	441.8	452.9	93.65-16.41	11.73	2032.56	
90.0	1.98	467.8	479.5	93.66-16.41	11.78	2032.56	
95.0	2.09	493.8	506.2	93.66-16.40	11.84	2032.56	
100.0	2.20	519.8	532.8	93.66-16.40	11.90	2032.23	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.01
Compartment descr: Water Ballast 06.01
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	64.4	66.0	117.60	10.15	0.07	17673.23
10.0	0.27	128.8	132.0	117.60	10.23	0.13	18284.29
15.0	0.40	193.2	198.0	117.60	10.29	0.20	18718.10
20.0	0.53	257.6	264.1	117.60	10.34	0.27	19058.20
25.0	0.66	322.0	330.1	117.60	10.38	0.33	19330.58
30.0	0.79	386.4	396.1	117.60	10.41	0.40	19554.99
35.0	0.92	450.8	462.1	117.60	10.44	0.46	19742.36
40.0	1.05	515.2	528.1	117.60	10.47	0.53	19898.22
45.0	1.17	579.7	594.1	117.60	10.49	0.59	20026.86
50.0	1.30	644.1	660.2	117.59	10.51	0.66	20130.25
55.0	1.43	708.5	726.2	117.57	10.53	0.72	20214.55
60.0	1.56	772.9	792.2	117.56	10.55	0.79	20279.14
65.0	1.69	837.3	858.2	117.55	10.56	0.85	20325.87
70.0	1.82	901.7	924.2	117.55	10.58	0.92	20356.36
75.0	1.95	966.1	990.2	117.54	10.59	0.98	20367.89
80.0	2.08	1030.5	1056.3	117.53	10.60	1.05	20370.93
85.0	2.21	1094.9	1122.3	117.53	10.61	1.11	20370.93
90.0	2.34	1159.3	1188.3	117.52	10.62	1.18	20370.93
95.0	2.47	1223.7	1254.3	117.52	10.62	1.24	20370.93
100.0	2.60	1288.1	1320.3	117.51	10.63	1.30	20370.93

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.02
Compartment descr: Water Ballast 06.02
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	64.4	66.0	117.60	-10.15	0.07	17673.23
10.0	0.27	128.8	132.0	117.60	-10.23	0.13	18284.29
15.0	0.40	193.2	198.0	117.60	-10.29	0.20	18718.10
20.0	0.53	257.6	264.1	117.60	-10.34	0.27	19058.20
25.0	0.66	322.0	330.1	117.60	-10.38	0.33	19330.58
30.0	0.79	386.4	396.1	117.60	-10.41	0.40	19554.99
35.0	0.92	450.8	462.1	117.60	-10.44	0.46	19742.36
40.0	1.05	515.2	528.1	117.60	-10.47	0.53	19898.22
45.0	1.17	579.7	594.1	117.60	-10.49	0.59	20026.86
50.0	1.30	644.1	660.2	117.59	-10.51	0.66	20130.25
55.0	1.43	708.5	726.2	117.57	-10.53	0.72	20214.55
60.0	1.56	772.9	792.2	117.56	-10.55	0.79	20279.14
65.0	1.69	837.3	858.2	117.55	-10.56	0.85	20325.87
70.0	1.82	901.7	924.2	117.55	-10.58	0.92	20356.36
75.0	1.95	966.1	990.2	117.54	-10.59	0.98	20367.89
80.0	2.08	1030.5	1056.3	117.53	-10.60	1.05	20370.93
85.0	2.21	1094.9	1122.3	117.53	-10.61	1.11	20370.93
90.0	2.34	1159.3	1188.3	117.52	-10.62	1.18	20370.93
95.0	2.47	1223.7	1254.3	117.52	-10.62	1.24	20370.93
100.0	2.60	1288.1	1320.3	117.51	-10.63	1.30	20370.93

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.05
Compartment descr: Water Ballast 06.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	105.3	107.9	117.30	5.53	2.80	2809.29
10.0	0.81	210.6	215.9	117.30	5.53	3.01	2809.29
15.0	1.22	315.9	323.8	117.30	5.53	3.21	2809.29
20.0	1.62	421.2	431.8	117.33	5.55	3.41	3003.41
25.0	2.01	526.6	539.7	117.38	5.58	3.61	3003.41
30.0	2.41	631.9	647.7	117.41	5.59	3.81	3003.41
35.0	2.80	737.2	755.6	117.43	5.60	4.01	3003.41
40.0	3.23	842.5	863.6	117.45	5.56	4.21	2254.91
45.0	3.66	947.8	971.5	117.46	5.52	4.42	2254.91
50.0	4.10	1053.1	1079.4	117.47	5.48	4.62	2254.91
55.0	4.53	1158.4	1187.4	117.48	5.45	4.83	2254.91
60.0	4.96	1263.7	1295.3	117.48	5.43	5.04	2254.91
65.0	5.40	1369.0	1403.3	117.49	5.41	5.25	2254.91
70.0	5.83	1474.4	1511.2	117.49	5.39	5.46	2254.91
75.0	6.23	1579.7	1619.2	117.50	5.41	5.67	3017.23
80.0	6.63	1685.0	1727.1	117.51	5.42	5.88	3017.23
85.0	7.02	1790.3	1835.0	117.51	5.44	6.09	3017.23
90.0	7.41	1895.6	1943.0	117.52	5.45	6.30	3017.23
95.0	7.81	2000.9	2050.9	117.52	5.46	6.50	3017.23
100.0	8.20	2106.2	2158.9	117.53	5.48	6.71	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.06
Compartment descr: Water Ballast 06.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	105.3	107.9	117.30	-5.53	2.80	2809.29
10.0	0.81	210.6	215.9	117.30	-5.53	3.01	2809.29
15.0	1.22	315.9	323.8	117.30	-5.53	3.21	2809.29
20.0	1.62	421.2	431.8	117.33	-5.55	3.41	3003.41
25.0	2.01	526.6	539.7	117.38	-5.58	3.61	3003.41
30.0	2.41	631.9	647.7	117.41	-5.59	3.81	3003.41
35.0	2.80	737.2	755.6	117.43	-5.60	4.01	3003.41
40.0	3.23	842.5	863.6	117.45	-5.56	4.21	2254.91
45.0	3.66	947.8	971.5	117.46	-5.52	4.42	2254.91
50.0	4.10	1053.1	1079.4	117.47	-5.48	4.62	2254.91
55.0	4.53	1158.4	1187.4	117.48	-5.45	4.83	2254.91
60.0	4.96	1263.7	1295.3	117.48	-5.43	5.04	2254.91
65.0	5.40	1369.0	1403.3	117.49	-5.41	5.25	2254.91
70.0	5.83	1474.4	1511.2	117.49	-5.39	5.46	2254.91
75.0	6.23	1579.7	1619.2	117.50	-5.41	5.67	3017.23
80.0	6.63	1685.0	1727.1	117.51	-5.42	5.88	3017.23
85.0	7.02	1790.3	1835.0	117.51	-5.44	6.09	3017.23
90.0	7.41	1895.6	1943.0	117.52	-5.45	6.30	3017.23
95.0	7.81	2000.9	2050.9	117.52	-5.46	6.50	3017.23
100.0	8.20	2106.2	2158.9	117.53	-5.48	6.71	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.07
Compartment descr: Water Ballast 06.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	117.60	16.44	2.81	2127.84
10.0	0.82	195.3	200.2	117.60	16.44	3.01	2127.84
15.0	1.23	292.9	300.2	117.60	16.44	3.21	2127.84
20.0	1.64	390.5	400.3	117.60	16.44	3.42	2127.84
25.0	2.05	488.2	500.4	117.60	16.44	3.62	2127.84
30.0	2.46	585.8	600.5	117.60	16.44	3.83	2127.84
35.0	2.87	683.5	700.5	117.60	16.44	4.03	2127.84
40.0	3.28	781.1	800.6	117.60	16.44	4.24	2127.84
45.0	3.69	878.7	900.7	117.60	16.44	4.44	2127.84
50.0	4.10	976.4	1000.8	117.60	16.44	4.65	2127.84
55.0	4.51	1074.0	1100.9	117.60	16.44	4.85	2127.84
60.0	4.92	1171.6	1200.9	117.60	16.44	5.06	2127.84
65.0	5.33	1269.3	1301.0	117.60	16.44	5.26	2127.84
70.0	5.74	1366.9	1401.1	117.60	16.44	5.47	2127.84
75.0	6.15	1464.6	1501.2	117.60	16.44	5.67	2127.84
80.0	6.56	1562.2	1601.3	117.60	16.44	5.88	2127.84
85.0	6.97	1659.8	1701.3	117.60	16.44	6.08	2127.84
90.0	7.38	1757.5	1801.4	117.60	16.44	6.29	2127.84
95.0	7.79	1855.1	1901.5	117.60	16.44	6.49	2127.84
100.0	8.20	1952.7	2001.6	117.60	16.44	6.70	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.08
 Compartment descr: Water Ballast 06.08
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
 Fore end at frame 162.00 (129.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	97.6	100.1	117.60-16.44	2.81	2127.84	
10.0	0.82	195.3	200.2	117.60-16.44	3.01	2127.84	
15.0	1.23	292.9	300.2	117.60-16.44	3.21	2127.84	
20.0	1.64	390.5	400.3	117.60-16.44	3.42	2127.84	
25.0	2.05	488.2	500.4	117.60-16.44	3.62	2127.84	
30.0	2.46	585.8	600.5	117.60-16.44	3.83	2127.84	
35.0	2.87	683.5	700.5	117.60-16.44	4.03	2127.84	
40.0	3.28	781.1	800.6	117.60-16.44	4.24	2127.84	
45.0	3.69	878.7	900.7	117.60-16.44	4.44	2127.84	
50.0	4.10	976.4	1000.8	117.60-16.44	4.65	2127.84	
55.0	4.51	1074.0	1100.9	117.60-16.44	4.85	2127.84	
60.0	4.92	1171.6	1200.9	117.60-16.44	5.06	2127.84	
65.0	5.33	1269.3	1301.0	117.60-16.44	5.26	2127.84	
70.0	5.74	1366.9	1401.1	117.60-16.44	5.47	2127.84	
75.0	6.15	1464.6	1501.2	117.60-16.44	5.67	2127.84	
80.0	6.56	1562.2	1601.3	117.60-16.44	5.88	2127.84	
85.0	6.97	1659.8	1701.3	117.60-16.44	6.08	2127.84	
90.0	7.38	1757.5	1801.4	117.60-16.44	6.29	2127.84	
95.0	7.79	1855.1	1901.5	117.60-16.44	6.49	2127.84	
100.0	8.20	1952.7	2001.6	117.60-16.44	6.70	2127.84	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.09
Compartment descr: Water Ballast 06.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	117.60	5.69	10.85	3017.23
10.0	0.22	58.9	60.3	117.60	5.69	10.91	3017.23
15.0	0.33	88.3	90.5	117.60	5.69	10.96	3017.23
20.0	0.44	117.7	120.7	117.60	5.69	11.02	3017.23
25.0	0.55	147.1	150.8	117.60	5.69	11.07	3017.23
30.0	0.66	176.6	181.0	117.60	5.69	11.13	3017.23
35.0	0.77	206.0	211.2	117.60	5.69	11.18	3017.23
40.0	0.88	235.4	241.3	117.60	5.69	11.24	3017.23
45.0	0.99	264.9	271.5	117.60	5.69	11.30	3017.23
50.0	1.10	294.3	301.7	117.60	5.69	11.35	3017.23
55.0	1.21	323.7	331.8	117.60	5.69	11.40	3017.23
60.0	1.32	353.2	362.0	117.60	5.69	11.46	3017.23
65.0	1.43	382.6	392.1	117.60	5.69	11.51	3017.23
70.0	1.54	412.0	422.3	117.60	5.69	11.57	3017.23
75.0	1.65	441.4	452.5	117.60	5.69	11.63	3017.23
80.0	1.76	470.9	482.6	117.60	5.69	11.68	3017.23
85.0	1.87	500.3	512.8	117.60	5.69	11.73	3017.23
90.0	1.98	529.7	543.0	117.60	5.69	11.79	3017.23
95.0	2.09	559.2	573.1	117.60	5.69	11.84	3017.23
100.0	2.20	588.6	603.3	117.60	5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.10
Compartment descr: Water Ballast 06.10
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	29.4	30.2	117.60	-5.69	10.85	3017.23
10.0	0.22	58.9	60.3	117.60	-5.69	10.91	3017.23
15.0	0.33	88.3	90.5	117.60	-5.69	10.96	3017.23
20.0	0.44	117.7	120.7	117.60	-5.69	11.02	3017.23
25.0	0.55	147.1	150.8	117.60	-5.69	11.07	3017.23
30.0	0.66	176.6	181.0	117.60	-5.69	11.13	3017.23
35.0	0.77	206.0	211.2	117.60	-5.69	11.18	3017.23
40.0	0.88	235.4	241.3	117.60	-5.69	11.24	3017.23
45.0	0.99	264.9	271.5	117.60	-5.69	11.30	3017.23
50.0	1.10	294.3	301.7	117.60	-5.69	11.35	3017.23
55.0	1.21	323.7	331.8	117.60	-5.69	11.40	3017.23
60.0	1.32	353.2	362.0	117.60	-5.69	11.46	3017.23
65.0	1.43	382.6	392.1	117.60	-5.69	11.51	3017.23
70.0	1.54	412.0	422.3	117.60	-5.69	11.57	3017.23
75.0	1.65	441.4	452.5	117.60	-5.69	11.63	3017.23
80.0	1.76	470.9	482.6	117.60	-5.69	11.68	3017.23
85.0	1.87	500.3	512.8	117.60	-5.69	11.73	3017.23
90.0	1.98	529.7	543.0	117.60	-5.69	11.79	3017.23
95.0	2.09	559.2	573.1	117.60	-5.69	11.84	3017.23
100.0	2.20	588.6	603.3	117.60	-5.69	11.90	3017.23

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.11
Compartment descr: Water Ballast 06.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.2	26.9	117.60	16.44	10.86	2127.84
10.0	0.22	52.4	53.7	117.60	16.44	10.91	2127.84
15.0	0.33	78.6	80.6	117.60	16.44	10.96	2127.84
20.0	0.44	104.8	107.4	117.60	16.44	11.02	2127.84
25.0	0.55	131.0	134.3	117.60	16.44	11.07	2127.84
30.0	0.66	157.2	161.1	117.60	16.44	11.13	2127.84
35.0	0.77	183.4	188.0	117.60	16.44	11.18	2127.84
40.0	0.88	209.6	214.8	117.60	16.44	11.24	2127.84
45.0	0.99	235.8	241.7	117.60	16.44	11.30	2127.84
50.0	1.10	262.0	268.5	117.60	16.44	11.35	2127.84
55.0	1.21	288.1	295.4	117.60	16.44	11.40	2127.84
60.0	1.32	314.3	322.2	117.60	16.44	11.46	2127.84
65.0	1.43	340.5	349.1	117.60	16.44	11.51	2127.84
70.0	1.54	366.7	375.9	117.60	16.44	11.57	2127.84
75.0	1.65	392.9	402.8	117.60	16.44	11.63	2127.84
80.0	1.76	419.1	429.6	117.60	16.44	11.68	2127.84
85.0	1.87	445.3	456.5	117.60	16.44	11.74	2127.84
90.0	1.98	471.5	483.3	117.60	16.44	11.79	2127.84
95.0	2.09	497.7	510.2	117.60	16.44	11.84	2127.84
100.0	2.20	523.9	537.0	117.60	16.44	11.90	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB06.12
Compartment descr: Water Ballast 06.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 132.00 (105.60 m)
Fore end at frame 162.00 (129.60 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	26.2	26.9	117.60	-16.44	10.86	2127.84
10.0	0.22	52.4	53.7	117.60	-16.44	10.91	2127.84
15.0	0.33	78.6	80.6	117.60	-16.44	10.96	2127.84
20.0	0.44	104.8	107.4	117.60	-16.44	11.02	2127.84
25.0	0.55	131.0	134.3	117.60	-16.44	11.07	2127.84
30.0	0.66	157.2	161.1	117.60	-16.44	11.13	2127.84
35.0	0.77	183.4	188.0	117.60	-16.44	11.18	2127.84
40.0	0.88	209.6	214.8	117.60	-16.44	11.24	2127.84
45.0	0.99	235.8	241.7	117.60	-16.44	11.30	2127.84
50.0	1.10	262.0	268.5	117.60	-16.44	11.35	2127.84
55.0	1.21	288.1	295.4	117.60	-16.44	11.40	2127.84
60.0	1.32	314.3	322.2	117.60	-16.44	11.46	2127.84
65.0	1.43	340.5	349.1	117.60	-16.44	11.51	2127.84
70.0	1.54	366.7	375.9	117.60	-16.44	11.57	2127.84
75.0	1.65	392.9	402.8	117.60	-16.44	11.63	2127.84
80.0	1.76	419.1	429.6	117.60	-16.44	11.68	2127.84
85.0	1.87	445.3	456.5	117.60	-16.44	11.74	2127.84
90.0	1.98	471.5	483.3	117.60	-16.44	11.79	2127.84
95.0	2.09	497.7	510.2	117.60	-16.44	11.84	2127.84
100.0	2.20	523.9	537.0	117.60	-16.44	11.90	2127.84

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.01
Compartment descr: Water Ballast 07.01
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.14	75.9	77.8	143.66	9.85	0.07	19593.60
10.0	0.27	151.8	155.6	143.68	9.94	0.14	20376.53
15.0	0.40	227.7	233.4	143.69	10.01	0.20	20944.61
20.0	0.54	303.6	311.1	143.70	10.07	0.27	21395.88
25.0	0.67	379.4	388.9	143.71	10.11	0.34	21770.71
30.0	0.80	455.3	466.7	143.71	10.15	0.40	22087.01
35.0	0.93	531.2	544.5	143.72	10.19	0.47	22356.47
40.0	1.05	607.1	622.3	143.73	10.22	0.53	22586.74
45.0	1.18	683.0	700.1	143.73	10.25	0.60	22784.32
50.0	1.31	758.9	777.8	143.72	10.27	0.66	22953.95
55.0	1.44	834.8	855.6	143.71	10.30	0.73	23101.67
60.0	1.57	910.7	933.4	143.71	10.32	0.79	23225.28
65.0	1.70	986.5	1011.2	143.70	10.34	0.86	23331.03
70.0	1.83	1062.4	1089.0	143.70	10.36	0.92	23418.90
75.0	1.96	1138.3	1166.8	143.70	10.37	0.99	23491.21
80.0	2.09	1214.2	1244.6	143.70	10.39	1.05	23553.40
85.0	2.22	1290.1	1322.3	143.69	10.40	1.12	23603.86
90.0	2.34	1366.0	1400.1	143.69	10.41	1.18	23654.83
95.0	2.47	1441.9	1477.9	143.69	10.42	1.25	23702.87
100.0	2.60	1517.8	1555.7	143.70	10.44	1.31	23745.01

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.02
Compartment descr: Water Ballast 07.02
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.14	75.9	77.8	143.66	-9.85	0.07	19593.60
10.0	0.27	151.8	155.6	143.68	-9.94	0.14	20376.53
15.0	0.40	227.7	233.4	143.69	-10.01	0.20	20944.61
20.0	0.54	303.6	311.1	143.70	-10.07	0.27	21395.88
25.0	0.67	379.4	388.9	143.71	-10.11	0.34	21770.71
30.0	0.80	455.3	466.7	143.71	-10.15	0.40	22087.01
35.0	0.93	531.2	544.5	143.72	-10.19	0.47	22356.47
40.0	1.05	607.1	622.3	143.73	-10.22	0.53	22586.74
45.0	1.18	683.0	700.1	143.73	-10.25	0.60	22784.32
50.0	1.31	758.9	777.8	143.72	-10.27	0.66	22953.95
55.0	1.44	834.8	855.6	143.71	-10.30	0.73	23101.67
60.0	1.57	910.7	933.4	143.71	-10.32	0.79	23225.28
65.0	1.70	986.5	1011.2	143.70	-10.34	0.86	23331.03
70.0	1.83	1062.4	1089.0	143.70	-10.36	0.92	23418.90
75.0	1.96	1138.3	1166.8	143.70	-10.37	0.99	23491.21
80.0	2.09	1214.2	1244.6	143.70	-10.39	1.05	23553.40
85.0	2.22	1290.1	1322.3	143.69	-10.40	1.12	23603.86
90.0	2.34	1366.0	1400.1	143.69	-10.41	1.18	23654.83
95.0	2.47	1441.9	1477.9	143.69	-10.42	1.25	23702.87
100.0	2.60	1517.8	1555.7	143.70	-10.44	1.31	23745.01

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.05
Compartment descr: Water Ballast 07.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 2.60 m above BL
Highest point 11.40 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	126.8	130.0	143.68	5.56	2.80	3413.82
10.0	0.81	253.6	260.0	143.68	5.56	3.01	3413.82
15.0	1.22	380.5	390.0	143.68	5.56	3.21	3413.82
20.0	1.62	507.3	520.0	143.72	5.57	3.41	3606.86
25.0	2.01	634.1	649.9	143.77	5.59	3.61	3606.86
30.0	2.41	760.9	779.9	143.80	5.61	3.81	3606.86
35.0	2.81	887.7	909.9	143.83	5.62	4.01	3606.86
40.0	3.23	1014.5	1039.9	143.84	5.58	4.21	2692.06
45.0	3.67	1141.4	1169.9	143.85	5.53	4.42	2692.06
50.0	4.10	1268.2	1299.9	143.85	5.49	4.62	2692.06
55.0	4.54	1395.0	1429.9	143.86	5.46	4.83	2692.06
60.0	4.98	1521.8	1559.9	143.86	5.43	5.04	2692.06
65.0	5.41	1648.6	1689.9	143.87	5.41	5.25	2692.06
70.0	5.85	1775.5	1819.8	143.87	5.39	5.47	2692.06
75.0	6.25	1902.3	1949.8	143.88	5.41	5.68	3620.67
80.0	6.64	2029.1	2079.8	143.89	5.43	5.89	3620.67
85.0	7.04	2155.9	2209.8	143.89	5.44	6.10	3620.67
90.0	7.43	2282.7	2339.8	143.90	5.46	6.31	3620.67
95.0	7.83	2409.5	2469.8	143.90	5.47	6.51	3620.67
100.0	8.80	2536.4	2599.8	143.95	5.48	6.72	144.57

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.06
Compartment descr: Water Ballast 07.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 2.60 m above BL
Highest point 11.40 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.41	126.8	130.0	143.68	-5.56	2.80	3413.82
10.0	0.81	253.6	260.0	143.68	-5.56	3.01	3413.82
15.0	1.22	380.5	390.0	143.68	-5.56	3.21	3413.82
20.0	1.62	507.3	520.0	143.72	-5.57	3.41	3606.86
25.0	2.01	634.1	649.9	143.77	-5.59	3.61	3606.86
30.0	2.41	760.9	779.9	143.80	-5.61	3.81	3606.86
35.0	2.81	887.7	909.9	143.83	-5.62	4.01	3606.86
40.0	3.23	1014.5	1039.9	143.84	-5.58	4.21	2692.06
45.0	3.67	1141.4	1169.9	143.85	-5.53	4.42	2692.06
50.0	4.10	1268.2	1299.9	143.85	-5.49	4.62	2692.06
55.0	4.54	1395.0	1429.9	143.86	-5.46	4.83	2692.06
60.0	4.98	1521.8	1559.9	143.86	-5.43	5.04	2692.06
65.0	5.41	1648.6	1689.9	143.87	-5.41	5.25	2692.06
70.0	5.85	1775.5	1819.8	143.87	-5.39	5.47	2692.06
75.0	6.25	1902.3	1949.8	143.88	-5.41	5.68	3620.67
80.0	6.64	2029.1	2079.8	143.89	-5.43	5.89	3620.67
85.0	7.04	2155.9	2209.8	143.89	-5.44	6.10	3620.67
90.0	7.43	2282.7	2339.8	143.90	-5.46	6.31	3620.67
95.0	7.83	2409.5	2469.8	143.90	-5.47	6.51	3620.67
100.0	8.80	2536.4	2599.8	143.95	-5.48	6.72	144.57

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.07
Compartment descr: Water Ballast 07.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.42	116.6	119.5	143.76	16.34	2.81	2436.17
10.0	0.83	233.2	239.1	143.79	16.35	3.02	2456.82
15.0	1.24	349.8	358.6	143.81	16.36	3.22	2476.14
20.0	1.66	466.5	478.1	143.82	16.37	3.43	2491.17
25.0	2.07	583.1	597.6	143.84	16.37	3.64	2502.88
30.0	2.48	699.7	717.2	143.85	16.38	3.84	2513.34
35.0	2.89	816.3	836.7	143.86	16.38	4.05	2521.53
40.0	3.30	932.9	956.2	143.88	16.39	4.25	2528.07
45.0	3.71	1049.5	1075.8	143.88	16.39	4.46	2533.96
50.0	4.12	1166.1	1195.3	143.89	16.40	4.66	2538.60
55.0	4.53	1282.8	1314.8	143.90	16.40	4.87	2542.52
60.0	4.93	1399.4	1434.4	143.91	16.40	5.08	2545.61
65.0	5.34	1516.0	1553.9	143.91	16.40	5.28	2547.71
70.0	5.75	1632.6	1673.4	143.92	16.41	5.48	2549.57
75.0	6.16	1749.2	1792.9	143.92	16.41	5.69	2550.47
80.0	6.57	1865.8	1912.5	143.93	16.41	5.89	2551.37
85.0	6.98	1982.4	2032.0	143.93	16.41	6.10	2552.05
90.0	7.38	2099.1	2151.5	143.94	16.41	6.30	2552.62
95.0	7.79	2215.7	2271.1	143.94	16.41	6.51	2552.91
100.0	8.20	2332.3	2390.6	143.94	16.42	6.71	2553.22

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.08
Compartment descr: Water Ballast 07.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.42	116.6	119.5	143.76-16.34	2.81	2436.17	
10.0	0.83	233.2	239.1	143.79-16.35	3.02	2456.82	
15.0	1.24	349.8	358.6	143.81-16.36	3.22	2476.14	
20.0	1.66	466.5	478.1	143.82-16.37	3.43	2491.17	
25.0	2.07	583.1	597.6	143.84-16.37	3.64	2502.88	
30.0	2.48	699.7	717.2	143.85-16.38	3.84	2513.34	
35.0	2.89	816.3	836.7	143.86-16.38	4.05	2521.53	
40.0	3.30	932.9	956.2	143.88-16.39	4.25	2528.07	
45.0	3.71	1049.5	1075.8	143.88-16.39	4.46	2533.96	
50.0	4.12	1166.1	1195.3	143.89-16.40	4.66	2538.60	
55.0	4.53	1282.8	1314.8	143.90-16.40	4.87	2542.52	
60.0	4.93	1399.4	1434.4	143.91-16.40	5.08	2545.61	
65.0	5.34	1516.0	1553.9	143.91-16.40	5.28	2547.71	
70.0	5.75	1632.6	1673.4	143.92-16.41	5.48	2549.57	
75.0	6.16	1749.2	1792.9	143.92-16.41	5.69	2550.47	
80.0	6.57	1865.8	1912.5	143.93-16.41	5.89	2551.37	
85.0	6.98	1982.4	2032.0	143.93-16.41	6.10	2552.05	
90.0	7.38	2099.1	2151.5	143.94-16.41	6.30	2552.62	
95.0	7.79	2215.7	2271.1	143.94-16.41	6.51	2552.91	
100.0	8.20	2332.3	2390.6	143.94-16.42	6.71	2553.22	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.09
Compartment descr: Water Ballast 07.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	34.9	35.8	143.41	5.69	10.86	3469.81
10.0	0.23	69.8	71.6	143.41	5.69	10.91	3469.81
15.0	0.34	104.7	107.4	143.41	5.69	10.97	3469.81
20.0	0.45	139.7	143.1	143.41	5.69	11.03	3469.81
25.0	0.57	174.6	178.9	143.41	5.69	11.08	3469.81
30.0	0.68	209.5	214.7	143.48	5.69	11.14	3620.67
35.0	0.79	244.4	250.5	143.55	5.69	11.20	3620.67
40.0	0.89	279.3	286.3	143.61	5.69	11.25	3620.67
45.0	1.00	314.2	322.1	143.65	5.69	11.31	3620.67
50.0	1.11	349.1	357.9	143.69	5.69	11.36	3620.67
55.0	1.22	384.1	393.7	143.72	5.69	11.42	3620.67
60.0	1.33	419.0	429.4	143.74	5.69	11.47	3620.67
65.0	1.44	453.9	465.2	143.76	5.69	11.53	3620.67
70.0	1.55	488.8	501.0	143.78	5.69	11.58	3620.67
75.0	1.66	523.7	536.8	143.79	5.69	11.64	3620.67
80.0	1.76	558.6	572.6	143.80	5.69	11.69	3620.67
85.0	1.87	593.5	608.4	143.82	5.69	11.75	3620.67
90.0	1.98	628.5	644.2	143.83	5.69	11.80	3620.67
95.0	2.09	663.4	680.0	143.84	5.69	11.85	3620.67
100.0	2.20	698.3	715.7	143.84	5.69	11.91	3620.67

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.10
Compartment descr: Water Ballast 07.10
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	34.9	35.8	143.41	-5.69	10.86	3469.81
10.0	0.23	69.8	71.6	143.41	-5.69	10.91	3469.81
15.0	0.34	104.7	107.4	143.41	-5.69	10.97	3469.81
20.0	0.45	139.7	143.1	143.41	-5.69	11.03	3469.81
25.0	0.57	174.6	178.9	143.41	-5.69	11.08	3469.81
30.0	0.68	209.5	214.7	143.48	-5.69	11.14	3620.67
35.0	0.79	244.4	250.5	143.55	-5.69	11.20	3620.67
40.0	0.89	279.3	286.3	143.61	-5.69	11.25	3620.67
45.0	1.00	314.2	322.1	143.65	-5.69	11.31	3620.67
50.0	1.11	349.1	357.9	143.69	-5.69	11.36	3620.67
55.0	1.22	384.1	393.7	143.72	-5.69	11.42	3620.67
60.0	1.33	419.0	429.4	143.74	-5.69	11.47	3620.67
65.0	1.44	453.9	465.2	143.76	-5.69	11.53	3620.67
70.0	1.55	488.8	501.0	143.78	-5.69	11.58	3620.67
75.0	1.66	523.7	536.8	143.79	-5.69	11.64	3620.67
80.0	1.76	558.6	572.6	143.80	-5.69	11.69	3620.67
85.0	1.87	593.5	608.4	143.82	-5.69	11.75	3620.67
90.0	1.98	628.5	644.2	143.83	-5.69	11.80	3620.67
95.0	2.09	663.4	680.0	143.84	-5.69	11.85	3620.67
100.0	2.20	698.3	715.7	143.84	-5.69	11.91	3620.67

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.11
Compartment descr: Water Ballast 07.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	31.3	32.0	144.00	16.44	10.85	2553.31
10.0	0.22	62.5	64.1	144.00	16.44	10.91	2553.35
15.0	0.33	93.8	96.1	144.00	16.44	10.96	2553.37
20.0	0.44	125.0	128.2	144.00	16.44	11.02	2553.40
25.0	0.55	156.3	160.2	144.00	16.44	11.07	2553.39
30.0	0.66	187.6	192.2	144.00	16.44	11.13	2553.41
35.0	0.77	218.8	224.3	144.00	16.44	11.18	2553.41
40.0	0.88	250.1	256.3	144.00	16.44	11.24	2553.41
45.0	0.98	281.3	288.4	144.00	16.44	11.29	2553.41
50.0	1.09	312.6	320.4	144.00	16.44	11.35	2553.41
55.0	1.20	343.8	352.4	144.00	16.44	11.40	2482.95
60.0	1.31	375.1	384.5	144.01	16.43	11.46	2476.73
65.0	1.42	406.4	416.5	144.02	16.43	11.51	2470.84
70.0	1.54	437.6	448.6	144.03	16.43	11.57	2470.69
75.0	1.65	468.9	480.6	144.04	16.42	11.62	2470.69
80.0	1.76	500.1	512.6	144.04	16.42	11.68	2470.69
85.0	1.87	531.4	544.7	144.05	16.42	11.73	2470.68
90.0	1.98	562.7	576.7	144.05	16.41	11.79	2470.68
95.0	2.09	593.9	608.8	144.06	16.41	11.84	2470.68
100.0	2.20	625.2	640.8	144.06	16.41	11.90	2470.26

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB07.12
Compartment descr: Water Ballast 07.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 162.00 (129.60 m)
Fore end at frame 198.00 (158.40 m)
Lowest point 10.80 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	31.3	32.0	144.00-16.44	10.85	2553.31	
10.0	0.22	62.5	64.1	144.00-16.44	10.91	2553.35	
15.0	0.33	93.8	96.1	144.00-16.44	10.96	2553.37	
20.0	0.44	125.0	128.2	144.00-16.44	11.02	2553.40	
25.0	0.55	156.3	160.2	144.00-16.44	11.07	2553.39	
30.0	0.66	187.6	192.2	144.00-16.44	11.13	2553.41	
35.0	0.77	218.8	224.3	144.00-16.44	11.18	2553.41	
40.0	0.88	250.1	256.3	144.00-16.44	11.24	2553.41	
45.0	0.98	281.3	288.4	144.00-16.44	11.29	2553.41	
50.0	1.09	312.6	320.4	144.00-16.44	11.35	2553.41	
55.0	1.20	343.8	352.4	144.00-16.44	11.40	2482.95	
60.0	1.31	375.1	384.5	144.01-16.43	11.46	2476.73	
65.0	1.42	406.4	416.5	144.02-16.43	11.51	2470.82	
70.0	1.54	437.6	448.6	144.03-16.43	11.57	2470.69	
75.0	1.65	468.9	480.6	144.04-16.42	11.62	2470.69	
80.0	1.76	500.1	512.6	144.04-16.42	11.68	2470.69	
85.0	1.87	531.4	544.7	144.05-16.42	11.73	2470.69	
90.0	1.98	562.7	576.7	144.05-16.41	11.79	2470.69	
95.0	2.09	593.9	608.8	144.06-16.41	11.84	2470.69	
100.0	2.20	625.2	640.8	144.06-16.41	11.90	2470.27	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.01
 Compartment descr: Water Ballast 08.01
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	41.7	42.7	167.60	-0.04	0.05	18296.48
10.0	0.21	83.3	85.4	167.62	-0.04	0.10	18460.65
15.0	0.31	125.0	128.1	167.64	-0.04	0.16	18602.79
20.0	0.41	166.6	170.8	167.65	-0.04	0.21	18747.65
25.0	0.51	208.3	213.5	167.67	-0.04	0.26	18886.46
30.0	0.62	250.0	256.2	167.68	-0.04	0.31	19028.45
35.0	0.72	291.6	298.9	167.69	-0.04	0.36	19107.27
40.0	0.82	333.3	341.6	167.70	-0.04	0.41	19146.09
45.0	0.92	374.9	384.3	167.71	-0.04	0.46	19185.53
50.0	1.02	416.6	427.0	167.72	-0.04	0.51	19202.69
55.0	1.13	458.2	469.7	167.73	-0.04	0.56	19202.69
60.0	1.23	499.9	512.4	167.73	-0.04	0.62	19202.69
65.0	1.33	541.6	555.1	167.74	-0.04	0.67	19202.69
70.0	1.43	583.2	597.8	167.74	-0.04	0.72	19202.69
75.0	1.53	624.9	640.5	167.74	-0.04	0.77	19202.69
80.0	1.63	666.5	683.2	167.75	-0.04	0.82	19202.69
85.0	1.73	708.2	725.9	167.75	-0.04	0.87	19202.69
90.0	1.84	749.9	768.6	167.75	-0.04	0.92	19202.69
95.0	1.94	791.5	811.3	167.75	-0.04	0.97	19202.69
100.0	2.60	833.2	854.0	167.59	-0.04	1.03	108.41

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.03
 Compartment descr: Water Ballast 08.03
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.24	9.3	9.5	163.72	13.47	0.13	99.10
10.0	0.44	18.5	19.0	163.85	13.59	0.24	132.34
15.0	0.62	27.8	28.4	164.02	13.65	0.34	158.60
20.0	0.78	37.0	37.9	164.19	13.70	0.43	181.27
25.0	0.93	46.3	47.4	164.34	13.74	0.51	201.90
30.0	1.07	55.5	56.9	164.47	13.77	0.59	221.08
35.0	1.20	64.8	66.4	164.60	13.80	0.67	238.71
40.0	1.33	74.0	75.9	164.71	13.83	0.75	254.98
45.0	1.45	83.3	85.3	164.82	13.86	0.82	270.13
50.0	1.57	92.5	94.8	164.91	13.89	0.89	284.47
55.0	1.68	101.8	104.3	165.00	13.91	0.95	298.27
60.0	1.79	111.0	113.8	165.08	13.93	1.02	311.65
65.0	1.90	120.3	123.3	165.15	13.96	1.08	324.75
70.0	2.01	129.5	132.7	165.22	13.98	1.14	337.65
75.0	2.11	138.8	142.2	165.29	14.00	1.21	349.99
80.0	2.21	148.0	151.7	165.35	14.02	1.27	362.40
85.0	2.31	157.3	161.2	165.40	14.04	1.32	374.89
90.0	2.41	166.5	170.7	165.45	14.07	1.38	387.13
95.0	2.51	175.8	180.2	165.50	14.09	1.44	398.68
100.0	2.60	185.0	189.6	165.55	14.11	1.49	410.01

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.04
 Compartment descr: Water Ballast 08.04
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.24	9.3	9.5	163.72-13.47	0.13	99.10	
10.0	0.44	18.5	19.0	163.85-13.59	0.24	132.34	
15.0	0.62	27.8	28.4	164.02-13.65	0.34	158.60	
20.0	0.78	37.0	37.9	164.19-13.70	0.43	181.27	
25.0	0.93	46.3	47.4	164.34-13.74	0.51	201.90	
30.0	1.07	55.5	56.9	164.47-13.77	0.59	221.08	
35.0	1.20	64.8	66.4	164.60-13.80	0.67	238.71	
40.0	1.33	74.0	75.9	164.71-13.83	0.75	254.98	
45.0	1.45	83.3	85.3	164.82-13.86	0.82	270.13	
50.0	1.57	92.5	94.8	164.91-13.89	0.89	284.47	
55.0	1.68	101.8	104.3	165.00-13.91	0.95	298.27	
60.0	1.79	111.0	113.8	165.08-13.93	1.02	311.65	
65.0	1.90	120.3	123.3	165.15-13.96	1.08	324.75	
70.0	2.01	129.5	132.7	165.22-13.98	1.14	337.65	
75.0	2.11	138.8	142.2	165.29-14.00	1.21	349.99	
80.0	2.21	148.0	151.7	165.35-14.02	1.27	362.40	
85.0	2.31	157.3	161.2	165.40-14.04	1.32	374.89	
90.0	2.41	166.5	170.7	165.45-14.07	1.38	387.13	
95.0	2.51	175.8	180.2	165.50-14.09	1.44	398.68	
100.0	2.60	185.0	189.6	165.55-14.11	1.49	410.01	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.07
Compartment descr: Water Ballast 08.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
Fore end at frame 222.00 (177.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.57	58.5	60.0	166.55	14.59	2.89	479.81
10.0	1.10	117.0	120.0	166.63	14.67	3.16	542.80
15.0	1.60	175.6	179.9	166.70	14.75	3.42	601.25
20.0	2.07	234.1	239.9	166.76	14.82	3.68	655.47
25.0	2.53	292.6	299.9	166.81	14.88	3.92	706.43
30.0	2.97	351.1	359.9	166.85	14.94	4.16	794.12
35.0	3.39	409.6	419.9	166.87	14.98	4.39	840.46
40.0	3.80	468.1	479.8	166.88	15.02	4.62	883.96
45.0	4.20	526.7	539.8	166.90	15.06	4.84	926.17
50.0	4.59	585.2	599.8	166.93	15.10	5.05	966.86
55.0	4.97	643.7	659.8	166.95	15.14	5.26	1005.15
60.0	5.35	702.2	719.8	166.97	15.17	5.47	1043.95
65.0	5.72	760.7	779.7	166.99	15.21	5.68	1080.02
70.0	6.09	819.2	839.7	167.01	15.24	5.88	1114.96
75.0	6.45	877.8	899.7	167.03	15.28	6.08	1148.80
80.0	6.81	936.3	959.7	167.05	15.31	6.27	1181.28
85.0	7.16	994.8	1019.7	167.07	15.34	6.47	1211.92
90.0	7.51	1053.3	1079.6	167.09	15.37	6.66	1241.66
95.0	7.86	1111.8	1139.6	167.11	15.40	6.85	1269.04
100.0	8.20	1170.3	1199.6	167.13	15.42	7.04	1296.32

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.08
Compartment descr: Water Ballast 08.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
Fore end at frame 222.00 (177.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.57	58.5	60.0	166.55-14.59	2.89	479.81	
10.0	1.10	117.0	120.0	166.63-14.67	3.16	542.80	
15.0	1.60	175.6	179.9	166.70-14.75	3.42	601.25	
20.0	2.07	234.1	239.9	166.76-14.82	3.68	655.47	
25.0	2.53	292.6	299.9	166.81-14.88	3.92	706.43	
30.0	2.97	351.1	359.9	166.85-14.94	4.16	794.12	
35.0	3.39	409.6	419.9	166.87-14.98	4.39	840.46	
40.0	3.80	468.1	479.8	166.88-15.02	4.62	883.96	
45.0	4.20	526.7	539.8	166.90-15.06	4.84	926.17	
50.0	4.59	585.2	599.8	166.93-15.10	5.05	966.86	
55.0	4.97	643.7	659.8	166.95-15.14	5.26	1005.15	
60.0	5.35	702.2	719.8	166.97-15.17	5.47	1043.95	
65.0	5.72	760.7	779.7	166.99-15.21	5.68	1080.02	
70.0	6.09	819.2	839.7	167.01-15.24	5.88	1114.96	
75.0	6.45	877.8	899.7	167.03-15.28	6.08	1148.80	
80.0	6.81	936.3	959.7	167.05-15.31	6.27	1181.28	
85.0	7.16	994.8	1019.7	167.07-15.34	6.47	1211.92	
90.0	7.51	1053.3	1079.6	167.09-15.37	6.66	1241.66	
95.0	7.86	1111.8	1139.6	167.11-15.40	6.85	1269.04	
100.0	8.20	1170.3	1199.6	167.13-15.42	7.04	1296.32	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.09
Compartment descr: Water Ballast 08.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
Fore end at frame 222.00 (177.60 m)
Lowest point 11.40 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.08	17.1	17.6	168.00	5.69	11.44	2413.78
10.0	0.16	34.2	35.1	168.00	5.69	11.48	2413.78
15.0	0.24	51.4	52.7	168.00	5.69	11.52	2413.78
20.0	0.32	68.5	70.2	168.00	5.69	11.56	2413.78
25.0	0.40	85.6	87.8	168.00	5.69	11.60	2413.78
30.0	0.48	102.7	105.3	168.00	5.69	11.64	2413.78
35.0	0.56	119.9	122.9	168.00	5.69	11.68	2413.78
40.0	0.64	137.0	140.4	168.00	5.69	11.72	2413.78
45.0	0.72	154.1	158.0	168.00	5.69	11.76	2413.78
50.0	0.80	171.2	175.5	168.00	5.69	11.80	2413.78
55.0	0.88	188.3	193.1	168.00	5.69	11.84	2413.78
60.0	0.96	205.5	210.6	168.00	5.69	11.88	2413.78
65.0	1.04	222.6	228.2	168.00	5.69	11.92	2413.78
70.0	1.12	239.7	245.7	168.00	5.69	11.96	2413.78
75.0	1.20	256.8	263.3	168.00	5.69	12.00	2413.78
80.0	1.28	274.0	280.8	168.00	5.69	12.04	2413.78
85.0	1.36	291.1	298.4	168.00	5.69	12.08	2413.78
90.0	1.44	308.2	315.9	168.00	5.69	12.12	2413.78
95.0	1.52	325.3	333.5	168.00	5.69	12.16	2413.78
100.0	1.60	342.5	351.0	168.00	5.69	12.20	2413.78

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.10
Compartment descr: Water Ballast 08.10
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
Fore end at frame 222.00 (177.60 m)
Lowest point 11.40 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.08	17.1	17.6	168.00	-5.69	11.44	2413.78
10.0	0.16	34.2	35.1	168.00	-5.69	11.48	2413.78
15.0	0.24	51.4	52.7	168.00	-5.69	11.52	2413.78
20.0	0.32	68.5	70.2	168.00	-5.69	11.56	2413.78
25.0	0.40	85.6	87.8	168.00	-5.69	11.60	2413.78
30.0	0.48	102.7	105.3	168.00	-5.69	11.64	2413.78
35.0	0.56	119.9	122.9	168.00	-5.69	11.68	2413.78
40.0	0.64	137.0	140.4	168.00	-5.69	11.72	2413.78
45.0	0.72	154.1	158.0	168.00	-5.69	11.76	2413.78
50.0	0.80	171.2	175.5	168.00	-5.69	11.80	2413.78
55.0	0.88	188.3	193.1	168.00	-5.69	11.84	2413.78
60.0	0.96	205.5	210.6	168.00	-5.69	11.88	2413.78
65.0	1.04	222.6	228.2	168.00	-5.69	11.92	2413.78
70.0	1.12	239.7	245.7	168.00	-5.69	11.96	2413.78
75.0	1.20	256.8	263.3	168.00	-5.69	12.00	2413.78
80.0	1.28	274.0	280.8	168.00	-5.69	12.04	2413.78
85.0	1.36	291.1	298.4	168.00	-5.69	12.08	2413.78
90.0	1.44	308.2	315.9	168.00	-5.69	12.12	2413.78
95.0	1.52	325.3	333.5	168.00	-5.69	12.16	2413.78
100.0	1.60	342.5	351.0	168.00	-5.69	12.20	2413.78

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.11
 Compartment descr: Water Ballast 08.11
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	19.1	19.6	167.50	15.97	10.86	1304.80
10.0	0.22	38.2	39.1	167.50	15.97	10.91	1312.98
15.0	0.33	57.3	58.7	167.50	15.98	10.97	1320.86
20.0	0.44	76.4	78.3	167.51	15.98	11.02	1328.63
25.0	0.55	95.4	97.8	167.51	15.99	11.08	1336.44
30.0	0.66	114.5	117.4	167.52	15.99	11.13	1343.76
35.0	0.77	133.6	137.0	167.52	16.00	11.19	1350.72
40.0	0.88	152.7	156.5	167.52	16.00	11.24	1357.80
45.0	0.99	171.8	176.1	167.53	16.01	11.30	1364.82
50.0	1.10	190.9	195.7	167.53	16.01	11.35	1371.90
55.0	1.21	210.0	215.2	167.54	16.02	11.41	1305.37
60.0	1.32	229.1	234.8	167.55	16.01	11.46	1304.66
65.0	1.43	248.1	254.4	167.56	16.01	11.52	1304.49
70.0	1.54	267.2	273.9	167.57	16.01	11.57	1310.21
75.0	1.65	286.3	293.5	167.58	16.01	11.63	1316.42
80.0	1.76	305.4	313.0	167.59	16.01	11.68	1322.28
85.0	1.87	324.5	332.6	167.60	16.01	11.74	1327.88
90.0	1.98	343.6	352.2	167.61	16.01	11.79	1333.51
95.0	2.09	362.7	371.7	167.61	16.01	11.85	1339.18
100.0	2.20	381.8	391.3	167.62	16.01	11.90	1344.48

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB08.12
 Compartment descr: Water Ballast 08.12
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 198.00 (158.40 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point 10.80 m above BL
 Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.11	19.1	19.6	167.50-15.97	10.86	1304.80	
10.0	0.22	38.2	39.1	167.50-15.97	10.91	1312.98	
15.0	0.33	57.3	58.7	167.50-15.98	10.97	1320.86	
20.0	0.44	76.4	78.3	167.51-15.98	11.02	1328.63	
25.0	0.55	95.4	97.8	167.51-15.99	11.08	1336.44	
30.0	0.66	114.5	117.4	167.52-15.99	11.13	1343.76	
35.0	0.77	133.6	137.0	167.52-16.00	11.19	1350.72	
40.0	0.88	152.7	156.5	167.52-16.00	11.24	1357.80	
45.0	0.99	171.8	176.1	167.53-16.01	11.30	1364.82	
50.0	1.10	190.9	195.7	167.53-16.01	11.35	1371.90	
55.0	1.21	210.0	215.2	167.54-16.02	11.41	1305.37	
60.0	1.32	229.1	234.8	167.55-16.01	11.46	1304.66	
65.0	1.43	248.1	254.4	167.56-16.01	11.52	1304.49	
70.0	1.54	267.2	273.9	167.57-16.01	11.57	1310.20	
75.0	1.65	286.3	293.5	167.58-16.01	11.63	1316.42	
80.0	1.76	305.4	313.0	167.59-16.01	11.68	1322.28	
85.0	1.87	324.5	332.6	167.60-16.01	11.74	1327.88	
90.0	1.98	343.6	352.2	167.61-16.01	11.79	1333.52	
95.0	2.09	362.7	371.7	167.61-16.01	11.85	1339.17	
100.0	2.20	381.8	391.3	167.62-16.01	11.90	1344.49	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.01
 Compartment descr: Water Ballast 09.01
 Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
 Fore end at frame 237.00 (189.60 m)
 Lowest point 0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	21.4	21.9	183.00	-0.00	0.07	3590.49
10.0	0.25	42.8	43.9	183.04	-0.00	0.13	4154.76
15.0	0.36	64.2	65.8	183.06	0.00	0.19	4636.72
20.0	0.47	85.6	87.7	183.08	0.00	0.24	5083.91
25.0	0.58	107.0	109.7	183.09	-0.00	0.30	5498.23
30.0	0.68	128.4	131.6	183.10	-0.00	0.36	5909.46
35.0	0.79	149.8	153.5	183.11	-0.00	0.41	6299.00
40.0	0.89	171.2	175.4	183.12	0.00	0.46	6690.25
45.0	0.98	192.6	197.4	183.12	-0.00	0.52	7061.80
50.0	1.08	214.0	219.3	183.13	-0.00	0.57	7426.54
55.0	1.17	235.3	241.2	183.13	0.00	0.62	7789.49
60.0	1.29	256.7	263.2	183.07	0.00	0.67	7277.17
65.0	1.41	278.1	285.1	182.98	0.00	0.72	7734.97
70.0	1.53	299.5	307.0	182.92	0.00	0.77	8169.05
75.0	1.64	320.9	329.0	182.86	0.00	0.83	8600.26
80.0	1.76	342.3	350.9	182.81	0.00	0.88	9023.50
85.0	1.87	363.7	372.8	182.77	0.00	0.94	9437.24
90.0	1.98	385.1	394.7	182.74	0.00	0.99	9852.97
95.0	2.27	406.5	416.7	182.83	-0.00	1.05	6077.87
100.0	2.60	427.9	438.6	182.90	-0.00	1.12	6749.26

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.07
Compartment descr: Water Ballast 09.07
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 237.00 (189.60 m)
Lowest point 2.60 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.87	30.9	31.7	183.69	10.43	3.06	107.50
10.0	1.57	61.8	63.3	183.68	10.62	3.44	146.57
15.0	2.18	92.7	95.0	183.67	10.78	3.79	183.45
20.0	2.74	123.6	126.7	183.67	10.91	4.11	218.64
25.0	3.25	154.5	158.3	183.66	11.03	4.40	252.41
30.0	3.74	185.4	190.0	183.66	11.13	4.69	284.85
35.0	4.20	216.3	221.7	183.65	11.23	4.96	316.58
40.0	4.64	247.2	253.3	183.65	11.32	5.21	347.85
45.0	5.07	278.1	285.0	183.65	11.40	5.46	378.16
50.0	5.49	309.0	316.7	183.64	11.47	5.71	408.09
55.0	5.89	339.9	348.4	183.64	11.54	5.94	438.02
60.0	6.47	370.8	380.0	183.56	11.72	6.18	136.59
65.0	7.03	401.6	411.7	183.50	11.89	6.42	155.54
70.0	7.57	432.5	443.4	183.44	12.04	6.67	174.52
75.0	8.08	463.4	475.0	183.40	12.18	6.92	192.78
80.0	8.57	494.3	506.7	183.36	12.30	7.17	211.13
85.0	9.05	525.2	538.4	183.33	12.42	7.42	228.94
90.0	9.51	556.1	570.0	183.30	12.52	7.67	246.07
95.0	9.96	587.0	601.7	183.28	12.62	7.91	263.29
100.0	10.40	617.9	633.4	183.26	12.72	8.16	279.88

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.08
Compartment descr: Water Ballast 09.08
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 237.00 (189.60 m)
Lowest point 2.60 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.87	30.9	31.7	183.69-10.43	3.06	107.50	
10.0	1.57	61.8	63.3	183.68-10.62	3.44	146.57	
15.0	2.18	92.7	95.0	183.67-10.78	3.79	183.45	
20.0	2.74	123.6	126.7	183.67-10.91	4.11	218.64	
25.0	3.25	154.5	158.3	183.66-11.03	4.40	252.41	
30.0	3.74	185.4	190.0	183.66-11.13	4.69	284.85	
35.0	4.20	216.3	221.7	183.65-11.23	4.96	316.58	
40.0	4.64	247.2	253.3	183.65-11.32	5.21	347.85	
45.0	5.07	278.1	285.0	183.65-11.40	5.46	378.16	
50.0	5.49	309.0	316.7	183.64-11.47	5.71	408.09	
55.0	5.89	339.9	348.4	183.64-11.54	5.94	438.02	
60.0	6.47	370.8	380.0	183.56-11.72	6.18	136.59	
65.0	7.03	401.6	411.7	183.50-11.89	6.42	155.54	
70.0	7.57	432.5	443.4	183.44-12.04	6.67	174.52	
75.0	8.08	463.4	475.0	183.40-12.18	6.92	192.78	
80.0	8.57	494.3	506.7	183.36-12.30	7.17	211.13	
85.0	9.05	525.2	538.4	183.33-12.42	7.42	228.94	
90.0	9.51	556.1	570.0	183.30-12.52	7.67	246.07	
95.0	9.96	587.0	601.7	183.28-12.62	7.91	263.29	
100.0	10.40	617.9	633.4	183.26-12.72	8.16	279.88	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.09
Compartment descr: Water Ballast 09.09
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 237.00 (189.60 m)
Lowest point 11.40 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.08	16.1	16.5	183.21	0.00	11.44	5617.51
10.0	0.16	32.3	33.1	183.21	0.00	11.48	5617.51
15.0	0.24	48.4	49.6	183.21	0.00	11.52	5617.51
20.0	0.32	64.5	66.2	183.21	0.00	11.56	5617.51
25.0	0.40	80.7	82.7	183.21	0.00	11.60	5617.51
30.0	0.48	96.8	99.2	183.21	0.00	11.64	5617.51
35.0	0.56	112.9	115.8	183.21	0.00	11.68	5617.51
40.0	0.64	129.1	132.3	183.21	0.00	11.72	5617.51
45.0	0.72	145.2	148.8	183.21	0.00	11.76	5617.51
50.0	0.80	161.3	165.4	183.21	0.00	11.80	5617.51
55.0	0.88	177.5	181.9	183.21	0.00	11.84	5617.51
60.0	0.96	193.6	198.5	183.21	0.00	11.88	5617.51
65.0	1.04	209.8	215.0	183.21	0.00	11.92	5617.51
70.0	1.12	225.9	231.5	183.21	0.00	11.96	5617.51
75.0	1.20	242.0	248.1	183.21	0.00	12.00	5617.51
80.0	1.28	258.2	264.6	183.21	0.00	12.04	5617.51
85.0	1.36	274.3	281.1	183.21	0.00	12.08	5617.51
90.0	1.44	290.4	297.7	183.21	0.00	12.12	5617.51
95.0	1.52	306.6	314.2	183.21	0.00	12.16	5617.51
100.0	1.60	322.7	330.8	183.21	0.00	12.20	5617.51

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.11
Compartment descr: Water Ballast 09.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 237.00 (189.60 m)
Lowest point 13.00 m above BL
Highest point 27.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.88	71.5	73.3	182.51	14.09	13.44	518.82
10.0	1.73	143.1	146.6	182.54	14.15	13.88	561.71
15.0	2.55	214.6	220.0	182.56	14.21	14.30	604.25
20.0	3.35	286.1	293.3	182.58	14.27	14.71	645.53
25.0	4.12	357.7	366.6	182.60	14.33	15.12	686.53
30.0	4.88	429.2	439.9	182.62	14.38	15.52	726.66
35.0	5.62	500.7	513.3	182.63	14.43	15.91	766.48
40.0	6.34	572.3	586.6	182.65	14.48	16.29	806.22
45.0	7.05	643.8	659.9	182.67	14.53	16.67	845.42
50.0	7.74	715.3	733.2	182.68	14.58	17.04	885.06
55.0	8.43	786.9	806.6	182.70	14.63	17.41	924.24
60.0	9.10	858.4	879.9	182.71	14.67	17.77	962.72
65.0	9.76	929.9	953.2	182.72	14.71	18.13	1001.15
70.0	10.40	1001.5	1026.5	182.74	14.75	18.48	1038.80
75.0	11.04	1073.0	1099.8	182.75	14.80	18.83	1076.37
80.0	11.67	1144.6	1173.2	182.76	14.84	19.18	1112.68
85.0	12.29	1216.1	1246.5	182.78	14.88	19.52	1149.05
90.0	12.91	1287.6	1319.8	182.79	14.91	19.86	1183.76
95.0	13.51	1359.2	1393.1	182.80	14.95	20.19	1216.52
100.0	14.70	1430.7	1466.5	182.74	14.97	20.53	518.46

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB09.12
Compartment descr: Water Ballast 09.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 237.00 (189.60 m)
Lowest point 13.00 m above BL
Highest point 27.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.88	71.5	73.3	182.51-14.09	13.44	518.86	
10.0	1.73	143.1	146.6	182.54-14.15	13.88	561.74	
15.0	2.55	214.6	220.0	182.56-14.21	14.30	604.29	
20.0	3.35	286.1	293.3	182.58-14.27	14.71	645.57	
25.0	4.12	357.7	366.6	182.60-14.33	15.12	686.57	
30.0	4.88	429.2	439.9	182.62-14.38	15.52	726.69	
35.0	5.62	500.7	513.3	182.63-14.43	15.91	766.50	
40.0	6.34	572.3	586.6	182.65-14.48	16.29	806.24	
45.0	7.05	643.8	659.9	182.67-14.53	16.67	845.41	
50.0	7.74	715.3	733.2	182.68-14.58	17.04	885.04	
55.0	8.43	786.9	806.5	182.70-14.63	17.41	924.25	
60.0	9.10	858.4	879.9	182.71-14.67	17.77	962.82	
65.0	9.76	929.9	953.2	182.72-14.71	18.13	1001.23	
70.0	10.40	1001.5	1026.5	182.74-14.75	18.48	1038.85	
75.0	11.04	1073.0	1099.8	182.75-14.80	18.83	1076.42	
80.0	11.67	1144.5	1173.1	182.76-14.84	19.18	1112.76	
85.0	12.29	1216.1	1246.5	182.78-14.88	19.52	1149.11	
90.0	12.91	1287.6	1319.8	182.79-14.91	19.86	1183.76	
95.0	13.51	1359.1	1393.1	182.80-14.95	20.19	1216.48	
100.0	14.70	1430.7	1466.4	182.74-14.97	20.53	464.06	

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB10.01
Compartment descr: Water Ballast 10.01
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 237.00 (189.60 m)
Fore end at frame 249.00 (199.20 m)
Lowest point -0.00 m above BL
Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.21	15.3	15.7	193.85	0.00	0.11	615.47
10.0	0.38	30.7	31.4	193.89	0.00	0.20	822.95
15.0	0.54	46.0	47.1	193.91	0.00	0.29	1000.93
20.0	0.69	61.3	62.8	193.93	0.00	0.37	1169.36
25.0	0.84	76.6	78.6	193.94	0.00	0.45	1330.31
30.0	0.98	92.0	94.3	193.95	0.00	0.53	1486.06
35.0	1.11	107.3	110.0	193.96	0.00	0.60	1638.69
40.0	1.24	122.6	125.7	193.97	0.00	0.67	1783.99
45.0	1.37	138.0	141.4	193.98	0.00	0.74	1929.24
50.0	1.49	153.3	157.1	193.98	0.00	0.81	2069.74
55.0	1.61	168.6	172.8	193.99	0.00	0.88	2209.36
60.0	1.73	183.9	188.5	193.99	0.00	0.94	2348.45
65.0	1.84	199.3	204.3	194.00	0.00	1.01	2485.85
70.0	1.96	214.6	220.0	194.00	0.00	1.07	2624.32
75.0	2.07	229.9	235.7	194.00	0.00	1.14	2759.40
80.0	2.18	245.3	251.4	194.01	0.00	1.20	2892.88
85.0	2.29	260.6	267.1	194.01	0.00	1.26	3028.43
90.0	2.39	275.9	282.8	194.01	0.00	1.32	3163.25
95.0	2.50	291.2	298.5	194.02	0.00	1.38	3290.90
100.0	2.60	306.6	314.2	194.02	0.00	1.44	3417.68

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB10.05
Compartment descr: Water Ballast 10.05
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 237.00 (189.60 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 2.60 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.66	51.0	52.3	194.08	4.18	2.93	550.98
10.0	1.27	102.1	104.6	194.08	4.32	3.25	649.45
15.0	1.86	153.1	156.9	194.08	4.44	3.56	741.32
20.0	2.42	204.1	209.2	194.09	4.55	3.85	826.83
25.0	2.97	255.1	261.5	194.09	4.65	4.14	906.47
30.0	3.50	306.2	313.8	194.09	4.74	4.42	982.11
35.0	4.01	357.2	366.1	194.09	4.82	4.70	1053.99
40.0	4.52	408.2	418.4	194.10	4.90	4.97	1121.55
45.0	5.01	459.2	470.7	194.10	4.97	5.24	1185.73
50.0	5.50	510.3	523.0	194.10	5.04	5.50	1247.44
55.0	5.99	561.3	575.3	194.10	5.11	5.76	1064.32
60.0	6.51	612.3	627.6	194.12	5.21	6.02	1118.39
65.0	7.02	663.3	679.9	194.14	5.30	6.27	1170.86
70.0	7.52	714.4	732.2	194.16	5.38	6.53	1223.35
75.0	8.02	765.4	784.5	194.17	5.46	6.79	1275.04
80.0	8.51	816.4	836.8	194.18	5.53	7.04	1327.19
85.0	8.99	867.5	889.1	194.19	5.60	7.29	1379.22
90.0	9.47	918.5	941.4	194.20	5.67	7.55	1431.57
95.0	9.94	969.5	993.7	194.21	5.73	7.80	1484.43
100.0	10.40	1020.5	1046.0	194.21	5.79	8.04	1535.88

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB10.06
Compartment descr: Water Ballast 10.06
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 237.00 (189.60 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 2.60 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.66	51.0	52.3	194.08	-4.18	2.93	550.95
10.0	1.27	102.1	104.6	194.08	-4.32	3.25	649.42
15.0	1.86	153.1	156.9	194.08	-4.44	3.56	741.29
20.0	2.42	204.1	209.2	194.09	-4.55	3.85	826.83
25.0	2.97	255.1	261.5	194.09	-4.65	4.14	906.44
30.0	3.50	306.2	313.8	194.09	-4.74	4.42	982.15
35.0	4.01	357.2	366.1	194.09	-4.82	4.70	1053.99
40.0	4.52	408.2	418.4	194.10	-4.90	4.97	1121.57
45.0	5.01	459.2	470.7	194.10	-4.97	5.24	1185.71
50.0	5.50	510.3	523.0	194.10	-5.04	5.50	1247.29
55.0	5.99	561.3	575.3	194.10	-5.10	5.76	1137.13
60.0	6.51	612.3	627.6	194.13	-5.20	6.02	1192.93
65.0	7.02	663.3	679.9	194.16	-5.28	6.27	1246.95
70.0	7.52	714.4	732.2	194.18	-5.36	6.53	1301.02
75.0	8.02	765.4	784.5	194.19	-5.43	6.79	1354.23
80.0	8.51	816.4	836.8	194.21	-5.50	7.04	1407.90
85.0	8.99	867.5	889.1	194.22	-5.57	7.29	1461.51
90.0	9.47	918.5	941.4	194.23	-5.63	7.55	1515.25
95.0	9.94	969.5	993.7	194.24	-5.70	7.80	1569.49
100.0	10.40	1020.5	1046.0	194.25	-5.75	8.04	1622.27

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB10.11
Compartment descr: Water Ballast 10.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 237.00 (189.60 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 13.00 m above BL
Highest point 26.90 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.97	46.6	47.8	193.63	10.63	13.49	167.80
10.0	1.89	93.3	95.6	193.64	10.70	13.96	190.80
15.0	2.77	139.9	143.4	193.65	10.76	14.42	214.04
20.0	3.61	186.6	191.2	193.67	10.82	14.86	237.72
25.0	4.42	233.2	239.0	193.68	10.88	15.30	261.61
30.0	5.21	279.9	286.8	193.69	10.94	15.72	286.45
35.0	5.96	326.5	334.7	193.70	10.99	16.13	312.46
40.0	6.69	373.1	382.5	193.72	11.05	16.53	340.05
45.0	7.39	419.8	430.3	193.73	11.10	16.92	369.21
50.0	8.07	466.4	478.1	193.74	11.16	17.30	399.58
55.0	8.73	513.1	525.9	193.76	11.21	17.67	432.05
60.0	9.37	559.7	573.7	193.77	11.27	18.04	466.29
65.0	9.99	606.3	621.5	193.78	11.32	18.39	502.58
70.0	10.60	653.0	669.3	193.79	11.38	18.74	540.63
75.0	11.19	699.6	717.1	193.81	11.43	19.09	581.07
80.0	11.76	746.3	764.9	193.82	11.49	19.42	623.46
85.0	12.31	792.9	812.7	193.83	11.54	19.75	667.95
90.0	12.86	839.6	860.5	193.84	11.59	20.08	714.73
95.0	13.38	886.2	908.4	193.85	11.65	20.40	763.50
100.0	13.90	932.8	956.2	193.86	11.70	20.71	814.03

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB10.12
Compartment descr: Water Ballast 10.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 237.00 (189.60 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 13.00 m above BL
Highest point 26.90 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.97	46.6	47.8	193.63	-10.63	13.49	167.80
10.0	1.89	93.3	95.6	193.64	-10.70	13.96	190.80
15.0	2.77	139.9	143.4	193.65	-10.76	14.42	214.04
20.0	3.61	186.6	191.2	193.67	-10.82	14.86	237.72
25.0	4.42	233.2	239.0	193.68	-10.88	15.30	261.61
30.0	5.21	279.9	286.8	193.69	-10.94	15.72	286.45
35.0	5.96	326.5	334.7	193.70	-10.99	16.13	312.46
40.0	6.69	373.1	382.5	193.72	-11.05	16.53	340.05
45.0	7.39	419.8	430.3	193.73	-11.10	16.92	369.21
50.0	8.07	466.4	478.1	193.74	-11.16	17.30	399.58
55.0	8.73	513.1	525.9	193.76	-11.21	17.67	432.05
60.0	9.37	559.7	573.7	193.77	-11.27	18.04	466.29
65.0	9.99	606.3	621.5	193.78	-11.32	18.39	502.58
70.0	10.60	653.0	669.3	193.79	-11.38	18.74	540.63
75.0	11.19	699.6	717.1	193.81	-11.43	19.09	581.07
80.0	11.76	746.3	764.9	193.82	-11.49	19.42	623.46
85.0	12.31	792.9	812.7	193.83	-11.54	19.75	667.95
90.0	12.86	839.6	860.5	193.84	-11.59	20.08	714.73
95.0	13.38	886.2	908.4	193.85	-11.65	20.40	763.50
100.0	13.90	932.8	956.2	193.86	-11.70	20.71	814.03

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB11.01
Compartment descr: Water Ballast 11.01
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 249.00 (199.20 m)
Fore end at frame 270.87 (216.70 m)
Lowest point -0.00 m above BL
Highest point 13.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.43	80.0	82.0	204.76	-0.00	0.84	381.47
10.0	2.31	160.0	164.0	204.96	-0.00	1.36	669.35
15.0	3.13	240.0	246.0	205.18	-0.00	1.81	609.18
20.0	4.13	319.9	327.9	205.55	-0.00	2.26	762.98
25.0	4.95	399.9	409.9	205.70	-0.00	2.72	1574.27
30.0	5.59	479.9	491.9	205.73	-0.00	3.14	1799.48
35.0	6.21	559.9	573.9	205.75	-0.00	3.54	2000.52
40.0	6.80	639.9	655.9	205.78	0.00	3.90	2180.26
45.0	7.37	719.9	737.9	205.79	0.00	4.26	2342.54
50.0	7.92	799.8	819.8	205.81	0.00	4.60	2483.41
55.0	8.46	879.8	901.8	205.82	0.00	4.93	2600.64
60.0	8.94	959.8	983.8	205.80	-0.00	5.24	2702.03
65.0	9.41	1039.8	1065.8	205.78	-0.00	5.54	2763.40
70.0	9.89	1119.8	1147.8	205.75	-0.00	5.84	2802.08
75.0	10.37	1199.8	1229.8	205.71	-0.01	6.13	2813.27
80.0	10.87	1279.7	1311.7	205.67	-0.01	6.41	2788.10
85.0	11.40	1359.7	1393.7	205.60	-0.01	6.69	2805.27
90.0	11.93	1439.7	1475.7	205.53	-0.01	6.96	2886.82
95.0	12.47	1519.7	1557.7	205.47	-0.01	7.23	2973.09
100.0	13.00	1599.7	1639.7	205.41	-0.01	7.51	3064.31

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB11.11
Compartment descr: Water Ballast 11.11
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 249.00 (199.20 m)
Fore end at frame 270.84 (216.67 m)
Lowest point 13.00 m above BL
Highest point 30.53 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.24	92.9	95.2	204.37	3.78	13.62	496.35
10.0	2.41	185.8	190.5	204.46	3.81	14.22	545.27
15.0	3.52	278.7	285.7	204.54	3.85	14.80	603.68
20.0	4.58	371.6	380.9	204.60	3.90	15.36	682.68
25.0	5.58	464.5	476.1	204.65	3.95	15.91	773.81
30.0	6.56	557.4	571.4	204.72	4.02	16.44	874.28
35.0	7.52	650.3	666.6	204.80	4.08	16.96	995.17
40.0	8.42	743.2	761.8	204.87	4.15	17.46	1124.58
45.0	9.27	836.1	857.0	204.93	4.22	17.96	1261.37
50.0	10.07	929.0	952.3	204.99	4.30	18.43	1401.87
55.0	10.84	1021.9	1047.5	205.05	4.37	18.89	1546.02
60.0	11.57	1114.9	1142.7	205.10	4.45	19.33	1696.39
65.0	12.27	1207.8	1237.9	205.16	4.52	19.76	1852.91
70.0	12.95	1300.7	1333.2	205.20	4.60	20.18	2015.77
75.0	13.60	1393.6	1428.4	205.25	4.67	20.58	2183.93
80.0	14.35	1486.5	1523.6	205.33	4.77	20.98	1917.80
85.0	15.19	1579.4	1618.8	205.44	4.89	21.38	2125.68
90.0	15.97	1672.3	1714.1	205.53	5.01	21.78	2336.76
95.0	16.72	1765.2	1809.3	205.62	5.12	22.18	2552.07
100.0	17.53	1858.1	1904.5	205.72	5.21	22.57	18.05

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: WB11.12
Compartment descr: Water Ballast 11.12
Contents : Ballast Water (BW, RHO = 1.025)

Extreme points of comp: Aft end at frame 249.00 (199.20 m)
Fore end at frame 270.84 (216.67 m)
Lowest point 13.00 m above BL
Highest point 30.53 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	1.21	95.8	98.2	204.16	-3.63	13.61	531.53
10.0	2.36	191.5	196.3	204.25	-3.66	14.20	581.44
15.0	3.46	287.3	294.5	204.33	-3.70	14.76	641.11
20.0	4.50	383.1	392.6	204.39	-3.75	15.31	720.29
25.0	5.49	478.8	490.8	204.45	-3.80	15.85	813.06
30.0	6.46	574.6	589.0	204.51	-3.86	16.38	913.78
35.0	7.41	670.4	687.1	204.59	-3.93	16.89	1037.28
40.0	8.31	766.1	785.3	204.66	-4.00	17.39	1169.89
45.0	9.16	861.9	883.4	204.72	-4.07	17.88	1310.34
50.0	9.97	957.7	981.6	204.79	-4.14	18.35	1455.64
55.0	10.74	1053.4	1079.8	204.85	-4.22	18.81	1604.27
60.0	11.47	1149.2	1177.9	204.90	-4.29	19.25	1759.27
65.0	12.18	1244.9	1276.1	204.96	-4.36	19.68	1920.92
70.0	12.85	1340.7	1374.2	205.01	-4.44	20.09	2088.97
75.0	13.51	1436.5	1472.4	205.05	-4.51	20.50	2262.78
80.0	14.25	1532.2	1570.5	205.13	-4.61	20.90	1892.50
85.0	15.11	1628.0	1668.7	205.25	-4.74	21.30	2106.29
90.0	15.93	1723.8	1766.9	205.36	-4.86	21.70	2323.67
95.0	16.70	1819.5	1865.0	205.45	-4.99	22.10	2545.41
100.0	17.53	1915.3	1963.2	205.56	-5.08	22.49	18.05

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: HF01
 Compartment descr: HF0 Storage Tank 1
 Contents : Heavy Fuel Oil (HF0, RHO = 0.991)

Extreme points of comp: Aft end at frame 72.00 (57.60 m)
 Fore end at frame 87.00 (69.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.40	53.1	52.6	63.60	5.69	2.80	1458.57
10.0	0.79	106.2	105.2	63.60	5.69	3.00	1458.57
15.0	1.19	159.2	157.8	63.60	5.69	3.20	1458.57
20.0	1.59	212.3	210.4	63.60	5.69	3.39	1458.57
25.0	1.98	265.4	263.0	63.60	5.69	3.59	1458.57
30.0	2.38	318.5	315.6	63.60	5.69	3.79	1458.57
35.0	2.78	371.6	368.2	63.60	5.69	3.99	1458.57
40.0	3.20	424.7	420.8	63.60	5.64	4.19	1106.72
45.0	3.64	477.7	473.4	63.60	5.59	4.39	1106.72
50.0	4.07	530.8	526.0	63.60	5.55	4.60	1106.72
55.0	4.51	583.9	578.6	63.60	5.52	4.81	1106.72
60.0	4.94	637.0	631.2	63.60	5.49	5.02	1106.72
65.0	5.38	690.1	683.9	63.60	5.47	5.23	1106.72
70.0	5.81	743.1	736.5	63.60	5.45	5.44	1106.72
75.0	6.22	796.2	789.1	63.60	5.46	5.65	1458.57
80.0	6.61	849.3	841.7	63.60	5.47	5.86	1458.57
85.0	7.01	902.4	894.3	63.60	5.48	6.07	1458.57
90.0	7.41	955.5	946.9	63.60	5.50	6.28	1458.57
95.0	7.80	1008.6	999.5	63.60	5.51	6.48	1458.57
100.0	8.20	1061.6	1052.1	63.60	5.52	6.69	1458.57

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: HF02
 Compartment descr: HF0 Storage Tank 2
 Contents : Heavy Fuel Oil (HF0, RHO = 0.991)

 Extreme points of comp: Aft end at frame 72.00 (57.60 m)
 Fore end at frame 90.00 (72.00 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL %	H m	VNET m3	MASS t	LCG m	TCG m	VCG m	FSM tm
5.0	0.40	63.7	63.1	64.80	-5.69	2.80	1750.29
10.0	0.79	127.4	126.2	64.80	-5.69	3.00	1750.29
15.0	1.19	191.1	189.4	64.80	-5.69	3.20	1750.29
20.0	1.59	254.8	252.5	64.80	-5.69	3.39	1750.29
25.0	1.98	318.5	315.6	64.80	-5.69	3.59	1750.29
30.0	2.38	382.2	378.7	64.80	-5.69	3.79	1750.29
35.0	2.78	445.9	441.9	64.80	-5.69	3.99	1750.29
40.0	3.20	509.6	505.0	64.80	-5.64	4.19	1328.06
45.0	3.64	573.3	568.1	64.80	-5.59	4.39	1328.06
50.0	4.07	637.0	631.2	64.80	-5.55	4.60	1328.06
55.0	4.51	700.7	694.4	64.80	-5.52	4.81	1328.06
60.0	4.94	764.4	757.5	64.80	-5.49	5.02	1328.06
65.0	5.38	828.1	820.6	64.80	-5.47	5.23	1328.06
70.0	5.81	891.8	883.7	64.80	-5.45	5.44	1328.06
75.0	6.22	955.5	946.9	64.80	-5.46	5.65	1750.29
80.0	6.61	1019.2	1010.0	64.80	-5.47	5.86	1750.29
85.0	7.01	1082.9	1073.1	64.80	-5.48	6.07	1750.29
90.0	7.41	1146.6	1136.2	64.80	-5.50	6.28	1750.29
95.0	7.80	1210.3	1199.4	64.80	-5.51	6.48	1750.29
100.0	8.20	1274.0	1262.5	64.80	-5.52	6.69	1750.29

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: HF03
 Compartment descr: HFO Storage Tank 3
 Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 90.00 (72.00 m)
 Fore end at frame 102.00 (81.60 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.42	41.7	41.3	76.61	5.36	2.81	993.42
10.0	0.83	83.4	82.7	76.61	5.36	3.02	993.42
15.0	1.25	125.1	124.0	76.61	5.36	3.22	993.42
20.0	1.65	166.8	165.3	76.63	5.40	3.43	1153.45
25.0	2.05	208.6	206.7	76.66	5.45	3.64	1153.45
30.0	2.44	250.3	248.0	76.68	5.48	3.84	1153.45
35.0	2.83	292.0	289.4	76.70	5.51	4.04	1153.45
40.0	3.26	333.7	330.7	76.70	5.47	4.24	851.76
45.0	3.69	375.4	372.0	76.71	5.43	4.44	851.76
50.0	4.12	417.1	413.4	76.71	5.40	4.65	851.76
55.0	4.56	458.8	454.7	76.72	5.38	4.86	851.76
60.0	4.99	500.5	496.0	76.72	5.35	5.07	851.76
65.0	5.42	542.3	537.4	76.72	5.34	5.28	851.76
70.0	5.86	584.0	578.7	76.72	5.32	5.49	851.76
75.0	6.25	625.7	620.1	76.73	5.34	5.70	1166.86
80.0	6.64	667.4	661.4	76.73	5.36	5.91	1166.86
85.0	7.03	709.1	702.7	76.74	5.38	6.12	1166.86
90.0	7.42	750.8	744.1	76.74	5.40	6.32	1166.86
95.0	7.81	792.5	785.4	76.74	5.41	6.53	1166.86
100.0	8.20	834.2	826.7	76.75	5.43	6.73	1166.86

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: HF04
Compartment descr: HFO Storage Tank 4
Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 90.00 (72.00 m)
Fore end at frame 102.00 (81.60 m)
Lowest point 2.60 m above BL
Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.42	41.7	41.3	76.61	-5.36	2.81	993.42
10.0	0.83	83.4	82.7	76.61	-5.36	3.02	993.42
15.0	1.25	125.1	124.0	76.61	-5.36	3.22	993.42
20.0	1.65	166.8	165.3	76.63	-5.40	3.43	1153.45
25.0	2.05	208.6	206.7	76.66	-5.45	3.64	1153.45
30.0	2.44	250.3	248.0	76.68	-5.48	3.84	1153.45
35.0	2.83	292.0	289.4	76.70	-5.51	4.04	1153.45
40.0	3.26	333.7	330.7	76.70	-5.47	4.24	851.76
45.0	3.69	375.4	372.0	76.71	-5.43	4.44	851.76
50.0	4.12	417.1	413.4	76.71	-5.40	4.65	851.76
55.0	4.56	458.8	454.7	76.72	-5.38	4.86	851.76
60.0	4.99	500.5	496.0	76.72	-5.35	5.07	851.76
65.0	5.42	542.3	537.4	76.72	-5.34	5.28	851.76
70.0	5.86	584.0	578.7	76.72	-5.32	5.49	851.76
75.0	6.25	625.7	620.1	76.73	-5.34	5.70	1166.86
80.0	6.64	667.4	661.4	76.73	-5.36	5.91	1166.86
85.0	7.03	709.1	702.7	76.74	-5.38	6.12	1166.86
90.0	7.42	750.8	744.1	76.74	-5.40	6.32	1166.86
95.0	7.81	792.5	785.4	76.74	-5.41	6.53	1166.86
100.0	8.20	834.2	826.7	76.75	-5.43	6.73	1166.86

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.03
Compartment descr: HFO Settling Tank P
Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 231.00 (184.80 m)
Fore end at frame 236.00 (188.80 m)
Lowest point 2.00 m above BL
Highest point 7.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.52	2.8	2.8	186.19	3.94	2.35	5.98
10.0	0.79	5.6	5.6	186.49	3.94	2.50	5.98
15.0	1.07	8.4	8.3	186.60	3.94	2.65	5.98
20.0	1.34	11.2	11.1	186.65	3.94	2.79	5.98
25.0	1.61	14.0	13.9	186.68	3.94	2.92	5.98
30.0	1.88	16.8	16.7	186.70	3.94	3.06	5.98
35.0	2.16	19.6	19.5	186.71	3.94	3.20	5.98
40.0	2.43	22.4	22.2	186.72	3.94	3.34	5.98
45.0	2.70	25.2	25.0	186.73	3.94	3.47	5.98
50.0	2.97	28.0	27.8	186.74	3.94	3.61	5.98
55.0	3.25	30.8	30.6	186.74	3.94	3.75	5.98
60.0	3.52	33.6	33.3	186.75	3.94	3.88	5.98
65.0	3.79	36.5	36.1	186.75	3.94	4.02	5.98
70.0	4.06	39.3	38.9	186.76	3.94	4.15	5.98
75.0	4.34	42.1	41.7	186.76	3.94	4.29	5.98
80.0	4.61	44.9	44.5	186.76	3.94	4.43	5.98
85.0	4.88	47.7	47.2	186.76	3.94	4.56	5.98
90.0	5.15	50.5	50.0	186.77	3.94	4.70	5.98
95.0	5.43	53.3	52.8	186.77	3.94	4.84	5.98
100.0	5.70	56.1	55.6	186.77	3.94	4.97	5.98

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.04
Compartment descr: HFO Settling Tank S
Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 232.00 (185.60 m)
Fore end at frame 236.00 (188.80 m)
Lowest point 2.10 m above BL
Highest point 7.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.47	2.2	2.2	186.80	-3.94	2.41	4.78
10.0	0.74	4.4	4.4	187.00	-3.94	2.56	4.78
15.0	1.01	6.7	6.6	187.07	-3.94	2.70	4.78
20.0	1.28	8.9	8.8	187.10	-3.94	2.83	4.78
25.0	1.55	11.1	11.0	187.12	-3.94	2.97	4.78
30.0	1.82	13.3	13.2	187.13	-3.94	3.11	4.78
35.0	2.09	15.6	15.4	187.14	-3.94	3.24	4.78
40.0	2.36	17.8	17.6	187.15	-3.94	3.38	4.78
45.0	2.63	20.0	19.8	187.16	-3.94	3.51	4.78
50.0	2.90	22.2	22.0	187.16	-3.94	3.65	4.78
55.0	3.17	24.4	24.2	187.16	-3.94	3.78	4.78
60.0	3.44	26.7	26.4	187.17	-3.94	3.92	4.78
65.0	3.71	28.9	28.6	187.17	-3.94	4.05	4.78
70.0	3.98	31.1	30.8	187.17	-3.94	4.19	4.78
75.0	4.25	33.3	33.0	187.17	-3.94	4.32	4.78
80.0	4.52	35.6	35.2	187.17	-3.94	4.46	4.78
85.0	4.79	37.8	37.4	187.18	-3.94	4.59	4.78
90.0	5.06	40.0	39.6	187.18	-3.94	4.73	4.78
95.0	5.33	42.2	41.9	187.18	-3.94	4.86	4.78
100.0	5.60	44.5	44.1	187.18	-3.94	5.00	4.78

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.05
Compartment descr: HFO Service Tank P
Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 231.00 (184.80 m)
Fore end at frame 236.00 (188.80 m)
Lowest point 2.00 m above BL
Highest point 7.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.52	1.9	1.9	186.19	6.13	2.35	1.77
10.0	0.79	3.7	3.7	186.49	6.13	2.50	1.77
15.0	1.07	5.6	5.6	186.60	6.13	2.65	1.77
20.0	1.34	7.5	7.4	186.65	6.13	2.79	1.77
25.0	1.61	9.3	9.3	186.68	6.13	2.92	1.77
30.0	1.88	11.2	11.1	186.70	6.13	3.06	1.77
35.0	2.16	13.1	13.0	186.71	6.13	3.20	1.77
40.0	2.43	15.0	14.8	186.72	6.13	3.34	1.77
45.0	2.70	16.8	16.7	186.73	6.13	3.47	1.77
50.0	2.97	18.7	18.5	186.74	6.13	3.61	1.77
55.0	3.25	20.6	20.4	186.74	6.13	3.75	1.77
60.0	3.52	22.4	22.2	186.75	6.13	3.88	1.77
65.0	3.79	24.3	24.1	186.75	6.13	4.02	1.77
70.0	4.06	26.2	25.9	186.76	6.13	4.15	1.77
75.0	4.34	28.0	27.8	186.76	6.13	4.29	1.77
80.0	4.61	29.9	29.6	186.76	6.13	4.43	1.77
85.0	4.88	31.8	31.5	186.76	6.13	4.56	1.77
90.0	5.15	33.6	33.3	186.77	6.13	4.70	1.77
95.0	5.43	35.5	35.2	186.77	6.13	4.84	1.77
100.0	5.70	37.4	37.1	186.77	6.13	4.97	1.77

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.06
Compartment descr: HFO Service Tank S
Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

Extreme points of comp: Aft end at frame 232.00 (185.60 m)
Fore end at frame 236.00 (188.80 m)
Lowest point 2.10 m above BL
Highest point 7.70 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.47	1.5	1.5	186.80	-6.13	2.41	1.42
10.0	0.74	3.0	2.9	187.00	-6.13	2.56	1.42
15.0	1.01	4.4	4.4	187.07	-6.13	2.70	1.42
20.0	1.28	5.9	5.9	187.10	-6.13	2.83	1.42
25.0	1.55	7.4	7.3	187.12	-6.13	2.97	1.42
30.0	1.82	8.9	8.8	187.13	-6.13	3.11	1.42
35.0	2.09	10.4	10.3	187.14	-6.13	3.24	1.42
40.0	2.36	11.9	11.7	187.15	-6.13	3.38	1.42
45.0	2.63	13.3	13.2	187.16	-6.13	3.51	1.42
50.0	2.90	14.8	14.7	187.16	-6.13	3.65	1.42
55.0	3.17	16.3	16.2	187.16	-6.13	3.78	1.42
60.0	3.44	17.8	17.6	187.17	-6.13	3.92	1.42
65.0	3.71	19.3	19.1	187.17	-6.13	4.05	1.42
70.0	3.98	20.7	20.6	187.17	-6.13	4.19	1.42
75.0	4.25	22.2	22.0	187.17	-6.13	4.32	1.42
80.0	4.52	23.7	23.5	187.17	-6.13	4.46	1.42
85.0	4.79	25.2	25.0	187.18	-6.13	4.59	1.42
90.0	5.06	26.7	26.4	187.18	-6.13	4.73	1.42
95.0	5.33	28.2	27.9	187.18	-6.13	4.86	1.42
100.0	5.60	29.6	29.4	187.18	-6.13	5.00	1.42

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: MGO
 Compartment descr: MGO Storage Tank
 Contents : Diesel Oil (DO, RHO = 0.9)

 Extreme points of comp: Aft end at frame 87.00 (69.60 m)
 Fore end at frame 90.00 (72.00 m)
 Lowest point 2.60 m above BL
 Highest point 10.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.40	10.6	9.6	70.80	5.69	2.80	264.93
10.0	0.79	21.2	19.1	70.80	5.69	3.00	264.93
15.0	1.19	31.8	28.7	70.80	5.69	3.20	264.93
20.0	1.59	42.5	38.2	70.80	5.69	3.39	264.93
25.0	1.98	53.1	47.8	70.80	5.69	3.59	264.93
30.0	2.38	63.7	57.3	70.80	5.69	3.79	264.93
35.0	2.78	74.3	66.9	70.80	5.69	3.99	264.93
40.0	3.20	84.9	76.4	70.80	5.64	4.19	201.02
45.0	3.64	95.5	86.0	70.80	5.59	4.39	201.02
50.0	4.07	106.2	95.5	70.80	5.55	4.60	201.02
55.0	4.51	116.8	105.1	70.80	5.52	4.81	201.02
60.0	4.94	127.4	114.7	70.80	5.49	5.02	201.02
65.0	5.38	138.0	124.2	70.80	5.47	5.23	201.02
70.0	5.81	148.6	133.8	70.80	5.45	5.44	201.02
75.0	6.22	159.2	143.3	70.80	5.46	5.65	264.93
80.0	6.61	169.9	152.9	70.80	5.47	5.86	264.93
85.0	7.01	180.5	162.4	70.80	5.48	6.07	264.93
90.0	7.41	191.1	172.0	70.80	5.50	6.28	264.93
95.0	7.80	201.7	181.5	70.80	5.51	6.48	264.93
100.0	8.20	212.3	191.1	70.80	5.52	6.69	264.93

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.07
Compartment descr: MD0 Storage P
Contents : Diesel Oil (DO, RHO = 0.9)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 228.00 (182.40 m)
Lowest point 2.00 m above BL
Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.18	1.7	1.6	179.27	9.57	2.09	6.54
10.0	0.35	3.4	3.1	179.29	9.58	2.18	6.54
15.0	0.53	5.2	4.7	179.29	9.58	2.26	6.54
20.0	0.70	6.9	6.2	179.29	9.58	2.35	6.54
25.0	0.88	8.6	7.8	179.30	9.58	2.44	6.54
30.0	1.05	10.3	9.3	179.30	9.58	2.53	6.54
35.0	1.23	12.1	10.9	179.30	9.58	2.61	6.54
40.0	1.40	13.8	12.4	179.30	9.58	2.70	6.54
45.0	1.58	15.5	14.0	179.30	9.58	2.79	6.54
50.0	1.75	17.2	15.5	179.30	9.58	2.88	6.54
55.0	1.93	19.0	17.1	179.30	9.58	2.96	6.54
60.0	2.10	20.7	18.6	179.30	9.58	3.05	6.54
65.0	2.28	22.4	20.2	179.30	9.58	3.14	6.54
70.0	2.45	24.1	21.7	179.30	9.58	3.23	6.54
75.0	2.63	25.9	23.3	179.30	9.58	3.31	6.54
80.0	2.80	27.6	24.8	179.30	9.58	3.40	6.54
85.0	2.98	29.3	26.4	179.30	9.58	3.49	6.54
90.0	3.15	31.0	27.9	179.30	9.58	3.58	6.54
95.0	3.33	32.8	29.5	179.30	9.58	3.66	6.54
100.0	3.50	34.5	31.0	179.30	9.58	3.75	6.54

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.08
Compartment descr: MDO Storage S
Contents : Diesel Oil (DO, RHO = 0.9)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 228.00 (182.40 m)
Lowest point 2.00 m above BL
Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.18	1.7	1.6	179.27	-9.57	2.09	6.54
10.0	0.35	3.4	3.1	179.29	-9.58	2.18	6.54
15.0	0.53	5.2	4.7	179.29	-9.58	2.26	6.54
20.0	0.70	6.9	6.2	179.29	-9.58	2.35	6.54
25.0	0.88	8.6	7.8	179.30	-9.58	2.44	6.54
30.0	1.05	10.3	9.3	179.30	-9.58	2.53	6.54
35.0	1.23	12.1	10.9	179.30	-9.58	2.61	6.54
40.0	1.40	13.8	12.4	179.30	-9.58	2.70	6.54
45.0	1.58	15.5	14.0	179.30	-9.58	2.79	6.54
50.0	1.75	17.2	15.5	179.30	-9.58	2.88	6.54
55.0	1.93	19.0	17.1	179.30	-9.58	2.96	6.54
60.0	2.10	20.7	18.6	179.30	-9.58	3.05	6.54
65.0	2.28	22.4	20.2	179.30	-9.58	3.14	6.54
70.0	2.45	24.1	21.7	179.30	-9.58	3.23	6.54
75.0	2.63	25.9	23.3	179.30	-9.58	3.31	6.54
80.0	2.80	27.6	24.8	179.30	-9.58	3.40	6.54
85.0	2.98	29.3	26.4	179.30	-9.58	3.49	6.54
90.0	3.15	31.0	27.9	179.30	-9.58	3.58	6.54
95.0	3.33	32.8	29.5	179.30	-9.58	3.66	6.54
100.0	3.50	34.5	31.0	179.30	-9.58	3.75	6.54

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.18
Compartment descr: MDO Service P
Contents : Diesel Oil (DO, RHO = 0.9)

Extreme points of comp: Aft end at frame 225.00 (180.00 m)
Fore end at frame 228.00 (182.40 m)
Lowest point 2.00 m above BL
Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.18	0.6	0.6	181.20	8.63	2.09	0.61
10.0	0.35	1.2	1.1	181.20	8.63	2.17	0.61
15.0	0.53	1.9	1.7	181.20	8.63	2.26	0.61
20.0	0.70	2.5	2.2	181.20	8.63	2.35	0.61
25.0	0.88	3.1	2.8	181.20	8.63	2.44	0.61
30.0	1.05	3.7	3.3	181.20	8.63	2.52	0.61
35.0	1.23	4.3	3.9	181.20	8.63	2.61	0.61
40.0	1.40	4.9	4.4	181.20	8.63	2.70	0.61
45.0	1.58	5.6	5.0	181.20	8.63	2.79	0.61
50.0	1.75	6.2	5.6	181.20	8.63	2.88	0.61
55.0	1.93	6.8	6.1	181.20	8.63	2.96	0.61
60.0	2.10	7.4	6.7	181.20	8.63	3.05	0.61
65.0	2.28	8.0	7.2	181.20	8.63	3.14	0.61
70.0	2.45	8.6	7.8	181.20	8.63	3.23	0.61
75.0	2.63	9.3	8.3	181.20	8.63	3.31	0.61
80.0	2.80	9.9	8.9	181.20	8.63	3.40	0.61
85.0	2.98	10.5	9.4	181.20	8.63	3.49	0.61
90.0	3.15	11.1	10.0	181.20	8.63	3.58	0.61
95.0	3.33	11.7	10.6	181.20	8.63	3.66	0.61
100.0	3.50	12.3	11.1	181.20	8.63	3.75	0.61

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.19
Compartment descr: MDO Service S
Contents : Diesel Oil (DO, RHO = 0.9)

Extreme points of comp: Aft end at frame 225.00 (180.00 m)
Fore end at frame 228.00 (182.40 m)
Lowest point 2.00 m above BL
Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.18	0.6	0.6	181.20	-8.63	2.09	0.61
10.0	0.35	1.2	1.1	181.20	-8.63	2.17	0.61
15.0	0.53	1.9	1.7	181.20	-8.63	2.26	0.61
20.0	0.70	2.5	2.2	181.20	-8.63	2.35	0.61
25.0	0.88	3.1	2.8	181.20	-8.63	2.44	0.61
30.0	1.05	3.7	3.3	181.20	-8.63	2.52	0.61
35.0	1.23	4.3	3.9	181.20	-8.63	2.61	0.61
40.0	1.40	4.9	4.4	181.20	-8.63	2.70	0.61
45.0	1.58	5.6	5.0	181.20	-8.63	2.79	0.61
50.0	1.75	6.2	5.6	181.20	-8.63	2.88	0.61
55.0	1.93	6.8	6.1	181.20	-8.63	2.96	0.61
60.0	2.10	7.4	6.7	181.20	-8.63	3.05	0.61
65.0	2.28	8.0	7.2	181.20	-8.63	3.14	0.61
70.0	2.45	8.6	7.8	181.20	-8.63	3.23	0.61
75.0	2.63	9.3	8.3	181.20	-8.63	3.31	0.61
80.0	2.80	9.9	8.9	181.20	-8.63	3.40	0.61
85.0	2.98	10.5	9.4	181.20	-8.63	3.49	0.61
90.0	3.15	11.1	10.0	181.20	-8.63	3.58	0.61
95.0	3.33	11.7	10.6	181.20	-8.63	3.66	0.61
100.0	3.50	12.3	11.1	181.20	-8.63	3.75	0.61

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: FW1
Compartment descr: Fresh Water P
Contents : Fresh Water (FW, RHO = 1)

Extreme points of comp: Aft end at frame 238.00 (190.40 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 16.40 m above BL
Highest point 23.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.29	11.0	11.0	194.80	4.81	16.55	61.41
10.0	0.58	21.9	21.9	194.80	4.81	16.69	61.41
15.0	0.87	32.9	32.9	194.80	4.81	16.84	61.41
20.0	1.16	43.8	43.8	194.80	4.81	16.98	61.41
25.0	1.45	54.8	54.8	194.80	4.81	17.13	61.41
30.0	1.74	65.8	65.8	194.80	4.81	17.27	61.41
35.0	2.03	76.7	76.7	194.80	4.81	17.42	61.41
40.0	2.32	87.7	87.7	194.80	4.81	17.56	61.41
45.0	2.61	98.7	98.7	194.80	4.81	17.71	61.41
50.0	3.68	109.6	109.6	194.82	4.84	17.92	38.07
55.0	4.06	120.6	120.6	194.84	4.86	18.14	38.07
60.0	4.43	131.5	131.5	194.85	4.88	18.35	38.07
65.0	4.80	142.5	142.5	194.86	4.90	18.55	38.07
70.0	5.17	153.5	153.5	194.87	4.91	18.75	38.06
75.0	5.54	164.4	164.4	194.88	4.93	18.95	38.06
80.0	5.91	175.4	175.4	194.89	4.94	19.15	38.07
85.0	6.29	186.4	186.4	194.90	4.95	19.35	38.07
90.0	6.66	197.3	197.3	194.90	4.95	19.54	38.07
95.0	7.03	208.3	208.3	194.91	4.96	19.74	38.07
100.0	7.40	219.2	219.2	194.91	4.97	19.93	38.07

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: FW2
Compartment descr: Fresh Water S
Contents : Fresh Water (FW, RHO = 1)

Extreme points of comp: Aft end at frame 238.00 (190.40 m)
Fore end at frame 249.00 (199.20 m)
Lowest point 16.40 m above BL
Highest point 23.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.29	11.0	11.0	194.80	-4.81	16.55	61.41
10.0	0.58	21.9	21.9	194.80	-4.81	16.69	61.41
15.0	0.87	32.9	32.9	194.80	-4.81	16.84	61.41
20.0	1.16	43.8	43.8	194.80	-4.81	16.98	61.41
25.0	1.45	54.8	54.8	194.80	-4.81	17.13	61.41
30.0	1.74	65.8	65.8	194.80	-4.81	17.27	61.41
35.0	2.03	76.7	76.7	194.80	-4.81	17.42	61.41
40.0	2.32	87.7	87.7	194.80	-4.81	17.56	61.41
45.0	2.61	98.7	98.7	194.80	-4.81	17.71	61.41
50.0	3.68	109.6	109.6	194.82	-4.84	17.92	38.07
55.0	4.06	120.6	120.6	194.84	-4.86	18.14	38.07
60.0	4.43	131.5	131.5	194.85	-4.88	18.35	38.07
65.0	4.80	142.5	142.5	194.86	-4.90	18.55	38.07
70.0	5.17	153.5	153.5	194.87	-4.91	18.75	38.06
75.0	5.54	164.4	164.4	194.88	-4.93	18.95	38.06
80.0	5.91	175.4	175.4	194.89	-4.94	19.15	38.07
85.0	6.29	186.4	186.4	194.90	-4.95	19.35	38.07
90.0	6.66	197.3	197.3	194.90	-4.95	19.54	38.07
95.0	7.03	208.3	208.3	194.91	-4.96	19.74	38.07
100.0	7.40	219.2	219.2	194.91	-4.97	19.93	38.07

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: FW3
 Compartment descr: Fresh Water C
 Contents : Fresh Water (FW, RHO = 1)

Extreme points of comp: Aft end at frame 242.00 (193.60 m)
 Fore end at frame 249.00 (199.20 m)
 Lowest point 19.73 m above BL
 Highest point 23.80 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.20	7.8	7.8	196.40	0.00	19.83	160.07
10.0	0.41	15.7	15.7	196.40	0.00	19.93	160.07
15.0	0.61	23.5	23.5	196.40	0.00	20.03	160.07
20.0	0.82	31.3	31.3	196.40	0.00	20.13	160.07
25.0	1.02	39.1	39.1	196.40	0.00	20.23	160.07
30.0	1.22	47.0	47.0	196.40	0.00	20.34	160.07
35.0	1.43	54.8	54.8	196.40	0.00	20.44	160.07
40.0	1.63	62.6	62.6	196.40	0.00	20.54	160.07
45.0	1.83	70.4	70.4	196.40	0.00	20.64	160.07
50.0	2.04	78.3	78.3	196.40	0.00	20.74	160.07
55.0	2.24	86.1	86.1	196.40	-0.00	20.85	160.07
60.0	2.44	93.9	93.9	196.40	-0.00	20.95	160.07
65.0	2.65	101.8	101.8	196.40	-0.00	21.05	160.07
70.0	2.85	109.6	109.6	196.40	-0.00	21.15	160.07
75.0	3.06	117.4	117.4	196.40	-0.00	21.25	160.07
80.0	3.26	125.2	125.2	196.40	-0.00	21.36	160.07
85.0	3.46	133.1	133.1	196.40	-0.00	21.46	160.07
90.0	3.67	140.9	140.9	196.40	-0.00	21.56	160.07
95.0	3.87	148.7	148.7	196.40	-0.00	21.66	160.07
100.0	4.07	156.5	156.5	196.40	0.00	21.76	160.07

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T10.01
 Compartment descr: Potable Water
 Contents : Fresh Water (FW, RHO = 1)

Extreme points of comp: Aft end at frame 242.00 (193.60 m)
 Fore end at frame 249.00 (199.20 m)
 Lowest point 16.40 m above BL
 Highest point 19.02 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	3.8	3.8	196.40	0.00	16.47	67.53
10.0	0.26	7.6	7.6	196.40	0.00	16.53	67.53
15.0	0.39	11.3	11.3	196.40	0.00	16.60	67.53
20.0	0.52	15.1	15.1	196.40	0.00	16.66	67.53
25.0	0.66	18.9	18.9	196.40	0.00	16.73	67.53
30.0	0.79	22.7	22.7	196.40	0.00	16.79	67.53
35.0	0.92	26.5	26.5	196.40	0.00	16.86	67.53
40.0	1.05	30.3	30.3	196.40	0.00	16.92	67.53
45.0	1.18	34.0	34.0	196.40	0.00	16.99	67.53
50.0	1.31	37.8	37.8	196.40	0.00	17.06	67.53
55.0	1.44	41.6	41.6	196.40	0.00	17.12	67.53
60.0	1.58	45.4	45.4	196.40	0.00	17.19	67.53
65.0	1.71	49.2	49.2	196.40	0.00	17.25	67.53
70.0	1.84	52.9	52.9	196.40	0.00	17.32	67.53
75.0	1.97	56.7	56.7	196.40	0.00	17.38	67.53
80.0	2.10	60.5	60.5	196.40	0.00	17.45	67.53
85.0	2.23	64.3	64.3	196.40	0.00	17.52	67.53
90.0	2.36	68.1	68.1	196.40	0.00	17.58	67.53
95.0	2.49	71.8	71.8	196.40	0.00	17.65	67.53
100.0	2.63	75.6	75.6	196.40	0.00	17.71	67.53

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.10
Compartment descr: Feed Water
Contents : Technical Water (TFW, RHO = 1)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 228.00 (182.40 m)
Lowest point 5.50 m above BL
Highest point 8.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	2.0	2.0	179.80	-9.33	5.57	10.35
10.0	0.30	4.0	4.0	179.80	-9.33	5.65	10.35
15.0	0.45	6.0	6.0	179.80	-9.33	5.72	10.35
20.0	0.60	8.0	8.0	179.80	-9.33	5.80	10.35
25.0	0.75	10.0	10.0	179.80	-9.33	5.87	10.35
30.0	0.90	12.0	12.0	179.80	-9.33	5.95	10.35
35.0	1.05	14.1	14.1	179.80	-9.33	6.02	10.35
40.0	1.20	16.1	16.1	179.80	-9.33	6.10	10.35
45.0	1.35	18.1	18.1	179.80	-9.33	6.17	10.35
50.0	1.50	20.1	20.1	179.80	-9.33	6.25	10.35
55.0	1.65	22.1	22.1	179.80	-9.33	6.32	10.35
60.0	1.80	24.1	24.1	179.80	-9.33	6.40	10.35
65.0	1.95	26.1	26.1	179.80	-9.33	6.47	10.35
70.0	2.10	28.1	28.1	179.80	-9.33	6.55	10.35
75.0	2.25	30.1	30.1	179.80	-9.33	6.62	10.35
80.0	2.40	32.1	32.1	179.80	-9.33	6.70	10.35
85.0	2.55	34.1	34.1	179.80	-9.33	6.77	10.35
90.0	2.70	36.1	36.1	179.80	-9.33	6.85	10.35
95.0	2.85	38.2	38.2	179.80	-9.33	6.92	10.35
100.0	3.00	40.2	40.2	179.80	-9.33	7.00	10.35

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.20
 Compartment descr: Bilge Water Settling
 Contents : Bilge Water (BIW, RHO = 1)

Extreme points of comp: Aft end at frame 234.00 (187.20 m)
 Fore end at frame 237.00 (189.60 m)
 Lowest point 2.30 m above BL
 Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.30	0.5	0.5	188.01	-0.73	2.50	0.61
10.0	0.45	1.0	1.0	188.20	-0.73	2.59	0.61
15.0	0.61	1.6	1.6	188.27	-0.72	2.67	0.61
20.0	0.76	2.1	2.1	188.30	-0.73	2.75	0.61
25.0	0.91	2.6	2.6	188.32	-0.73	2.83	0.61
30.0	1.06	3.1	3.1	188.34	-0.73	2.90	0.61
35.0	1.22	3.6	3.6	188.34	-0.72	2.98	0.61
40.0	1.37	4.2	4.2	188.35	-0.73	3.06	0.61
45.0	1.52	4.7	4.7	188.36	-0.73	3.13	0.61
50.0	1.67	5.2	5.2	188.36	-0.73	3.21	0.61
55.0	1.83	5.7	5.7	188.37	-0.73	3.29	0.61
60.0	1.98	6.2	6.2	188.37	-0.73	3.36	0.61
65.0	2.13	6.8	6.8	188.37	-0.73	3.44	0.61
70.0	2.28	7.3	7.3	188.37	-0.73	3.52	0.61
75.0	2.44	7.8	7.8	188.37	-0.73	3.59	0.61
80.0	2.59	8.3	8.3	188.38	-0.73	3.67	0.61
85.0	2.74	8.8	8.8	188.38	-0.73	3.74	0.61
90.0	2.89	9.4	9.4	188.38	-0.73	3.82	0.61
95.0	3.05	9.9	9.9	188.38	-0.73	3.90	0.61
100.0	3.20	10.4	10.4	188.38	-0.73	3.97	0.61

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.22
 Compartment descr: Bilge Water
 Contents : Bilge Water (BIW, RHO = 1)

Extreme points of comp: Aft end at frame 231.00 (184.80 m)
 Fore end at frame 237.00 (189.60 m)
 Lowest point 1.20 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.05	1.5	1.5	186.86	-3.31	1.23	135.09
10.0	0.11	3.0	3.0	186.86	-3.31	1.25	135.09
15.0	0.16	4.6	4.6	186.86	-3.31	1.28	135.09
20.0	0.21	6.1	6.1	186.86	-3.31	1.31	135.09
25.0	0.26	7.6	7.6	186.86	-3.31	1.33	135.09
30.0	0.32	9.1	9.1	186.86	-3.31	1.36	135.09
35.0	0.37	10.7	10.7	186.86	-3.31	1.38	135.09
40.0	0.42	12.2	12.2	186.86	-3.31	1.41	135.09
45.0	0.48	13.7	13.7	186.86	-3.31	1.44	135.09
50.0	0.53	15.2	15.2	186.86	-3.31	1.46	135.09
55.0	0.58	16.8	16.8	186.86	-3.31	1.49	135.09
60.0	0.63	18.3	18.3	186.86	-3.31	1.52	135.09
65.0	0.69	19.8	19.8	186.86	-3.31	1.54	135.09
70.0	0.74	21.3	21.3	186.86	-3.31	1.57	135.09
75.0	0.79	22.8	22.8	186.86	-3.31	1.60	135.09
80.0	0.85	24.4	24.4	186.86	-3.31	1.62	118.12
85.0	0.91	25.9	25.9	186.88	-3.30	1.65	95.02
90.0	0.99	27.4	27.4	186.92	-3.29	1.68	65.28
95.0	1.10	28.9	28.9	186.98	-3.25	1.71	16.64
100.0	1.40	30.5	30.5	187.07	-3.20	1.74	0.11

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.21
 Compartment descr: Sludge
 Contents : Sludge (SLU, RHO = 2.38)

Extreme points of comp: Aft end at frame 231.00 (184.80 m)
 Fore end at frame 237.00 (189.60 m)
 Lowest point 1.20 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.05	1.5	3.6	186.86	3.31	1.23	321.52
10.0	0.11	3.0	7.2	186.86	3.31	1.25	321.52
15.0	0.16	4.6	10.9	186.86	3.31	1.28	321.52
20.0	0.21	6.1	14.5	186.86	3.31	1.31	321.52
25.0	0.26	7.6	18.1	186.86	3.31	1.33	321.52
30.0	0.32	9.1	21.7	186.86	3.31	1.36	321.52
35.0	0.37	10.7	25.4	186.86	3.31	1.38	321.52
40.0	0.42	12.2	29.0	186.86	3.31	1.41	321.52
45.0	0.48	13.7	32.6	186.86	3.31	1.44	321.52
50.0	0.53	15.2	36.2	186.86	3.31	1.46	321.52
55.0	0.58	16.8	39.9	186.86	3.31	1.49	321.52
60.0	0.63	18.3	43.5	186.86	3.31	1.52	321.52
65.0	0.69	19.8	47.1	186.86	3.31	1.54	321.52
70.0	0.74	21.3	50.7	186.86	3.31	1.57	321.52
75.0	0.79	22.8	54.4	186.86	3.31	1.60	321.52
80.0	0.85	24.4	58.0	186.86	3.31	1.62	281.12
85.0	0.91	25.9	61.6	186.88	3.30	1.65	226.15
90.0	0.99	27.4	65.2	186.92	3.29	1.68	155.37
95.0	1.10	28.9	68.9	186.98	3.25	1.71	39.61
100.0	1.40	30.5	72.5	187.07	3.20	1.74	0.27

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.23
 Compartment descr: Clean F.O. Drain Tank P
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 220.00 (176.00 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	0.3	0.3	176.80	4.37	0.05	0.71
10.0	0.20	0.5	0.5	176.80	4.37	0.10	0.71
15.0	0.30	0.8	0.8	176.80	4.37	0.15	0.71
20.0	0.40	1.1	1.1	176.80	4.37	0.20	0.71
25.0	0.50	1.4	1.4	176.80	4.37	0.25	0.71
30.0	0.60	1.6	1.6	176.80	4.37	0.30	0.71
35.0	0.70	1.9	1.9	176.80	4.37	0.35	0.71
40.0	0.80	2.2	2.2	176.80	4.37	0.40	0.71
45.0	0.90	2.5	2.5	176.80	4.37	0.45	0.71
50.0	1.00	2.7	2.7	176.80	4.37	0.50	0.71
55.0	1.10	3.0	3.0	176.80	4.37	0.55	0.71
60.0	1.20	3.3	3.3	176.80	4.37	0.60	0.71
65.0	1.30	3.6	3.6	176.80	4.37	0.65	0.71
70.0	1.40	3.8	3.8	176.80	4.37	0.70	0.71
75.0	1.50	4.1	4.1	176.80	4.37	0.75	0.71
80.0	1.60	4.4	4.4	176.80	4.37	0.80	0.71
85.0	1.70	4.7	4.7	176.80	4.37	0.85	0.71
90.0	1.80	4.9	4.9	176.80	4.37	0.90	0.71
95.0	1.90	5.2	5.2	176.80	4.37	0.95	0.71
100.0	2.00	5.5	5.5	176.80	4.37	1.00	0.71

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.24
 Compartment descr: Dirty Oil Drain Tank P
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 220.00 (176.00 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	0.3	0.3	176.80	2.62	0.05	0.71
10.0	0.20	0.5	0.5	176.80	2.62	0.10	0.71
15.0	0.30	0.8	0.8	176.80	2.62	0.15	0.71
20.0	0.40	1.1	1.1	176.80	2.62	0.20	0.71
25.0	0.50	1.4	1.4	176.80	2.62	0.25	0.71
30.0	0.60	1.6	1.6	176.80	2.62	0.30	0.71
35.0	0.70	1.9	1.9	176.80	2.62	0.35	0.71
40.0	0.80	2.2	2.2	176.80	2.62	0.40	0.71
45.0	0.90	2.5	2.5	176.80	2.62	0.45	0.71
50.0	1.00	2.7	2.7	176.80	2.62	0.50	0.71
55.0	1.10	3.0	3.0	176.80	2.62	0.55	0.71
60.0	1.20	3.3	3.3	176.80	2.62	0.60	0.71
65.0	1.30	3.6	3.6	176.80	2.62	0.65	0.71
70.0	1.40	3.8	3.8	176.80	2.62	0.70	0.71
75.0	1.50	4.1	4.1	176.80	2.62	0.75	0.71
80.0	1.60	4.4	4.4	176.80	2.62	0.80	0.71
85.0	1.70	4.7	4.7	176.80	2.62	0.85	0.71
90.0	1.80	4.9	4.9	176.80	2.62	0.90	0.71
95.0	1.90	5.2	5.2	176.80	2.62	0.95	0.71
100.0	2.00	5.5	5.5	176.80	2.62	1.00	0.71

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.25
 Compartment descr: Dirty Oil Drain Tank S
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 220.00 (176.00 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	0.3	0.3	176.80	-0.87	0.05	0.71
10.0	0.20	0.5	0.5	176.80	-0.87	0.10	0.71
15.0	0.30	0.8	0.8	176.80	-0.87	0.15	0.71
20.0	0.40	1.1	1.1	176.80	-0.87	0.20	0.71
25.0	0.50	1.4	1.4	176.80	-0.87	0.25	0.71
30.0	0.60	1.6	1.6	176.80	-0.87	0.30	0.71
35.0	0.70	1.9	1.9	176.80	-0.87	0.35	0.71
40.0	0.80	2.2	2.2	176.80	-0.87	0.40	0.71
45.0	0.90	2.5	2.5	176.80	-0.87	0.45	0.71
50.0	1.00	2.7	2.7	176.80	-0.87	0.50	0.71
55.0	1.10	3.0	3.0	176.80	-0.87	0.55	0.71
60.0	1.20	3.3	3.3	176.80	-0.87	0.60	0.71
65.0	1.30	3.6	3.6	176.80	-0.87	0.65	0.71
70.0	1.40	3.8	3.8	176.80	-0.87	0.70	0.71
75.0	1.50	4.1	4.1	176.80	-0.87	0.75	0.71
80.0	1.60	4.4	4.4	176.80	-0.87	0.80	0.71
85.0	1.70	4.7	4.7	176.80	-0.87	0.85	0.71
90.0	1.80	4.9	4.9	176.80	-0.87	0.90	0.71
95.0	1.90	5.2	5.2	176.80	-0.87	0.95	0.71
100.0	2.00	5.5	5.5	176.80	-0.87	1.00	0.71

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.26
 Compartment descr: Clean F.O. Drain Tank S
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 220.00 (176.00 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	0.3	0.3	176.80	-2.62	0.05	0.71
10.0	0.20	0.5	0.5	176.80	-2.62	0.10	0.71
15.0	0.30	0.8	0.8	176.80	-2.62	0.15	0.71
20.0	0.40	1.1	1.1	176.80	-2.62	0.20	0.71
25.0	0.50	1.4	1.4	176.80	-2.62	0.25	0.71
30.0	0.60	1.6	1.6	176.80	-2.62	0.30	0.71
35.0	0.70	1.9	1.9	176.80	-2.62	0.35	0.71
40.0	0.80	2.2	2.2	176.80	-2.62	0.40	0.71
45.0	0.90	2.5	2.5	176.80	-2.62	0.45	0.71
50.0	1.00	2.7	2.7	176.80	-2.62	0.50	0.71
55.0	1.10	3.0	3.0	176.80	-2.62	0.55	0.71
60.0	1.20	3.3	3.3	176.80	-2.62	0.60	0.71
65.0	1.30	3.6	3.6	176.80	-2.62	0.65	0.71
70.0	1.40	3.8	3.8	176.80	-2.62	0.70	0.71
75.0	1.50	4.1	4.1	176.80	-2.62	0.75	0.71
80.0	1.60	4.4	4.4	176.80	-2.62	0.80	0.71
85.0	1.70	4.7	4.7	176.80	-2.62	0.85	0.71
90.0	1.80	4.9	4.9	176.80	-2.62	0.90	0.71
95.0	1.90	5.2	5.2	176.80	-2.62	0.95	0.71
100.0	2.00	5.5	5.5	176.80	-2.62	1.00	0.71

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.27
 Compartment descr: F0 Overflow Tank
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 214.00 (171.20 m)
 Fore end at frame 219.00 (175.20 m)
 Lowest point 2.00 m above BL
 Highest point 5.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.18	1.2	1.2	173.20	-0.87	2.09	1.79
10.0	0.35	2.4	2.4	173.20	-0.87	2.17	1.79
15.0	0.53	3.6	3.6	173.20	-0.87	2.26	1.79
20.0	0.70	4.8	4.8	173.20	-0.87	2.35	1.79
25.0	0.88	6.0	6.0	173.20	-0.87	2.44	1.79
30.0	1.05	7.2	7.2	173.20	-0.87	2.52	1.79
35.0	1.23	8.4	8.4	173.20	-0.88	2.61	1.79
40.0	1.40	9.6	9.6	173.20	-0.87	2.70	1.79
45.0	1.58	10.8	10.8	173.20	-0.88	2.79	1.79
50.0	1.75	12.0	12.0	173.20	-0.88	2.88	1.79
55.0	1.93	13.2	13.2	173.20	-0.88	2.96	1.79
60.0	2.10	14.4	14.4	173.20	-0.88	3.05	1.79
65.0	2.28	15.6	15.6	173.20	-0.88	3.14	1.79
70.0	2.45	16.8	16.8	173.20	-0.88	3.22	1.79
75.0	2.63	18.0	18.0	173.20	-0.88	3.31	1.79
80.0	2.80	19.2	19.2	173.20	-0.87	3.40	1.79
85.0	2.98	20.4	20.4	173.20	-0.88	3.49	1.79
90.0	3.15	21.6	21.6	173.20	-0.88	3.58	1.79
95.0	3.33	22.8	22.8	173.20	-0.88	3.66	1.79
100.0	3.50	24.0	24.0	173.20	-0.88	3.75	1.79

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.28
 Compartment descr: Cool F.W. Drain Tk.
 Contents : Miscellaneous (MIS, RHO = 1)

Extreme points of comp: Aft end at frame 220.00 (176.00 m)
 Fore end at frame 222.00 (177.60 m)
 Lowest point -0.00 m above BL
 Highest point 2.00 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.10	0.3	0.3	176.80	0.87	0.05	0.71
10.0	0.20	0.5	0.5	176.80	0.87	0.10	0.71
15.0	0.30	0.8	0.8	176.80	0.87	0.15	0.71
20.0	0.40	1.1	1.1	176.80	0.87	0.20	0.71
25.0	0.50	1.4	1.4	176.80	0.87	0.25	0.71
30.0	0.60	1.6	1.6	176.80	0.87	0.30	0.71
35.0	0.70	1.9	1.9	176.80	0.87	0.35	0.71
40.0	0.80	2.2	2.2	176.80	0.87	0.40	0.71
45.0	0.90	2.5	2.5	176.80	0.87	0.45	0.71
50.0	1.00	2.7	2.7	176.80	0.87	0.50	0.71
55.0	1.10	3.0	3.0	176.80	0.87	0.55	0.71
60.0	1.20	3.3	3.3	176.80	0.87	0.60	0.71
65.0	1.30	3.6	3.6	176.80	0.87	0.65	0.71
70.0	1.40	3.8	3.8	176.80	0.87	0.70	0.71
75.0	1.50	4.1	4.1	176.80	0.87	0.75	0.71
80.0	1.60	4.4	4.4	176.80	0.87	0.80	0.71
85.0	1.70	4.7	4.7	176.80	0.87	0.85	0.71
90.0	1.80	4.9	4.9	176.80	0.87	0.90	0.71
95.0	1.90	5.2	5.2	176.80	0.87	0.95	0.71
100.0	2.00	5.5	5.5	176.80	0.87	1.00	0.71

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.11
 Compartment descr: Black Water
 Contents : Gray Water (GWA, RHO = 1)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
 Fore end at frame 228.00 (182.40 m)
 Lowest point 5.50 m above BL
 Highest point 8.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	0.9	0.9	180.00	4.16	5.58	0.90
10.0	0.30	1.9	1.9	180.00	4.16	5.65	0.90
15.0	0.45	2.8	2.8	180.00	4.16	5.73	0.90
20.0	0.60	3.7	3.7	180.00	4.16	5.80	0.90
25.0	0.75	4.6	4.6	180.00	4.16	5.88	0.90
30.0	0.90	5.6	5.6	180.00	4.16	5.95	0.90
35.0	1.05	6.5	6.5	180.00	4.16	6.03	0.90
40.0	1.20	7.4	7.4	180.00	4.16	6.10	0.90
45.0	1.35	8.3	8.3	180.00	4.16	6.18	0.90
50.0	1.50	9.3	9.3	180.00	4.16	6.25	0.90
55.0	1.65	10.2	10.2	180.00	4.16	6.33	0.90
60.0	1.80	11.1	11.1	180.00	4.16	6.40	0.90
65.0	1.95	12.0	12.0	180.00	4.16	6.48	0.90
70.0	2.10	13.0	13.0	180.00	4.16	6.55	0.90
75.0	2.25	13.9	13.9	180.00	4.16	6.63	0.90
80.0	2.40	14.8	14.8	180.00	4.16	6.70	0.90
85.0	2.55	15.7	15.7	180.00	4.16	6.78	0.90
90.0	2.70	16.7	16.7	180.00	4.16	6.85	0.90
95.0	2.85	17.6	17.6	180.00	4.16	6.93	0.90
100.0	3.00	18.5	18.5	180.00	4.16	7.00	0.90

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.13
 Compartment descr: Gray Water
 Contents : Gray Water (GWA, RHO = 1)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
 Fore end at frame 228.00 (182.40 m)
 Lowest point 5.50 m above BL
 Highest point 8.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	1.5	1.5	180.00	5.91	5.57	4.19
10.0	0.30	3.1	3.1	180.00	5.91	5.65	4.19
15.0	0.45	4.6	4.6	180.00	5.91	5.73	4.19
20.0	0.60	6.2	6.2	180.00	5.91	5.80	4.19
25.0	0.75	7.7	7.7	180.00	5.91	5.88	4.19
30.0	0.90	9.3	9.3	180.00	5.91	5.95	4.19
35.0	1.05	10.8	10.8	180.00	5.91	6.03	4.19
40.0	1.20	12.3	12.3	180.00	5.91	6.10	4.19
45.0	1.35	13.9	13.9	180.00	5.91	6.18	4.19
50.0	1.50	15.4	15.4	180.00	5.91	6.25	4.19
55.0	1.65	17.0	17.0	180.00	5.91	6.33	4.19
60.0	1.80	18.5	18.5	180.00	5.91	6.40	4.19
65.0	1.95	20.1	20.1	180.00	5.91	6.48	4.19
70.0	2.10	21.6	21.6	180.00	5.91	6.55	4.19
75.0	2.25	23.2	23.2	180.00	5.91	6.63	4.19
80.0	2.40	24.7	24.7	180.00	5.91	6.70	4.19
85.0	2.55	26.2	26.2	180.00	5.91	6.78	4.19
90.0	2.70	27.8	27.8	180.00	5.91	6.85	4.19
95.0	2.85	29.3	29.3	180.00	5.91	6.93	4.19
100.0	3.00	30.9	30.9	180.00	5.91	7.00	4.19

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.14
 Compartment descr: NO.1 M/E & A/E L.O. STORAGE TANK
 Contents : Lubricating Oil (LO, RHO = 0.9)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
 Fore end at frame 228.00 (182.40 m)
 Lowest point 8.50 m above BL
 Highest point 11.40 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	1.8	1.6	180.00	4.81	8.57	6.51
10.0	0.29	3.6	3.2	180.00	4.81	8.65	6.51
15.0	0.44	5.4	4.8	180.00	4.81	8.72	6.51
20.0	0.58	7.2	6.4	180.00	4.81	8.79	6.51
25.0	0.73	9.0	8.1	180.00	4.81	8.86	6.51
30.0	0.87	10.7	9.7	180.00	4.81	8.94	6.51
35.0	1.01	12.5	11.3	180.00	4.81	9.01	6.51
40.0	1.16	14.3	12.9	180.00	4.81	9.08	6.51
45.0	1.30	16.1	14.5	180.00	4.81	9.15	6.51
50.0	1.45	17.9	16.1	180.00	4.81	9.23	6.51
55.0	1.60	19.7	17.7	180.00	4.81	9.30	6.51
60.0	1.74	21.5	19.3	180.00	4.81	9.37	6.51
65.0	1.88	23.3	20.9	180.00	4.81	9.44	6.51
70.0	2.03	25.1	22.6	180.00	4.81	9.52	6.51
75.0	2.17	26.9	24.2	180.00	4.81	9.59	6.51
80.0	2.32	28.6	25.8	180.00	4.81	9.66	6.51
85.0	2.46	30.4	27.4	180.00	4.81	9.73	6.51
90.0	2.61	32.2	29.0	180.00	4.81	9.81	6.51
95.0	2.76	34.0	30.6	180.00	4.81	9.88	6.51
100.0	2.90	35.8	32.2	180.00	4.81	9.95	6.51

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.15

Compartment descr: NO.2 M/E & A/E L.O. STORAGE TANK

Contents : Lubricating Oil (LO, RHO = 0.9)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
Fore end at frame 225.00 (180.00 m)
Lowest point 5.50 m above BL
Highest point 8.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	1.2	1.1	178.80	-5.25	5.58	7.72
10.0	0.30	2.5	2.2	178.80	-5.25	5.65	7.72
15.0	0.45	3.7	3.3	178.80	-5.25	5.73	7.72
20.0	0.60	4.9	4.4	178.80	-5.25	5.80	7.72
25.0	0.75	6.2	5.6	178.80	-5.25	5.88	7.72
30.0	0.90	7.4	6.7	178.80	-5.25	5.95	7.72
35.0	1.05	8.6	7.8	178.80	-5.25	6.03	7.72
40.0	1.20	9.9	8.9	178.80	-5.25	6.10	7.72
45.0	1.35	11.1	10.0	178.80	-5.25	6.18	7.72
50.0	1.50	12.3	11.1	178.80	-5.25	6.25	7.72
55.0	1.65	13.6	12.2	178.80	-5.25	6.33	7.72
60.0	1.80	14.8	13.3	178.80	-5.25	6.40	7.72
65.0	1.95	16.1	14.4	178.80	-5.25	6.48	7.72
70.0	2.10	17.3	15.6	178.80	-5.25	6.55	7.72
75.0	2.25	18.5	16.7	178.80	-5.25	6.63	7.72
80.0	2.40	19.8	17.8	178.80	-5.25	6.70	7.72
85.0	2.55	21.0	18.9	178.80	-5.25	6.78	7.72
90.0	2.70	22.2	20.0	178.80	-5.25	6.85	7.72
95.0	2.85	23.5	21.1	178.80	-5.25	6.93	7.72
100.0	3.00	24.7	22.2	178.80	-5.25	7.00	7.72

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: T09.09
 Compartment descr: Used LO
 Contents : Lubricating Oil (LO, RHO = 0.9)

Extreme points of comp: Aft end at frame 222.00 (177.60 m)
 Fore end at frame 228.00 (182.40 m)
 Lowest point 5.50 m above BL
 Highest point 8.50 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.15	2.0	1.8	179.80	9.33	5.57	9.31
10.0	0.30	4.0	3.6	179.80	9.33	5.65	9.31
15.0	0.45	6.0	5.4	179.80	9.33	5.72	9.31
20.0	0.60	8.0	7.2	179.80	9.33	5.80	9.31
25.0	0.75	10.0	9.0	179.80	9.33	5.87	9.31
30.0	0.90	12.0	10.8	179.80	9.33	5.95	9.32
35.0	1.05	14.1	12.7	179.80	9.33	6.02	9.32
40.0	1.20	16.1	14.5	179.80	9.33	6.10	9.32
45.0	1.35	18.1	16.3	179.80	9.33	6.17	9.32
50.0	1.50	20.1	18.1	179.80	9.33	6.25	9.32
55.0	1.65	22.1	19.9	179.80	9.33	6.32	9.32
60.0	1.80	24.1	21.7	179.80	9.33	6.40	9.32
65.0	1.95	26.1	23.5	179.80	9.33	6.47	9.32
70.0	2.10	28.1	25.3	179.80	9.33	6.55	9.32
75.0	2.25	30.1	27.1	179.80	9.33	6.62	9.32
80.0	2.40	32.1	28.9	179.80	9.33	6.70	9.32
85.0	2.55	34.1	30.7	179.80	9.33	6.77	9.32
90.0	2.70	36.1	32.5	179.80	9.33	6.85	9.32
95.0	2.85	38.2	34.3	179.80	9.33	6.92	9.32
100.0	3.00	40.2	36.1	179.80	9.33	7.00	9.32

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: R03.01
 Compartment descr: Dry Tank
 Contents : Dry Tank (DRY, RHO = 0)

Extreme points of comp: Aft end at frame 63.00 (50.40 m)
 Fore end at frame 65.00 (52.00 m)
 Lowest point -0.00 m above BL
 Highest point 2.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.13	0.4	0.4	51.20	0.88	0.06	0.00
10.0	0.26	0.7	0.7	51.20	0.88	0.13	0.00
15.0	0.39	1.1	1.1	51.20	0.88	0.19	0.00
20.0	0.52	1.4	1.4	51.20	0.88	0.26	0.00
25.0	0.65	1.8	1.8	51.20	0.88	0.32	0.00
30.0	0.78	2.1	2.1	51.20	0.88	0.39	0.00
35.0	0.91	2.5	2.5	51.20	0.88	0.45	0.00
40.0	1.04	2.9	2.9	51.20	0.88	0.52	0.00
45.0	1.17	3.2	3.2	51.20	0.88	0.58	0.00
50.0	1.30	3.6	3.6	51.20	0.88	0.65	0.00
55.0	1.43	3.9	3.9	51.20	0.88	0.71	0.00
60.0	1.56	4.3	4.3	51.20	0.88	0.78	0.00
65.0	1.69	4.6	4.6	51.20	0.88	0.84	0.00
70.0	1.82	5.0	5.0	51.20	0.87	0.91	0.00
75.0	1.95	5.4	5.4	51.20	0.87	0.97	0.00
80.0	2.08	5.7	5.7	51.20	0.88	1.04	0.00
85.0	2.21	6.1	6.1	51.20	0.88	1.10	0.00
90.0	2.34	6.4	6.4	51.20	0.88	1.17	0.00
95.0	2.47	6.8	6.8	51.20	0.88	1.23	0.00
100.0	2.60	7.1	7.1	51.20	0.88	1.30	0.00

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

Compartment ident: R11.02
 Compartment descr: Dry Tank
 Contents : Dry Tank (DRY, RHO = 0)

Extreme points of comp: Aft end at frame 249.00 (199.20 m)
 Fore end at frame 252.00 (201.60 m)
 Lowest point -0.00 m above BL
 Highest point 3.60 m above BL

FILL	H	VNET	MASS	LCG	TCG	VCG	FSM
%	m	m3	t	m	m	m	tm
5.0	0.16	0.6	0.6	200.40	0.00	0.08	0.00
10.0	0.31	1.3	1.3	200.40	0.00	0.16	0.00
15.0	0.47	1.9	1.9	200.40	0.00	0.23	0.00
20.0	0.62	2.6	2.6	200.40	0.00	0.31	0.00
25.0	0.78	3.2	3.2	200.40	0.00	0.39	0.00
30.0	0.93	3.8	3.8	200.40	0.00	0.47	0.00
35.0	1.09	4.5	4.5	200.40	0.00	0.54	0.00
40.0	1.25	5.1	5.1	200.40	0.00	0.62	0.00
45.0	1.40	5.8	5.8	200.40	0.00	0.70	0.00
50.0	1.56	6.4	6.4	200.40	0.00	0.78	0.00
55.0	1.71	7.0	7.0	200.40	0.00	0.86	0.00
60.0	1.87	7.7	7.7	200.40	0.00	0.93	0.00
65.0	2.02	8.3	8.3	200.40	0.00	1.01	0.00
70.0	2.18	9.0	9.0	200.40	0.00	1.09	0.00
75.0	2.33	9.6	9.6	200.40	0.00	1.17	0.00
80.0	2.49	10.2	10.2	200.40	0.00	1.25	0.00
85.0	2.66	10.9	10.9	200.40	-0.00	1.32	0.00
90.0	2.91	11.5	11.5	200.37	-0.00	1.40	0.00
95.0	3.24	12.2	12.2	200.34	0.00	1.49	0.00
100.0	3.60	12.8	12.8	200.31	-0.00	1.59	0.00

FILL	filling degree	%
H	height	m
VNET	net volume	m3
MASS	weight of load	t
LCG	x-coordinate of center of gravity	m
TCG	y-coordinate of center of gravity	m
VCG	z-coordinate of center of gravity	m
FSM	free surf. mom/real	tm

5.3.3 GZ correction tables

Compartment ident: T09.20
Compartment descr: Bilge Water Settling
Contents : Bilge Water (BIW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	1	1	1	1
40.0	0	0	0	1	1	1	2
50.0	0	0	0	1	1	1	2
60.0	0	0	0	1	1	1	2
80.0	0	0	0	1	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.22
Compartment descr: Bilge Water
Contents : Bilge Water (BIW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	12	14	14	14	12	10	8
40.0	19	23	23	22	19	16	13
50.0	21	25	25	24	21	18	14
60.0	21	26	25	24	21	18	14
80.0	17	21	21	19	17	14	11
98.0	3	3	3	3	2	2	2

Compartment ident: WB01.03
Compartment descr: Water Ballast 01.03
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	296	482	540	530	489	426	348
40.0	458	654	697	676	617	533	430
50.0	550	693	715	688	626	537	430
60.0	600	699	696	657	593	505	399
80.0	522	565	546	501	437	359	269
98.0	74	90	96	90	82	67	51

Compartment ident: WB01.11
Compartment descr: Water Ballast 01.11

Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	27	56	89	133	196	267	362
40.0	27	56	89	133	200	331	627
50.0	27	56	89	133	200	331	677
60.0	27	56	89	133	200	331	627
80.0	27	56	89	133	196	267	362
98.0	23	29	32	32	31	29	27

Compartment ident: WB01.12

Compartment descr: Water Ballast 01.12

Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	27	56	89	133	196	267	362
40.0	27	56	89	133	200	331	627
50.0	27	56	89	133	200	331	677
60.0	27	56	89	133	200	331	627
80.0	27	56	89	133	196	267	362
98.0	23	29	32	32	31	29	27

Compartment ident: WB02.05

Compartment descr: Water Ballast 02.05

Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	308	612	898	1058	1137	1169	1135
40.0	441	825	1203	1561	1737	1749	1655
50.0	453	870	1286	1604	1766	1776	1677
60.0	450	888	1277	1505	1625	1644	1560
80.0	444	777	915	971	981	964	934
98.0	123	128	125	117	104	90	73

Compartment ident: WB02.06

Compartment descr: Water Ballast 02.06

Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	304	595	820	962	1049	1103	1091
40.0	438	871	1300	1639	1763	1749	1640
50.0	483	960	1439	1752	1854	1820	1690
60.0	517	1026	1457	1694	1781	1740	1608
80.0	458	851	1017	1088	1107	1093	1031
98.0	126	130	128	121	113	107	105

Compartment ident: WB02.07
Compartment descr: Water Ballast 02.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	147	329	540	756	864	879	835
40.0	238	513	827	1063	1185	1192	1123
50.0	267	581	883	1077	1183	1194	1123
60.0	289	623	870	1008	1081	1087	1023
80.0	324	522	605	633	630	609	582
98.0	81	82	80	74	66	56	45

Compartment ident: WB02.08
Compartment descr: Water Ballast 02.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	114	203	272	323	364	403	428
40.0	198	363	501	614	669	675	646
50.0	228	425	595	689	719	705	658
60.0	254	481	633	693	701	671	612
80.0	296	436	481	489	474	438	388
98.0	72	74	72	67	60	51	41

Compartment ident: WB02.09
Compartment descr: Water Ballast 02.09
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	415	509	531	507	458	390	309
40.0	561	802	818	773	694	589	465
50.0	569	839	853	806	723	614	484
60.0	564	801	817	772	693	588	464
80.0	418	507	527	503	454	387	306
98.0	64	65	61	56	50	41	31

Compartment ident: WB02.10
Compartment descr: Water Ballast 02.10
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	407	496	517	495	447	381	301
40.0	547	783	800	757	679	577	456
50.0	556	821	836	790	709	602	475
60.0	553	785	801	757	679	577	455
80.0	415	502	520	497	448	381	301
98.0	64	65	61	56	50	41	31

Compartment ident: WB02.11
Compartment descr: Water Ballast 02.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	284	356	378	369	338	292	235
40.0	364	563	587	561	508	437	349
50.0	365	588	611	583	527	452	361
60.0	364	552	577	552	500	429	343
80.0	298	359	375	364	332	285	228
98.0	47	48	47	42	37	30	23

Compartment ident: WB02.12
Compartment descr: Water Ballast 02.12
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	294	385	411	399	364	314	252
40.0	370	613	644	616	559	480	384
50.0	393	643	673	643	583	501	401
60.0	416	621	649	621	562	481	384
80.0	322	393	409	394	359	308	246
98.0	50	50	48	44	39	32	24

Compartment ident: WB03.01
Compartment descr: Water Ballast 03.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	289	387	407	394	359	310	249
40.0	347	562	595	572	520	447	360
50.0	357	581	614	590	536	461	371
60.0	350	544	581	559	509	439	353
80.0	265	342	363	358	333	289	235
98.0	44	47	47	44	40	33	27

Compartment ident: WB03.02
Compartment descr: Water Ballast 03.02
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	242	311	330	324	299	260	212
40.0	338	513	543	523	476	410	329
50.0	354	547	574	551	500	430	345
60.0	357	532	558	534	485	416	333
80.0	275	357	376	363	331	285	229
98.0	40	44	46	42	38	33	27

Compartment ident: WB03.05
Compartment descr: Water Ballast 03.05
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1064	1530	1693	1775	1754	1621	1409
40.0	1076	2083	2739	2875	2773	2526	2175
50.0	1011	2180	2885	3016	2904	2643	2274
60.0	972	2158	2743	2880	2779	2533	2181
80.0	1021	1511	1735	1803	1757	1622	1411
98.0	231	236	226	206	179	151	121

Compartment ident: WB03.06
Compartment descr: Water Ballast 03.06
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	167	315	431	527	600	654	666
40.0	167	346	553	749	909	1014	1049
50.0	173	357	571	788	946	1058	1088
60.0	179	367	575	773	914	992	1025
80.0	179	349	470	547	604	640	637
98.0	71	82	85	84	79	71	64

Compartment ident: WB03.07
Compartment descr: Water Ballast 03.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	274	594	965	1355	1656	1910	2088
40.0	345	730	1186	1758	2442	2842	2982
50.0	360	752	1208	1802	2443	2853	3003
60.0	366	759	1215	1756	2214	2581	2760
80.0	368	758	1040	1204	1310	1397	1518
98.0	153	165	165	156	142	125	106

Compartment ident: WB03.08
Compartment descr: Water Ballast 03.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	233	431	600	743	871	1003	1192
40.0	322	634	943	1259	1600	1910	2078
50.0	347	693	1055	1447	1836	2076	2189
60.0	360	731	1131	1560	1856	2023	2088
80.0	368	757	1032	1183	1263	1296	1294
98.0	153	165	165	156	142	125	104

Compartment ident: WB03.09

Compartment descr: Water Ballast 03.09

Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB03.10

Compartment descr: Water Ballast 03.10

Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB03.11

Compartment descr: Water Ballast 03.11

Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	287	353	370	358	326	280	224
40.0	359	551	576	550	498	427	341
50.0	356	577	600	573	519	445	356
60.0	353	549	574	549	497	427	341
80.0	280	349	369	358	326	281	226
98.0	45	46	44	40	36	29	23

Compartment ident: WB03.12
Compartment descr: Water Ballast 03.12
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	287	357	375	364	332	285	229
40.0	368	564	588	562	508	436	348
50.0	368	593	616	588	531	455	363
60.0	366	566	590	564	510	437	349
80.0	284	357	378	366	334	288	231
98.0	46	47	45	41	36	31	24

Compartment ident: WB04.01
Compartment descr: Water Ballast 04.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1862	2127	2084	1923	1687	1394	1056
40.0	2880	3231	3125	2870	2512	2071	1565
50.0	3018	3365	3252	2985	2611	2151	1623
60.0	2866	3207	3096	2843	2487	2049	1546
80.0	1836	2062	2021	1863	1634	1348	1018
98.0	249	247	234	218	185	150	113

Compartment ident: WB04.02
Compartment descr: Water Ballast 04.02
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1671	1908	1878	1736	1526	1263	959
40.0	2629	2987	2897	2664	2334	1927	1459
50.0	2788	3140	3040	2794	2446	2017	1524
60.0	2678	3016	2915	2679	2345	1933	1459
80.0	1792	1972	1926	1774	1556	1281	965
98.0	248	246	241	213	182	147	110

Compartment ident: WB04.07
Compartment descr: Water Ballast 04.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1242	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB04.08
Compartment descr: Water Ballast 04.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1242	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB04.09
Compartment descr: Water Ballast 04.09
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB04.10
Compartment descr: Water Ballast 04.10
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB04.11
Compartment descr: Water Ballast 04.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	289	358	378	366	333	287	230
40.0	368	566	590	563	509	437	349
50.0	368	593	616	588	531	455	363
60.0	368	566	590	563	509	437	349
80.0	289	358	378	366	333	287	230
98.0	46	47	45	41	37	30	23

Compartment ident: WB04.12
Compartment descr: Water Ballast 04.12
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	289	358	378	366	333	287	230
40.0	368	566	590	563	509	437	349
50.0	368	593	616	588	531	455	363
60.0	368	566	590	563	509	437	349
80.0	289	358	378	366	333	287	230
98.0	46	47	45	41	37	30	23

Compartment ident: WB05.01
 Compartment descr: Water Ballast 05.01
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1874	2128	2084	1922	1686	1392	1053
40.0	2900	3250	3142	2885	2525	2081	1571
50.0	3044	3391	3276	3007	2630	2166	1634
60.0	2895	3234	3124	2864	2506	2064	1556
80.0	1853	2081	2039	1880	1648	1359	1026
98.0	251	250	236	220	187	152	114

Compartment ident: WB05.02
 Compartment descr: Water Ballast 05.02
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1740	1978	1943	1795	1576	1303	987
40.0	2731	3086	2989	2747	2406	1985	1501
50.0	2891	3242	3135	2879	2519	2076	1567
60.0	2767	3109	3002	2757	2412	1988	1500
80.0	1828	2021	1976	1822	1596	1316	992
98.0	251	249	243	216	185	149	111

Compartment ident: WB05.05
 Compartment descr: Water Ballast 05.05
 Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	519	956	1167	1274	1400	1541	1592
40.0	390	806	1380	2067	2481	2619	2583
50.0	390	843	1501	2268	2671	2792	2734
60.0	399	959	1611	2185	2566	2685	2629
80.0	521	1027	1302	1450	1531	1589	1619
98.0	179	193	191	179	162	141	118

Compartment ident: WB05.06
Compartment descr: Water Ballast 05.06
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	492	968	1245	1397	1485	1549	1588
40.0	477	1029	1632	2183	2560	2674	2613
50.0	390	924	1562	2274	2666	2783	2721
60.0	390	814	1463	2104	2480	2611	2571
80.0	494	880	1097	1239	1393	1515	1553
98.0	179	193	191	179	162	140	113

Compartment ident: WB05.07
Compartment descr: Water Ballast 05.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1241	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB05.08
Compartment descr: Water Ballast 05.08
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1242	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB05.09
 Compartment descr: Water Ballast 05.09
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB05.10
 Compartment descr: Water Ballast 05.10
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB05.11
 Compartment descr: Water Ballast 05.11
 Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	287	352	369	357	325	279	224
40.0	358	549	573	548	496	425	340
50.0	353	574	598	571	517	443	354
60.0	351	547	572	547	496	425	340
80.0	279	348	367	357	325	280	225
98.0	45	45	43	40	36	29	22

Compartment ident: WB05.12
Compartment descr: Water Ballast 05.12
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	287	356	375	364	331	285	228
40.0	368	564	588	561	508	435	348
50.0	368	593	616	588	531	455	363
60.0	366	566	590	563	510	437	349
80.0	283	357	378	366	334	288	231
98.0	47	47	45	41	36	31	24

Compartment ident: WB06.01
Compartment descr: Water Ballast 06.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1874	2128	2084	1922	1686	1392	1053
40.0	2900	3250	3142	2885	2525	2081	1571
50.0	3044	3391	3276	3007	2630	2166	1634
60.0	2895	3234	3124	2864	2506	2064	1556
80.0	1853	2081	2039	1880	1648	1359	1026
98.0	251	250	236	220	187	152	114

Compartment ident: WB06.02
Compartment descr: Water Ballast 06.02
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1740	1978	1943	1795	1576	1303	987
40.0	2731	3086	2989	2747	2406	1985	1501
50.0	2891	3242	3135	2879	2519	2076	1567
60.0	2767	3109	3002	2757	2412	1988	1500
80.0	1828	2021	1976	1822	1596	1316	992
98.0	251	249	243	216	185	149	111

Compartment ident: WB06.05
Compartment descr: Water Ballast 06.05
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	519	956	1167	1274	1400	1541	1592
40.0	390	806	1380	2067	2481	2619	2583
50.0	390	843	1501	2268	2671	2792	2734
60.0	399	959	1611	2185	2566	2685	2629
80.0	521	1027	1302	1450	1531	1589	1619
98.0	179	193	191	180	162	141	118

Compartment ident: WB06.06
Compartment descr: Water Ballast 06.06
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	492	968	1245	1397	1485	1549	1588
40.0	477	1029	1632	2183	2560	2674	2613
50.0	390	924	1562	2274	2666	2783	2721
60.0	390	814	1463	2104	2480	2611	2571
80.0	494	880	1097	1239	1393	1515	1553
98.0	179	193	191	179	162	140	113

Compartment ident: WB06.07
Compartment descr: Water Ballast 06.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1242	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB06.08
Compartment descr: Water Ballast 06.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	368	753	1007	1151	1242	1313	1384
40.0	368	760	1216	1723	2109	2275	2279
50.0	368	760	1216	1808	2237	2397	2390
60.0	368	760	1216	1723	2109	2275	2279
80.0	368	753	1007	1151	1242	1313	1384
98.0	144	157	156	147	134	117	99

Compartment ident: WB06.09
Compartment descr: Water Ballast 06.09
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB06.10
Compartment descr: Water Ballast 06.10
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	382	464	482	461	416	354	280
40.0	515	732	747	705	633	537	424
50.0	521	767	780	736	660	560	441
60.0	515	732	747	705	633	537	424
80.0	382	464	482	461	416	354	280
98.0	59	59	56	51	46	38	29

Compartment ident: WB06.11
Compartment descr: Water Ballast 06.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	289	358	378	366	333	287	230
40.0	368	566	590	563	509	437	349
50.0	368	593	616	588	531	455	363
60.0	368	566	590	563	509	437	349
80.0	289	358	378	366	333	287	230
98.0	46	47	45	41	37	30	23

Compartment ident: WB06.12
Compartment descr: Water Ballast 06.12
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	289	358	378	366	333	287	230
40.0	368	566	590	563	509	437	349
50.0	368	593	616	588	531	455	363
60.0	368	566	590	563	509	437	349
80.0	289	358	378	366	333	287	230
98.0	46	47	45	41	37	30	23

Compartment ident: WB07.01
Compartment descr: Water Ballast 07.01
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	2183	2496	2447	2259	1983	1639	1242
40.0	3370	3795	3673	3375	2956	2438	1843
50.0	3534	3954	3823	3511	3072	2532	1912
60.0	3357	3769	3641	3345	2927	2413	1822
80.0	2154	2424	2377	2194	1924	1588	1201
98.0	293	291	275	256	218	177	134

Compartment ident: WB07.02
Compartment descr: Water Ballast 07.02
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1977	2258	2224	2057	1808	1497	1137
40.0	3099	3531	3426	3152	2763	2282	1728
50.0	3284	3710	3595	3305	2895	2388	1806
60.0	3154	3563	3445	3167	2774	2288	1728
80.0	2111	2331	2277	2099	1845	1518	1144
98.0	298	294	277	261	220	177	130

Compartment ident: WB07.05
Compartment descr: Water Ballast 07.05
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	623	1147	1398	1525	1676	1846	1911
40.0	465	962	1650	2478	2980	3150	3110
50.0	465	1009	1798	2721	3212	3362	3295
60.0	479	1151	1933	2625	3088	3235	3170
80.0	626	1231	1565	1744	1844	1915	1952
98.0	213	237	230	217	196	171	143

Compartment ident: WB07.06
Compartment descr: Water Ballast 07.06
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	596	1175	1509	1692	1797	1873	1921
40.0	572	1235	1968	2637	3092	3233	3161
50.0	465	1104	1874	2742	3220	3363	3290
60.0	465	970	1749	2529	2993	3155	3110
80.0	597	1055	1317	1488	1674	1828	1881
98.0	213	232	230	217	197	170	138

Compartment ident: WB07.07
Compartment descr: Water Ballast 07.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	432	892	1203	1382	1497	1591	1679
40.0	438	907	1454	2066	2534	2734	2738
50.0	439	910	1457	2167	2682	2874	2866
60.0	440	911	1458	2062	2524	2724	2728
80.0	441	902	1205	1376	1483	1568	1654
98.0	172	187	186	176	160	140	118

Compartment ident: WB07.08
Compartment descr: Water Ballast 07.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	428	870	1173	1347	1457	1546	1636
40.0	436	897	1426	2015	2479	2683	2693
50.0	438	903	1438	2120	2630	2826	2824
60.0	439	907	1446	2036	2479	2680	2688
80.0	441	901	1202	1372	1475	1553	1624
98.0	172	187	186	176	160	140	118

Compartment ident: WB07.09
Compartment descr: Water Ballast 07.09
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	452	551	573	547	494	420	332
40.0	612	869	886	836	750	637	502
50.0	622	910	925	872	782	664	523
60.0	616	869	885	836	750	636	501
80.0	455	551	572	546	493	419	331
98.0	70	70	67	61	55	45	34

Compartment ident: WB07.10
Compartment descr: Water Ballast 07.10
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	452	551	573	547	494	420	332
40.0	612	869	886	836	750	637	502
50.0	622	910	925	872	782	664	523
60.0	616	869	885	836	750	636	501
80.0	455	551	572	546	493	419	331
98.0	70	70	67	61	55	45	34

Compartment ident: WB07.11
Compartment descr: Water Ballast 07.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	345	425	446	431	393	338	270
40.0	433	664	694	663	600	514	411
50.0	429	695	724	691	625	536	428
60.0	427	663	692	662	599	514	411
80.0	338	421	444	431	393	339	272
98.0	54	55	52	48	43	35	27

Compartment ident: WB07.12
Compartment descr: Water Ballast 07.12
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	345	428	451	437	398	343	275
40.0	441	677	706	674	610	523	417
50.0	441	712	739	705	638	546	436
60.0	440	679	708	676	612	524	419
80.0	341	429	453	440	401	345	277
98.0	55	56	54	49	43	37	28

Compartment ident: WB08.01
Compartment descr: Water Ballast 08.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1368	1464	1402	1277	1106	900	665
40.0	2175	2264	2149	1950	1687	1370	1011
50.0	2273	2361	2241	2033	1758	1427	1052
60.0	2158	2245	2132	1936	1676	1360	1004
80.0	1352	1450	1395	1266	1097	893	660
98.0	177	173	163	152	128	104	79

Compartment ident: WB08.03
Compartment descr: Water Ballast 08.03
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	36	80	103	109	105	96	82
40.0	50	107	138	144	139	128	110
50.0	56	109	136	143	138	126	109
60.0	60	103	123	131	127	117	102
80.0	53	70	76	77	75	71	63
98.0	10	10	10	9	8	7	5

Compartment ident: WB08.04
Compartment descr: Water Ballast 08.04
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	26	42	50	55	57	56	51
40.0	38	67	87	96	95	89	79
50.0	43	77	98	105	102	95	83
60.0	48	85	101	105	102	93	81
80.0	54	75	80	79	73	64	54
98.0	13	14	12	10	8	7	6

Compartment ident: WB08.07
Compartment descr: Water Ballast 08.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	120	265	452	639	826	967	1029
40.0	160	341	566	888	1208	1388	1455
50.0	174	375	622	925	1219	1391	1455
60.0	187	401	663	907	1108	1268	1330
80.0	210	423	552	624	667	698	743
98.0	81	86	85	80	72	63	53

Compartment ident: WB08.08
Compartment descr: Water Ballast 08.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	106	207	302	380	444	510	608
40.0	146	286	428	584	758	918	1005
50.0	160	317	478	657	852	979	1042
60.0	174	345	524	714	853	941	981
80.0	198	392	510	571	599	606	601
98.0	83	92	91	87	80	68	57

Compartment ident: WB08.09
Compartment descr: Water Ballast 08.09
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	248	286	285	265	234	195	149
40.0	373	444	434	402	354	294	225
50.0	393	463	453	419	369	307	234
60.0	373	444	434	402	354	294	225
80.0	248	286	285	265	234	195	149
98.0	35	35	33	31	26	21	16

Compartment ident: WB08.10
Compartment descr: Water Ballast 08.10
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	248	286	285	265	234	195	149
40.0	373	444	434	402	354	294	225
50.0	393	463	453	419	369	307	234
60.0	373	444	434	402	354	294	225
80.0	248	286	285	265	234	195	149
98.0	35	35	33	31	26	21	16

Compartment ident: WB08.11
Compartment descr: Water Ballast 08.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	193	244	259	252	231	200	162
40.0	229	373	397	382	348	300	242
50.0	228	389	412	397	361	312	251
60.0	228	369	393	379	345	298	241
80.0	187	236	250	245	226	196	158
98.0	31	31	30	27	25	20	16

Compartment ident: WB08.12
Compartment descr: Water Ballast 08.12
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	183	233	247	243	223	194	157
40.0	230	368	392	377	344	297	239
50.0	233	389	412	396	360	311	250
60.0	233	373	395	380	346	299	241
80.0	189	243	258	250	228	197	159
98.0	32	33	33	30	27	23	19

Compartment ident: WB09.01
Compartment descr: Water Ballast 09.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	561	679	674	625	550	457	348
40.0	945	1062	1019	931	811	665	499
50.0	1031	1111	1058	964	838	685	511
60.0	1013	1062	1011	920	798	650	483
80.0	689	696	659	597	518	419	307
98.0	98	93	91	81	68	54	38

Compartment ident: WB09.07
Compartment descr: Water Ballast 09.07
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	41	95	174	298	467	648	763
40.0	65	145	256	436	707	928	1079
50.0	75	168	295	490	750	972	1119
60.0	45	149	292	491	717	938	1077
80.0	39	85	171	302	440	595	748
98.0	25	29	29	27	44	85	92

Compartment ident: WB09.08
Compartment descr: Water Ballast 09.08
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	35	66	95	125	158	180	190
40.0	56	107	135	148	153	152	159
50.0	56	74	85	96	109	129	170
60.0	22	43	63	85	113	153	214
80.0	34	68	101	135	159	176	189
98.0	28	34	35	33	30	28	24

Compartment ident: WB09.09
Compartment descr: Water Ballast 09.09
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	415	440	422	384	333	271	200
40.0	631	661	629	571	495	402	297
50.0	659	688	655	595	515	419	310
60.0	631	661	629	571	495	402	297
80.0	415	440	422	384	333	271	200
98.0	60	66	64	56	48	39	28

Compartment ident: WB09.11
Compartment descr: Water Ballast 09.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	115	247	407	583	820	1134	1578
40.0	144	306	509	797	1228	1852	2364
50.0	157	335	556	867	1364	1896	2384
60.0	171	364	601	934	1375	1782	2163
80.0	198	419	649	821	961	1111	1268
98.0	95	119	129	131	131	126	122

Compartment ident: WB09.12
Compartment descr: Water Ballast 09.12
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	108	216	332	467	617	803	1036
40.0	135	272	420	597	843	1178	1574
50.0	149	299	463	660	918	1236	1632
60.0	162	326	505	719	952	1224	1510
80.0	188	379	561	688	768	825	881
98.0	119	141	141	135	124	110	94

Compartment ident: WB10.01
Compartment descr: Water Ballast 10.01
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	188	286	305	294	268	231	186
40.0	301	440	452	428	385	328	260
50.0	341	458	467	442	396	336	264
60.0	354	441	446	420	376	318	249
80.0	277	305	300	279	246	205	157
98.0	45	45	42	38	34	27	19

Compartment ident: WB10.05
Compartment descr: Water Ballast 10.05
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	155	352	598	820	1035	1179	1253
40.0	206	456	788	1251	1585	1741	1771
50.0	228	500	860	1289	1639	1791	1811
60.0	219	516	883	1227	1515	1672	1701
80.0	240	489	668	793	881	958	1037
98.0	98	110	110	105	97	86	73

Compartment ident: WB10.06
Compartment descr: Water Ballast 10.06
Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	131	247	354	447	517	579	663
40.0	182	347	494	646	826	978	1053
50.0	193	361	530	714	891	1003	1060
60.0	196	386	580	766	881	950	992
80.0	233	447	561	618	638	633	610
98.0	99	109	107	102	93	80	64

Compartment ident: WB10.11
 Compartment descr: Water Ballast 10.11
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	43	93	159	260	449	742	1116
40.0	61	134	230	381	662	1136	1543
50.0	72	158	273	455	747	1101	1510
60.0	85	186	323	520	749	998	1326
80.0	114	249	375	461	524	581	661
98.0	63	70	71	67	62	55	47

Compartment ident: WB10.12
 Compartment descr: Water Ballast 10.12
 Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	39	78	118	165	221	297	412
40.0	56	111	170	238	329	479	746
50.0	66	131	199	278	384	560	797
60.0	77	152	230	322	445	596	756
80.0	102	201	300	367	407	436	462
98.0	66	75	75	71	66	59	51

Compartment ident: WB11.01
 Compartment descr: Water Ballast 11.01
 Contents : Ballast Water (BW, RHO = 1.025)

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FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	150	350	598	897	1273	1633	1893
40.0	375	758	1120	1498	1855	2146	2365
50.0	426	865	1328	1701	1962	2159	2293
60.0	465	943	1416	1744	1932	2035	2080
80.0	486	924	1171	1301	1360	1361	1307
98.0	190	217	213	205	189	166	140

Compartment ident: WB11.11
Compartment descr: Water Ballast 11.11
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	125	278	492	863	1698	2651	3325
40.0	206	463	834	1505	2504	3599	4498
50.0	259	584	1059	1795	2600	3545	4350
60.0	318	726	1300	1941	2543	3171	3731
80.0	407	950	1362	1624	1744	1788	1868
98.0	155	174	176	170	158	142	125

Compartment ident: WB11.12
Compartment descr: Water Ballast 11.12
Contents : Ballast Water (BW, RHO = 1.025)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	119	236	362	511	675	861	1120
40.0	190	370	554	761	1036	1464	2072
50.0	236	455	677	924	1225	1653	2236
60.0	285	550	811	1062	1356	1729	2102
80.0	311	604	895	1088	1157	1152	1165
98.0	243	273	245	236	223	194	156

Compartment ident: T09.07
Compartment descr: MDO Storage P
Contents : Diesel Oil (DO, RHO = 0.9)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1	2	3	4	5	5	7
40.0	1	2	4	6	8	9	11
50.0	1	2	4	6	8	10	12
60.0	1	2	4	6	8	10	12
80.0	1	2	4	5	6	7	8
98.0	1	1	1	1	1	1	1

Compartment ident: T09.08
Compartment descr: MDO Storage S
Contents : Diesel Oil (DO, RHO = 0.9)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1	2	4	5	6	7	8
40.0	1	2	4	6	8	10	12
50.0	1	2	4	6	8	10	12
60.0	1	2	4	6	7	9	11
80.0	1	2	3	4	5	5	7
98.0	1	1	1	1	1	1	1

Compartment ident: MGO
Compartment descr: MGO Storage Tank
Contents : Diesel Oil (DO, RHO = 0.9)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	46	84	102	111	121	133	137
40.0	35	72	122	182	218	230	227
50.0	35	75	132	200	235	246	241
60.0	35	84	142	193	226	237	232
80.0	46	90	115	128	135	141	143
98.0	16	17	17	16	14	12	10

Compartment ident: T09.18
Compartment descr: MDO Service P
Contents : Diesel Oil (DO, RHO = 0.9)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	1	1	1	1
40.0	0	0	0	1	1	1	2
50.0	0	0	0	1	1	1	3
60.0	0	0	0	1	1	1	2
80.0	0	0	0	1	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.19
Compartment descr: MDO Service S
Contents : Diesel Oil (DO, RHO = 0.9)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	1	1	1	1
40.0	0	0	0	1	1	1	2
50.0	0	0	0	1	1	1	3
60.0	0	0	0	1	1	1	2
80.0	0	0	0	1	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: FW1
Compartment descr: Fresh Water P
Contents : Fresh Water (FW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	11	22	35	45	48	55	68
40.0	10	14	22	37	63	101	149
50.0	7	10	18	34	60	109	166
60.0	7	14	21	30	52	99	156
80.0	7	14	22	32	43	54	82
98.0	5	6	6	7	6	6	7

Compartment ident: FW2
Compartment descr: Fresh Water S
Contents : Fresh Water (FW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	11	22	35	45	48	50	56
40.0	10	14	18	27	44	73	111
50.0	7	11	14	24	41	74	125
60.0	7	14	22	30	40	67	117
80.0	7	14	22	31	41	50	60
98.0	5	6	6	6	5	5	5

Compartment ident: FW3
Compartment descr: Fresh Water C
Contents : Fresh Water (FW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	28	51	62	67	69	70	67
40.0	28	57	89	109	116	114	106
50.0	28	57	92	116	122	119	110
60.0	28	57	89	109	116	114	106
80.0	28	51	62	67	69	70	67
98.0	8	9	9	8	7	6	5

Compartment ident: T10.01
Compartment descr: Potable Water
Contents : Fresh Water (FW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	12	20	23	25	25	25	23
40.0	12	24	35	41	42	40	36
50.0	12	24	37	43	44	41	37
60.0	12	24	35	41	42	40	36
80.0	12	20	23	25	25	25	23
98.0	3	3	3	3	3	2	2

Compartment ident: T09.11
Compartment descr: Black Water
Contents : Gray Water (GWA, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	1	1	1	1	2
40.0	0	0	1	1	1	2	3
50.0	0	0	1	1	1	2	4
60.0	0	0	1	1	1	2	3
80.0	0	0	1	1	1	1	2
98.0	0	0	0	0	0	0	0

Compartment ident: T09.13
Compartment descr: Gray Water
Contents : Gray Water (GWA, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1	1	2	3	4	4	5
40.0	1	1	2	4	5	8	9
50.0	1	1	2	4	5	8	10
60.0	1	1	2	4	5	8	9
80.0	1	1	2	3	4	4	5
98.0	0	0	0	0	0	0	0

Compartment ident: HF01
Compartment descr: HF0 Storage Tank 1
Contents : Heavy Fuel Oil (HF0, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	252	464	564	612	666	731	756
40.0	191	396	673	1000	1199	1265	1247
50.0	191	411	729	1101	1295	1353	1324
60.0	195	465	780	1062	1247	1304	1276
80.0	252	498	633	705	746	774	789
98.0	87	94	93	87	79	69	57

Compartment ident: HF02
Compartment descr: HF0 Storage Tank 2
Contents : Heavy Fuel Oil (HF0, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	302	598	759	846	895	929	946
40.0	281	605	980	1314	1531	1593	1552
50.0	230	543	923	1365	1594	1658	1617
60.0	230	480	859	1249	1485	1559	1532
80.0	286	514	644	729	819	899	931
98.0	105	113	111	105	95	82	66

Compartment ident: HF03

Compartment descr: HFO Storage Tank 3

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	199	366	450	494	549	607	627
40.0	147	304	526	797	958	1012	999
50.0	147	321	576	872	1028	1076	1054
60.0	153	369	621	839	986	1032	1011
80.0	202	396	501	556	587	609	620
98.0	69	74	73	69	62	54	45

Compartment ident: HF04

Compartment descr: HFO Storage Tank 4

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	178	348	455	514	549	576	596
40.0	181	392	605	809	955	1003	984
50.0	147	351	590	843	995	1043	1023
60.0	147	307	556	790	923	976	964
80.0	192	337	418	471	529	569	576
98.0	69	74	73	69	62	54	43

Compartment ident: T09.03

Compartment descr: HFO Settling Tank P

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1	2	3	5	7	9	11
40.0	1	2	3	5	8	13	21
50.0	1	2	3	5	8	13	23
60.0	1	2	3	5	8	13	21
80.0	1	2	3	5	7	9	11
98.0	1	1	1	1	1	1	1

Compartment ident: T09.04

Compartment descr: HFO Settling Tank S

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	1	2	3	4	5	7	9
40.0	1	2	3	4	6	10	17
50.0	1	2	3	4	6	10	18
60.0	1	2	3	4	6	10	17
80.0	1	2	3	4	6	7	9
98.0	1	1	1	1	1	1	1

Compartment ident: T09.05

Compartment descr: HFO Service Tank P

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	1	1	2	2	3	5
40.0	0	1	1	2	2	4	8
50.0	0	1	1	2	2	4	8
60.0	0	1	1	2	2	4	8
80.0	0	1	1	2	2	3	5
98.0	0	0	0	0	0	0	0

Compartment ident: T09.06

Compartment descr: HFO Service Tank S

Contents : Heavy Fuel Oil (HFO, RHO = 0.991)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	1	1	1	2	3	4
40.0	0	1	1	1	2	3	6
50.0	0	1	1	1	2	3	6
60.0	0	1	1	1	2	3	6
80.0	0	1	1	1	2	3	4
98.0	0	0	0	0	0	0	0

Compartment ident: T09.23
Compartment descr: Clean F.O. Drain Tank P
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	0	1	1	1
40.0	0	0	0	1	1	1	1
50.0	0	0	0	1	1	1	1
60.0	0	0	0	1	1	1	1
80.0	0	0	0	0	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.24
Compartment descr: Dirty Oil Drain Tank P
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	0	1	1	1
40.0	0	0	0	1	1	1	1
50.0	0	0	0	1	1	1	1
60.0	0	0	0	1	1	1	1
80.0	0	0	0	0	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.25
Compartment descr: Dirty Oil Drain Tank S
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	0	1	1	1
40.0	0	0	0	1	1	1	1
50.0	0	0	0	1	1	1	1
60.0	0	0	0	1	1	1	1
80.0	0	0	0	0	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.26
Compartment descr: Clean F.O. Drain Tank S
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	0	1	1	1
40.0	0	0	0	1	1	1	1
50.0	0	0	0	1	1	1	1
60.0	0	0	0	1	1	1	1
80.0	0	0	0	0	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.27
Compartment descr: FO Overflow Tank
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	1	1	2	2	3	3
40.0	0	1	1	2	2	4	6
50.0	0	1	1	2	2	4	7
60.0	0	1	1	2	2	4	6
80.0	0	1	1	2	2	3	3
98.0	0	0	0	0	0	0	0

Compartment ident: T09.28
Compartment descr: Cool F.W. Drain Tk.
Contents : Miscellaneous (MIS, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	0	0	0	0	1	1	1
40.0	0	0	0	1	1	1	1
50.0	0	0	0	1	1	1	1
60.0	0	0	0	1	1	1	1
80.0	0	0	0	0	1	1	1
98.0	0	0	0	0	0	0	0

Compartment ident: T09.21
Compartment descr: Sludge
Contents : Sludge (SLU, RHO = 2.38)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	38	43	42	38	34	28	21
40.0	48	54	53	49	43	36	28
50.0	44	52	52	49	43	36	28
60.0	38	47	48	45	40	34	26
80.0	20	26	27	26	24	21	16
98.0	2	3	3	2	2	2	2

Compartment ident: T09.10
Compartment descr: Feed Water
Contents : Technical Water (TFW, RHO = 1)

FILL %	10 deg tm	20 deg tm	30 deg tm	40 deg tm	50 deg tm	60 deg tm	70 deg tm
20.0	2	4	5	6	7	8	8
40.0	2	4	6	8	11	13	15
50.0	2	4	6	9	12	14	15
60.0	2	4	6	9	12	14	15
80.0	2	4	6	7	8	8	9
98.0	1	1	1	1	1	1	1

5.4 Flooding angle curves

Flooding angle curves are shown for following trims:

-2.0m, -1.0m, 0.0m +1.0m, +2.0m, +3.0m, +4.0m, +5.0m, +6.0m,

All unprotected openings used in intact stability calculations are presented.

OPENING IMMERSION

TRIM=-2.0m

P6047/M

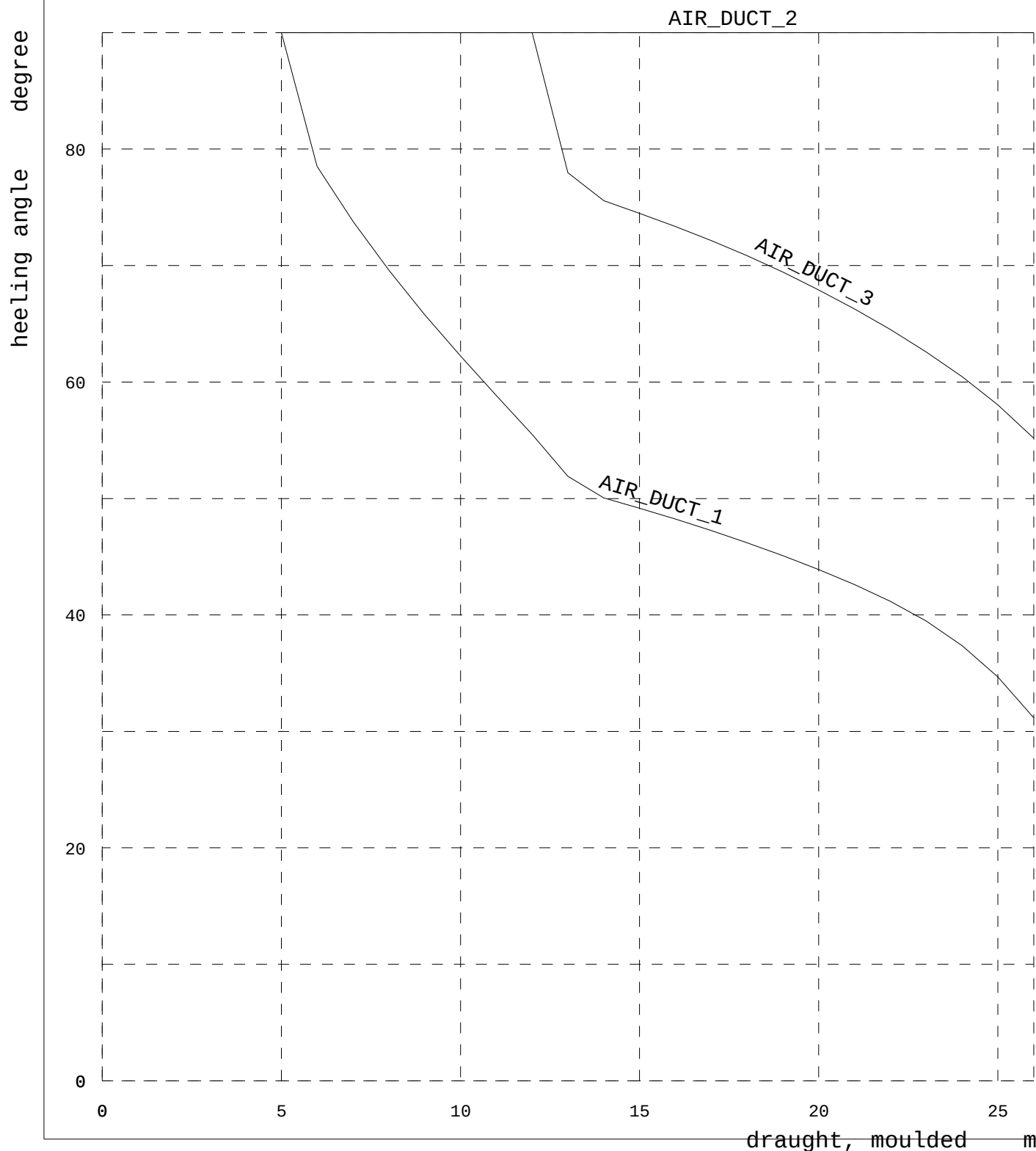
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.26

USER TH



OPENING IMMERSION

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P6047/M

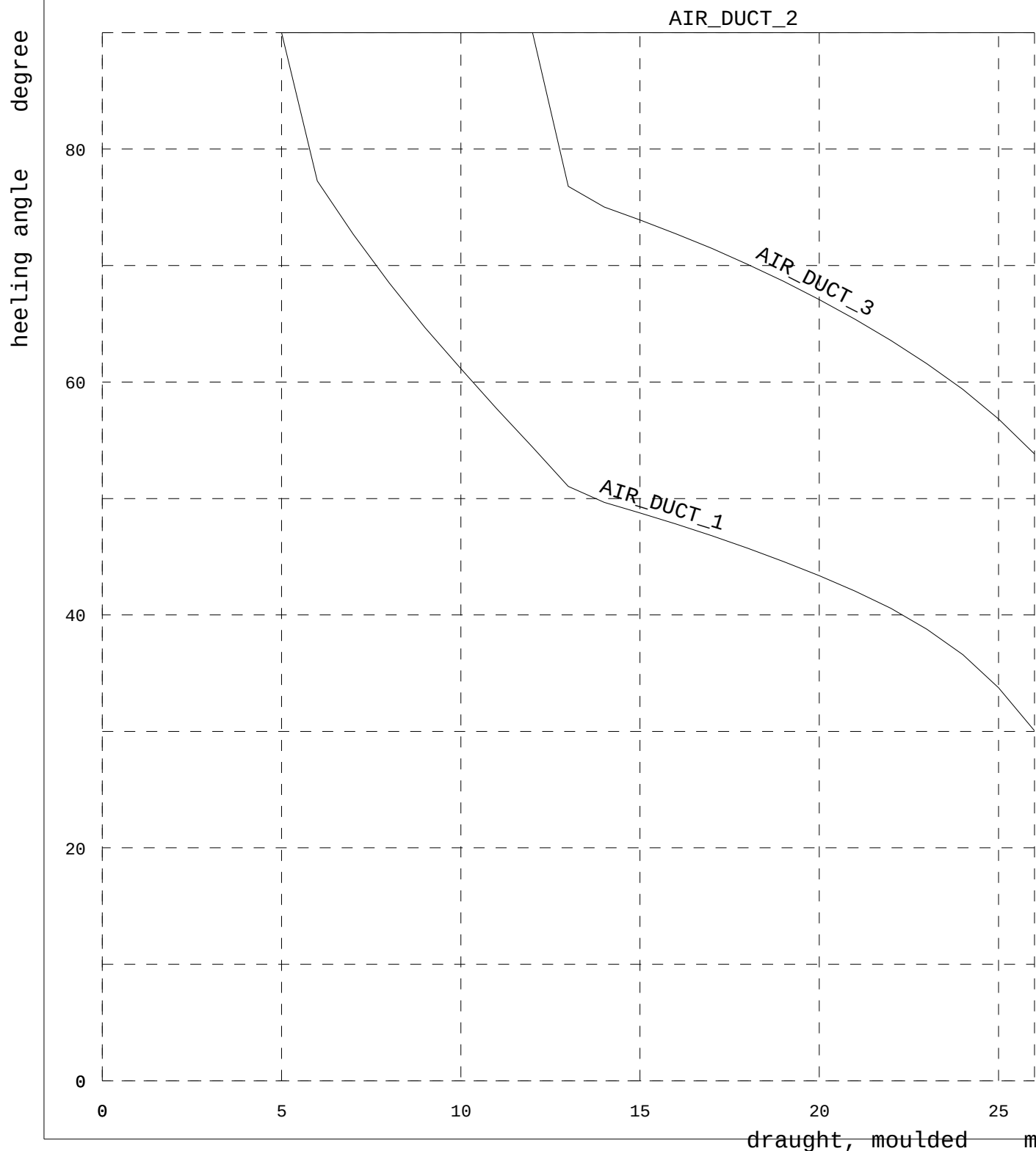
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.24

USER TH



OPENING IMMERSION

TRIM=0.0m

P6047/M

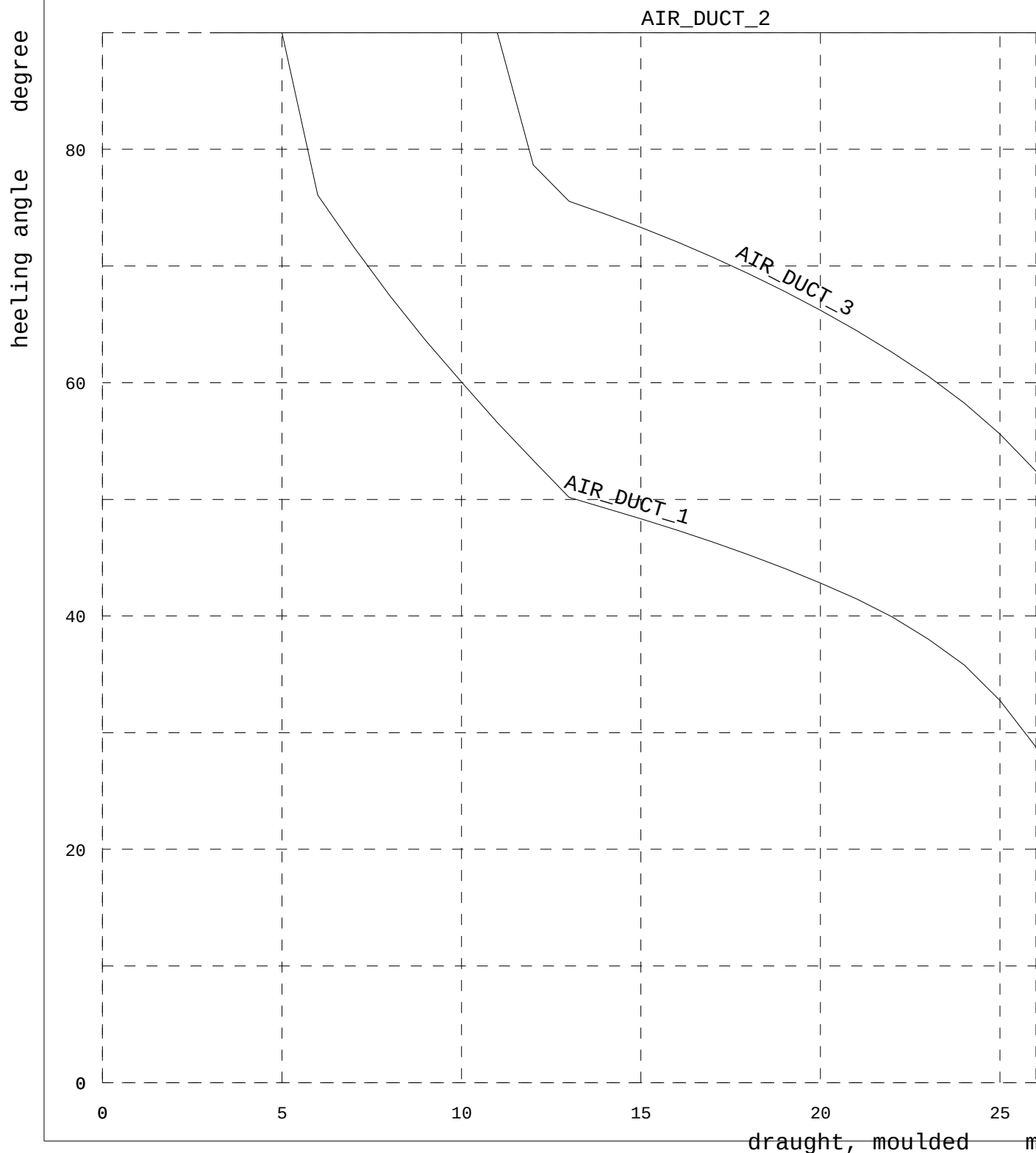
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 14.36

USER TH



OPENING IMMERSION

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P6047/M

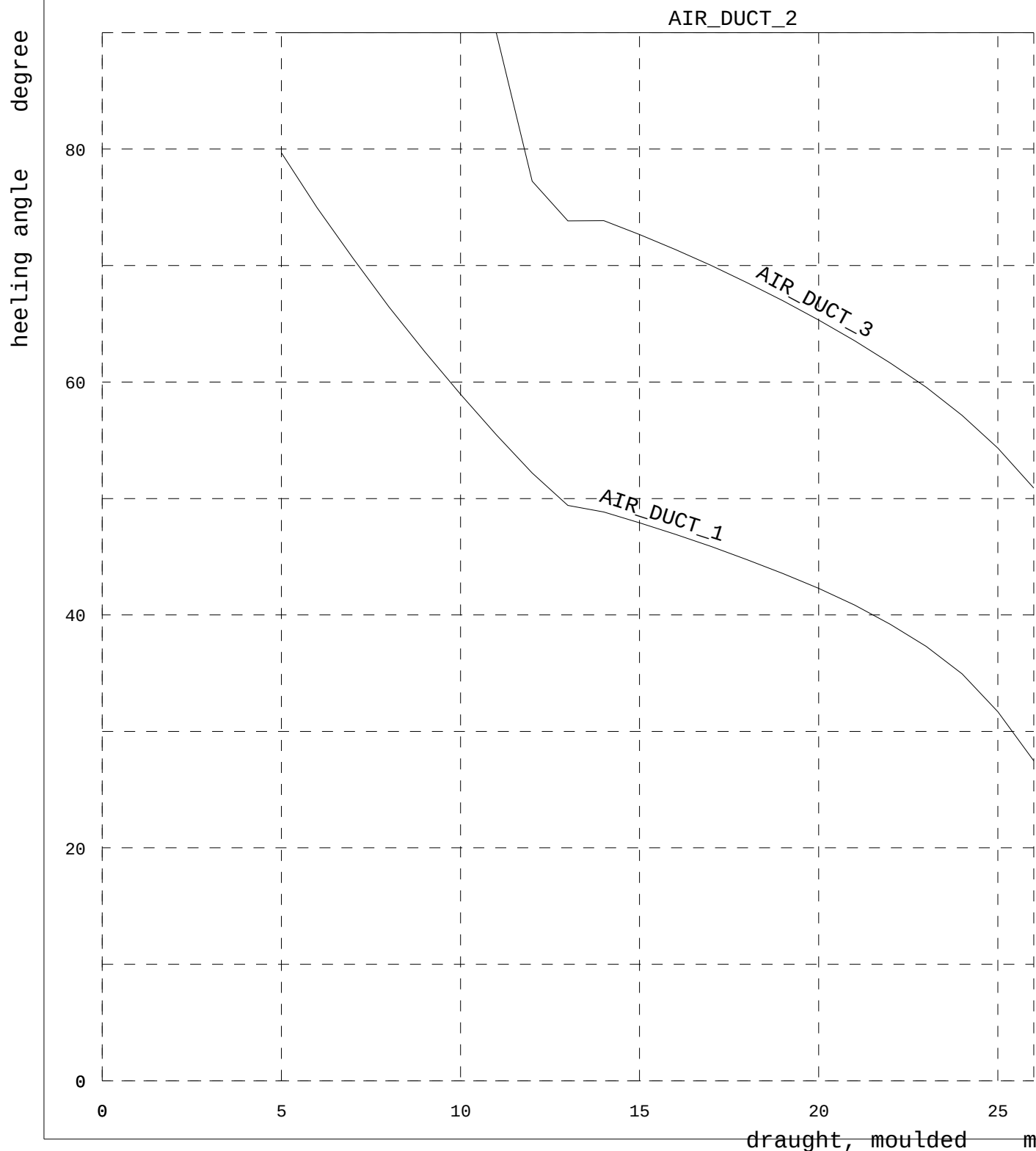
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DATE 2008-04-24

RHO=1.025

TIME 15.11

USER TH



OPENING IMMERSION

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P6047/M

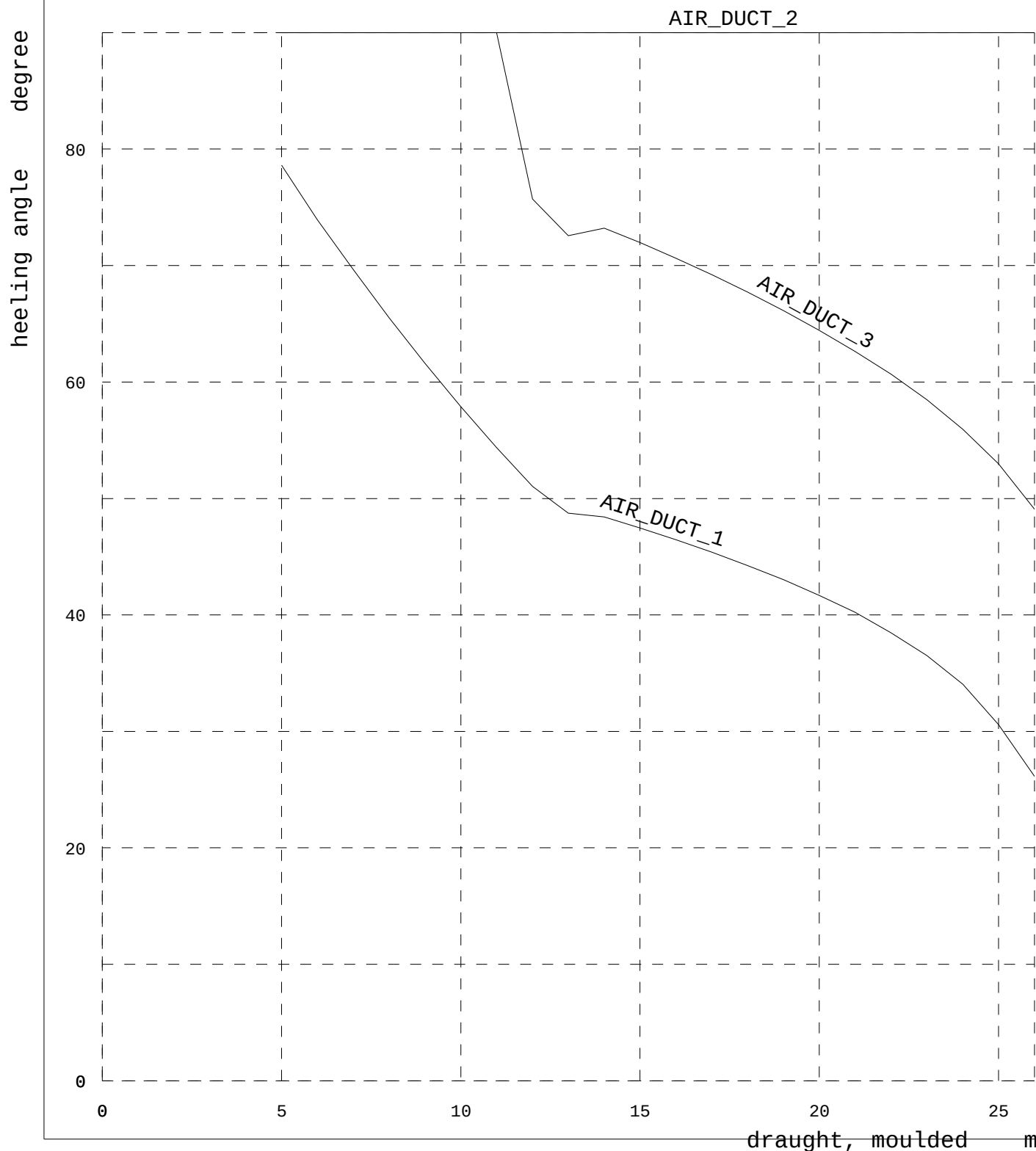
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.14

USER TH



OPENING IMMERSION

TRIM=+3.0m

P6047/M

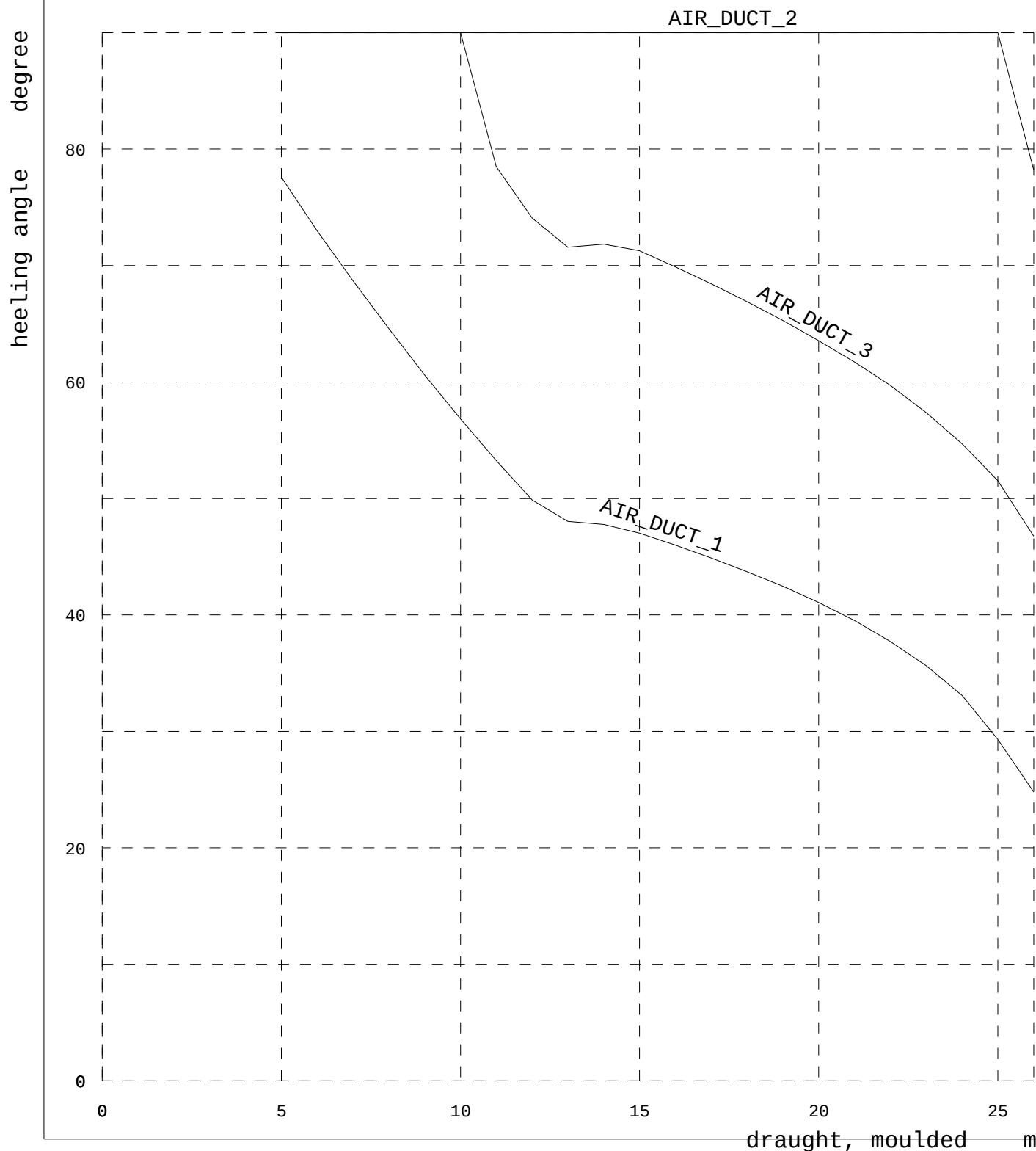
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.16

USER TH



OPENING IMMERSION

TRIM=+4.0m

P6047/M

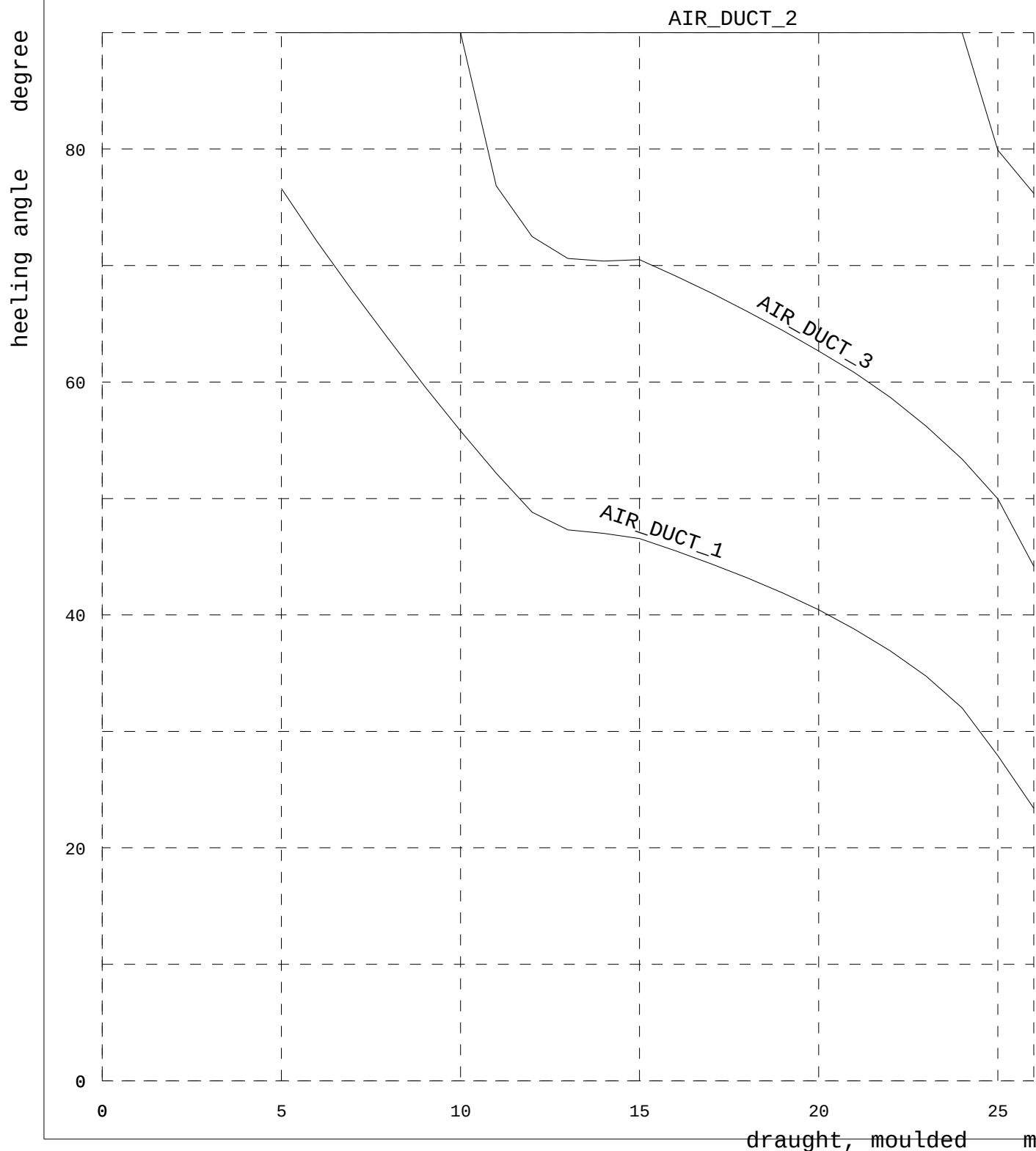
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.17

USER TH



OPENING IMMERSION

TRIM=+5.0m

P6047/M

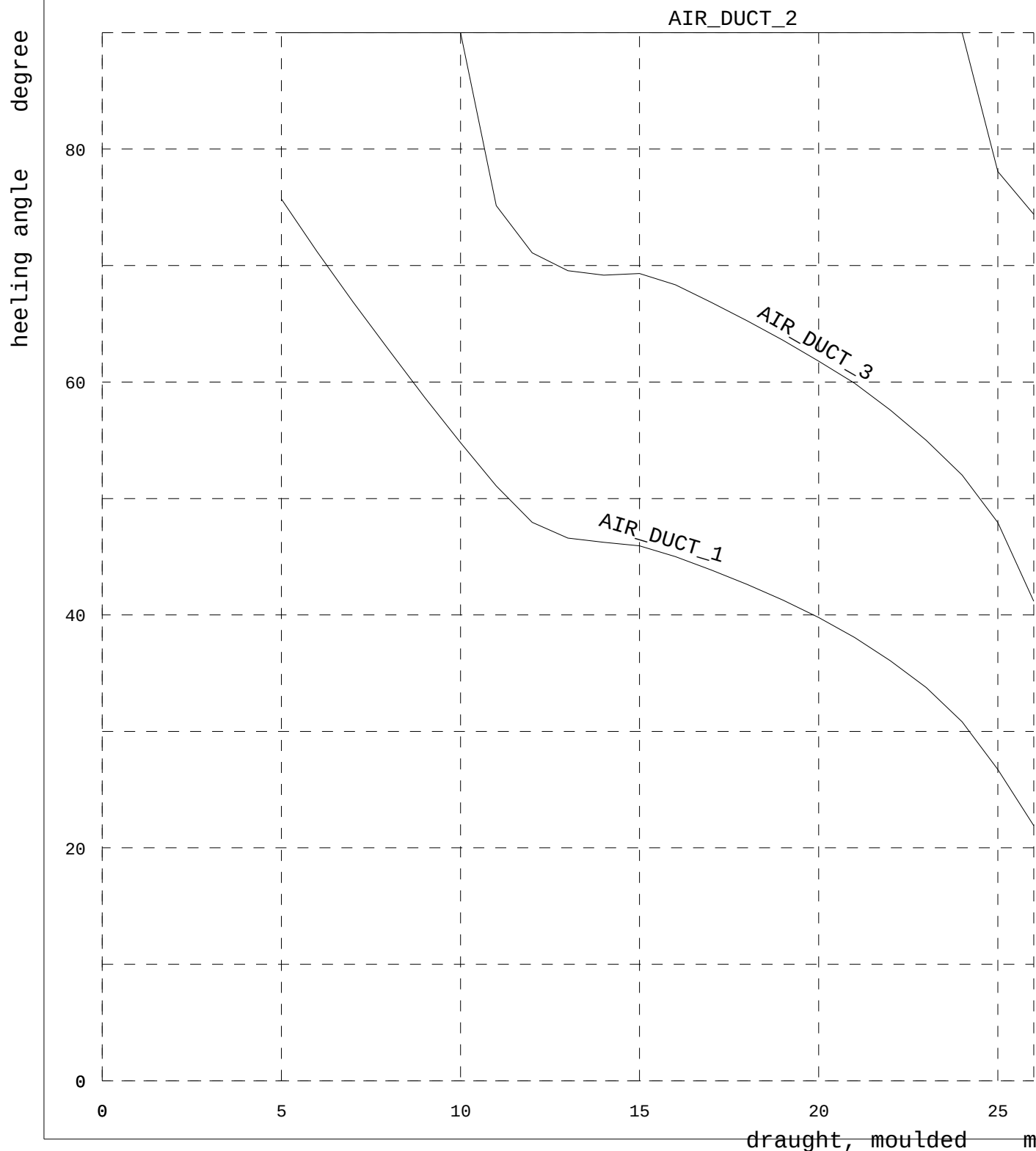
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.19

USER TH



OPENING IMMERSION

TRIM=+6.0m

P6047/M

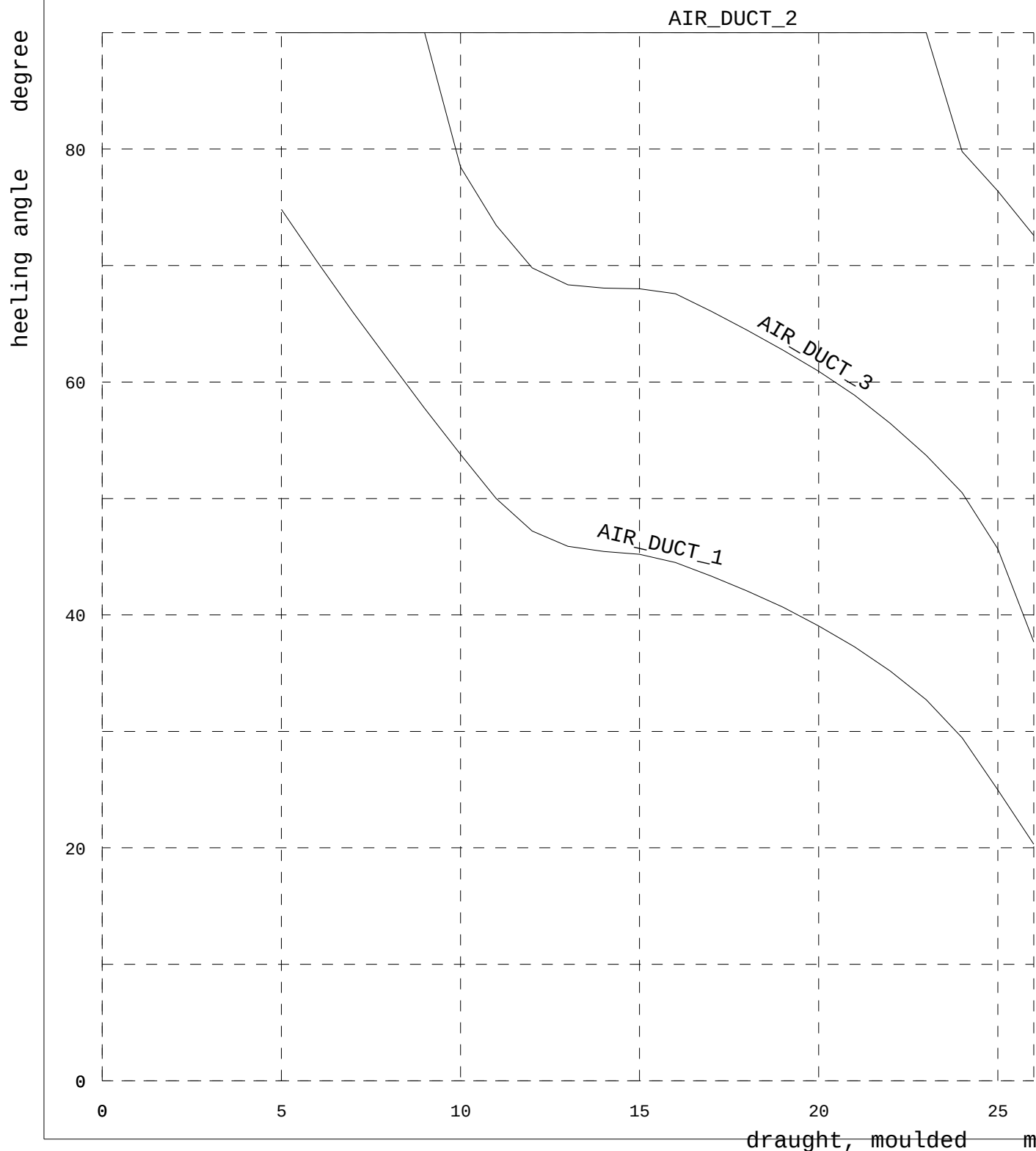
STABHULL

DATE 2008-04-24

RHO=1.025

TIME 15.20

USER TH

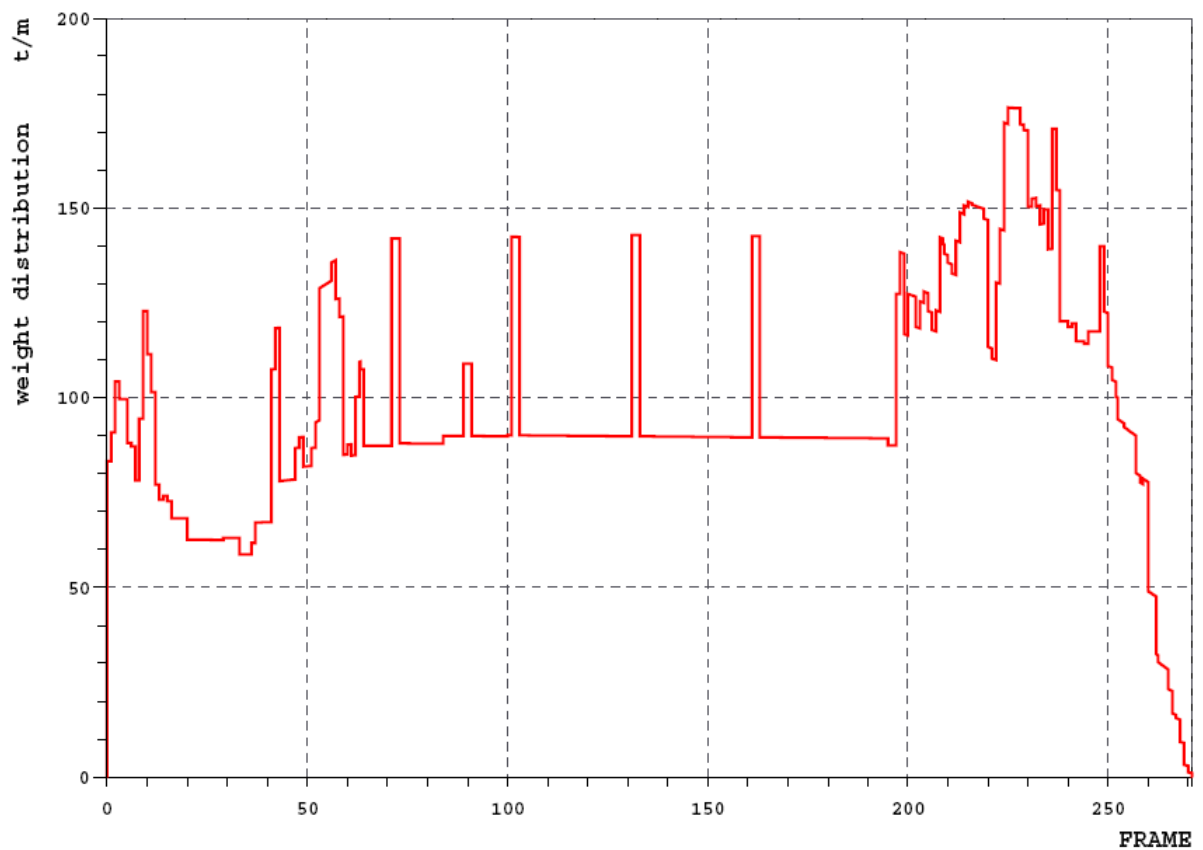


5.5 Lightship particulars

Estimation of lightweight and centre of gravity:

	Weight (t)	LCG (m)	TCG (m)	VCG (m)
Lightship	20871.4	113.44	-0.01	10.41

Lightship weight distribution is shown in figure below. Weight distribution in table format is shown in a table. Lightweight does not include fixed ballast.



Lightship weight distribution

Lightship weight distribution

FR #	WD t/m	FR #	WD t/m	FR #	WD t/m	FR #	WD t/m	FR #	WD t/m	FR #	WD t/m
0	84.80	50	81.56	100	90.12	150	89.68	200	127.36	250	120.94
1	92.47	51	84.78	101	142.36	151	89.67	201	124.19	251	109.36
2	105.89	52	91.50	102	142.35	152	89.66	202	123.81	252	105.72
3	105.87	53	126.91	103	142.34	153	89.65	203	122.61	253	95.42
4	101.19	54	127.55	104	90.08	154	89.64	204	125.12	254	94.69
5	101.18	55	128.19	105	90.08	155	89.63	205	124.74	255	93.01
6	89.63	56	133.64	106	90.07	156	89.63	206	119.67	256	92.18
7	88.68	57	134.28	107	90.06	157	89.62	207	120.18	257	91.46
8	96.03	58	124.12	108	90.05	158	89.61	208	139.43	258	80.92
9	124.35	59	119.41	109	90.04	159	89.60	209	139.05	259	78.49
10	124.34	60	87.61	110	90.03	160	89.59	210	136.25	260	77.78
11	111.34	61	87.71	111	90.02	161	142.58	211	133.87	261	48.30
12	101.33	62	100.19	112	90.01	162	142.57	212	140.05	262	47.59
13	76.87	63	109.44	113	90.01	163	142.57	213	147.46	263	29.89
14	73.95	64	107.49	114	90.00	164	89.55	214	149.40	264	29.11
15	73.94	65	87.36	115	89.99	165	89.55	215	150.30	265	28.34
16	72.56	66	87.35	116	89.98	166	89.54	216	149.92	266	22.63
17	68.05	67	87.34	117	89.97	167	89.53	217	149.20	267	16.39
18	68.04	68	87.33	118	89.96	168	89.52	218	148.82	268	15.16
19	68.04	69	87.32	119	89.95	169	89.51	219	148.44	269	8.99
20	68.03	70	87.31	120	89.94	170	89.50	220	145.53	270	2.97
21	62.39	71	141.93	121	89.93	171	89.49	221	111.70		
22	62.38	72	141.93	122	89.93	172	89.48	222	129.05		
23	62.38	73	141.92	123	89.92	173	89.48	223	143.11		
24	62.37	74	87.91	124	89.91	174	89.47	224	171.15		
25	62.36	75	87.90	125	89.90	175	89.46	225	175.17		
26	62.35	76	87.89	126	89.89	176	89.45	226	175.04		
27	62.34	77	87.88	127	89.88	177	89.44	227	174.99		
28	62.33	78	87.87	128	89.87	178	89.43	228	174.94		
29	62.84	79	87.86	129	89.86	179	89.42	229	170.66		
30	62.83	80	87.85	130	89.86	180	89.41	230	169.18		
31	62.82	81	87.85	131	142.85	181	89.40	231	151.16		
32	62.81	82	87.84	132	142.84	182	89.40	232	151.28		
33	62.81	83	87.83	133	142.83	183	89.39	233	149.35		
34	58.41	84	89.93	134	89.82	184	89.38	234	148.29		
35	58.40	85	89.92	135	89.81	185	89.37	235	148.29		
36	61.48	86	89.91	136	89.80	186	89.36	236	169.54		
37	66.86	87	89.90	137	89.79	187	89.35	237	169.59		
38	66.87	88	89.89	138	89.78	188	89.34	238	153.40		
39	66.88	89	109.01	139	89.78	189	89.33	239	118.86		
40	66.89	90	109.00	140	89.77	190	89.32	240	118.81		
41	107.21	91	108.99	141	89.76	191	89.32	241	118.34		
42	117.98	92	89.86	142	89.75	192	89.31	242	118.29		
43	118.06	93	89.85	143	89.74	193	89.30	243	113.61		
44	77.85	94	89.84	144	89.73	194	89.29	244	113.56		
45	77.95	95	89.83	145	89.72	195	89.28	245	116.17		
46	78.05	96	89.82	146	89.71	196	87.38	246	116.16		
47	86.39	97	89.81	147	89.70	197	127.38	247	116.15		
48	89.23	98	89.80	148	89.70	198	138.36	248	138.66		
49	89.33	99	89.79	149	89.69	199	137.97	249	138.67		

Appendix 1. Noble denton stability criteria

- Noble Denton Stability criteria
- KG-T limit curves including Noble Denton criteria

Noble Denton stability criteria

Noble Denton criteria is only for reference for captain in this manual.

Following Noble Denton are shown:

ND-RANGE

Intact range of stability, defined as the range between 0° degrees heel or trim and the angle at which the righting arm (GZ) becomes negative, shall not be less than 36 degrees.

Alternatively, if maximum amplitudes of motion for a specific towage or voyage can be derived from model tests or motion response calculations, the intact range of stability shall be not less than;

$$(20 + 0.8\theta)$$

where θ = the maximum amplitude of roll or pitch caused by the design sea state, plus the static wind heel or trim caused by the design wind, in degrees.

ND-WO

For the intact condition, the area under the righting moment curve, shall be not less than 40% in excess of the area under the wind overturning arm curve. The areas shall be bounded by 0° heel or trim, and the second intercept of the righting and wind overturning arm curves, or the down flooding angle, whichever is less.

ND-DWNFLD

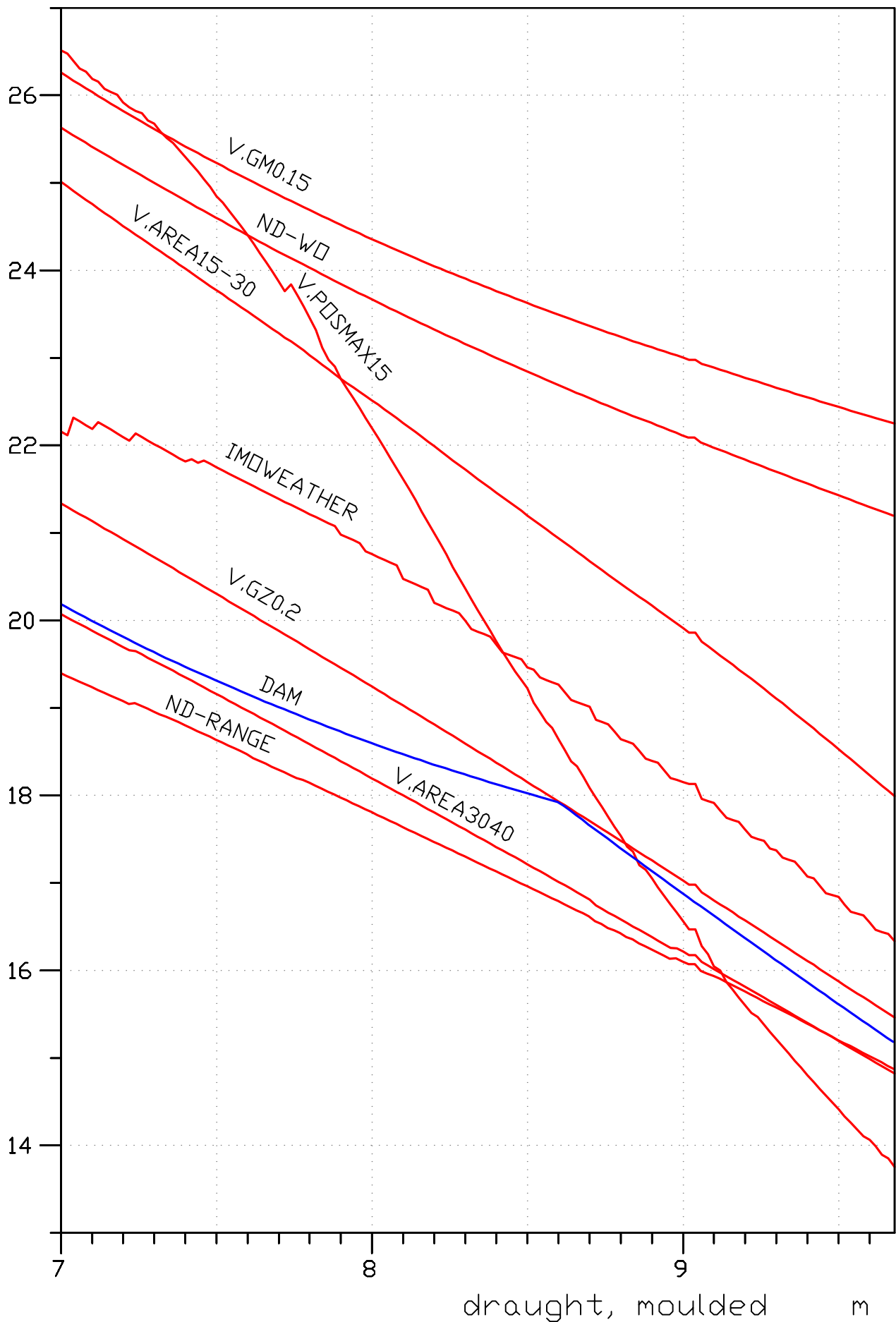
Any opening giving an angle of down flooding less than 20 degrees, or $(\theta + 5)$ degrees, where θ is the angle as defined in "Range of stability (ND-RANGE)", shall be closed and watertight, or protected by automatic closures in operable condition.

KG-T limit curves including Noble Denton criteria

KG-T limit curves for each criterion (including Noble Denton criteria) are shown in next pages. Limit curve from the damage stability calculations is also included (criterion dam). Limit curves are shown for trims -2m, -1m, 0m, 1m and 2m.

KG-T Summary, Trim -2
No effect of cargo buoyancy

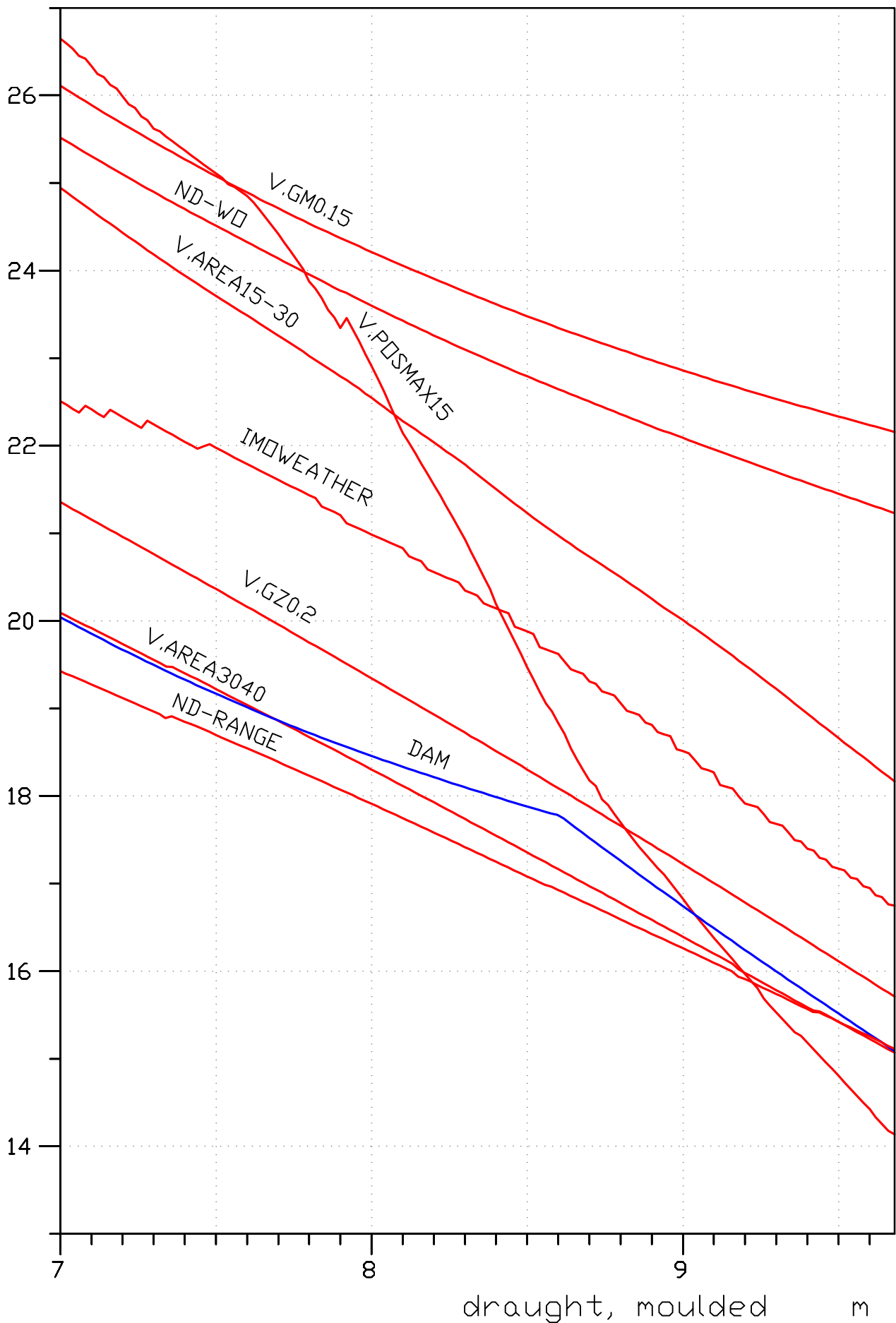
Proj	P60470
Date	2011-06-30
Time	9.55
Sign	MJA



KG-T Summary, Trim -1

No effect of cargo buoyancy

Proj	P60470
Date	2011-06-30
Time	9.53
Sign	MJA



KG-T Summary, Trim 0

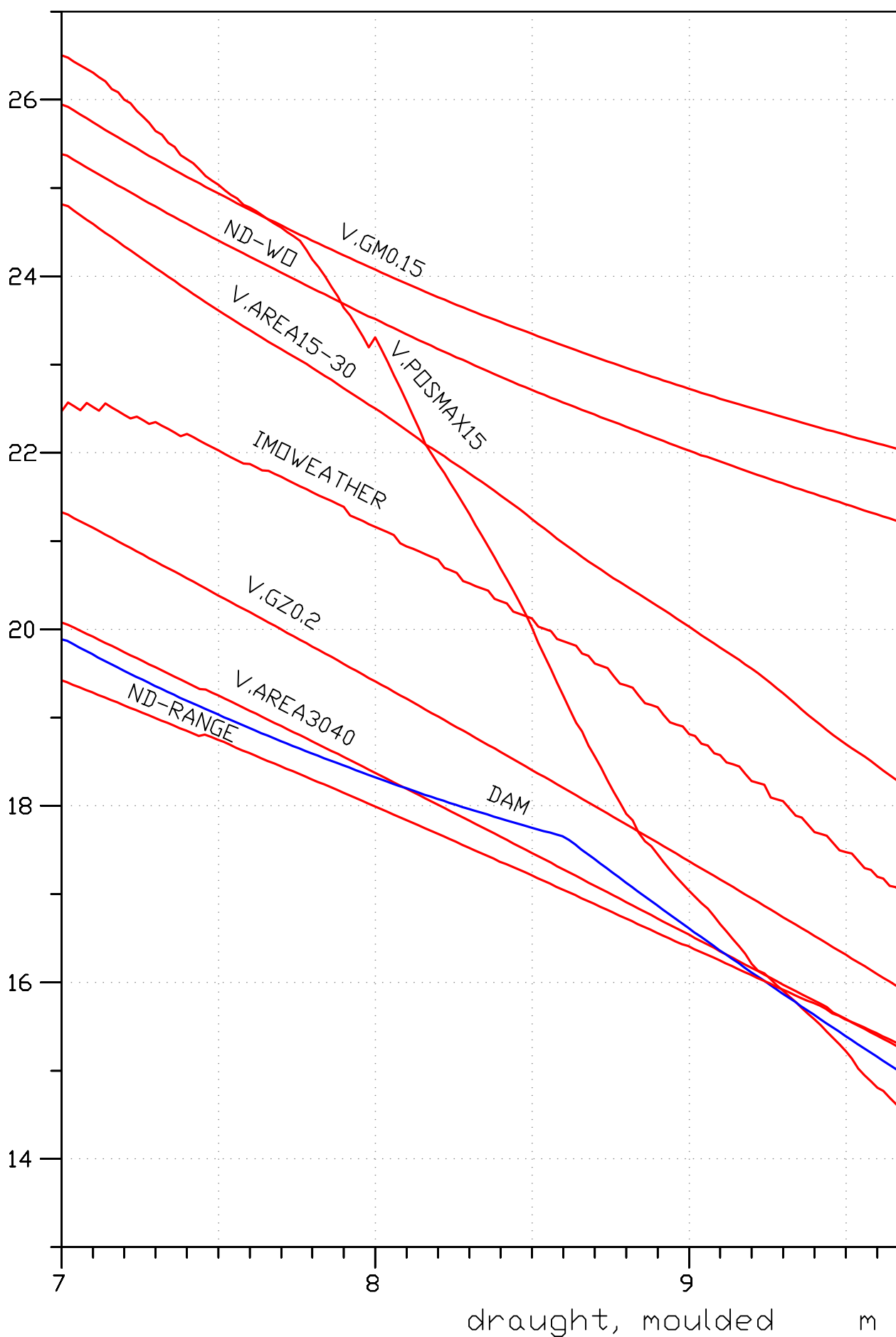
No effect of cargo buoyancy

Proj P60470

Date 2011-06-30

Time 9.48

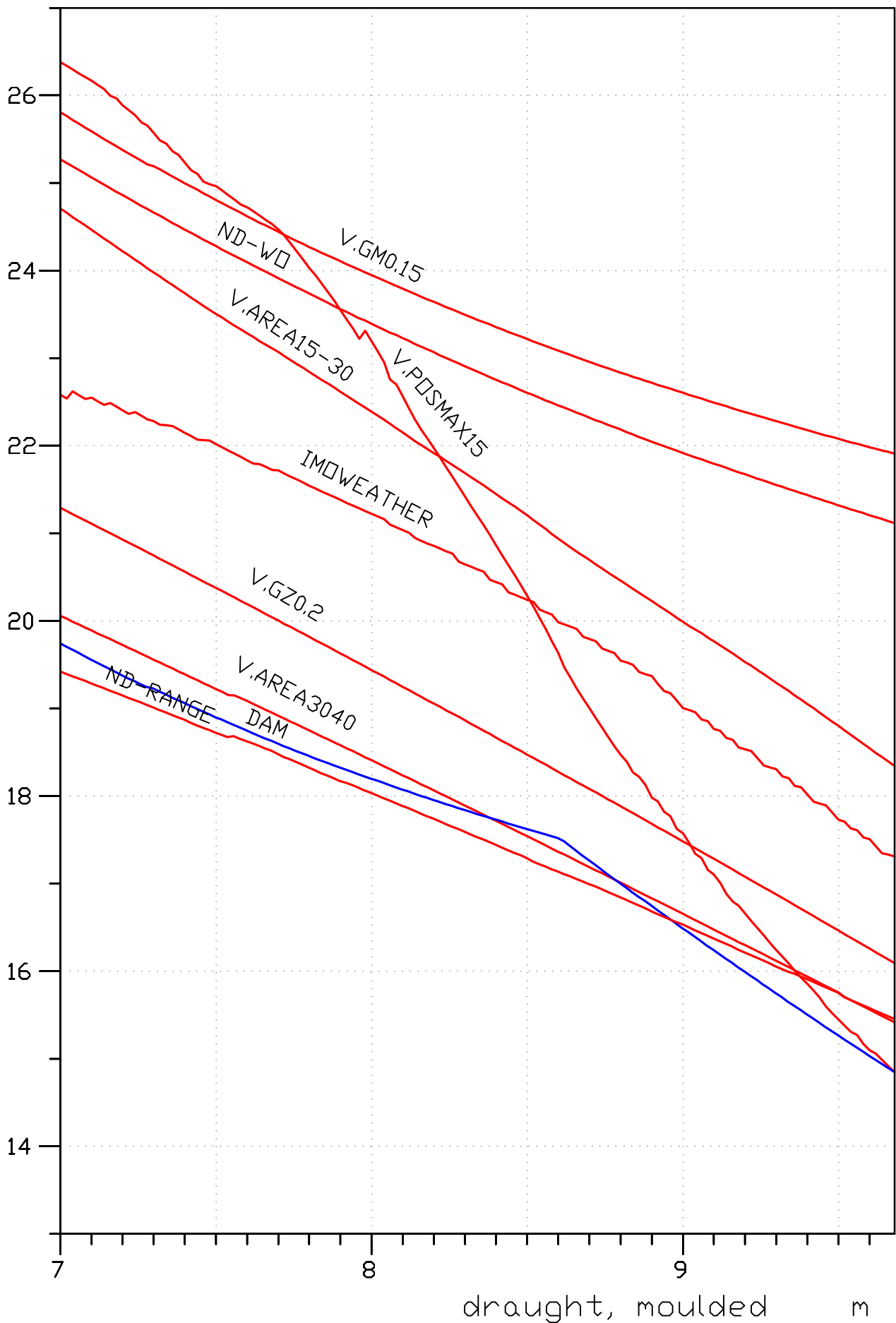
Sign MJA



KG-T Summary, Trim 1

No effect of cargo buoyancy

Proj	P60470
Date	2011-06-30
Time	9.51
Sign	MJA



KG-T Summary, Trim 2
No effect of cargo buoyancy

Proj	P60470
Date	2011-06-30
Time	9.55
Sign	MJA

